Organization of the Petroleum Exporting Countries

2026

World Oil Outlook 2050

20th edition

2026 World Oil Outlook 2050

Organization of the Petroleum Exporting Countries

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Foreword

It is vital that the world maintains a long-term focus on oil and energy outlooks. While short-term dynamics, such as geopolitical developments, often dominate the headlines, it is equally important to remain focused on the key trends driving our longer-term future.

This message is central to the 20th edition of the World Oil Outlook (WOO), which succeeds in building an all-encompassing picture that takes on board challenges and opportunities, as well as uncertainties and realities. The WOO 2026 data and analysis offer readers a veritable wealth of information, but overall, there are perhaps six key high-level takeaways.

First, many governments around the world continue to undergo a re-evaluation of energy policy frameworks, as populations continue to voice concern regarding energy security and its incompatibility with recent approaches, particularly those related to net-zero targets. Indeed, there is an increasing realization of the need to balance the elements of energy security, emissions reductions and sustainable development.

Second, global primary energy demand will continue to expand. From now until 2050, we see it increasing by 23%, with growth almost entirely coming from developing countries. This will be driven by economic growth, expanding populations, ever-increasing urbanization, new energy-intensive industries, the rapid growth of data centres, and the need to bring energy to the billions that still go without.

Third, all energies will be required to meet this demand. The future is not one in which the world can choose some energies while disregarding others. The scale of humanity's energy consumption means that we need to embrace all available energy sources.

Indeed, just as our energy history was one of additions – a fact that was particularly evident in 2025, when oil, gas, coal and renewables all reached record demand levels – our energy future will be too.

In this regard, policymakers everywhere must appreciate what each energy source offers to the world. For example, oil and its associated petroleum products are vital for almost every aspect of the global economy and our daily lives. This can be viewed in a series of videos the OPEC Secretariat has produced over the past year, all of which can be found on our website.

Against this backdrop, we see oil demand retaining the largest share in the energy mix and reaching 124 million barrels a day by 2050, with no peak in oil demand on the horizon.

Furthermore, the importance of an inclusive energy future is highlighted in the overall expansion of all energy sources, with the exception of coal. In fact, it is renewables that see the largest overall growth, with many OPEC Member Countries making significant investments in this sector.

Fourth, in another important area of investment for Member Countries, the WOO underscores the importance of continued efforts to reduce emissions. This includes technologies such as carbon capture, utilization and storage, direct air capture, as well as frameworks such as the circular carbon economy.

Fifth, and a point I have highlighted on many occasions, is the overall requirement for major investments in all energies and all technologies. This cannot be emphasized enough. For oil alone, we see the need for investment of $17.7 trillion from 2026 to 2050.

And sixth, the WOO is a key part of OPEC's voice on our evolving energy futures, offering comprehensive, transparent and objective views, and providing a platform for discussion with all stakeholders.

What these points highlight is that there is no credible way to address all the challenges and embrace all the opportunities before us without utilizing all available energy sources and all relevant technologies, and with energy market stability as a cornerstone for the huge investments required.

The launch of the WOO 2026 is another tremendous achievement for all those involved. The expertise, commitment and togetherness that have driven this publication, and which continue to push the boundaries of our research, are key defining elements of this Organization. I am deeply grateful to all of my colleagues for their contributions to this truly collaborative effort. It reflects teamwork at its best.

Haitham Al Ghais
Secretary General

Executive Summary

The global economic and energy landscapes remain dynamic, with uncertainty accompanied by emerging opportunities

The World Oil Outlook (WOO) 2026 sets out OPEC's long-term views on the evolving energy trends and establishes a comprehensive framework to assess the outlook for the global energy future. The objective of this Outlook is to provide a realistic, impartial and data-driven assessment of this future, one that reflects both the uncertainties and opportunities facing the economic and energy landscapes, and one that remains highly relevant for policymakers, industry and other stakeholders.

The recalibration of energy policy priorities continues

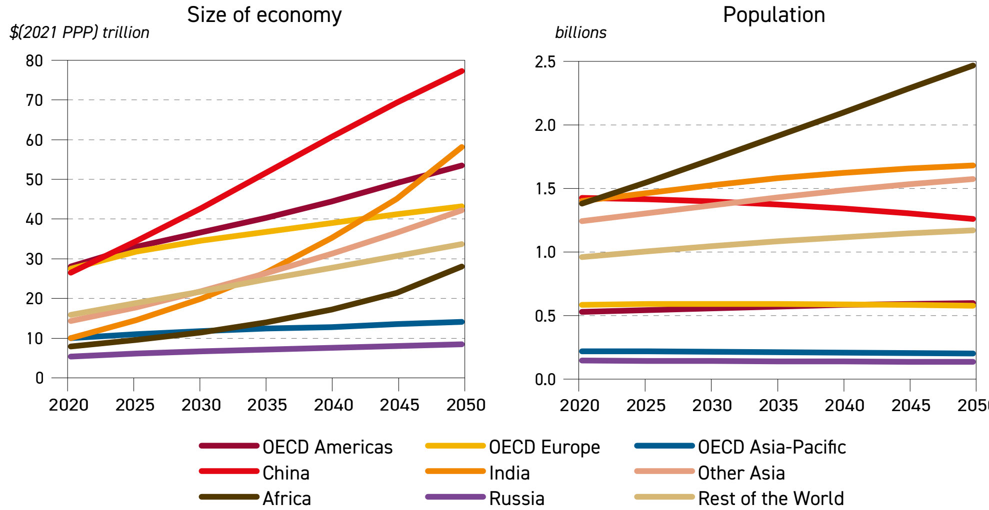
Energy policy frameworks across major economies are undergoing a broad recalibration as governments work to balance the elements of energy security, emissions reductions and sustainable development. Rather than discarding stated objectives, authorities are adjusting their delivery pathways to account for capital requirements and infrastructure constraints in modernizing energy systems. For example, advanced economies are increasingly reevaluating their approach to climate targets to protect industrial competitiveness and ensure baseline energy reliability, while major emerging economies are adopting dual strategies, increasing the deployment of renewables and maintaining conventional energy sources to meet rapidly expanding energy demand.

Demographics and economic growth drive future energy demand

Global demographic expansion and sustained economic growth remain the main engines driving future energy demand. The world's population is set to rise by 1.4 billion to nearly 9.7 billion by 2050, the working-age population is set to increase by 751 million to more than 6.1 billion and the urban population is expected to grow by 1.7 billion, largely in developing regions.

The global economy is projected to expand on average by 2.9% per annum (p.a.), more than doubling in size from $177 trillion in 2025 to $359 trillion in 2050 (on a 2021 PPP basis). Global average GDP per capita is set to increase from $21,500 to $37,100 (on a 2021 PPP basis) over the same period, with emerging markets and developing economies accounting for most of this expansion, as well as the associated energy demand.

Size of major economies and population trends, 2020–2050

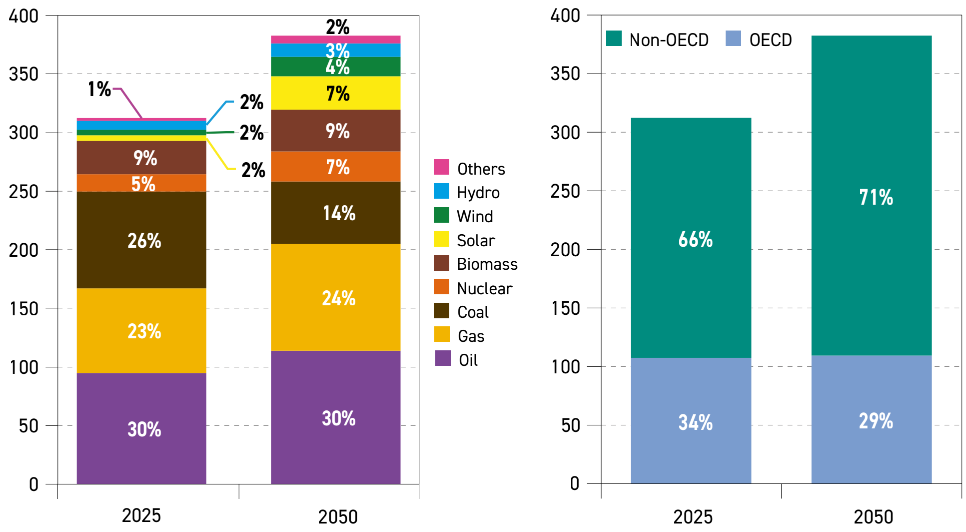


Source: OPEC.

Global primary energy demand is set to rise by 23% to 2050

Global primary energy demand is set to increase from around 312 million barrels of oil equivalent per day (mboe/d) in 2025 to nearly 383 mboe/d in 2050. This is an increase of around 23% over the outlook period, or 0.8% p.a. on average. The growth will come almost entirely from developing countries and regions, led by India, Other Asia, the Middle East, Africa and Latin America. At the same time, primary energy demand in developed countries is expected to increase only marginally, or in some cases decline.

Total primary energy demand by fuel and region, 2025 and 2050, mboe/d



Source: OPEC.

Demand for all energy sources increases, except for coal

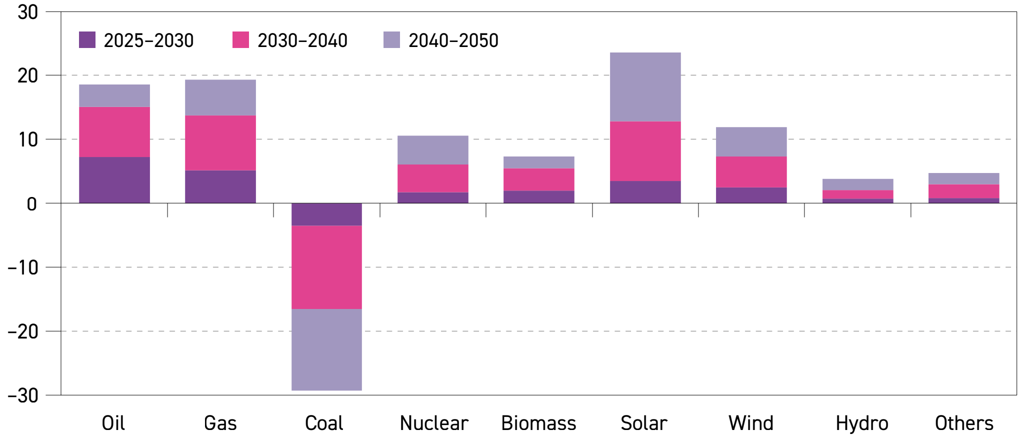
Demand for all primary fuels is set to increase in the period to 2050, except for coal. Demand for renewable energy, including biomass, solar, wind, hydro and other sources, is expected to record the strongest combined growth of 51.3 mboe/d. Solar and wind account for most of this expansion, driven mainly by declining generation costs and policy support. However, grid constraints and rising integration costs remain the key challenges.

Demand for oil and gas is also expected to increase strongly, in line with the need for reliable and affordable energy. Oil demand is expected to rise by 18.6 mboe/d to 2050, while natural gas rises by 19.3 mboe/d. After a long period of stagnation, nuclear energy is likely to see significant growth, rising by 10.5 mboe/d within the outlook period. At the same time, demand for coal is expected to drop by 29.3 mboe/d, due to unfavourable energy and climate policies and its substitution by other fuels, especially in the power sector.

Oil retains the largest energy mix share by 2050, while oil and gas combined are at around 54%

Oil is set to maintain the largest share in the energy mix over the entire forecast period. It is projected to be at just below 30% by 2050. The combined share of oil and gas is expected to be at around 54% at the end of the forecast period. At the same time, the share of renewables in the energy mix increases to around 26% in 2050, up by ten percentage points (pp) from around 15% seen in 2025.

Growth in primary energy demand by fuel, 2025–2050, mboe/d

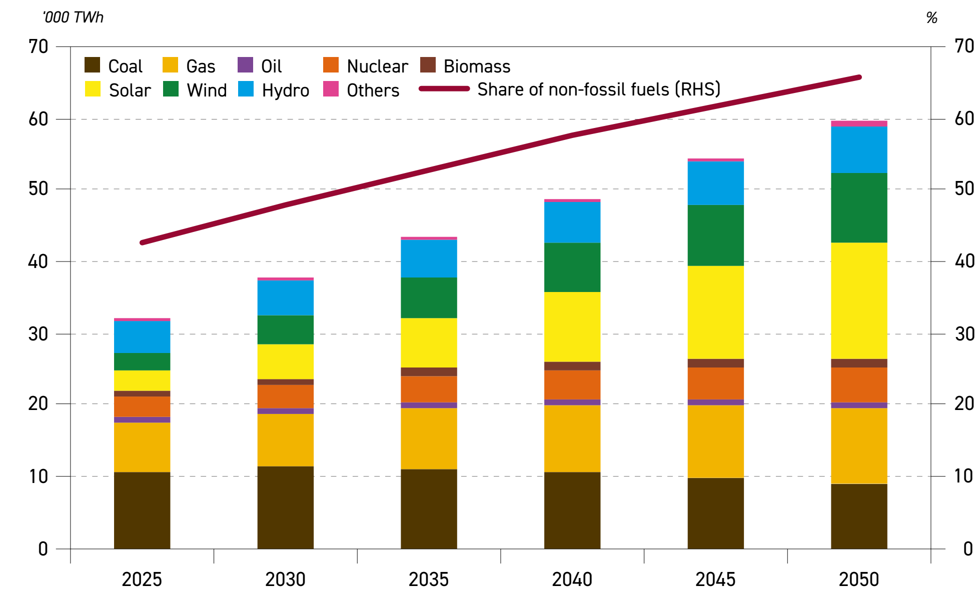


Source: OPEC.

Electricity generation set to rise by over 85%, supported by expansion in developing countries

Total electricity generation is expected to increase from around 32,000 terawatt hours (TWh) in 2025 to around 59,500 TWh in 2050. This expansion is supported by demand growth in the residential/commercial sector, industry, transport and data centres. Around 75% of this growth is anticipated to come from developing countries, with almost 60% alone from developing countries in Asia. By far the largest increase in the generation mix is projected for wind and solar, expanding from around 5,400 TWh in 2025 to around 26,000 TWh in 2050.

Global electricity generation by fuel, 2025–2050



Source: OPEC.

Global oil demand projected to see robust growth, reaching 124 mb/d by 2050

Supported by recent policy shifts and steady economic growth, global oil demand is set for continued robust growth in the medium and long term. Demand is set to reach 113.3 million barrels per day (mb/d) by 2030 and 124.1 mb/d in 2050. This translates into demand growth of 19 mb/d over the entire outlook period. The primary driver for this is non-OECD countries, where demand increases by 7.4 mb/d between 2025 and 2030, and 26.9 mb/d between 2025 and 2050. OECD demand is expected to increase by around 0.7 mb/d in the period to 2030 but decline in the long term. The OECD decline is almost 8 mb/d between 2025 and 2050.

Long-term oil demand by region, mb/d

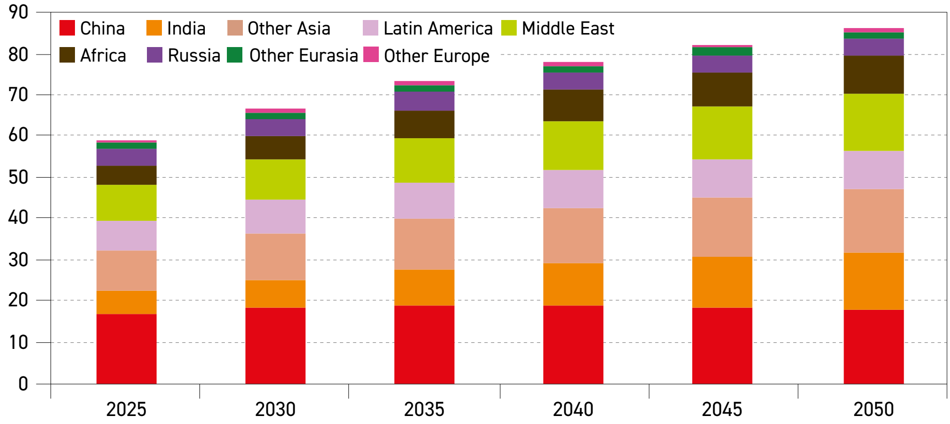
	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	25.4	25.9	25.8	24.9	23.6	22.4	–3.0
OECD Europe	13.4	13.7	13.3	12.3	11.2	10.2	–3.3
OECD Asia-Pacific	7.1	7.1	6.8	6.4	5.9	5.5	–1.6
OECD	**45.9**	**46.7**	**45.9**	**43.6**	**40.8**	**38.0**	**–7.9**
China	16.9	18.1	18.9	18.8	18.4	18.0	1.1
India	5.6	7.1	8.8	10.4	12.1	13.8	8.1
Other Asia	9.8	11.3	12.5	13.4	14.3	15.1	5.3
Latin America	6.9	7.8	8.6	9.2	9.5	9.7	2.8
Middle East	8.9	9.9	11.0	12.0	12.9	13.6	4.7
Africa	4.9	5.7	6.5	7.4	8.3	9.2	4.3
Russia	4.0	4.3	4.4	4.3	4.3	4.2	0.2
Other Eurasia	1.3	1.5	1.6	1.6	1.7	1.7	0.4
Other Europe	0.8	0.9	1.0	1.0	0.9	0.9	0.1
Non-OECD	**59.2**	**66.6**	**73.1**	**78.1**	**82.3**	**86.1**	**26.9**
World	**105.1**	**113.3**	**119.0**	**121.7**	**123.1**	**124.1**	**19.0**

Source: OPEC.

India leads incremental long-term oil demand growth

The primary sources of long-term oil demand growth are India, Other Asia, the Middle East, Africa and Latin America. Demand in these regions is set to increase by 25.2 mb/d between 2025 and 2050, with India alone adding 8.1 mb/d. At the same time, oil demand in Other Asia, the Middle East and Africa is set to increase by 5.3 mb/d, 4.7 mb/d and 4.3 mb/d, respectively. Oil demand in Latin America is projected to increase by 2.8 mb/d and China by 1.1 mb/d over the same period.

Non-OECD oil demand by region, 2025–2050, mb/d



Source: OPEC.

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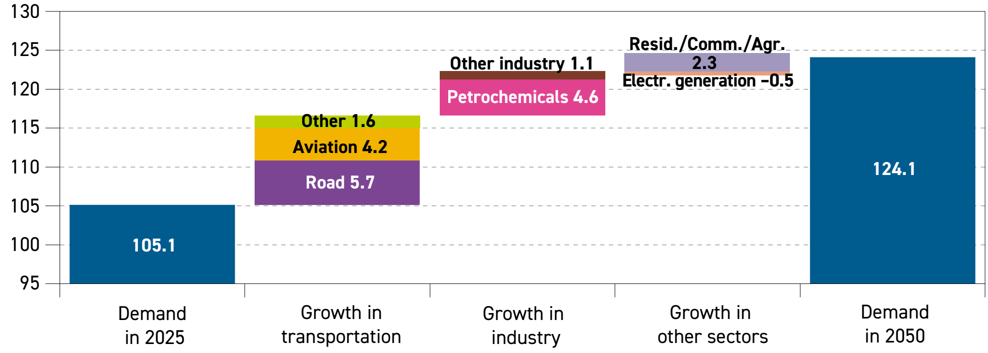
Road transport, petrochemicals and aviation are central to future oil demand growth

From a sectoral perspective, oil demand growth out to 2050 is set to be driven by road transportation, aviation and petrochemicals, with increases of 5.7 mb/d, 4.2 mb/d and 4.6 mb/d, respectively. A modest demand increase is also expected in other sectors, such as 'other transportation', 'other industry' and residential/commercial/agricultural. The only sector where demand for oil is expected to see a minor decline over the long term is electricity generation.

Global oil demand growth by sector, 2025–2050, mb/d

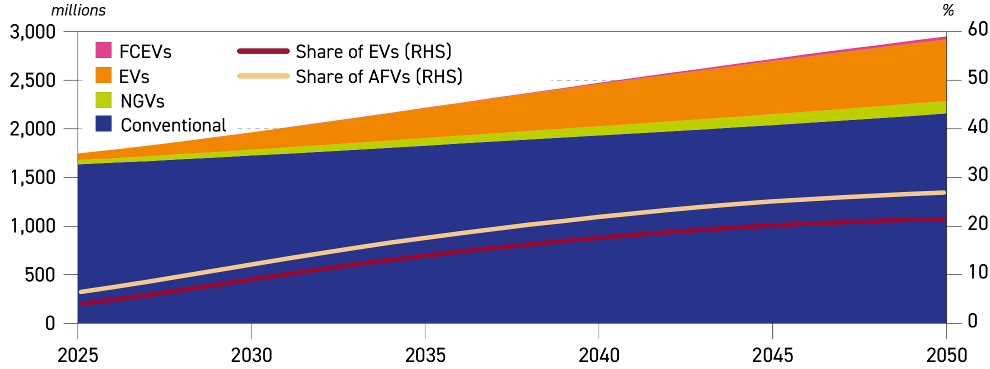


Source: OPEC.

Global vehicle fleet to reach almost three billion by 2050

Demand growth in road transportation comes on the back of a large expansion in the global vehicle fleet, particularly in developing countries. The global vehicle fleet is expected to increase from 1.75 billion in 2025 to around three billion in 2050, with the fastest growth expected in the segment of electric vehicles (EVs). Nevertheless, internal combustion engine (ICE)-based vehicles are set to continue to dominate the global fleet. They are expected to account for around 73% of the total in 2050.

Global fleet composition, 2025–2050*



*Conventional vehicles include gasoline, diesel, LPG and HEVs.
Source: OPEC.

Majority of future demand growth is for middle distillates and light products

Largely reflecting sectoral oil demand trends, light refined products and middle distillates are expected to drive most of the future increase, while heavy products are set to witness only modest changes due to regulatory constraints and ongoing oil substitution by alternative energy sources. As a result, major long-term demand growth is expected for jet/kerosene (4.2 mb/d), gasoil/diesel (3.8 mb/d), liquefied petroleum gas (LPG)/ethane (3.5 mb/d), naphtha (3.2 mb/d) and gasoline (2.4 mb/d).

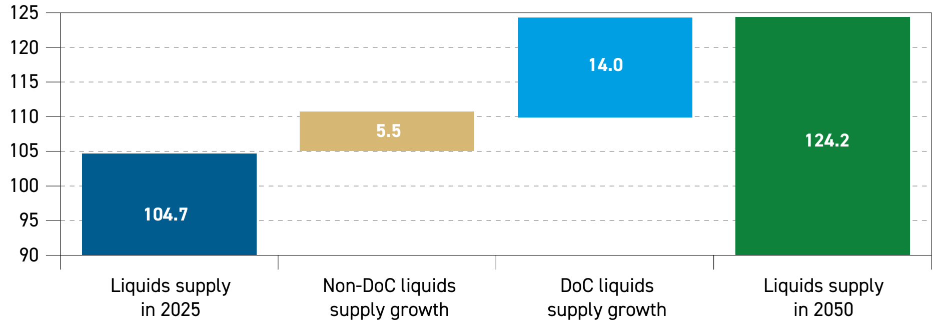
Non-DoC liquids supply growth to slow, with US tight crude likely to have peaked

In the medium term to 2030, non-Declaration of Cooperation (DoC) liquids supply is projected to grow by about 4.1 mb/d to 58.2 mb/d, meeting around half the expected oil demand increase. Supply growth is primarily driven by Brazil, Qatar, the US, Argentina and Canada, as well as a number of newer producers in non-DoC Africa. The latest analysis indicates that US tight crude supply likely peaked in 2025, at just over 9 mb/d, resulting in only modest total US liquids supply growth of 0.4 mb/d until 2030, before a plateau thereafter.

DoC liquids supply market share to increase after non-DoC supply peaks in the 2030s

By the 2030s, non-DoC supply is projected to peak and experience a long plateau at around 60 mb/d. Canada and Argentina stand out among non-DoC producers as countries where supply is forecast to continue to increase even in the long term. By 2050, non-DoC liquids supply is projected to average 59.6 mb/d, up by 5.5 mb/d from 2025.

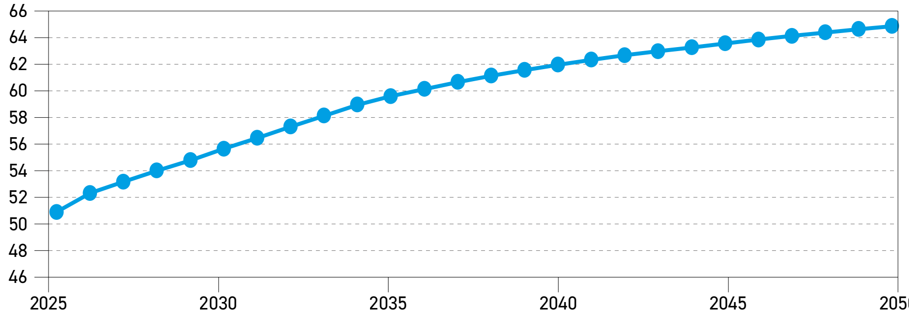
Composition of global liquids supply growth, 2025–2050, mb/d



Source: OPEC.

With oil demand projected to grow to 2050, the relative role of DoC liquids supply rises significantly, increasing from 50.6 mb/d in 2025 to 64.5 mb/d by 2050. This implies an increase in market share from 48% in 2025 to 52% in 2050.

DoC total liquids supply, mb/d

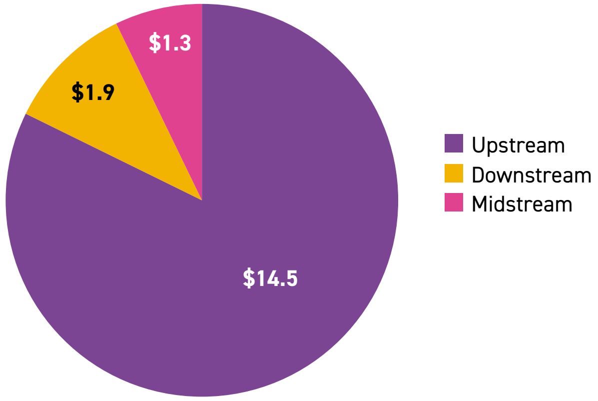


Source: OPEC.

Oil investment of $17.7 trillion required to meet demand in the long term

Total cumulative oil investments to meet demand requirements add up to $17.7 trillion (US$2026) for the period 2026–2050. For the upstream, investment requirements are assessed at $14.5 trillion, adjusted upwards slightly on a p.a. basis from the 2025 Outlook due to higher projected long-term supply and a shift towards higher-cost resources. Downstream needs are projected at $1.9 trillion, and midstream at $1.3 trillion. This means that average total p.a. investment requirements are now seen at over $700 billion.

Cumulative oil-related investment requirements by segment, 2026–2050, $(2026) trillion

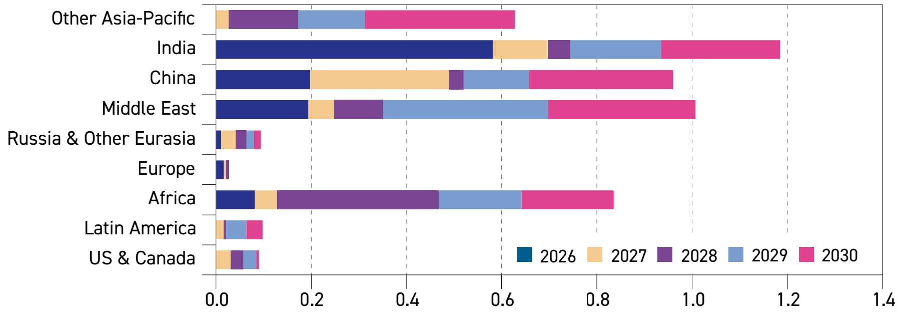


Source: OPEC.

Asia-Pacific, Middle East and Africa lead medium-term refinery capacity additions

The medium-term outlook sees around 4.9 mb/d of capacity additions coming online. The bulk of this growth is set to be commissioned in the Asia-Pacific (2.8 mb/d), the Middle East (1 mb/d) and Africa (0.8 mb/d). This represents around 94% of the global additions to 2030. The global annual average rate of capacity additions for the period 2026–2030 is estimated at 1 mb/d.

Distillation capacity additions from existing projects, 2026–2030, mb/d

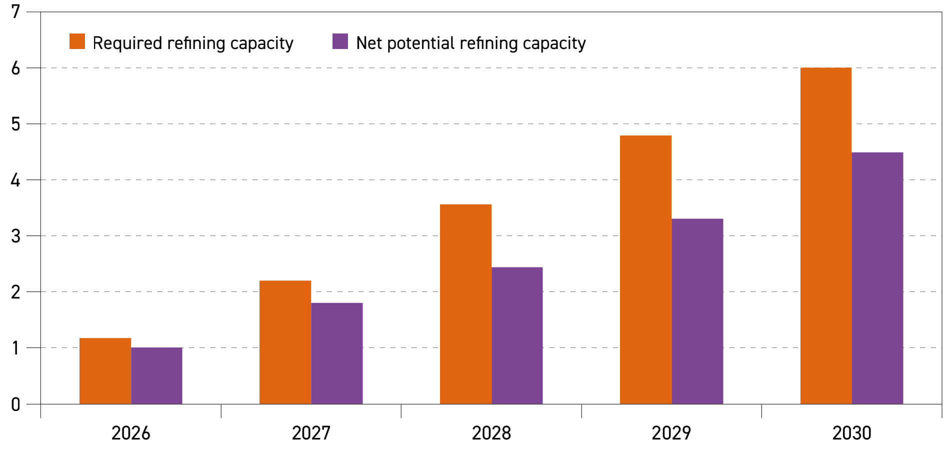


Source: OPEC.

Downstream market in the medium term projected to gradually tighten

At the global level, the projected evolution of net incremental refining capacity and required refining capacity shows a tightening market throughout the medium term, relative to 2025. The tightening market is set to result in an increase in global utilization rates, rising from 80.8% in 2025 to around 82.7% by 2030.

Additional global cumulative refinery crude runs, net potential and required**, mb/d*



* *Net potential: based on expected distillation capacity additions and closures.*

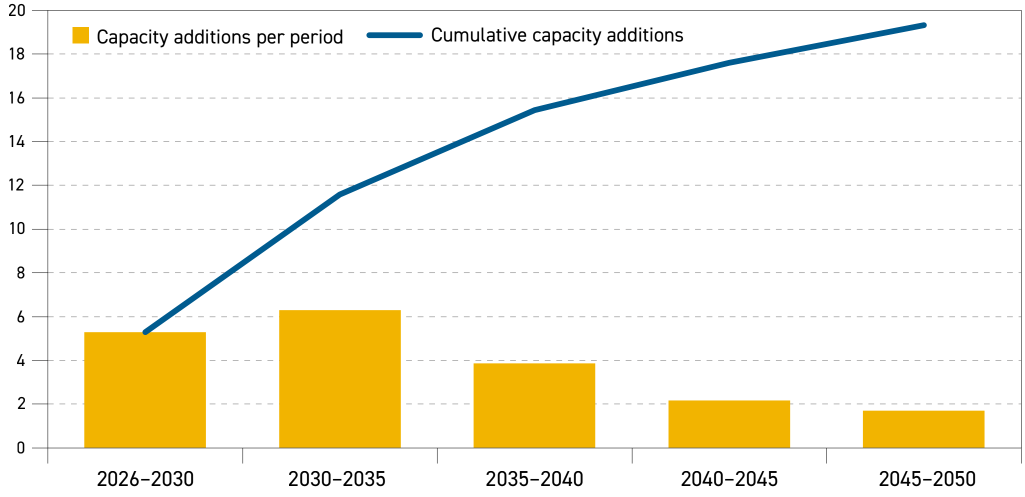
** *Required: based on projected demand increases, assuming no change in refined products trade pattern.*

Source: OPEC.

Around 80% of long-term refining capacity additions are set to materialize before 2040

Global refining additions between 2026 and 2050 are projected at 19.3 mb/d. Capacity additions between 2026 and 2030 are estimated at 5.3 mb/d (including creep capacity). Refining capacity is set to increase by 6.3 mb/d in the period 2030–2035, which is supported by rising demand in most developing regions. The additions decline thereafter and are estimated at around 3.9 mb/d in the 2035–2040 period, followed by 2.2 mb/d in the 2040–2045 period and 1.7 mb/d in the 2045–2050 period. This mirrors the slowdown in demand growth towards the end of the outlook.

Long-term distillation capacity additions by period and cumulative, 2026–2050, mb/d



Source: OPEC.

Global oil trade set to rise by almost 25% by 2050

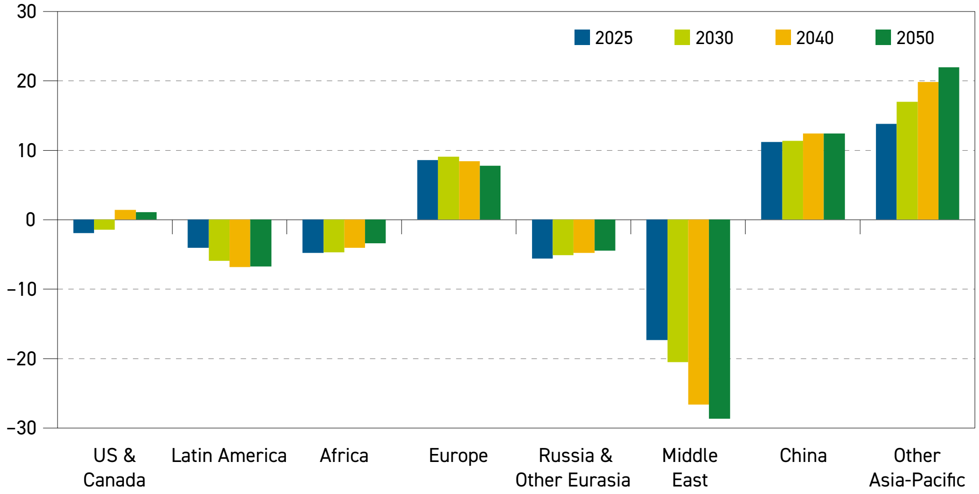
In line with the projected increase in oil demand, the interregional exports of crude, condensate and products is anticipated to rise, as regions balance their demand needs with the availability of supply. In 2030, global oil trade is projected to reach almost 62 mb/d, up from 55 mb/d in 2025. In the following decade, further growth is expected to drive trade volumes to just under 68 mb/d by 2040. Between 2040 and 2050, growth is set to slow, a reflection of slowing oil demand growth and expanding refinery capacity in developing countries that is anticipated to reduce volumes available for export. By 2050, global crude, condensate and product exports are expected to total around 69 mb/d.

The largest share of global oil movements, at 67%, was dedicated to crude and condensate trade in 2025. It is anticipated that this figure will expand to over 69% by 2050.

Middle East and Asia-Pacific remain central to crude and condensate trade

Net importers can be clearly identified as Europe, China and Other Asia-Pacific. The latter was the largest net importer in 2025 at 13.8 mb/d, with volumes projected to grow significantly over the outlook period and reach close to 22 mb/d by 2050. China's net imports were estimated at 11.2 mb/d in 2025, and while the long-term trend is not as robust as Other Asia-Pacific, its net imports are expected to rise to 12.4 mb/d in 2050. Taken as a whole, the Asia-Pacific region is by far the largest importer of crude and condensate, both today and through to mid-century.

Regional net crude and condensate imports by region, mb/d



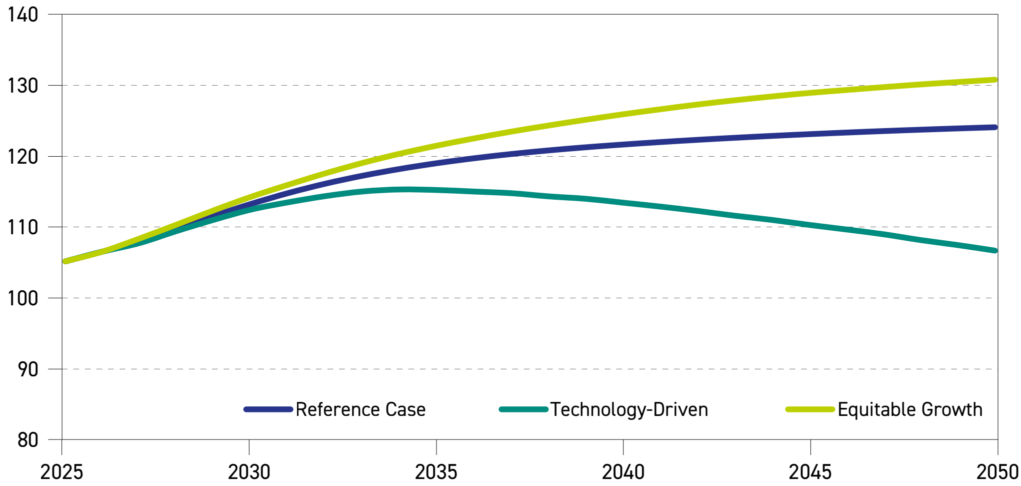
Source: OPEC.

Latin America, Africa, Russia & Other Eurasia and the Middle East were all net exporters in 2025 and are set to remain so through to 2050, albeit with different export trajectories. The Middle East was the largest net exporter in 2025 at 17.4 mb/d and its role as an exporter is projected to expand extensively to reach 28.7 mb/d in 2050. Notably, the US & Canada region is expected to switch from a minor net exporter to a minor net importer over the long term.

Scenarios highlight the large range of uncertainty facing the global energy system

Two alternative scenarios to the WOO Reference Case, namely the 'Technology-Driven Scenario' (TDS) and the 'Equitable Growth Scenario' (EGS), are useful to explore the range of uncertainty facing the global energy system. Compared to the Reference Case, the TDS assumes faster investments in advanced energy technologies. These investments result in significant fuel substitution and energy efficiency improvements, leading to lower primary energy demand and a changed energy mix. Global oil demand in this scenario gradually departs from the Reference Case and starts declining after 2035. Nonetheless, it remains in the range of 110 mb/d to 115 mb/d for most of the forecast period, while declining slightly to below 107 mb/d by 2050. This is around 17 mb/d lower *vis-à-vis* oil demand in the Reference Case in 2050.

Global liquids demand in the Reference Case and in alternative scenarios, 2025–2050, mb/d



Source: OPEC.

Compared to the Reference Case, the key component of the EGS is moderately faster economic growth in developing countries, coupled with efforts to improve electricity access and further eradicate energy poverty. This scenario results in higher long-term energy demand, in general, and oil, in particular. Oil demand in this scenario reaches 121 mb/d by 2035 and continues growing to 131 mb/d in 2050. This is almost 7 mb/d higher than in the Reference Case by 2050. The largest potential for higher oil demand in the EGS is in the industrial, residential and road transportation sectors, each in the range of up to 2 mb/d by 2050.

Introduction

This year's World Oil Outlook (WOO) reflects a long-term perspective for energy and oil markets that is not much different from the one we saw one year ago. To put it simply, the Organization retains its long-held view that all energy sources will be required to meet the world's needs for the foreseeable future.

The year 2025 witnessed strong demand growth for all major sources of energy. Oil, gas and coal demand reached new record-high levels, continuing to provide 80% of the world's energy needs. Counter to views heard from certain quarters, they remain vital to populations around the world today and will continue to be so in the future.

Renewable energy sources grew strongly too, a trend we continue to monitor closely. Indeed, this year's WOO contains a more detailed breakdown and analysis of renewables, reflecting a recognition of the significant expansion of intermittent renewable energy sources, such as solar and wind. Besides this, the Outlook also identifies a renewed interest in nuclear energy.

These trends, coupled with persistent demand for 'baseload' electricity generated from other sources of energy including natural gas and coal, reflect the continued strong growth in electricity demand, not least due to Artificial Intelligence and data centres.

This Outlook also identifies and analyses ongoing shifts in energy-related policy. In many parts of the world, a continued and persistent refocusing on the significance of energy security and affordability – aligned with sustainability – can be observed, compared to a previously-seen more singular focus on the latter.

At last year's COP30 in Brazil, a reorientation towards a more pragmatic and realistic approach to the energy trilemma was also noted, after many submissions of renewed nationally determined contributions (NDCs) were delayed. This was against a general backdrop of an arguably widening diversity of opinions and approaches.

Many energy companies too, including some of the world's largest oil and gas producers, are re-orienting themselves towards a focus on oil and gas, retreating somewhat from previous remakes as 'energy solution providers'. As a result, these companies' investments in renewables are being scaled back, while making more ambitious investments on oil and gas investments.

More generally, some other warning signals have been flashing, heralding a rethink towards more realistic solutions to the world's energy challenges. A widespread realization is sinking in that grassroots support and funding are lacking for overly ambitious subsidies for energy transitions. Moreover, there have been repeated instances of temporary energy shortages and price spikes related to an over-reliance on still.

The WOO remains an outlook focused on the medium and long term and the specific challenges that entails. Current issues related to geopolitical tensions and market volatility are seen as short-term challenges, and hence, are addressed in other OPEC publications.

The WOO thus remains a useful tool to understand future energy pathways. It underscores OPEC's commitment to transparently laying out the assumptions, logic and thinking behind the projections. Underpinned by solid quantitative and detailed analysis, as well as alternative scenarios, it serves as a sound basis for dialogue and discussions.

Key assumptions

Key takeaways

- The global population is projected to grow by over 1.4 billion to 2050, increasing from around 8.2 billion in 2025 to almost 9.7 billion. Almost all of this growth will be in non-OECD countries.
- The working-age population is expected to increase by 751 million and surpass 6.1 billion by 2050.
- Around 6.5 billion people are expected to reside in urban environments, an expansion of 1.7 billion between 2025 and 2050.
- The global economy is expected to remain resilient, with gross domestic product (GDP) growth between 2025 and 2050 anticipated to increase at an average rate of 2.9% p.a. Non-OECD economies are expected to see stronger growth, averaging 3.6% p.a., while the OECD is projected to expand at a steady 1.5% p.a.
- In absolute terms, the global economy is set to more than double in size, rising from $177 trillion in 2025 to $359 trillion in 2050 (2021 PPP).
- Global average GDP per capita is expected to increase from $21,500 in 2025 to $37,100 in 2050 (2021 PPP).
- Global energy policies are increasingly shaped by the interplay of energy security, economic development and climate commitments. While multilateral climate cooperation remains active, progress has been incremental. Across major economies, national priorities are driving a range of approaches, from accelerated hydrocarbons development and expanded export capacity to the rapid scaling of renewables.
- Technological trends will continue to shape future energy sector expectations. However, this Outlook assumes incremental improvements in efficiency, performance and costs, and excludes any sudden technological breakthroughs.
- Across the transportation sector, efficiency gains in ICEs and hybrid powertrains continue to extend fleet life and lower emissions in road transport, while alternative fuels and propulsion systems, including sustainable aviation fuels, liquefied natural gas (LNG), methanol, ammonia and hydrogen, face varying degrees of scalability and commercialization challenges in the aviation and maritime segments.

The Reference Case in this Outlook is structured around a set of assumptions that together frame the medium- and long-term outlook across all major energy sources. These assumptions span four closely linked areas: evolving population and demographic trends, diverse economic growth patterns, the direction and stringency of energy and climate policies, as well as steady progression of technological innovation and efficiency gains across energy systems.

1.1 Population and demographics

Over recent decades, life expectancy has risen in almost all regions, albeit at a varying pace. This improvement has been supported by expanded access to basic health services, improved disease prevention, better nutrition and the provision of safer water and sanitation services. Positively, this progress has transformed living conditions for billions of people, reducing mortality at younger ages, steadily extending life expectancy and benefitting long-term economic development. More recently, rapid advances in digital health technologies and data analytics have further strengthened health systems by improving health assessments, clinical diagnosis and service delivery.

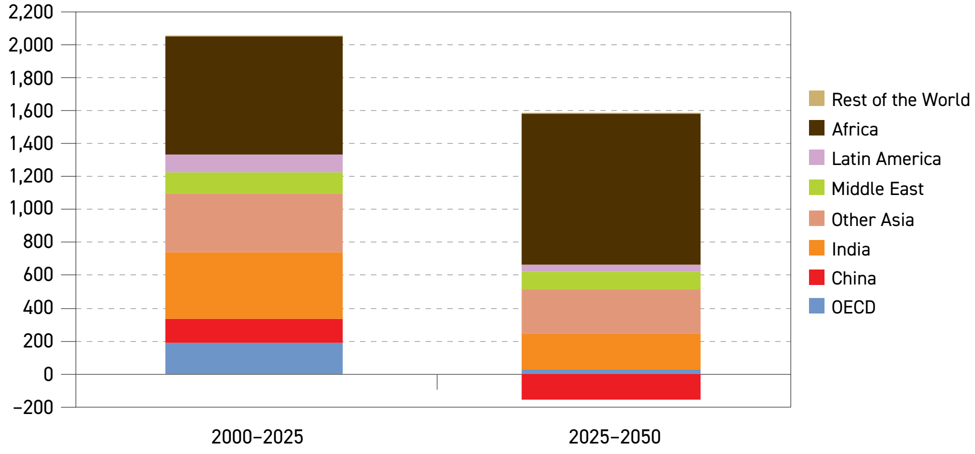
Against this backdrop, the global population is projected to rise significantly from approximately 8.2 billion in 2025 to about 9.7 billion by 2050 (Table 1.1), according to the latest World Population Prospects by the United Nations Department of Economic and Social Affairs (UNDESA). This net increase of more than 1.4 billion people represents an increase of about 17% in the global population. An expected decline in total fertility rates means that the increase between 2025 and 2050 is smaller than the previous 25-year period. This reflects a gradual moderation in global population growth (Figure 1.1).

Table 1.1
World population by region, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	543	558	571	582	592	599	56
OECD Europe	590	590	589	587	583	577	–13
OECD Asia-Pacific	217	215	212	209	205	202	–15
OECD	**1,350**	**1,363**	**1,372**	**1,378**	**1,380**	**1,377**	**28**
China	1,416	1,398	1,373	1,343	1,306	1,260	–156
India	1,464	1,525	1,579	1,623	1,656	1,680	216
Other Asia	1,304	1,367	1,428	1,484	1,533	1,575	270
Latin America	513	527	540	549	555	558	45
Middle East	291	315	337	359	379	398	107
Africa	1,550	1,727	1,910	2,096	2,282	2,467	917
Russia	144	142	140	138	137	136	–8
Other Eurasia	152	157	161	165	169	173	21
Other Europe	48	47	45	44	42	41	–7
Non-OECD	**6,882**	**7,206**	**7,513**	**7,799**	**8,060**	**8,287**	**1,405**
World	**8,232**	**8,569**	**8,885**	**9,177**	**9,440**	**9,664**	**1,433**

Source: UN.

Figure 1.1
World population growth, 2000–2025 *versus* 2025–2050, millions

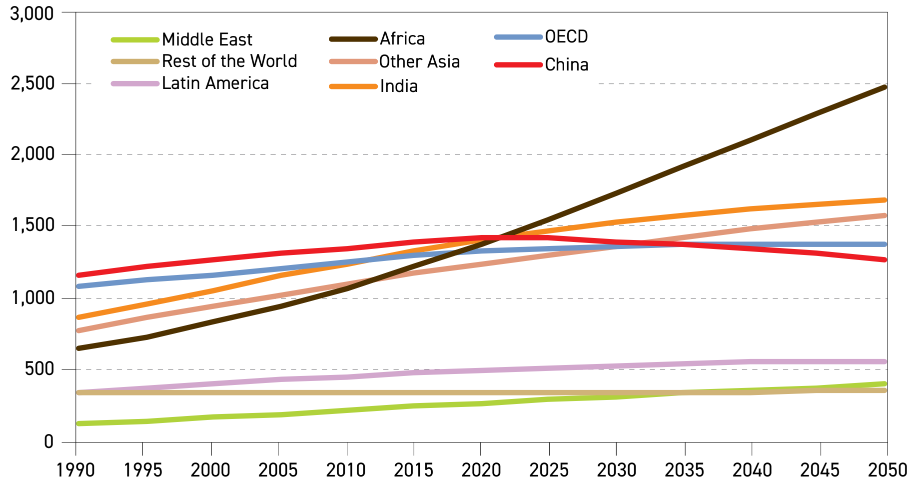


Source: UN.

Using the regional classifications contained in Annex B of this Outlook (see Annex B), the population data underscores a clear contrast between relatively flat population trends in advanced economies and vigorous expansion elsewhere. In the OECD, below-replacement level birth rates and rising life expectancy are reducing population increases and accelerating population ageing. Consequently, the region is projected to record only modest population growth of around 28 million, increasing from about 1.35 billion in 2025 to close to 1.38 billion by 2050. Growth in OECD Americas, however, is expected to more than compensate for the slight declines anticipated elsewhere within the OECD.

Compared to modest growth in the OECD region, non-OECD countries are expected to see substantial population increases through to mid-century (Figure 1.2). This is driven by

Figure 1.2
World population by region, 1990–2050, millions



Source: UN.

higher birth rates, improved life expectancy and youthful populations, even as growth rates gradually moderate in the long term. Between 2025 and 2050, the non-OECD population rises from about 6.9 billion to roughly 8.3 billion people, an increase of just over 1.4 billion. By 2050, the region accounts for around 86% of the world's population.

The starkest development across the major regions is Africa's population trend. The continent's population, already larger than that of either India or China, continues to rise steadily, as illustrated in Figure 1.2. With a projected net addition of around 917 million people by 2050, Africa's share of the total global population rises to slightly more than 25% by the end of the outlook period.

India, now the world's most populous country, is on a more moderate demographic growth trajectory. Its population is projected to expand by nearly 216 million between 2025 and 2050, rising from around 1.46 billion to about 1.68 billion. This continued growth is sustained by a youthful age structure and marked differences in fertility across states and other subnational regions, even as the national average gradually declines towards replacement level.

Meanwhile China's demographic profile has entered a new phase of measured demographic adjustment. It is set to witness a net drop of approximately 156 million by 2050, falling from around 1.42 billion in 2025 to 1.26 billion people in 2050. This shift reflects the cumulative legacy of earlier population policies, persistently low fertility rates that are well below replacement levels, as well as evolving socioeconomic conditions. Despite the relaxation of birth restrictions, including the introduction of a three-child policy in 2021, birth rates remain low. This transition is the most pronounced among major economies and represents a significant departure from China's past population growth.

Other Asia, encompassing a diverse group of emerging and developing economies, is expected to experience robust population growth. Its population is projected to rise by about 270 million between 2025 and 2050, reflecting a relatively young population profile and only a gradual drop in fertility rates across several countries.

The Middle East is set to experience continued population growth, with an increase of 107 million people between 2025 and 2050. This steady rise stems from relatively youthful demographic profiles in several countries and ongoing improvements in healthcare and living standards that continue to lift life expectancy, even though fertility rates are gradually declining from earlier high levels.

Russia is projected to see a modest population decline of eight million by 2050. This trend is accompanied by an ageing population and a contracting labour force, driven mainly by persistently low fertility rates and limited net migration. Combined, these contribute to a higher share of older age groups in the overall population. Despite these shifts, the country maintains a generally stable demographic base.

Other Eurasia is projected to add 21 million people over the period to 2050, reflecting younger population profiles in several countries. Higher birth rates are set to help preserve a demographic structure that supports future labour-force availability and maintains a relatively large share of children and young adults in the total population.

Other Europe is experiencing ageing combined with low fertility, which results in slow or slightly negative population growth and a projected decline of seven million people by 2050.

Longer life expectancy alongside fewer births is steadily increasing the share of older people while narrowing the size of the working-age cohort. These changes reflect Other Europe's progress in health and life expectancy while pointing to a more mature population profile over time.

1.1.1 Working-age population

By 2050, the global working-age population (defined as individuals between 15 and 64 years old) is set to reach a total of 6.1 billion (Table 1.2). This reflects an addition of around 751 million people to the world's potential workforce compared to 2025, supporting a growing economy. Across the world, regional patterns of working-age population growth vary considerably, creating a clear demographic differentiation between economies managing more mature workforce transitions and those absorbing large numbers of new labour market entrants.

Table 1.2
Working population (age 15 to 64) by region, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	355	361	366	369	372	372	17
OECD Europe	378	373	365	355	344	333	–46
OECD Asia-Pacific	135	131	126	119	113	109	–26
OECD	**868**	**865**	**857**	**843**	**828**	**813**	**–55**
China	987	972	930	859	807	745	–241
India	1,002	1,053	1,092	1,118	1,133	1,134	132
Other Asia	854	903	945	979	1,007	1,027	173
Latin America	347	357	364	366	364	359	12
Middle East	194	213	229	242	255	264	70
Africa	892	1,018	1,150	1,286	1,423	1,559	667
Russia	94	93	92	90	86	82	–11
Other Eurasia	98	101	104	107	108	108	11
Other Europe	31	30	28	27	25	24	–7
Non-OECD	**4,498**	**4,740**	**4,934**	**5,073**	**5,208**	**5,304**	**805**
World	**5,366**	**5,605**	**5,791**	**5,916**	**6,036**	**6,117**	**751**

Source: UN.

The working-age share of the global population is projected to remain broadly steady throughout the outlook period, shifting modestly from around 65% in 2025 to 63% in 2050. The difference is a reflection of the overall global population rising faster than the working-age population, alongside an expanding proportion of the population being over 64 years of age.

The OECD working-age population is projected to gradually drop from about 868 million in 2025 to around 813 million in 2050, a net reduction of roughly 55 million people that brings the working-age share down from around 64% to approximately 59%. This transition signals an evolving shift in labour supply dynamics across the OECD region, with significant implications for sectoral composition and the relative energy intensity of economic output, as well as the way technological progress reshapes labour-intensive activities and production processes.

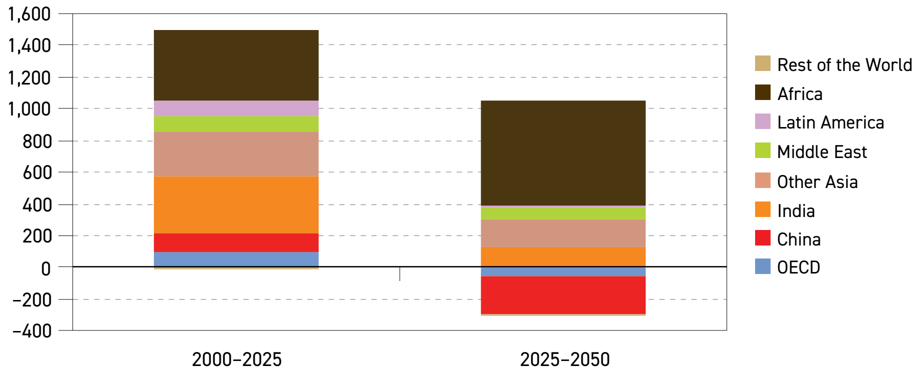
Within the OECD, only the OECD Americas region is projected to record an increase in the working-age population, rising by about 17 million. This is driven primarily by net immigration that offsets natural decreases. By contrast, OECD Europe and OECD Asia-Pacific are projected to experience notable downward adjustments. OECD Europe's working-age population is projected to drop by 46 million, representing a 12% reduction, which may require adjustments in the industrial sector, as well as pension system design. OECD Asia-Pacific faces a similarly severe contraction, with its working-age population expected to decline by 26 million, or approximately 20%. This reflects the effects of advanced ageing in Japan and South Korea, as well as ongoing demographic shifts in Australia.

At the same time, in many OECD countries the statutory and effective retirement age is expected to rise towards 67–70 years or beyond by 2050, implying that an increasing share of people will remain economically active well into their late 60s and 70s.

Non-OECD economies are projected to be the main source of growth in the global working-age population. They are set to add about 805 million people by 2050, with people aged 15–64 representing around 64% of the region's total population. This is despite a growing share of the population transitioning out of the working-age segment in several large emerging markets. By mid-century, the non-OECD working-age population will account for roughly 87% of the world's total working-age population.

Within the non-OECD, Africa is the largest contributor to global working-age population growth. The region's working-age population is expected to increase by around 667 million between 2025 and 2050 (Figure 1.3), substantially raising Africa's share of the global working-age total. Elsewhere, Other Asia is projected to add approximately 173 million working-age individuals over the same period, supporting further expansion in manufacturing, services and the primary sector.

Figure 1.3
Working population (age 15 to 64) growth, 2000–2025 *versus* 2025–2050, millions



Source: UN.

India's working-age population is projected to grow by about 132 million, a marked moderation compared with the roughly 367 million added between 2000 and 2025. This reflects a transition to a more mature demographic phase. In contrast, the Middle East is projected to record a substantial increase in its working-age population, rising by about 70 million. Finally, Latin

America is expected to add roughly 12 million, in line with a more advanced but still heterogeneous demographic transition across the region.

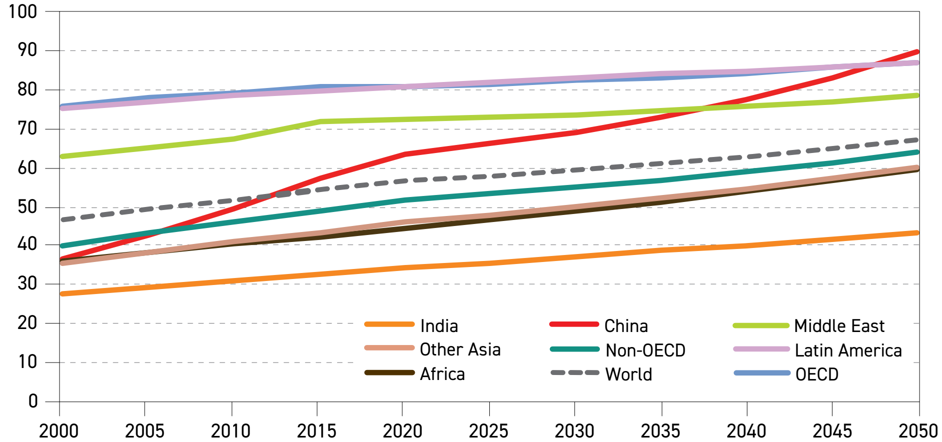
Other Eurasia is projected to record moderate growth of around 11 million in its working-age population over the period. Russia and Other Europe are projected to see reductions of roughly 11 million and seven million, respectively, slightly reducing their share of the global labour force.

China's working-age population is projected to drop from approximately 987 million in 2025 to around 745 million by 2050. This is a decline of around 241 million. This shift follows growth of 337 million from 2000 to 2025, marking a notable transition from a period of workforce expansion to a new phase of demographic recalibration in labour supply.

1.1.2 Urbanization

Urbanization is the principal pathway to expanding modern energy access and is fundamentally linked to rising incomes, structural economic change and evolving patterns of sectoral energy demand. In 2025, about 58% of the global population lived in urban areas, up from 47% in 2000, and the share is projected to reach roughly 67% by 2050 (Figure 1.4). This year's projections incorporate data from the UN's World Urbanization Prospects 2025 report, which now implements a harmonized international framework that consistently classifies settlements across national boundaries into three typologies, namely cities, towns or rural areas.

Figure 1.4
Urbanization rate for selected regions, 2000–2050, %



Source: UN.

The 2025 revision introduces the 'Degree of Urbanization' methodology. This is a spatially consistent approach endorsed by the UN Statistical Commission that maps entire territories onto a population grid and applies uniform density and contiguity thresholds. In this framework, 'cities' are contiguous areas with at least 1,500 inhabitants per square kilometre and a combined population of 50,000 or more, 'towns' are intermediate density zones with at least 300 inhabitants per square kilometre, while 'rural areas' fall below 300 inhabitants per square kilometre. Using a single method across geographies strengthens comparability and provides a clearer view of how settlement change influences energy systems, while also aligning with wider analytical efforts that examine the full spectrum of urban settlements, from towns and cities to larger metropolitan areas.

In this regard, it is worth noting that the scale of urbanization in highly populated areas is shifting. In 2025, the number of megacities with populations of ten million or more was estimated at 33, but this figure is projected to rise to 37 by mid-century. Jakarta is currently the world's most populous agglomeration with nearly 43 million residents, followed by Dhaka at around 37 million and Tokyo at roughly 33 million. By 2050, Dhaka and Jakarta are expected to surpass 50 million residents, underscoring the growing concentration of people in a handful of very large urban regions and the need for resilient energy and transport systems.

Across regions, urbanization levels are high and generally continuing to rise. In 2025, the OECD and Latin America were the most urbanized large groupings, with 81% and 82%, respectively, reflecting mature urban landscapes. Within the OECD, both the Americas and the Asia-Pacific exhibit particularly advanced urbanization. Building on these already elevated levels, further increases are expected, with the OECD approaching about 87% and Latin America reaching around 87% by mid-century.

Urbanization in the Middle East is above the global average and continues to advance, with its urban share set to rise from 73% in 2025 to around 78% by 2050.

China's urban transition has been exceptional. From roughly 36% at the start of the century, China reached about 66% in 2025. Maintaining this momentum, the country has continued to urbanize as the economy matures, and its urban share is projected to approach 90% by mid-century.

Table 1.3
Urban population by region, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	438	455	472	490	507	525	86
OECD Europe	468	473	478	482	485	487	19
OECD Asia-Pacific	192	191	190	189	187	186	–7
OECD	**1,099**	**1,120**	**1,140**	**1,160**	**1,179**	**1,197**	**98**
China	939	966	999	1,038	1,084	1,131	191
India	522	568	611	652	690	723	201
Other Asia	626	685	747	812	878	944	318
Latin America	418	436	452	465	476	484	66
Middle East	212	232	251	271	291	312	100
Africa	720	843	980	1,132	1,296	1,472	752
Russia	108	108	108	108	109	110	2
Other Eurasia	87	90	93	96	100	103	16
Other Europe	28	28	28	27	27	27	–1
Non-OECD	**3,662**	**3,956**	**4,269**	**4,603**	**4,951**	**5,306**	**1,644**
World	**4,761**	**5,076**	**5,410**	**5,763**	**6,130**	**6,503**	**1,742**

Source: UN.

India is on a lower and more measured path. With an urbanization rate of 36% in 2025, it is among the lowest of the regions assessed. By 2050, this figure is set to rise to around 43%.

Other Asia and Africa display broadly similar trajectories over the projection period. Other Asia stands at 48% in 2025 and is expected to reach 60% by mid-century, while Africa stands at 46% in 2025 and is projected to also reach 60% by mid-century.

The implications of these trajectories are evident in absolute terms (Table 1.3). The global urban population is forecast to increase from about 4.76 billion in 2025 to around 5.08 billion in 2030, an increase of roughly 320 million people. Over the same period, the OECD contribution is relatively modest, rising from about 1.1 billion to around 1.12 billion. Overall, the global urban population is set to reach justover 6.5 billion by 2050, up nearly 1.75 billion from 2025.

1.1.3 Migration

The movement of people internationally is an important demographic mechanism to consider over the long term. For countries experiencing slower population growth or population ageing, particularly those in OECD regions, net migration can help to offset lower birth rates and support the size of the working-age population. However, the resulting economic benefits depend heavily on successful labour-market integration, and net migration may be affected by a tightening of migration policies in some countries.

Net migration represents the population change attributable to cross border movement and is assessed by the UN as the difference between its medium variant demographic projection and a hypothetical zero migration counterfactual scenario. In aggregate, as shown in Table 1.4, the OECD region is expected to gain approximately 12.3 million net migrants between 2025 and 2030, and a cumulative total of just under 63 million to 2050.

Table 1.4
Net migration by region, millions

	2025–2030	2030–2035	2035–2040	2040–2045	2045–2050	2025–2050
OECD Americas	7.1	7.0	7.0	7.1	7.1	35.3
OECD Europe	3.4	4.1	3.9	3.8	3.7	19.1
OECD Asia-Pacific	1.8	1.6	1.6	1.6	1.6	8.3
OECD	**12.3**	**12.8**	**12.5**	**12.5**	**12.5**	**62.6**
China	–1.2	–1.1	–1.1	–1.0	–1.0	–5.3
India	–2.1	–1.9	–1.9	–1.9	–1.9	–9.8
Other Asia	–8.8	–5.8	–5.8	–5.5	–5.2	–31.2
Latin America	–1.6	–1.3	–1.0	–0.9	–0.9	–5.8
Middle East	2.4	0.0	–0.3	–0.3	–0.3	1.5
Africa	–2.8	–3.5	–3.3	–3.8	–4.3	–17.8
Russia	1.0	1.5	1.5	1.6	1.7	7.4
Other Eurasia	1.4	–0.4	–0.4	–0.3	–0.3	0.0
Other Europe	–0.4	–0.3	–0.3	–0.3	–0.3	–1.6
Non-OECD	**–12.3**	**–12.8**	**–12.5**	**–12.5**	**–12.5**	**–62.6**

Source: UN.

Projected migration dynamics remain subject to significant uncertainty linked to policy choices, geopolitical stability and economic convergence between regions. Changes in immigration policy frameworks, border management practices, regional integration agreements and labour-market regulation can all substantially alter migration volumes and composition.

1.2 Economic growth

The global economy continues to demonstrate encouraging resilience and expand robustly. This trend is forecast to extend across the medium- and long-term horizon.

Steady growth in the US, India and China is projected to remain a key driver of global economic expansion, particularly in the medium term. In the long term, economies in Africa and Other Asia are forecast to contribute steadily to the global economic growth dynamic, while emerging and major developed economies are anticipated to see more normalized growth trends. Combined with steady growth in the Eurozone and Japan, this is expected to provide additional support to the global outlook. At the same time, Brazil and Russia are also expected to sustain stable growth momentum. This broad-based and balanced growth dynamic is supported by expectations for normalized inflation levels in the medium to long term, supportive financial conditions and improvements in global trade going forward.

This positive dynamic materializes despite the current period of increasing fragmentation and ever-evolving geopolitical dynamics. In times of uncertainties and structural global economic changes, it is important to understand the interactions between global economic trends and energy markets. Gaining a clear understanding of the key trends, including both challenges and opportunities, is vital. Against this backdrop, this overview provides valuable insights into expectations for the medium- to long-term trajectories of the global economy.

1.2.1 Current situation and short-term growth

Building on the steady growth momentum observed in 2025, global economic growth is set to continue to grow at around 3% in the short term. Despite some volatility, primarily driven by US trade policies, trade partners' policy responses, as well as geopolitical developments, the 2025 global economic growth trend was relatively steady. Trade-related uncertainty in the first half of 2025 was followed by more certainty in the second half of the year, with major trade agreements supporting the growth dynamic. However, trade-related uncertainties remain, especially following the US Supreme Court decision in February 2026 to ban the US Administration's use of tariffs under the International Economic Emergency Powers Act.

Overall, strong performances in the US, India and China again contributed significantly to global economic growth in 2025. This is expected to continue in the short term. Elsewhere, the Eurozone and Japan continued to expand steadily, albeit at relatively lower economic growth rates, while Brazil and Russia also sustained steady growth trends.

In 2026, the fiscal boost in the US in the first half of the year – including tax cuts and a boost from the government reopening after the shutdown in the fourth quarter of 2025 – is anticipated to underpin growth. Fiscal and policy support measures are also set to gather pace in other major economies, including China, Japan and parts of the Eurozone, while India is projected to sustain strong growth amid continuing fiscal support and expectations of improving trade relations with the US.

1.2.2 Medium-term economic growth

Economic growth is expected to remain well supported over the medium term with several key trends continuing to shape the global economic outlook. That said, against a backdrop of gradually improving global trade, elements of economic fragmentation are expected to persist. Furthermore, this fragmentation could be exacerbated by persistent inflationary pressures, particularly in advanced economies such as the US, the Eurozone and the UK, while some countries may also increasingly seek to build more integrated domestic or regional supply chains, in contrast to existing globalized structures.

Consequently, relatively less intertwined global supply chains may lead to less integrated financial markets. This environment is likely to be accompanied by somewhat tighter financial conditions, which, together with elevated global debt levels and continued fiscal support in some economies, could weigh on the pace of economic expansion in the years ahead, although still allowing for positive global growth.

Moreover, as governments navigate fiscal pressures, rising debt obligations and generally expanding budgets, among other factors, the dynamics they face may encourage a broader review of revenue strategies. This could possibly drive adjustments to taxes and other duties across various regions, a trend that is already materializing in some parts of the global economy.

Encouragingly, advancements in artificial intelligence (AI) and robotics have demonstrated the ability to boost productivity growth and help offset some of these challenges. While the expected continued shortage of skilled labour, further impacted by stagnant or shrinking working-age populations in many major economies, could temper some of these productivity gains, it also presents opportunities for the accelerated adoption of automation and workforce innovation.

Emerging and developing economies are projected to outgrow advanced economies in the medium term, albeit some of these economies may see growth rates levelling off, or moderating gradually, as their domestic economies mature. This is particularly the case for China. More moderate growth trajectories reflect a natural progression towards more sustainable and balanced growth paths.

Overall, the medium-term global growth outlook will be shaped by a number of key factors, the effects of which will also influence the early years of the long-term outlook. These major factors include inflation, monetary policy, debt-related challenges, geopolitical fragmentation and global productivity.

Inflation is expected to witness a continued gradual decline in the coming years after a temporary rise in the short term and then normalize towards the end of the medium-term period. While global inflation stood at more than 4.0% in 2024, it dropped slightly to around 3.5% in 2025, and the level is expected to slow further and stand at around 2.5% at the end of the medium-term period. In the US and the Eurozone, in particular, inflation is forecast to moderate and normalize over the medium-term period. However, the average inflation level may remain above the targeted 2% level. In Japan, inflation is forecast to remain higher compared to its OECD peer-economies in the short term. However, it is then set to approach the 2.0% target level towards the end of the medium-term period.

In China, which is currently experiencing a relatively low inflationary trend, consumer price inflation is forecast to gradually rise from its current low level to just above 1.0% towards the end of the medium-term period. India's inflationary dynamic has recently seen a retraction, and it is forecast to mean-revert to normalized levels of up to 4.0% towards the end of the medium term. Brazil and Russia, in particular, continue to face more persistent inflationary pressures. While inflation trends have eased, rates are likely to remain elevated, potentially at around 4.0% in Brazil and 5.0% in Russia at the end of the medium-term horizon.

These trajectories will consequently impact monetary policy and interest rate expectations. Therefore, monetary policies are expected to remain in a relatively more accommodative manner in the major OECD economies. The exception is Japan, which is expected to continue on a policy path of monetary tightening, given relatively stronger inflationary pressures and the expectation of ongoing expansionary fiscal policies. Elsewhere, China's anticipated low inflation path provides its central bank with ongoing monetary policy flexibility, while India's retraction in inflation points towards continued monetary policy accommodation. Similarly, Brazil, and especially Russia, are anticipated to ease their monetary policy stance further over the medium term.

In connection with inflation and interest rate trends, **debt-related** challenges in various economies will also need to be closely monitored. While global debt levels seem to have been relatively well absorbed to date, expanding debt levels across the world may pose a challenge going forward. Based on the latest findings by the Institute of International Finance, global debt surged by over $26 trillion in the first three quarters of 2025, leading to a fresh high of nearly $346 trillion. The new records for global debt levels are driven largely by government borrowing and debt in both mature and emerging markets.

Elevated debt levels are increasingly leading to fiscal constraints in several economies, including some Eurozone economies and Russia. Similar pressures could also potentially emerge in Brazil over the medium term. Other major economies, including the US, China and Japan, may also face rising fiscal limitations and should therefore be closely monitored too. Similar to the previous Outlook, however, no major dislocation from this debt-related situation is assumed moving forward.

In times of high debt, governments are often faced with the need to increase revenues, mostly via raising levies on goods and income, as well as on assets, and through capital gains, property, high incomes, and corporate profits to manage debt costs. While such measures can weigh on near-term economic activity, they also serve to strengthen fiscal foundations and restore investor confidence over time.

Amid fluid **geopolitical developments**, this Outlook remains dynamic. Growth momentum is expected to remain resilient, with geopolitics playing a limited role beyond the current underlying assumptions. That said, resolving conflicts and strengthening multilateral cooperation could support both higher regional and global growth. Regardless, building on recent developments, it appears that the trend for **global fragmentation** could continue. In particular, global trade is projected to continue fragmenting and become more regionally segmented among the three main trading hubs: the US-centred trade region of the Americas, dominated by North America; the European region, with its dominant forces of Germany, France and the UK; and the Asian region, centred around China, India and Japan.

Productivity, a key factor in economic growth, has recently improved in advanced economies after declining in the years leading up to the pandemic. This trend is possibly due to the latest advancements in AI and related technologies. While forecasts still expect only modest productivity gains, this area may provide some upside for current growth projections, as the effective adoption of AI, robotics and other new technologies could significantly boost productivity in both industrial production and services. This trend is expected to continue at a gradual pace, though major uncertainties remain. This is why such developments are not fully reflected in the medium-term outlook.

Taking these growth assumptions and developments into consideration, the global economic growth forecast stands at around 3.2% for most of the medium-term period. While this can be considered a robust growth trend, there is further upside to these annual growth assumptions. This is dependent on some key factors. Among these are solutions to geopolitical conflicts, which could also lead to some relief in global inflationary levels, and hence, a further continuation of more accommodative monetary policies. Moreover, continued productivity gains – most likely supported by advancements in digitalization, including AI – could lead to additional progress in the global growth dynamic.

The remainder of this section examines, in more detail, the medium-term economic growth prospects of each region or major economy. As such, it focuses on the OECD Americas, OECD Europe, OECD Asia-Pacific, China, India, Other Asia, Latin America, the Middle East, Africa, Russia, Other Eurasia, and Other Europe.

Across the **OECD** economies, OECD Americas is anticipated to lead growth, driven largely by the US. Elsewhere, OECD Europe is anticipated to rebound modestly and to gain momentum towards the middle of the medium term. OECD Asia-Pacific is expected to improve over the medium-term period too, albeit seeing a lower growth level than the other two OECD regions. This is primarily due to the low growth dynamic in the region’s largest economy, Japan. The OECD is expected to see overall growth of 1.8% in 2026, a level that is expected to be maintained over the medium term.

As mentioned, growth in **OECD Americas** is expected to be mainly driven by the US. However, trade developments between the US, Canada and Mexico will also play an important role in the region’s economic growth. For example, the US-Mexico-Canada trade agreement, scheduled to be extended in mid-2026, will be vital. Similarly, the US-China trade agreement that expires in October 2026 is crucial. While uncertainties persist at the time of writing, no major trade dislocation that could affect the growth dynamic in OECD Americas is anticipated.

Over the medium term, economic growth in OECD Americas is set to remain within the range of 2.0% to 2.1%. However, potential upward inflationary pressure from tariffs, along with high US debt levels, could strain growth momentum, particularly if interest rates do not ease. In contrast, gains in AI-related productivity and rising investments in this area could provide some upside to growth in the region.

Growth in **OECD Europe** is expected to remain well supported, although it may be impacted by continued challenges in the industrial sector, in combination with uncertainties related to US-trade relations and fiscal constraints in most major economies. While inflation has retracted and is forecast to gradually slow over the medium-term period, uncertainties remain. The conflicts in Eastern Europe and the Middle East, as well as their continued effect on Europe’s energy supplies and energy prices, may continue to have an impact on economic development. More positively, however, fiscal support in Germany and rising defence spending plans in most economies, in combination with continued accommodative monetary policies, are expected to support the region's growth trend. In addition, as inflation is expected to recede over the medium term, rising real incomes in the region are likely to provide further support. Against this backdrop, growth is expected to maintain a sound level of around 1.6% throughout the period, with a peak of 1.7% expected in the middle of the medium term.

In **OECD Asia-Pacific**, Japan is expected to see economic growth on the lower side as its economy is anticipated to remain relatively stagnant over the medium term amid continued monetary tightening. In 2026, however, iwith both Australia and New Zealand forecasting a rebound, the region's growth is set to lift substantially to 1.5%. Furthermore, the region's major trading partner, China, is forecast to maintain a healthy growth level in the medium term. For OECD Asia-Pacific, growth is set to remain at 1.5% in the early part of the medium term before settling at around the 1.4–1.5% range over the remainder of the medium-term period.

Medium-term growth in **non-OECD** countries remains relatively strong, with diverse growth trends. China, India and Other Asia, in particular, comprise the main growth engines. Latin America, the Middle East, Africa and the Eastern European economies all show lower, albeit robust, growth trends too. The non-OECD is expected to see growth of around 4.2% over the medium-term period, more than twice the growth dynamic anticipated in OECD economies.

China is expected to record a growth rate of 4.5% in the early years of the medium term before remaining relatively stable over the remainder of the medium term. While challenges in external trade are expected to remain in the medium term, domestic demand is set to gradually pick up, with initiatives to support local consumption in the 15th Five-Year Plan (FYP). Furthermore, the central government is anticipated to counterbalance any deviation from the government's growth targets. Growth is expected to ease modestly to around 4.3% by the end of the period.

India's growth is projected to see a relatively stable growth trajectory over the medium term, after an anticipated drop from exceptionally high growth in 2025. The economy is set to benefit from the country's population growth, a rising middle class and continued government support. This includes investing in major infrastructure projects over the medium term. An ongoing normalization of inflation over the medium term and the expectation of accommodative monetary policy is set to provide further support to medium-term growth. India's growth is set to reach around 6.6% in the early years of the medium term, before moderating slightly to 6.5% thereafter and maintaining a similar average level over the remainder of the period.

Other Asia is expected to see relatively robust medium-term growth, fuelled by dynamic economies in this group of countries. In the early years of the medium term, growth is expected at around 4.2% before easing slightly to 4.1% thereafter. Growth is then set to accelerate over the remainder of the period, reaching an overall average of 4.2%.

In **Latin America**, the two major economies, Brazil and Argentina, will likely shape the economic growth pattern. Brazil is expected to slightly decelerate at the beginning of the medium-term period, following very high growth levels in 2024 and 2025. This is due to the country needing to manage its sovereign debt situation. However, Brazil is anticipated to rebound towards the end of the medium-term period, lifting growth for the region. A similar growth trend is also seen in Argentina. The country has demonstrated some recent improvement but still has to deal with a number of fiscal challenges, at least at the beginning of the medium term. Nonetheless, the administration's introduction of a number of policies, including those aimed at stabilizing the currency exchange rate, may further accelerate the recovery. Overall, considering the country's high debt levels and limited fiscal space in which to manoeuvre, Argentina is anticipated to maintain stable growth over the medium term. Growth in Latin America is expected at around 2.3% in the early years of the medium term, rising to 2.6% thereafter, before accelerating to 2.8% at the end of the period. The average growth rate over the medium-term period stands at 2.5%.

In the **Middle East**, medium-term growth is expected to see levels of growth of around 3.1% over the medium term, following a strong appreciation from 2025. OPEC Member Country economies are set to provide strong support to the region's growth dynamic. Similarly, **Africa** is forecast to accelerate and maintain an average growth rate of around 3.8% over the medium-term period. This follows growth of 3.7% in 2025. Both regions' growth dynamics are set to be supported by expanding regional trade, population growth, a consequent rise in domestic demand, an expansion of the middle class and ongoing steady commodity demand.

Russia has overcome numerous external challenges related to its economy in recent years. As a result, the country's growth trend is forecast to appreciate throughout the medium-term period. Following growth of 1.3% in the early years of the medium term, the growth dynamic is forecast to appreciate to an average growth level of 1.4% for the remainder of the period. **Other Eurasia** is forecast to slow slightly, falling from a robust 4.3% in the early years of the medium term to a still healthy 4.0% by the end of the period. Elsewhere, economic growth in **Other Europe** is set to accelerate slightly, increasing from 2.3% in the early years of the medium term to 2.4% by the end of the medium-term period.

1.2.3 Long-term economic growth

Productivity gains remain a key source of resilience and continued growth in the global economy. AI is expected to become one of the primary drivers of long-term productivity growth after 2030. Its adoption is set to expand to large-scale integration for industrial processes and it is also expected to become more deeply embedded in the services sector. Manufacturing, logistics and high-skill service sectors are expected to benefit first, with more broader gains to follow as capabilities diffuse across economies. However, these productivity gains will not be evenly distributed. They are likely to initially be concentrated in countries with strong digital infrastructure and abundant capital, before eventually expanding into emerging markets.

In addition, the distributional and structural effects of AI adoption will be complex. Hardware and infrastructure constraints, for example, remain a critical factor that could potentially limit wider adoption across various sectors. Labour markets will also undergo an adjustment as AI expands into roles previously resistant to automation, particularly in analytical and service-based functions. While many tasks requiring adaptability and physical presence will remain human-dominated, mid-level management and coordination roles are likely to be affected.

Global inflationary trends are also expected to see some shifts in the period through 2050. The generally low and stable inflation regime of the past two decades was enabled primarily by the cost benefits of globally integrated production and coordinated monetary policy. However, with supply chain redundancies and parallel industrial capacity, combined with regionally segmented production, an increase in baseline costs for production is expected. This will place upward pressure on prices, potentially leading to higher interest rates from central banks as they respond. This likely inflationary shift, along with a subsequent rise in borrowing costs, may place fiscal expansion under tighter constraints. Debt levels are forecast to remain elevated following continued fiscal stimulus and ongoing infrastructure expenditure across many economies. Potentially rising interest rates may increase debt servicing burdens, further limiting governments' ability to finance investment or respond to economic shocks. Advanced economies are likely to retain market access, but many emerging markets will face tighter financing conditions.

The international currency system is expected to retain the US dollar at its centre, albeit with gradual changes. The dollar continues to benefit from the depth of US capital markets and global confidence; however, its embedded role in trade and finance is anticipated to gradually diminish, particularly in regional and bilateral transactions. Alternative payment systems, such as central bank digital currencies, the continued development of cryptocurrencies and the wider use of local-currency settlement arrangements, are set to generate new channels that could slightly reduce reliance on the dollar. However, as global trade continues to expand and accelerate in non-strategic goods, the role of the dollar is expected to remain central to the system, with no other single currency emerging to rival it through 2050.

Structural economic conditions are expected to affect regions unevenly over the period to 2050. Factors such as existing infrastructural capacity, geography, trade linkages, and the degree of regional integration are set to play a critical role in shaping long-term growth.

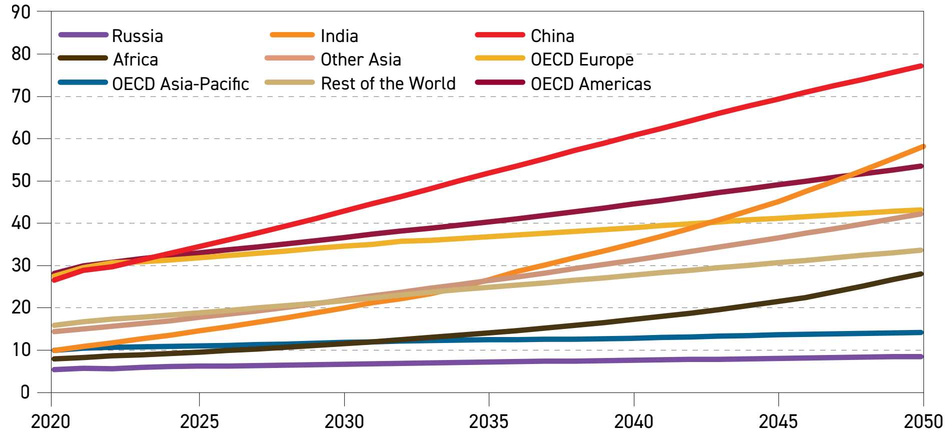
Overall, global GDP is expected to expand at an average rate of 2.9% p.a. from 2025 to 2050, with growth easing slightly from 2.9% p.a. between 2030 and 2040 to 2.7% p.a. from 2040 to 2050 (Table 1.5 and Figure 1.5). In absolute terms, the global economy is expected to more than double in size, rising from $177 trillion in 2025 to $359 trillion in 2050 (in 2021 PPP).

Table 1.5
Long-term annual GDP growth rate (in real terms), % p.a.

	2025–2030	2030–2040	2040–2050	2025–2050
OECD Americas	2.1	2.0	1.9	1.9
OECD Europe	1.6	1.2	1.0	1.2
OECD Asia-Pacific	1.5	0.8	1.0	1.0
OECD	**1.8**	**1.5**	**1.4**	**1.5**
China	4.4	3.6	2.4	3.3
India	6.5	5.8	5.1	5.7
Other Asia	4.2	3.6	3.1	3.5
Latin America	2.5	2.2	1.7	2.1
Middle East	3.1	3.0	2.3	2.7
Africa	3.8	4.1	5.0	4.4
Russia	1.4	1.4	1.1	1.3
Other Eurasia	4.0	2.8	2.2	2.8
Other Europe	2.4	0.9	1.0	1.3
Non-OECD	**4.2**	**3.7**	**3.3**	**3.6**
World	**3.2**	**2.9**	**2.7**	**2.9**

Source: OPEC.

Figure 1.5
GDP of major economies, 2020–2050, $(2021 PPP) trillion



Source: OPEC.

OECD economies are expected to retain significant leverage within the global system and remain well positioned to extract gains from technological advancement. However, the mature nature of these economies and their weakening demographic dynamics are expected to constrain growth, holding the OECD average at 1.5% p.a. from 2025 to 2050. Overall, the region is anticipated to follow a steady but mildly decelerating trend.

In contrast, **non-OECD** economies are expected to record stronger growth, supported by more favourable demographic conditions and the scope for catch-up gains from comparatively lower initial income levels at the start of the projection period. As a result, non-OECD growth is projected at 3.6% p.a. from 2025 to 2050. Levels are expected to gradually moderate over time from 3.7% p.a. between 2030 and 2040 to 3.3% p.a. from 2040 to 2050, as an increasing number of economies approach more mature stages of development.

OECD Americas is expected to be the fastest-growing region within the OECD, although growth is projected to moderate gradually over time. Output is expected to expand at around 2% p.a. between 2030 and 2040 before decelerating slightly to 1.9% p.a. from 2040 to 2050. This results in an average growth rate of 1.9% p.a. over the period between 2025 and 2050. This expansion will continue to be driven by an expanding working-age population and supported by immigration and technological advancements. Within the region, trade tensions are expected to ease, allowing economic integration to continue. However, Canada and Mexico will likely continue diversifying trade relationships outside of the region.

OECD Europe is expected to expand at an average rate of 1.2% p.a. from 2025 to 2050, reflecting a gradually decelerating growth trend. Economic growth is projected at 1.2% p.a. during the period from 2030 to 2040, before slowing to 1.0% p.a. between 2040 and 2050. The services sector is expected to remain the primary driver of economic expansion, while the industrial base and weakening demographics are anticipated to slow the overall growth potential. Although inflation and interest rates are likely to stay relatively elevated in the short term, they are expected to ease gradually over time, making them an insignificant contributor to the deceleration in the period after 2030. The region is set to continue to face increasing competition from other parts of the world, particularly in terms of its industrial sector, but deeper economic integration within Europe is expected to offset part of this drag.

OECD Asia-Pacific is expected to follow a more uneven growth path, with its industrial sector continuing to face competitive pressure from China and other manufacturing hubs. Growth is projected to slow to 0.8% p.a. over the period from 2030 to 2040 before picking up to 1.0% p.a. from 2040 to 2050. This results in an average growth rate of 1.0% p.a. from 2025 to 2050. This modest acceleration reflects China's transition towards a higher-income growth model and a gradual easing in its growth rates, which is expected to support stronger activity elsewhere in the region. It is likely that technological progress and productivity gains will help offset the country's demographic decline but they are unlikely to fully reverse it.

Latin America is expected to expand at an average rate of 2.1% p.a. from 2025 to 2050, supported by contributions from agriculture and mining, alongside a growing domestic services sector. Growth is projected at 2.2% p.a. over the period from 2030 to 2040, before slowing to approximately 1.7% p.a. from 2040 to 2050, as demographic trends increasingly weigh on regional momentum. Brazil is expected to remain the region's main centre of economic activity, supported by its large agricultural base and ongoing structural reforms that support longer-term growth. Other economies, including Chile and Argentina, have the scope to achieve relatively stronger growth through 2040, although demographic headwinds are expected to constrain regional expansion towards 2050.

In the **Middle East**, growth is expected to average 2.7% p.a. from 2025 to 2050. This is supported by a growing population, ongoing diversification efforts, a continued expansion of the industrial sector, alongside strengthening trade links with India and China. Growth is projected at 3.0% p.a. during the period from 2030 to 2040 before moderating to 2.3% p.a. from 2040 to 2050, reflecting base effects after higher economic growth in the previous decade and rising competition from other regions.

Africa is expected to expand at an average rate of 4.4% p.a. from 2025 to 2050, driven by a rapidly growing working-age population. The continent is projected to remain the only major region to record accelerating growth, with expansion rising from 4.1% p.a. from 2030 to 2040 to 5.0% p.a. over the period from 2040 to 2050. Strong global demand for commodities, a gradually expanding industrial base and a growing domestic services sector are expected to be the main drivers of this trend. Key challenges will include strengthening global integration and trade linkages, containing inflationary pressures, attracting foreign investment and improving currency stability across major economies.

India is expected to remain the world's fastest-growing major economy throughout the outlook period, expanding at an average rate of 5.7% p.a. from 2025 to 2050. Growth is projected at 5.8% p.a. during the period from 2030 to 2040, before easing to 5.1% p.a. from 2040 to 2050, reflecting base effects. Strong infrastructure spending and sustained government support for key sectors, including technology, pharmaceuticals and telecommunications, are expected to further entrench India's role as a global manufacturing hub. A large and growing population, including an expanding middle class, is set to support domestic demand, which is expected to be further reinforced by improvements in education and labour-market incentives.

Geoeconomic fragmentation is expected to reshape India's external trade environment, requiring a more focused approach to managing trade relationships across major partners.

At the same time, given India's size, scale and growing role in global supply-chain diversification, it is well positioned to continue deepening economic integration, even within a more fragmented global system. The primary policy challenge will likely remain improving labour-market conditions and expanding employment opportunities to sustain the strong growth momentum.

China's economic growth is expected to decelerate gradually as the economy matures and demographic constraints become more binding. Growth is projected at 3.6% p.a. from 2030 to 2040 before slowing to 2.4% p.a. over the period from 2040 to 2050, resulting in an average growth rate of 3.3% p.a. from 2025 to 2050. The slowdown is expected to be driven primarily by demographics and base effects, rather than a loss of industrial capacity. China is set to remain a major global manufacturing and industrial hub, while continuing its move up the value chain towards higher-technology production. The country has strong positions in renewables, EVs, advanced chips and AI-related industries. Lower-tech manufacturing is expected to begin shifting to regional partners, primarily Other Asia and increasingly India, as labour costs rise and trading partners diversify supply chains.

A key structural challenge will remain the transition towards a more consumption-driven growth model. High household savings continue to constrain domestic demand, reflecting limited social safety nets. Expanding social benefits could support consumption over time, but this will need to be balanced against China's global export and manufacturing dominance. In practice, progress towards higher consumption is likely to occur gradually as higher-value manufacturing raises incomes rather than through a rapid policy shift. Despite gradual progress in global diversification efforts, China's dominance in critical inputs, such as rare earth minerals, is projected to continue through 2050. While China's share in rare earths is expected to decline, it is still set to retain the largest share of the sector.

In **Other Asia**, growth is expected to remain relatively strong at an average rate of 3.5% p.a. from 2025 to 2050. The region is expected to benefit from ongoing shifts in global manufacturing and industrial activity, with large and dynamic economies such as Vietnam, Indonesia, the Philippines and Malaysia, absorbing a greater share of production as supply chains continue to diversify. Growth is projected at 3.6% p.a. during the period from 2030 to 2040 before moderating to 3.1% p.a. from 2040 to 2050, with favourable demographic trends supporting growth.

Russia is expected to record relatively steady economic growth over the long term, despite an ongoing demographic decline weighing on potential output. Growth is projected at 1.4% p.a. during the period from 2030 to 2040 and 1.1% p.a. from 2040 to 2050, resulting in an average of 1.3% p.a. from 2025 to 2050. Inflationary pressures are expected to ease over time, while structural constraints will likely persist, although this is expected to be partly offset by productivity gains and deeper economic ties with China and India.

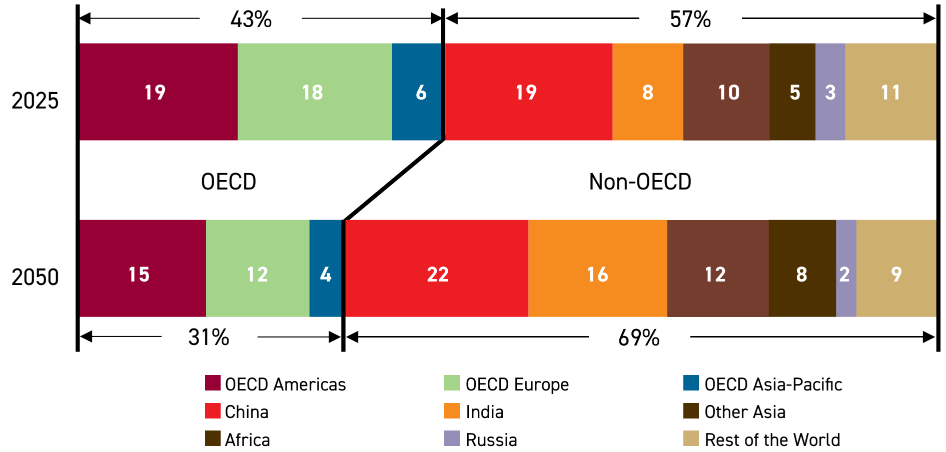
Other Eurasia is expected to expand at an average rate of 2.8% p.a. from 2025 to 2050, following a mildly decelerating growth profile. Growth is projected at 2.8% p.a. during the period from 2030 to 2040 before easing to 2.2% p.a. from 2040 to 2050. This trajectory is set to be supported by external demand, as well as rising commodity demand and improving labour productivity.

Other Europe is expected to follow a growth trajectory broadly in line with OECD Europe, reflecting close economic ties and a slightly declining population. Growth is projected at 0.9%

p.a. during the period from 2030 to 2040, before edging up to 1.0% p.a. from 2040 to 2050. This results in an average growth rate of 1.3% p.a. from 2025 to 2050. The modest acceleration in the latter period is expected to be driven by gradual gains in labour productivity.

With higher growth rates in non-OECD economies, the region's share of the global economy is expected to rise from roughly 57% in 2025 to 69% in 2050 (Figure 1.6). India is projected to double its share, increasing from 8% in 2025 to 16% in 2050, corresponding to an absolute increase of around $44 trillion. China's share is expected to grow from 19% to 22%, adding approximately $43 trillion to the global economy. Africa will also see an expansion in its share of the global economy from 5% in 2025 to 8% in 2050, adding almost $19 trillion in absolute terms.

Figure 1.6
Distribution of global GDP, 2025 and 2050, %



Source: OPEC.

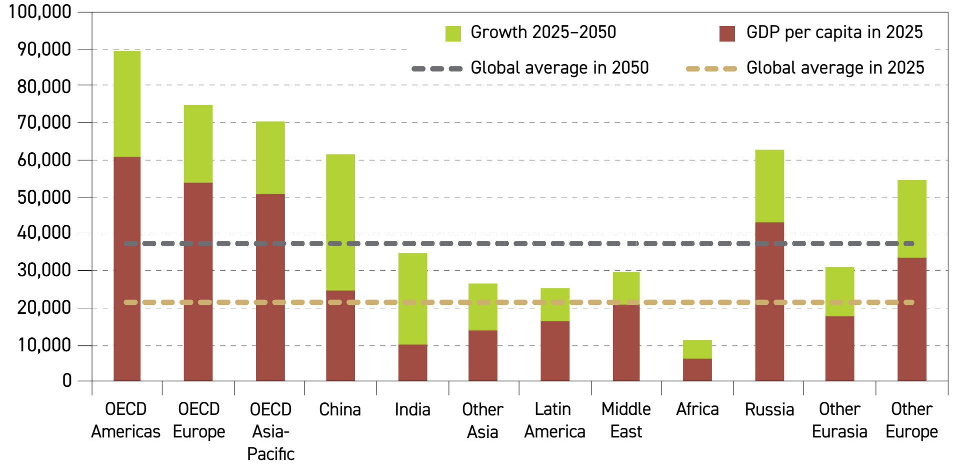
In terms of GDP per capita (Figure 1.7), advanced economies are expected to widen the gap with emerging economies, reflecting a combination of continued growth and declining populations. However, GDP per capita growth in China is expected to outpace that of the OECD economies, while India is expected to grow broadly in line with the OECD.

Across the OECD economies, OECD Americas is expected to outpace the other regions, expanding by $28,500 to reach $89,400 by 2050 and remaining the highest globally. OECD Europe is expected to reach $74,800, while OECD Asia-Pacific is set to reach $70,100 by 2050.

India's GDP per capita is projected to rise from $10,000 in 2025 to $34,600 in 2050, more than tripling over the period. The global average GDP per capita is expected to increase from $21,500 in 2025 to $37,100 in 2050 (2021 PPP).

Outside of India and China, GDP per capita in Other Asia is expected to roughly double from 2025 to 2050, reaching $26,800 and surpassing Latin America. Africa is expected to reach a GDP per capita of $11,400 by 2050 while the Middle East is anticipated to reach $29,800. Russia's GDP per capita is set to remain slightly above China at $62,600 by the end of the forecast period.

Figure 1.7
Real GDP per capita in 2025 and 2050, $(2021 PPP)



Source: OPEC.

1.3 Energy policies

Energy policies are pivotal drivers of both the medium- and long-term energy landscape. Recent developments highlight how national approaches are increasingly diverging rather than converging around a single energy path. This outlook undertakes a comprehensive examination of energy policies, evaluating their viability through their progress, potential implementation challenges and time frames, and the need to balance energy security, affordability, access and sustainability.

1.3.1 Climate action and negotiations

As the world enters the second half of this critical decade, Parties to the Paris Agreement face increasing pressure to accelerate climate action and close gaps in climate ambition, implementation and finance. At the same time, the geopolitical landscape continues to become more complex, with far-reaching implications for international cooperation.

A major development has been the renewed withdrawal of the US from global climate governance. In early 2025, the US government initiated a year-long process to withdraw from the Paris Agreement. In early 2026, it signalled its intention to disengage from a broad range of UN institutions, including the United Nations Framework Convention on Climate Change (UNFCCC) and the Intergovernmental Panel on Climate Change (IPCC), as reflected by its absence from the latest negotiation sessions. These decisions were accompanied by significant national policy reversals, including the cancellation of clean energy incentives, the suspension of renewable energy projects, the rollback of environmental regulations and steps to eliminate the Environmental Protection Agency (EPA)'s authority to regulate greenhouse gas (GHG) emissions.

Such shifts raised concerns about the implications for collective climate efforts, international finance and the pace of implementation in other economies. For instance, the Green Climate Fund (GCF), the world's largest dedicated climate fund, has already confirmed that the United States has withdrawn from the Fund and vacated its Board seat, renewing concerns about the adequacy and predictability of funding. This withdrawal removes a major contributor and influential shareholder from the GCF, adding to uncertainty over future climate finance flows to developing countries. The removal of US financial and technical contributions has also heightened uncertainty about scientific cooperation.

In addition, the broader policy environment continues to be shaped by multiple headwinds. The EU communicated its third round of nationally determined contributions (NDCs) in November 2025. This contained significant delays and adjusted elements to its 2035 ICE regulation. Elsewhere, national debates in major economies, such as Canada and Australia, have also led to delays or revisions in climate mitigation action. Negotiations on an internationally legally binding instrument on plastics (INC), as well as on emissions from international shipping and other UN environmental and sectoral processes, have not reached consensus and stalled, necessitating additional negotiation sessions. Developing countries have also raised concerns about emerging unilateral trade restricting measures, particularly the EU's Carbon Border Adjustment Mechanism (CBAM), considering the potential adverse economic impacts and the need for coherence with the principles of equity and common but differentiated responsibilities and respective capabilities (CBDR-RC).

These dynamics contributed to a challenging context for COP30 in Belém, Brazil, last year. Negotiations were intense and focused on sensitive issues, including developed countries' climate finance obligations under Article 9.1 of the Paris Agreement, climate-related trade restrictive unilateral measures, just transition pathways, the mitigation ambition gap, response measures, the integration of scientific assessments into decision-making, and matters relating to the Global Stocktake (GST) process. Adaptation featured prominently at COP30, while the Brazilian Presidency also sought to advance a process towards sectoral and thematic roadmaps, including voluntary approaches to energy transitions and deforestation. Discussions on these roadmaps remained inconclusive, as developing countries argued that many proposed approaches undermined the core principles of multilateralism and were inconsistent with realities. Brazil also launched the Tropical Forests Forever Facility, aimed at providing long-term financial support for the protection of tropical forests, with initial pledges amounting to $6.7 billion.

The Belém Political Package was adopted, comprising decisions across 18 negotiating items. Central among these is the Global Mutirão Decision: uniting humanity in a global mobilization against climate change, which draws on a Brazilian concept of collective effort and mutual support, and places an emphasis on implementation, adaptation finance, just transitions and trade-related considerations. In particular, Parties launched the Global Implementation Accelerator and the Belém Mission to 1.5°C, aimed at strengthening support for implementing NDCs and National Adaptation Plans (NAPs), with progress to be reviewed at COP31.

The *2025 NDC Synthesis Report* was also considered in Belém, noting that by the close of COP30, 119 Parties representing about 74% of global emissions had submitted new or updated NDCs. These submissions reflected incremental progress, but they also indicated that the collective ambition gap remains significant, underscoring the need to address ambition, implementation and finance gaps. By early 2026, 135 Parties had submitted new or updated NDCs, covering around 78% of global emissions. These include contributions from major economies; however, their assessments show that collective efforts continue to fall short of pathways consistent with the Paris Agreement's long-term goals.

On adaptation, Parties sought to finalize a framework for monitoring progress under the Global Goal on Adaptation (GGA). Negotiations proved challenging, particularly with regard to means of implementation. A consolidated set of adaptation indicators was adopted. However, some delegations expressed concerns about the scientific robustness and measurability of certain indicators. These issues are likely to be considered in 2026, noting that adopted Conference of the Parties (COP) decisions cannot be reopened.

Doubts were further raised about the achievement of the Glasgow commitment to double adaptation finance by 2025, considering declining finance trends in recent years. On the other hand, estimates indicate developing countries' adaptation needs could reach $365 billion annually by 2035. Thus, the Parties eventually agreed to a new target to triple annual climate adaptation finance by 2035, although the baseline year for this tripling has yet to be specified.

Progress was made on loss and damage, as well as on finance-related issues. This includes the further operationalization of the Fund for Responding to Loss and Damage through new guidance and the initiation of funding requests under the Barbados Implementation Modalities. The Baku to Belém roadmap, developed jointly by the COP29 and COP30 Presidencies, outlined proposed actions to support the mobilization of $1.3 trillion annually by 2035. The Parties did not formally adopt the roadmap, but they agreed to consider advancing its recommendations. Negotiations under Article 9.1 proved difficult, resulting in the establishment of a two-year work programme to clarify public finance obligations. A separate dialogue was also initiated on Article 2.1(c) to enhance coherence in aligning financial flows with climate goals.

Negotiations on enhanced mitigation efforts by 2030 and just transition pathways continued to reflect long-standing divergences among Parties. The mitigation work programme and its link to global stocktake outcomes remained contentious. Yet, the Parties launched a Belém Mechanism for Just Global Transition aimed at strengthening cooperation and capacity building, as well as the Technology Implementation Program to support technological priorities in developing countries. On trade-related issues, the final decision mandated a three-year dialogue on climate and trade in cooperation with the World Trade Organization (WTO) and other relevant institutions, culminating in a synthesis report in 2028.

COP30 also adopted operational guidance for the Article 6.4 mechanism, enabling progress towards market-based cooperation. Key elements included enhanced transparency requirements, capacity-building support, deadlines for transitioning Clean Development Mechanism (CDM) projects and early financing needed to support the mechanism's eventual self-sustainability. These steps are intended to ensure environmental integrity and facilitate broader participation once the mechanism becomes fully operational.

Overall, COP30 was expected to be a critical milestone for advancing the Paris Agreement's implementation, including a follow-up on the first global stocktake adopted at COP28 in Dubai and the new collective quantified goal on climate finance agreed at COP29 in Baku. It convened amid increased geopolitical fragmentation and logistical hurdles. While negotiations revealed deep divisions on several substantive issues, COP30 still demonstrated that international cooperation for climate action remains possible.

Through extensive pre-sessional engagement, including intersessional negotiations in Bonn and a series of Presidency-led consultations, efforts were made to foster trust and encourage collaboration. Momentum was also supported in 2025 through an enhanced Climate Week platform, with events held in Panama City in May and Addis Ababa in September aiming to bridge technical negotiations and on-the-ground implementation.

Beyond the UNFCCC process, BRICS leaders, meeting in Rio de Janeiro in July 2025, reaffirmed the importance of a fair, rules-based multilateral trading system, highlighting that hydrocarbons will continue to play a key role in the energy mix of emerging economies and that transitions should respect national circumstances and technological neutrality. They highlighted Article 6 of the Paris Agreement as a mechanism to mobilize investment and reiterated the need for accessible and affordable finance. The G20 Summit, hosted by South Africa in November 2025, endorsed the outcomes of COP30 and stressed the importance of national just transition pathways as enablers of sustainable development, while also underscoring energy security as a critical dimension of economic stability. In December 2025, the seventh UN Environment Assembly (UNEA) in Nairobi adopted several resolutions, including on the environmentally sound management of critical minerals and metals.

Looking ahead to COP31 in Antalya, Türkiye, scheduled to take place in November 2026, engagement by Parties will be essential to ensure that international cooperation continues in an equitable and balanced manner, aligned with national circumstances, development priorities and the principles of the Convention and the Paris Agreement. Australia will provide the President of Negotiations, while central to this year's negotiations are likely to be the delivery of predictable and scaled-up finance, trade-related matters and issues deferred from earlier sessions, such as discussions on a potential institutional reform of the UNFCCC to preserve its credibility and inclusiveness.

1.3.2 Regional energy policies

Across the world's major energy consuming economies, energy policy frameworks are being reset as governments reconcile security of supply, affordability, infrastructure delivery, and long-term sustainability commitments. Rather than discarding stated objectives, authorities are recalibrating delivery pathways to reflect the capital realities of expanding and modernizing energy systems.

This recalibration reflects the recognition that energy system shifts involve substantial capital outlays, long asset lives and close links to employment, competitiveness and regional development. Policy adjustments adopted during the outlook period under review point to a growing recognition of these realities among decision makers. The resulting landscape is not uniform: countries, regions and economic blocs are pursuing pathways aligned with their resource endowments, technological capabilities, political constraints and strategic positioning. Such diversity shapes global energy balances and influences the speed at which sectoral developments are likely to materialize.

The US: strengthening domestic energy supply and regulatory reform

In the US, energy policy has shifted towards a stronger emphasis on domestic hydrocarbon production, particularly oil and natural gas and its associated infrastructure, while narrowing the scope of federal support for some renewable technologies. Executive actions have accelerated permitting and expanded leasing across federal lands and waters. New guidance has also introduced a faster, more predictable process for LNG export approvals, reinforcing an export-led growth agenda and related midstream investment.

The One Big Beautiful Bill Act (OBBBA), signed into law on 4 July 2025, and formally titled An Act to provide for reconciliation pursuant to title II of House Concurrent Resolution 14, has emerged as the central energy related statute since last Outlook. The legislation simultaneously expands federal support for domestic hydrocarbon production and scales back incentives for renewable energy and electrification, thereby reshaping the US fiscal and regulatory framework for energy investment.

Specific changes include the federal EV purchase credit ceasing for acquisitions after 30 September 2025, residential incentives for thermal insulation and heat pumps expiring at year end 2025, and the hydrogen production credit being phased out by December 2027.

The Department of Energy later cancelled more than $13 billion in previously allocated but unspent clean energy funds, returning them to the US Treasury. This further narrowed the federal funding pipeline for renewable and clean energy programmes established under the previous Administration's Inflation Reduction Act.

In October 2025, the Department of the Interior finalized the Federal Lands Energy Development Rule. This targets the upstream oil and gas sector and aims to accelerate crude oil and natural gas production on federal lands and waters. The rule shortened review timelines and expanded eligible acreage, providing clearer multi-year visibility for producers and supporting a sustained increase in domestic output.

Following passage of the One Big Beautiful Bill Act, relevant federal agencies issued new permitting guidance, unlocking several large scale upstream and midstream projects. Initial data from the fourth quarter of 2025 indicates a measurable uptick in drilling permits and infrastructure investment commitments, signalling renewed confidence among domestic producers. In parallel, LNG export approvals accelerated from late 2025 onward.

In January 2026, the Federal Energy Regulatory Commission issued new guidance under the American Natural Gas Export Enhancement Act. This targets the midstream and LNG export sector and aims to expand natural gas pipeline and terminal capacity. This measure streamlined approval processes for new export facilities and storage projects, with the stated goal of strengthening the role of the US in international gas supply.

Beyond oil and gas, the administration acted to preserve existing dispatchable generation capacity. In February 2026, the US President signed an Executive Order, entitled, Strengthening United States National Defense With America's Beautiful Clean Coal Power Generation Fleet. This directs the Department of Energy to support the continued operation of coal fired power plants as a matter of grid reliability and national security. By the end of 2025, the Department had already preserved more than 17 gigawatts (GW) of coal-fired electricity generation through earlier policy measures.

Rising domestic content mandates, increasing from 45% US-sourced components in 2025, to 50% in 2026 and 55% from 2027, now apply to solar modules, battery packs, critical minerals and energy equipment. These requirements have narrowed the pool of eligible suppliers, raised some near term capital costs and contributed to longer project lead times.

The EU: energy security and affordability move to the forefront while climate ambition recalibrated

The EU maintains its formal commitment to long-term energy and climate objectives yet faces practical implementation constraints. These relate to infrastructure readiness, regulatory processes and investment requirements, which may influence the pace of deployment and alignment between planned and actual timelines. The Clean Industrial Deal and Affordable Energy Action Plan constitute the overarching policy architecture, with both explicitly recognizing that energy security considerations are influencing the pace and sequencing of developments.

The implementation of this policy framework proceeded through 2025 without substantive statutory amendment, reflecting a widening gap between policy ambition and practical delivery capacity. The policy focus increasingly shifted towards removing project-level bottlenecks and improving administrative processes, rather than introducing new headline targets. This signals a move from target setting towards near-term implementation and risk management.

Looking ahead, the European Commission's 2026 Work Programme, published in October 2025 under the title 'Europe's Independence Moment', schedules an Electrification Strategy, an Energy Security Package and a Heating and Cooling initiative for the first quarter of 2026, with new EU targets for the post-2030 period and an Energy Omnibus simplification package to follow later in the year. The programme reinforces the Commission's emphasis on completing the internal energy market, strengthening infrastructure and improving energy affordability as central policy priorities for the period ahead.

The Revised Renewable Energy Directive (RED III) mandates 42.5% renewable energy in gross final consumption by 2030, up from 32% under RED II, with an aspirational goal of 45%. The directive establishes streamlined permitting, with indicative time limits of 12 to 18 months. It also introduces binding targets for the use of renewable fuels in transport and industry, notably through strengthened minimum shares for renewable fuels of non-biological origin (RFNBOs), such as green hydrogen and e-fuels. Nevertheless, member state experience through 2025 shows that subsequent approvals, including grid-connection assessments, environmental reviews and local authority clearances, often remain lengthy. This effectively extends total development cycles beyond stated timelines.

The EU's 2035 prohibition on new ICE vehicles remains legally in force despite sustained debate among member states and manufacturers, and it continues to guide investment planning across the automotive value chain.

Political discussions throughout 2025 explored the role of e-fuels, possible flexibilities and alternative stringency paths. However, no formal legislative amendments were adopted, preserving the current legal framework while maintaining a degree of regulatory uncertainty for manufacturers' long-term investment decisions.

The FuelEU Maritime Regulation imposes rising operational compliance requirements, with a 2% greenhouse-gas-intensity reduction in 2025 that increases to 80% by 2050. Early-2025 operational experience indicates higher compliance costs and emerging differences in competitive exposure, with some shipping operators adjusting fuel-sourcing strategies and redeploying vessel schedules to minimize exposure to regulated routes, particularly on short-sea and intra-European services.

The ReFuelEU Aviation regulation establishes binding Sustainable Aviation Fuel (SAF) blending mandates, targeting a 70% SAF share by 2050. Industry assessments point to significant implementation challenges, including scalability limits, persistent cost differentials and currently limited public financial support. The scheduled 2027 policy review has therefore emerged as a key milestone for stakeholders seeking to refine timelines and align SAF requirements with airline financial capacity and fuel-supply availability.

The Energy Performance of Buildings Directive continues to advance, but its roll-out is occurring amid elevated construction and renovation costs, tighter household budgets and administrative complexity at national and local levels. This has led to several member states sequencing measures more gradually and prioritizing the most cost-effective segments of the building stock.

The UK: balancing upstream resilience with accelerated low-emission electricity deployment

The UK is pursuing a parallel energy policy that seeks to balance the commercial viability of upstream oil and gas with the expansion of renewable energy. The Energy Profits Levy, which is currently legislated to apply until 31 March 2030, adds to fiscal costs for North Sea producers, with a particularly strong impact on higher-cost and marginal field developments.

The Clean Power 2030 Action Plan provides the main statutory framework for electricity system development, building on the existing Contracts for Difference (CfD) scheme, which procures renewable-based generation through competitive auctions that also incorporate industrial-policy objectives. Allocation Round 7 in August 2025 introduced a 'Clean Industry Bonus' for offshore wind projects demonstrating UK-sourced supply-chain investment. This marked a shift from a purely cost-minimization focus towards greater emphasis on supply-chain security and domestic manufacturing capacity.

Great British Energy continued its operational build-up during 2025 without major strategic amendments. It is focusing on floating offshore wind, grid-scale battery energy storage systems (BESS) and long-duration electricity storage (LDES) technologies to support grid adequacy and renewable integration. The Clean Power 2030 Action Plan anchors the government's commitment to a substantially renewables-based generation mix by 2030, while grid stability is to be managed through evolving market design and capacity mechanisms.

In transport and heating, the previous government's September 2023 decision to defer the phase-out date for new ICE vehicle sales to 2035 and to reverse proposed restrictions on new gas-boiler installations remained in force in 2025. These decisions reflect a recognition of near-term constraints on electrification and the practical considerations around upgrading vehicle fleets and heating infrastructure.

Japan: deepening nuclear restarts as low-emission power expands

Japan's energy policy continues to operate within well-known structural constraints, but in 2025 it shifted further towards the concrete implementation of the Green Transformation (GX) and S+3E approach, which balances Safety, Energy Security, Economic Efficiency and Environment. The Seventh Strategic Energy Plan, approved in February 2025 alongside the GX2040 Vision, establishes an integrated framework for Japan's long term energy mix, investment priorities and a low emissions pathway to 2040.

By 2040, the plan targets renewables at 40–50% of electricity generation, nuclear at about 20% and thermal generation at 30–40%, so that low emission sources together could account for around 70% of the power mix. This structure keeps LNG as a key fuel for system flexibility and energy security, while offshore wind targets of 10 GW by 2030 and 30–45 GW by 2040 signal steady, incremental growth, rather than any dramatic shift in project pipelines.

In 2025, Japan also pushed forward several enabling measures crucial for mobilizing private investment. Hydrogen and ammonia value chains advanced as anchor projects were identified and prepared for certification, and renewable support schemes continued shifting from fixed Feed-in Tariffs (FIT) towards Feed-in Premiums (FIP) and Contracts for Difference, bringing remuneration schemes closer to international practice and supporting more cost efficient deployment. These moves support Japan's wider goals around a low-emission industrial sector and a more flexible power system operation.

Supportive market and infrastructure frameworks progressed as well. GX legislation establishes a domestic emissions trading scheme to launch in fiscal year 2026, complementing the country's growth-oriented carbon pricing initiative. Amendments to the Renewable Energy Sea Area Utilisation Act, passed in June 2025, extended offshore wind development rights into the exclusive economic zone (EEZ), unlocking new sites for larger scale and hybrid projects.

Nuclear policy took a significant step forward with local authorization for the restart of the Kashiwazaki-Kariwa plant, and TEPCO's subsequent restart of Unit 6 in early 2026, marking its first nuclear reactor return to commercial operation since 2011. These steps reaffirm nuclear's ongoing role in providing low-emission baseload power under stricter safety standards.

At the same time, structural challenges continue to influence the pace and shape of the transition. Public concerns over nuclear safety affect restart timelines and potential lifetime extensions, Japan's mountainous and densely populated terrain limits options for large scale renewables, and regional transmission bottlenecks hinder the movement of variable renewable output from resource rich areas to high demand centres. In this context, achieving the 2040 targets in the Seventh Strategic Energy Plan will depend on steady progress in permitting, grid upgrades and the roll-out of technology over the next decade.

South Korea: safeguarding power security while developing low-emission fuels

South Korea is addressing rising electricity demand, driven primarily by AI data centres, semiconductor expansion and broader electrification, through a strategy that prioritizes reliable capacity alongside clean energy sources.

The 11th Basic Plan for Electricity Supply and Demand (2024–2038), finalized in February 2025, targets emission-free generation of approximately 71% by 2038, up from about 39% in 2023, and explicitly incorporates demand growth from high-load sectors. It relies on expanded nuclear output, faster deployment of renewables and additional flexibility options to handle peak loads and sustain reserve margins, while retaining gas-fired plants as an important balancing resource that underpins continued LNG imports in the medium term. Site selection and preparation work for new nuclear reactors and other strategic facilities is progressing, with a bidding process for host locations launched in early 2026 and site finalization expected by 2027.

Market mechanisms for low-emission fuels are advancing, though not without adjustments. The Clean Hydrogen Portfolio Standard (CHPS) and the ammonia co-firing framework build on South Korea's LNG procurement experience to provide clearer price signals and contract structures for low-emission hydrogen and ammonia supply. The inaugural CHPS tender in 2024 targeted 6,500 GWh of low-emission hydrogen-to-power capacity but awarded only around 750 GWh to a single project, reflecting investor caution regarding delivery risks and evolving policy design. A follow-on tender of about 3,000 GWh, originally envisaged for 2025, was cancelled in October 2025 to further calibrate ammonia co-firing rules and ensure consistency with coal phase-out objectives.

Efforts to reduce emissions in aviation moved forward with the September 2025 release of the National Sustainable Aviation Fuel Blending Roadmap. It introduces a mandatory 1% SAF blend for international departing flights from 2027, rising to 3–5% by 2030 and 7–10% by 2035, with exact 2030 and 2035 levels to be confirmed in 2026 and 2029. The roadmap supports South Korea's 2050 carbon-neutrality goal by fostering a predictable domestic market for SAF production, logistics and associated infrastructure. From 2028, international carriers must ensure that at least 90% of their annual jet fuel uplift at South Korean airports complies with the SAF blending requirement, anchoring demand at domestic hubs and encouraging the development of a local SAF supply chain.

China: strengthening supply security through large-scale investments and digital grid

China's energy policy framework supports sustained economic growth through a combination of fiscal measures and the parallel development of conventional and low-emission energy infrastructure, ensuring diversified sources to meet rising electricity demand. As the country moves into the 15th FYP period (2026–2030), recent policy documents emphasize energy security, technological self-reliance and digitalization, with advanced manufacturing, clean energy and AI identified as key drivers of what policymakers describe as "new quality productive forces".

Renewable deployment continues under the 14th FYP but is increasingly supported by an expanded national emissions trading system (ETS) and demand-side mechanisms. Guidelines released in August 2025 set out a phased approach: the ETS will cover all major high-emitting industrial sectors by 2027 and begin introducing absolute emissions caps in industries with relatively stable output, before transitioning to a mature system by 2030 with a mix of free and auctioned allowances. This moves beyond the current intensity-based, free-allocation approach towards more effective price signals for emissions reduction.

Complementary draft regulations from the National Development and Reform Commission would introduce binding minimum renewable-electricity consumption shares for major industrial and commercial users. This shifts the focus from supply-side capacity additions to obligations on end-users to procure renewable power and thereby improve integration.

Grid and system policies prioritize absorbing more than 200 GW of new wind and solar capacity annually between 2025 and 2027, while targeting renewable utilization rates above 90% in key regions. This reflects the recognition that recent capacity additions have outpaced network capabilities in some provinces and highlights the need for continued investment in ultra-high-voltage transmission, inter-provincial corridors and utility-scale storage. In the medium term, coal and natural gas remain important for providing flexibility and maintaining system reserve margins, even as their share in generation is expected to decline over time.

Looking ahead to the 15th FYP, the State Grid Corporation of China announced plans in January 2026 to increase fixed-asset investment by around 40% compared with the previous plan period. This now reaches a level of about 4 trillion yuan (around US$574 billion) between 2026 and 2030, averaging roughly US$115 billion per year. The programme prioritizes digital and AI-enabled grid infrastructure, including integrated computing-power networks, intelligent substations and 'AI Plus' applications for system operation and demand management. It also plans to expand cross-provincial and cross-regional transmission capacity, which is expected to increase by more than 30% from end-2025 levels.

These investments are intended to strengthen security of supply, improve operational efficiency and facilitate the reliable integration of high levels of variable renewable generation sources, while maintaining necessary flexibility from conventional sources during the transition.

India: advancing hydrocarbon self-reliance while scaling up other power capacity

India's energy policy pursues a dual track; balancing an expansion of domestic oil and gas output with rapid growth in other capacity to meet rising electricity demand and climate goals.

The Oilfield Regulatory and Development (Amendment) Act, enacted in March 2025, has reshaped upstream licensing to improve project economics, while the tenth Open Acreage Licensing Policy round offered 25 blocks across 13 sedimentary basins, including 19 offshore deep water areas, with bidding extended through December 2025 in response to cautious international interest.

On the power side, non-hydrocarbon capacity (excluding large hydropower) reached over 200 GW by October 2025, and total renewable capacity – including large hydropower – was close to 251 GW, supported by about $22.3 billion in clean energy investment. This acceleration reflects more supportive policy frameworks, lower equipment costs and expanded domestic manufacturing.

At the same time, transmission congestion has become a binding constraint in some states, particularly where grid expansion lags renewable capacity build-out. Meeting the country's 500 GW non hydrocarbon target by 2030 will require substantial additional investment in transmission corridors and energy storage. Storage needs are estimated at roughly 15–20% of variable renewable capacity, which moderates deployment speed and introduces execution risk to stated targets.

ASEAN: advancing cross-border energy integration to strengthen regional energy security

Regional energy cooperation among the Association of Southeast Asian Nations (ASEAN) member states continues to strengthen, as cross border power links take on greater importance for energy security and day-to-day system reliability.

The ASEAN Power Grid (APG) and the Lao PDR-Thailand-Malaysia-Singapore Power Integration Project (LTMS PIP) moved from initial pilot trades into steady commercial operation during 2025. Multidirectional trading arrangements expanded in early 2026, supported by upgraded interconnection planning and Singapore's firm target to import up to 6 GW of low carbon electricity by 2035. This creates a defined long-term market for export oriented hydropower and solar projects in neighbouring countries.

National policy frameworks are being brought into line with this regional architecture. Vietnam's revised Power Development Plan 8 (PDP8), updated in April 2025, pairs higher wind and solar targets with mandatory storage requirements and clearer grid connection rules to improve integration and project sequencing.

In the Philippines, the fourth round of the Green Energy Auction (GEA-4) concluded in November 2025. It awarded more than 10 GW of new renewable capacity, including integrated renewable plus storage systems, through competitive tenders and standardized transmission contracts.

Malaysia's National Energy Transition Roadmap (NETR), together with the Large Scale Solar 6 (LSS6) programme and the new Solar Accelerated Transition Action Programme (Solar ATAP) launched in January 2026, links domestic grid readiness and corporate offtake with potential exports via the APG. This brings national investment plans and emerging opportunities in regional power trade closer together.

1.3.3 Energy policies by sector

Road transport

While EV sales have continued to rise in recent years, the global road transport fleet remains overwhelmingly dominated by ICE vehicles. Governments around the world continue to back electrification, yet policy approaches in the three largest markets have begun to diverge, with some measures becoming less supportive at the margin.

In the US, legislation adopted in July 2025 allowed the main EV purchase tax credit under Section 30D, along with credits for commercial and used clean vehicles, to expire for vehicles acquired after 30 September 2025. This makes new EV purchases noticeably less attractive for many buyers, especially when combined with stricter domestic content rules.

Across the EU, the 2035 phase out regulation for new ICE cars and vans formally remains in place, but political discussions on e-fuels and Euro 7 standards led the European Commission on 15 December 2025 to propose amendments that would preserve a role for certain hybrid and combustion engine vehicles after 2035, while leaving the overall emissions reduction target unchanged. This provides some additional clarity on potential flexibility but still leaves manufacturers with a demanding long-term regulatory framework.

In the UK, the decision to push back the ICE phase-out to 2035 has strengthened expectations that conventional vehicles will keep a meaningful share of the market over the medium term, while the reversal of planned restrictions on new gas-fired boilers points to a slower shift in residential heating as well.

In contrast, China has established itself as the world's largest EV market and exporter, driven by lower battery and vehicle costs and consistent policy backing on both the supply and demand sides. At the same time, recent industrial policy signals indicate that China is widening its strategic focus across a broader range of high technology sectors, which may limit the scope for additional, EV-specific incentives compared with the past decade.

Against this evolving policy landscape, major vehicle manufacturers have adjusted, rather than abandoned, their earlier full electrification targets, extending some deadlines and broadening their line-ups to include more hybrid and ICE options. These changes reflect an assessment that consumer demand, charging networks and critical materials supply chains still require further development before any rapid, full-scale shift across the global fleet becomes feasible.

Aviation

The aviation sector continues to balance objectives related to climate change with the practical realities of fuel cost and availability. The Carbon Offsetting and Reduction Scheme for International Aviation (CORSIA), administered by the International Civil Aviation Organization (ICAO), offers a multilateral framework for monitoring, reporting and offsetting emissions, but provides airlines with only modest direct financial encouragement to switch from conventional jet fuel to SAF. National and regional policies have therefore become the primary drivers of SAF adoption.

Across the US, legislation adopted in July 2025 reduced the maximum SAF production credit under the federal Clean Fuel Production Credit (Section 45Z) from $1.75 to $1.00 per gallon for fuels produced at qualifying facilities, while extending the incentive through 2029. This change has dampened expectations for a rapid expansion of new SAF capacity, as project economics have become less favourable relative to earlier assumptions.

Within the EU, the ReFuelEU Aviation regulation, in force since January 2025, sets mandatory SAF blending requirements that began at 2% of fuel supplied at EU airports in 2025 and rises in steps to 70% by 2050, including a dedicated sub mandate for synthetic e-fuels. Early operational experience in 2025 has highlighted the persistent cost gap between SAF and conventional jet fuel, as well as practical constraints on scaling production quickly enough to meet the more ambitious targets in later years.

Among other key markets, several SAF-related measures have remained broadly stable. The UK's SAF Mandate, effective from 2025, requires fuel suppliers to meet a 2% SAF share in total UK jet fuel supply, rising to 10% in 2030 and 22% in 2040. Japan has established a policy objective of a 10% SAF share in fuel supplied for international flights departing from Japanese airports by 2030. And China continues to support pilot programmes and non binding consumption goals as part of its wider low-emission aviation and carbon market initiatives.

Industry assessments indicate that conventional petroleum-based jet fuel will continue to meet the large majority of global aviation fuel demand through 2030, with SAF remaining a modest, albeit gradually increasing component.

Maritime

The maritime sector operates within a layered regulatory environment where regional initiatives often move ahead of global agreements as regional measures are implemented more quickly than emerging global rules. The International Maritime Organization (IMO) has developed a Net Zero Framework that would introduce mandatory fuel greenhouse gas intensity standards and a two tier carbon pricing and reward mechanism for international shipping. The package gained approval in principle at the Marine Environment Protection Committee's 83rd session in April 2025.

However, an extraordinary Marine Environment Protection Committee (MEPC) meeting in October 2025 ended without formal adoption of the associated legal amendments, following procedural concerns and differing positions among member states. Final agreement has therefore been deferred to 2026 and a definitive timeline for entry into force has yet to be confirmed.

In the meantime, shipping companies are expected to comply with an increasingly diverse set of regional requirements and a growing number of region specific regulations. The EU's inclusion of maritime transport in its Emissions Trading System (EU ETS), together with the FuelEU Maritime regulation, effective from January 2025, applies carbon pricing and GHG intensity limits to voyages involving EU ports, creating a regional benchmark that does not yet have an equivalent global framework. This creates overlapping compliance obligations for operators serving multiple markets and can lead to overlapping and sometimes duplicative compliance obligations for operators serving multiple markets.

Given the current mix of evolving rules and the high upfront costs of zero emission vessels and fuels, shipowners are prioritizing near-term investments in proven efficiency measures and transitional fuels such as biofuels and LNG, while large-scale deployment of fully zero-emission ships remains limited.

1.4 Technology and innovation

Technology continues to play a central role in the energy sector and underpins several key assumptions in this year's long-term outlook. Established technologies drive efficiency gains and operational improvements in the short and medium term, while emerging solutions are reflected in longer-term projections as they move closer to commercial deployment. Recent advances in AI and digitalization are already visible across oil and gas exploration, production, power systems and end-use sectors, supporting more efficient and data-driven operations.

In this Outlook, technological progress is assumed to continue along an evolutionary path, with actual contributions shaped by the speed and scale of deployment over the outlook period. The subsections that follow review the main developments by sector and by fuel.

1.4.1 Focus on transportation technologies

Road transportation

Technological progress in road transport continues to improve both conventional ICE platforms and electrification pathways. In passenger vehicles, high-efficiency hybrid powertrains and synthetic drop-in fuels are extending the useful life of existing ICE fleets, with thermal efficiencies now exceeding 48% through advanced turbocharging, ultra-high-pressure injection and variable valve timing. These improvements reduce fuel consumption and enable lower life-cycle emissions when paired with synthetic fuels produced from renewable hydrogen and captured carbon dioxide (CO_2).

Freight, hydrogen and natural gas engines have reached a more advanced stage of deployment. Pilot and early commercial projects indicate that hydrogen ICEs can match diesel performance in heavy duty applications, with peak efficiencies around 44%, while modern natural gas engines deliver comparable power with lower pollutant emissions. Diesel technology is also evolving, with improved after-treatment systems and more sophisticated control and diagnostic tools, including selected AI applications, helping to enhance fuel economy and comply with stringent nitrogen oxides (NO_x) standards.

Air transportation

Engine manufacturers are achieving steady efficiency gains through new turbofan designs that improve thermal performance and aerodynamics, reducing fuel burn by around 10–15% in the latest-generation aircraft. SAF production pathways, including alcohol-to-jet and power-to-liquid processes are expanding, and are now used in commercial flights, typically in blends of up to 50% with conventional jet fuel and in line with current technical standards. It should be noted, however, that mandated SAF shares in total fuel use remain relatively low.

Hydrogen and electric propulsion solutions remain at the demonstration stage, with progress concentrated in short-haul and smaller aircraft, where hydrogen combustion and hybrid-electric systems are undergoing flight testing and early certification.

Maritime transportation

In maritime shipping, fuel and engine innovation is progressing around alternative fuel options and incremental upgrades to existing fleets. Methanol dual-fuel engines have moved from pilot projects to early commercial deployment, allowing vessels to operate on conventional fuels while retaining the capability to use lower-carbon alternatives as supply chains expand.

Orders for LNG-fuelled vessels remain robust, supported by improvements in engine design and onboard fuel management systems that help reduce emissions relative to traditional marine fuels. Ammonia-fuelled engines are advancing through bench and marine validation tests, alongside port-level bunkering pilots and emerging safety frameworks. Hydrogen applications are being tested in short-sea shipping and ferry segments through fuel cell and combustion demonstrators.

Digital tools, including AI-assisted route optimization and performance monitoring, together with wind-assisted propulsion systems, can deliver fuel savings on retrofitted vessels, with typical estimates at around 5–10% depending on route and operating conditions.

1.4.2 Technologies by fuel

Oil

AI and digital tools are now standard features in upstream operations, supporting real-time decision-making and operational optimization across the value chain. In drilling, human-supervised autonomous systems are being introduced to handle routine tasks such as weight-on-bit control, rotary speed adjustments, mud-flow regulation and trajectory corrections, while human oversight is retained for critical decisions. These systems improve safety, consistency and efficiency, particularly in directional and horizontal drilling and in complex formations such as shale and deepwater.

Advanced well completions further enhance recovery through high-precision designs, Internet of Things (IoT)-enabled downhole sensors and inflow-control valves that enable zonal isolation, real-time flow monitoring and active reservoir management. Digital reservoir management integrates real-time data, advanced analytics and physics-based models to create continuously updated digital twins, improving forecasting accuracy and development planning.

In parallel, some drilling rigs are shifting from diesel systems to grid-, gas- or storage-based power, which reduces local emissions and supports closer integration with modern control and automation systems.

Natural gas

Recent advances in gas-fired power and infrastructure have improved both efficiency and flexibility. New turbine materials and designs support higher firing temperatures and more complete combustion, lowering NO_x emissions.

Ceramic-matrix-composite components and next-generation lean-premixed or catalytic burners perform well with high-hydrogen blends, while dry low emissions (DLE) retrofits extend load ranges on existing units.

At the plant level, modular and containerized gas systems are increasingly deployed near load centres such as data centres and industrial microgrids, often paired with renewables and storage to reach combined-heat-and-power efficiencies of around 80%. Digitalization of the gas supply chain through IoT sensors enables real-time leak detection and predictive maintenance, cutting downtime and operating costs. Looking ahead, hydrogen-ready turbines and compact carbon-capture pilots are preparing existing gas assets for lower-emission operation while preserving their role in enabling system flexibility.

Coal

Innovation in coal-fired power focuses on incremental, system integrated upgrades that improve performance and reduce emissions. AI platforms analyze real-time data from boilers, turbines and emissions control systems to optimize heat rates and lower pollutants. Ammonia co-firing has emerged as a practical transitional option, with demonstrations such as the planned 20% share at Japan's Hekinan plant showing that existing units can co-fire ammonia while maintaining stable combustion and controlled emissions. Advanced multi pollutant control systems that combine sorbents, scrubbers and filtration in modular units are now deployed at commercial scale in parts of Other Asia.

Modular small-scale gasification systems, with capacities up to around 50 megawatts (MW), are being piloted for remote industrial clusters, producing synthesis gas for power, chemicals or hydrogen with good potential for carbon capture integration. Automated unmanned inspection technologies, including drones and robotic crawlers, support predictive maintenance and help limit fugitive emissions as plants become more flexible and more closely managed through digital tools.

Nuclear

Nuclear technology has entered a more dynamic phase, with renewed momentum in small modular reactors (SMRs) and early progress in fusion. Clearer regulatory pathways and improved access to financing since 2025 have accelerated next-generation designs. In the US, the NuScale Power SMR design has been certified at 77 megawatts electric (MWe) and has received around $1.6 billion in federal support. The UK continues to advance Rolls-Royce's 470 MWe SMR programme, with first units targeted for construction before the end of the decade.

Fusion has achieved important technical milestones, including repeated net-energy-gain experiments at the US National Ignition Facility. Commercial efforts such as Helion's planned 50 MW project in Washington State, backed by a power-purchase agreement with Microsoft, and progress by Commonwealth Fusion Systems, TAE Technologies and General Fusion, point towards demonstration-scale plants in the early 2030s.

Solar

Solar technology continues its rapid move from laboratory to commercial deployment. Perovskite–silicon tandem cells now offer higher efficiencies than conventional modules, while lightweight and flexible panels are opening new rooftop and non-traditional surface applications. Printed perovskite films are undergoing outdoor testing, and dual-use solutions such as floating photovoltaic (PV) and agrivoltaics are gaining traction where land-use constraints are significant.

Digitalization and system integration are reshaping deployment. AI-supported optimization, autonomous drones and dry-cleaning robots reduce maintenance costs and water use. The solar-plus-storage model is expanding in isolated or disturbance-prone networks, supported by grid-forming inverters that help maintain voltage and frequency. Pilot projects are also testing direct solar-to-hydrogen pathways through photoelectrochemical reactors as a complement to conventional electrolysis.

Wind

Wind-technology development has shifted from turbine scaling to performance optimization through digitalization, advanced materials and better grid integration. Fleet-wide forecasting and optimization platforms developed by major manufacturers use high-resolution meteorological and turbine data to improve availability and reduce operating costs. Automation via drones and robotic crawlers supports blade and tower inspection, while thermoplastic composite blades and advanced erosion-resistant coatings extend component lifetimes. Silicon-carbide converters and grid-forming controls allow wind farms to contribute more actively to system stability.

Hydrogen

Electrolysis technology has seen gains in efficiency, cost and modularity. Anion-exchange-membrane units are emerging as an intermediate option between alkaline and proton-exchange-membrane designs, while non-precious-metal catalysts reduce reliance on scarce materials. High-temperature solid-oxide electrolysers have reached electrical efficiencies approaching 90% when coupled with industrial waste heat.

Transport and storage solutions are diversifying. Liquid organic hydrogen carriers offer compatibility with existing logistics, and modular ammonia-cracking units near ports enable reconversion close to demand centres. Downstream, solid-oxide and direct-ammonia fuel cells are entering early commercial use in maritime applications, backup power and industrial microgrids. Multi-vector hydrogen hubs that combine electrolysis, storage, carbon capture and co-located renewables are progressively embedding hydrogen into industrial clusters and power systems.

Energy demand

Key takeaways

- Global primary energy demand is projected to increase from more than 312 mboe/d in 2025 to around 383 mboe/d in 2050, an increase of around 23% over the outlook period, or 0.8% p.a. on average.

- Primary energy demand growth to 2050 is almost entirely concentrated in developing regions (non-OECD), while energy demand in developed countries (OECD) is expected to remain broadly flat, with its share of global energy demand declining from 34.3% in 2025 to 28.6% in 2050. Demand for all primary fuels, except coal, is set to increase over the outlook period.

- Demand for renewables (biomass, solar, wind, hydro and 'others') is forecast to increase from around 48 mboe/d in 2025 to just above 99 mboe/d in 2050. This growth is mainly driven by the expansion of solar and wind, which by 2050 increases by 23.6 mboe/d and 11.9 mboe/d, respectively.

- Oil and gas demand is expected to show a strong increase over the outlook period, in line with the need for reliable and affordable energy. Between 2025 and 2050, oil and gas demand are expected to increase by 18.6 mboe/d and 19.3 mboe/d, respectively.

- Despite a marginal decline, oil is projected to maintain the largest share in the energy mix, accounting for just under 30% in 2050. The combined share of oil and gas in the energy mix is expected to remain around 54% by 2050.

- Nuclear energy is expected to rise from 15 mboe/d in 2025 to 25.5 mboe/d in 2050, underlining its growing role in electricity systems.

- From 82.6 mboe/d in 2025, demand for coal is expected to drop to 53.3 mboe/d in 2050 due to unfavourable energy and climate policies, as well as the penetration of other fuels.

- Total electricity generation is expected to increase by more than 85% over the outlook period, growing from just above 32,000 TWh in 2025 to nearly 59,500 TWh in 2050. Around 75% of this growth is projected to come from developing countries, with more than half of global incremental electricity generation expected to materialize in developing Asia.

- The largest increase in the generation mix is projected for solar and wind, rising from around 5,400 TWh in 2025 to nearly 26,000 TWh by 2050. This accounts for the majority of the incremental global electricity supply.

- The share of non-hydrocarbons in global electricity generation is projected to rise from around 43% in 2025 to nearly 66% by 2050. Alongside strong growth in solar and wind, there is also a steady expansion in nuclear and hydropower and a structural decline in coal-fired generation.

- Per capita energy consumption continues to vary dramatically across the globe. Many people, particularly in the non-OECD, continue to be underserved by the energy sector, highlighting the urgency of alleviating energy poverty and expanding energy access.

This chapter discusses medium- and long-term primary energy demand by different fuels, sectors and major regions and countries. The key assumptions provided in Chapter 1, such as demographic and economic developments, as well as long-term energy technology trends and the evolution of energy policies, are taken into consideration in these projections. This chapter also focuses on the implications of energy demand trends on energy poverty and energy access in the Reference Case.

2.1 Major trends in energy demand

More than a decade after the Paris Agreement was adopted, the global effort to address ambitious climate goals has entered a more complex phase, with the pace of energy pathways varying widely across regions and sectors. At the same time, global energy demand continues to rise across all sectors. Meeting this growing demand necessitates the continued use of all available energy sources, particularly in emerging sectors like AI and data centres, where electricity requirements are expanding rapidly and continuously.

In the years following 2015, energy policy was often shaped by expectations of a straightforward, steady shift away from traditional fuels, supported by falling technology costs and expanding policy commitments. Over time, however, rigidities within energy systems and the continued reliance on established fuels to meet rising energy demand became increasingly evident. While the energy system demonstrated commendable flexibility in overcoming the combined effects of the COVID-19 pandemic, the energy crisis of 2022, supply chain issues and geopolitical uncertainty linked to conflicts and surging energy costs, these pressures exposed underlying vulnerabilities. These developments also reinforced the fact that global energy systems adapt more effectively through improvements and additional energy types rather than a rapid replacement of existing energy sources.

In a positive development, governments are increasingly recognizing that energy security, affordability and system resilience must remain a priority alongside emissions reductions. As a result, several advanced economies prudently began to focus more on securing a reliable and cost-stable energy supply.

In the US, for example, a stronger focus by the government on domestic resource development and adjustments to earlier support frameworks has shifted priorities towards supply security and cost stability. Elsewhere, Europe has embarked on a recalibration of energy security and affordability considerations alongside its decarbonization objectives, resulting in the region developing a heightened sense of awareness towards industrial competitiveness. At the same time, support for upstream oil and gas investment and LNG infrastructure has remained present in several markets, reflecting concerns about supply adequacy and price volatility.

China continues to prioritize economic resilience while maintaining its position as the largest market for renewable capacity expansion. It has accelerated its deployment of renewables, demonstrating manufacturing scale and supply-chain leadership in doing so. Across developing economies, particularly in Asia, Africa and Latin America, energy demand is increasing rapidly and becoming an important driver of the global energy expansion, with India also emerging as a key player. Across these regions, policy approaches continue to balance economic growth with energy access needs, with many countries using and even expanding their use of hydrocarbons while gradually expanding alternative energy pathways.

These trends are also reflected in international discussions. For instance, at the COP30 in Belém, Brazil, negotiations underlined the widening gap between ambition and implementation, with an increasing emphasis on adaptation finance, just transition pathways and national circumstances. Similarly, the G20 reaffirmed the importance of energy security and economic stability alongside climate goals, highlighting that transition pathways will vary among economies and depend on their development priorities and available resources.

This momentum is increasingly reflected in overall energy demand trends. Global energy demand continued to expand in 2025 across all sectors, with electricity consumption accounting for a significant share of overall growth. While the deployment of renewables remains strong in absolute numbers, particularly in terms of installed capacity additions, it is nevertheless being challenged by growing system integration requirements.

Indeed, the ever-rising share of renewables in the power generation mix in some countries is increasingly responsible for seasonal imbalances due to occasional periods of low wind and solar generation. These periods pose significant challenges to power systems, as they require substantial storage capacity and/or power imports. Furthermore, limited transfer capacity has increased congestion and forced renewable energy generators to curtail their output at times.

Against this backdrop, many market participants have become increasingly sceptical about the feasibility of ambitious narratives that call for one-sided solutions. Instead, demand for most fuels actually increased in 2025, including for oil, gas, coal and nuclear, all of which reached record high levels. Oil demand remains robust, particularly in terms of transport (including road transportation and aviation) and the petrochemicals sector. The strongest growth in oil demand continues to be seen in developing economies, where mobility and industrial demand continue to expand. Although the deployment of EVs is growing, adoption remains uneven across regions, thus limiting the displacement of oil in road transportation. In addition, aviation demand continues to expand alongside global mobility and trade activity.

Similar trends are reflected in growing natural gas demand. This can be attributed to its role in the power sector as a flexible source of generation that complements intermittent renewables, while also supporting demand within industry, transport and the residential sector. Meanwhile, global coal use remains important in developing economies, particularly in the power generation sector and energy-intensive industries where alternatives remain constrained by cost and infrastructure. In some regions, however, including parts of the US, periods of high gas prices and extreme weather have driven increases in coal-fired generation.

In recent years, nuclear energy demand has regained momentum, marking a significant recovery from the post-2011 slowdown. In 2025, global generation reached record levels, emphasizing the role of nuclear energy in supporting system stability amid rising demand for electricity. While capacity additions remain more measured than the rapid expansion of renewables, the strategic value of nuclear energy has become increasingly evident as power systems confront tighter supply margins and more complex reliability requirements.

Many countries are advancing life-extension programmes for existing reactors to preserve this important source of baseload capacity. Meanwhile, new construction activity is concentrated in Asia, where most reactors under development are located. China continues to lead global expansion, supported by large reactor construction plans and additional project approvals in 2025.

Against this evolving policy and market backdrop, global primary energy demand continues to expand over the outlook period. As shown in Table 2.1, global primary energy demand is expected to increase from 312.3 mboe/d in 2025 to 382.7 mboe/d by 2050, an expansion of more than 70 mboe/d. This represents an increase of around 23% over the outlook period, or an average of 0.8% p.a.

Table 2.1
World primary energy demand by fuel type, 2025–2050

	Levels *mboe/d*						Growth *mboe/d*	Growth *% p.a.*	Fuel share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	94.9	102.2	107.4	110.0	111.9	113.5	18.6	0.7	30.4	29.7
Gas	72.0	77.1	81.9	85.7	88.9	91.3	19.3	1.0	23.0	23.8
Coal	82.6	79.1	72.9	66.1	59.5	53.3	–29.3	–1.7	26.4	13.9
Nuclear	15.0	16.7	18.8	21.0	23.3	25.5	10.5	2.2	4.8	6.7
Renewables	47.8	57.3	67.9	78.6	88.8	99.1	51.3	3.0	15.3	25.9
Biomass	*28.4*	*30.4*	*32.3*	*33.9*	*34.9*	*35.7*	*7.3*	*0.9*	*9.1*	*9.3*
Solar	*4.9*	*8.4*	*12.7*	*17.7*	*23.0*	*28.5*	*23.6*	*7.3*	*1.6*	*7.4*
Wind	*4.7*	*7.2*	*9.8*	*12.1*	*14.3*	*16.6*	*11.9*	*5.1*	*1.5*	*4.3*
Hydro	*7.7*	*8.4*	*9.1*	*9.8*	*10.6*	*11.5*	*3.8*	*1.6*	*2.5*	*3.0*
*Others**	*2.1*	*2.9*	*4.0*	*5.1*	*6.0*	*6.8*	*4.7*	*4.8*	*0.7*	*1.8*
Total	**312.3**	**332.4**	**348.9**	**361.3**	**372.4**	**382.7**	**70.4**	**0.8**	**100**	**100**

* *Includes geothermal, tidal and wave.*
Source: OPEC.

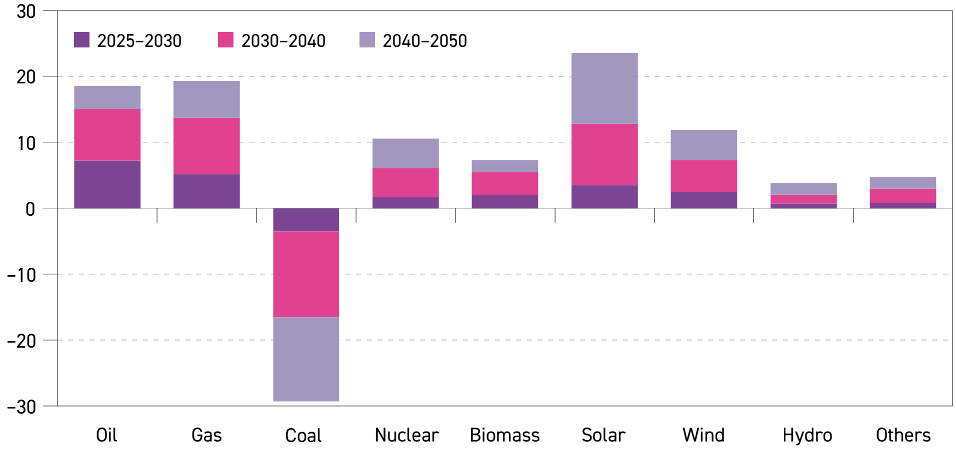
The growth rate is expected to decelerate towards the latter part of the outlook period, mainly due to demographic stabilization in several regions. This slowdown can also be attributed in part to the growing share of other renewables, which have higher efficiency than traditional fuels when viewed from the perspective of primary energy demand. This means that the slowdown is much less pronounced in final energy demand, measured after energy transformation.

Demand for all primary energy sources is expected to increase, except for coal, as shown in Figure 2.1. Coal demand is projected to decline significantly, mainly due to unfavourable energy and climate policies, as well as the increasing deployment of alternative fuels, especially in the power generation sector. As a result, global coal demand is expected to drop by more than 29 mboe/d, falling from nearly 83 mboe/d in 2025 to just above 53 mboe/d by 2050.

In contrast, oil and gas demand is expected to increase over the outlook period. Oil remains resilient, maintaining its position as the largest source of primary energy demand globally, seeing only marginal changes to its overall share in the energy mix. Global oil demand is expected to rise by 18.6 mboe/d to reach 113.5 mboe/d by 2050. Growth is concentrated in road transportation, petrochemicals and aviation, where alternatives remain limited or costly.

At the same time, global natural gas demand is projected to strengthen its long-term position in the energy mix, largely replacing coal and addressing critical supply gaps. Demand will be

Figure 2.1
Growth in primary energy demand by fuel type, 2025–2050, mboe/d



Source: OPEC.

supported by the availability of vast resources, competitive economics and its lower-carbon footprint. Natural gas will also play an increasingly important role in the power generation sector. Demand for natural gas is set to rise from 72 mboe/d in 2025 to 91.3 mboe/d in 2050, with its share in the energy mix increasing slightly to reach 23.8% by 2050, thereby overtaking coal.

Nuclear energy demand was at a low point following the Fukushima accident in Japan in 2011. Since that period, demand has gradually recovered, mainly due to its role as a stable baseload that helps maintain power system reliability. It also receives additional support from favourable energy security policies.

Moving forward, demand is expected to continue to rise from 15 mboe/d to 25.5 mboe/d by 2050. This will mainly be driven by the developing Asian economies, where most reactors under construction are located, particularly in China. Despite these promising trends, the outlook for nuclear deployment remains uncertain. Ultimately, the sector's future prospects will be influenced by the strength of policy and market support, the ability of nuclear energy to respond to rapidly rising power demand and which new reactor technologies reach commercial scale first.

Demand for renewable energy, including biomass, solar, wind, hydro and other sources (including geothermal, tidal and wave energy), is expected to record the strongest collective growth over the outlook period. Total renewables demand is projected to more than double, rising from around 48 mboe/d in 2025 to just above 99 mboe/d by 2050, with an average growth rate of about 3% p.a.

Demand for solar and wind account for most of this expansion. Solar demand is expected to increase from just below 5 mboe/d in 2025 to 28.5 mboe/d by 2050, making it the fastest-growing energy source, adding roughly 24 mboe/d, or about 7% p.a. Demand for wind power is also expected to rise significantly, increasing from 4.7 mboe/d to 16.6 mboe/d by 2050. The deployment of both technologies is mainly driven by declining generation costs and policy support. However, intermittency, grid constraints and rising integration costs remain the main challenges.

Biomass was the largest renewable energy source in 2025 and is expected to maintain this position through to 2050. Demand for biomass is expected to increase from 28.4 mboe/d to around 35.7 mboe/d over the outlook period, adding more than 7 mboe/d, supported by biofuel mandates in the road transportation and aviation sectors, the expansion of advanced biofuels and select industrial applications.

Demand for hydro is expected to expand at a slower but stable pace, particularly in developing regions where untapped potential remains. It sees an increase of just around 4 mboe/d, from 7.7 mboe/d in 2025 to 11.5 mboe/d by 2050. Nevertheless, hydropower's share of global electricity generation is expected to decline from around 14% to just above 11% in 2050, as other energy sources ramp up faster and most viable hydropower resources have already been exploited.

The category of 'others', which includes geothermal, tidal and wave energy, remains a small part of the global energy mix but continues to grow at a fast rate. Combined demand for these sources is projected to increase from above 2 mboe/d in 2025 to 6.8 mboe/d by 2050.

The energy mix is expected to experience large changes in the long term. The share of renewables will increase notably, rising from 15.3% in 2025 to nearly 26% in 2050. Nonetheless, all fuels will be needed to satisfy global energy demand growth. This is especially the case for oil and gas, whose combined share in the energy mix is expected to stay around 54% by 2050.

When examining primary energy demand growth from a regional perspective, as shown in Table 2.2 and Figure 2.2, growth is expected to be concentrated in non-OECD economies. Overall, non-OECD demand is anticipated to rise by 68.1 mboe/d between 2025 and 2050, corresponding to an average annual growth rate of about 1.2% p.a. Over the same period, demand in the OECD is expected to increase only marginally, namely by 2.3 mboe/d.

Regional energy demand growth remains robust across most non-OECD regions. The only exceptions to this pattern are China and Russia. In China, demand is projected to peak in the mid-2030s at 85.5 mboe/d before gradually easing to about 80 mboe/d by 2050. This moderation is mainly due to a maturing economy, rising energy efficiency and a declining population. Over the same period, Russia's energy demand is expected to decrease from 17.1 mboe/d in 2025 to 15.7 mboe/d by 2050, largely due to demographic contraction and efficiency gains in energy-intensive sectors.

In contrast, India emerges as the main driver of global energy demand growth. The country's primary energy demand is projected to double from just below 23 mboe/d in 2025 to 45.6 mboe/d by 2050, an increase of around 23 mboe/d, representing the largest absolute growth of any country or region. This is mainly due to an increasing population, economic growth, expanding urbanization and industrial initiatives, as well as rapidly rising electricity use across residential, commercial and industrial sectors.

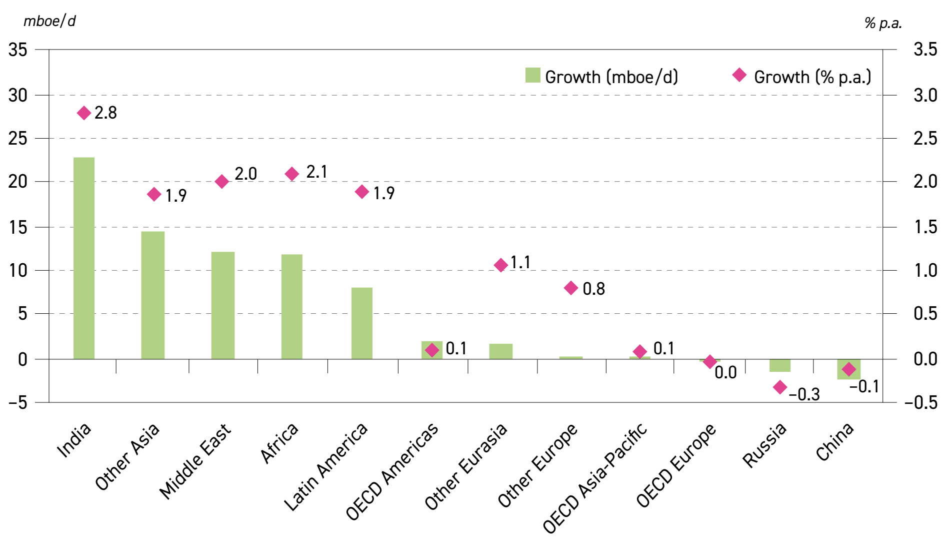
Other Asia is also expected to see strong demand growth of around 14.5 mboe/d, increasing from 24.8 mboe/d in 2025 to 39.4 mboe/d by 2050. This is mainly driven by continued economic expansion, rising industrial activity, improving energy access and strong growth in electricity consumption across fast-developing economies in the region.

Table 2.2
Total primary energy demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	56.9	58.6	59.3	59.7	59.8	59.1	2.2	0.1	18.2	15.4
OECD Europe	32.7	33.2	33.1	32.7	32.5	32.4	–0.3	0.0	10.5	8.5
OECD Asia–Pacific	17.5	18.0	18.3	18.3	18.2	18.0	0.5	0.1	5.6	4.7
OECD	**107.1**	**109.8**	**110.7**	**110.8**	**110.5**	**109.5**	**2.3**	**0.1**	**34.3**	**28.6**
China	82.3	85.0	85.5	83.9	81.9	80.0	–2.2	–0.1	26.3	20.9
India	22.9	27.1	31.4	35.8	40.6	45.6	22.7	2.8	7.3	11.9
Other Asia	24.8	28.2	31.6	34.6	37.2	39.4	14.5	1.9	8.0	10.3
Latin America	13.8	15.9	17.9	19.6	20.9	22.0	8.2	1.9	4.4	5.8
Middle East	18.9	21.6	24.5	26.8	29.1	31.1	12.2	2.0	6.1	8.1
Africa	17.5	19.5	21.8	24.2	26.6	29.3	11.8	2.1	5.6	7.7
Russia	17.1	16.9	16.6	16.3	16.0	15.7	–1.3	–0.3	5.5	4.1
Other Eurasia	5.8	6.2	6.7	7.1	7.4	7.6	1.8	1.1	1.9	2.0
Other Europe	2.0	2.1	2.2	2.3	2.3	2.4	0.4	0.8	0.6	0.6
Non-OECD	**205.1**	**222.6**	**238.2**	**250.5**	**261.9**	**273.2**	**68.1**	**1.2**	**65.7**	**71.4**
World	**312.3**	**332.4**	**348.9**	**361.3**	**372.4**	**382.7**	**70.4**	**0.8**	**100**	**100**

Source: OPEC.

Figure 2.2
Growth in primary energy demand by region, 2025–2050



Source: OPEC.

The Middle East is expected to rise from slightly below 19 mboe/d in 2025 to just above 31 mboe/d by 2050, increasing by 12.2 mboe/d over the long term. This expansion is mainly driven by population growth, rising living standards and continued economic diversification. Furthermore, increasing electricity demand for cooling, petrochemicals and water desalination supports growth in both electricity and fuel use, while the availability of abundant, low-cost oil and gas continues to shape the region's energy mix even with the Middle East's ongoing expansion of renewable energy capacity.

The outlook also sees Africa emerging as a centre of long-term energy demand growth, with total demand projected to increase from 17.5 mboe/d in 2025 to 29.3 mboe/d by 2050, an increase of 11.8 mboe/d. This growth is mainly driven by rapid population expansion, accelerating urbanization and gradually rising income levels, all of which increase demand for modern energy services. At the same time, energy demand in Latin America is projected to increase by 8.2 mboe/d, reaching 22 mboe/d in 2050.

Other Eurasia is expected to record moderate demand growth, rising from 5.8 mboe/d in 2025 to 7.6 mboe/d by 2050. This is driven by economic development and population trends in countries such as Kazakhstan, Azerbaijan and Georgia, alongside continued efficiency improvements that moderate growth. At the same time, Other Europe sees limited primary energy demand growth, increasing from 2 mboe/d to 2.4 mboe/d. This is supported by an expansion in services and transport, while efficiency gains again moderate overall demand growth.

Across the OECD, primary energy demand is projected to increase from just above 107 mboe/d in 2025 to 109.5 mboe/d by 2050. In OECD Americas, demand is expected to increase from almost 57 mboe/d to just above 59 mboe/d, as efficiency gains offset economic growth. In OECD Europe, demand is expected to ease from 32.7 mboe/d to 32.4 mboe/d, mainly due to strong efficiency gains and ongoing electrification across end-use sectors. In the OECD Asia-Pacific, demand is expected to increase slightly from 17.5 mboe/d in 2025 to 18 mboe/d in 2050, as ageing populations, high efficiency standards and advanced technology deployment continue to moderate energy use.

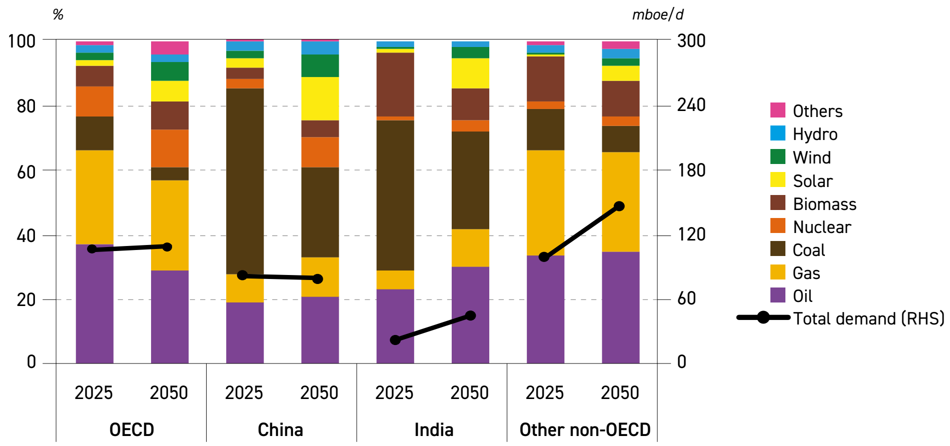
2.2 Energy demand by major regions

This section analyses regional trends in primary energy demand over the long term. It also explores the evolution of the energy mix in relation to regional economic development and major energy policies. The focus of this section is on major regions, such as the OECD and non-OECD, as well as on two influential major economies: China and India.

Figure 2.3 shows the projected evolution of primary energy demand and tne energy mix by major region between 2025 and 2050, for the OECD, China, India and Other non-OECD. Although several regions display common trends, such as a gradual diversification of their energy mixes and a rising contribution from non-fossil sources, the pace and composition of change differ significantly across regions.

In the OECD, primary energy demand is expected to rise only slightly over the outlook period, with the share of non-hydrocarbon energy sources increasing significantly in the energy mix. At the same time, the share of coal is expected to decline substantially, mainly due to a shift towards renewable energy, strict climate policies and the decommissioning of older power plants. Natural gas demand is also expected to decline slightly. Demand for nuclear power is projected to increase, driven mainly by policy measures and energy security considerations, although nuclear plans in OECD countries remain diverse, ranging from capacity and lifetime extensions to nuclear being phased-out. Meanwhile, the share of oil in the energy mix is expected to decline over time, predominantly due to energy policies aimed at improving energy efficiency and replacing oil, as well as slower economic growth and an ageing population.

Figure 2.3
Energy mix and primary energy demand in selected regions, 2025–2050



Source: OPEC.

In contrast, China's primary energy demand declines slightly over the outlook period, while oil demand is simultaneously expected to increase slightly, driven by growth in petrochemicals, freight transport and industrial activity. Coal demand declines significantly, in line with updated NDCs to reduce GHG emissions to between 7% and 10% below peak emission levels by 2035, with coal's role expected to shift from baseload towards providing flexibility and system support for solar and wind. At the same time, natural gas demand increases as part of China's efforts to strengthen energy security, improve system reliability and diversify supply.

Nuclear energy demand is expected to increase in both the OECD and China, driven mainly by policy support and energy security. China's trajectory is clear and it is expected to reach between 110 and 150 GW of installed capacity by 2035. The visible difference between the OECD and China is related to the share of oil in the energy mix, which remains higher in the OECD than in China.

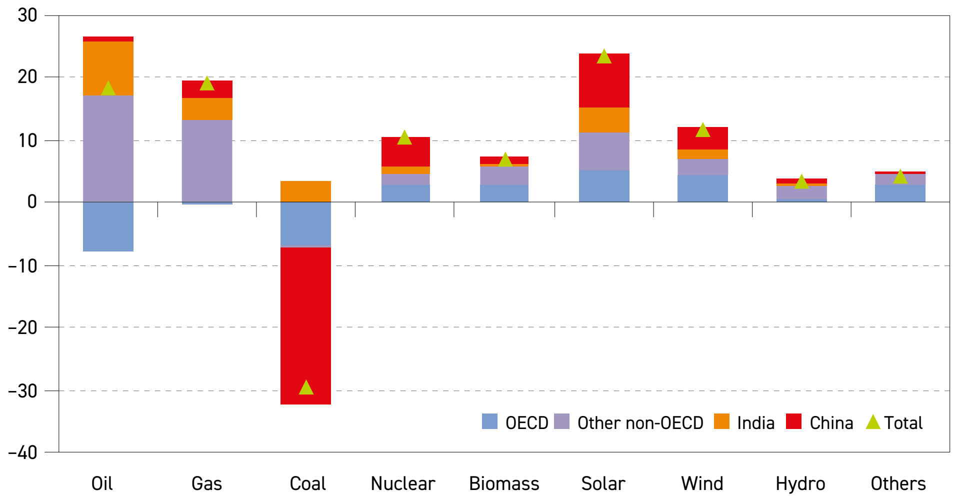
India's energy demand for all types of fuel is expected to expand, driven by the need to support its economic growth and expanding population. There is a small increase in the share of non-hydrocarbon energy sources in the energy mix, albeit much less pronounced than in the OECD and China. In 2025, India achieved its goal of increasing the share of non-hydrocarbons in power generation to 50% of its installed electricity capacity, and its efforts to support renewable deployment are expected to continue moving forward.

As renewable generation remains intermittent and thus cannot reliably meet the country's growing electricity needs alone, gas and coal will remain a central part of power generation and increase gradually for most of the outlook period. At the same time, nuclear capacity is expected to increase as well, driven by India's plans to increase nuclear capacity to 22 GW by 2031 and 2032, up from around 7.5 GW currently in operation and 6 GW under construction. Looking more long term, India has also set an ambitious target to reach 100 GW of nuclear power capacity by 2047.

Similar to India, other non-OECD countries are experiencing a sustained rise in energy demand across all major fuel types, driven by rapid population growth, continued urbanization and increasing industrial activity. While the deployment of renewable energy is accelerating in areas where related policy, financing and infrastructure frameworks allow, hydrocarbons remain a key component of the energy mix, reflecting considerations of affordability, existing infrastructure and the need for reliable supply.

Figure 2.4 summarizes the incremental growth in primary energy demand over the outlook period and highlights the regional sources of this change. Growth in oil demand is driven mainly by the non-OECD, particularly Asia and the Middle East, although declining demand in OECD countries offsets part of this increase. Coal displays the most obvious regional divergence, driven by a substantial decline in China alongside smaller expected declines in OECD economies.

Figure 2.4
Growth in primary energy demand by fuel type and region, 2025–2050, mboe/d



Source: OPEC.

Natural gas demand is expected to rise across all regions, except in OECD countries, where it declines slightly, with the largest increases coming from other non-OECD economies. Nuclear energy demand is expected to increase in all regions, led mainly by China, where new capacity additions continue to support the provision of stable, low-carbon baseload electricity. Hydropower and biomass expand more gradually, with growth distributed across developing regions where resource potential remains available; nevertheless, OECD countries are expected to drive the adoption of advanced biomass technologies, including biofuels and biogas. Overall, the most prominent energy demand trend of note relates to the rapid expansion in the use of solar and wind energy, in addition to other sources such as geothermal, tidal and wave energy, which together constitute the largest contribution to primary energy demand growth. This expansion is being driven primarily by China, India and OECD countries.

2.2.1 OECD

Table 2.3 shows that primary energy demand across the OECD is expected to grow by around 2.3 mboe/d over the outlook period, increasing from just above 107 mboe/d in 2025 to 110.8 mboe/d by the early 2040s. Demand then plateaus throughout the 2040s before easing slightly to around 109.5 mboe/d by 2050. While the absolute size of energy demand varies only slightly, the individual energy sources that form the energy mix are expected to experience large changes. In particular, the share of energy sources that are not hydrocarbons is expected to rise from around 23% to around 39%. Renewables demand increases by more than 15 mboe/d over the outlook period. Solar and wind account for the majority of this expansion, alongside smaller but noticeable increases in demand for biomass and other renewable sources.

The outlook for the OECD remains driven by policy and investment uncertainty, particularly in OECD Americas and especially in the US, where recent policy shifts and the country's withdrawal from the Paris Agreement have exacerbated uncertainty about long-term energy planning and investment decisions.

Table 2.3
OECD primary energy demand by fuel type, 2025–2050

	Levels *mboe/d*						Growth *mboe/d*	Growth *% p.a.*	Fuel share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	39.9	40.3	39.3	36.9	34.3	31.8	–8.1	–0.9	37.3	29.0
Gas	30.9	31.2	31.4	31.4	31.2	30.7	–0.2	0.0	28.8	28.0
Coal	11.6	10.5	8.6	7.1	5.8	4.5	–7.0	–3.7	10.8	4.2
Nuclear	9.8	10.2	10.7	11.2	11.8	12.5	2.7	1.0	9.1	11.4
Renewables	14.9	17.6	20.7	24.2	27.4	30.0	15.1	2.8	13.9	27.4
Biomass	7.3	7.7	8.3	9.0	9.5	10.0	2.7	1.3	6.8	9.1
Solar	1.8	2.8	4.0	5.3	6.5	6.9	5.1	5.4	1.7	6.3
Wind	2.1	2.8	3.5	4.3	5.2	6.2	4.1	4.4	2.0	5.6
Hydro	2.4	2.5	2.6	2.7	2.8	2.8	0.4	0.6	2.3	2.6
Others	1.3	1.8	2.3	2.9	3.5	4.1	2.8	4.7	1.2	3.7
Total	107.1	109.8	110.7	110.8	110.5	109.5	2.3	0.1	100	100

Source: OPEC.

Primary energy demand for oil across the OECD is expected to increase over the medium-term period before declining overall by 8.1 mboe/d by 2050. The latter is due to rising energy efficiency, increasing EV penetration and a gradual substitution of oil in transport, buildings and some parts of the industrial sector.

The strongest decline is expected in OECD Europe, at 3.4 mboe/d, reflecting a relatively pronounced contraction relative to its lower base, followed closely by OECD Americas, where demand drops by about 3.2 mboe/d, representing a more moderate decline relative to its larger base. The OECD Asia-Pacific records a more moderate drop from a lower base, at around 1.5 mboe/d, though this is still notable in proportional terms. The variation in the rate of decline reflects differences in energy policies and their effectiveness in each region, as well as deviations in evolving efficiency improvement rates.

Demand for natural gas in the OECD is expected to remain stable throughout the outlook period, starting and ending the period at around 31 mboe/d. This is mainly due to the role of natural gas in providing flexibility and maintaining power system reliability as variable renewables expand across power systems. In OECD Americas, gas remains vital across multiple sectors and is expected to increase by 0.5 mboe/d between 2025 and 2050, supported by its role in power generation and industry. The OECD Asia-Pacific sees limited growth of 0.1 mboe/d, amid gas remaining important for the power sector's ongoing transition away from coal. Meanwhile, gas demand in OECD Europe is projected to decline from 7.3 mboe/d in 2025 to 6.4 mboe/d in 2050.

Coal demand across the OECD is projected to decline over the outlook period, dropping by 7 mboe/d from 11.6 mboe/d in 2025 to 4.5 mboe/d by 2050. This decline is driven mainly by shifts in the power sector, where coal-fired generation is increasingly displaced by natural gas, renewables and, in some markets, nuclear power. Regionally, the largest decline is expected in OECD Americas at around 3.1 mboe/d, although the pace of this fall remains uncertain given recent US policy shifts and ongoing considerations regarding system reliability. These factors could slow the closure of coal-fired power plants in the medium term. Elsewhere, OECD Europe and the OECD Asia-Pacific are projected to decline by around 2.4 mboe/d and 1.6 mboe/d, respectively. However, there remains a potential upside for coal demand in the event of electricity shortages, and particularly if the expansion of renewables and/or nuclear energy proceeds at a slower-than-expected pace in countries across these regions.

Demand for nuclear energy is expected to increase by 2.7 mboe/d to reach about 12.5 mboe/d by 2050, supported mainly by concerns related to energy security. Growth is led by the OECD Asia-Pacific, reflecting the strongest increase among OECD regions, where reactor restarts in Japan and continued policy support are reinforcing nuclear energy's role in underpinning energy security and system stability.

In OECD Europe, demand increases moderately, supported by selective new-build projects aimed at maintaining system adequacy amid coal capacity declining and the penetration of renewables rising. In OECD Americas, nuclear demand is also expected to increase modestly, with policy increasingly focused on sustaining existing capacity, extending reactor lifetimes and promoting new nuclear development. This is supported by recently signed project orders in the US and expanded international cooperation initiatives.

Biomass remains the largest renewable energy source within the OECD and is expected to maintain this position through to 2050. Demand increases gradually from more than 7 mboe/d in 2025 to 10 mboe/d by 2050, supported by biofuel-blending mandates, sustainability frameworks and its continued use in select industrial and heating applications. OECD Europe leads this expansion, while OECD Americas also records steady increases. The OECD Asia-Pacific contributes more moderate growth, reflecting its smaller starting base and more limited resource availability.

Demand for solar energy is set to increase by more than 5 mboe/d over the outlook period, increasing from 1.8 mboe/d to nearly 7.0 mboe/d. This growth is mainly driven by cost-reduction expectations in generation and supportive policies that encourage both rooftop and utility-scale systems, while national clean-energy strategies sustain expansion in several advanced economies. In the US, solar deployment continues to rise, although medium-term prospects are influenced by changes in tax incentives and policy uncertainty. Overall, the pace of solar growth in the OECD is increasingly shaped by system-level constraints, including grid connection delays, land-use limitations and the growing need for baseload capacity as intermittent generation increases.

Wind energy is also expected to increase from just above 2 mboe/d to more than 6 mboe/d in 2050, adding more than 4 mboe/d. Around half of this growth is expected in OECD Europe, driven by energy policy support, expanded offshore targets and faster permitting. The OECD Asia-Pacific also contributes to this growth, as offshore wind becomes more central to long-term energy planning, while growth in OECD Americas continues but remains restricted due to policy uncertainty. Overall, wind deployment faces constraints that have significantly slowed the pace of development, including grid bottlenecks, rising costs and capital constraints regarding regulatory and permitting conditions.

Hydropower is expected to add around 0.4 mboe/d. This is mainly driven by modernization, as well as the updating of existing assets to increase their lifetimes, rather than large new developments. Most of this increase occurs in OECD Americas and OECD Europe, supported by regulatory frameworks that prioritize reliability and operational flexibility. Growth in the OECD Asia-Pacific remains limited, especially as increasing hydrological variability continues to create operational uncertainty.

Other energy sources, including geothermal, tidal and wave energy are set to increase modestly from 1.3 mboe/d in 2025 to 4.1 mboe/d in 2050, with geothermal accounting for most of this growth. This expansion is driven by advances in next-generation geothermal technologies and supported by improvements in drilling and exploration technologies and targeted research and development programmes across several OECD countries. Tidal and wave technologies are advancing modestly and are expected to remain largely confined to pilot and demonstration projects until costs decline and technical performance improves.

2.2.2 Non-OECD

Total primary energy demand in the non-OECD is expected to rise from just above 205 mboe/d in 2025 to 273.2 mboe/d by 2050 (Table 2.4). This represents an increase of more than 68 mboe/d, with an average growth rate of 1.2 % p.a. India accounts for the largest share of this increase, while Other Asia, the Middle East and Africa also contribute significantly. Most of this demand growth is due to population growth, improving living standards, reduced energy poverty and increasing industrialization.

Across the non-OECD, energy systems continue to expand while gradually integrating new technologies in line with national capabilities and policy priorities. Energy pathways therefore remain diverse, as countries balance immediate economic growth, domestic resource availability and energy security needs, with the need to reduce emissions, alongside access to technologies that can help achieve this. In this context, the non-OECD energy mix is expected to evolve considerably. The share of hydrocarbons is expected to decline from more than 81% in 2025 to 70% by 2050, mainly due to an anticipated decline in coal demand driven by policies in China.

The largest increase is expected for oil, which is projected to grow from 55 mboe/d in 2025 to around 82 mboe/d in 2050, representing an increase of 26.7 mboe/d over the long term. The major drivers here are population growth and an increasing middle class, as well as rising energy access in many developing countries. This is reflected in significant increases in demand from the transportation and industrial sectors, particularly in petrochemicals.

Table 2.4
Non-OECD primary energy demand by fuel type, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Fuel share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	**55.0**	**61.9**	**68.1**	**73.1**	**77.6**	**81.7**	**26.7**	**1.6**	**26.8**	**29.9**
Gas	**41.0**	**45.9**	**50.6**	**54.2**	**57.6**	**60.6**	**19.5**	**1.6**	**20.0**	**22.2**
Coal	**71.0**	**68.6**	**64.3**	**59.0**	**53.8**	**48.8**	**–22.2**	**–1.5**	**34.6**	**17.9**
Nuclear	**5.2**	**6.5**	**8.1**	**9.8**	**11.5**	**13.0**	**7.8**	**3.8**	**2.5**	**4.8**
Renewables	**32.9**	**39.7**	**47.2**	**54.4**	**61.4**	**69.1**	**36.2**	**3.0**	**16.0**	**25.3**
Biomass	*21.2*	*22.7*	*24.0*	*25.0*	*25.4*	*25.7*	*4.6*	*0.8*	*10.3*	*9.4*
Solar	*3.0*	*5.6*	*8.8*	*12.4*	*16.5*	*21.5*	*18.5*	*8.1*	*1.5*	*7.9*
Wind	*2.6*	*4.4*	*6.3*	*7.8*	*9.1*	*10.5*	*7.8*	*5.7*	*1.3*	*3.8*
Hydro	*5.3*	*5.9*	*6.5*	*7.1*	*7.8*	*8.6*	*3.4*	*2.0*	*2.6*	*3.2*
Others	*0.8*	*1.2*	*1.6*	*2.1*	*2.5*	*2.7*	*1.9*	*5.0*	*0.4*	*1.0*
Total	**205.1**	**222.6**	**238.2**	**250.5**	**261.9**	**273.2**	**68.1**	**1.2**	**100**	**100**

Source: OPEC.

Despite some countries' policy measures, growth may be moderated but not reversed. Furthermore, rising energy access is expected to support demand growth for LPG, especially in Sub-Saharan Africa. Consequently, the share of oil in the energy mix of the non-OECD is expected to increase from 26.8% to almost 30% by 2050, displacing coal as the fuel with the highest share in the energy mix in 2025.

Gas is the second largest source of growth after oil. It is expected to increase from 41 mboe/d in 2025 to 60.6 mboe/d by 2050, with its share rising from 20% to 22.2%. This rise is driven by policy efforts aimed at improving air quality, expanding gas infrastructure and facilitating a shift away from coal-fired power generation. Large-scale initiatives such as city gas distribution programmes in India, LNG terminal expansions across Asia, and gas-to-power strategies in the Middle East and Africa are set to contribute to this trend. Furthermore, some non-OECD regions, such as the Middle East and Africa, will benefit from having substantial gas resources available at cheap cost.

Coal is the only fuel that is expected to see a decline over the outlook period, with demand set to drop from 71 mboe/d in 2025 to around 49 mboe/d by 2050, a fall of 1.5% p.a. This shift is driven primarily by China's medium- and longer-term transition from coal, shaped by the priorities of the 15th FYP, updated NDC targets and a rising share of solar and wind within the country's power generation system.

Elsewhere, Russia's coal demand is also expected to decline by 1.3 mboe/d due to the replacement of coal by natural gas and nuclear power in electricity generation, along with efficiency gains and policies aimed at reducing emissions. In contrast, coal demand is expected to increase in India and Other Asia due to a continued reliance on coal for secure and affordable baseload generation.

Demand for nuclear is also expected to grow, increasing from slightly above 5 mboe/d in 2025 to 13 mboe/d by 2050, an expansion of 3.8% p.a. China is projected to lead this expansion under its long-term nuclear plan, while new-build programmes in the Middle East and expansions in some Asian economies are anticipated to reflect the growing importance of stable electricity supply within evolving energy systems. The development of advanced reactors and early SMR frameworks is expected to reinforce the prominence of nuclear energy within broader energy security and diversification plans.

Renewables are projected to grow over the outlook period, rising from just below 33 mboe/d in 2025 to above 69 mboe/d by 2050. Biomass is expected to remain the largest renewable energy source within the non-OECD throughout the outlook period, increasing from 21.2 mboe/d in 2025 to 25.7 mboe/d in 2050, an expansion of 0.8% p.a.

In Africa, which accounts for the largest share of biomass use, demand is expected to expand strongly, reflecting its large potential. This is despite ongoing constraints related to finance and infrastructure. Growth in other regions, including India, Other Asia and Latin America, is generally seen as moderate to steady, supported by biofuel blending policies, sustainability initiatives and a gradual shift towards modern biomass solutions and waste-to-energy conversion.

Solar is expected to record the fastest growth in the non-OECD, expanding from 3 mboe/d in 2025 to 21.5 mboe/d in 2050. This is an average growth rate of 8.1% p.a. China's manufacturing scale and competitive project economics are expected to support non-OECD growth, while India is anticipated to accelerate deployment through national programmes and rooftop schemes. Other Asia and the Middle East are also projected to expand their solar capacity under long-term diversification plans.

Wind power is expected to grow substantially as well, rising from 2.6 mboe/d in 2025 to 10.5 mboe/d in 2050. This is an average annual growth rate of 5.7%. China is projected to lead both onshore and offshore deployment, while Other Asia and India are also anticipated to advance wind energy, particularly with regard to industrial development. In the Middle East and Latin America, wind is expected to rise moderately. Supply-chain pressures, particularly around critical minerals, are expected to remain an important variable in deployment timelines across all regions.

Elsewhere, although the increase in hydro energy is set to be smaller than that of solar and wind, demand is nevertheless projected to expand, albeit mainly in Asia, Latin America and Africa, where resource potential remains significant. Demand is expected to increase from 5.3 mboe/d in 2025 to 8.6 mboe/d in 2050, or 2% p.a. on average. Policies promoting energy access, regional integration and system reliability will support this expansion. At the same time, hydrological variability and environmental constraints are likely to limit new large-scale projects and shift hydro stakeholders' attention to updating and upgrading efficiency at existing facilities.

The renewables subcategory, 'others', is expected to remain small in absolute terms but gradually increase its contribution over the outlook period. Demand is projected to rise from 0.8 mboe/d in 2025 to 2.7 mboe/d by 2050, reflecting the development of enhanced geothermal systems, expanded resource exploration and the continued pilot deployment of marine energy projects. Although this subcategory's overall share in 2050 remains limited at only 1%.

The following section provides more details concerning energy demand trends in China and India, which, as already mentioned, will have a significant impact on the global energy landscape due to the size of their economies and populations.

China

The primary energy demand outlook for China, shown in Table 2.5, indicates a modest increase from just over 82 mboe/d in 2025 to more than 85 mboe/d in the early to mid-2030s. This is supported by continued industrial activity and steady economic performance. Beyond this period, demand is projected to ease gradually, returning to around 80 mboe/d by 2050, as population growth levels off, energy efficiencies improve and the economy continues to shift to less energy-intensive activities.

Table 2.5
China primary energy demand by fuel type, 2025–2050

	Levels *mboe/d*						Growth *mboe/d*	Growth *% p.a.*	Fuel share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	15.7	16.8	17.5	17.4	17.0	16.6	0.9	0.2	19.1	20.8
Gas	7.2	8.4	9.2	9.6	9.8	9.9	2.7	1.3	8.7	12.3
Coal	47.3	43.7	38.5	32.9	27.4	22.3	-25.0	-3.0	57.5	27.9
Nuclear	2.6	3.5	4.6	5.7	6.7	7.5	5.0	4.4	3.1	9.4
Renewables	9.5	12.6	15.7	18.3	20.9	23.7	14.2	3.7	11.5	29.6
Biomass	*2.9*	*3.4*	*3.8*	*4.0*	*4.2*	*4.3*	*1.4*	*1.5*	*3.6*	*5.4*
Solar	*2.2*	*3.5*	*4.9*	*6.6*	*8.5*	*10.6*	*8.4*	*6.5*	*2.6*	*13.3*
Wind	*1.9*	*3.1*	*4.2*	*4.7*	*5.1*	*5.5*	*3.5*	*4.2*	*2.4*	*6.8*
Hydro	*2.4*	*2.6*	*2.8*	*3.0*	*3.1*	*3.3*	*0.9*	*1.3*	*2.9*	*4.1*
Others	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*5.0*	*0.0*	*0.0*
Total	**82.3**	**85.0**	**85.5**	**83.9**	**81.9**	**80.0**	**-2.2**	**-0.1**	**100**	**100**

Source: OPEC

Coal remains important throughout the outlook period, but its role is expected to gradually diminish. Although significant new coal-fired capacity has been added in recent years – mainly high-efficiency units replacing older plants – the rapid growth of wind and solar is anticipated to reduce coal utilization over time. China's coal demand, around 47 mboe/d in 2025, is expected to decline gradually in the medium term before falling more sharply to around 22 mboe/d by 2050, reflecting a total drop of 25 mboe/d over the outlook period. While coal use is expected to fall, it is set to continue to provide backup during periods of low renewable output.

In contrast, oil demand is projected to increase from about 16 mboe/d in 2025 to around 17.5 mboe/d in the late 2030s, before moderating slightly, while remaining one of the largest contributors to global oil demand through to 2050. Growth is concentrated in petrochemicals and aviation, where alternatives remain limited. Looking at road transport, although EV adoption continues to expand, the number of conventional ICE vehicles is projected to remain above current levels – rising from around 330 million to over 380 million by 2040 before gradually declining – thereby helping to sustain oil demand in this sector.

China's natural gas demand is expected to gain importance in the energy mix. Demand increases from over 7 mboe/d in 2025 to just below 10 mboe/d by 2050, supported by its role in balancing the electricity grid as wind and solar capacity expands. Additional growth is also anticipated in transport, particularly amid the increased use of LNG-fuelled trucks. Moreover, nuclear energy is projected to triple, increasing from 2.6 mboe/d in 2025 to 7.5 mboe/d by 2050. Recently, China reportedly has 35 reactors under construction and 60 reactors in operation, supported by relatively low development costs, an experienced workforce and economies of scale achieved by constructing multiple reactors at one site. Moreover, China also benefits from less stringent regulatory issues, limited local opposition to nuclear projects, and a sophisticated supply chain supported by lower labour costs.

Demand for renewables is expected to rise from 9.5 mboe/d in 2025 to 23.7 mboe/d by 2050, raising the share of non-hydrocarbons in the energy mix from around 15% to 39%. Solar is expected to record the strongest growth, adding 8.4 mboe/d. This increase is supported by China's goal of achieving peak emissions by 2030, as well as strong industrial policy and manufacturing scale, which have significantly lowered renewable technology costs in the country and accelerated both domestic deployment and global market penetration.

Wind energy is also expected to expand steadily, rising from just below 2 mboe/d to 5.5 mboe/d by 2050. This expansion is driven by continued onshore development in northern and western provinces, as well as a simultaneous scaling-up of offshore capacity along coastal regions. This is supported by longer-term clean power targets that address combined wind and solar capacity, which is expected to grow from around 1,800 GW in 2025 to 3,600 GW by 2035.

Biomass demand is projected to rise at a moderate pace, adding around 1.4 mboe/d between 2025 and 2050, reflecting wider industrial applications and progress in advanced biofuels. Hydropower and the subcategory of 'others' are expected to increase modestly, adding around 1 mboe/d over the outlook period, while continuing to support seasonal balancing as renewable penetration rises. This role is supported by expanded storage capacity, with targets exceeding 120 GW by 2030.

India

India's primary energy demand is projected to nearly double over the outlook period, rising from just below 23 mboe/d in 2025 to over 45.5 mboe/d by 2050 (Table 2.6), making it the single largest contributor to global energy demand growth over the long term. This expansion is driven by population growth, a rapidly expanding middle class and sustained economic and industrial development.

Demand for all fuels is expected to increase. The largest growth is expected for oil, with demand rising from 5.3 mboe/d in 2025 to 13.8 mboe/d by 2050, an increase of 8.5 mboe/d. Growth is supported mainly by transportation and petrochemicals, as well as to a lesser extent by residential and commercial use, including LPG for cooking. Oil's share in India's energy mix is expected to rise to over 30% by 2050, up from around 23% in 2025.

Natural gas demand is anticipated to expand from 1.3 mboe/d in 2025 to 5.2 mboe/d by 2050. This reflects policy support for gas as part of efforts to decarbonize power generation and accommodate a rising share of intermittent renewables. Additional demand stems from the transportation sector, where the number of Natural Gas Vehicles (NGVs) continues to rise, as well as from expanding gas distribution networks in major consuming regions, which support residential and commercial use while reducing reliance on traditional biomass. The outlook aligns with plans to support domestic gas production over the medium and long term.

Table 2.6
India primary energy demand by fuel type, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Fuel share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	5.3	6.8	8.3	10.1	12.0	13.8	8.5	3.9	23.3	30.3
Gas	1.3	1.8	2.3	3.1	4.0	5.2	3.8	5.6	5.8	11.3
Coal	10.7	12.0	12.9	13.3	13.7	14.1	3.3	1.1	46.8	30.8
Nuclear	0.3	0.4	0.6	0.9	1.2	1.5	1.2	6.7	1.3	3.2
Renewables	5.2	6.2	7.2	8.5	9.7	11.1	5.8	3.0	22.8	24.3
Biomass	*4.4*	*4.6*	*4.7*	*4.8*	*4.7*	*4.6*	*0.2*	*0.2*	*19.3*	*10.1*
Solar	*0.3*	*0.7*	*1.3*	*2.0*	*3.0*	*4.2*	*3.9*	*11.0*	*1.4*	*9.3*
Wind	*0.2*	*0.5*	*0.8*	*1.2*	*1.4*	*1.5*	*1.3*	*8.3*	*0.9*	*3.2*
Hydro	*0.3*	*0.4*	*0.4*	*0.5*	*0.6*	*0.7*	*0.4*	*3.6*	*1.3*	*1.6*
Others	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*	*0.0*
Total	**22.9**	**27.1**	**31.4**	**35.8**	**40.6**	**45.6**	**22.7**	**2.8**	**100**	**100**

Source: OPEC.

Coal, meanwhile, remains necessary to India's power system throughout much of the outlook. Demand is expected to rise from around 11 mboe/d in 2025 to above 14 mboe/d by 2050, an addition of 3.3 mboe/d. This is mainly due to strong growth in electricity demand and industrial activity. Over time, as renewable capacity expands and efficiency improves, coal's growth moderates. Nevertheless, the scale and pace of electricity demand means that coal is expected to continue providing a substantial share of India's power generation.

Solar and wind record the strongest growth in India's power generation sector. Starting from around 0.5 mboe/d in 2025, demand more than doubles by 2030 and rises through the 2030s as costs decline, grid access improves and domestic manufacturing benefits from targeted policy support. By 2040, demand for solar and wind combined is expected to reach 3.2 mboe/d before rising to 5.7 mboe/d by 2050. Solar remains the dominant contributor, supported by large-scale deployment in both utility-scale and distributed applications, while wind expands steadily, particularly where resource quality and grid conditions allow.

Nuclear energy is expected to expand gradually from around 0.3 mboe/d in 2025 to about 1.5 mboe/d by 2050. As previously mentioned, this growth is in line with the country's plans to raise its nuclear capacity to 22 GW by 2032. India has also set an ambitious target to reach 100 GW of nuclear power capacity by 2047, as part of its initiative, 'Viksit Bharat @2047'. With several reactors under construction, India continues to advance its nuclear programme as part of efforts to diversify its power mix.

Hydropower is also anticipated to increase, growing from roughly 0.3 mboe/d in 2025 to 0.7 mboe/d by 2050. Although growth is moderated by the country's resource base and permitting constraints, hydropower continues to support system balance.

Biomass maintains a substantial role in India's energy system. It is expected to expand by 0.2 mboe/d from 4.4 mboe/d in 2025 to 4.6 mboe/d by 2030, before remaining relatively stable through 2050. Its evolution is shaped by improvements in feedstock management, the increasing use of modern bioenergy technologies, such as biogas and compressed biogas, as well as the continued roll-out of biofuel blending programmes. However, biomass also remains associated with energy access challenges, as some rural and low-income households still rely on traditional biomass for cooking despite the expansion of access to LPG, which is expected to gradually reduce this dependence.

2.3 Energy demand by fuel

2.3.1 Oil

Global oil demand remained robust in 2025, increasing by 1.1 mboe/d compared to 2024 (when expressed on an energy content basis). This was driven primarily by the road transportation, aviation and shipping sectors, alongside resilient industrial and petrochemical consumption. Demand growth was concentrated in developing regions and countries, particularly Other Asia, as well as China, India, the Middle East and Africa, where economic expansion, urbanization and rising living standards continue to translate into greater mobility, trade and industrial activity.

Although these drivers are expected to remain strong in the medium term, tightening efficiency standards, EV penetration and the early stages of fuel substitution in some sectors are moderating overall demand growth. Looking ahead, the long-term outlook reveals a clear divergence between OECD and non-OECD regions, in line with differences in economic growth, demographics and policy direction.

Overall, oil demand is expected to remain resilient, especially in non-OECD economies, where demand is projected to increase in the transportation, residential and industrial sectors. As shown in Table 2.7, global oil demand is projected to rise from just below 95 mboe/d in 2025 to 113.5 mboe/d by 2050, with non-OECD growth more than offsetting OECD declines.

Across the non-OECD, oil demand is expected to rise by about 27 mboe/d over the outlook period, driven by population growth and rising urbanization, as well as continued robust economic growth. With the bulk of demand growth concentrated in some regions, India emerges as the single largest contributor to growth, accounting for around one-third of the total increase in global oil demand. The country's demand is expected to increase by 8.5 mboe/d, driven by rapid industrialization, population growth and economic development.

China's oil demand, the largest in the non-OECD region in 2025, is expected to see an increase of around 1 mboe/d. Demand is expected to rise modestly in the 2030s before slightly declining to 16.6 mboe/d in 2050, as efficiency gains, electrification and economic rebalancing take effect, while petrochemicals and heavy transport continue to drive consumption.

Demand in Other Asia is expected to increase by nearly 5 mboe/d, largely supported by ongoing industrialization and urban development, while the Middle East is anticipated to see a similar increase due to strong growth in petrochemical and transport activity. Elsewhere, Africa adds more than 4 mboe/d over the outlook period, driven by rapid population growth and rising vehicle ownership, while demand in Latin America increases by just under 3 mboe/d. In contrast, Russia and Other Eurasia and Other Europe show little net change over the period, as slower economic growth and efficiency improvements largely offset incremental demand.

Table 2.7
Oil demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	21.2	21.4	21.2	20.3	19.1	18.0	–3.2	–0.7	22.4	15.9
OECD Europe	12.2	12.3	11.8	10.8	9.7	8.7	–3.4	–1.3	12.8	7.7
OECD Asia-Pacific	6.6	6.5	6.3	5.9	5.5	5.0	–1.5	–1.1	6.9	4.4
OECD	**39.9**	**40.3**	**39.3**	**36.9**	**34.3**	**31.8**	**–8.1**	**–0.9**	**42.1**	**28.0**
China	15.7	16.8	17.5	17.4	17.0	16.6	0.9	0.2	16.5	14.6
India	5.3	6.8	8.3	10.1	12.0	13.8	8.5	3.9	5.6	12.2
Other Asia	9.3	10.6	11.7	12.6	13.4	14.1	4.9	1.7	9.8	12.4
Latin America	6.3	7.1	7.9	8.4	8.8	9.1	2.8	1.5	6.6	8.0
Middle East	8.1	9.1	10.2	11.2	12.2	13.1	5.0	1.9	8.5	11.5
Africa	4.6	5.4	6.2	7.0	7.9	8.7	4.1	2.5	4.9	7.7
Russia	3.6	3.9	3.9	3.9	3.8	3.8	0.1	0.2	3.8	3.3
Other Eurasia	1.2	1.4	1.5	1.5	1.6	1.6	0.3	1.0	1.3	1.4
Other Europe	0.8	0.9	0.9	0.9	0.9	0.9	0.1	0.3	0.8	0.8
Non-OECD	**55.0**	**61.9**	**68.1**	**73.1**	**77.6**	**81.7**	**26.7**	**1.6**	**57.9**	**72.0**
World	**94.9**	**102.2**	**107.4**	**110.0**	**111.9**	**113.5**	**18.6**	**0.7**	**100**	**100**

Source: OPEC.

In the OECD, oil demand is expected to show limited growth in the medium term, reaching 40.3 mboe/d by 2030, before declining to just under 32 mboe/d in 2050 – an overall decline in demand of more than 8 mboe/d over the full outlook period. These trends reflect rising competition from other fuels such as gas, nuclear and renewables, as well as increasing electrification, particularly in road transport through the growing adoption of EVs. Apart from developments in policy and technology, oil demand in the OECD is also set to be negatively affected by an ageing population and relatively stagnant long-term economic growth.

In OECD Americas, oil demand is expected to increase over the medium term from 21.2 mboe/d in 2025, before gradually declining to 18 mboe/d by 2050, a drop of 3.2 mboe/d. Robust petrochemical activity, freight transport and aviation continue to lend support. However, the pace of the shift away from oil remains uncertain and will depend on future developments in US climate and energy policies. In OECD Europe, oil demand sees a more pronounced decline of 3.4 mboe/d by 2050, driven by policies that include the EU's 2030 targets and 2040 ambitions, tighter vehicle standards and expanding biofuel mandates. The OECD Asia-Pacific region follows a more modest decline trajectory due to much lower base demand in 2025, with demand expected to decline by 1.5 mboe/d to 2050. This is mainly due to efficiency improvements and energy policy developments. However, oil continues to play a major role in freight, industry and aviation across the region.

Further details about regional and sectoral oil demand trends, as well as demand for specific refined products, are available in Chapter 3. However, it is important to note that the numbers presented in Chapter 3 are not directly comparable with those shown in this section. There are two main reasons for this. First, Chapter 2 uses energy equivalent units (mboe/d) to allow for a comparison between the different primary fuel types. In other chapters, however, oil is expressed in volumetric units of mb/d. Second, the definition of oil in Chapter 2 is different from that used in chapters 3 through 6. While Chapter 2 deals with primary energy sources, other chapters consider the outlook for all liquid fuels. In this sense, biofuels are considered biomass in this chapter, coal-to-liquids (CTLs) as coal and gas-to-liquids (GTLs) as gas, but they are all part of the liquids outlook in chapters 3 through 6.

2.3.2 Natural gas

Following market disruptions caused by the conflict in Eastern Europe during 2022 and 2023, and the recovery observed in 2024, global natural gas demand continued to rise in 2025 thanks to industrial expansion and rising electricity demand in Asia, the Middle East and OECD Americas. Although the pace of growth in natural gas demand slowed in 2025, gas-fired power generation remained important, often helping to balance the increasing share of renewable energy in electricity systems. This role is expected to become more important as coal-fired capacity continues to be retired in several regions, increasing the reliance on flexible generation sources.

On the supply side, investment in new gas and LNG infrastructure remains strong, driven primarily by expansion in the US and Qatar. In OECD Americas, the US is leading midstream capacity growth with projects such as Golden Pass, Blackstone Phase II, Port Arthur and Rio Grande, along with Louisiana LNG and the Corpus Christi 8-9 midstream units, while Canada has entered the export market with LNG Canada. In the Middle East, supply growth is centred on the expansion of Qatar's fields, with increased capacity also expected in the UAE and Oman later this decade. Elsewhere, Africa could see new supplies from Nigeria and the Republic of the Congo, with Mozambique potentially adding LNG capacity by the end of the decade. In other regions, new capacity from Australia and Mexico could help offset declining output from older facilities in Malaysia and Indonesia.

Looking ahead, as shown in Table 2.8 and Figure 2.5, global natural gas demand is expected to rise from 72 mboe/d in 2025 to more than 77 mboe/d by 2030, with growth concentrated almost entirely in non-OECD economies. China, India, Other Asia and the Middle East account for most of the increase. This reflects expanding industrial activity and the expanding role of gas in power generation, where it increasingly substitutes for coal. OECD demand, by contrast, shows only limited growth in the medium term.

In the long term, global gas demand is expected to continue rising to more than 91 mboe/d by 2050, an addition of over 19 mboe/d from 2025 levels. This expansion is driven by non-OECD regions, while OECD demand plateaus and gradually declines after 2040 due to efficiency improvements and tighter emissions frameworks. Policy shifts in some markets, including the US, introduce near-term variability. Nevertheless, gas remains a widely available and flexible fuel, supporting power systems with rising renewable shares while meeting growing industrial demand.

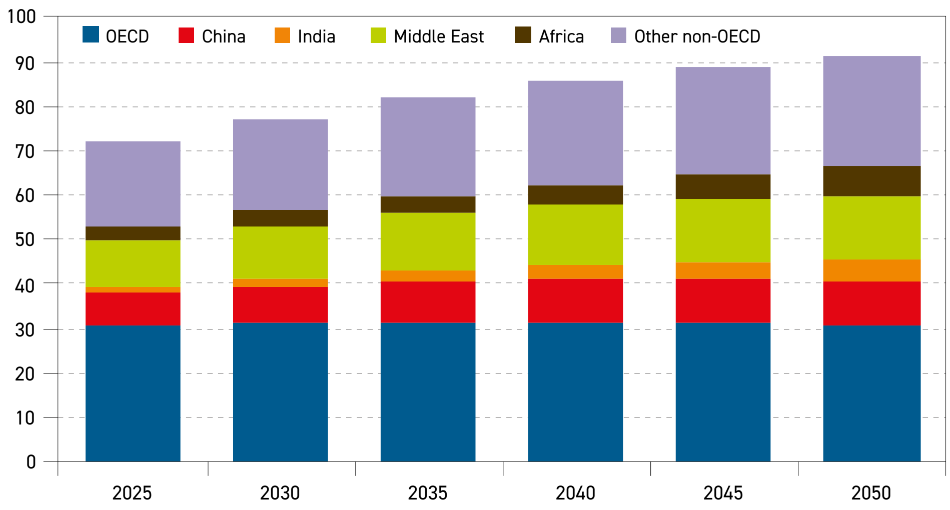
From a regional perspective, the Middle East is expected to see the largest increase, with demand rising by 4 mboe/d between 2025 and 2050. This growth is driven by expanding electricity generation supported by new gas power plants coming online, petrochemical development and the availability of competitive domestic gas resources. Another driver is the growing use of LNG as a bunkering fuel, as it accounts for more than a quarter of

Table 2.8
Natural gas demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	19.9	20.3	20.6	20.8	20.8	20.5	0.5	0.1	27.7	22.4
OECD Europe	7.3	7.2	7.0	6.9	6.7	6.4	–0.8	–0.5	10.1	7.1
OECD Asia-Pacific	3.7	3.7	3.7	3.8	3.8	3.8	0.1	0.1	5.1	4.2
OECD	**30.9**	**31.2**	**31.4**	**31.4**	**31.2**	**30.7**	**–0.2**	**0.0**	**43.0**	**33.6**
China	7.2	8.4	9.2	9.6	9.8	9.9	2.7	1.3	10.0	10.8
India	1.3	1.8	2.3	3.1	4.0	5.2	3.8	5.6	1.8	5.7
Other Asia	4.9	5.8	6.7	7.4	7.9	8.1	3.2	2.0	6.9	8.9
Latin America	2.5	3.0	3.5	4.0	4.6	5.1	2.7	3.0	3.4	5.6
Middle East	10.3	11.7	12.8	13.5	14.1	14.4	4.0	1.3	14.4	15.7
Africa	3.0	3.5	4.1	4.8	5.6	6.5	3.6	3.2	4.1	7.1
Russia	8.7	8.4	8.1	7.9	7.6	7.4	–1.3	–0.6	12.0	8.1
Other Eurasia	2.8	3.0	3.3	3.5	3.6	3.6	0.8	1.0	4.0	4.0
Other Europe	0.3	0.3	0.4	0.4	0.4	0.4	0.1	1.2	0.4	0.5
Non-OECD	**41.0**	**45.9**	**50.6**	**54.2**	**57.6**	**60.6**	**19.5**	**1.6**	**57.0**	**66.4**
World	**72.0**	**77.1**	**81.9**	**85.7**	**88.9**	**91.3**	**19.3**	**1.0**	**100**	**100**

Source: OPEC.

Figure 2.5
Natural gas demand by region, 2025–2050, mboe/d



Source: OPEC.

the new-build ship order book. Several countries – including Saudi Arabia, IR Iran, the UAE and Iraq – have announced plans to increase gas production, which also supports efforts to reduce crude oil use in power generation.

China's gas demand is set to increase to meet growing demand in the industrial, power and transportation sectors, with demand rising from above 7 mboe/d in 2025 to nearly 10 mboe/d by 2050. Furthermore, gas is expected to play a role in substituting coal and balancing generation from renewables. India's demand is expected to expand more rapidly, adding 3.8 mboe/d, supported by urbanization, air-quality initiatives, expanding gas distribution networks and its greater use across the residential, commercial and transportation sectors.

Gas demand in Other Asia is expected to rise from 4.9 mboe/d in 2025 to 8.1 mboe/d by 2050. Industrial expansion and a gradual substitution of coal are anticipated to support this increase. At the same time, price sensitivity is expected to remain a defining feature of the region, with several countries likely to adjust LNG imports in response to market volatility, as observed during previous supply disruptions.

Africa is also set to record robust growth, with gas demand more than doubling from 3 mboe/d in 2025 to 6.5 mboe/d by 2050. Rising electricity demand, expanding industrial activity and efforts to replace traditional biomass in residential use are expected to drive this expansion, while the region's continued development of resources and investments in infrastructure are likely to support broader gas penetration across the continent.

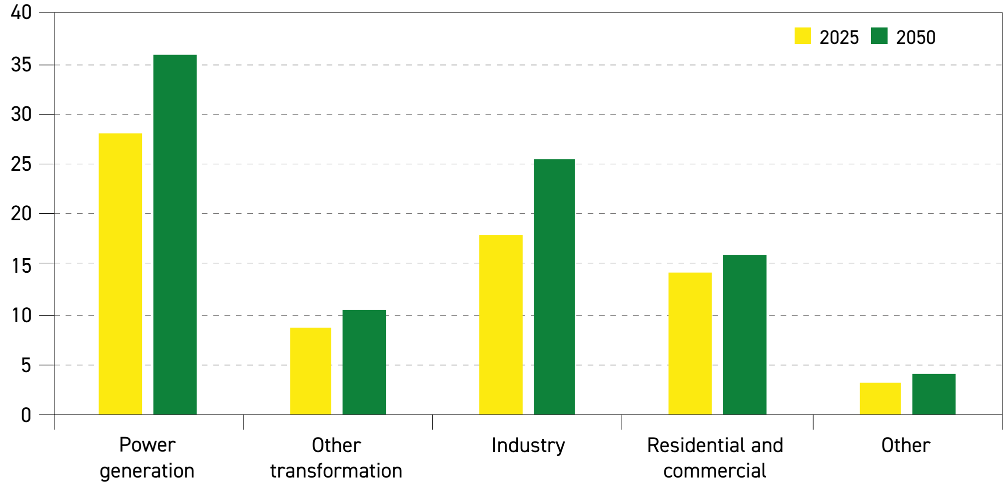
In contrast, Russia is projected to see a decline in gas demand of around 1.3 mboe/d between 2025 and 2050, reflecting demographic trends and efficiency gains. Other Eurasia is expected to post modest growth of around 0.8 mboe/d over the same period, supported by domestic resource availability and industrial development, while Other Europe is projected to remain stable, with gas continuing to replace coal in parts of the power generation sector and supporting electrification efforts.

In the OECD, gas demand is anticipated to remain stable throughout the 2030s, hovering around 31 mboe/d, before easing slightly to less than 31 mboe/d by 2050. In OECD Europe demand is expected to drop by around 0.8 mboe/d, shaped by tightening emissions policies, cleaner heating solutions and accelerated electrification in the residential and industry sectors. Efforts to substitute gas with renewables, nuclear power and renewable gases, such as biomethane, are anticipated to moderate long-term demand.

In contrast, demand in OECD Americas is expected to edge up from just below 20 mboe/d in 2025 to 20.5 mboe/d by 2050, supported by strong electricity demand – including the expansion of data centres – and industrial activity. Gas-fired generation is anticipated to continue playing a balancing role, driven by a large domestic resource base and established infrastructure. The OECD Asia-Pacific is projected to record only modest growth of around 0.1 mboe/d from 2025–2050, reflecting balancing needs as the penetration of renewables increases.

Figure 2.6 presents a breakdown of global natural gas demand from a sectoral perspective. Global gas demand is expected to increase across all sectors over the outlook period. This expansion is supported by the relatively lower emissions profile of gas compared to coal, certain policy frameworks favouring cleaner fuels, as well as the continued use of gas in power generation and industry. Abundant and cost-competitive supply, expanding LNG trade

Figure 2.6
Global natural gas demand by sector, 2025 and 2050, mboe/d



Source: OPEC.

and the development of CCUS are also all expected to support gas use in power, industry and blue hydrogen production.

The power generation sector remains the largest source of natural gas demand. Standing at 28.1 mboe/d in 2025, this figure is expected to increase to 36 mboe/d by 2050. Growth is driven by rising electricity demand and the need for flexible capacity, particularly in developing regions, while trends in the OECD remain mixed due to fuel substitution and efficiency gains.

Industrial demand for natural gas is also expected to increase strongly, rising from 18 mboe/d in 2025 to 25.4 mboe/d by 2050, supported by coal substitution in energy-intensive industries and rising petrochemical feedstock demand. Of note, CCUS is expected to facilitate emissions reductions in hard-to-abate processes.

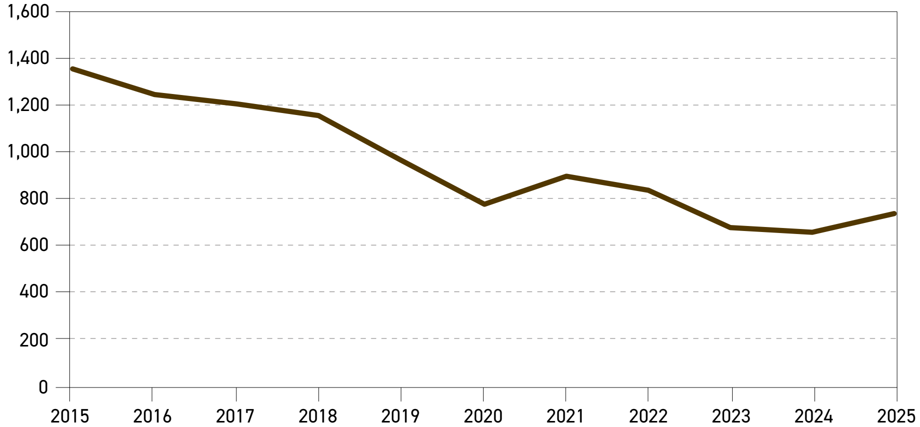
Demand in other transformation sectors is expected to increase from 8.7 mboe/d in 2025 to 10.3 mboe/d in 2050, largely due to rising hydrogen production. Residential and commercial demand grows more moderately, increasing from 14.2 mboe/d to 15.8 mboe/d over the outlook period, as declining consumption in developed economies is offset by expanding access and gasification in developing regions. Finally, demand for other uses – including transportation – is projected to rise from 3 mboe/d in 2025 to 3.8 mboe/d by 2050. Growth is driven primarily by the increased use of LNG in freight transport and shipping, alongside the continued expansion of gas pipeline transport, supporting broader energy trade.

2.3.3 Coal

In 2025, global coal demand recorded another year of growth, albeit at a slower pace than in 2024, to reach a new global high. Growth is supported primarily by developing economies – led by China, India and Other Asia – where rising electricity demand and energy security considerations continue to support coal's use.

At the same time, the phase-out of coal in advanced economies remains uneven. In the US, coal-fired power generation rose for the first time in around five years, as plant closures were paused, output increased during periods of extreme weather, including storms and freezing conditions, and in response to higher natural gas prices. However, coal power generation levels are still much lower than in 2015 (Figure 2.7). In OECD Europe, it should be noted that weaker renewable energy output can result in a temporary increase in coal-fired generation to maintain system balance, despite the overall decline in coal use.

Figure 2.7
Coal-fired electricity generation in the US, 2015–2025, TWh



Source: Ember.

Looking ahead, as shown in Table 2.9 and Figure 2.8, global coal demand is expected to ease slightly in the medium term. Declines in China and OECD Europe are expected to outweigh increases in India and parts of Other Asia. The power sector remains the main driver of these shifts, accounting for most of the global coal demand. Over the longer term, demand is expected to decline further, reaching 53.3 mboe/d by 2050 – a reduction of more than 29.3 mboe/d over the outlook period. This decline is concentrated in the OECD and China, while India, select Asian regions and Africa record overall increases.

Across the OECD, coal demand is expected to decline steadily over the outlook period, supported by climate policy, electrification trends and the expanding role of gas, nuclear and renewables in power generation. In OECD Americas, coal demand is projected to fall from 4.5 mboe/d in 2025 to 1.4 mboe/d by 2050. However, there is also a high degree of uncertainty in this respect, as the US continues to focus on energy security and expedite the development of its national resources.

OECD Europe is expected to see a reduction in coal demand by 2.4 mboe/d between 2025 and 2050, due to climate policy objectives – including the EU's 2040 emissions target – and accelerated renewable and nuclear energy deployment. The OECD Asia-Pacific follows a more moderate path, declining from 4.1 mboe/d in 2025 to 2.5 mboe/d in 2050, shaped by legacy coal infrastructure and varied national approaches to power system and energy security.

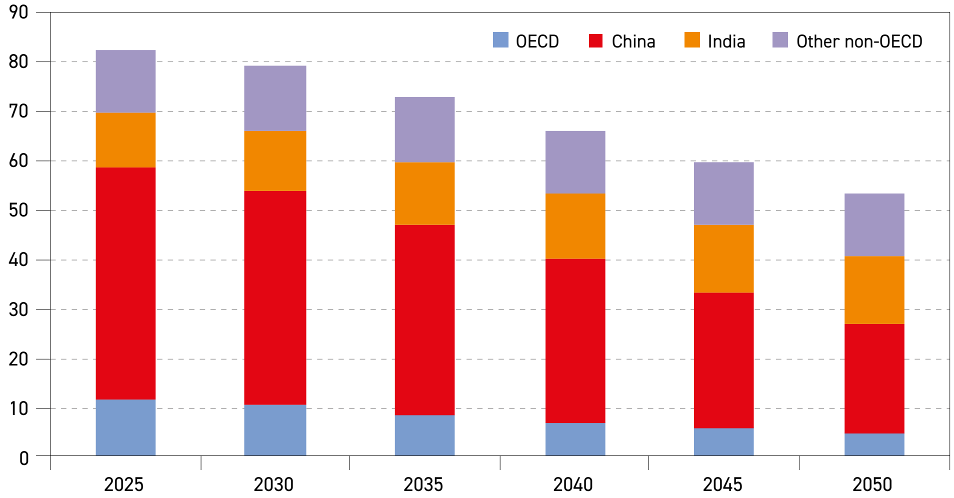
Within the non-OECD and globally, China represents the main source of long-term decline. Over the medium term, coal demand is projected to drop by 3.6 mboe/d to 43.7 mboe/d in 2030. This is mainly due to efforts to achieve a peak in coal consumption by around

Table 2.9
Coal demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	4.5	4.3	3.4	2.6	1.9	1.4	–3.1	–4.5	5.4	2.6
OECD Europe	3.0	2.3	1.7	1.2	0.9	0.6	–2.4	–6.0	3.7	1.2
OECD Asia-Pacific	4.1	3.8	3.5	3.3	3.0	2.5	–1.6	–2.0	5.0	4.7
OECD	**11.6**	**10.5**	**8.6**	**7.1**	**5.8**	**4.5**	**–7.0**	**–3.7**	**14.0**	**8.5**
China	47.3	43.7	38.5	32.9	27.4	22.3	–25.0	–3.0	57.3	41.9
India	10.7	12.0	12.9	13.3	13.7	14.1	3.3	1.1	13.0	26.4
Other Asia	5.9	6.2	6.4	6.6	6.8	6.9	1.0	0.6	7.1	12.9
Latin America	0.4	0.4	0.4	0.4	0.4	0.4	0.1	0.6	0.5	0.8
Middle East	0.1	0.1	0.1	0.1	0.1	0.1	0.0	0.5	0.1	0.2
Africa	2.1	2.2	2.2	2.2	2.3	2.3	0.2	0.4	2.5	4.3
Russia	3.0	2.8	2.5	2.3	2.0	1.7	–1.3	–2.2	3.7	3.3
Other Eurasia	1.1	1.1	1.0	1.0	1.0	0.9	–0.2	–0.8	1.3	1.7
Other Europe	0.3	0.3	0.2	0.1	0.1	0.1	–0.3	–6.3	0.4	0.1
Non-OECD	**71.0**	**68.6**	**64.3**	**59.0**	**53.8**	**48.8**	**–22.2**	**–1.5**	**86.0**	**91.5**
World	**82.6**	**79.1**	**72.9**	**66.1**	**59.5**	**53.3**	**–29.3**	**–1.7**	**100**	**100**

Source: OPEC.

Figure 2.8
Coal demand by region, 2025–2050, mboe/d



Source: OPEC.

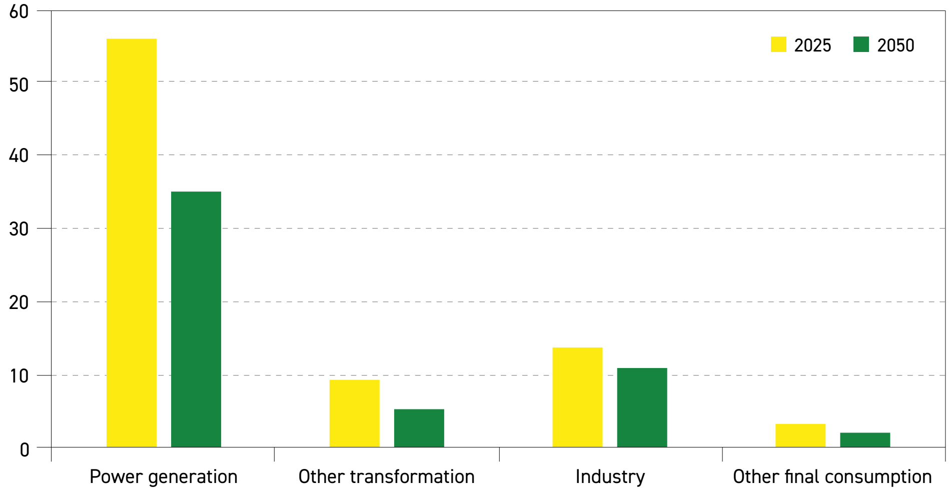
2030. Thereafter, demand is set to decline to 22.3 mboe/d by 2050. This reduction reflects a continued expansion of wind, solar and nuclear capacity, alongside broader emissions targets and efficiency improvements in newer coal-fired power plants, which require less coal compared to older units.

In contrast, coal demand in India is projected to rise in the coming decades, increasing from 10.7 mboe/d in 2025 to 13.3 mboe/d by the early 2040s, before growth slows and demand approaches just above 14 mboe/d by 2050. Economic expansion, infrastructure development and rising electricity demand drive this increase, although clean energy policies and air-quality measures are expected to moderate coal demand over time.

Other Asia is expected to record a modest increase of 1 mboe/d over the long term, supported by industrial expansion and strong electricity demand. In Africa, coal demand rises by 0.2 mboe/d, reflecting expanding access to electricity and industrial development. Latin America shows limited change (+0.1 mboe/d), as hydropower, gas and renewables dominate power generation. The Middle East remains largely coal-free, with demand staying minimal at around 0.1 mboe/d, while Russia, Other Eurasia and Other Europe experience declines of 1.3 mboe/d, 0.2 mboe/d and 0.3 mboe/d, respectively. This is driven by efficiency improvements and the increasing competitiveness of gas and nuclear energy.

From a sectoral perspective, global coal demand in the power generation sector reached around 56 mboe/d in 2025. Over the outlook period, however, demand in this sector is expected to drop to around 35 mboe/d by 2050. The bulk of this reduction comes in China, where coal demand in power generation is expected to decline by around 46% over the outlook period, accounting for nearly three-quarters of the global decline. OECD regions further reinforce this downward trend, as coal is gradually replaced by alternative energy sources – particularly in the US, which records the largest reduction among advanced economies. These declines are only partly offset by rising coal demand in some non-OECD countries, including India and some other Asian countries, where electricity demand continues to expand.

Figure 2.9
Global coal demand by sector, 2025 and 2050, mboe/d



Source: OPEC.

The industrial sector represented the second largest source of coal demand in 2025, amounting to approximately 13.9 mboe/d. Coal demand for the industrial sector is concentrated in China. India also accounts for a large share of coal demand in industry, driven by its expanding industrial base. Coal demand in the industrial sector is mainly shaped by opposing forces, where emerging economies increase demand to fuel their economic growth and industrialization processes, while mature economies drive efficiency gains and substitution with alternative energy sources, reinforcing downward pressure on coal demand.

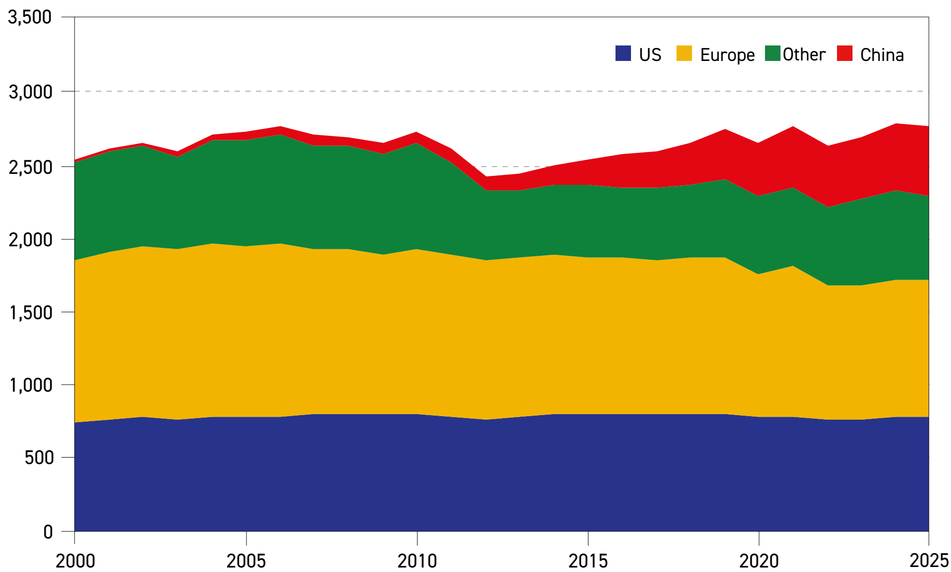
As a result, global coal demand is expected to decline by 2.8 mboe/d in the industrial sector to reach 11 mboe/d in 2050. This represents a smaller reduction than the decline observed in the power generation sector, as it reflects the technical complexity and costs associated with switching from coal to alternative energy sources, as well as constraints related to industrial processes and competitiveness.

Aside from the power generation and industrial sectors, coal demand remains comparatively limited. Demand associated with other transformation activities, including processing and own use, is forecast to drop from 9.3 mboe/d in 2025 to 5.3 mboe/d by 2050. As for other final consumption, coal demand is expected to decrease from 3.4 mboe/d in 2025 to 2.1 mboe/d in 2050.

2.3.4 Nuclear

After the Fukushima disaster in 2011, several advanced economies reduced their reliance on nuclear power. For example, Germany completed the shutdown of its remaining reactors in 2023. Over the same period, the centre of nuclear expansion shifted towards Asia, especially China, where rising electricity demand, air-quality concerns and energy security priorities supported strong nuclear expansion (Figure 2.10).

Figure 2.10
Nuclear generation by region, TWh



Source: Ember.

Nuclear energy is now in one of its strongest positions in decades, with around 415 operating reactors providing more than 379 GW of capacity worldwide. Although total operating capacity declined marginally in 2025 due to retirements exceeding new additions – including several reactor closures in Belgium as part of its earlier phase-out policy – the broader outlook remains supportive.

Indeed, underlining how nuclear policy in some advanced economies continues to adjust in response to evolving considerations regarding energy security and supply, Belgium has since decided to extend the operational lifespan of its remaining reactors to 2035, and has even reopened the possibility of new nuclear development. Japan represents another example of this shift in thinking. Before the Fukushima disaster of 2011, Japan's installed nuclear capacity stood at an impressive 47 GW. While it is unlikely that these levels will be reached again, Japan's nuclear industry has nevertheless gained momentum in recent years. Japan currently operates 14 reactors in total, representing approximately 12.6 GW of installed capacity. Unit 6 at the Kashiwazaki-Kariwa Nuclear Power Plant, one of the world's largest nuclear facilities, restarted in early 2026, bringing back online 1.3 GW of the plant's total capacity of 8.2 GW.

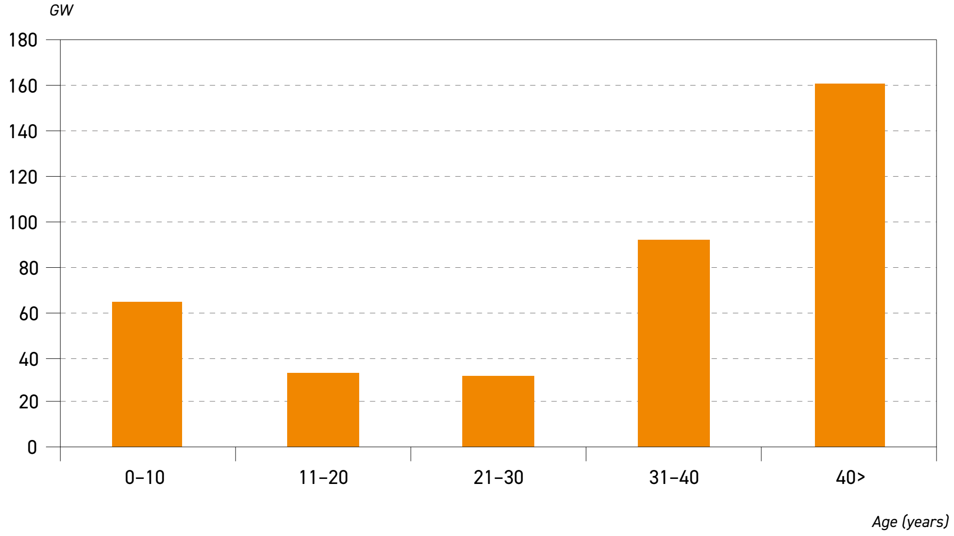
Looking ahead, the momentum is undeniable. The global project pipeline features 72 reactors – representing roughly 75 GW of installed capacity – currently under construction. However, activity remains uneven across regions. China remains at the forefront, leading new development and accounting for 35 of these reactors, while advancing standardized designs to shorten build times. Additional projects of note that are progressing can be found in Russia, Türkiye, India, Bangladesh and parts of the Middle East, where nuclear power is expected to strengthen energy diversification and long-term supply stability.

Within the OECD, developments are more selective. In 2025, OECD Americas did not have any reactors under construction, while new-build projects have advanced only cautiously in OECD Europe, with some countries, including the UK, Slovakia and Hungary, currently having reactors under construction, despite both regions having some of the highest nuclear shares in electricity generation. At the same time, rising baseload requirements – particularly from data centres and increasingly electrifying industrial sectors – are renewing interest in advanced reactor designs and SMRs. This suggests that nuclear development in the medium term could expand beyond traditionally active regions.

Alongside new construction, lifetime extensions are expected to remain supportive of nuclear supply. Many reactors in OECD countries are moving from initial forty-year licences towards sixty years of operation (Figure 2.11), supported by upgraded safety systems, improved monitoring technologies and the growing value of firm capacity as coal power plants retire and renewables increase. Technical assessments are also examining the feasibility of extending operations out to between 70 and 80 years in some cases. If realized, these extensions would sustain nuclear generation without the financial and logistical requirements associated with new builds, thus further supporting the role of nuclear energy in systems where stable output is increasingly valuable amid the rising penetration of renewables.

Policy support drives much of this evolving landscape. Across several OECD countries, long-term electricity policies increasingly identify nuclear energy as being essential in meeting climate objectives while maintaining affordability and system adequacy. Investment mechanisms aimed at reducing financial risk for large-scale projects, together with clearer regulatory pathways for SMRs, are adding to confidence in the sector. In OECD Europe, nuclear

Figure 2.11
Nuclear net power generation capacity globally by age



Source: IAEA.

power continues to supply a significant share of electricity generation. France maintains one of the highest nuclear shares globally, while several countries in Central Europe and Northern Europe rely substantially on nuclear energy as part of their energy security strategies.

Outside the OECD, nuclear expansion is primarily driven by the need to diversify supply and meet rapidly rising electricity demand. At the same time, the expansion of AI-related data centres, characterized by continuous, concentrated and high-volume electricity use, is reinforcing the appeal of nuclear energy as a dependable baseload source capable of supporting emerging digital infrastructure. Taken together, lifetime extensions, new construction, supportive policy frameworks and rising electricity demand are expected to position nuclear as a durable component of future power systems. Across many regions, nuclear energy is increasingly complementing growing renewable capacity by providing stable output in systems facing higher variability.

Table 2.10 and Figure 2.12 present the long-term outlook for nuclear demand. Globally, nuclear energy is projected to increase from 15 mboe/d in 2025 to 25.5 mboe/d by 2050. Around 75% of this growth is expected to materialize in the non-OECD, primarily led by China, which is projected to triple its nuclear demand from 2.6 mboe/d in 2025 to around 7.5 mboe/d by 2050. The country's expansion of nuclear energy is supported by its rising electricity demand and continued emphasis on energy security. China is also advancing SMR development, which could support longer-term nuclear expansion.

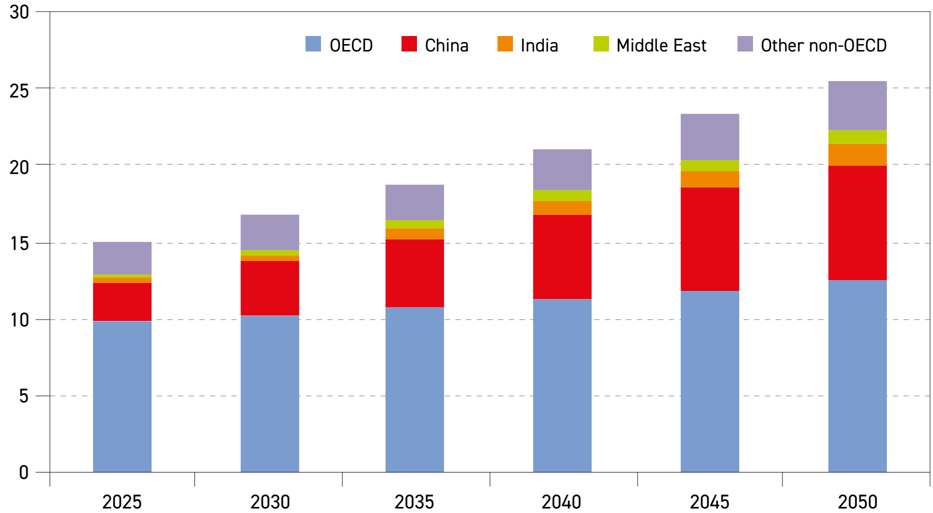
In India, nuclear demand is projected to quintuple over the outlook period, rising from 0.3 mboe/d in 2025 to 1.5 mboe/d by 2050. The country is targeting 100 GW of nuclear capacity by 2047 as part of the 'Viksit Bharat @2047 initiative', including the development of a fast breeder reactor programme and thorium-based initiatives, both of which could shape its longer-term nuclear trajectory.

Table 2.10
Nuclear energy demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	4.8	4.9	5.0	5.2	5.4	5.7	0.8	0.6	32.3	22.2
OECD Europe	3.4	3.4	3.6	3.7	3.8	4.0	0.6	0.7	22.7	15.8
OECD Asia-Pacific	1.6	1.9	2.1	2.4	2.6	2.8	1.2	2.4	10.4	11.0
OECD	**9.8**	**10.2**	**10.7**	**11.2**	**11.8**	**12.5**	**2.7**	**1.0**	**65.4**	**49.0**
China	2.6	3.5	4.6	5.7	6.7	7.5	5.0	4.4	17.2	29.5
India	0.3	0.4	0.6	0.9	1.2	1.5	1.2	6.7	1.9	5.8
Other Asia	0.2	0.3	0.3	0.3	0.3	0.3	0.1	1.2	1.6	1.2
Latin America	0.1	0.1	0.2	0.2	0.2	0.2	0.1	1.9	0.8	0.8
Middle East	0.3	0.4	0.5	0.6	0.8	0.9	0.6	5.2	1.7	3.6
Africa	0.0	0.0	0.0	0.1	0.1	0.1	0.0	1.8	0.3	0.3
Russia	1.1	1.2	1.3	1.4	1.6	1.7	0.5	1.5	7.7	6.5
Other Eurasia	0.4	0.4	0.5	0.5	0.6	0.6	0.3	2.2	2.4	2.5
Other Europe	0.1	0.2	0.2	0.2	0.2	0.2	0.1	2.2	1.0	1.0
Non-OECD	**5.2**	**6.5**	**8.1**	**9.8**	**11.5**	**13.0**	**7.8**	**3.8**	**34.6**	**51.0**
World	**15.0**	**16.7**	**18.8**	**21.0**	**23.3**	**25.5**	**10.5**	**2.2**	**100**	**100**

Source: OPEC.

Figure 2.12
Nuclear energy demand by region, 2025–2050, mboe/d



Source: OPEC.

Nuclear demand in the Middle East is projected to increase from around 0.3 mboe/d in 2025 to 0.9 mboe/d by 2050, supported by multi-reactor programmes and efforts to reduce reliance on hydrocarbons in power generation. Africa is expected to record modest growth, as South Africa maintains its existing fleet and Egypt progresses towards commissioning new units. Latin America and Eurasia are projected to see incremental increases through targeted additions and the optimization of existing plants, while Russia is set to maintain stable-to-slightly higher demand supported by ongoing modernization and its export-oriented reactor programme.

In the OECD, nuclear demand is expected to rise from 9.8 mboe/d in 2025 to 12.5 mboe/d by 2050. The OECD Asia-Pacific leads this increase, supported by Japan's reactor restarts and South Korea's continued policy backing. OECD Europe is expected to see growth as part of lifetime extensions and selective new-build decisions, aimed at offsetting coal retirements and supporting grid stability as wind and solar shares expand. In OECD Americas, the US remains the largest nuclear producer globally, operating nearly 94 reactors with around 97 GW of capacity. Looking ahead, demand is projected to grow modestly on the back of lifetime extensions and improved operational performance, which sustains output amid rising electricity consumption, including from the expanding digital sector.

An additional and increasingly relevant driver of nuclear interest is the rapid expansion of AI and hyperscale data centres, which require continuous and highly reliable electricity supply. Recently, major technology companies including Amazon, Google, Microsoft and Meta have announced potential nuclear-linked power deals ranging from roughly 300 MW to more than 10 GW. Many proposals involve SMRs or dedicated nuclear supply agreements aligned with data centre demand profiles. Although most projects remain at an early stage, they signal a broader shift in electricity market dynamics. Nuclear energy is increasingly linked not only to traditional power planning, but also to digital infrastructure expansion. Over time, this connection could improve investor confidence, support advanced reactor deployment and reinforce the role of nuclear energy as electricity systems adapt to sustained demand growth.

2.3.5 Renewables

This section focuses on renewable energy demand, including biomass, solar, wind, hydropower and other sources, such as geothermal, tidal and wave energy. As already discussed, renewables are projected to record the largest collective growth in demand over the outlook period. Total renewable demand is expected to more than double from around 48 mboe/d in 2025 to just above 99 mboe/d by 2050, reflecting an average growth rate of about 3% p.a. As a result, the share of renewables in the global energy mix is projected to increase from just over 15% in 2025 to nearly 26% by 2050.

Solar

Global solar power capacity additions reached a record level of approximately 510 GW in 2025 marking another year of exceptional expansion. China remained by far the largest contributor, adding over 315 GW, more than half the global total. The US followed, adding around 35 GW, while significant additions were also recorded in Brazil, Germany, Saudi Arabia, and other emerging markets in the Middle East and Latin America. At the end of 2025, total global capacity was at more than 2,390 GW.

After years of extremely rapid expansion, however, growth has begun to slow in many of the largest markets. The sheer scale of recent additions has begun to weigh on system economics.

The widespread deployment of solar has led to a sharp drop in wholesale electricity prices during peak production hours, increased grid congestion and higher levels of curtailment. In Brazil, for example, curtailment of large-scale solar power has reached significant levels in some regions, at times approaching one-fifth of total production. Similar pressures have emerged in parts of China and many European markets, where the number of hours of zero or negative pricing increased significantly.

The long-term outlook for solar energy by region is highlighted in Table 2.11 and Figure 2.13. Demand is expected to rise from 4.9 mboe/d in 2025 to 28.5 mboe/d by 2050, an increase of 23.6 mboe/d over the outlook period. Solar expansion is mainly driven by non-OECD regions, led by China, although growth continues across the OECD, particularly in OECD Americas. Over time, demand shifts more clearly towards emerging non-OECD markets, especially India, Other Asia and the Middle East. Additionally, China's earlier large-scale buildout continues to account for a substantial share of global installed capacity, and it maintains a dominant position in manufacturing. Rising electricity demand in these regions underpins most of the additional growth to 2050.

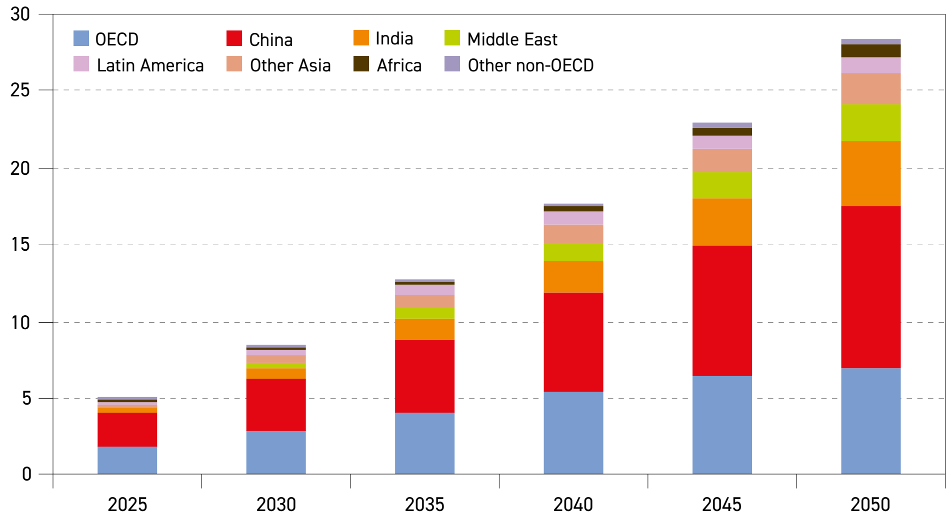
At the same time, uncertainty remains elevated over the outlook period. Solar deployment is highly vulnerable to supply chain conditions, including critical minerals and module costs, while rising grid integration and system balancing costs are expected to play an important role in shaping the pace and scale of future demand.

Table 2.11
Solar energy demand by region, 2025–2050

	Levels *mboe/d*						Growth *mboe/d*	Growth *% p.a.*	Share *%*	
	2025	**2030**	**2035**	**2040**	**2045**	**2050**	**2025–2050**	**2025–2050**	**2025**	**2050**
OECD Americas	0.7	1.2	1.9	2.8	3.5	3.7	2.9	6.6	15.3	12.9
OECD Europe	0.7	0.9	1.2	1.5	1.8	2.0	1.4	4.4	14.2	7.2
OECD Asia-Pacific	0.4	0.6	0.9	1.0	1.1	1.2	0.8	4.5	8.1	4.2
OECD	**1.8**	**2.8**	**4.0**	**5.3**	**6.5**	**6.9**	**5.1**	**5.4**	**37.7**	**24.3**
China	2.2	3.5	4.9	6.6	8.5	10.6	8.4	6.5	44.7	37.3
India	0.3	0.7	1.3	2.0	3.0	4.2	3.9	11.0	6.4	14.9
Other Asia	0.2	0.5	0.8	1.2	1.5	2.0	1.8	10.0	3.8	7.1
Latin America	0.2	0.4	0.6	0.7	0.9	1.0	0.8	6.8	3.9	3.5
Middle East	0.1	0.3	0.7	1.2	1.7	2.4	2.3	14.8	1.6	8.4
Africa	0.0	0.1	0.2	0.4	0.6	0.8	0.8	12.6	0.9	2.9
Russia	0.0	0.0	0.0	0.0	0.0	0.0	0.0	6.1	0.1	0.1
Other Eurasia	0.0	0.0	0.1	0.1	0.2	0.3	0.3	11.4	0.4	1.1
Other Europe	0.0	0.0	0.1	0.1	0.1	0.1	0.1	6.4	0.5	0.4
Non-OECD	**3.0**	**5.6**	**8.8**	**12.4**	**16.5**	**21.5**	**18.5**	**8.1**	**62.3**	**75.7**
World	**4.9**	**8.4**	**12.7**	**17.7**	**23.0**	**28.5**	**23.6**	**7.3**	**100**	**100**

Source: OPEC.

Figure 2.13
Solar energy demand by region, 2025–2050, mboe/d



Source: OPEC.

China remains the single largest contributor, with solar demand projected to increase from 2.2 mboe/d in 2025 to 10.6 mboe/d by 2050. Growth is expected to moderate as grid constraints, curtailment risks and market saturation in some provinces become more visible. Nevertheless, strong industrial policy and manufacturing scale continue to drive deployment, supporting China's position in terms of domestic expansion and global supply chains.

Elsewhere in the non-OECD, the largest share of additional solar demand is expected to come from India, followed by Other Asia, the Middle East and Africa. Other Asia is projected to add around 1.8 mboe/d between 2025 and 2050, while the Middle East adds 2.3 mboe/d, supported by large-scale utility projects and diversification strategies, particularly in Saudi Arabia and the UAE. Africa and Latin America each add 0.8 mboe/d over the outlook period, reflecting improving project economics and expanding electricity access.

Within the OECD, solar demand is expected to grow, albeit at a more moderate pace, increasing from 1.8 mboe/d in 2025 to 6.9 mboe/d by 2050. OECD Americas accounts for the largest share of this, rising from 0.7 mboe/d to 3.7 mboe/d. This is supported by favourable project economics, corporate power procurement and expanding electricity demand from digital infrastructure. OECD Europe grows from 0.7 mboe/d to 2.0 mboe/d, as solar complements mature wind fleets and supports electrification objectives, though deployment is increasingly shaped by grid congestion, land-use constraints and system-balancing needs. The OECD Asia-Pacific is set to increase from 0.4 mboe/d to 1.2 mboe/d, reflecting supportive policy frameworks but slower growth in more saturated markets.

Wind

Global wind power recorded strong growth in 2025, with installations reaching around 160 GW, driven overwhelmingly by onshore wind, which accounted for 93% of new capacity. This expansion is primarily led by Asia, particularly China, which alone added more than

119 GW of capacity in a single year for the first time. Although at a much smaller scale, India also reached record additions of more than 6 GW, positioning it to become the third largest onshore wind market in 2025. Led by Germany, Europe registered strong expansion as well, with new additions reaching an all-time high of about 14 GW. Elsewhere, Türkiye emerged as a fast-growing market, with large projects progressing that use Chinese turbines. In the US, wind installations reached more than 6 GW, reflecting a modest rebound following several years of decline.

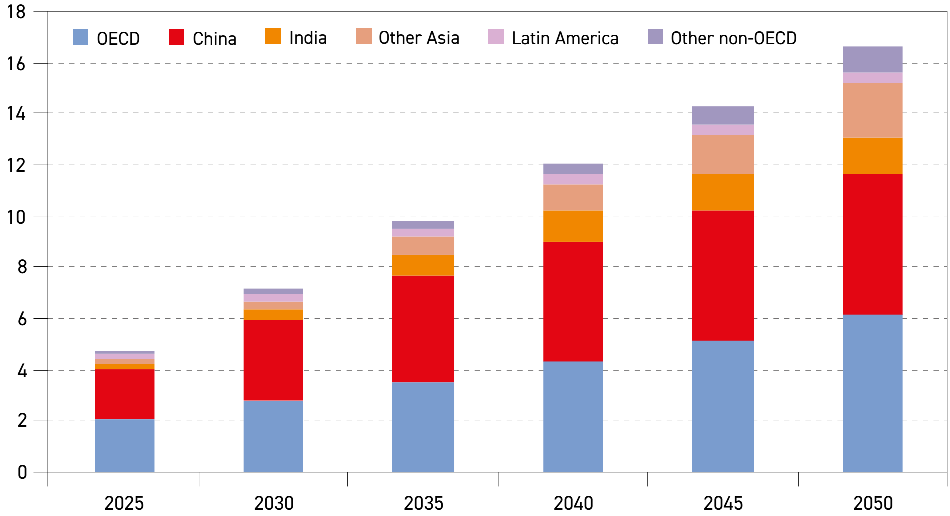
Table 2.12 and Figure 2.14 present the long-term outlook for wind energy demand by region. Global primary energy demand from wind is anticipated to rise from around 4.7 mboe/d in 2025 to 16.6 mboe/d by 2050, amounting to an average growth rate of 5.1% p.a. Growth is expected to be driven by the non-OECD, particularly China, India and Other Asia, which remain the largest markets over both the medium and long term. At the same time, deployment is expected to increase across all OECD regions.

In the OECD, wind demand is expected to triple over the outlook period, increasing from just above 2 mboe/d in 2025 to 6.2 mboe/d by 2050. OECD Europe accounts for the largest share of this increase, with demand rising from just above 1 mboe/d to more than 3 mboe/d. Recent permitting reforms are expected to continue to support project approvals, particularly in established markets such as Germany, the UK and the Netherlands, while Poland emerges as an additional contributor. Nevertheless, ageing infrastructure, grid congestion and the pace of network expansions are anticipated to influence the speed of deployment over the remainder of the outlook period.

Table 2.12
Wind energy demand by region, 2025–2050

	Levels *mboe/d*						Growth *mboe/d*	Growth *% p.a.*	Share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	0.9	1.2	1.4	1.6	1.8	2.1	1.1	3.2	19.9	12.6
OECD Europe	1.1	1.4	1.8	2.1	2.6	3.1	2.0	4.4	22.3	18.5
OECD Asia-Pacific	0.1	0.2	0.4	0.5	0.8	1.0	0.9	9.5	2.1	6.0
OECD	**2.1**	**2.8**	**3.5**	**4.3**	**5.2**	**6.2**	**4.1**	**4.4**	**44.4**	**37.0**
China	1.9	3.1	4.2	4.7	5.1	5.5	3.5	4.2	40.8	32.8
India	0.2	0.5	0.8	1.2	1.4	1.5	1.3	8.3	4.3	8.9
Other Asia	0.1	0.3	0.6	1.1	1.6	2.1	2.0	12.0	2.6	12.8
Latin America	0.3	0.3	0.3	0.4	0.4	0.4	0.2	2.0	5.5	2.6
Middle East	0.0	0.0	0.1	0.1	0.1	0.2	0.1	11.9	0.2	0.9
Africa	0.1	0.1	0.1	0.2	0.2	0.3	0.2	6.4	1.2	1.7
Russia	0.0	0.0	0.1	0.1	0.1	0.2	0.2	13.5	0.2	1.5
Other Eurasia	0.0	0.0	0.0	0.0	0.1	0.1	0.1	9.1	0.2	0.6
Other Europe	0.0	0.0	0.1	0.1	0.1	0.2	0.2	8.5	0.5	1.1
Non-OECD	**2.6**	**4.4**	**6.3**	**7.8**	**9.1**	**10.5**	**7.8**	**5.7**	**55.6**	**63.0**
World	**4.7**	**7.2**	**9.8**	**12.1**	**14.3**	**16.6**	**11.9**	**5.1**	**100**	**100**

Figure 2.14
Wind energy demand by region, 2025–2050, mboe/d



Source: OPEC.

In OECD Americas, wind demand is projected to rise from just under 1 mboe/d in 2025 to 2.1 mboe/d by 2050. Growth is supported by strong resource potential and competitive cost fundamentals. However, the outlook is increasingly shaped by policy uncertainty in the US. The 'OBBBA' has rolled back several clean energy provisions introduced under the previous administration, creating uncertainty around project economics and potentially affecting investment timing. As a result, growth is expected to continue but with more volatility than previously anticipated.

In the OECD Asia-Pacific, wind demand begins from a smaller base but is expected to increase steadily from around 0.1 mboe/d to close to 1 mboe/d by mid-century. Expansion is expected to be supported by streamlined environmental approvals, updated repowering frameworks and improved utilization of constrained coastal and mountainous sites, particularly in mature markets where land availability and grid access remain limiting factors.

By contrast, growth in non-OECD markets is stronger. Non-OECD wind demand rises from 2.6 mboe/d in 2025 to 10.5 mboe/d by 2050. China remains the largest market and is the major contributor over the outlook period, with demand projected to rise from just under 2 mboe/d in 2025 to 5.5 mboe/d by 2050. Growth moderates over time as expansion becomes more geographically diversified, with increasing contributions from other fast-growing markets, such as India and Other Asia.

India's wind demand is projected to rise from around 0.2 mboe/d in 2025 to 1.5 mboe/d by 2050, supported by growing electricity needs and continued renewable targets. However, transmission constraints and grid congestion may influence the pace of expansion, while declining solar and battery costs may redirect some investment. Wind power demand is also anticipated to increase in Other Asia (+2 mboe/d), representing the second largest increase among non-OECD regions.

Elsewhere, including Latin America, the Middle East, Africa, Russia, Other Eurasia and Other Europe, wind demand is expected to increase in the range of 0.1–0.2 mboe/d over the outlook period. Growth is set to be supported by key markets such as Brazil, Saudi Arabia, Egypt and South Africa, alongside smaller expansions in parts of Eastern Europe and Central Asia as countries seek to diversify electricity supply and strengthen energy security.

Hydro

Hydropower remains the largest source of renewable electricity globally and continues to play a central role in many countries' power generation sectors. Following drought conditions in 2023 that restrained output in several major markets, generation rebounded in 2024 and stayed strong throughout 2025 as conditions improved and new capacity entered operation. It reached a level of around 4,600 TWh in 2025. Around 18 GW of new capacity was added in 2025, led mainly by China and the broader Asia-Pacific region, with additional contributions from Latin America and Africa. Total installed capacity was estimated at around 1,300 GW at the end of 2025.

Additionally, storage use continues to expand alongside conventional hydropower, supporting its position as the largest source of long-duration electricity storage in many systems. By the end of 2025, total pure pumped-storage hydropower capacity had reached 160 GW.

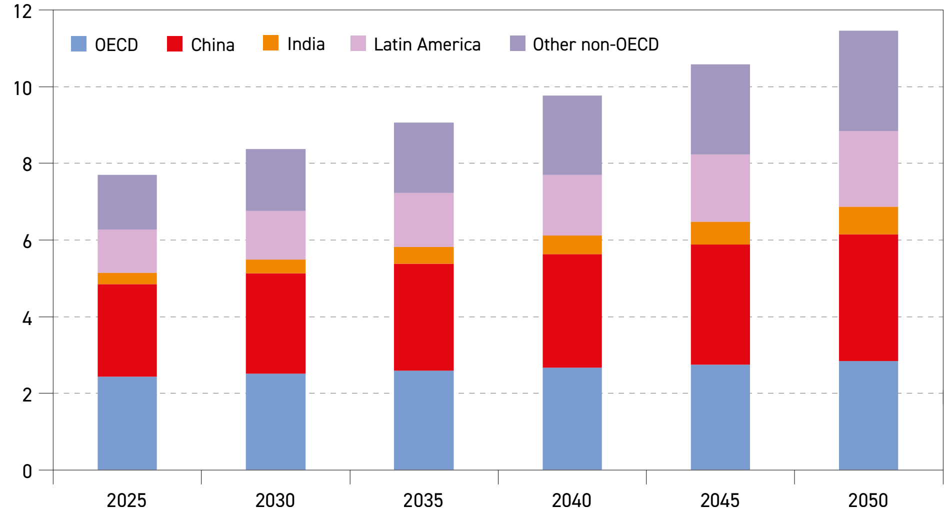
In the Reference Case, hydropower demand is estimated to increase from 7.7 mboe/d in 2025 to 11.5 mboe/d by 2050, representing growth of 3.8 mboe/d, of which 3.4 mboe/d is expected to be located in the non-OECD (Table 2.13 and Figure 2.15). Expansion is mainly concentrated in developing countries, while developed regions are projected to mainly rely on modernization, including upgrade and efficiency gains, rather than large-scale new developments.

Table 2.13
Hydro energy demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025-2050	2025-2050	2025	2050
OECD Americas	1.2	1.3	1.3	1.3	1.4	1.4	0.2	0.6	15.8	12.4
OECD Europe	1.0	1.1	1.1	1.1	1.2	1.2	0.2	0.6	13.3	10.3
OECD Asia-Pacific	0.2	0.2	0.2	0.2	0.2	0.2	0.0	0.6	2.6	2.0
OECD	**2.4**	**2.5**	**2.6**	**2.7**	**2.8**	**2.8**	**0.4**	**0.6**	**31.6**	**24.7**
China	2.4	2.6	2.8	3.0	3.1	3.3	0.9	1.3	31.4	28.9
India	0.3	0.4	0.4	0.5	0.6	0.7	0.4	3.6	3.9	6.3
Other Asia	0.6	0.6	0.7	0.8	1.0	1.1	0.5	2.7	7.2	9.4
Latin America	1.1	1.3	1.4	1.6	1.8	2.0	0.9	2.3	14.6	17.2
Middle East	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.4	0.3
Africa	0.3	0.4	0.5	0.6	0.7	0.8	0.5	3.9	3.8	6.6
Russia	0.3	0.3	0.4	0.4	0.4	0.4	0.1	1.1	4.3	3.8
Other Eurasia	0.1	0.1	0.2	0.2	0.2	0.2	0.1	1.5	1.7	1.7
Other Europe	0.1	0.1	0.1	0.1	0.1	0.1	0.0	1.5	1.1	1.1
Non-OECD	**5.3**	**5.9**	**6.5**	**7.1**	**7.8**	**8.6**	**3.4**	**2.0**	**68.4**	**75.3**
World	**7.7**	**8.4**	**9.1**	**9.8**	**10.6**	**11.5**	**3.8**	**1.6**	**100**	**100**

Source: OPEC.

Figure 2.15
Hydro energy demand by region, 2025–2050, mboe/d



Source: OPEC.

China is expected to remain the major contributor to growth, with hydropower demand rising from 2.4 mboe/d in 2025 to 3.3 mboe/d by 2050. This increase reflects new projects, the optimization of major river basin systems and the continued expansion of pumped storage. Hydropower is also projected to play an important role in supporting large-scale wind and solar integration while strengthening domestic supply security. At the same time, environmental considerations and permitting processes are expected to influence the pace of future dam construction.

Hydropower demand is also anticipated to increase in India (+0.4 mboe/d), Other Asia (+0.5 mboe/d), Latin America (+0.9 mboe/d) and Africa (+0.5 mboe/d) over the outlook period. This is supported by rising electricity demand and the continued development of the remaining resource potential.

In OECD regions, hydropower demand is projected to increase modestly, with OECD Americas and OECD Europe each rising by around 0.2 mboe/d, while OECD Asia-Pacific remains broadly unchanged. This growth is driven by upgrades and efficiency improvements, with few major new installations expected to come online. Nevertheless, long-term output is likely to remain sensitive to hydrological variability, and climate-related water availability continues to represent an important source of uncertainty across several regions.

Biomass

Biomass is one of the most important components of the global energy mix and continues to be the largest renewable energy source in primary energy terms. It supports power generation, heating, industrial feedstocks and, particularly in households in emerging economies, energy use and access. The outlook reflects two contrasting dynamics: in advanced economies, deployment is increasingly shaped by sustainability certification, lifecycle-emissions criteria and waste-to-energy strategies, while in developing regions biomass gradually shifts from traditional uses towards more efficient and modern bioenergy services.

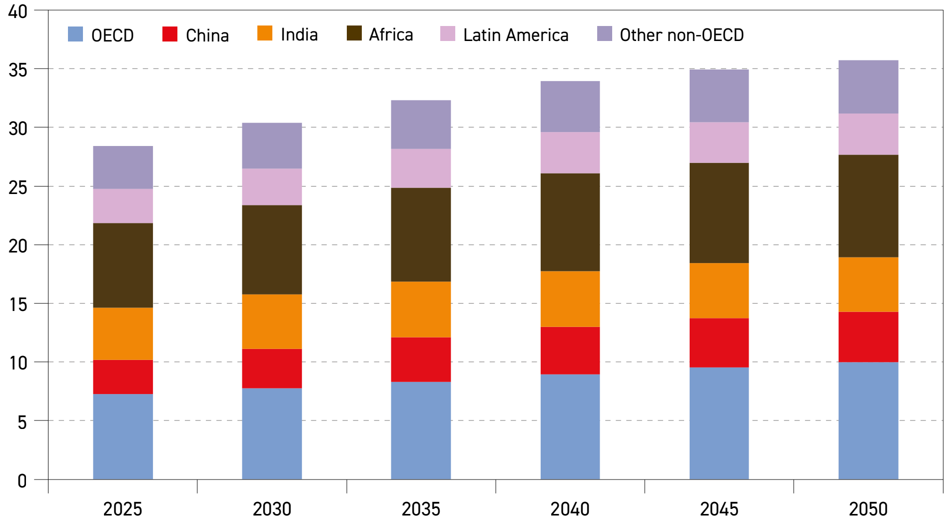
As shown in Table 2.14 and Figure 2.16, global biomass demand is expected to rise from 28.4 mboe/d in 2025 to 35.7 mboe/d by 2050, representing an average growth rate of about 0.9% p.a. Although this growth is significant, the limited availability of biomass represents a potential challenge. This is associated with competition for arable land between food and biomass production. Additionally, challenges related to biomass and/or waste collection often make the overall process more expensive and less competitive than traditional fuels.

Table 2.14
Biomass energy demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	3.1	3.2	3.4	3.7	3.9	4.0	1.0	1.1	10.8	11.3
OECD Europe	3.6	3.9	4.2	4.6	4.9	5.2	1.6	1.5	12.6	14.5
OECD Asia-Pacific	0.6	0.6	0.7	0.7	0.8	0.8	0.2	1.0	2.1	2.2
OECD	**7.3**	**7.7**	**8.3**	**9.0**	**9.5**	**10.0**	**2.7**	**1.3**	**25.6**	**27.9**
China	2.9	3.4	3.8	4.0	4.2	4.3	1.4	1.5	10.4	12.0
India	4.4	4.6	4.7	4.8	4.7	4.6	0.2	0.2	15.6	13.0
Other Asia	3.1	3.2	3.3	3.5	3.5	3.5	0.4	0.5	10.7	9.8
Latin America	2.9	3.1	3.3	3.5	3.5	3.5	0.6	0.7	10.2	9.7
Middle East	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.9	0.1	0.1
Africa	7.2	7.6	8.0	8.3	8.5	8.8	1.6	0.8	25.4	24.6
Russia	0.2	0.2	0.3	0.3	0.3	0.4	0.2	2.2	0.8	1.1
Other Eurasia	0.1	0.2	0.2	0.2	0.2	0.3	0.1	2.7	0.5	0.7
Other Europe	0.2	0.3	0.3	0.3	0.4	0.4	0.2	2.0	0.8	1.1
Non-OECD	**21.2**	**22.7**	**24.0**	**25.0**	**25.4**	**25.7**	**4.6**	**0.8**	**74.4**	**72.1**
World	**28.4**	**30.4**	**32.3**	**33.9**	**34.9**	**35.7**	**7.3**	**0.9**	**100**	**100**

Source: OPEC.

Figure 2.16
Biomass energy demand by region, 2025–2050, mboe/d



Source: OPEC.

In the OECD, demand is expected to increase from 7.3 mboe/d in 2025 to 10 mboe/d by 2050. OECD Europe records the largest gain, rising from 3.6 mboe/d to 5.2 mboe/d, supported by stricter sustainability rules and a wider use of forest residues and agricultural by-products. OECD Americas adds 1 mboe/d, mainly driven by biofuel blending mandates and waste-to-energy projects. The OECD Asia-Pacific shows more moderate growth of about 0.2 mboe/d, reflecting improvements in feedstock use and gradual upgrades of heating systems.

Across non-OECD regions, patterns differ. In China, biomass demand is expected to increase from around 3 mboe/d in 2025 to more than 4 mboe/d by 2050, supported by agricultural residue use and expanding waste to energy capacity. Other Asia rises from about 3.1 mboe/d to 3.5 mboe/d, largely linked to industrial demand. In India, biomass use is expected to grow modestly from 4.4 mboe/d in 2025 to 4.6 mboe/d by 2050, as declining traditional consumption is broadly offset by industrial applications and progress in modern biofuels.

Latin America is expected to add 0.6 mboe/d, mainly driven by established biofuel industries and strong agricultural resources. Africa remains the largest consumer globally, accounting for around 25% of total biomass demand, with demand set to rise from 7.2 mboe/d in 2025 to about 8.8 mboe/d by 2050. This reflects a gradual shift from traditional biomass use towards cleaner fuels, alongside increased deployment of modern biomass in industry and power generation.

In other regions, including Russia, the Middle East, Other Eurasia and Other Europe, biomass contributes only modestly to global growth. In Russia, demand rises slightly, mainly driven by industrial heat use and waste-to-energy applications. Other Eurasia sees gradual gains supported by using agricultural byproducts as local energy sources. In the Middle East, biomass demand remains limited but increases slowly as part of early-stage energy diversification efforts. Meanwhile, Other Europe experiences steady, gradual growth linked to a greater use of local biomass resources and the ongoing modernization of district heating systems.

Others

Global demand for geothermal, tidal and wave energy remains small compared to other primary energy sources. However, it is projected to increase over the outlook period, rising from just over 2 mboe/d in 2025 to nearly 7 mboe/d by 2050, as shown in Table 2.15 and Figure 2.17. These technologies provide firm and dispatchable supply, complementing the expansion of variable renewables. Their development is largely determined by resource availability, upfront investment costs and long project timelines, resulting in more targeted and policy-driven growth than that seen in solar or wind.

Looking forward, geothermal, tidal and wave technologies are expected to face constraints. This is due to high capital requirements, exploration and resource risks, lengthy permitting processes and grid connection challenges that collectively limit rapid expansion. Marine technologies remain at an early stage of commercial development, while geothermal projects depend heavily on suitable geological conditions. At the same time, improvements in drilling techniques, enhanced geothermal systems and turbine performance are expected to gradually strengthen project viability.

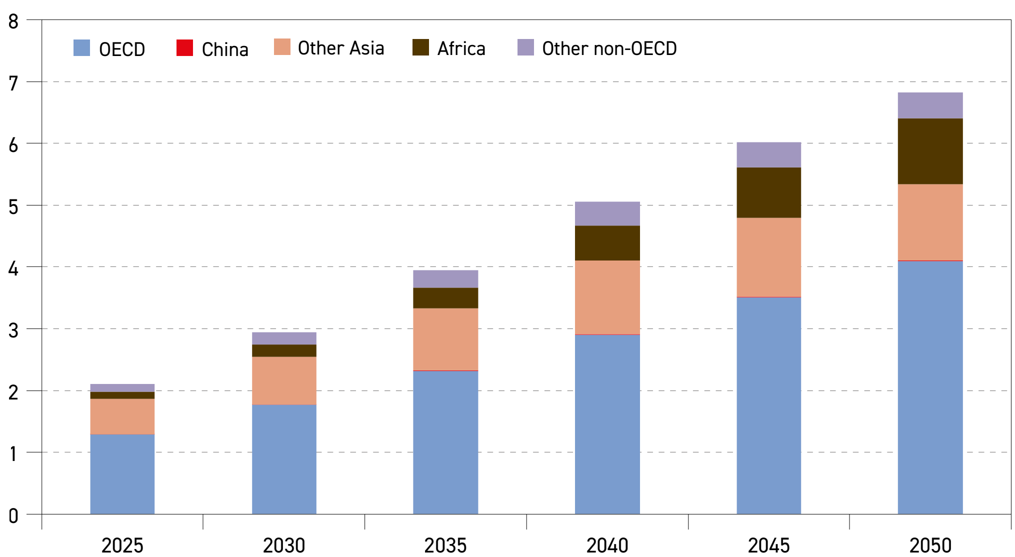
Geothermal capacity reached 16 GW globally in 2025, alongside pilot projects in enhanced geothermal systems and tidal-stream installations. Marine energy witnessed growth of

Table 2.15
Other energy demand by region, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	0.5	0.7	1.1	1.5	1.9	2.4	1.9	6.4	24.0	34.5
OECD Europe	0.5	0.7	0.8	0.9	1.0	1.1	0.6	3.3	23.2	15.9
OECD Asia-Pacific	0.3	0.4	0.5	0.5	0.6	0.6	0.4	3.2	14.1	9.5
OECD	**1.3**	**1.8**	**2.3**	**2.9**	**3.5**	**4.1**	**2.8**	**4.7**	**61.2**	**59.9**
China	0.0	0.0	0.0	0.0	0.0	0.0	0.0	5.0	0.2	0.2
India	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Asia	0.6	0.8	1.0	1.2	1.3	1.2	0.7	3.1	27.0	18.1
Latin America	0.1	0.2	0.2	0.3	0.3	0.3	0.2	3.7	5.6	4.3
Middle East	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Africa	0.1	0.2	0.3	0.6	0.8	1.1	0.9	9.3	5.4	15.6
Russia	0.0	0.0	0.0	0.1	0.1	0.1	0.1	10.4	0.5	1.8
Other Eurasia	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0	11.3	0.0	0.1
Non-OECD	**0.8**	**1.2**	**1.6**	**2.1**	**2.5**	**2.7**	**1.9**	**5.0**	**38.8**	**40.1**
World	**2.1**	**2.9**	**4.0**	**5.1**	**6.0**	**6.8**	**4.7**	**4.8**	**100**	**100**

Source: OPEC.

Figure 2.17
Other energy demand by region, 2025–2050, mboe/d



Source: OPEC.

around 0.5 GW, reflecting growing policy and investor interest. Targeted support measures, including auctions, risk-sharing mechanisms and long-term contracts, are expected to support further gradual deployment. As electricity systems seek reliable capacity to complement expanding wind and solar generation, these technologies are projected to play a role in selected resource-rich and isolated markets, particularly where energy security and supply diversification are priorities.

The OECD accounts for the majority of global demand throughout the outlook period. OECD demand is expected to rise from around 1.3 mboe/d in 2025 to just over 4 mboe/d by 2050, led by OECD Americas, where geothermal power and heat expand alongside the pilot deployment of enhanced geothermal systems. In OECD Europe, geothermal heating networks and early-stage tidal and wave projects support decarbonization and system flexibility objectives. The OECD Asia-Pacific sees more modest increases, concentrated in countries with favourable geothermal resources and isolated or island-based power systems.

In non-OECD regions, Other Asia and Africa drive most of the increase, reflecting the expansion of geothermal power in resource-rich countries and the gradual emergence of marine technologies in select coastal and island systems. Africa's contribution rises steadily, as geothermal supports grid stability and energy security in fast-growing power systems. In contrast, contributions from other non-OECD regions remain very limited, as national energy strategies continue to prioritize larger-scale renewable projects and conventional generation.

2.4 Final electricity consumption and generation

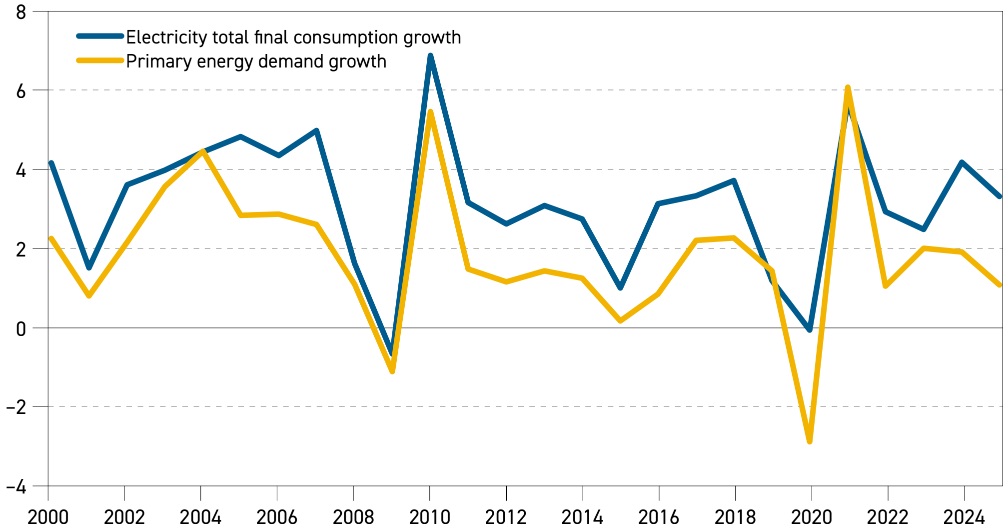
This section discusses recent trends related to total final consumption (TFC) for electricity and electricity generation. It also provides a long-term outlook at the global level and for major regions. It is worth noting that TFC in this section refers to electricity delivered to the sectors of final consumption and does not include own energy use in the transformation sector, nor transmission and distribution losses.

In 2025, electricity TFC recorded strong growth once again, rising by around 3.3% or roughly 850 TWh. This growth is somewhat lower than the 4.2% recorded in 2024, but remains significant when compared to historical levels. The main driver of the strong growth during these two years was rising demand in the industrial, transportation and residential sectors, as well as soaring demand stemming from AI and data centres. At the regional level, the largest incremental demand was recorded in Asia (including China and India), the Middle East and the US.

As presented in Figure 2.18, electricity TFC growth has been rising steadily and at a faster pace than primary energy demand over the past three decades. This is the result of several factors, including the increasing electrification of economies, supported by energy policies, as well as rising electricity access in developing countries. Furthermore, rising power generation efficiency (notably in coal- and gas-fired plants), as well as the broader deployment of renewable electricity, primarily from wind and solar, support this trend.

In total, electricity TFC rose by 115% from 2000 to 2025, while primary energy demand increased by around 59%. On an annual basis, TFC has been rising by around 3.2% p.a. since 2000, while primary energy demand has grown by 1.9% p.a. over the same period. The difference between electricity TFC and primary energy demand growth became particularly evident in 2024 and 2025, with the gap widening. Electricity consumption grew more than twice as fast as primary energy demand during these years.

Figure 2.18
Global primary energy demand and electricity total final consumption growth, 2000–2025, %



Source: OPEC.

Looking forward, this trend is projected to continue in the medium and long term. Global electricity TFC is expected to increase by more than 85% over the forecast period, rising from around 27,000 TWh in 2025 to just above 50,500 TWh in 2050 (Table 2.16 and Figure 2.19). This translates to an overall increase of around 23,500 TWh, representing average annual growth of about 2.5% p.a.

More than half of global incremental electricity TFC is expected to materialize in non-OECD Asia. Electricity consumption in China alone is set to increase by almost 6,500 TWh in the period to 2050, supported by further electrification of the manufacturing sector and EV deployment, as well as demand from the residential sector and data centres/AI.

Electricity consumption in India is expected to more than triple over the outlook period, rising by around 3,700 TWh between 2025 and 2050. Demand growth in this country is predominantly driven by the residential and commercial sectors, especially for air conditioning, on the back of rising disposable income and improved electricity access. In other non-OECD Asian countries, final electricity consumption growth is projected to increase by more than 3,500 TWh over the same period. Similarly to India, rising income supports expanding electricity consumption in the residential sector.

Electricity consumption growth in the Middle East is expected to almost double over the outlook period, reaching levels close to 2,500 TWh in 2050. In addition to demand from the residential, commercial and industrial sectors, water desalination is also a significant demand driver in this region. Even stronger growth is projected in Africa, where consumption is set to almost triple, increasing from around 800 TWh in 2025 to around 2,300 TWh in 2050. Rising energy access and electrification in the residential and commercial sectors are the major growth drivers.

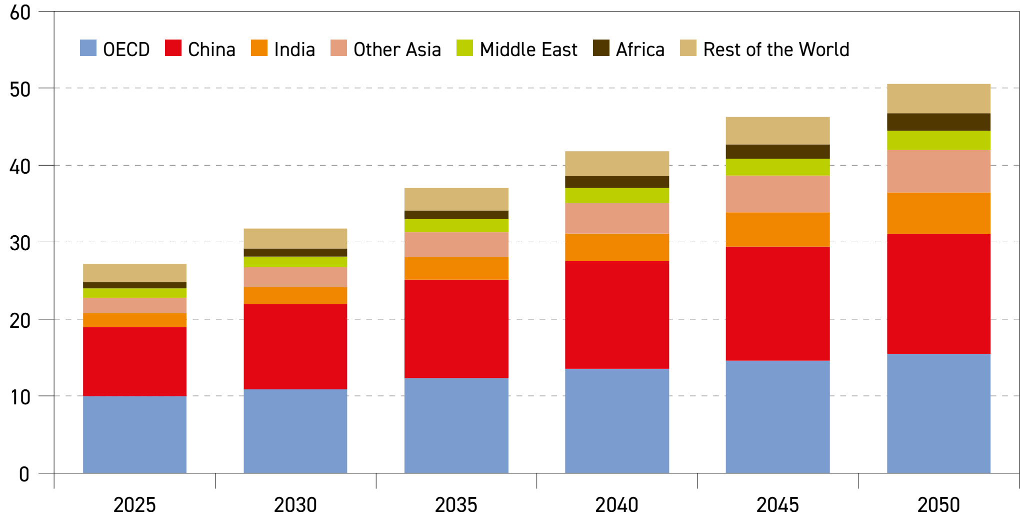
Turning to the OECD, electricity TFC in several regions has been stable or even declined in recent years. However, this trend is expected to change in the medium and long term. The growth here will also be strong, totalling around 5,600 TWh over the forecast period, with electricity demand reaching almost 15,500 TWh by 2050. The largest share of this growth, almost 40%, is accounted for by the US. The strong expansion of data centres and AI in the US is the major growth driver. Other OECD regions are also set to see strong electricity

Table 2.16
Global total final electricity consumption by region, 2025–2050

	Levels *TWh*						Growth *TWh*	Growth *% p.a.*	Share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
OECD Americas	5,147	5,637	6,377	6,960	7,427	7,815	2,669	1.7	18.9	15.5
OECD Europe	3,028	3,353	3,844	4,315	4,746	5,170	2,142	2.2	11.1	10.2
OECD Asia-Pacific	1,785	1,912	2,121	2,293	2,426	2,524	740	1.4	6.6	5.0
OECD	**9,960**	**10,901**	**12,342**	**13,568**	**14,599**	**15,510**	**5,550**	**1.8**	**36.6**	**30.7**
China	9,048	11,052	12,826	13,954	14,789	15,520	6,472	2.2	33.3	30.7
India	1,752	2,242	2,878	3,622	4,468	5,421	3,669	4.6	6.4	10.7
Other Asia	2,020	2,540	3,208	3,969	4,761	5,539	3,518	4.1	7.4	11.0
Latin America	1,038	1,214	1,457	1,725	1,988	2,225	1,186	3.1	3.8	4.4
Middle East	1,204	1,430	1,693	1,954	2,209	2,457	1,253	2.9	4.4	4.9
Africa	809	978	1,206	1,487	1,832	2,253	1,444	4.2	3.0	4.5
Russia	834	829	824	818	810	800	–34	–0.2	3.1	1.6
Other Eurasia	352	384	422	465	509	551	199	1.8	1.3	1.1
Other Europe	160	174	195	221	254	292	132	2.4	0.6	0.6
Non-OECD	**17,218**	**20,841**	**24,710**	**28,216**	**31,620**	**35,057**	**17,840**	**2.9**	**63.4**	**69.3**
World	**27,177**	**31,743**	**37,052**	**41,784**	**46,219**	**50,567**	**23,390**	**2.5**	**100**	**100**

Source: OPEC.

Figure 2.19
Global total final electricity consumption by region, 2025–2050, '000 TWh



Source: OPEC.

consumption growth, supported by increasing electrification in the residential, commercial and industrial sectors, as well as a rising number of EVs.

Table 2.17 and Figure 2.20 show the total final consumption by major sectors in 2025 and 2050 (own use and transmission losses are not included). It is important to note that consumption in all sectors increases during the forecast period. The most significant contributor to the growth is the residential and commercial sector, with consumption rising by around 9,300 TWh in the period to 2050. This is due to a rising middle class in developing countries, as well as rising electricity access and affordability.

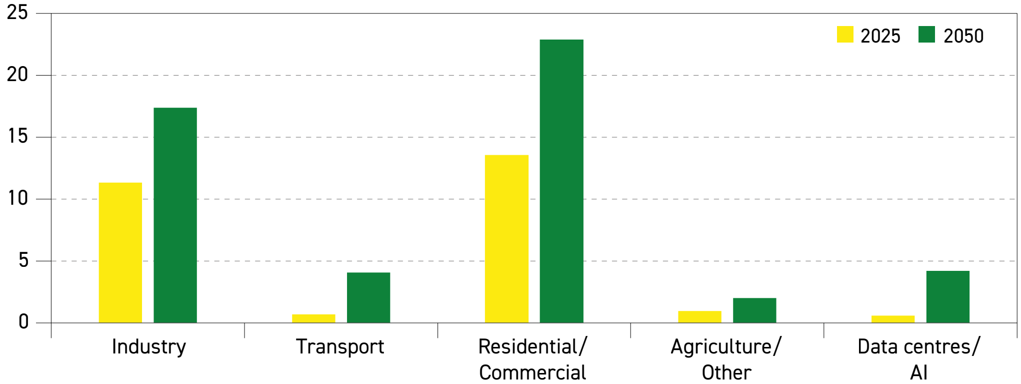
In developed regions, part of the growth in this sector is the substitution of traditional heating fuels (including wood, coal and gas) with electricity in the residential sector (e.g. heat pumps). At the same time, industrial electricity consumption is expected to increase by around 6,000 TWh in the period to 2050, driven by economic development and coal and gas substitution. However, despite this large increase in absolute terms, the combined share of the residential,

Table 2.17
Global total final electricity consumption by sector, 2025–2050

	Levels *TWh*						Growth *TWh*	Growth *% p.a.*	Share *%*	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Industry	11,339	12,560	13,720	14,899	16,126	17,384	6,045	1.7	41.7	34.4
Transport	702	1,298	2,058	2,823	3,482	4,075	3,373	7.3	2.6	8.1
Resid./Comm.	13,578	15,589	17,576	19,347	21,102	22,897	9,320	2.1	50.0	45.3
Agriculture/Other	956	1,146	1,352	1,562	1,779	2,003	1,046	3.0	3.5	4.0
Data centres/AI	602	1,150	2,346	3,152	3,730	4,208	3,606	8.1	2.2	8.3
Total	**27,177**	**31,743**	**37,052**	**41,784**	**46,219**	**50,567**	**23,390**	**2.5**	**100**	**100**

Source: OPEC.

Figure 2.20
Global total final electricity consumption by sector, 2025 and 2050, '000 TWh



* *Total final consumption does not include own use and transmission/distribution losses and therefore is not equal to electricity generation.*

Source: OPEC.

commercial and industrial sectors in total final electricity consumption is set to decline from 92% in 2025 to 80% in 2050.

This is due to strong consumption growth expected from sectors such as data centres and AI, as well as the transportation sector (especially electric mobility). Electricity TFC stemming from data centres and AI is projected to grow by around 3,600 TWh over the outlook period, reaching around 4,200 TWh by 2050. Consequently, the share of data centres and AI increases from around 2% in 2025 to 8% in 2050. Similarly, the transportation sector is expected to see a significant expansion, with electricity consumption growing by roughly 3,400 TWh, mostly due to the deployment of EVs. The share of the transportation sector in overall final electricity consumption is expected to reach roughly 8% in 2050, up from 3% in 2025.

Electricity generation

This section focuses on electricity generation, which includes TFC, transmission and distribution losses, as well as electricity consumed in various transformation processes and own use in the energy sector.

Mirroring the electricity consumption pattern, generation has risen significantly over the last 25 years. Moreover, the generation mix has also witnessed significant changes, especially in recent years. While generation from almost all fuels has increased, their respective growth rates differ significantly.

The combined share of coal- and gas-fired generation was estimated at around 63% in 2010, for example, but this then witnessed a gradual decline to a level of 55% in 2025. However, despite this declining share in the mix, when reviewing absolute figures, it becomes apparent that generation from coal and gas increased significantly over the last 15 years, namely by around 4,000 TWh.

At the same time, the share of solar and wind power increased substantially. From only 2% in 2010, the combined share of solar and wind rose to 17% in 2025 (an increase of 5,000 TWh), reflecting strong policy support and declining generation costs. However, despite this impressive growth in renewable electricity, increases in wind and solar generation at the global level have so far represented only energy additions, rather than substitutions.

Moving forward, global electricity generation is projected to rise from around 32,000 TWh in 2025 to more than 59,000 TWh in 2050, an increase of 27,000 TWh (Table 2.18 and Figure 2.21). More than half of this addition is expected to come from solar power (+13,700 TWh), supported by declining generation costs and supportive energy policies. Despite current hurdles, wind generation is also set to increase, rising by nearly 7,000 TWh. Hydropower is expected to see a significant increase over the long term too, rising by 2,200 TWh. Strong support for nuclear energy is expected to help to increase related generation to nearly 4,900 TWh in 2050, up from 2,900 TWh in 2025. Nuclear energy is likely to become an important pillar of supply security, especially given the rising share of intermittent generation sources, such as wind and solar.

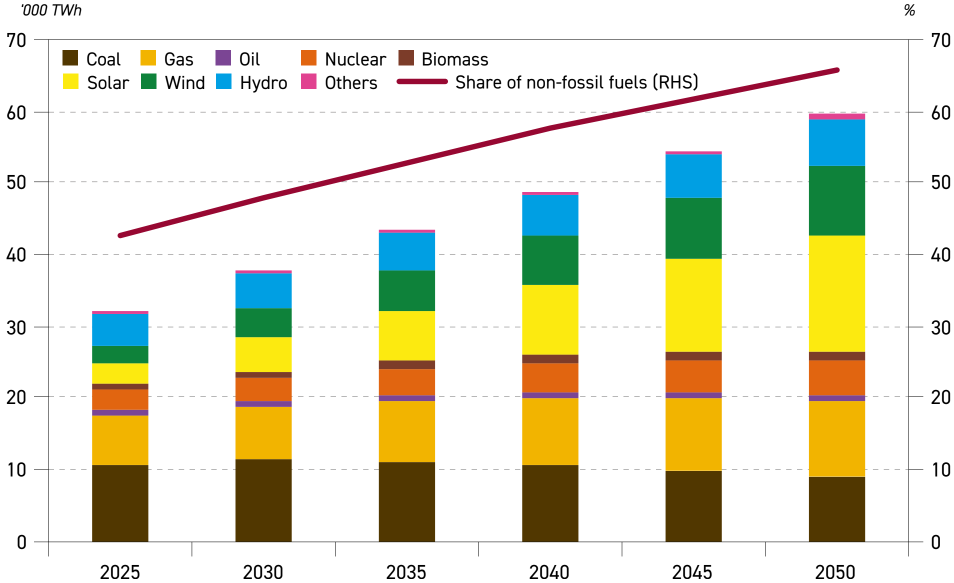
Another significant contribution to the future increase in electricity generation is forecast to come from gas-fired plants, which add almost 4,000 TWh between 2025 and 2050. Gas-fired generation is also supported by climate policies seeking to replace coal in the generation mix. Competitive gas prices in regions such as the US also help to expand

Table 2.18
Global electricity generation by fuel, 2025–2050

	Levels TWh						Growth TWh	Growth % p.a.	Fuel share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Coal	10,758	11,221	10,946	10,469	9,873	8,904	–1,854	–0.8	33.6	15.0
Gas	6,782	7,616	8,694	9,479	10,200	10,728	3,946	1.9	21.2	18.0
Oil	781	792	776	739	685	620	–161	–0.9	2.4	1.0
Nuclear	2,860	3,194	3,588	4,016	4,461	4,875	2,015	2.2	8.9	8.2
Renewables	10,836	14,770	19,352	24,099	29,079	34,347	23,511	4.7	33.8	57.8
Biomass	*816*	*936*	*1,052*	*1,144*	*1,222*	*1,294*	*478*	*1.9*	*2.5*	*2.2*
Solar	*2,667*	*4,613*	*7,103*	*9,974*	*13,077*	*16,355*	*13,688*	*7.5*	*8.3*	*27.5*
Wind	*2,746*	*4,171*	*5,693*	*7,002*	*8,275*	*9,628*	*6,882*	*5.1*	*8.6*	*16.2*
Hydro	*4,458*	*4,855*	*5,251*	*5,662*	*6,128*	*6,644*	*2,186*	*1.6*	*13.9*	*11.2*
Others	*149*	*196*	*253*	*317*	*376*	*426*	*278*	*4.3*	*0.5*	*0.7*
World	**32,017**	**37,593**	**43,356**	**48,803**	**54,298**	**59,474**	**27,457**	**2.5**	**100**	**100**

Source: OPEC.

Figure 2.21
Global electricity generation by fuel, 2025–2050



Source: OPEC.

gas-fired generation in the long term. At the same time, coal-fired generation on a global scale is expected to decline by nearly 2,000 TWh over the outlook period, driven by energy policies targeting CO_2 emissions, mostly in China and OECD countries. Nevertheless, coal-fired generation is likely to remain an important part of the mix in many developing countries over the entire forecast period.

Based on the trends described above, the share of electricity generated from non-hydrocarbons is projected to increase by around 23 pp, rising from around 43% in 2025 to almost 66% in 2050.

Rising electricity generation has put significant pressure on electricity grids. This is visible in many regions, including in the US and many developed countries in Europe. In many US states, demand from data centres and AI is rising faster than the capability of grids to accommodate demand. This is why some technology companies are planning to build their data centres off-grid and combine them with their own generation capacities.

Furthermore, rising intermittent generation from solar and wind sources requires additional grid investments. This means the expansion of transmission and distribution lines, as well as the enhancement and modernization of existing networks. In addition, further investments will be needed to improve grid management in response to variability associated with rising wind and solar generation.

Finally, growth in electric mobility requires the enhancement and expansion of related charging infrastructure. All of these factors will lead to rising investment needs in electricity networks in the medium and long term. These investments will likely increase average electricity bills for final consumers, which could once again highlight energy affordability issues in the years to come.

2.5 Energy intensity and consumption per capita

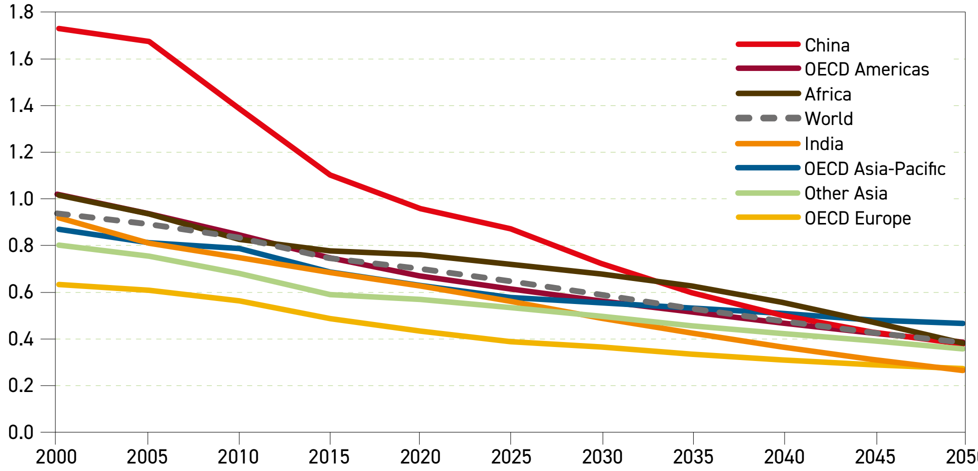
This section explores measures that help provide additional insights on the use of energy in different regions of the world. The first to be covered is energy intensity, which can help to understand the energy efficiency of an economy. For this purpose, the ratio of primary energy per unit of GDP (adjusted for purchasing power parity) is used.

Although this is a simplification of a complex topic, energy intensity is often used as a proxy for efficiency and is one of the indicators utilized to measure progress against the UN's Sustainable Development Goal (SDG) 7, which seeks to: 'Ensure access to affordable, reliable, sustainable and modern energy for all'. It is logical that with an increase in energy efficiency, the energy intensity of an economy decreases, and that without such efficiency improvements, energy demand in the future would be much higher, *ceteris paribus*. Improving energy efficiency also has the advantage of helping to address energy security, energy affordability and environmental concerns in parallel.

The challenge is that other factors can influence energy intensity that are unrelated to energy efficiency. For instance, a service-based economy is typically less-energy intensive than one based on manufacturing. Moreover, other factors such as climate and behavioural aspects of the population also impact this measure.

Figure 2.22 shows the development of energy intensity since 2000 and provides an indication of trends up to 2050 based on the outlook for primary energy demand presented in this chapter and the GDP outlook from Chapter 1. Between 2000 and 2025, the global economy grew by a factor of around 2.3, while global energy demand increased by a factor of almost 1.6. This resulted in a significant drop in energy intensity in all major regions. Looking ahead, it is evident that this trend will continue as the link between energy use and GDP is set to further weaken.

Figure 2.22
Evolution and projections of energy intensity in major regions, 2000–2050, boe/$1,000 (PPP 2021)



Source: OPEC.

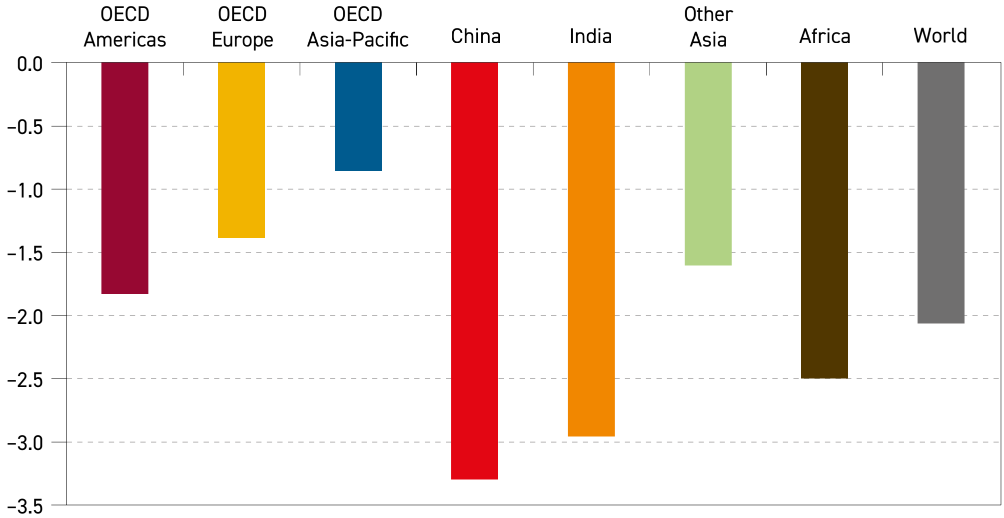
A continuation of this trend would reflect further efficiency improvements in energy technology and the implementation of energy policies that are in place across the globe. Furthermore, as is evident in the long-term outlook for energy, electricity is anticipated to grow faster than primary energy demand. This has important implications, as the deployment of renewable energy sources (e.g. solar and wind) helps to reduce transformation losses and therefore improves energy intensity.

As shown in Figure 2.23, the expected rate of energy intensity improvement is set to vary significantly across regions and countries. Overall, the world's energy intensity is projected to improve by an average rate of around 2.1% p.a. between 2025 and 2050.

Given China's dramatic historical improvement, and expectations of what is to come, it is worth highlighting that the government has included a specific target for the energy intensity of its economy in the 15th FYP covering the period from 2026 to 2030. Coupled with China's rapid increase in renewable energy, as highlighted earlier in this chapter, the country is expected to see an average annual improvement rate of 3.3%.

As the fastest growing economy in this outlook, India's GDP is expected to continue expanding at a faster rate than the country's energy demand. High-level government policies and programmes that support improved energy efficiency in industry and across other areas of the economy will also help improve energy intensity going forward. For example, in mid-2025, the government announced a new scheme to support the financing of energy-efficiency projects by businesses, which will be implemented by the Bureau of Energy Efficiency. The Bureau also continues to expand its energy efficiency labelling programme for appliances, with the mandatory labelling of evaporative coolers to be introduced by mid-2026. The latter represents an important target area due to the anticipated rise in electricity use for cooling. In the long term, close to a 3.0% average annual rate of improvement is expected for India.

Figure 2.23
Average annual rate of improvement in energy intensity, 2025–2050, %



Source: OPEC.

The underlying factors at play for a continent as large and diverse as Africa are complex. The slower decline rate in energy intensity seen in the past does not necessarily mean that energy efficiency has not been improving. African countries are at varying stages of economic development and industrialization, while hundreds of millions of people still lack access to reliable electricity. As a result, improved energy access and affordability, combined with growing industrial production and improved mobility, could more than offset the impact of higher efficiency in a number of the countries in the region. This will likely moderate the decline in the region's energy intensity over the next 10 to 15 years, while a larger decline is expected in the second half of the forecast period.

Overall, Africa's energy intensity is expected to improve at an average rate of 2.5% p.a. – in line with a rapidly expanding economy that will continue to grow faster than its energy demand. Africa's rate is anticipated to be broadly similar to the average rate for all non-OECD regions.

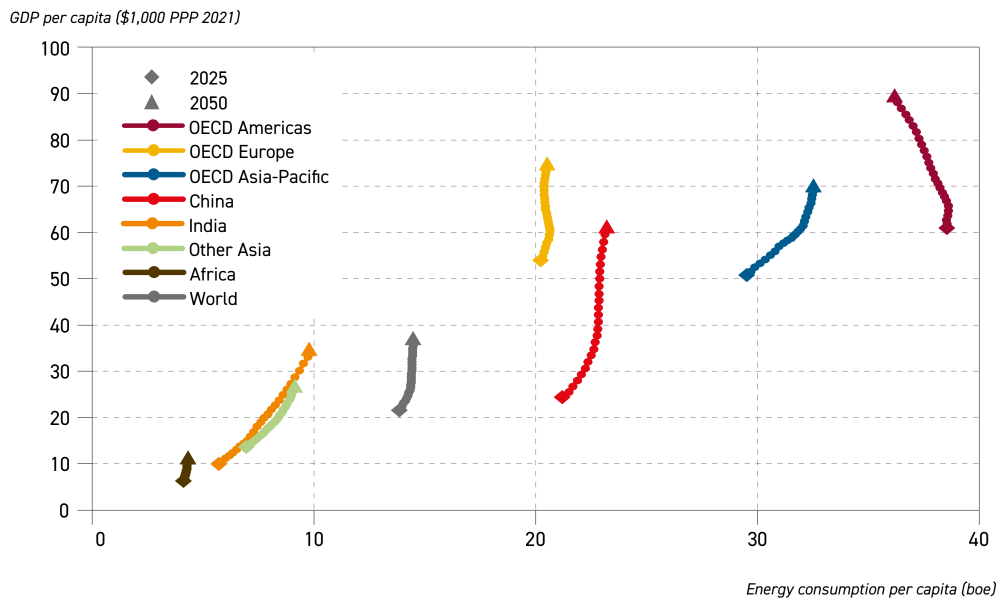
Other Asia is expected to see an average annual improvement rate of 1.6% going forward, slower than the global average, although it is also worth noting that the region starts from a relatively lower intensity level of around 0.5 boe/$1,000 in 2025.

OECD countries have a long history of policy efforts to improve energy efficiency. These continue today and will support an ongoing drop in energy intensity between 2025 and 2050, albeit with different rates seen across each of the OECD sub-regions. OECD Americas is set to see an average improvement rate of 1.8% but the region is starting from a higher intensity level than others. In contrast, OECD Europe has already established a lower energy intensity level, and improvements are forecast to be at a slower average rate of 1.4% going forward.

Figure 2.24 plots energy consumption per capita against GDP per capita over the outlook period for select regions. This figure highlights that per capita energy consumption is markedly different across the world's regions. Despite past and future increases in energy consumption in developing countries, the persistent gap between OECD and non-OECD per capita energy consumption acutely underlines the need for a strong multilateral and collaborative approach to tackle energy poverty and increase energy access. Failure to do so will continue to see billions of people across the globe remaining underserved by the energy sector and their energy consumption remaining low over the long term.

The current disparity between regions becomes clear when considering that OECD Americas consumed more than 38 boe per capita in 2025. Elsewhere, even though India's energy use increases significantly, it is still only expected to reach fewer than 10 boe per capita by 2050.

Figure 2.24
Per capita GDP and energy consumption, 2025–2050



Source: OPEC.

Meanwhile, China's per capita energy consumption grew by almost 15 boe between 2000 and 2025. However, between 2025 and 2050, the country's increase will be only 2 boe per capita. In contrast, over the same period, China's GDP per capita will increase significantly, supported by a growing economy and a declining population. China's energy consumption pattern is the starkest example of the impact of a structural change in the energy mix, and does not mean that consumers will use less or the same amount of energy as before. As this section considers the consumption of primary energy, China's massive shift away from the use of coal and large increase in renewable energy sources provides a more efficient supply of energy to consumers.

In contrast to India and China, Africa's increase in both energy consumption and GDP per capita are much more limited. Despite being home to over 19% of the world's population in 2025, the continent's share of global energy demand was only 6%. The latest report tracking progress of SDG7 has estimated that 666 million people continue to go without access to electricity and 2.1 billion remain without access to clean cooking. Most of these people are in Africa and the impact of this lack of access is reflected in the continent's energy consumption per capita value of 4.1 boe in 2025. As energy consumption per capita rises by only 0.2 boe between 2025 and 2050, the continent clearly continues to be underserved with energy.

Figure 2.24 also illustrates the de-linking between economic and energy demand growth for OECD Americas and OECD Europe. The populations of both are expected to see much larger GDP per capita by 2050, with lower or similar levels of energy consumption per capita. Again, it is worth noting the structural change in the energy mix for these two regions, with coal use dropping over the long term and an even larger increase in renewable energy use, which is more efficient from the primary energy demand side. In contrast, the OECD Asia-Pacific is expected to continue to see an increase in both per capita GDP and energy consumption.

Although an oversimplification of a complex topic, if one were to consider an average individual in the world, that individual would be expected to increase their energy consumption by 4.5% between 2025 and 2050, while their contribution to GDP would increase by 72%. This showcases the enormous discrepancy between economic output and the necessary energy supply, emphasizing a fragile and uncertain system. In reality, a multitude of factors impact primary energy demand and mean that an understanding of the issue at the country and regional level is vital.

Oil demand

Key takeaways

- The increased focus on energy security and energy affordability has shifted the energy policy landscape across the globe. This is reflected in policy adjustments and reversals, which are expected to be supportive of oil demand in the medium and long term.

- Recent energy policy changes in the US, the EU and many other parts of the world support oil demand in these regions. In many cases, these shifts reflect the reversal, delay or cancellation of previously ambitious targets and commitments aimed at reducing oil demand, through substitution or improved efficiency across various sectors in the medium and long term.

- Oil demand is set to increase from 105.1 mb/d in 2025 to 113.3 mb/d in 2030, on the back of economic growth in the non-OECD region and recent policy shifts in OECD countries.

- Non-OECD oil demand is projected to increase by 7.4 mb/d between 2025 and 2030, while OECD demand is set to increase by 0.7 mb/d over the same period.

- In the long term, oil demand is projected to increase by 19 mb/d to reach 124.1 mb/d in 2050.

- Oil demand in the non-OECD is projected to increase by almost 27 mb/d over the modelling horizon, which will be partly offset by an expected OECD demand decline of nearly 8 mb/d.

- The largest demand increments are set to come from India, Other Asia, the Middle East, Africa and Latin America. Combined demand growth in these regions is estimated at more than 25 mb/d between 2025 and 2050. India is the single largest contributor to oil demand growth, adding 8.1 mb/d over the outlook period.

- China's oil demand is set to increase from 16.9 mb/d in 2025 to nearly 19 mb/d in 2035. This is followed by a gradual decline to 18 mb/d in 2050.

- From a sectoral perspective, the largest long-term incremental demand growth is expected in road transportation (+5.7 mb/d), petrochemicals (+4.6 mb/d) and aviation (+4.2 mb/d).

- The global vehicle fleet is projected to rise from 1.75 billion in 2025 to around three billion in 2050. The share of EVs in the fleet is set to increase from 4% in 2025 to 21.5% in 2050. At the same time, ICE vehicles remain dominant in the global fleet over the entire outlook period, with an expected share of 73% in 2050.

- From a product perspective, demand in all segments increases. This is led by jet/kerosene (+4.2 mb/d), gasoil/diesel (+3.8 mb/d), ethane/LPG (+3.5 mb/d), naphtha (+3.2 mb/d) and gasoline (+2.4 mb/d).

This Outlook confirms the main trends of last year's publication. The emerging shift in energy transition narratives has continued over the past 12 months, increasingly recognizing the realities of global energy systems. Several regions and countries have expressed the need for a more balanced approach that would include not only issues related to emissions reductions, but also those related to energy security, availability and energy affordability.

Furthermore, recent geopolitical tensions have seen major energy consumers rethink their positioning in the global energy market. Energy security has become increasingly important and has resulted in different strategies, depending on specific national circumstances. One thing is clear – the global energy landscape has become more complex.

On the international scene, the withdrawal of the US from the Paris Agreement, as well as from some UN bodies, means that climate change issues in some regions are likely to be revisited. A set of newly adopted policies and regulations in recent months clearly confirms this notion. The repercussions of the US withdrawal will continue to be felt on the international scene and could affect further negotiations and result in spillover effects.

In Europe, climate change remains high on the agenda, with clear announced commitments to policy frameworks such as the 'Green Deal' and related emissions reductions targets. However, some policies at the EU level have already been readjusted and/or softened relative to their initial versions, thus recognizing market realities and overall sentiment.

This is partly the result of high energy prices in Europe in recent years, which have led to an increased focus on energy affordability for final consumers and also affected the international competitiveness of European industry. Lower than expected EV sales in Europe in recent years, alongside policymakers coming under pressure to soften ambitious transportation policies and deadlines, are also apparent. Against this backdrop, European policymakers are increasingly willing to accept a rebalancing of their priorities related to decarbonization efforts *versus* energy security and affordability.

China reveals a different picture. The structural changes in the Chinese economy have reflected the country's efforts to move towards high-technology and high-value-added production. This also includes impressive progress related to renewable energy (solar and wind) and EVs. Chinese technological progress and international competitiveness has helped to increase the share of renewable energy in the country's energy mix in recent years. Furthermore, strong EV sales (including a large share of plug-in hybrids) have already significantly impacted the composition of vehicle sales in the country and this will likely continue in the future. Chinese EV producers are also penetrating international car markets, with rising exports across Asia, Africa and Europe.

Energy and oil demand in other developing regions is increasing steadily, driven by positive demographic trends, a rising middle class and expanding economies. This applies to India, Other Asia, the Middle East, Africa, and Latin America. These areas are likely to see expanding passenger and commercial vehicle fleets, rising demand for air travel and increasing industrial development (including petrochemicals), all of which will unlock significant oil demand growth in the medium and long term. In some developing regions, such as Africa, oil will play an important role in providing access to modern energy services, thus replacing the traditional use of biomass.

Future oil demand trajectories are shaped not only by economic growth and energy policy frameworks, but also by the availability, cost and performance of current and emerging energy technologies. In many cases, these technologies could affect oil demand growth in some sectors in the future.

For instance, technological progress can improve fuel efficiency and support the substitution of oil within the broader energy mix. However, many alternative technologies involve additional costs as well as many practical limitations, which could restrict their ability to fully replace oil. At the same time, certain technological innovations could reinforce or even expand oil use. This includes advanced petrochemical products that can substitute traditional materials while lowering overall costs, as well as carbon capture, utilization, and storage (CCUS) technologies that can reduce the carbon footprint of oil across various sectors, potentially supporting its continued role in the energy system. One example of this can be seen in mobile CCUS devices that can help reduce emissions from trucks and ships.

This chapter analyzes oil demand trends across the main regions, sectors, and major product categories, identifying the key factors expected to shape demand through 2050. It does so by weighing likely regional socio-economic developments against anticipated energy policy frameworks and technological advancements. Each of these elements involves its own inherent uncertainty. After accounting for these uncertainties, the analysis indicates strong and sustained growth in global oil demand through 2050 in the medium and long term, with no evidence of a peak occurring within the outlook horizon.

3.1 Oil demand outlook by region

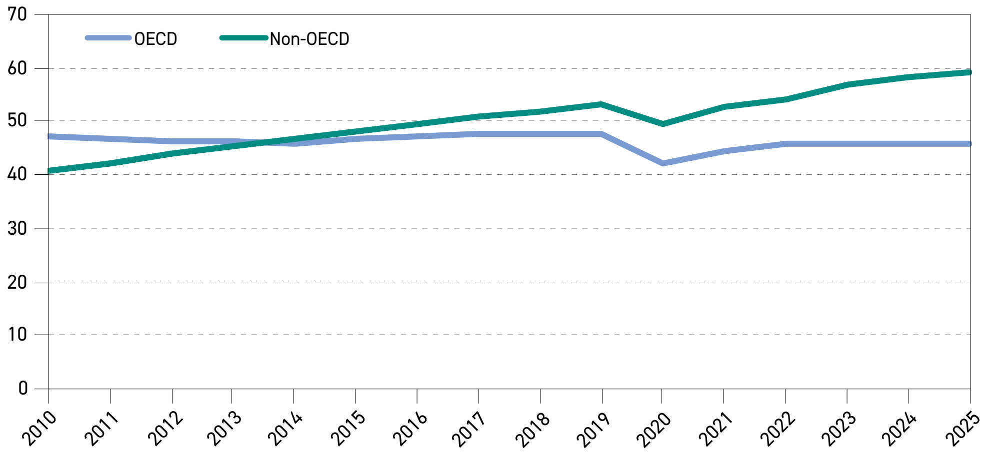
In 2025, global oil demand rose by 1.3 mb/d to reach 105.1 mb/d. This growth was driven primarily by developing regions. Demand increased in China (+0.2 mb/d), Other Asia (+0.3 mb/d), Africa (+0.2 mb/d) and Latin America (+0.1 mb/d), while it remained stable in other developing regions. As a result, total non-OECD demand climbed to 59.2 mb/d in 2025, up from 58 mb/d in 2024. Meanwhile, OECD oil demand edged up slightly to 45.9 mb/d, supported by rising demand in the US (+0.2 mb/d) but partly offset by declines in the OECD Asia-Pacific region (–0.1 mb/d).

From a sectoral perspective, oil demand growth in 2025 was primarily driven by petrochemicals, road transportation and aviation, with a smaller contribution coming from the residential and commercial sector. Overall, these trends reflect the continuation of patterns observed in previous years. OECD demand remains on a prolonged plateau, while growth is concentrated in developing countries.

Looking back, global oil demand rose from around 87.7 mb/d in 2010 to 105.1 mb/d in 2025, an increase of 17.4 mb/d. This growth came exclusively from non-OECD regions (mostly developing countries), where oil demand increased by 18.6 mb/d over the period (Figure 3.1). This increase was partially offset by a marginal decline in the OECD of 1.2 mb/d.

Non-OECD demand growth was strongly driven by increasing population and economic growth in developing countries. This was reflected in trends such as rising urbanization, expanding middle classes and increasing industrialization. The single largest contribution to non-OECD demand growth in this period came from China, where oil demand rose by almost 8 mb/d, accounting for almost half of all non-OECD demand growth from 2010 to 2025. Other

Figure 3.1
Oil demand by region, 2010–2025, mb/d



Source: OPEC.

notable demand increments materialized in Other Asia, India, the Middle East and Africa. The road transportation, aviation and industrial sectors accounted for most of the additional demand in non-OECD countries.

In the OECD, except for during the COVID-19 pandemic in 2020 and 2021, oil demand remained relatively stable at around 46–47 mb/d between 2010 and 2025. Demand has been on an upward trajectory since the pandemic to hit its previous heights, reaching almost 46 mb/d in 2025.

OECD oil demand trends have reflected two opposing forces. Downward pressure has come from energy policies and fuel substitution, as many countries have sought to reduce oil consumption through increased energy efficiency and the adoption of alternative fuels across multiple sectors. This has included a rising rate of substitution in the road transportation, residential and industrial sectors. These measures were largely motivated by climate-change policies, particularly following the adoption of the Paris Agreement in 2015.

In contrast, upward support for oil demand has been provided by economic growth and the related expansion of the petrochemical industry, the aviation sector and resilient oil consumption in road transportation. Furthermore, along with rising concerns about energy security and affordability, several OECD countries have begun readjusting their energy policies in recent years, delaying or even rolling back some measures targeting oil in their energy mixes. The result of these trends has been a prolonged oil demand plateau in the OECD.

This Outlook assumes that these demand trends are extended in the medium term, supported by further economic growth and an evolving policy landscape. As outlined in Chapter 1, GDP in non-OECD economies is projected to grow at an average annual rate of 4.2% in the period to 2030, while OECD economies are expected to expand by 1.8% p.a. during the same period.

Another important factor worth closely monitoring is that of the evolving energy policy landscape. On the international front, the recent COP30 meeting in Brazil delivered progress on enhanced mitigation, the global stocktake process, adaptation finance and market mechanisms, among others. However, collective ambition remains insufficient to meet the Paris Agreement's long-term goals, and major challenges remain regarding financial obligations, equity and the issue of just transitions. Furthermore, the withdrawal of the US from the Paris Agreement may negatively affect future climate negotiations. The US also announced its potential disengagement from certain UN institutions, notably the UNFCCC and the IPCC.

In the US, the new administration has introduced changes affecting support mechanisms for renewable energy, EVs and fuel efficiency standards. It also revoked so-called 'endangerment finding' in early 2026, which was the scientific and legal basis for the regulation of greenhouse gas (GHG) emissions in the US. This finding, established in 2009, allowed the Environmental Protection Agency (EPA) to establish rules to limit emissions from different sectors, including transportation and power generation. It was also the basis to establish EV incentives. Collectively, these policy developments are likely to result in higher projected US oil demand growth.

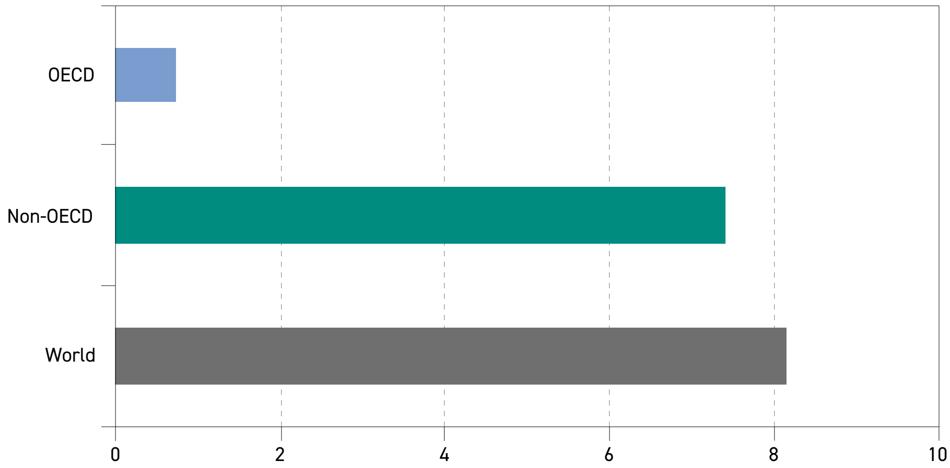
In Europe, there are growing indications that energy affordability and security of supply have moved to the forefront of the policy agenda. Rising energy and electricity prices in European markets have raised concerns regarding the international competitiveness of European industry. In addition, the deployment of renewable energy and EVs has been slower than initially expected. This is why there are already signs of policymakers delaying and reconsidering their previously adopted policies and targets. Some first steps have already been taken in this regard, including the recent adjustment of the EU's 2040 emissions reduction target and formal proposals to potentially dilute the planned 2035 phase-out of ICE vehicle sales.

This renewed and heightened emphasis on energy security and affordability in developed economies is likely to generate spillover effects across many developing regions. At the same time, geopolitics and escalating global tensions create uncertainty regarding the future scale and reliability of international (financial) support for developing countries. In many developing countries, this support and funding are critical for the expansion of renewable energy capacity and related infrastructure.

For China, it is likely to develop somewhat differently. Although it will remain one of the most important sources of incremental oil demand over the medium term, its overall demand increase is somewhat lower compared to previous years. This reflects the country's faster penetration of EVs and its development of related charging infrastructure, as well as continued oil substitution in several sectors. The country is also expected to sustain a high rate of growth in adding renewable electricity generation capacity to meet growing demand for electricity and substitute coal in the power sector (discussed in Chapter 2).

Consequently, this Outlook expects robust global oil demand growth in the period to 2030. Global oil demand is expected to reach levels of 113.3 mb/d in 2030, representing growth of over 8 mb/d from the base year (Figure 3.2). Non-OECD countries, especially in the Asia-Pacific, are expected to be the primary driver of oil demand growth. Overall non-OECD oil demand is set to rise from around 59.2 mb/d in 2025 to 66.6 mb/d in 2030, an increase of almost 7.5 mb/d.

Figure 3.2
Incremental oil demand by region, 2025–2030, mb/d



Source: OPEC.

The OECD region is projected to continue growing marginally, reaching levels of around 46.7 mb/d in 2030, up by 0.7 mb/d from the base year. While this growth is relatively modest – in the range of 0.1 mb/d to 0.2 mb/d p.a. – it adds to the overall demand increase rather than offsetting part of non-OECD growth.

Beyond 2030, two distinct trends are expected to emerge and confirm the views from previous outlooks. In OECD countries, oil demand is projected to reach a peak and gradually decline over the long term. This trajectory reflects expectations that the impact of recent policy shifts, as mentioned above, will moderate over time, while structural factors such as efficiency improvements and fuel substitution will likely gain more momentum.

In the transportation sector, the increasing penetration of EVs and rising fuel efficiency of ICE vehicles in the road fleet is set to reduce demand for gasoline and diesel. In maritime transport, the growing use of LNG as a bunker fuel may weigh on fuel oil demand growth. In aviation, the expanding deployment of biofuels and synthetic fuels is, to some extent, projected to curb growth in oil consumption.

In the industrial and residential sectors, oil is expected to face intensifying competition from electricity and natural gas, further contributing to a gradual decline in demand across OECD economies. In the residential sector, oil-based heating systems are likely to be replaced by alternatives such as heat pumps and gas-based heating systems. Finally, OECD countries are set to see declining oil demand in power generation, reflecting rising competition from other energy sources such as gas, nuclear and renewables. Apart from policy and technological developments, OECD oil demand will also be negatively affected by an ageing population and relatively low long-term economic growth.

Consequently, OECD oil demand is set to decline from 46.7 mb/d in 2030 to around 43.6 mb/d in 2040, before reaching 38 mb/d in 2050 (Table 3.1). This represents an overall demand drop of almost 8 mb/d from 2025–2050.

Table 3.1
Long-term oil demand by region, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	25.4	25.9	25.8	24.9	23.6	22.4	–3.0
OECD Europe	13.4	13.7	13.3	12.3	11.2	10.2	–3.3
OECD Asia-Pacific	7.1	7.1	6.8	6.4	5.9	5.5	–1.6
OECD	**45.9**	**46.7**	**45.9**	**43.6**	**40.8**	**38.0**	**-7.9**
China	16.9	18.1	18.9	18.8	18.4	18.0	1.1
India	5.6	7.1	8.8	10.4	12.1	13.8	8.1
Other Asia	9.8	11.3	12.5	13.4	14.3	15.1	5.3
Latin America	6.9	7.8	8.6	9.2	9.5	9.7	2.8
Middle East	8.9	9.9	11.0	12.0	12.9	13.6	4.7
Africa	4.9	5.7	6.5	7.4	8.3	9.2	4.3
Russia	4.0	4.3	4.4	4.3	4.3	4.2	0.2
Other Eurasia	1.3	1.5	1.6	1.6	1.7	1.7	0.4
Other Europe	0.8	0.9	1.0	1.0	0.9	0.9	0.1
Non-OECD	**59.2**	**66.6**	**73.1**	**78.1**	**82.3**	**86.1**	**26.9**
World	**105.1**	**113.3**	**119.0**	**121.7**	**123.1**	**124.1**	**19.0**

Source: OPEC.

In contrast, as shown in Table 3.1, non-OECD oil demand continues to expand to 2050. It increases steadily over the entire outlook period, albeit with signs of a growth slowdown in some subregions. Non-OECD oil demand is set to increase by almost 27 mb/d between 2025 and 2050, driven by rising populations and urbanization, as well as continued robust economic growth. The latter will be reflected in an increasing passenger vehicle fleet and expanding aviation sector, which will provide mobility for an increasing middle class in the non-OECD.

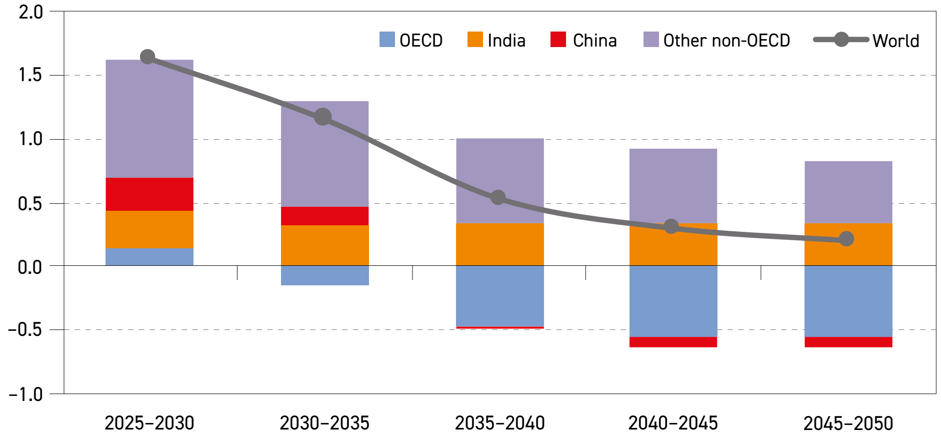
Moreover, continued economic development is expected to result in an expanding commercial vehicle fleet, including a growing number of trucks, buses and agricultural machinery, most of which remain primarily reliant on oil products. In addition, robust demand for petrochemical products, the ongoing shift from the traditional use of biomass to LPG, as well as the expansion of maritime trade are all projected to support sustained regional oil demand growth.

Overall projected OECD demand losses between 2025 and 2050 will be more than offset by demand growth in non-OECD regions. Consequently, global oil demand is set to increase by 19 mb/d over the outlook period, reaching 124.1 mb/d in 2050.

Figure 3.3 shows annual average oil demand increments by region and globally. What is clear is that global oil demand growth experiences a gradual slowdown over the entire period. The average annual growth in the period to 2030 is estimated at around 1.6 mb/d. However, this declines to 0.5 mb/d between 2035 and 2040, before dropping further to 0.2 mb/d in the last five years of the outlook.

This is a result of declining OECD demand, as well as slowing demand in China after 2030. China's demand growth is set to turn negative after 2040. In addition, oil demand growth in many non-OECD regions is likely to see a gradual deceleration, including in Other Asia, the Middle East and Latin America, based on slower economic growth, rising fuel substitution and increasing efficiency. At the same time, oil demand growth in India is projected to remain steady over the entire outlook period at around 0.3 mb/d p.a. Similarly, oil demand growth in Africa is expected to stay at around 0.2 mb/d p.a. between 2025 and 2050. This reflects dynamic economic development, rising urbanization and industrialization, the substitution of biomass by LPG and a rapid expansion of transport services.

Figure 3.3
Average annual oil demand growth by region, 2025–2050, mb/d



Source: OPEC.

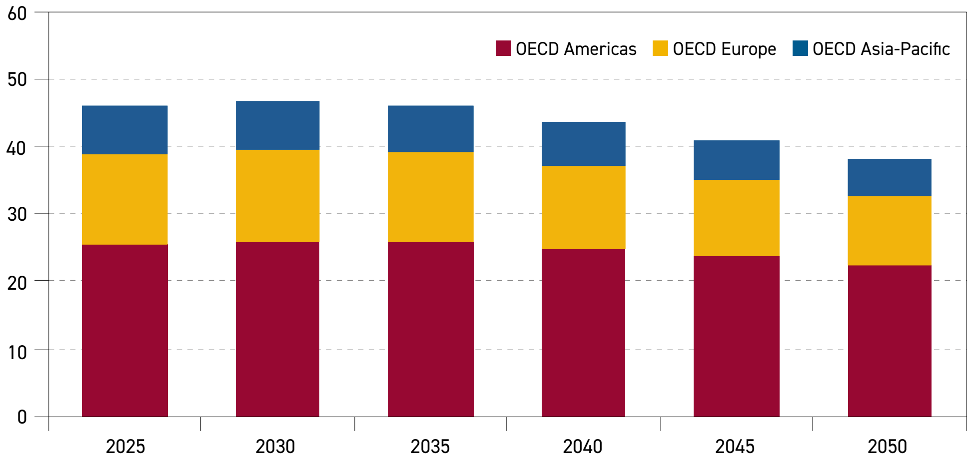
3.1.1 OECD

The oil demand trajectory in OECD countries is closely linked to climate, energy and industrial policies and their subsequent implementation. It has become clear that previously adopted policies related to emissions reductions were overly ambitious in their scope and pace of implementation. Consequently, this has resulted in gradual readjustments and delays in some of these policies. In some cases, regions have witnessed complete U-turns related to their initial policy frameworks. In addition, the evolving geopolitical landscape and rising uncertainty is bringing energy security, energy independence and energy affordability considerations increasingly into focus. This is likely to result in different approaches across OECD countries, which will affect oil demand trends too.

Oil demand is set to witness modest growth in the period to 2030. Total OECD oil demand is set to reach around 46.7 mb/d by 2030, up by 0.7 mb/d relative to 2025 (Figure 3.4). This growth is projected to come mostly from the OECD Americas, up by around 0.5 mb/d. Economic growth, in combination with less strict policies related to oil consumption in the US, are the primary drivers of this growth.

Oil demand in OECD Europe is also expected to see marginal growth in the period to 2030, reaching 13.7 mb/d. This is mostly based on a more optimistic view regarding expected demand in the road transportation sector. Although marginal, this increase represents a major shift from the widely accepted view of an imminent oil demand peak in this region. Instead, due to shifts in policies and slower-than-expected substitution of oil by other energy sources, this Outlook maintains the view of a prolonged plateau over the medium term. Finally, oil demand in the OECD Asia-Pacific is set to hover around 7.1 mb/d in the period to 2030.

Figure 3.4
OECD oil demand by region, 2025–2050, mb/d



Source: OPEC.

In the longer term, OECD oil demand is projected to start declining. It drops from 46.7 mb/d in 2030 to 43.6 mb/d in 2040 and further to 38 mb/d in 2050. This represents a decline of almost 8 mb/d, or 17%, over the entire outlook period. The oil demand peak in the OECD, and the gradual drop, thereafter, reflects the implementation of energy policies aimed at energy efficiency improvements and oil substitution, but also sluggish economic growth and an ageing population.

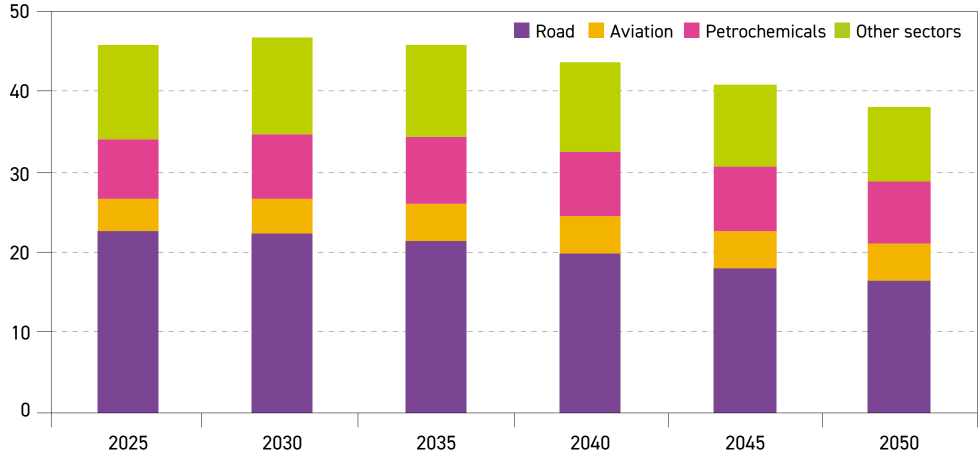
Nevertheless, there is a visible impact of recent energy policy delays and reversals across the OECD region. This year's Outlook sees OECD oil demand in 2040 around 1.3 mb/d higher when compared to last year's projection. The upside revision is projected to narrow somewhat by the end of the outlook period but remains significant at around 0.8 mb/d in 2050. Slower-than-expected EV penetration is one of the major reasons for this upside revision throughout the long term.

The largest long-term demand drop is projected for OECD Europe, at 3.3 mb/d between 2025 and 2050. This represents a drop of around 24% over the entire outlook period. Meanwhile, OECD Americas sees a somewhat lower demand drop of 3.0 mb/d in absolute terms, and 12% in relative terms. The lowest overall demand decline is projected for the OECD Asia-Pacific at 1.6 mb/d. However, due to a much lower base demand in 2025, this decline is comparable to OECD Europe in relative terms, as it represents around 23% of base demand.

Figure 3.5 provides further insights on the long-term OECD demand outlook from a sectoral perspective. The road transportation sector sees the largest decline in the outlook period. From around 22.7 mb/d in 2025, demand in road transportation is set to decrease to 16.3 mb/d in 2050, a decline of 6.4 mb/d. This is due to the rising penetration of EVs within the road transportation fleet, as well as rising efficiency improvements in conventional vehicles.

Fuel efficiency improvements are driven by a combination of policy frameworks, regulations and technological developments. In this respect, the strictest policies within the OECD region

Figure 3.5
OECD oil demand by sector, 2025–2050, mb/d



Source: OPEC.

exist in the EU. According to the current regulation, the corporate average emissions standard for new cars sold within each calendar year stands at 93.6 g CO_2/km. This standard is valid from 2025–2029 but should be tightened to 49.5 g CO_2/km from 2030–2034. This is almost impossible to achieve without a significant share of EVs, which count as zero emission vehicles.

However, due to increasing opposition from European carmakers and the broader public, the European Commission has proposed amending the regulation related to the fleet-wide efficiency target. The Commission recently proposed shifting the strict annual targets to a three-year average (2025–2027), a process that would facilitate the banking and borrowing of credits, thereby providing some flexibility to carmakers in this period. It remains to be seen, however, whether these targets can be reached. This Outlook assumes that these EU emissions targets will be softened further, as carmakers will not be able to achieve them fully and within the current timeline.

In OECD Americas, the most significant development relates to the incumbent administration's move to revoke the so-called 'endangerment finding', which was the scientific and legal basis for regulations of GHG emissions in the US. The finding, established in 2009, allowed the EPA to establish rules to limit emissions from different sectors including transportation and power generation. It was also the basis to establish EV incentives. In 2025, the incumbent administration also proposed rolling back fuel economy standards (CAFE standards) for the period to 2031. All of these developments are likely to support oil demand in the road transportation sector, which is why oil demand in the road transportation sector in OECD Americas has been revised up relative to last year's projections. In Japan, the programme to improve the energy efficiency of vehicles – known as the Top Runner Programme – is targeting fuel efficiency of 25.4 km/l by 2030.

Despite some of the above-mentioned rollbacks, it is assumed that efficiency improvements will gradually progress and contribute to declining long-term oil demand in the road transportation sector. The historical experience in OECD countries, however, has shown that part of this improvement can be offset by consumers' shift to larger vehicles (especially SUVs); hence, the net effect is typically lower than expected, albeit still significant.

The second reason for declining oil demand in the road transportation sector relates to the growing penetration of EVs. Although recent EV sales in the OECD have lagged, causing a downward revision to initial expectations and projections, the share of EV sales will continue to rise. They are projected to erode the share of ICE vehicles, especially in the long term.

Bearing in mind the considerable uncertainties regarding future support for EVs, the total number of vehicles in the OECD is projected to increase by around 113 million between 2025 and 2050, while the number of EVs is set to increase by 222 million over the same period. However, the replacement of ICE vehicles is largely focused on the segment of passenger vehicles. ICE vehicles in the commercial vehicle segment that have higher average fuel consumption per vehicle are expected to keep growing. Therefore, the net oil demand effect is set to be less than the simple proportion related to the decline in the number of ICE vehicles.

The displacement of ICE passenger vehicles from OECD roads is primarily expected to affect gasoline demand. As presented in Figure 3.6, this product (including both oil-based gasoline and ethanol) is projected to gradually decline from nearly 14 mb/d in 2025 to 10.2 mb/d by 2050. The increased penetration of EVs is also expected to partly affect diesel demand (including oil-based diesel and biodiesel), though a larger part of its decline is linked to a combination of developments in the residential sector, industry and marine transport. The combined effect is that OECD demand for diesel is set to decline by almost as much as gasoline, namely by 3.6 mb/d between 2025 and 2050.

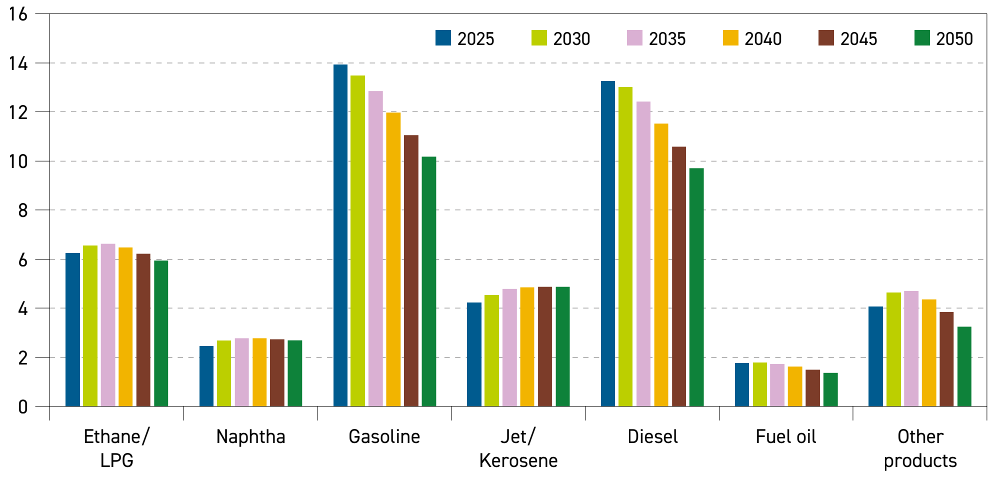
In addition to road transportation, oil demand in 'other sectors' is expected to see a significant decline of 2.4 mb/d over the outlook period. This drop relates mostly to the residential and commercial sector due to the reduction in oil use for heating, as well as better insulation in buildings. The rising share of heat pumps and/or gas-based heating in the residential sector is anticipated to largely replace oil in the residential sector. Oil demand is also set to decline in the industrial sector (excluding petrochemicals), where rising efficiency and substitution by electricity and natural gas will play an important role. In the longer term, the deployment of low-carbon hydrogen can also play a role in displacing oil demand in industry.

Elsewhere, OECD oil demand in the petrochemical sector is expected to rise from around 7.6 mb/d in 2025 to 8.4 mb/d in 2035. This reflects rising demand in the US, which is largely based on increasing volumes of ethane/LPG from unconventional oil and gas production. However, demand in the petrochemical sector is subsequently projected to decline gradually in the OECD. This is largely based on slower economic growth, as well as higher competition from other regions, especially China. Petrochemical oil demand in 2050 is projected at 7.6 mb/d in 2050, which is almost identical to demand in the base year.

Aviation is the only sector in the OECD where demand is expected to keep increasing over the entire outlook period. Demand is projected to increase from 3.9 mb/d in 2025 to 4.8 mb/d in 2050, which partly offsets losses in other sectors. It is important to note that oil demand in the aviation sector will increasingly be covered by some SAF, which include biofuels and synthetic fuels. While the contribution of SAF in the short term is almost zero, it is expected to increase in the long term, reaching levels of around 0.6 mb/d in 2050. This growth is based mostly on regulations and related mandates established by various OECD governments to achieve emissions reductions.

Figure 3.6 shows OECD oil demand trends by main product category. The largest changes are projected for gasoline and diesel, as these are linked to the road transportation sector. As already described, this is the sector that is projected to see the largest drop in demand over the entire outlook period.

Figure 3.6
OECD oil demand by product, 2025–2050, mb/d



Source: OPEC.

As for gasoline demand (including both oil-based gasoline and ethanol), demand is projected to decline gradually throughout the outlook from 13.9 mb/d in 2025 to just below 12 mb/d in 2040 and then further to 10.2 mb/d in 2050. This translates to a demand drop of 3.7 mb/d over the modelling horizon. The decline is based on rising fuel efficiency in most OECD regions, as well rising EV penetration. Diesel demand (including oil-based diesel and biodiesel) is also set to decline throughout the outlook period. Demand is estimated at 13.3 mb/d in 2025, before dropping to 11.5 mb/d by 2040 and then 9.7 mb/d in 2050. This represents a decline of 3.6 mb/d.

At the same time, demand for jet/kerosene (includes oil-based jet/kerosene and SAF) is projected to increase throughout the outlook period, which is in line with developments in the aviation sector already highlighted. Jet/kerosene demand is projected at 4.9 mb/d in 2050, up by 0.6 mb/d from 2025. It is important to note that this incremental demand for jet/kerosene will be largely covered by SAF, consisting of both bio jet fuel and synthetic jet fuel, and will largely be driven by policy and mandates.

OECD naphtha demand is almost entirely linked to the petrochemical sector. Demand in the OECD is projected to increase from 2.4 mb/d in 2025 to around 2.8 mb/d in 2035, before inching down to 2.7 mb/d by the end of the outlook period. Ethane/LPG demand is also largely linked to the petrochemical sector (especially in the OECD Americas) but also to other sectors such as the residential, industrial and road transportation sectors. Ethane/LPG demand is projected to increase from 6.2 mb/d in 2025 to 6.6 mb/d in 2035, partly driven by rising US supply and increasing petrochemical demand (mostly ethane-based demand in the US). However, demand for these products is projected to decline gradually to just below 6 mb/d by 2050, in line with falling demand in the petrochemical industry and other sectors.

After an initial period of stagnation, fuel oil demand is projected to decline in the long term. Fuel oil demand is projected at 1.4 mb/d in 2050, down from 1.8 mb/d in 2025. Most of this decline can be attributed to lower demand in the power generation sector, with the exception of electricity generation within the refinery gates. In marine transport, some substitution by other fuels – such as LNG and methanol – is possible, albeit to a much lesser extent.

Demand for 'other products' is expected to drop from 4.1 mb/d in 2025 to 3.2 mb/d in 2050, representing a decline of roughly 0.8 mb/d over the outlook period. This reflects lower demand for bitumen and lubricants across OECD countries. This is in line with slower industrial growth over the long term.

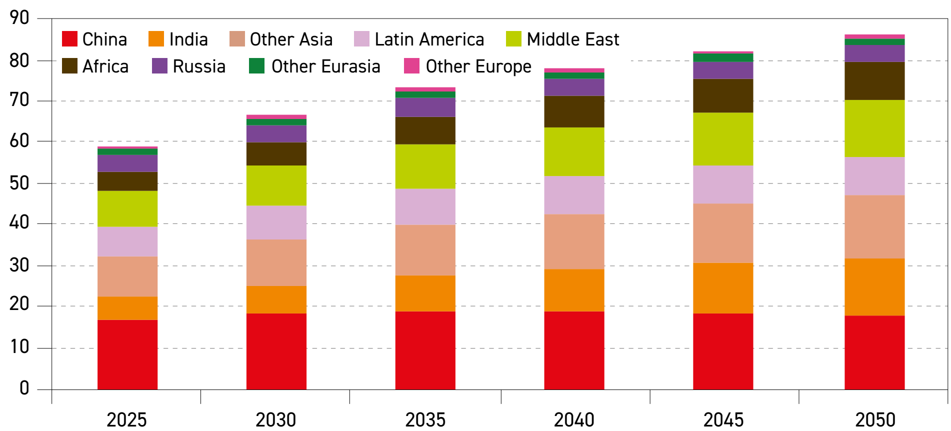
3.1.2 Non-OECD

For decades, the major driver of global oil demand growth has been non-OECD regions, predominantly Asian countries. Total non-OECD oil demand rose from 40.6 mb/d in 2010 to 59.2 mb/d in 2025, representing growth of nearly 19 mb/d. Around 70% of this growth materialized in Asian developing countries, led by China. Other regions, such as the Middle East, Africa and Latin America, also contributed to this growth but to a much lesser extent.

Oil demand trends in the non-OECD have shifted gradually in recent years, with China's oil demand growth showing signs of slowing, in line with the country's demographic and economic developments. At the same time, oil demand in other regions, especially India, Other Asia, the Middle East and Africa, is growing faster, reflecting rising populations, economic development and improving energy access.

As presented in Figure 3.7, this Outlook sees oil demand in the non-OECD expanding at strong growth rates in the medium term, supported by rapid industrialization, demand from the petrochemical sector and further growth in road transportation and aviation. Rising energy access also supports oil demand growth. After 2030, demand growth remains robust, but will start to decelerate in line with slower economic and demographic growth in

Figure 3.7
Non-OECD oil demand by region, 2025–2050, mb/d

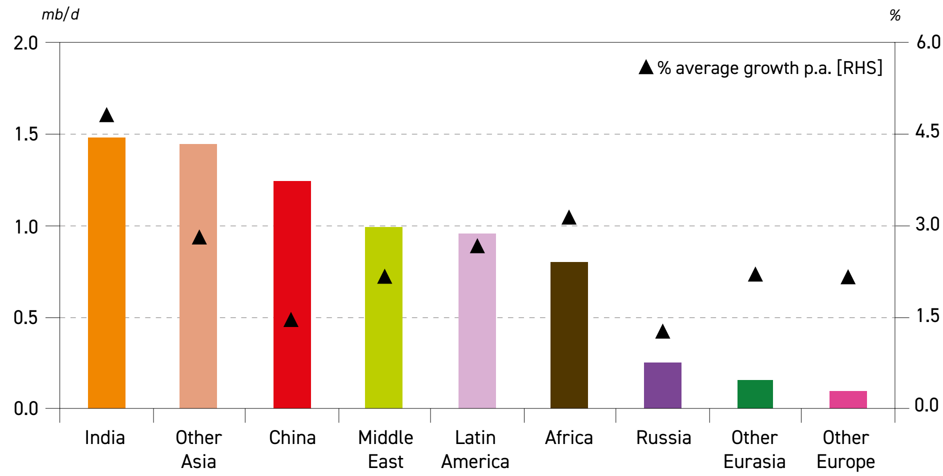


Source: OPEC.

some non-OECD regions, as well as rising energy efficiency and substitution by other fuels. Total non-OECD oil demand is projected to reach 66.6 mb/d in 2030, up by 7.4 mb/d relative to 2025. In the long term, non-OECD oil demand is projected to reach 78.1 mb/d in 2040 and 86.1 mb/d in 2050. This represents an increase of almost 27 mb/d.

Figure 3.8 shows non-OECD oil demand growth by major subregion over the period to 2030. India is projected to be the main source of demand growth, with an increase of 1.5 mb/d in this period. This represents average annual oil demand growth of 4.8% p.a. over this period. The main contributor to this growth is the road transportation sector, in line with the country's rising vehicle fleet, and supported by expanding road infrastructure. The petrochemical sector in India will also see increasing demand, a reflection of expanding refining capacity. Demand in the residential sector is projected to increase over the outlook period to 2030 as well, mostly linked to LPG use for cooking. Finally, the aviation sector is set to witness significant growth over this period too, in line with an expanding aviation fleet.

Figure 3.8
Non-OECD regional oil demand growth, 2025–2030



Source: OPEC.

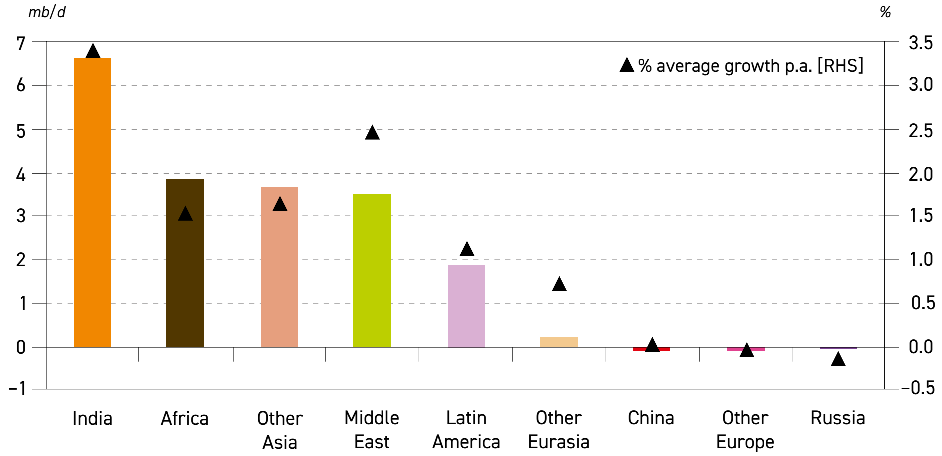
Similar trends are projected in Other Asia (including Indonesia, the Philippines, Malaysia and Vietnam), where oil demand is projected to increase by 1.4 mb/d between 2025 and 2030. Due to a higher demand base level, the average annual growth in this period is calculated at around 2.8% p.a. The major driver of this growth is rapid urbanization leading to higher road transportation and aviation sector demand. Demand also rises in the petrochemical sector.

Upon examining specific drivers of China's oil demand growth, there are significant differences to other developing regions in Asia. China's oil demand has seen robust growth for more than 20 years. From only around 4.7 mb/d in 2000, demand reached almost 17 mb/d in 2025. Due to structural changes, however, including slowing population and economic growth, and a shift to less energy-intensive industry, China's oil demand growth is projected to already show signs of deceleration in the medium term. Despite this, it should be noted that China is still set to be the third largest contributor to global demand growth over this period. China's oil demand is projected to increase by around 1.2 mb/d, which is significant growth, albeit with a lower average growth rate of 1.4% p.a. due to the high base level of demand.

Although smaller, the contribution of other non-OECD regions to medium-term oil demand growth is still significant. Oil demand in each of the Middle East and Latin America is projected to rise by 1 mb/d by 2030. At the same time, African oil demand is expected to grow by 0.8 mb/d. Road transportation, petrochemicals and aviation are the main drivers of this demand growth. Other regions in the non-OECD are also set to see some oil demand growth. Cumulatively, oil demand in Russia, Other Eurasia and Other Europe is projected to increase by around 0.5 mb/d in the period to 2030.

Figure 3.9 shows oil demand growth between 2030 and 2050, with a somewhat altered picture compared to the medium term. The overall increase in non-OECD oil demand is projected at 19.5 mb/d in this period. The single largest driver of oil demand growth over this period is India, with demand projected to increase by an impressive 6.6 mb/d. Similar to the medium term, oil demand is driven by several sectors, including road transportation, petrochemicals, aviation and the residential sector.

Figure 3.9
Non-OECD regional oil demand growth, 2030–2050



Source: OPEC.

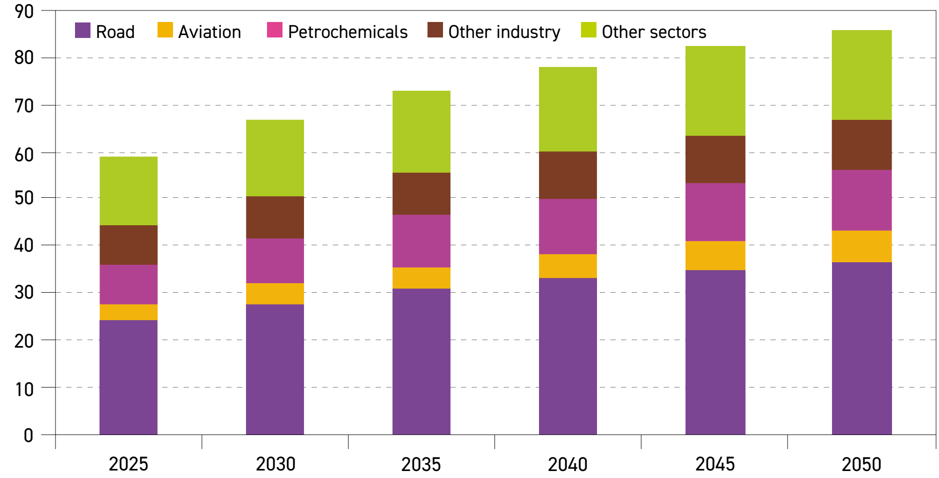
The second-largest oil demand growth is projected for Other Asia at 3.8 mb/d, followed closely by the Middle East at 3.7 mb/d and Africa at 3.5 mb/d. Rising urbanization, industrialization and improving energy access are the key drivers. In Latin America, oil demand is projected to increase by 1.9 mb/d between 2030 and 2050.

While China was one of the main contributors to growth in the period to 2030, this changes in the longer term. China's oil demand is projected to increase from around 18.1 mb/d in 2030 to a peak close to 19 mb/d in 2035, followed by a gradual decline to 18 mb/d in 2050. Structural changes in China's economy, as well as the rising substitution of oil in the road transportation sector, are the main reasons behind this trend. Oil demand changes in the remaining non-OECD regions between 2030 and 2050 period are minimal, consisting of a minor increase in

Other Eurasia and a minor decline in Russia, while demand in Other Europe is set to stay largely flat over this period.

Figure 3.10 shows non-OECD oil demand broken down by major sector. The road transportation sector shows the largest incremental demand growth over the outlook period. Demand in this sector is projected to increase by more than 12 mb/d between 2025 and 2050, which represents around 45% of total non-OECD demand growth. This is in line with an expected large expansion of the non-OECD vehicle fleet, which more than doubles over this period. Despite the addition of a considerable number of EVs, the majority of new vehicles are expected to use diesel, gasoline and, to some extent, LPG (more details on this are provided in section 3.2.1).

Figure 3.10
Non-OECD oil demand by sector, 2025–2050, mb/d



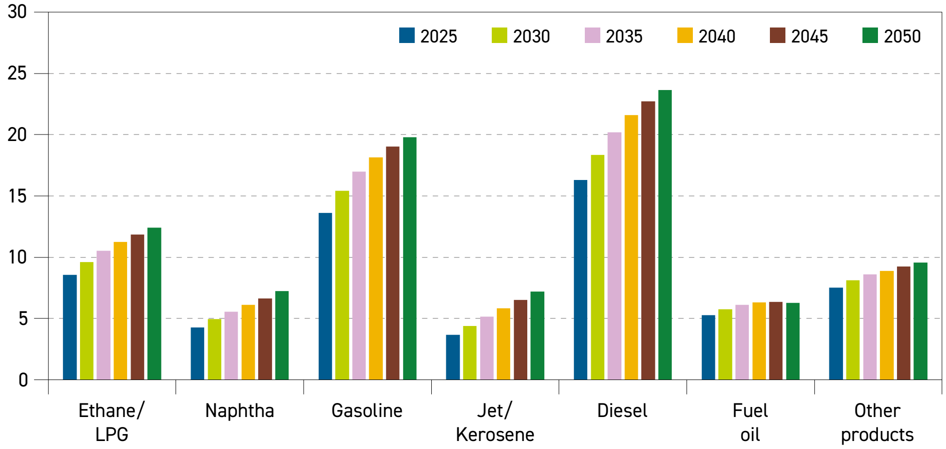
Source: OPEC.

The second largest increment over the long term comes from the petrochemical sector. Rising petrochemical demand mirrors economic development in the non-OECD and expected higher demand for consumer goods and materials in the industrial sector. From around 8.5 mb/d in 2025, oil demand in the petrochemical sector is projected to reach 13.2 mb/d in 2050. Most of this additional demand is set to materialize in Asian developing countries (including India and China), as well as the Middle East.

Oil demand in the aviation sector is forecast to grow strongly over the long term, rising by around 3.3 mb/d. In relative terms, demand in this sector doubles between 2025 and 2050. An expanding middle class in the non-OECD leads to a rising need for air travel, supported by growing aviation fleets and expanding infrastructure. Residential and commercial demand growth is projected at around 2.9 mb/d, reflecting rising energy access across developing countries. Rising industrial output is also set to increase oil demand by around 2.1 mb/d over the modelling horizon, including higher own use (e.g. in the refining sector).

Figure 3.11 provides details related to oil demand from the perspective of major refined products. Demand for all products is projected to increase over the outlook period. The largest increase is projected for transportation fuels, with diesel demand expected to increase by 7.3 mb/d between 2025 and 2050. Diesel demand also benefits from rising consumption in the agricultural sector. The second largest increment is seen for gasoline, with demand expanding by 6.2 mb/d over the same period, in line with the rising passenger vehicle fleet.

Figure 3.11
Non-OECD oil demand by product, 2025–2050, mb/d



Source: OPEC.

Largely due to the expansion of the aviation sector, jet/kerosene demand is projected to almost double from 3.7 mb/d in 2025 to 7.2 mb/d in 2050. At the same time, rising demand for petrochemicals is the major driver of ethane/LPG and naphtha demand growth, with demand projected to increase by 3.8 mb/d and 3.0 mb/d, respectively. A smaller portion of LPG demand growth can also be attributed to higher demand in the road transportation sector. Demand for 'other products' (e.g. bitumen, lubricants and waxes) is projected to increase by 2.1 mb/d to reach 9.6 mb/d in 2050, while fuel oil demand is expected to increase by 1 mb/d.

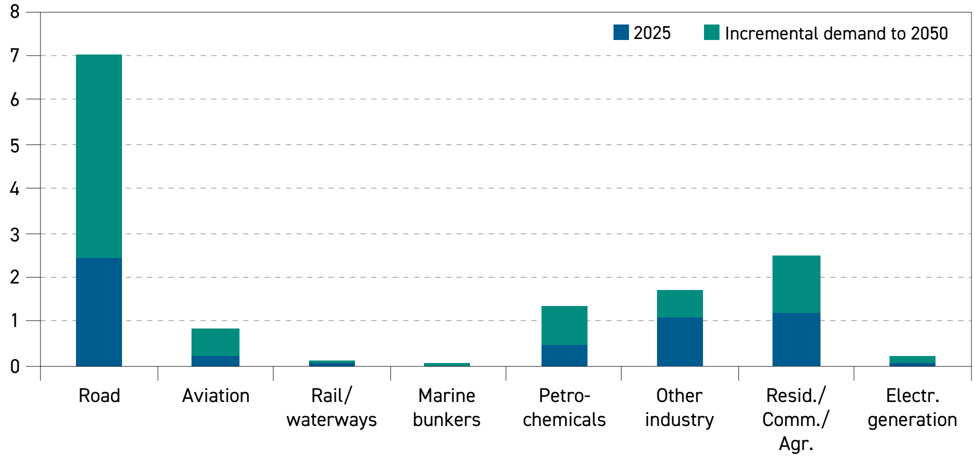
India

This Outlook sees India as the largest single contributor to global energy and oil demand growth over the modelling horizon, driven by significant population growth, economic development and rising urbanization. Over the long term, India's population is projected to increase by 216 million to reach nearly 1.7 billion by 2050.

This has direct implications for the working population in India, which is set to increase from 1 billion in 2025 to 1.13 billion in 2050. In addition, the urban population in India is projected to increase by over 200 million and reach around 720 million by 2050. India's economic growth is projected at a strong 6.5% p.a. in the medium term. Although growth is expected to decline thereafter, it is set to remain above 5% throughout the outlook period. The average annual GDP growth over the entire outlook period is calculated at 5.7% p.a.

Figure 3.12 shows oil demand in India in 2025 and 2050 by major sector. In total, oil demand is anticipated to increase by 8.1 mb/d, growing from 5.6 mb/d in 2025 to 13.8 mb/d in 2050.

Figure 3.12
Oil demand by sector in India, 2025, and incremental demand to 2050, mb/d



Source: OPEC.

By far the largest demand increase is expected in the road transportation sector, where demand is projected to increase by 4.6 mb/d, in line with the rising passenger vehicle fleet in the country. The passenger car fleet is projected to increase to 246 million in 2050 (excluding two-wheelers), up from only 56 million currently. The Outlook assumes that EV penetration in India remains limited over the modelling horizon. In addition, commercial vehicles used over the long term are expected to remain predominantly ICE, thus supporting oil demand growth. The commercial fleet in India is also set to grow strongly and is projected to almost quadruple, rising from 25 million in 2025 to 99 million in 2050.

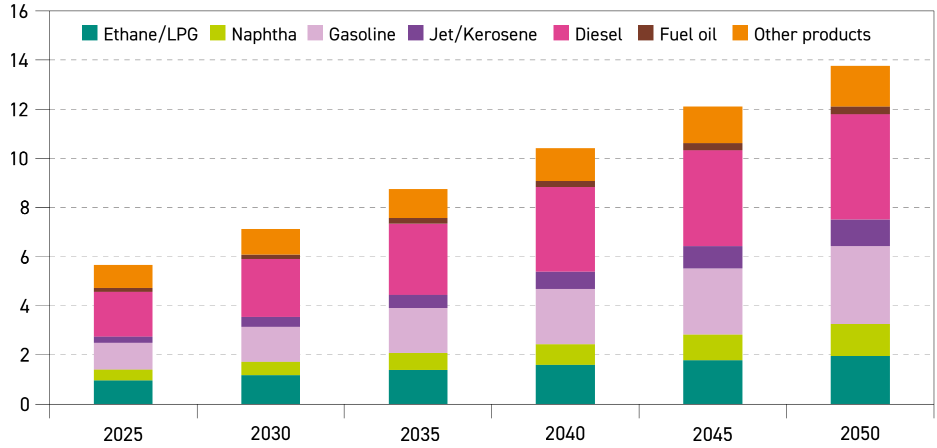
Combined demand growth in the residential/commercial and agricultural sector is anticipated to reach 1.3 mb/d over the outlook period. India has significant residential oil demand, largely reflecting LPG use in the household sector, which is mostly for cooking. Rising energy access in India will support LPG use in the long term. In addition, India's agricultural sector is also set to witness a significant increase in energy use, including oil use in farming infrastructure and agricultural machinery.

India's petrochemical demand is projected to increase by around 0.9 mb/d, representing an increase of 175% by 2050. This is in line with rising demand for consumer plastics, fertilizers and various other plastic-based materials. A variety of petrochemical projects are scheduled to come online in the coming years, not only further enhancing oil demand, but also placing India at the forefront of petrochemical project development in Asia.

Demand in the aviation sector in India is set to quadruple over the modelling horizon, growing from around 0.2 mb/d in 2025 to 0.8 mb/d in 2050. This in line with population growth, a rising middle class and higher mobility needs. A key prerequisite for this development is the expected expansion of aviation infrastructure in the medium and long term. The number of operational airports is expected to increase strongly, expanding from around 160 in 2025 to between 350 and 400 by 2047, according to the government's vision. Oil demand growth is also expected in 'other industry', reaching 1.7 mb/d by 2050, up by 0.6 mb/d from 2025.

Figure 3.13 highlights India's oil demand outlook by major product. Diesel/gasoil is the fuel with the highest share in the mix in 2025 and its position is set to remain unchanged over the outlook period. From a base of around 1.8 mb/d, diesel/gasoil demand is expected to rise to 4.3 mb/d by 2050, an increase of almost 2.5 mb/d. Road transportation is the main contributor to this growth (especially freight transportation), with support from other sectors too, including industry and agriculture.

Figure 3.13
Oil demand in India by product, 2025–2050, mb/d



Source: OPEC.

Gasoline demand is also projected to increase strongly, in line with the country's rising passenger car fleet. It is assumed that conventional ICE cars remain the backbone of India's car fleet throughout the outlook period. From around 1.1 mb/d in 2025, gasoline demand is set to almost triple over the outlook period to reach 3.1 mb/d in 2050.

India's petrochemical sector is expected to be almost the sole driver of the country's increasing naphtha demand. Moving from around 0.4 mb/d in 2025, naphtha demand is set to increase to 1.3 mb/d by 2050. India's petrochemical sector also contributes to ethane/LPG demand growth. Additionally, LPG use is widespread in India's residential sector and the country is projected to see higher demand for this product going forward. Overall, ethane/LPG demand is projected to double between 2025 and 2050 to reach nearly 2 mb/d.

Jet/kerosene demand is set to increase from only 0.3 mb/d in 2025 to 1.1 mb/d by 2050, in line with the country's expanding aviation sector. Finally, demand for 'other products' is forecast to rise by 0.7 mb/d from 2025 to reach 1.7 mb/d in 2050. This reflects the country's growing use of different products such as asphalt, bitumen, petcoke, lubricants and waxes. Demand is mostly from energy-intensive industries like cement, as well as the construction industry, both of which are expanding in India. In addition, some of the above-mentioned products are a by-product of refinery operations and are partly used as refinery fuels.

Finally, fuel oil is the only refined product expected to only grow slightly over the modelling period, increasing by some 0.2 mb/d by 2050. This is attributable to India's lack of major international bunkering hubs, as well as limited demand growth from the power generation sector.

China

China's economy is projected to continue expanding throughout the outlook period. Due to demographic trends and structural changes, however, economic growth is set to slow modestly, as discussed in Chapter 1. Furthermore, China's economy is increasingly moving away from energy-intensive industries towards more advanced industrial sectors. This includes services and innovative products with greater value-added, such as (electric) cars, electronic devices, advanced technology equipment and various appliances. The growing EV industry is expected to have a significant impact on the composition of China's road transportation fleet.

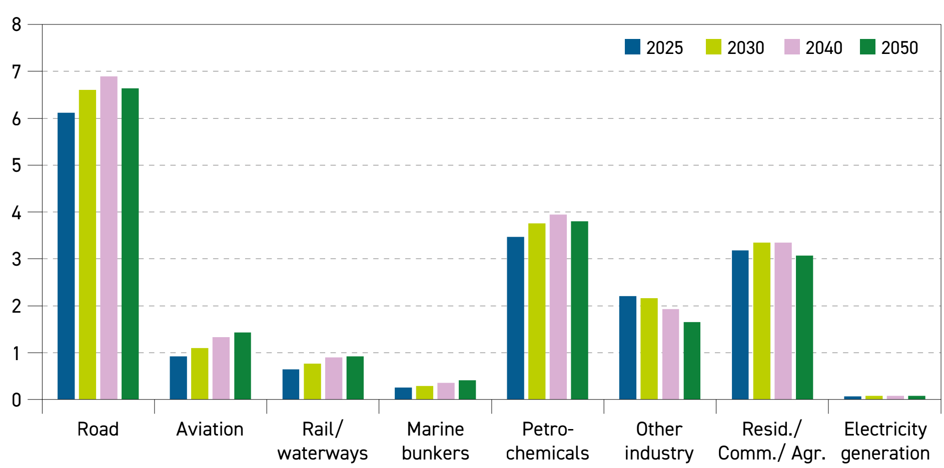
All of these factors are set to have a profound impact on China's oil demand in the near and long term. Oil demand growth in this country is set to slow significantly in the coming years and eventually reach a plateau around 2035, followed by a moderate decline from 2040 onwards.

In detail, this Outlook projects that China's oil demand increases from just under 17 mb/d in 2025 to almost 19 mb/d in 2035. In the period after 2035, oil demand is projected to see a gradual decline before reaching 18 mb/d in 2050.

Road transportation is by far the most important sector for China's oil demand, accounting for around 36% of total oil demand in 2025 (Figure 3.14). It has been the major driver of China's oil demand in recent decades. However, due to gradual shifts in China's car fleet and an increasing substitution of ICE vehicles by alternatives such as new energy vehicles (NEVs), oil demand in this sector is set to experience a gradual slowdown. It increases from 6.1 mb/d in 2025 to 6.9 mb/d in 2040, largely supported by freight transportation. Thereafter, demand is projected to start inching down towards 6.6 mb/d in 2050. It is important to note that a significant share of NEV vehicles in China consist of plug-in hybrids, which still consume gasoline.

At the same time, China's oil demand in the aviation sector is projected to increase from around 0.9 mb/d in 2025 to 1.4 mb/d in 2050. This continuous increase reflects rising demand for national and international mobility. Typically, international flights capture the larger portion of aviation oil demand, but China's growing domestic market is also set to play

Figure 3.14
Oil demand in China by sector, 2025–2050, mb/d



Source: OPEC.

an important role and support future demand too. Oil demand in the rail/waterway sector and marine bunkers is projected to increase by 0.3 mb/d and 0.1 mb/d, respectively. Local tourism and the transportation of goods drive this demand growth, with the latter specifically related to the domestic waterways sector.

The shift in China's economy to value-added products will also lead to higher demand for advanced materials. Consequently, oil demand in the petrochemical sector is projected to increase, rising from 3.5 mb/d in 2025 to 3.9 mb/d in 2040, followed by a minor decline by the end of the modelling period.

Oil demand in 'other industry' is projected to drop from around 2.2 mb/d in 2025 to 1.6 mb/d in 2050. This reflects progress in shifting away from heavy and energy-intensive industries in China, as well as rising energy efficiency. Furthermore, the substitution of oil by other fuels, including natural gas and electricity, also underpins this trend.

The combined demand in the residential, commercial and agricultural sector is set to increase in the medium term, mostly driven by rising demand from the agricultural sector. From around 3.2 mb/d in 2025, demand climbs to 3.4 mb/d in 2030. Combined demand across this sector then gradually declines to levels around 3.1 mb/d in 2050, slightly lower than the base year. Progress in the substitution of oil by electricity and gas is set to curb oil demand in this sector in the long term.

Looking into major product categories, gasoline demand accounts for the largest share in the product mix over the entire outlook period. Gasoline demand is set to increase from around 3.9 mb/d in 2025 to around 4.6 mb/d in 2035 and then remain around that level by the end of the outlook period. This stagnation is due to the rising share of EVs in the passenger fleet. As already mentioned, however, a significant part of these vehicles are likely to be plug-in hybrids, which also consume gasoline. With respect to diesel/gasoil demand in China, demand is projected to grow slightly in the period to 2035, followed by a gradual decline towards 2050.

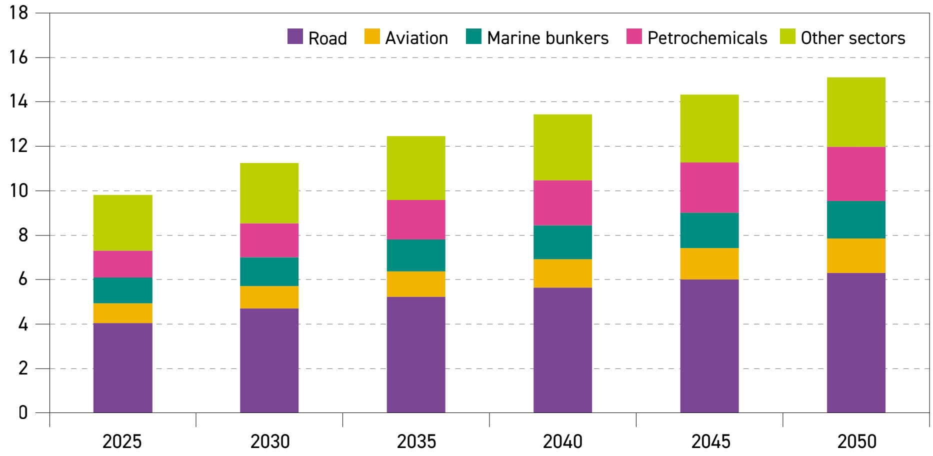
Overall, diesel/gasoil demand is expected to decline by around 0.1 mb/d between 2025 and 2050. Demand is supported by a rising fleet of commercial vehicles and supplemented by growing demand in agriculture and domestic waterways. Towards the end of the period, however, rising electrification of the commercial fleet is likely to negatively impact diesel demand. Driven by developments in the aviation sector, jet/kerosene demand is set to experience corresponding growth, while naphtha demand is set to follow the expansion in the petrochemical sector, as China's petrochemical industry largely uses naphtha as a feedstock. Other refined products are projected to see a temporary demand increase, primarily during the medium term, followed by stable or marginally declining demand in the long term.

Other non-OECD regions

Another important region for global oil demand growth is Other Asia, a region that groups together developing countries in Asia. Population growth here is estimated at around 270 million over the outlook period to reach almost 1.6 billion in 2050. The working population is set to expand by around 170 million in the same period, with the urban population rising by around 320 million. Economic growth is estimated at around 4.2% p.a. on average out to 2030 but then declines gradually over the long term. That said, at an average growth rate of 3.1% p.a. between 2040 and 2050, the economic growth in this region is set to remain strong.

All these factors contribute to rising oil demand in Other Asia, as shown in Figure 3.15. Demand in all sectors, except electricity generation, increases over the modelling horizon. Similar to India, the sector with the largest demand increase is the road transportation sector, where demand is expected to rise by 2.2 mb/d between 2025 and 2050. This growth is driven by an expanding car and commercial vehicle fleet, which is expected to remain dominated by ICE vehicles. Elsewhere, demand in the region's aviation sector is set to increase by 0.7 mb/d over the outlook period, while demand in the marine bunkers sector rises by 0.5 mb/d to reach 1.7 mb/d in 2050. Singapore's bunkering port is the largest of its kind and, in combination with other large bunkering ports in the region, will act as the main driver of this sector's demand growth. Economic development is set to drive demand in the petrochemical sector too, which is expected to double from 1.2 mb/d in 2025 to 2.4 mb/d in 2050. Aggregate demand in 'other sectors' (including the industrial, residential and commercial sectors) is estimated at around 2.5 mb/d in 2025 and is likely to increase to 3.1 mb/d by 2050.

Figure 3.15
Oil demand in 'Other Asia' by sector, 2025–2050, mb/d



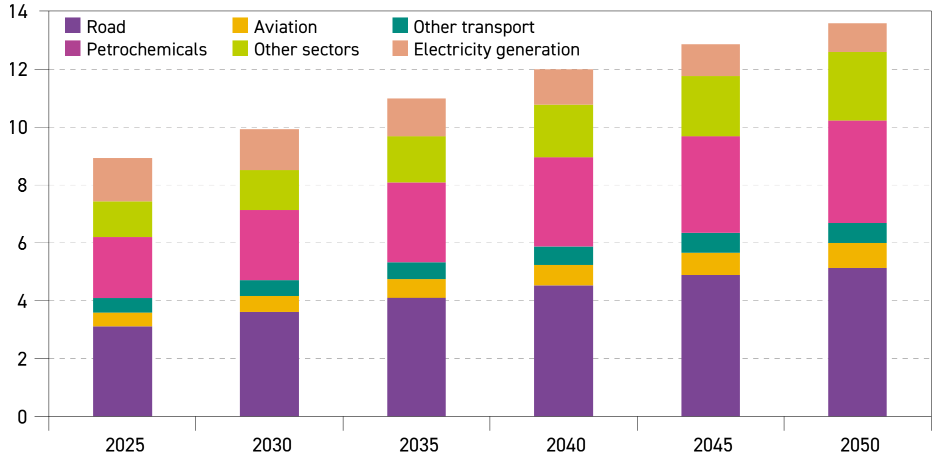
Source: OPEC.

Rising population and economic growth will also support strong oil demand growth in two other regions, the Middle East and Africa.

In the Middle East (Figure 3.16), oil demand is projected to increase by 4.7 mb/d over the outlook period. This is largely front-loaded and is expected to decelerate after 2040, as regional economies mature and population growth slows. The largest demand growth is expected in the road transportation sector and petrochemicals.

Driven by an expansion in the ICE passenger and commercial vehicle fleets, demand in the road transportation sector is anticipated to rise by around 2 mb/d and reach 5.1 mb/d in 2050. In the petrochemical sector, oil demand increases from 2.1 mb/d in 2025 to 3.5 mb/d in 2050, driven by an expansion in refining capacity and petrochemical plants. A wave of new petrochemical projects in the region (more details on this are provided in Chapter 5 and section 3.2.3) and the availability of cost-competitive feedstock provide a significant advantage for this industry, particularly when compared to other regions, thus helping to drive related oil demand higher.

Figure 3.16
Oil demand in the Middle East by sector, 2025–2050, mb/d



Source: OPEC.

Oil demand growth is also expected in the aviation sector, rising from around 0.5 mb/d in 2025 to nearly 0.9 mb/d in 2050. The Middle East serves as an important transport hub between Asia, Europe and Africa and this is projected to expand even further in the long term. Demand in maritime transport is projected to increase too, reaching 0.7 mb/d in 2050, up by 0.2 mb/d over the outlook period. Strong demand growth is projected in the industrial sector, with a large portion of this growth attributable to energy use in the oil and gas industry. The only sector in which demand is expected to decline is power generation, where oil is replaced by other fuels. Demand in this sector declines from 1.5 mb/d in 2025 to 1 mb/d in 2050. This is in line with the official policies of several Middle Eastern countries, as it is economical and frees up significant volumes of oil for export.

Overall oil demand growth in Africa is estimated at about 4.3 mb/d over the outlook period. From 4.9 mb/d in 2025, oil demand is projected to increase to 9.2 mb/d in 2050. The largest share of the increase can be attributed to the road transportation sector, where demand is projected to grow by 1.7 mb/d. This reflects an expected expansion of the commercial and passenger vehicle fleets, especially in Sub-Saharan Africa. Rising demand for air travel in Africa also leads to higher oil demand in the aviation sector too, which is expected to triple between 2025 and 2050. In 2050, demand in the aviation sector is estimated at roughly 0.7 mb/d. Economic growth and rising urbanization is also set to lead to higher demand in the petrochemical and industrial sectors, increasing by 0.3 mb/d and 0.5 mb/d, respectively.

Moreover, expanding sea transport is expected to lead to higher demand in the marine transportation sector, albeit the increase will likely remain limited due to a lack of major bunkering hubs. Marine bunker demand is set to rise by 0.2 mb/d to 0.3 mb/d in 2050. Elsewhere, robust growth is likely to emerge in the residential sector in line with rising energy access. For example, the region's growing use of LPG in the household sector is expected to reduce the traditional use of biomass, particularly in Sub-Saharan Africa. Consequently, oil demand in the residential sector doubles from around 0.5 mb/d in 2025 to nearly 1.0 mb/d in 2050. Finally, oil will also see higher demand in the power generation sector, with this growth largely attributable to own consumption in the energy sector and decentralized generation.

From around 0.5 mb/d in 2025, oil demand for power generation is expected to increase to 0.8 mb/d in 2050.

Oil demand in Latin America is set to grow significantly in the outlook period, albeit at a significantly lower rate relative to Africa and the Middle East. Total oil demand in Latin America is projected to increase from just under 7 mb/d in 2025 to 9.7 mb/d in 2050.

Like other regions, the road transportation sector is forecast to be the major growth driver. Demand is projected to increase by 0.8 mb/d over the outlook period to reach 4.5 mb/d in 2050. This development is supported by increased car ownership and expanding road infrastructure. The aviation sector is projected to increase by 0.4 mb/d between 2025 and 2050, while demand in marine transport rises by 0.3 mb/d. Elsewhere, demand in the residential sector is projected to double, rising from 0.4 mb/d in 2025 to 0.8 mb/d in 2050, while demand in the agricultural sector is expected to rise by almost 0.3 mb/d. In addition, industrial oil demand (including petrochemicals) is set to increase by 0.3 mb/d, while 'other sectors' will likely only see a minor increase over the long term.

Oil demand growth in the remaining non-OECD regions is limited. Demand in Russia is set to increase from around 4 mb/d in 2025 to 4.4 mb/d in 2035, followed by a gradual decline towards 4.2 mb/d in 2050. This will be offset by a modest increase from 1.3 mb/d in 2025 to 1.7 mb/d in 2050 in Other Eurasia. Finally, demand in Other Europe is set to expand only slightly, inching up by 0.1 mb/d over the outlook period.

3.2 Oil demand outlook by sector

This section provides an overview of the oil demand outlook by major sector, with Table 3.2 providing a summary of major sectors at the global level. It is clear that oil demand in the transportation sector (road, aviation, marine bunkers and railway & domestic waterways) represents the backbone of global oil demand. For example, the transportation sector accounted for almost 60% of global oil demand in 2025, and the outlook sees this share remaining relatively stable throughout the forecast period. Oil demand in this sector is expected to grow by 11.5 mb/d in the outlook period, which also accounts for 60% of the overall demand growth, as shown in Figure 3.17.

The second largest sector underpinning oil demand growth is the industrial sector, which accounted for nearly 30 mb/d in 2025. This sector is dominated by petrochemicals, where oil is predominantly used as a feedstock in the production of various products. Against this backdrop, demand for materials and plastic is set to increase strongly in the medium and long term, supported by economic growth and rising demand for consumer goods.

This trend explains why petrochemicals is expected to be the major driver underpinning oil demand growth in the industrial sector. Oil demand in 'other industry' also increases, albeit this growth is likely to remain limited due to continuous energy efficiency improvements and substitution by other fuels. Demand in the industrial sector is projected to increase by around 5.7 mb/d, ultimately reaching 35.6 mb/d in 2050.

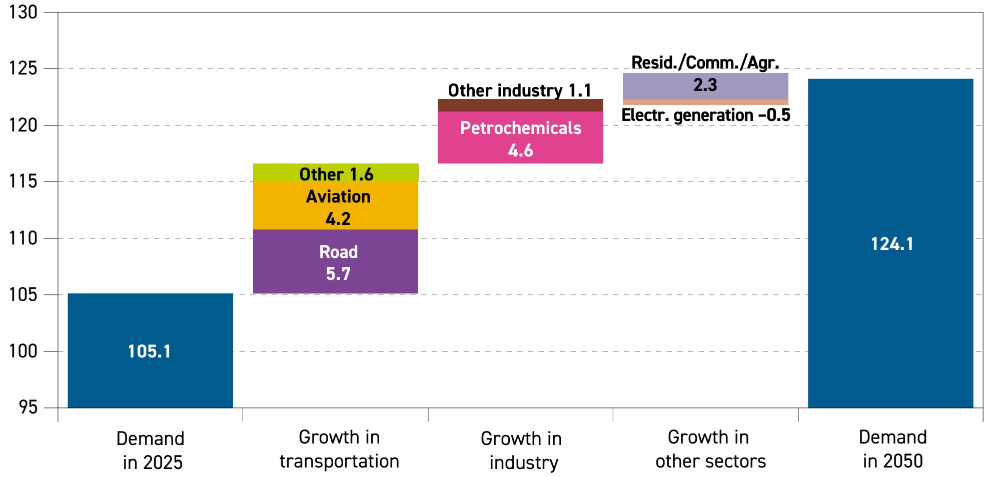
The remaining oil demand (around 15% in 2025) can be attributed to several sectors, including residential/commercial/agriculture and electricity generation. Oil demand trends in these sectors, however, are linked to very specific regional circumstances. At the global level,

Table 3.2
Global oil demand by sector, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
Road	47.0	49.9	51.9	52.6	52.7	52.7	5.7
Aviation	7.2	8.3	9.3	10.1	10.8	11.5	4.2
Rail/waterways	1.9	2.1	2.2	2.3	2.3	2.3	0.3
Marine bunkers	3.9	4.3	4.6	4.8	5.0	5.2	1.2
Transportation	**60.1**	**64.7**	**68.1**	**69.8**	**70.8**	**71.5**	**11.5**
Petrochemicals	16.1	18.0	19.4	20.1	20.5	20.7	4.6
Other industry	13.8	14.5	14.9	15.0	14.9	14.9	1.1
Industry	**29.9**	**32.5**	**34.3**	**35.1**	**35.4**	**35.6**	**5.7**
Res./Comm./Agric.	11.2	12.0	12.7	13.1	13.3	13.5	2.3
Electricity generation	4.0	4.0	3.9	3.8	3.6	3.5	–0.5
Other uses	**15.1**	**16.0**	**16.6**	**16.8**	**16.9**	**17.0**	**1.8**
World	**105.1**	**113.3**	**119.0**	**121.7**	**123.1**	**124.1**	**19.0**

Source: OPEC.

Figure 3.17
Global oil demand growth by sector, 2025–2050, mb/d



Source: OPEC.

oil demand in the residential/commercial/agricultural sector is projected to increase from 11.2 mb/d in 2025 to 13.5 mb/d in 2050. In contrast, demand in power generation is projected to drop, falling by around 0.5 mb/d over the outlook period to about 3.5 mb/d in 2050.

In Figure 3.17, oil demand from an incremental point of view is presented, alongside the major contributing sectors to oil demand over the outlook period. The data demonstrates that the largest contributors are road transportation, petrochemicals and aviation, whose combined demand is projected to increase by 14.5 mb/d between 2025 and 2050, accounting for more than 75% of overall demand growth.

Demand in road transportation is projected to increase from 47 mb/d in 2025 to 52.7 mb/d in 2050, an increase of 5.7 mb/d. This growth is due to an expansion of the passenger and commercial vehicle fleet, especially in developing countries. Oil demand growth in the road transportation sector is initially strong but it is then expected to slow towards the end of the outlook period, when it is projected to plateau at around 52.7 mb/d from 2040 onwards.

As already mentioned, demand growth is supported by an expanding vehicle fleet, with the global ICE fleet expected to increase by more than 500 million between 2025 and 2050. At the same time, fuel efficiency gains – partly supported by the hybridization of conventional ICE vehicles – will limit these gains.

In particular, significant support to future road transportation oil demand is expected to come from the rising number of commercial vehicles. At the global level, an increase of more than 360 million vehicles is anticipated between 2025 and 2050. Moreover, and in contrast to passenger cars, alternative vehicles (mainly powered by electricity and natural gas) are set to account for a much smaller share of this segment. In this context, commercial vehicles are set to be the primary source of expanding oil demand in this sector.

The oil demand increase in the petrochemical sector is estimated at 4.6 mb/d, rising from 16.1 mb/d in 2025 to 20.7 mb/d in 2050. This demand growth is due to the rising need for a variety of petrochemical products, including construction materials and plastics, chemicals, packaging, fertilizers and many others. Plastics are not only widely used as consumer goods, they are also important components in the production of cars, aircraft and electronics. In addition, they remain crucial for the renewable industry (e.g. PV panels and windfarms). The largest share of this growth is set to materialize in Asia and the Middle East, where petrochemical capacities are set to set expand significantly.

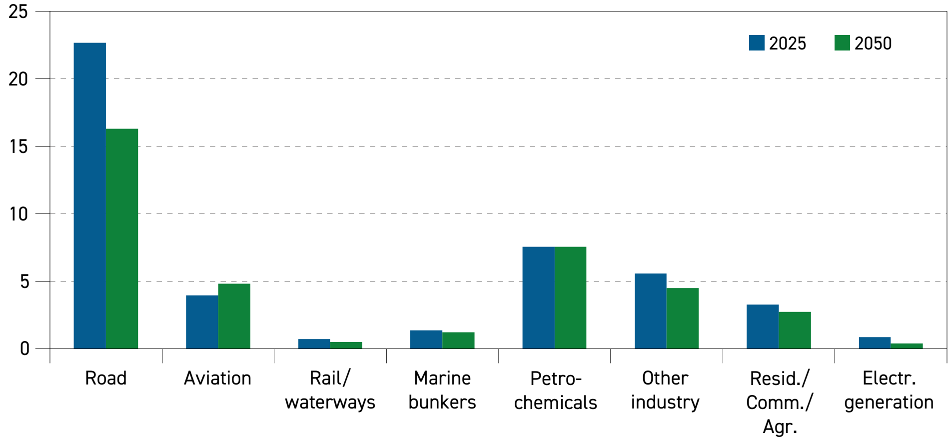
Oil in the petrochemicals sector is used in two different ways. Firstly, a smaller amount of oil demand in this sector is used as fuel in various thermal processes. For this purpose, natural gas, which is often cheaper, represents strong competition for oil. Secondly, a larger amount of oil is used as a feedstock, consisting of naphtha, ethane and LPG. Although some substitution in this process is possible with natural gas and biomass – meaning that these products will likely increase their share in this industry – oil-based products are much more suitable for a wider range of requirements.

The aviation sector is projected to witness demand growth of 4.2 mb/d to reach 11.5 mb/d in 2050. This growth materializes as air travel increases rapidly and comes irrespective of continuously improving engine efficiency and aircraft optimization. Rising oil demand for air travel is expected mostly in developing countries in Asia, the Middle East and Africa, but also in OECD regions. Demand growth in non-OECD Asia is projected at 1.8 mb/d, accounting for more than 40% of global growth. A significant share of aviation's oil demand growth is set to be met by a rising supply of SAFs (including biofuels and synthetic fuels), which is driven predominantly by mandates.

Demand increments in other sectors are more modest, partly due to higher competition from other energy sources, mainly natural gas and electricity. This applies to 'other transportation', 'other industry', as well as the residential/commercial/agricultural sector. A demand increase of 1.6 mb/d in 'other transportation' is largely driven by rising demand in the marine bunker sector amid rising interregional trade volumes. Demand in 'other industry' is projected to grow by 1.1 mb/d, mostly in developing countries, and in many cases is linked to own consumption in the energy sector. The increase of 2.3 mb/d in the residential/commercial/agricultural sectors is mainly attributable to oil's growing use in households in developing countries (including clean cooking), which, in contrast, is partly offset by declining demand in developed countries as oil is replaced by other fuels. Oil consumption also expands in the agricultural sector, in line with increased use of diesel-based machinery.

Figure 3.18 and Figure 3.19 show major sectoral oil demand changes in the OECD and non-OECD separately.

Figure 3.18
OECD oil demand by sector, 2025 and 2050, mb/d



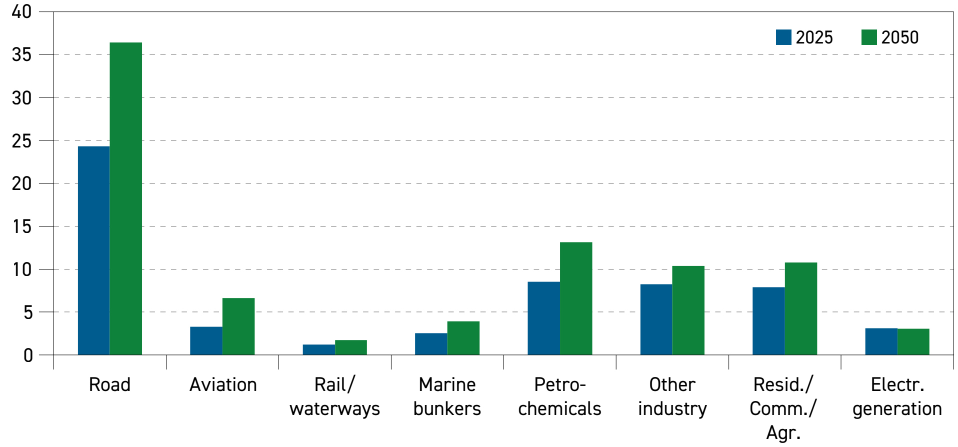
Source: OPEC.

Within the outlook horizon, OECD oil demand is projected to decline in all sectors with the exception of aviation. Declining oil demand is due to fuel competition and the increasing substitution by natural gas and electricity. In many sectors, substitution has been linked to policies and subsidy schemes that aim to reduce oil use. For instance, oil-based heating is expected to be increasingly replaced by electricity-driven heat pumps in OECD regions, especially in Europe. In other regions, gas-based heating systems are likely to replace oil. It is likely, however, that the speed of substitution in the long term will depend on the development and implementation of energy policies.

In addition, substitution also takes place in the 'other industry' sector, where oil is likely to be increasingly replaced by natural gas, electricity and potentially by hydrogen in the latter part of the outlook period. The more efficient use of fuel in industry will also help to reduce oil consumption over the outlook period. Moreover, oil demand in the electricity generation sector is set to decline significantly due to rising competition from other sources, with oil in the future likely only being used to provide backup power or utilized in emergencies.

In the non-OECD, demand in all sectors is expected to increase, except for oil use in electricity generation, which is projected to stay largely flat between 2025 and 2050. Competition between oil products and alternative fuels will be present in most sectors and areas, but the fast-growing overall energy needs in developing countries is expected to accommodate all available energy sources within the outlook period.

Figure 3.19
Non-OECD oil demand by sector, 2025 and 2050, mb/d



Source: OPEC.

The sector with the largest growth is the road transportation sector, driven by an expanding ICE fleet in the passenger and commercial vehicle sector. Although significant, the rising share of EVs in some subregions is not set to be sufficient to offset growth in the ICE vehicle fleet. Strong growth is expected in the industrial, petrochemical and aviation sectors too, while demand in other sectors is expected to be less pronounced but still significant.

3.2.1 Road transportation

Road transportation is the most important sector for global oil demand, accounting for 45% in 2025. This is why developments in this sector have far-reaching implications for global and regional demand trends in both the medium and long term.

Several important factors determine future oil demand trends in this sector, with vehicle fleet size being the most important. Population growth, in turn, in combination with economic development, influence the size of the vehicle parc. In this regard, improving socio-economic conditions in developing countries support new vehicle sales and generally lead to higher car ownership. Economic development also results in an expansion of commercial vehicle fleets. Against this backdrop, in line with strong population and economic growth, the outlook sees a robust increase in the global vehicle fleets over the long term.

At the same time, the composition of the global vehicle fleet is subject to continuous structural changes. Vehicles with alternative fuels already enjoy a considerable share of new sales, especially EVs, but also those utilizing other fuels like natural gas and hydrogen. Looking at current trends, the major uncertainty is related to the speed of electrification in the passenger and commercial vehicle fleet in the medium and long term.

Finally, ongoing fuel efficiency improvements also represent an important parameter determining oil demand in the road transportation sector. Fuel efficiency improvements are often the result of efforts to improve fuel economics but are also strongly related to regulations aiming to improve fuel efficiency over time. Normally, changes in fuel efficiency are evolutionary and not sudden and come following the development of technologies.

However, historical trends have also shown that some fuel efficiency improvements are typically offset by changing consumer behaviour.

This sub-section sets out to capture the interplay of what are often diverging trends, assessing their most likely impact on regional and global oil demand in the road transportation sector.

Vehicle stock

New vehicle sales in most major markets increased slightly in 2025 to reach a level of nearly 98 million worldwide. Slightly more than one third of new vehicle sales, or more than 34 million vehicles were registered in China last year, of which almost 90% were accounted for by passenger vehicles. China's car market is currently undergoing significant changes amid the rising penetration of EVs. The country is currently by far the largest market for EVs in absolute and relative terms. More than 13 million EVs were sold in China last year. However, a considerable share of EV sales was accounted for by plug-in hybrids, which still partly consume gasoline.

A considerable share of the global vehicle market was accounted for by OECD Americas and OECD Europe. They constituted around 17% and 14% of new sales globally in 2025.

In OECD Americas, led by the US, new sales were above 16 million in 2025. However, the share of EVs is much more modest, with less than two million units sold last year. At the same time, the market in OECD Americas (mostly due to the US market) has a much higher share of larger cars and SUVs, which leads to significantly higher fuel demand per vehicle relative to other regions.

In OECD Europe, vehicle sales increased by around 2.5% in 2025. Sales of EVs increased too, especially in the second half of 2025. Around one third of passenger car sales in OECD Europe related to EVs in 2025.

In OECD Asia-Pacific, new car sales were estimated at just under five million in 2025. The share of EVs in this region is still relatively low, making up less than 10% of sales last year. However, this region has a large share of mild hybrids, especially in Japan. This makes the car fleet more efficient compared to other regions. It is expected that this market will remain gasoline-based for many years to come.

In developing countries, India was the leader with around 5.5 million new car sales during 2025. This is still far below sales levels in China, but India is expected to see a rapid increase in car sales in the years to come.

Table 3.3 shows the development of the global vehicle fleet over the outlook period. From around 1.5 billion in 2025, the global passenger car fleet is expected to rise to over 2.3 billion in 2050, representing an increase of around 840 million cars.

Most of this growth is set to materialize in non-OECD Asia, totalling around 570 million units. In China, the passenger car fleet is projected to increase by around 200 million, followed by India with an addition of 190 million and Other Asia with almost 180 million over the outlook period. Other developing regions in the non-OECD (the Middle East, Africa and Latin America) are projected to add around 230 million units.

Table 3.3
Number of passenger cars by region, 2025–2050, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	289	300	313	323	328	330	41
OECD Europe	273	277	281	284	288	292	19
OECD Asia-Pacific	111	107	104	99	95	91	–20
OECD	**673**	**684**	**698**	**707**	**712**	**714**	**41**
China	342	413	481	529	550	544	203
India	56	80	113	154	200	246	190
Other Asia	97	120	151	187	228	275	178
Russia	38	38	38	37	36	35	–4
Other non-OECD	260	297	340	388	440	494	233
Non-OECD	**793**	**948**	**1,123**	**1,294**	**1,454**	**1,594**	**801**
World	**1,466**	**1,632**	**1,821**	**2,002**	**2,166**	**2,307**	**841**

Source: OPEC.

At the same time, car ownership in OECD regions has mostly reached maturity, with limited expansion over the outlook period. In OECD Americas, the fleet is projected to rise by around 40 million between 2025 and 2050, followed by OECD Europe, where the fleet is expected to expand by some 20 million units. In the OECD Asia-Pacific, car fleet growth is anticipated to be negative, contracting by some 20 million units, a reflection of demographic trends.

Despite strong growth in the non-OECD region, car ownership per 1,000 people will still remain far below OECD levels. The only exception is China, where car ownership per 1,000 people is set to come close to levels in the OECD Asia-Pacific at 430. This is still below OECD Americas, where car ownership per 1,000 people is estimated at around 550 throughout the outlook period. At the same time, car ownership in India, Other Asia and other developing non-OECD countries is forecast to increase strongly over the outlook period. However, the numbers are set to stay far below 200 vehicles per 1,000 people.

Table 3.4 shows the development of the global commercial fleet over the outlook period. It is expected to increase from 280 million in 2025 to nearly 650 million units in 2050. Again, the major drivers are developing countries, largely in Asia, where the commercial fleet is set to increase by 170 million, which is almost half of the global increment. Other non-OECD regions are set to add another 115 million vehicles over the same timeframe.

The number of commercial vehicles in the OECD is also set to increase, albeit at a significantly lower rate. Overall additions are estimated at around 70 million, which is around 20% of the global increment over the outlook period. OECD Americas and OECD Europe are forecast to expand their commercial fleet by more than 30 million units each, while the OECD Asia-Pacific region is projected to add six million units.

Vehicle fleet composition

The electrification of road transportation has the most potential to significantly change the future global fleet composition and thus impact future oil demand. Therefore, monitoring

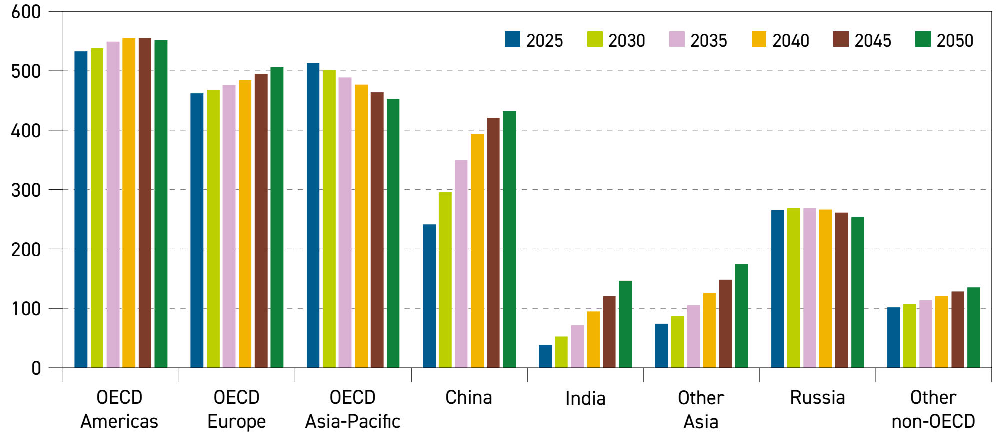
Table 3.4
Number of commercial vehicles by region, 2025–2050, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	43	47	53	60	68	77	33
OECD Europe	46	52	58	65	72	79	33
OECD Asia-Pacific	26	27	28	29	30	32	6
OECD	**115**	**126**	**139**	**154**	**170**	**187**	**72**
China	35	44	54	65	75	85	50
India	25	34	46	61	79	99	74
Other Asia	33	42	52	62	73	83	50
Russia	6	6	6	6	6	7	1
Other non-OECD	68	82	100	123	151	183	115
Non-OECD	**167**	**207**	**258**	**318**	**385**	**457**	**291**
World	**282**	**333**	**397**	**472**	**555**	**645**	**363**

Source: OPEC.

technology developments across regions, the investment plans of car manufacturers, infrastructure investments, as well as consumer behaviour regarding the acceptance and cost competitiveness of EVs, is a regular feature of oil demand modelling.

Figure 3.20
Passenger vehicle ownership by region, passenger vehicles per 1,000 people



Source: OPEC.

New EV sales in 2025 were estimated at almost 21 million units, an increase of almost four million cars relative to 2024. Similar to last year, by far the largest share of these sales materialized in China, where almost 13 million EVs were sold. This represented almost 50% of total new car sales in China in 2025. At the same time, EV sales in China accounted for more than 60% of global EV sales in 2025.

EV penetration in the global vehicle fleet is still relatively low, albeit there are significant regional differences. In China, for instance, it has reached around 11% and in OECD Europe it was estimated at around 7% in 2025. Other regions, however, show much lower EV penetration rates. The EV penetration rate in OECD Americas was estimated at around 3% in 2025, while the OECD Asia-Pacific region was only at 1%. Finally, EV penetration rates in developing non-OECD regions, excluding China, were at 1% or lower in 2025.

Against this backdrop, the two key questions to ask are: how fast could this situation change in the years to come, and can the electrification of road transportation progress to levels that might materially impact oil demand?

EV sales are strongly influenced by policy targets. Recently, several medium- and long-term targets that are directly or indirectly related to EVs have been delayed or, in some regions, entirely abolished. This is the case in two major EV markets, the US and the EU. In the US, the Trump administration revoked the so-called 'endangerment finding', which was the scientific and legal basis for regulations of GHG emissions in the US. The finding, established in 2009, allowed the EPA to establish rules in order to limit emissions from different sectors, including transportation and power generation. It also served as the basis to establish EV incentives and tighten CAFE standards. In addition, the OBBBA has eliminated penalties for CAFE standards violations. Finally, the incumbent administration has also challenged California's authority to set independent emission standards.

In late 2025, the European Commission presented a plan to modify the ban on new ICE vehicles from 2035, with emission reduction targets by 2035 being revised from 100% to 90%. The remaining 10% could be compensated for by using low-carbon steel and sustainable fuels. This is effectively a shift away from the 100% zero-emission vehicle policy previously adopted. Effectively, the proposal by the European Commission sees a significantly lower target for battery EVs (BEVs) compared to the 100% share in new sales from 2035 onwards.

Furthermore, several major carmakers have failed to make the necessary investments in EV production capacities, while others were forced to conduct impairment charges due to lower-than-expected EV sales. For instance, in early 2026, Stellantis announced a massive impairment charge in line with the readjustment of its EV strategy. Mercedes has also recalibrated its strategy and moved away from its previous ambitious electrification strategy and will adopt a more demand-based approach. General Motors announced a readjustment of its EV strategy too, in reaction to recent policy changes in the US, including the abolishment of tax credits for consumers purchasing EVs. The only exception in this context is China, where EV production capacities are supporting EV sales not only in China, but are also leading to rising EV exports abroad.

Based on these developments, this Outlook assumes continued progress in future EV sales and a rising share in the global vehicle fleet (Table 3.5). Nevertheless, the global EV fleet in this Outlook has been revised down modestly, mostly in developed countries. This is partly offset by a revision to the upside in China, based on recent developments. The global EV fleet is projected to rise from around 70 million vehicles in 2025 to 634 million in 2050. By far the largest addition to the EV fleet is expected in China and OECD regions.

In China alone, the EV fleet is projected to grow by more than 200 million, reaching levels above 240 million in 2050. In OECD Europe, the fleet is projected to expand by 95 million and in OECD Americas by almost 110 million. Considerable growth is expected in Other Asia too, where the fleet is projected to rise by 52 million, and in India, which is expected to increase by 27 million.

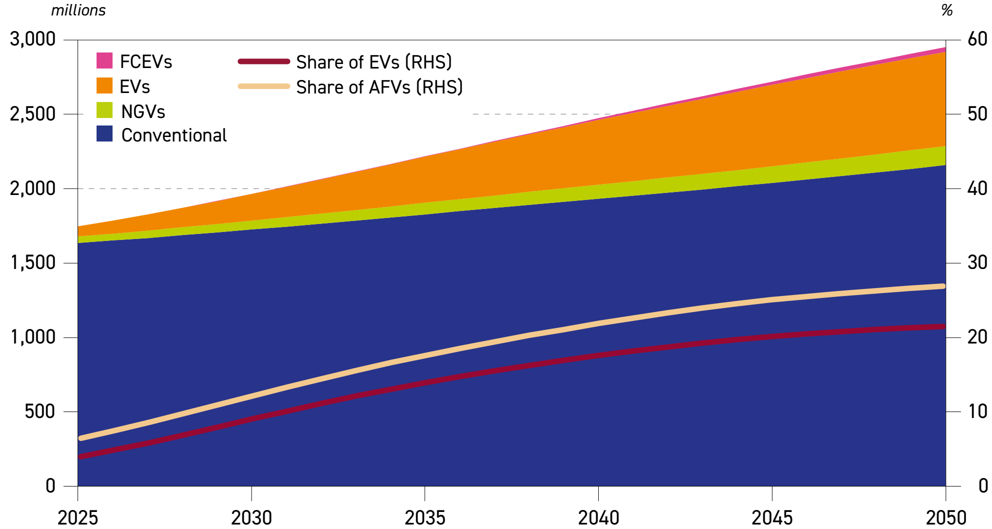
The fleet of two other alternative vehicle types is also expected to expand, but at a much slower rate compared to EVs (Figure 3.21). The global fleet of NGVs is currently estimated at just above 40 million in 2025 and is concentrated in several Asian and Latin American countries. The NGV fleet is projected to almost triple and reach nearly 130

Table 3.5
Number of electric vehicles by region, 2025–2050, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	9	26	50	75	97	109	100
OECD Europe	20	49	76	93	107	115	95
OECD Asia-Pacific	1	5	12	20	26	29	27
OECD	**31**	**80**	**138**	**188**	**230**	**252**	**222**
China	37	87	137	180	215	241	204
India	1	3	7	14	20	28	27
Other Asia	1	5	16	29	42	53	52
Russia	0	0	2	3	4	6	6
Other non-OECD	1	3	10	21	35	55	54
Non-OECD	**39**	**98**	**172**	**247**	**317**	**382**	**343**
World	**70**	**179**	**309**	**435**	**547**	**634**	**565**

Source: OPEC.

Figure 3.21
Global fleet composition*, 2025–2050



* *Conventional vehicles include gasoline, diesel, LPG and hybrid electric vehicles.*

Source: OPEC.

million by 2050, mostly in Asian countries. The majority of this increase is accounted for by passenger cars.

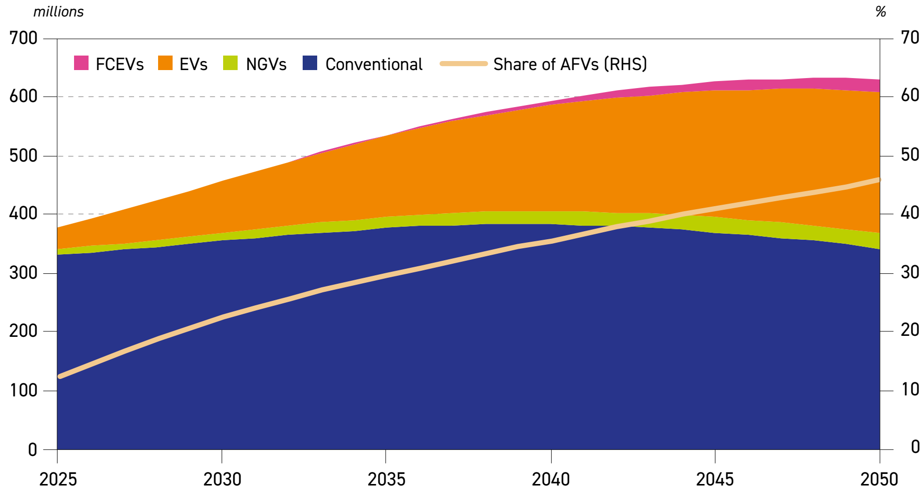
Hydrogen-based fuel cell electric vehicles (FCEV) are expected to remain a niche market over the entire outlook period. FCEVs are in most cases not competitive relative to EVs, which is partly linked to the limited availability of hydrogen, its high costs and underdeveloped infrastructure. In total, the number of FCEVs globally is projected to reach around 30 million by 2050.

Figure 3.21 shows the global fleet composition in the long term. The alternative fuel vehicle (AFV) fleet is forecast to grow fast, reaching almost 800 million units by 2050, an increase of 680 million. At the same time, the fleet of conventional ICE vehicles (including mild hybrids) is set to increase by around 520 million to reach 2.2 billion. The growth of the conventional fleet is driven by developing countries, in both the passenger and commercial segments.

Consequently, the share of the conventional segment in the global fleet is set to decline gradually due to the penetration of EVs and other alternative vehicle types. Even in 2050, however, conventional vehicles are project to account for more than 70% of the global fleet. This underlines that, due to the large existing base of ICE vehicles in the global fleet, the transition to alternative powertrains will likely take many decades, not years. Moreover, the large share of ICE vehicles in the global fleet will make oil demand in this sector resilient to any sudden change.

Figure 3.22 shows the vehicle fleet composition in China in the outlook period, which differs significantly from the global average. In 2025, almost 50% of passenger car sales in China were accounted for by EVs. As already mentioned, however, many of these are in the category of plug-in hybrids and thus do not fully replace corresponding gasoline demand (assuming

Figure 3.22
China fleet composition*, 2025–2050



* *Conventional vehicles include gasoline, diesel, LPG and hybrid electric vehicles.*
Source: OPEC.

these cars were ICE vehicles). In addition, the share of EVs in China is much lower when it comes to commercial vehicles – around 20% – and even lower when it comes to heavy-duty vehicles.

Furthermore, car ownership in China is still far below levels in developed countries. This means that China is likely to remain a fast-growing market in the medium and long term. The total fleet in China is projected to increase to 630 million units in 2050, up from almost 380 million in 2025. As shown in Figure 3.22, the increase in EV sales will significantly change the composition of the vehicle fleet in China over the long term. Their number is projected to increase from just under 40 million in 2025 to 240 million in 2050. Consequently, the share of EVs reaches almost 40% of the total fleet by the end of the outlook period.

At the same time, although the share of conventional ICE vehicles is projected to decline, their number is projected to stay above current levels throughout the outlook period. The ICE fleet is projected to increase from 330 million to above 380 million around 2040, before declining gradually towards 340 million by the end of the outlook period.

Adding to the complexity of estimating future oil demand in this sector is the fact that, within the group of 'conventional' vehicles, the share of commercial vehicles (with their higher demand per vehicle) is set to slightly increase, while the fuel efficiency of the passenger vehicle fleet is expected to improve, and the average mileage driven per vehicle is expected to slightly decline. Moreover, some diesel substitution by natural gas in the segment of commercial vehicles is also set to play a role, though its long-term impact is anticipated to be rather limited. These changes will have implications for the evolution of oil demand in the road transportation sector in China.

Table 3.6
Oil demand in the road transportation sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	13.7	13.7	13.4	12.7	11.9	11.2	–2.5
OECD Europe	6.4	6.1	5.6	5.0	4.4	3.8	–2.6
OECD Asia-Pacific	2.6	2.5	2.2	1.9	1.6	1.3	–1.3
OECD	**22.7**	**22.3**	**21.2**	**19.6**	**17.9**	**16.3**	**–6.4**
China	6.1	6.6	6.9	6.9	6.8	6.6	0.5
India	2.4	3.3	4.2	5.2	6.1	7.0	4.6
Other Asia	4.0	4.7	5.2	5.6	6.0	6.3	2.2
Latin America	3.7	4.1	4.4	4.6	4.6	4.5	0.8
Middle East	3.1	3.6	4.1	4.5	4.9	5.1	2.0
Africa	2.6	2.9	3.3	3.6	4.0	4.3	1.7
Russia	1.3	1.4	1.4	1.4	1.3	1.3	0.0
Other Eurasia	0.5	0.6	0.7	0.7	0.7	0.7	0.2
Other Europe	0.4	0.5	0.5	0.5	0.5	0.5	0.0
Non-OECD	**24.3**	**27.7**	**30.7**	**32.9**	**34.8**	**36.4**	**12.1**
World	**47.0**	**49.9**	**51.9**	**52.6**	**52.7**	**52.7**	**5.7**

Source: OPEC.

Considering all the critical parameters affecting oil demand in road transportation, including the above-mentioned changes in the size and structure of the vehicle fleet, consumer driving habits and assumed fuel efficiency changes, Table 3.6 provides a summary of projected regional demand.

Compared to other sectors at the global level, the road transportation sector is projected to see the largest oil demand growth in the outlook period, increasing from 47 mb/d in 2025 to 52.7 mb/d in 2050. This growth is front-loaded, in line with the development of the global fleet, and is expected to remain stable over the last decade of the outlook. This is due to growth in the ICE fleet being offset by the growing penetration of EVs and other AFVs, as well as the rising fuel efficiency of conventional cars, including an increasing share of mild hybrids.

Of note, regional trends are expected to diverge, with three distinctive pathways. Oil demand in developing countries, excluding China, is forecast to increase strongly in the outlook period. In this context, combined demand growth in India, Other Asia, the Middle East, Africa and Latin America is estimated at almost 11.5 mb/d between 2025 and 2050. This reflects rising car ownership in these regions and an expanding ICE fleet. In addition, an expansion of the commercial fleet will also be supportive of oil demand growth in these regions.

Demand in India alone is projected to increase by 4.6 mb/d in this period. It reaches 7 mb/d in 2050, overtaking China after 2045 to become the second largest region in terms of road transportation oil demand. At the same time, oil demand growth in China is set to increase from 6.1 mb/d in 2025 to almost 7 mb/d in 2035, followed by a gradual decline to 6.6 mb/d in 2050. This is in line with structural changes in China's vehicle fleet and the country's rising share of EVs. However, China will still make up almost 13% of global road transportation oil demand in 2050.

Finally, demand growth in developed regions (OECD) is likely to see a gradual decline throughout the modelling horizon. Limited growth of the vehicle fleet, combined with rising fuel efficiency and the penetration of EVs, will contribute to this trend. Accordingly, total OECD road transportation oil demand is expected to decline from 22.7 mb/d in 2025 to 16.3 mb/d in 2050.

3.2.2 Aviation

According to the International Air Transport Association (IATA), air travel measured in revenue passenger kilometres (RPK) increased by more than 5.3% in 2025 relative to 2024. This expansion was driven mostly by an increase in international demand and, to a lesser extent, domestic demand. The overall passenger load factor (PLF) increased slightly to 83.6%, representing a new record for full-year traffic. According to the IATA, current growth rates are now aligned with historical growth patterns, with the industry having rebounded from the downturn prompted by COVID-19.

RPK growth in 2025 was uneven across different regions. The above-mentioned air travel increase was predominantly driven by the Asia-Pacific and the Middle East, which expanded at a higher rate than the global average. Europe, on the other hand, recorded a similar growth level to the global average of 5.3%, while air traffic in North America only marginally increased in 2025.

The outlook for the aviation sector is bullish, with strong growth expected in most regions. According to the ICAO, global passenger traffic is expected to increase at a compound annual growth rate (CAGR) of 3.2% in the period to 2043. This will require a considerable increase in the size of the global aircraft fleet, as well as related infrastructure and services. While several challenges are likely to remain in this regard, oil demand is nevertheless set to increase over the entire forecast period, in line with the expectations outlined.

As shown in Table 3.7, oil demand in the aviation sector at the global level is projected to increase from 7.2 mb/d in 2025 to 11.5 mb/d in 2050. This represents an increase of 4.2 mb/d, or growth of almost 60% over base demand in 2025. The aviation sector is the third-largest contributor to future incremental demand, after the road transportation and petrochemical sectors. Initially strong, growth in demand is projected to slow moderately towards the end of the outlook period.

Table 3.7
Oil demand in the aviation sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	2.0	2.1	2.3	2.3	2.4	2.5	0.5
OECD Europe	1.3	1.5	1.6	1.6	1.6	1.6	0.2
OECD Asia-Pacific	0.6	0.7	0.7	0.8	0.8	0.8	0.2
OECD	**3.9**	**4.3**	**4.6**	**4.7**	**4.8**	**4.8**	**0.9**
China	0.9	1.1	1.2	1.3	1.4	1.4	0.5
India	0.2	0.3	0.4	0.6	0.7	0.8	0.6
Other Asia	0.9	1.0	1.2	1.3	1.4	1.6	0.7
Latin America	0.3	0.4	0.5	0.6	0.6	0.7	0.4
Middle East	0.5	0.5	0.6	0.7	0.8	0.9	0.4
Africa	0.2	0.3	0.4	0.5	0.7	0.8	0.5
Russia	0.2	0.2	0.2	0.3	0.3	0.3	0.1
Other Eurasia	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Other Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Non-OECD	**3.3**	**4.0**	**4.7**	**5.4**	**6.0**	**6.6**	**3.3**
World	**7.2**	**8.3**	**9.3**	**10.1**	**10.8**	**11.5**	**4.2**

Source: OPEC.

The majority of aviation oil demand growth is projected to materialize in non-OECD countries, where demand rises by 3.3 mb/d between 2025 and 2050. Demand is supported by population growth, economic development and rising demand for air travel. Other Asia, India, China and Africa are each projected to grow between 0.5 mb/d and 0.7 mb/d in the outlook period. At the same time, Latin America and the Middle East are each expected to see growth of 0.4 mb/d. In other non-OECD subregions, demand is set to be either stable or only grow marginally due to declining population growth and/or capacity constraints.

In the OECD, oil demand in the aviation sector is projected to increase by 0.9 mb/d. OECD Americas is expected to increase by 0.5 mb/d, while OECD Europe and the OECD Asia-Pacific each increase by 0.2 mb/d. It is worth noting that oil demand growth in all OECD regions is front-loaded, before it marginally increases or remains stable over the last decade of the outlook period. This is not only due to slower economic growth and population growth decline, but also due to an expected increase in fuel efficiency in this sector.

Policy developments, however, represent a source of considerable uncertainty in this sector. For example, several policies and initiatives at both national and international levels aim to reduce CO_2 emissions in this sector. This also includes initiatives by the ICAO and IATA.

The Long-Term Aspirational Goal (LTAG) of the ICAO sets an overarching target, while specific steps to achieve this goal were outlined by the IATA in 'The Net Zero Roadmap'. These documents discuss a potential substitution of jet kerosene by other fuels, including SAF, electricity and hydrogen, and define them as the main strategy to reach net-zero targets. The ICAO also recently reaffirmed its commitments to the LTAG and confirmed a collective "vision of a 5% reduction of CO_2 emissions in international aviation by 2030 through the use of aviation cleaner energies".

At the national level, many countries have already adopted obligatory mandates for SAF in the medium and long term, predominantly in Europe. The current EU target is set to increase the share of SAF in jet fuel to 6% in 2030 and 34% in 2040.

However, given current geopolitical uncertainties and an-ever increasing focus on security of supply and energy affordability, it is questionable whether these strategies and initiatives are feasible, or, if they are, in what timeframe they can be achieved. The IATA has noted that SAF accounted for up to 0.7% of total global aviation consumption in 2025, with the SAF production rate slowing. The major hurdle is cost, with SAF remaining up to five times costlier than conventional jet fuel.

Finally, several electric aircraft models are under development but are not likely to be available soon. Once deployed, these planes are unlikely to cover long-distance flights and it is likely that their use will remain limited. Consequently, the potential for a substitution of jet fuel by electricity remains limited within the timeframe of this Outlook.

3.2.3 Petrochemicals

As already noted, the petrochemical industry is set to be the second largest driver of oil demand growth over the outlook period, accounting for almost one quarter of total incremental oil demand globally. Currently, more than 15% of total oil demand is attributable to petrochemicals, and this share is expected to increase to nearly 17% by 2050. Demand in the petrochemical sector is ultimately driven by an increasing need for end-use products, including plastics, textiles, electronics, solvents, coatings, adhesives, fertilizers and pharmaceuticals. These end-use products create demand for basic petrochemicals, which in turn creates demand for petrochemical feedstocks.

The oil demand outlook in the petrochemical industry through 2050 will be shaped by a range of structural trends and evolving challenges. On the supportive side, robust growth for petrochemical products is expected to be underpinned by economic growth, population expansion and rising income levels, as well as the continued diversification of end-use applications. Petrochemical derivatives are increasingly embedded across a broad spectrum of industries and technologies, including in renewable energy systems – such as wind turbine components and PV panels, EVs, and the construction sector. These growing applications are expected to provide a solid foundation for long-term growth in petrochemical demand.

Nevertheless, this growth potential is expected to be partly affected by regulatory measures and policy initiatives aimed at addressing environmental concerns. These include commitments to reduce the sector's carbon intensity and the application of stricter emissions standards. In addition, policies that encourage increased recycling rates, promote circular economy principles and restrict certain uses of single-use plastics are all likely to impact oil demand growth in this sector. The gradual proliferation of bioplastics and alternative materials, along with improvements in product design and waste management systems, are also expected to enhance material efficiency and reduce the need for petrochemical raw materials. However, while these developments may slow the pace of demand growth, they are not expected to fundamentally alter the structural role of petrochemicals in the global economy, particularly in developing regions where demand fundamentals remain strong.

Table 3.8

Oil demand in the petrochemical sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	4.2	4.4	4.5	4.4	4.2	4.0	–0.3
OECD Europe	1.5	1.6	1.7	1.7	1.6	1.6	0.1
OECD Asia-Pacific	1.9	2.1	2.2	2.2	2.1	2.0	0.2
OECD	**7.6**	**8.2**	**8.4**	**8.3**	**7.9**	**7.6**	**0.0**
China	3.5	3.8	3.9	3.9	3.9	3.8	0.3
India	0.5	0.6	0.8	1.0	1.2	1.3	0.9
Other Asia	1.2	1.5	1.8	2.0	2.3	2.4	1.2
Latin America	0.3	0.3	0.4	0.4	0.5	0.5	0.2
Middle East	2.1	2.4	2.8	3.1	3.3	3.5	1.4
Africa	0.1	0.1	0.2	0.2	0.3	0.4	0.3
Russia	0.9	1.0	1.0	1.0	1.1	1.1	0.2
Other Eurasia	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Other Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Non-OECD	**8.5**	**9.8**	**11.0**	**11.8**	**12.6**	**13.2**	**4.6**
World	**16.1**	**18.0**	**19.4**	**20.1**	**20.5**	**20.7**	**4.6**

Source: OPEC.

Considering these factors and uncertainties, Table 3.8 shows global and regional oil demand outlooks in this sector over the forecast period. At the global level, oil demand in the petrochemical sector is projected to increase by 4.6 mb/d, rising from 16.1 mb/d in 2025 to 20.7 mb/d in 2050. Almost all growth is set to take place in non-OECD economies, particularly in the Asia-Pacific and the Middle East. In contrast, offsetting demand trends across OECD regions imply overall demand growth in developed economies being close to zero between 2025 and 2050. From around 7.6 mb/d, demand in these regions is set to increase to 8.4 mb/d through 2035, before peaking and then gradually declining to around 7.6 mb/d by 2050.

As a standalone region, the Middle East is projected to be the largest contributor to global oil demand in the petrochemical sector, with an increment of around 1.4 mb/d throughout the forecast period. The region continues to boost the petrochemical industry to add value to its raw materials, leveraging the availability of a wide range of feedstocks like ethane, LPG, naphtha and refinery dry gas at competitive price levels.

Oil demand in this region is expected to expand by more than 0.3 mb/d between 2025 and 2030, driven by several petrochemical projects that are mainly located in Saudi Arabia, the UAE and Qatar.

In Saudi Arabia, Aramco and TotalEnergies awarded engineering, procurement and construction contracts worth $11 billion to seven firms for the construction of a giant petrochemical complex in Jubail in June 2023. The project is expected to be operational by 2027 and reach full capacity in 2028. Additionally, Aramco and Sabic are proceeding with the first crude oil-to-chemicals petrochemical plant in Yanbu, which will be integrated with an existing refinery. The country is also set to increase its propylene production capacity by around 1.2 million tonnes per annum (mtpa) by 2029. Advanced Petrochemicals has also completed the construction of its major expansion facility at Jubail, which is expected to reach full capacity this year. In addition, other major investments are also expected to materialize in the country in the medium term, including the expansion of the Propane Dehydrogenation (PDH) unit of Al Waha Petrochemicals in Jubail and NATPET's new PDH project in Yanbu.

In Qatar, QatarEnergy and Chevron Phillips Chemical have started a $6 billion project in Ras Laffan Industrial City. This will include the region's largest ethane cracker, with an ethylene production capacity of 2.1 mtpa. Meanwhile, in the UAE, ADNOC and Borealis are developing a significant steam cracker project, Borouge 4. With a capacity of 1.4 mtpa, it is expected to be completed in the first half of the medium-term period, in addition to a simultaneous expansion of the existing steam cracker at the same location.

It is worth noting that the production of basic petrochemicals and their derivatives in the Middle East is heavily export oriented. This is due to relatively limited domestic demand and the strategic position of regional producers, which makes them ideally situated to supply global markets at the crossroads of major regions like the US, Europe and the Asia-Pacific. Furthermore, in developed regions, where traditional competitors are located, there is a trend towards closing older plants, providing an opportunity for Middle Eastern players to increase their exports. Therefore, the region is expected to develop numerous other petrochemical projects in the long term, further stimulating oil demand growth in this sector.

In the Asia-Pacific (including India and China), overall petrochemical oil demand is projected to increase by around 2.4 mb/d between 2025 and 2050, out of which 1.2 mb/d is expected to materialize in Other Asia, 0.9 mb/d in India and 0.3 mb/d in China. This growth is set to be driven by an increasing need for end-use products, including plastics, vinyl, textiles, synthetic materials and chemicals. This in turn comes on the back of economic expansion and population growth, supported by the region's well-established processing and manufacturing industries.

These regions have ambitious plans to expand their petrochemical production capacity in the medium term to improve self-sufficiency and reduce their dependency on petrochemical product imports. From 2026–2030, new petrochemical capacity additions are expected to total around 30 mt for ethylene, 25 mt for propylene, and 17 mt for aromatics, driven by numerous world-scale projects currently at various stages of development. The overwhelming majority of this capacity is oil-based. These regions are also supported by projects integrated into refining facilities, aimed at capturing more value from refining streams like naphtha, LPG and dry gas. It is also worth noting that a small portion of these additions is likely to be offset by closures that are due to occur, predominantly in China.

In China, major projects include a joint venture between SABIC and the Fujian Energy and Petrochemical Group that started construction of the SABIC Fujian Petrochemical Complex in February. With an expected ethylene capacity of 1.8 mtpa, the project is scheduled to be commissioned in 2027. In 2025, CSPC, a joint venture between CNOOC and Shell, approved the construction of a third ethylene cracker at its Daya Bay site in Huizhou, Guangdong Province. This new cracker is set to add 1.6 mtpa of ethylene capacity, increasing the site's total to 3.8 mtpa. In addition, significant PDH capacity additions are set to come online out to 2030, primarily concentrated in coastal industrial hubs such as Guangxi, Fujian, Zhejiang, Jiangsu and Shandong.

In India, increasing agricultural activities, a strong automobile market and infrastructure development is set to keep demand for petrochemical derivatives continuously growing. To cater to this expanding demand and to limit reliance on products imports, major companies in India are investing massively in the petrochemical industry, protected by import tariffs imposed by the Indian government. Key projects include Hindustan Petroleum's steam crackers that are expected to start up in 2026. BPCL's Bina Refinery has also announced a new project that will add 1.1 mtpa of capacity by 2027.

Elsewhere, olefins and aromatics capacities are likely to be expanded, such as in Indonesia and Taiwan, to reduce dependence on derivative imports. Against this backdrop, the petrochemical industry in Other Asia is anticipated to grow rapidly over the medium term, supported by strong economic growth. A major capacity increase in Other Asia involves a project announced by Hengyi, in Brunei, for 2 mtpa of paraxylene, scheduled to be commissioned in 2030. Beyond 2030, many petrochemical projects are scheduled to start up in the Asia-Pacific, thus supporting long-term oil demand growth in this sector.

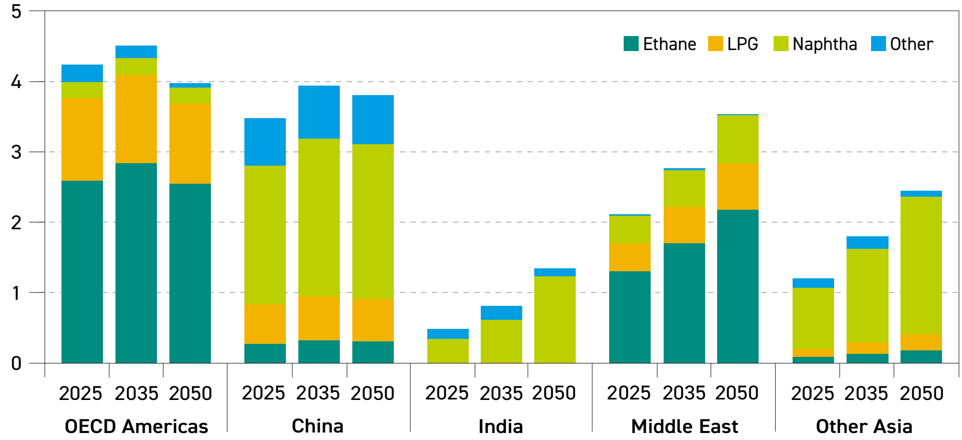
Africa is set to see its oil demand in the petrochemical sector increase by 0.3 mb/d between 2025 and 2050, the bulk of which is set to materialize beyond the medium-term horizon. However, some projects are scheduled to commence operations before 2030, such as the 0.75 mtpa PDH unit in Ain Sokhna, Egypt, and the 0.55 mtpa PDH unit in Arzew, Algeria.

Petrochemical projects in Russia and Other Eurasia, especially for ethylene and propylene, are also expected to drive oil demand in these two regions. Combined oil demand in the petrochemical sector is set to increase by more than 0.2 mb/d over the forecast period. Finally, the same level of demand growth is expected in Latin America, while no growth is projected in Other Europe.

In the developed regions, demand in all OECD regions is expected to increase over the period to 2035 but decline thereafter, leaving total OECD demand in 2050 essentially unchanged compared to its 2025 level. Driven by abundant supplies of ethane at competitive prices, several large ethane-based steam cracker projects are under construction in the US, with the majority expected to come online in the medium term. Beyond this wave of additions, any further expansion is considered unlikely, as the growth of low-cost feedstock availability is expected to end. As a result, oil demand in the petrochemical sector in OECD Americas is projected to peak around 2035 at 4.5 mb/d. Subsequently, oil demand in this region's petrochemical sector is expected to drop by 0.5 mb/d between 2035 and 2050.

From the perspective of major petrochemical feedstocks, Figure 3.23 presents oil demand in the petrochemical sector by feedstock in major regions. Demand for petrochemical feedstocks mirrors growth in demand for basic petrochemicals. Feedstocks include ethane, LPG, naphtha and other petroleum products. Altogether, they make up oil demand for the petrochemical sector.

Figure 3.23
Regional demand in the petrochemical sector by product, 2025–2050, mb/d



Source: OPEC.

The structure of petrochemical feedstock demand differs across regions. Feedstock demand in OECD Americas and the Middle East is dominated by ethane, while all Asian regions primarily use naphtha. These regional disparities are primarily due to the cost advantages of feedstocks in each region.

At the global level, in 2025, naphtha accounted for the largest share of total demand in this industry at 40%, followed by ethane (26%) and LPG (21%). In the long term, naphtha is expected to also provide the largest incremental demand throughout the outlook period, at almost 3.2 mb/d, given the significant increase in naphtha-based steam cracker capacity in Asian countries. As a result, naphtha is set to increase its share to 47% in 2050.

Ethane used for petrochemicals is set to grow significantly in the medium term, driven by abundant ethane from US shale supply and supported by low prices in the US and the Middle East. Ethane demand is projected to increase by an average annual rate of 2% between 2025 and 2030. Thereafter, growth for this feedstock is anticipated to slow to a rate of 0.7% p.a. between 2030 and 2050, mainly due to a decline in US shale supply that partly offsets continued growth in the Middle East. Ethane demand is forecast to increase by around 1.1 mb/d throughout the forecast period, growing from 4.2 mb/d in 2025 to 5.2 mb/d by 2050. The share of ethane is anticipated to remain flat throughout the forecast period at just above 25%.

Elsewhere, LPG demand is set to grow by more than 0.7 mb/d between 2025 and 2030. In the medium term, it is expected to grow slightly from almost 3.4 mb/d in 2025 to more than 4 mb/d by 2035, driven by demand growth for propylene to feed new PDH units in the Asia-Pacific.

After 2035, however, it is set to enter a prolonged plateau until the end of the long-term period. This is mainly due to an increasing amount of propylene production set to come from naphtha-based steam crackers and fluid catalytic cracking (FCC) units. Consequently, the share of LPG in the feedstock mix is set to slightly decline to around 20% by 2050.

3.2.4 Other sectors

Demand for oil in the residential/commercial/agricultural sector is projected to show different regional trends over the outlook period. In the OECD, demand is expected to decline gradually, shaped by shifting climate policies and strengthening efficiency standards. Rising competition from other fuels is another reason for oil demand declining in the residential and commercial sector. Key structural developments within this sector involve continued adoption of alternative technologies in the residential sector, especially based on electricity. These developments are often widely supported by energy polices and related subsidies and incentive schemes. The broader direction towards the replacement of oil in the OECD residential/commercial sector is expected to persist. At the same time, however, diesel will most likely remain of high importance for agricultural activities in rural and off-grid locations, where electrified alternatives are less practical.

Of note, recent concerns related to energy affordability have led to a readjustment of some policies that aimed to replace oil and gas in the residential sector. For instance, in 2023, the German government introduced a law to ban the installation of oil and gas heating systems from 2024. However, the current German government has now amended its legislation and will continue to allow households to keep using oil and gas in the future. Further policy delays and adjustments in other countries are possible, with implications for oil demand.

As shown in Table 3.9, this Outlook sees OECD oil demand in the residential/commercial/agricultural sector remaining stable over the period to 2030 at around 3.3 mb/d, followed by a gradual decline to 2.7 mb/d in 2050. This decline is projected to materialize mostly in OECD Europe and OECD Asia-Pacific. Demand in OECD Americas is projected to remain mostly flat throughout the outlook period.

By contrast, oil consumption in the residential/commercial/agricultural sector across non-OECD economies is seen increasing significantly in the medium and long term. Demand will be driven mostly by India, Africa and Latin America and is related to rising energy access with oil products such as LPG replacing biomass for cooking purposes. In addition, oil is well suited to regions with limitations in natural gas and electricity infrastructure. In India, especially, there are policy-driven efforts to expand the availability of LPG to further support growth in the residential sector.

Moreover, the agricultural sector is projected to remain an important driver for oil demand, resulting from the rising mechanization of this sector across developing economies. This will mostly affect diesel demand, with diesel used for irrigation systems, tractors and transport infrastructure for food distribution.

As illustrated in Table 3.9, oil demand in the residential/commercial/agricultural sector in non-OECD regions is forecast to increase by almost 3 mb/d over the outlook period. Demand in India is projected to rise by 1.3 mb/d, followed by Latin America (+0.8 mb/d) and Africa (+0.7 mb/d). A minor increase is projected for the Middle East and Other Asia, which each rise by 0.1 mb/d. China's demand is expected to increase from 3.2 mb/d in 2025 to around 3.4 mb/d in 2030, followed by a gradual decline to 3.1 mb/d in 2050. This is due to the advancement of the country's electrification.

Table 3.9
Oil demand in the residential/commercial/agricultural sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	1.4	1.4	1.4	1.4	1.3	1.3	0.0
OECD Europe	1.3	1.4	1.4	1.3	1.2	1.1	–0.2
OECD Asia-Pacific	0.6	0.5	0.5	0.4	0.3	0.3	–0.3
OECD	**3.3**	**3.3**	**3.2**	**3.1**	**2.9**	**2.7**	**–0.5**
China	3.2	3.4	3.4	3.3	3.2	3.1	–0.1
India	1.2	1.5	1.8	2.0	2.3	2.5	1.3
Other Asia	0.9	1.0	1.0	1.1	1.1	1.0	0.1
Latin America	0.7	0.9	1.1	1.2	1.3	1.5	0.8
Middle East	0.5	0.5	0.6	0.6	0.6	0.6	0.1
Africa	0.6	0.8	0.9	1.0	1.2	1.4	0.7
Russia	0.5	0.5	0.5	0.5	0.5	0.5	0.0
Other Eurasia	0.2	0.2	0.2	0.2	0.2	0.2	0.0
Other Europe	0.0	0.1	0.1	0.1	0.1	0.1	0.0
Non-OECD	**7.9**	**8.8**	**9.5**	**10.0**	**10.4**	**10.8**	**2.9**
World	**11.2**	**12.0**	**12.7**	**13.1**	**13.3**	**13.5**	**2.3**

Source: OPEC.

Global economic growth remains the major driver for international trade and global shipping, and consequently, this will affect the global marine bunkering sector over the outlook period. Oil is projected to remain the backbone of this sector. However, there are increasing efforts to reduce reliance on oil and introduce alternative fuels.

At the international level, these efforts are supported by the IMO, which has proposed the 'IMO Net-Zero Framework'. This outlines new fuel standards and a pricing mechanism for shipowners, both aimed at reducing future emissions. The proposed target was a reduction in emissions in the range of 20–30% by 2030 and 70–80% by 2040. In October 2025, however, several countries (including the US) opposed this plan and the adoption of the framework was postponed.

Despite these delays, it is assumed that efforts to reduce emissions in the shipping sector will continue shaping the sector's demand patterns. However, more pragmatic approaches are increasingly feasible. These include the expanding use of LNG but also onboard carbon capture and storage.

LNG currently offers the most likely alternative to oil. The growing use of LNG is supported by an expanding global bunkering network and the increasing availability of dual-fuel engines, making it a feasible option for operators seeking to meet regulatory standards without compromising vessel performance. In addition, the rising availability of LNG at competitive prices is also helping to accelerate the substitution. In fact, an expected increase in LNG use is the main reason why oil demand growth in maritime transportation is well below the rate of global trade expansion. However, the relatively long lifetime of vessels will make this process rather gradual. Indeed, it is expected to take decades rather than years for the penetration of LNG vessels to be large enough to displace a part of oil demand.

Moreover, it will likely take even longer for other alternative fuels to significantly penetrate the bunkering sector. Often considered options include biofuels, ammonia, hydrogen and their derivatives, such as e-ammonia, e-methanol and e-methane. All these options, however, face challenges. These include limited availability, high production costs, safety concerns and a lack of infrastructure. Ammonia, for example, raises toxicity and has handling concerns, while its low energy content limits its suitability for long-distance voyages. While hydrogen produces no emissions at the point of use, it is energy-intensive to produce, difficult to store and lacks the required infrastructure for broad application in the shipping sector. Similarly, biofuels, such as biodiesel and ethanol, have not yet found a significant role in marine fuel consumption, as reflected in the data.

Consequently, oil demand in the marine bunkers sector is set to increase from 3.9 mb/d in 2025 to 5.2 mb/d in 2050. This increase is exclusively driven by non-OECD regions, while growth in OECD countries is set to be negative.

In the non-OECD, oil demand is likely to grow by around 1.4 mb/d between 2025 and 2050. The majority of the growth is seen in Other Asia (+0.5 mb/d), driven by major international ports in Singapore, Malaysia and Indonesia. Demand is also projected to increase in the Middle East (+0.2 mb/d), Africa (+0.2 mb/d) and Latin America (+0.3 mb/d). China's demand for marine bunkers is projected to increase by 0.1 mb/d over the outlook period.

Table 3.10
Oil demand in the marine bunkers sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	0.3	0.4	0.4	0.4	0.4	0.4	0.1
OECD Europe	0.7	0.7	0.7	0.7	0.6	0.5	–0.2
OECD Asia-Pacific	0.3	0.3	0.3	0.3	0.3	0.3	0.0
OECD	**1.4**	**1.4**	**1.4**	**1.4**	**1.3**	**1.2**	**–0.1**
China	0.3	0.3	0.3	0.4	0.4	0.4	0.1
India	0.0	0.0	0.1	0.1	0.1	0.1	0.0
Other Asia	1.2	1.3	1.4	1.5	1.6	1.7	0.5
Latin America	0.2	0.3	0.4	0.4	0.5	0.5	0.3
Middle East	0.5	0.5	0.6	0.6	0.7	0.7	0.2
Africa	0.1	0.1	0.2	0.2	0.2	0.3	0.2
Russia	0.1	0.2	0.2	0.1	0.1	0.1	0.0
Other Eurasia	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Europe	0.1	0.1	0.2	0.2	0.1	0.1	0.0
Non-OECD	**2.6**	**2.9**	**3.2**	**3.5**	**3.7**	**3.9**	**1.4**
World	**3.9**	**4.3**	**4.6**	**4.8**	**5.0**	**5.2**	**1.2**

Source: OPEC.

In the OECD, oil demand growth in this sector is forecast to be mixed. In OECD Americas, a minor increase is projected, while demand in the OECD Asia-Pacific is set to remain stable. In OECD Europe, demand is expected to decline by 0.2 mb/d in the outlook period. This is likely to be due to substitution by other fuels, especially LNG.

Oil demand in the 'other industry' segment at the global level is projected to show limited growth in the first decade of the outlook, before entering a phase of stagnation in the long term. Oil consumption is set to rise from 13.8 mb/d in 2025 to 14.9 mb/d by 2035, before remaining largely stable until the end of the outlook period. However, oil demand trends in 'other industry' follow different regional trends.

In the OECD, oil demand in terms of 'other industry' is projected to decline by around 1.1 mb/d over the outlook period. Demand in OECD Americas is seen falling by 0.5 mb/d, followed by OECD Europe at 0.4 mb/d. Demand in the OECD Asia-Pacific is forecast to decline slightly between 2025 and 2050. This decline is due to policies aimed at replacing oil with other fuels. Although many industries are projected to increase their rate of electrification, replacement by electricity has limitations, particularly in hard-to-abate industrial segments. Residual fuel demand will largely remain concentrated in hard-to-abate industry segments, where commercially viable alternatives to oil are limited. In some regions, such as the US, the oil demand decline in 'other industry' will largely be due to substitution by highly competitive natural gas.

Oil consumption in the non-OECD is sent to increase by around 2.1 mb/d over the outlook period, reflecting continuous economic development and industrialization. Most of this growth is expected to materialize in the Middle East, India, Other Asia and Africa, which expand by 1 mb/d, 0.6 mb/d, 0.5 mb/d and 0.5 mb/d, respectively. At the same time, oil demand in China's 'other industry' sector is projected to decline by around 0.6 mb/d. This is due to structural

Table 3.11
Oil demand in the 'other industry' sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	3.0	3.1	3.2	3.0	2.8	2.5	–0.5
OECD Europe	1.8	1.9	1.9	1.7	1.6	1.3	–0.4
OECD Asia-Pacific	0.8	0.7	0.6	0.6	0.6	0.7	–0.1
OECD	**5.6**	**5.8**	**5.7**	**5.3**	**4.9**	**4.5**	**–1.1**
China	2.2	2.2	2.1	1.9	1.8	1.7	–0.6
India	1.1	1.2	1.2	1.3	1.5	1.7	0.6
Other Asia	0.9	1.0	1.1	1.1	1.2	1.4	0.5
Latin America	1.2	1.3	1.4	1.4	1.4	1.3	0.1
Middle East	0.7	0.8	1.0	1.2	1.5	1.7	1.0
Africa	0.7	0.9	1.0	1.1	1.2	1.2	0.5
Russia	0.8	0.9	0.8	0.8	0.8	0.7	–0.1
Other Eurasia	0.4	0.5	0.5	0.5	0.6	0.6	0.1
Other Europe	0.1	0.1	0.2	0.1	0.1	0.1	0.0
Non-OECD	**8.2**	**8.8**	**9.3**	**9.6**	**10.0**	**10.4**	**2.1**
World	**13.8**	**14.5**	**14.9**	**15.0**	**14.9**	**14.9**	**1.1**

Source: OPEC.

changes and a shift to less energy-intensive industry, as well as rising electrification. Other non-OECD regions are projected to see minor changes over the outlook period.

Oil demand related to rail and domestic waterways is currently estimated at just under 2 mb/d in 2025, with this figure representing the smallest portion of the transportation sector. China and the US account for more than 50% of global demand in this sector, while the remainder is divided across all other regions.

In the case of China, the bulk of the demand in this sector (around 90%) is linked to the world's largest inland waterway network, which has over 110,000 km of navigable rivers, lakes and canals. Moreover, there are several ongoing projects to expand this network, including the Yangtze River Economic Belt Development, revitalizing the Beijing-Hangzhou Grand Canal and expanding the Pearl River system. At the same time, the railway system in China is largely electrified, with no significant related oil demand.

Looking forward, oil demand in this sector in China is expected to increase even further, reaching 0.9 mb/d in 2050, up from 0.6 mb/d in 2025 (Table 3.12). Other Asia, India and Latin America are also regions with significant domestic waterway traffic, which is expected to expand in the future. In addition, these regions still have some diesel-based railway traffic, with this situation expected to be maintained in the future. Accordingly, oil demand is set to increase by 0.1 mb/d in each of these three regions.

In the OECD, oil demand in the railway and domestic waterways sector is projected to decline over the outlook period. In the railway sector, this is due to rising electrification in remote areas, as well as increasing engine efficiency. Oil demand is also set to decline in the OECD domestic waterways sector, reflecting rising engine efficiency and increasing substitution

Table 3.12
Oil demand in the rail and domestic waterways sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	0.4	0.4	0.4	0.4	0.4	0.4	–0.1
OECD Europe	0.2	0.2	0.1	0.1	0.1	0.1	–0.1
OECD Asia-Pacific	0.1	0.1	0.1	0.1	0.1	0.0	–0.1
OECD	**0.7**	**0.7**	**0.7**	**0.6**	**0.6**	**0.5**	**–0.2**
China	0.6	0.8	0.9	0.9	0.9	0.9	0.3
India	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Other Asia	0.2	0.2	0.2	0.2	0.3	0.3	0.1
Latin America	0.1	0.1	0.1	0.1	0.2	0.2	0.1
Middle East	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Africa	0.1	0.1	0.1	0.1	0.1	0.0	0.0
Russia	0.1	0.2	0.2	0.1	0.1	0.1	0.0
Other Eurasia	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Non-OECD	**1.2**	**1.4**	**1.6**	**1.7**	**1.7**	**1.8**	**0.5**
World	**1.9**	**2.1**	**2.2**	**2.3**	**2.3**	**2.3**	**0.3**

Source: OPEC.

by other fuels. This especially relates to the rising use of LNG in domestic inland waterway vessels in the US. Consequently, OECD oil demand in this sector is forecast to decline from 0.7 mb/d in 2025 to 0.5 mb/d in 2050.

Table 3.13 provides an overview of oil demand in the electricity generation sector over the outlook period. Global demand is expected to drop from 4 mb/d in 2025 to 3.5 mb/d in 2050, a decline of 0.5 mb/d. This stems from different regional trends.

The region with the highest oil demand in the power generation sector has traditionally been the Middle East. In 2025, oil demand for power generation was estimated at around 1.5 mb/d. However, many countries in this region are making efforts to replace oil with other fuels, including renewables (mostly solar), natural gas and nuclear. Currently, countries such as Saudi Arabia and the UAE are constructing large solar PV plants and have ambitious plans in the medium and long term. Consequently, oil demand in this region is set to decline to around 1 mb/d 2050, a drop of 0.5 mb/d.

At the same time, the outlook expects an increase of 0.3 mb/d in oil demand for power generation in Africa. Oil-based power plants are mostly used to cover peak demand periods and to provide supply in remote areas with no access to power networks. Oil plays an important role in emergency power supply (e.g. diesel-based generators). Elsewhere, a minor increase in oil demand for power generation is projected in India and Latin America, which is in line with rising electricity demand.

In OECD regions, oil demand in power generation is projected to decline gradually, falling from 0.9 mb/d in 2025 to 0.4 mb/d in 2050. This is due to the further phasing out of oil-fired generation and oil's substitution with other fuels. Oil-fired plants that have been used to cover power demand peaks are increasingly being replaced by natural gas-based plants. One important segment in the OECD where oil will still be used in the long term is for emergency power supply.

Table 3.13
Oil demand in the electricity generation sector by region, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
OECD Americas	0.4	0.3	0.3	0.3	0.2	0.2	–0.2
OECD Europe	0.2	0.2	0.2	0.2	0.2	0.2	–0.1
OECD Asia-Pacific	0.3	0.2	0.2	0.1	0.1	0.1	–0.2
OECD	**0.9**	**0.8**	**0.7**	**0.6**	**0.5**	**0.4**	**–0.5**
China	0.1	0.1	0.1	0.1	0.1	0.1	0.0
India	0.1	0.1	0.1	0.2	0.2	0.2	0.1
Other Asia	0.5	0.5	0.5	0.5	0.5	0.4	–0.1
Latin America	0.4	0.4	0.4	0.4	0.5	0.5	0.1
Middle East	1.5	1.4	1.3	1.2	1.1	1.0	–0.5
Africa	0.5	0.5	0.6	0.7	0.7	0.8	0.3
Russia	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Other Eurasia	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Europe	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Non-OECD	**3.1**	**3.2**	**3.2**	**3.2**	**3.1**	**3.1**	**0.0**
World	**4.0**	**4.0**	**3.9**	**3.8**	**3.6**	**3.5**	**–0.5**

Source: OPEC.

3.3 Oil demand outlook by product

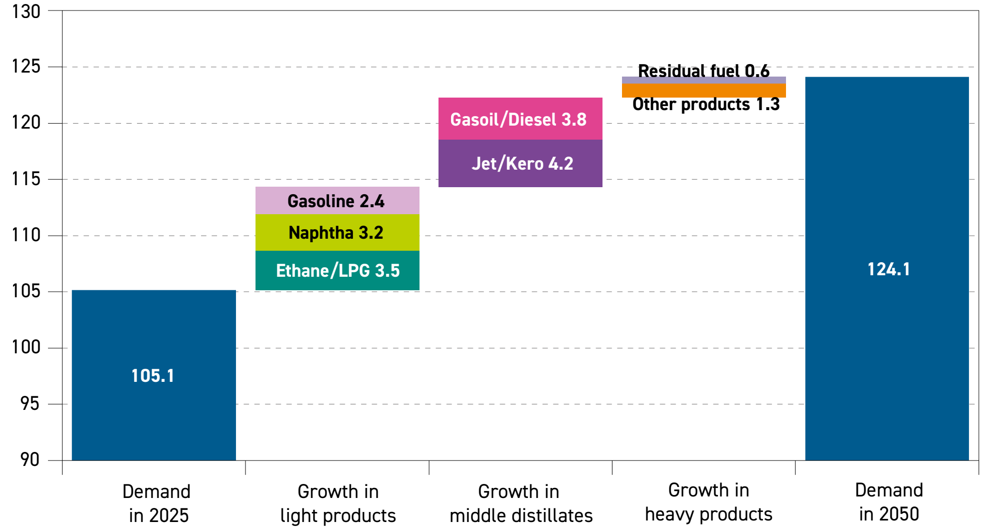
Based on the regional and sectoral future oil demand trends outlined, this section provides a detailed analysis from the perspective of major refined product categories. Global oil demand is projected to increase by 19 mb/d from 2025–2050. Demand for all refined products is set to increase too, as shown in Table 3.14. This growth will be driven predominantly by light distillates and middle distillates, while the increase in demand for heavy products is forecast to be modest, mostly due to their increasing rate of substitution.

Table 3.14
Global oil demand by product, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
Ethane/LPG	14.8	16.1	17.2	17.7	18.1	18.3	3.5
Naphtha	6.7	7.6	8.3	8.9	9.4	9.9	3.2
Gasoline	27.5	28.9	29.8	30.1	30.1	30.0	2.4
Light products	**49.0**	**52.7**	**55.3**	**56.7**	**57.5**	**58.2**	**9.2**
Jet/kero	7.9	8.9	9.9	10.7	11.4	12.1	4.2
Gasoil/diesel	29.6	31.4	32.6	33.1	33.3	33.3	3.8
Middle distillates	**37.5**	**40.3**	**42.5**	**43.8**	**44.7**	**45.4**	**7.9**
Residual fuel	7.0	7.5	7.9	7.9	7.9	7.6	0.6
Other products	11.6	12.8	13.3	13.2	13.1	12.8	1.3
Heavy products	**18.6**	**20.3**	**21.2**	**21.2**	**20.9**	**20.5**	**1.8**
World	**105.1**	**113.3**	**119.0**	**121.7**	**123.1**	**124.1**	**19.0**

Source: OPEC.

Figure 3.24
Demand growth by product category between 2025 and 2050, mb/d



Source: OPEC.

In the light products sector, the largest increase is expected for ethane/LPG, with demand increasing by 3.5 mb/d (Table 3.14 and Figure 3.24). This is also supported by rising ethane supply in some regions, such as OECD Americas and the Middle East, where this product represents the main feedstock for ethylene production. LPG is also used in the petrochemical sector and in the industrial and residential sectors. LPG is an important fuel for providing access to clean cooking, thus reducing the traditional use of biomass in Asia, Africa and Latin America. In OECD Europe and the OECD Asia-Pacific, ethane/LPG growth is likely to remain limited due to sufficient alternatives, especially electricity.

Naphtha demand is set to increase by 3.2 mb/d over the outlook period, reaching almost 10 mb/d by 2050. This increase is primarily linked to the petrochemical sector in developing countries, supported by rising populations and continuous economic development. The largest increase in naphtha demand is expected in China, India and Other Asia.

Gasoline demand is projected to increase by 2.4 mb/d over the outlook period, supported by the increase in the global ICE vehicle fleet. Global gasoline demand is projected to reach 30 mb/d by 2050. However, there are divergent regional trends related to this fuel that depend highly on EV adoption and fuel-efficiency developments. Gasoline demand is forecast to increase mostly in developing regions in Asia (excluding China), the Middle East, Africa and Latin America. In these regions, EV deployment will likely remain limited over the period to 2050. Demand growth in these regions is set to be offset by declines in other regions, including China and the OECD (especially OECD Europe and OECD Americas). EV deployment and rising fuel efficiency are the major reasons for the expected decline in gasoline demand in the long term.

Middle distillates, including jet/kerosene and gasoil/diesel, are projected to see continuous growth between 2025 and 2050. Jet/kerosene demand is projected to increase by 4.2 mb/d, mostly driven by the expanding aviation sector in developing countries, particularly in Asia, the Middle East, Africa and Latin America. Part of this demand increase will be covered by SAF, especially in OECD regions.

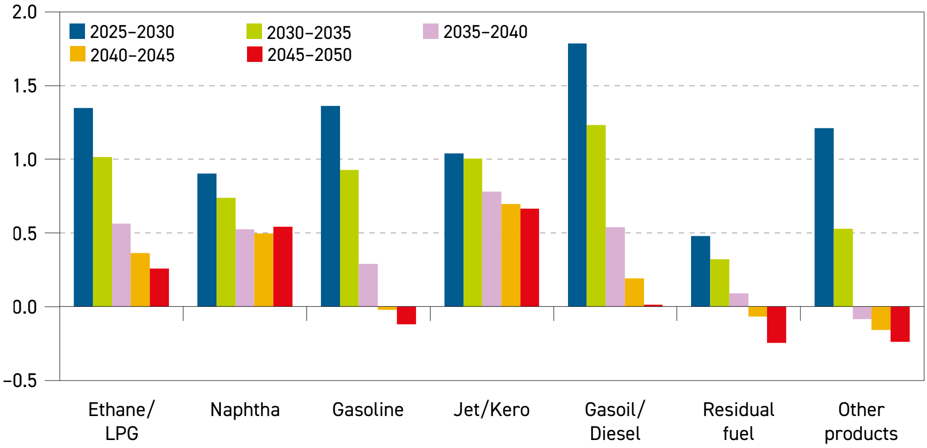
Gasoil/diesel demand is projected to increase from 29.6 mb/d in 2025 to 33.3 mb/d in 2050, which represents the second largest absolute increase among all refined products, a gain of 3.8 mb/d over the outlook period. Diesel has an important role in commercial transport, industry, as well as agriculture. Like gasoline, the largest share of the projected demand growth is set to materialize in developing regions, notably India, Africa, Latin America and Other Asia. This is set to be offset by an expected decline in developed regions, where substitution by other fuels will limit demand for diesel.

Figure 3.25 highlights these evolving dynamics, showing a slowdown in gasoline and diesel growth after 2030, while products like ethane/LPG, naphtha and jet fuel continue to expand.

In the heavy segment, demand for other products is set to increase by 1.3 mb/d, reaching 12.8 mb/d in 2050. This reflects economic and industrial development in non-OECD regions, which leads to higher demand for products like bitumen, petroleum coke, lubricants and waxes.

In particular, bitumen consumption is expected to remain robust as governments in Asia and Africa continue to prioritize road and transportation projects. At the same time, lubricants are anticipated to retain their importance in industrial machinery and automotive applications. In this context, the strongest growth is forecast to occur in non-OECD markets, where manufacturing and vehicle production are on the rise.

Figure 3.25
Growth in global oil demand by product, mb/d



Source: OPEC.

Finally, demand for fuel oil is projected to increase by 0.6 mb/d. From around 7 mb/d in 2025, fuel oil demand is projected to reach 7.9 mb/d in 2035. This will be followed by a period of stagnation, before demand for fuel oil drops to around 7.6 mb/d in 2050. This demand increase mostly stems from the marine bunkers sector in developing countries in Asia and the Middle East. However, this will be offset by gradual declines in developed regions. At the same time, fuel oil use in the power generation sector is projected to decline gradually, mostly in developed countries and the Middle East. This is due to energy policies and regulations, as well as oil's substitution by other fuels, which in many cases are cost competitive.

Liquids supply

Key takeaways

- Non-DoC liquids supply is set to increase from 54.2 mb/d in 2025 to 58.2 mb/d in 2030, meeting around half the expected oil demand increase. Supply growth in this period is primarily driven by Brazil, Qatar, the US, Argentina and Canada, as well as a number of smaller, newer producers in non-DoC Africa.

- A reassessment of the outlook for US tight crude now indicates that tight crude supply may have peaked in 2025, at just over 9 mb/d. However, total US liquids supply is expected to continue growing, albeit by a modest 0.4 mb/d until 2030, before plateauing thereafter. This is supported by a continued expansion in NGLs supply and a small projected increase in biofuels.

- From the early 2030s, non-DoC supply overall is projected to peak and enter a long plateau at around 60 mb/d. Canada and Argentina stand out among non-DoC producers, with supply from both countries set to continue increasing even beyond 2030. By 2050, non-DoC liquids supply is projected to average 59.6 mb/d, up by about 5.5 mb/d from 2025.

- Broken down into types of liquids, long-term dynamics observed in previous WOOs remain unchanged. From 2025–2030, crude, NGLs and other liquids supply (including refinery processing gains) each contribute around one-third of incremental non-DoC supply. In the long term, however, growth in NGLs, other liquids, and biofuels is partly offset by a decline in US tight and other crude sources.

- With oil demand projected to grow to 2050, the relative role of DoC liquids supply increases significantly, rising from 50.6 mb/d in 2025 to 55.3 mb/d in 2030 and thereafter to 64.5 mb/d by 2050. This implies an increase in market share from 48% in 2025 to 52% in 2050.

- Total cumulative oil-related investments required to meet projected demand amount to $17.7 trillion (in US$2026) for the period 2026–2050. Upstream investment requirements on an average basis have been adjusted slightly upwards due to higher projected supply in the long term and a shift towards higher-cost resources. They are now assessed at $14.5 trillion, or $581 billion per annum, on average.

- The lowered trajectory for US supply means that other non-DoC resources will be tapped, including notably Canadian oil sands, oil from frontier areas, such as Argentina, and new producers in Africa. Meanwhile, downstream needs are projected at $1.9 trillion, and midstream at $1.3 trillion. This means that average total oil-related p.a. investment requirements are now seen at $707 billion.

This chapter describes the outlook for liquids supply from 2025–2050. As in previous Outlooks, the medium-term projections for 2025–2030 and the longer-term outlook are discussed separately due to the different methodologies employed. The medium-term view relies on a bottom-up approach, identifying upstream project start-ups, their progress and the underlying decline in mature fields, while the long-term outlook is based on an assessment of the available resource base and other factors. US and other tight oil are also modelled and discussed separately, as are non-crude liquids.

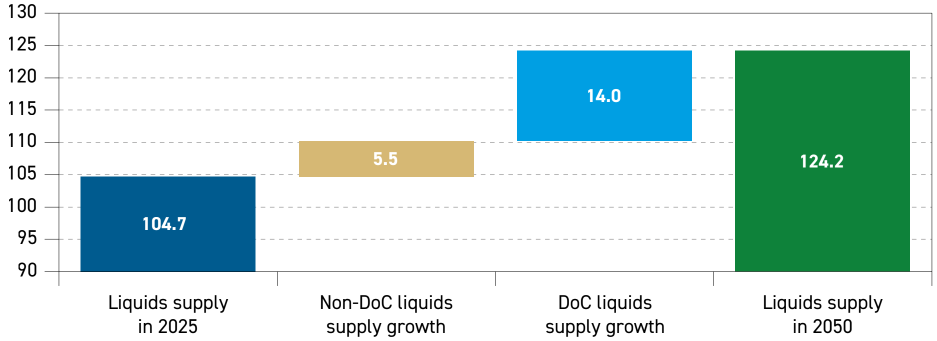
4.1 Global liquids supply outlook

The liquids supply outlook is shaped by the dual imperatives of meeting continued demand growth and navigating persistent economic, fiscal and geopolitical uncertainty. DoC participants have played a central role in maintaining market stability through measured and proactive policy responses.

In the medium-term period, non-DoC supply is projected to grow from 54.2 mb/d in 2025 to 58.2 mb/d in 2030, or by about 4.1 mb/d. This is primarily driven by Brazil, Qatar, Argentina and Canada, as well as a number of smaller, newer producers in non-DoC Africa. Meanwhile, following a reassessment of US tight crude growth potential, the analysis indicates that US total liquids supply is set to expand by only 0.4 mb/d in this period, with output approaching a plateau around 2030.

Heading into the 2030s, non-DoC supply growth is projected to peak and experience a long plateau at around 60 mb/d. As a result, against the backdrop of continued growth in oil demand, the onus is on DoC liquids supply to keep increasing. Volumes are projected to rise from 50.6 mb/d in 2025 to 64.5 mb/d by 2050 (Figure 4.1).

Figure 4.1
Composition of global liquids supply growth, 2025–2050, mb/d



Source: OPEC.

4.2 Drivers of medium- and long-term liquids supply

By far the most significant new development on the liquids supply horizon is a projected US supply peak coming into view. Uncertain economics, a gradual exhaustion of prime acreage and companies prioritizing shareholder returns and capital discipline related to growth, have resulted in a reduction in US tight crude supply investment, in particular. Medium-term growth in other US supply areas, including most notably in unconventional natural gas liquids (NGLs), but also modest increases in offshore and Alaskan crude, as well as biofuels, results in total US supply peaking around 2030.

As a result, while the trajectory for medium-term oil demand is not so different from previous editions of this Outlook, the relative importance of DoC vs. non-DoC producers, and the composition of sources of non-DoC liquids supply growth in this period, have changed. For example, while the US was seen as the primary contributor of supply growth in the WOO 2025's medium-term period from 2024–2030 (adding nearly 25% of non-DoC growth), in this year's Outlook, its contribution shrinks to a mere 9%, or 0.4 mb/d of 4.1 mb/d (Table 4.1).

Table 4.1
Long-term global liquids supply outlook, mb/d

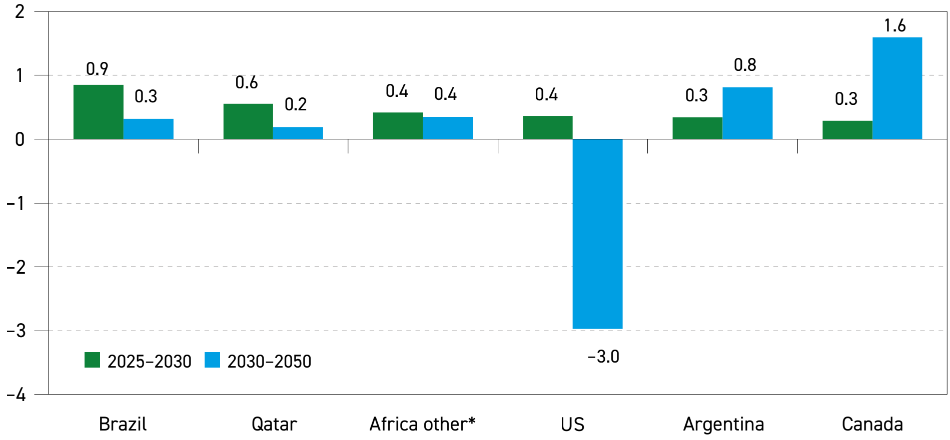
	2025	2030	2035	2040	2045	2050	Change 2025–2050
US	22.2	22.6	22.0	21.1	20.3	19.6	–2.6
of which: tight oil	*15.1*	*15.7*	*15.9*	*15.7*	*15.3*	*14.8*	*–0.3*
Canada	6.1	6.4	6.7	7.0	7.4	8.0	1.9
of which: oil sands	*3.4*	*3.7*	*3.9*	*4.2*	*4.7*	*5.3*	*1.9*
OECD Americas	28.3	28.9	28.7	28.1	27.7	27.6	–0.7
OECD Europe	3.6	3.7	3.7	3.8	3.8	3.9	0.3
OECD Asia-Pacific	0.4	0.4	0.4	0.4	0.4	0.5	0.1
Latin America	7.5	9.6	11.0	11.5	11.7	11.6	4.0
Middle East	2.0	2.6	2.8	2.9	2.9	3.0	1.0
Africa	2.3	2.8	3.0	3.1	3.1	3.0	0.7
China	4.6	4.6	4.5	4.5	4.5	4.4	–0.3
India	0.8	0.8	0.8	0.9	0.8	0.8	0.0
Other Asia	1.6	1.6	1.5	1.4	1.4	1.3	–0.3
Other Eurasia	0.4	0.3	0.3	0.3	0.3	0.3	–0.1
Other Europe	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Processing gains	2.5	2.8	3.0	3.2	3.3	3.4	0.8
Non-DoC liquids	**54.2**	**58.2**	**59.9**	**60.1**	**60.0**	**59.6**	**5.5**
DoC liquids	**50.6**	**55.3**	**59.3**	**61.6**	**63.2**	**64.5**	**14.0**
World	**104.7**	**113.6**	**119.1**	**121.7**	**123.2**	**124.2**	**19.5**

Source: OPEC.

Meanwhile, there are some bright spots elsewhere for non-DoC supply. In sub-Saharan Africa, numerous countries offshore in West and Southwest Africa, as well as Uganda and Mozambique in the east, have had a string of major discoveries, and in many cases will bringing major new fields onstream in the coming years (see Chapter 8 for more). Latin American supply, driven by Brazil and Argentina, among others, is projected to remain the greatest regional contributor to non-DoC liquids supply growth in both the medium and long term (Figure 4.2).

What has not changed compared to previous Outlooks, despite the downward-revised view on the US, is that non-DoC liquids supply essentially peaks and plateaus from the early 2030s, namely at around 60 mb/d (Figure 4.3). With demand growth to continue well into the 2040s, this means that the relative role of DoC liquids supply increases, rising from 50.6 mb/d in 2025 to 64.5 mb/d by 2050. This implies an increase in market share from 48% in 2025 to 52% in 2050. It further underlines the significance of the DoC in providing stability and reassurance to global energy markets.

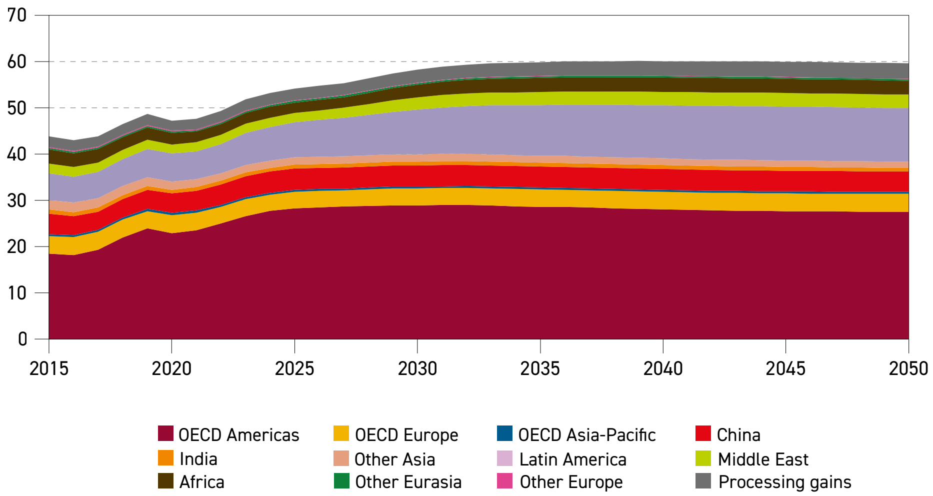
Figure 4.2
Select contributors to non-DoC total liquids production change, 2025–2030 and 2030–2050, mb/d



* *Africa other includes Cameroon, the Ivory Coast, Namibia, Niger, Senegal, Uganda and smaller producers, and excludes African DoC producers.*

Source: OPEC.

Figure 4.3
Long-term non-DoC liquids supply outlook, mb/d

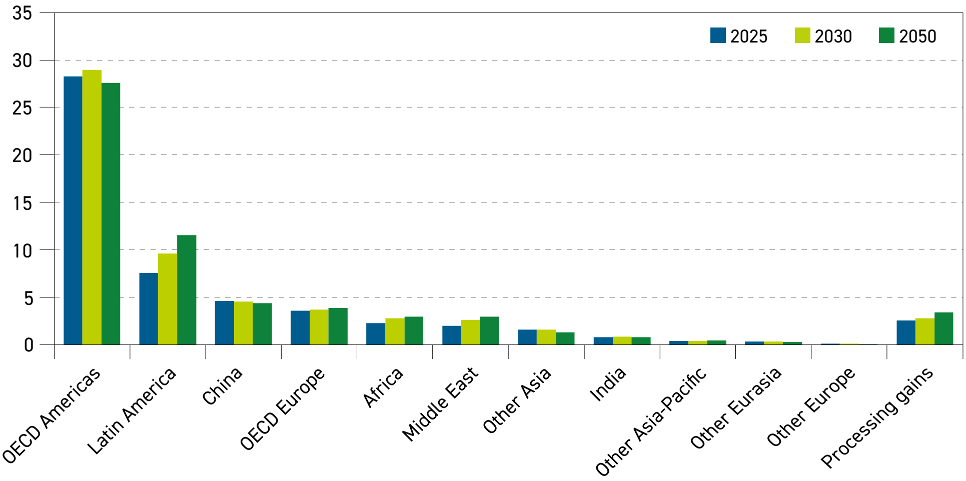


Source: OPEC.

4.3 Breakdown of liquids supply outlook by main regions

Regionally, as in previous Outlooks, Latin America is the most important contributor to both medium-term and long-term non-DoC liquids supply growth. In the medium term, liquids supply from the region is projected to expand from 7.5 mb/d in 2025 to 9.6 mb/d in 2030, making up just over 50% of the total incremental supply from non-DoC countries. Major contributors to this increase are Brazil, Argentina and others. Beyond the medium term, supply from Latin America is projected to continue to grow, rising to 11.6 mb/d by 2050 (Figure 4.4).

Figure 4.4
Non-DoC supply outlook by region, mb/d



Source: OPEC.

Other regional contributors to medium-term non-DoC supply growth are the OECD Americas (both the US and Canada; Mexico is included in the DoC grouping), the Middle East and Africa, each with around 15% of total non-DoC supply growth in this period. Growth elsewhere is essentially flat, with minimal declines in China and Other Asia.

Viewed over the entire 2025–2050 period, non-DoC Latin America's contribution is even more striking, adding 4 mb/d, or nearly 75% of total non-DoC incremental supply. The Middle East, Africa and OECD Europe also add volumes. By contrast, OECD Americas and to a lesser extent Other Asia and China, are expected to experience a supply decline.

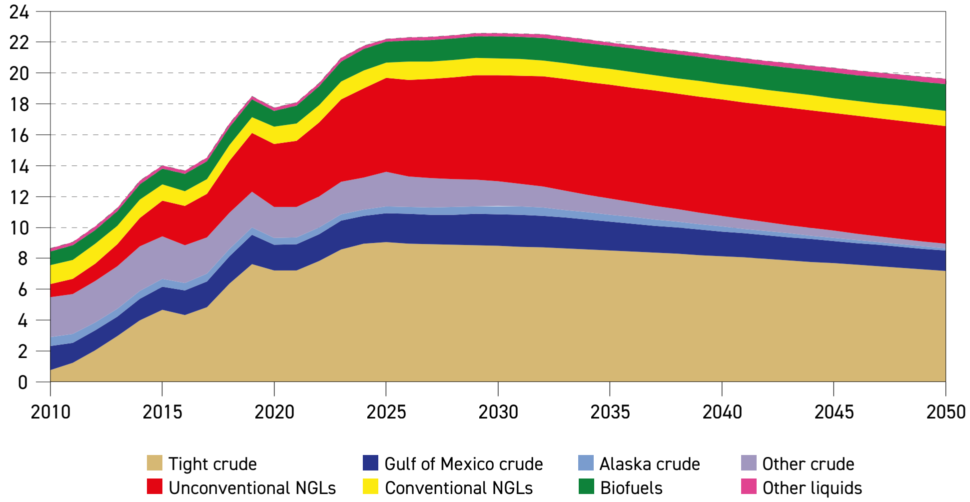
US

The US remains by far the world's largest producer of crude oil and other liquids. However, a reassessment of the outlook for US tight crude has led to this Outlook bringing its projected peak forward. US tight crude may now have peaked in 2025, at just over 9 mb/d, whereas the WOO 2025 saw continued growth to a peak of around 9.7 mb/d in 2030. This is largely due to a reassessment of the outlook for tight crude output from the Permian Basin and, to a lesser extent, the Niobrara and Eagle Ford (more on the details and outlook for US tight oil supply can be found in Section 4.4.1).

Overall, however, US liquids supply is set to keep increasing until 2030, with continued growth in NGLs supply, as well as a modest projected rise in biofuels. While total crude supply declines due to falling tight crude output, this is partly offset by increases in production from the Gulf of Mexico (GoM) and Alaska, even while conventional crude oil in most other onshore areas declines (Figure 4.5).

The US total liquids supply is set to rise from 22.2 mb/d in 2025 to 22.6 mb/d in 2030. Total crude supply is set to decline by 0.6 mb/d in this period, of which 0.3 mb/d is tight

Figure 4.5
US total liquids production outlook, mb/d



Source: OPEC.

crude. Supply from the offshore GoM is projected to rise from 1.9 mb/d to 2.1 mb/d by 2030, which would represent a new all-time high (and also a peak). With comparatively attractive economics, also *vis-à-vis* maturing onshore tight crude, and a spate of recent discoveries and ample infrastructure in place upon which to draw, the area is experiencing a continued renaissance.

Major fields that came onstream recently and will continue to ramp up output include the 75 thousand barrels per day (tb/d) Ballymore, 100 tb/d Shenandoah, 80 tb/d Whale and 60 tb/d Salamanca projects. New fields slated to start production in the medium term include BP's 100 tb/d Kaskida field, due to come online in 2028. In an increasingly common trend, and in an effort to streamline costs, the company has also copied Kaskida's design for a new standalone platform, Tiber-Guadalupe, with 80 tb/d due to come onstream in 2029. The project includes six wells in the Tiber field and a two-well tieback from the Guadalupe field. Combined, the fields are estimated to have recoverable resources of around 350 mboe from the initial phase, but additional wells could be drilled in future phases. Tiber-Guadalupe will be BP's second project in the GoM to produce oil from an ultra-high-pressure reservoir, another development likely to help unlock future resources.

In November 2025, the US government announced a five-year offshore leasing programme for 2026–2031, to replace the previous administration's schedule. Ambitiously, the new scheme was to include 34 lease sales through 2031, with some 1.27 billion acres being offered – compared to a previous programme of just three lease sales until 2029.

Importantly, the areas included would go well beyond the western GoM and existing producing areas in Alaska, and would include most of California, the eastern GoM, adjacent to Florida, and parts of the Chukchi and Beaufort Seas in Alaska.

All of this is part of the current administration's efforts to make more acreage available and streamline the permitting processes. Indeed, in a first sale held in December 2025, for the GoM, major international oil companies (IOCs), including BP, Chevron and Shell, as well as US independents, showed great interest, with 30 companies putting in bids for 181 blocks. This also followed the current administration slashing royalty rates.

Other lease sales may follow; however, outside of the core western GoM areas, local opposition is almost guaranteed. For the eastern GoM, the administration had already backed down from its original proposal, limiting areas on offer to only part of the new, eastern area that would be opened up. This sub-section would provide a 100-mile buffer from the coast of western Florida, where the tourism industry, among others, has declared its opposition.

In California, which has not seen any new leases since the 1980s, when local resistance emerged following a severe oil spill in the late 1960s, it is considered unlikely that the proposed six offshore sales scheduled for 2027–2030 will be realized. The Atlantic Coast was not included in the proposed new lease sale programme, after considerable criticism when it was first floated.

As a result, outside of the GoM and Alaska, which is expected to see something of a revival in production in the medium term, most importantly with the start-up of the large 180 tb/d Willow field, scheduled for around 2029, projections indicate a continued steady decline in other conventional crude production.

The picture for US NGLs is brighter. From 7.1 mb/d in 2025, an all-time high, total volumes are projected to rise to 8.0 mb/d in 2030 and keep rising modestly thereafter. Supply then plateaus at about 8.6 mb/d from around the early 2040s. Growth is overwhelmingly driven by unconventional NGLs from shale and tight plays, which already make up around 85% of total NGLs volumes. Unconventional NGLs supply growth, in turn, is essentially driven by expanding natural gas supply for both domestic consumption and LNG exports, as well as a trend towards 'gassier' oil from shale plays. Based on prevailing geopolitical trends, shifting trade flows and additional potential demand from data centres, among other factors, there is a potential upside to the natural gas supply outlook, and hence to NGLs supply.

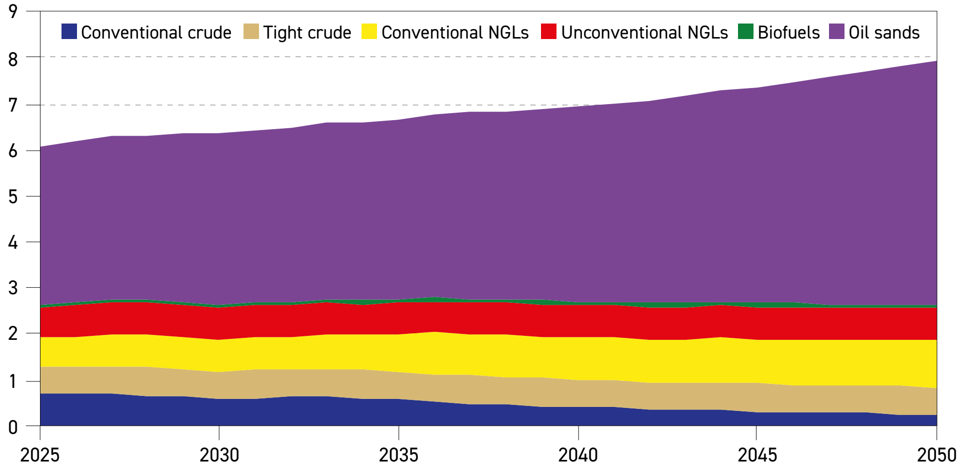
US biofuels, meanwhile, are projected to increase from 1.3 mb/d in 2025 to 1.7 mb/d in 2050, while other liquids – kerogen, methyl tert-butyl ether (MTBE) and some synthetic aviation fuel – are set to see a modest increase from 0.2 mb/d in 2025 to 0.3 mb/d in 2050.

Canada

Canada's liquids supply makes the strongest contribution to non-DoC totals over the long term, characterized by steady, slow growth. The country's total liquids supply is expected to increase from 6.1 mb/d in 2025 to 6.4 mb/d in 2030 and further to 8 mb/d by 2050 (Figure 4.6). Its huge resource base, attractive long-term economics, supportive political climate, growing offtake capacity and the market for medium and heavy sour crude grades (not least, its large neighbour to the south), make for major incentives for continued growth.

Liquids supply growth is overwhelmingly centred on oil sands, which in total increase from 3.4 mb/d in 2025 (raw bitumen and upgraded 'synthetic' crude combined) to 3.7 mb/d in 2030 and 5.3 mb/d by 2050.

Figure 4.6
Canada total liquids production outlook, mb/d



Source: OPEC.

Of note, strong demand for Canada's crude grades – which many Midwestern and Gulf refineries in the US are configured to take – as well as increased pipeline export capacity, have helped to boost exports and foster continued supply growth. Many of the larger oil sands producers report continued advances in maintaining low breakeven levels, with the potential to reduce them even further. Technical efficiency gains in production also continue to be achieved – for example, through the increased use of solvents in bitumen production, in combination with steam injection.

Perhaps most importantly, however, export capacities have been increased, while further expansion plans are also under development. The Trans Mountain pipeline expansion, which came online in 2025, has seen exports to US West Coast refineries and to Asian buyers increase meaningfully. The pipeline's capacity was increased from 300 tb/d to 890 tb/d. Now its operators are considering boosting capacity further, with another 85–90 tb/d mooted by using drag-reducing agents. A further expansion by 2030, perhaps adding another 300 tb/d, is also being considered. Given the continued risk of heightened trade volatility, additional outlets that enable exports outside of North America are increasingly attractive.

For example, in late 2025, plans for a 1 mb/d crude pipeline from near Edmonton to the Port of Prince Rupert in northern British Columbia were floated. This would represent a more northerly route than the Trans Mountain pipeline, which ends further south near Vancouver. It is, however, quite similar to the route proposed for the Northern Gateway pipeline, plans for which were scrapped in 2016. To this end, the federal government signed a Memorandum of Understanding with Alberta, also in late 2025, with a view to commencing construction in 2029.

Obstacles to this project remain, however, as a tanker ban in northern British Columbia's waters would have to be lifted. The federal government also insists that the pipeline is privately funded, which has attracted criticism from oil companies. By contrast, the Trans Mountain expansion only moved forward when the federal government bought it in 2018. Indeed, the federal government has suggested that it could designate the project as one of 'national interest', thus affording it preferential treatment in the approval process. It has also hinted that it could relax national oil- and gas-related emissions rules, even while it remains committed to the now rebranded Oil Sands Alliance CCUS project in Alberta (formerly called the Pathways Alliance).

Meanwhile, pipeline operator Enbridge has announced plans for a further expansion of its Mainline network leading south to the US, perhaps to boost it by another 150 tb/d over the coming years. Industry sources estimate that a list of smaller pipeline expansion plans could add up to nearly 1 mb/d by the mid-2030s, which might be sufficient even without the new pipeline to the west coast.

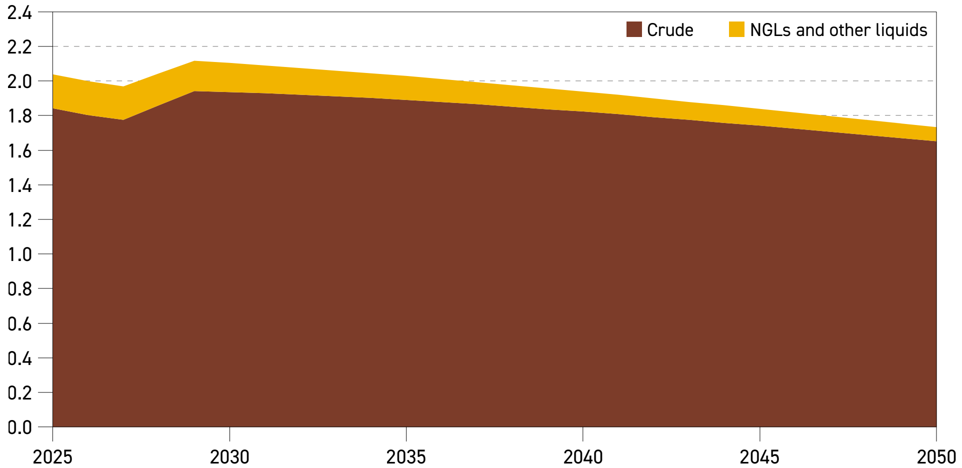
Outside of Alberta's oil sands, offshore Canada's East coast, the West White Rose project is set to come onstream in the first half of 2026, adding another 75 tb/d to conventional crude output. Meanwhile, the larger Bay du Nord field has been delayed again, with a final investment decision expected in around 2027, and oil to start flowing in 2031. Capacity is still envisaged to be 180 tb/d.

Nevertheless, Canada's crude supply is set to decline gradually, falling from 1.3 mb/d in 2025 to 0.9 mb/d in 2050. By contrast, and similar to the US, NGLs supply is projected to increase steadily, rising from 1.3 mb/d in 2025 to 1.4 mb/d in 2030 and further to 1.7 mb/d in 2050.

Norway

Norway's liquids supply is projected to keep growing in the medium term, but its relative contribution to incremental non-DoC production in this period is diminished. The country's total supply is projected to rise from 2 mb/d in 2025 to 2.1 mb/d in 2030 (Figure 4.7). Several smaller tie-backs are expected to boost total capacity, but the single largest contribution in this period comes from the large Yggdrasil cluster that is being developed and expanded by Aker-BP. It is the largest new development, following the start-up in 2025 of the 200 tb/d Johan Castberg field, and previously, the super-giant Johan Sverdrup field, which is now producing around 750 tb/d.

Figure 4.7
Norway total liquids production outlook, mb/d



Source: OPEC.

Regarding the latter, in 2025, Equinor and partners sanctioned the third phase of Johan Sverdrup, which aims to increase recoverable volumes and extend the production capacity life of the field. The goal is to raise recovery rates from 66% at present, more than the already-high Norwegian average of 47%, to as high as 75%. The companies that are active in Norway's waters remain among the most advanced in terms of utilizing Enhanced Oil Recovery (EOR), as well as other techniques to augment field lifetimes and boost recovery rates.

With the help of such techniques, Norway's production is projected to experience a long, very slow decline throughout the 2030s and 2040s. In the absence of major new discoveries, Norwegian production is projected to fall to 1.7 mb/d by 2050.

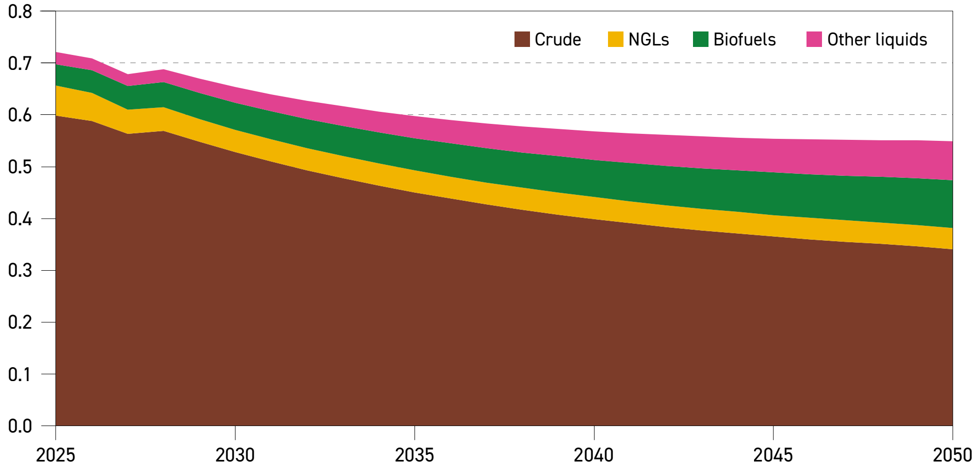
UK

As highlighted in previous Outlooks, the prospects for UK liquids supply are not overly positive, as seen in Figure 4.8. The current government has kept in place the Energy Profits Levy (EPL), the so-called 'windfall tax' introduced by the previous government in 2022, which most companies active in the UK consider to be prohibitive to developing new oil and gas assets.

In theory, smaller developments tied back to existing field infrastructure should still be encouraged, but the development of greenfields has largely dried up for the time being (the EPL runs until 2030 but may well be reviewed again beforehand). One major exception is the Rosebank field, which, after numerous delays, is set to see production start in 2027 and should add around 65 tb/d.

The situation in the UK has also led to significant consolidation, aimed at improving operating efficiencies and reducing costs. For instance, in 2025, TotalEnergies merged its UK assets with the Neo Next group (including Repsol, among others), to form the Neo Next+ joint venture, which as a result is set to become the UK North Sea's largest player. Previously, Eni combined its UK offshore assets with those of Ithaca Energy, and Shell and Equinor partnered in a joint venture called 'Adura'.

Figure 4.8
UK total liquids production outlook, mb/d



Source: OPEC.

UK total liquids supply is projected to stay flat at 0.7 mb/d in the 2025–2030 period, as despite crude oil output shrinking, this is largely offset by rising volumes of biofuels and synthetic aviation fuel. In the long term, however, UK liquids supply is set to decline to 0.5 mb/d by 2050.

Brazil

Brazil's liquids supply is set to continue its expansion and is projected to be the second-largest contributor to incremental non-DoC supply over the 2025–2050 period. Based on its impressive list of pre-salt, ultra-deepwater projects, crude supply is set to grow from 3.7 mb/d in 2025 to 4.4 mb/d in 2030 (Figure 4.9). Several major floating production, storage and offloading (FPSO) vessels started operations in 2025, with volumes still ramping up in some instances. These include the Equinor-operated 220 tb/d Bacalhau, stages six and seven of the huge Buzios field, each with 180 tb/d nameplate capacity, and the 180 tb/d Mero 4 project. All are operated by Petrobras.

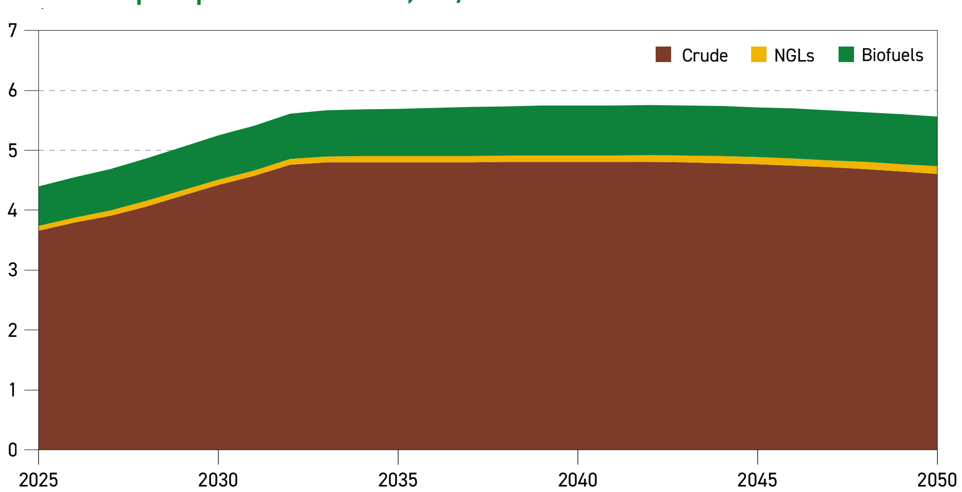
Medium-term supply growth in turn will be supported by the expected start-up of Buzios stages 8–11, each with around 200 tb/d capacity, Atapu-2 with 225 tb/d, ExxonMobil's Bacalhau Norte with 220 tb/d, Shell's 100 tb/d Gato de Mato, and the 225 tb/d Sepia-2 in 2030, among others. In total, nearly 2 mb/d of gross new crude production capacity is due to come online from 2026–2030, reflecting the scale and productivity of the pre-salt resource base.

Meanwhile, in August 2025, BP announced a huge find in the Bumerangue Block in ultra-deep waters in the Santos Basin. It was reportedly the company's largest discovery in 25 years, since Shah Deniz in the Caspian. The well was drilled around 400 km offshore, in water depths of 2.4 km, with the total well extending to 5.9 km. The discovery is estimated to have as much as eight billion barrels of liquids in place.

4

Frontier areas elsewhere in Brazil also remain in focus. In northeastern Brazil, the Foz de Amazonas area attracted many bids in a first round held in the summer of 2025. Petrobras, in particular, has shown great interest and is in the process of drilling first exploratory wells,

Figure 4.9
Brazil total liquids production outlook, mb/d



Source: OPEC.

including the so-called 'Norte de Amapa-1' or 'Morpho-1' well. The Foz de Amazonas region is considered one of Brazil's most promising untapped frontier areas, with geological proximity to the Guyana-Suriname Basin.

Until significant new resources are tapped, Brazil's potential for supply growth beyond the medium term is more limited. Nonetheless, the country is still set to see crude supply rise to a plateau of 4.8 mb/d. Combined with some small NGLs volumes and a limited increase in ethanol output, total Brazilian liquids supply is expected to rise to a peak of around 5.8 mb/d in the early 2040s, before declining again modestly to 5.6 mb/d in 2050.

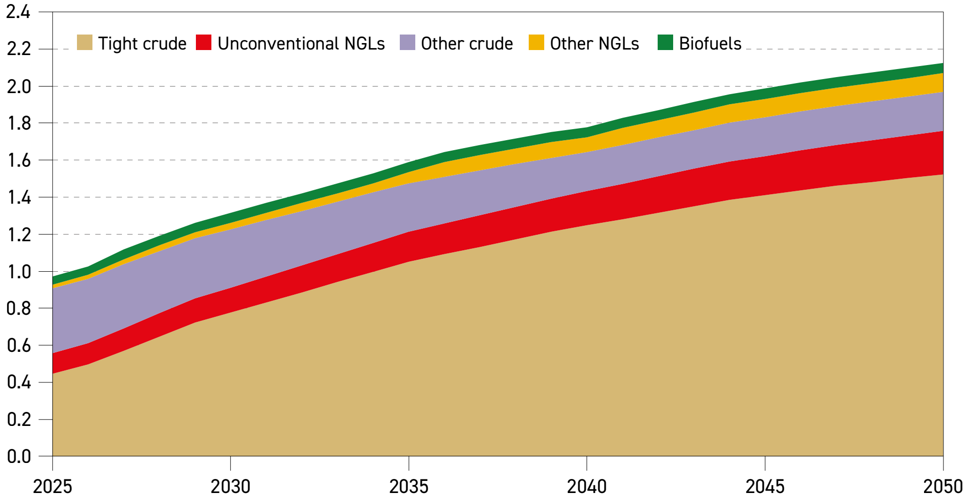
Argentina

Argentina continues to emerge as a significant contributor to non-DoC liquids supply growth. Tight crude production growth in the Vaca Muerta in the Neuquén Basin remains the growth driver, and if anything, exceeded expectations in 2025. Total liquids supply is set to rise from 1 mb/d in 2025 to 1.3 mb/d in 2030 (Figure 4.10). Crucially, offtake capacity from the landlocked basin is being increased. In 2025 and 2026, capacity on the existing Oldelval pipeline is being raised by 200 tb/d to 540 tb/d, with potential for a further increase to 700 tb/d.

From 2026–2029, multiple stages of the Vaca Muerta Oil Sur (VMOS) crude pipeline could add another 700 tb/d in total. This pipeline will run to the Punta Colorada terminal on the Atlantic Coast, adding 180 tb/d in a first stage in 2026, increasing to 550 tb/d in 2027, with capacity reportedly rising to the full 700 tb/d through the addition of pumping stations.

Besides much-needed offtake and export capacity, companies active in the Vaca Muerta continue to learn and apply lessons learned in US tight oil, with faster drilling, better proppants and longer laterals all serving to improve efficiencies and reduce costs. Improved economic and fiscal stability also serve to underpin continued growth in Argentina's liquids supply. Beyond the medium term, projections are for continued growth to 2.1 mb/d by 2050.

Figure 4.10
Argentina total liquids production outlook, mb/d



Source: OPEC.

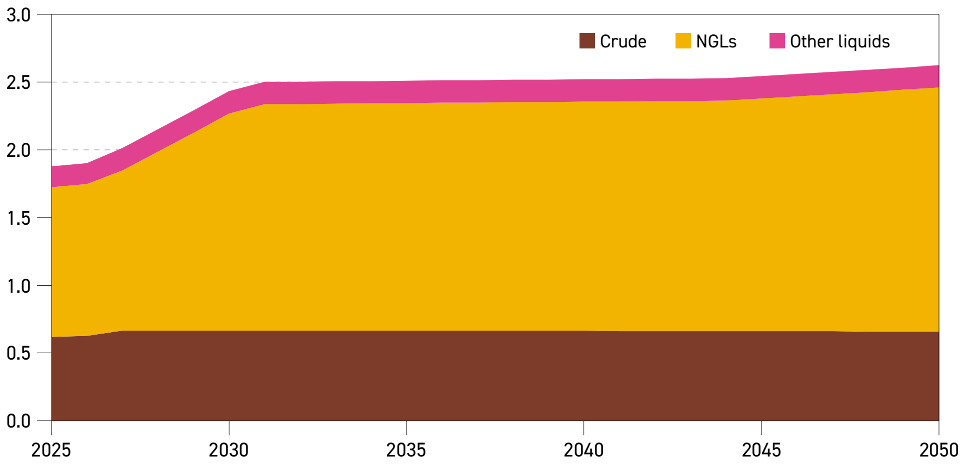
Other Latin America

In other Latin America, Suriname is on track to become a new producer when the TotalEnergies-operated 220 tb/d GranMorgu project starts up in 2028. This will likely be followed by other projects, if all goes well. In the Falkland Islands, after many years of deliberation, the Sea Lion oil field was given the go-ahead in late 2025. Projected to see 50 tb/d of production, it is set to come onstream from 2028 and use an FPSO to target some 170 mb of resources. First oil was discovered offshore in the Falkland Islands in 2010, and other discoveries exist, including the Casper and Darwin assets. Successful delivery of the first phase of Sea Lion could lead to the development of a second phase, targeting another 150 mb, and potentially make the development of other assets more likely.

Qatar

In Qatar, the major expansion of the country's natural gas and LNG capacity, centred on the huge North Field, has been delayed into 2028 from 2027. Nonetheless, a ramp-up in related liquids is projected to boost NGLs supply from 1.1 mb/d in 2025 to 1.6 mb/d in 2030. Crude production capacity is also set to increase modestly in this period, rising to 0.7 mb/d, as output at the Al-Shaheen field increases to 380 tb/d. In sum, total liquids production is projected to rise from 1.9 mb/d in 2025 to 2.4 mb/d in 2030, and creep up to 2.6 mb/d by 2050 (Figure 4.11).

Figure 4.11
Qatar total liquids production outlook, mb/d



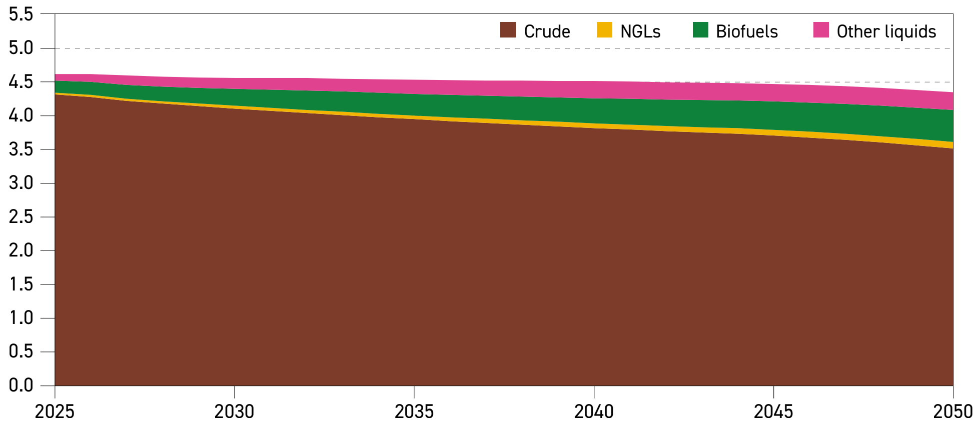
Source: OPEC.

China

Following a concerted campaign to boost domestic oil and gas production that was set in place in 2019 ('Seven-Year Action Plan to Enhance Oil and Gas Exploration'), and which included significant tax breaks and other incentives, Chinese liquids supply reversed its previous decline and expanded from 4.1 mb/d in 2019 to 4.6 mb/d in 2025. The country's three oil majors brought onstream new production in far western basins like Xinjiang and Ordos, as well as offshore, mostly in the Bohai Bay area. At the same time, EOR techniques continue to be deployed and refined on some of the country's long-established large, mature fields, including Daqing, Shengli, Liaohe and Henan.

This Outlook's projections suggest that China will achieve a stabilization in liquids supply, with output steady at 4.6 mb/d throughout the medium-term period. It then follows a long, tailing-off plateau thereafter. However, this masks the fact that a projected decline in crude oil output is partly offset by assumed growth in unconventionals, including modest increases in CTLs, as well as synthetic aviation fuel, with combined output rising from 0.2 mb/d in 2025 to 0.5 mb/d in 2050. Biofuels are also set to increase from 0.1 mb/d in 2025 to 0.3 mb/d in 2050. As such, Chinese total liquids supply is expected to decline only minimally in the long term, falling from 4.6 mb/d in the medium-term period to 4.4 mb/d in 2050 (Figure 4.12).

Figure 4.12
China total liquids production outlook, mb/d

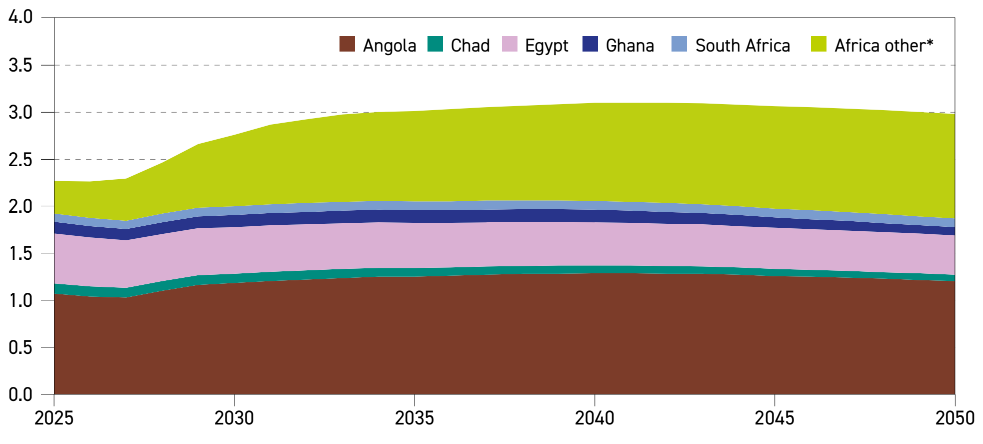


Source: OPEC.

Non-DoC Africa

Liquids supply from non-DoC African countries is projected to rise from 2.3 mb/d in 2025 to 2.8 mb/d in 2030. Output is set to grow in both established producers, such as Angola, and

Figure 4.13
Non-DoC Africa total liquids production outlook, mb/d



* *Africa other includes Cameroon, the Ivory Coast, Namibia, Niger, Senegal, Uganda and smaller producers, and excludes African DoC producers*

Source: OPEC.

a range of smaller ones including Ghana, the Ivory Coast and Senegal. Some new producers will also emerge in the coming years, including Uganda, Mozambique and Namibia. Looking at the long-term horizon, output is set to increase modestly, reaching a peak of 3.1 mb/d from the mid-2030s before declining minimally by 2050 to average 3.0 mb/d (Figure 4.13). (For more details on this region and relevant supply trends, see Chapter 8).

4.4 Breakdown of liquids supply by type of liquids

When breaking down the outlook for total non-DoC supply into different liquid types, trends seen in past Outlooks are unchanged. Of medium-term non-DoC supply growth projected at 4.1 mb/d, around one-third each is expected to come from incremental crude, NGLs and other liquids supply (including processing gains). In the long term, however, as US tight crude and other crude sources go into decline, of the total projected increase in non-DoC supply, growth is driven by NGLs and other liquids (+2.7 mb/d each), biofuels (+2.0 mb/d) and refinery processing gains (+0.8 mb/d). Meanwhile, crude supply is set to decline by 2.8 mb/d over the entire outlook period (Table 4.2).

Table 4.2
Non-DoC liquids supply outlook by type, mb/d

	2025	2030	2035	2040	2045	2050	Change 2025–2050
Crude	33.0	34.4	34.4	33.2	31.9	30.2	–2.8
NGLs	11.2	12.8	13.4	13.7	13.8	13.9	2.7
Biofuels	3.1	3.6	4.1	4.5	4.8	5.1	2.0
Other liquids	4.3	4.7	5.1	5.6	6.2	7.0	2.7
Global refinery processing gains	2.5	2.8	3.0	3.2	3.3	3.4	0.8
Total non-DoC liquids	**54.2**	**58.2**	**59.9**	**60.1**	**60.0**	**59.6**	**5.5**

Source: OPEC.

4.4.1 Non-DoC tight oil: US and other developments

With US tight crude now likely to have peaked, the focus in tight oil is increasingly shifting towards the potential in Argentina. Of total projected long-term supply growth of 1.1 mb/d in non-DoC, Argentina is estimated to make up 1.2 mb/d, with strong output gains seen in the Vaca Muerta formation in the Neuquén Basin. Added to assumed Chinese tight oil growth of 0.2 mb/d, this more than offsets a decline of 0.3 mb/d in the US over the period until 2050 (Table 4.3). Tight oil production elsewhere, such as in Canada and other smaller producers, is expected to be flat in the long term. Meanwhile, the pattern seen in total non-DoC crude and NGLs is mirrored, with growth in unconventional NGLs of 1.8 mb/d by 2050 partly offset by a decline in non-DoC tight crude of 0.6 mb/d.

Exploration activities and technology development continue in various regions around the world. While it has long been known that very large resource concentrations of tight oil exist in numerous countries, development has only taken place in a handful, a situation that may change in the future.

US tight crude, after recovering from a post-pandemic dip of 7.2 mb/d (2020 average), may now have peaked in 2025 at just over 9.0 mb/d. In reality, this is driven by a peak in the

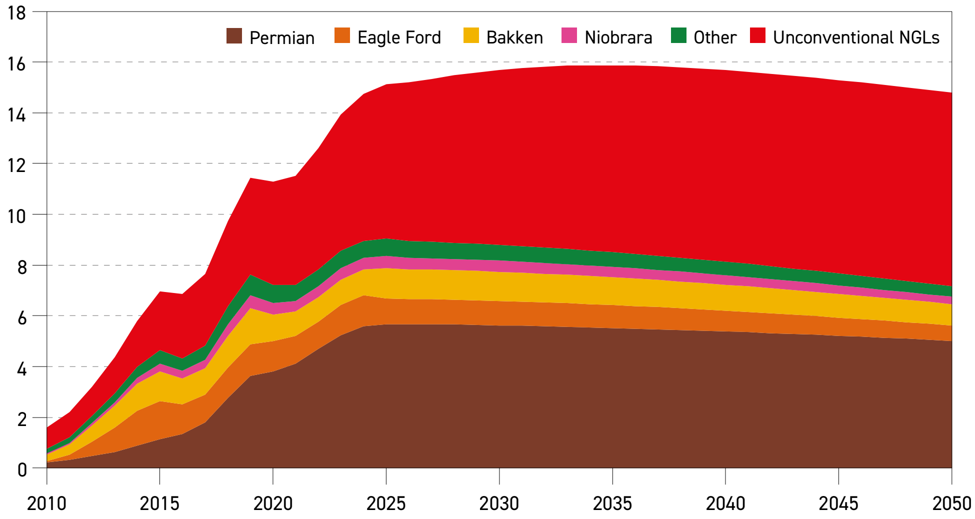
Table 4.3
Non-DoC tight oil production outlook, mb/d

	2025	2030	2035	2040	2045	2050	Change 2025–2050
US	15.1	15.7	15.9	15.7	15.3	14.8	–0.3
Canada	1.3	1.3	1.3	1.3	1.3	1.3	0.0
Argentina	0.6	0.9	1.2	1.4	1.6	1.8	1.2
China	0.3	0.3	0.4	0.4	0.4	0.5	0.2
Total tight oil	**17.3**	**18.2**	**18.8**	**18.8**	**18.7**	**18.4**	**1.1**

Source: OPEC.

Permian Basin, which straddles the Texas/New Mexico border and makes up nearly two-thirds of total US tight crude production. The other key basins, the Bakken, Eagle Ford and Niobrara, had already peaked in the years prior to the pandemic (Figure 4.14). The peak in production is not due to an exhaustion of resources, or a lack of innovation, but rather due to lower investment, more cautionary producer behaviour and consolidation among companies active in the US shale industry. In effect, shale is entering a new period of later-life maturity.

Figure 4.14
US tight oil production breakdown by source, mb/d



Source: OPEC.

As part of this evolution, efficiency gains continue to be made. According to findings by Rystad Energy, drilling pads continue to get larger, and the speed of drilling and fracking wells is still rising, even while ever-longer laterals are beginning to unlock Tier 3–4 acreage. By way of illustration, according to Rystad, an average pad held 4.7 wells in 2025, double the number in 2017. In addition, a typical rig can now drill twice as far as it could in 2017, while ultra-long laterals of more than 14,000 feet are increasingly common in large, contiguous acreage.

Consolidation among tight oil producers is also a big factor, but this cuts both ways. On the one hand, there are fewer smaller dynamic, risk-taking companies involved, who in the original days of shale reinvested a huge share of their cashflow in expanding production. On the other, the larger companies that now control the bulk of tight oil production have considerably more capital and are much less, or not at all, dependent upon outside capital to invest.

For instance, ExxonMobil, now one of the largest producers in the Permian, is targeting growth to 2.5 mboe/d by 2030, in part making use of some of its own in-house technologies. Exxon reportedly hopes to raise recovery rates from 10% to 20% in the coming years, among other things using patented lightweight proppants. By contrast, Chevron is curbing spending in the Permian in a shift towards maximizing cashflow. So, in a sense, consolidation is arguably leading to peak tight oil at a lower level, but a plateau sustained for longer – which is reflected in this Outlook. Consolidation also continues, with Devon Energy and Coterra Energy agreeing to merge in February of this year, thus forming a new shale heavyweight with around 2.0 mboe/d of combined production.

Meanwhile, the supply of US unconventional NGLs produced from tight and shale formations continues to grow. In the medium term, the supply of these barrels is projected to rise from 6.1 mb/d in 2025 to 6.9 mb/d in 2030. After slowing somewhat, it is then expected to rise to 7.6 mb/d by 2050. Growth remains driven by rising natural gas supply to meet rising domestic consumption and a steep expansion in LNG export capacity. As a result, total US NGLs is set to rise from 7.1 mb/d in 2025 to 8.6 mb/d in 2050.

While US tight oil will continue to make up the bulk of global supplies, at over 80%, Argentina sticks out as another huge success story (see details in Section 4.3 for more). From 0.6 mb/d in 2025, supply is projected to rise to 0.9 mb/d in 2030 and further to 1.8 mb/d by 2050. Canada, already the second-largest tight oil producer after the US, is projected to see flat supply throughout the forecast period at 1.3 mb/d. China, with a current supply of 0.3 mb/d, is estimated to see modest growth to 0.5 mb/d by 2050.

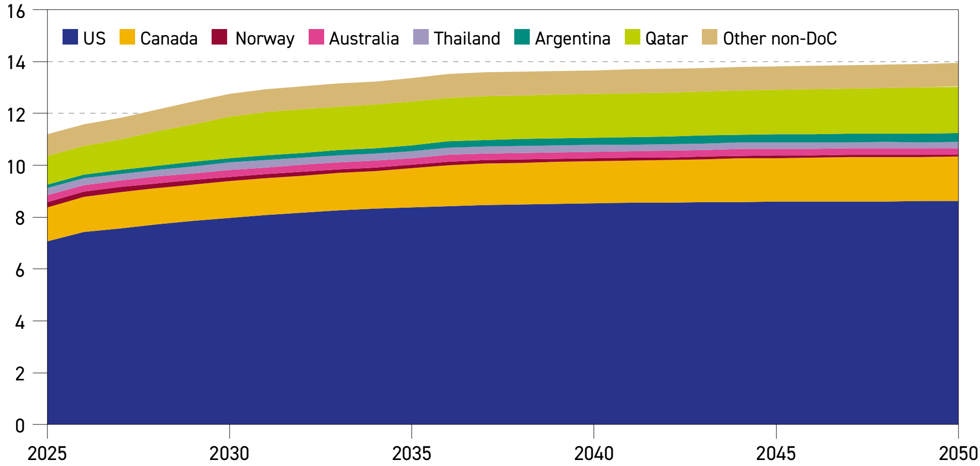
4.4.2 Other non-crude liquids supply

NGLs remain the most important part of non-DoC supply growth. From 11.2 mb/d in 2025, total volumes are projected to rise by 2.7 mb/d to 13.9 mb/d in 2050, even while crude oil supply declines. The US alone is expected to make up more than 50% of this, driven by strong growth in unconventional NGLs from shale formations. Qatar is set to add another 0.7 mb/d in the long term, with total supply rising to 1.8 mb/d, while Canada is expected see growth of 0.4 mb/d from 2025–2050, to reach 1.7 mb/d. These three countries make up the bulk of incremental non-DoC NGLs supply (Figure 4.15).

Biofuels supply in non-DoC countries is projected to rise from 3.1 mb/d in 2025 to 5.1 mb/d in 2050, with growth relatively evenly split between ethanol and biodiesel. Mandates and recycling requirements still drive growth in many countries, although overall volumes were revised slightly lower. Biojet, which is included in the biodiesel category, is projected to rise to a modest 0.2 mb/d by 2050, as costs remain very high (Table 4.4).

In 2025, the European Union's ReFuelEU Aviation rule came into force, mandating initially modest volumes of aviation fuel to be replaced by alternatives. In the EU too, the 'RED III' will gradually raise the targets for advanced biofuels in the transportation sector. Elsewhere, key

Figure 4.15
Non-DoC NGLs production outlook, mb/d



Source: OPEC.

Table 4.4
Long-term non-DoC biofuels and other liquids production outlook, mb/d

	2025	2030	2035	2040	2045	2050	Change 2025–2050
Fuel ethanol	1.9	2.1	2.4	2.6	2.7	2.8	0.9
Biodiesel	1.2	1.4	1.7	2.0	2.2	2.3	1.1
of which: biojet	*0.0*	*0.0*	*0.0*	*0.1*	*0.1*	*0.2*	*0.1*
Total biofuels	**3.1**	**3.6**	**4.1**	**4.5**	**4.8**	**5.1**	**2.0**
Canadian oil sands	3.4	3.7	3.9	4.2	4.7	5.3	1.9
Gas-to-liquids (GTL)	0.2	0.2	0.2	0.2	0.2	0.2	0.0
Coal-to-liquids (CTL)	0.2	0.3	0.3	0.3	0.3	0.3	0.1
Synthetic aviation fuel	0.0	0.1	0.3	0.5	0.6	0.8	0.8
Other*	0.4	0.4	0.3	0.3	0.3	0.3	–0.1
Total 'Other liquids'	**4.3**	**4.7**	**5.1**	**5.6**	**6.2**	**7.0**	**2.7**
Non-DoC total	**7.4**	**8.3**	**9.2**	**10.1**	**11.0**	**12.1**	**4.7**

* *Including kerogen, extra-heavy crude, MTBE and other refinery additives.*
Source: OPEC.

emerging consumers, India and Brazil, both expanded ethanol and biodiesel blending in their respective markets, while Indonesia raised biodiesel mandates.

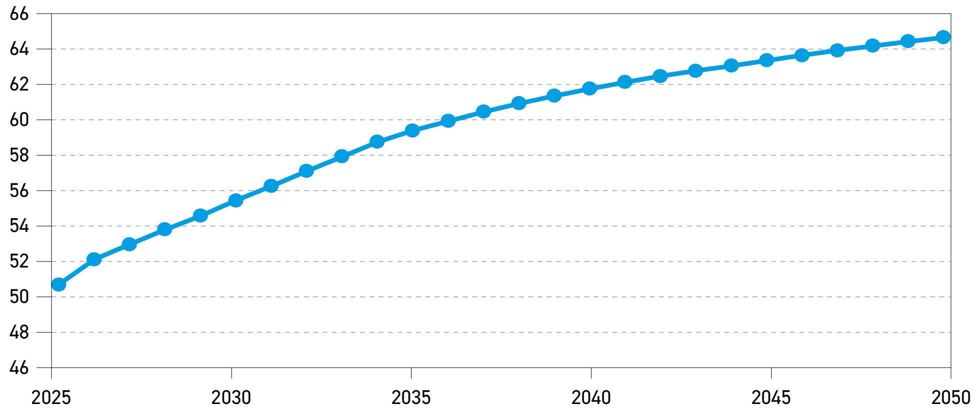
Other liquids in the non-DoC are projected to grow by a similar overall volume to biofuels, rising from 4.3 mb/d in 2025 to 7.0 mb/d in 2050. The bulk of this is projected growth in Canadian oil sands, which are expected to increase to 5.3 mb/d by the end of the Outlook horizon. Synthetic aviation fuel (included in 'other liquids', as opposed to biological-origin 'biojet', which is included in the biofuels category) is another source of growth, rising to 0.8 mb/d by 2050. Despite its still very high cost to produce, compared to oil-based aviation fuel, numerous countries now have mandates in place, and the assumption is that these will at least partially be fulfilled.

In the EU, based on the aforementioned ReFuelEU Aviation rule, 2% of aviation fuel must now be blended in, whether with biojet or synthetic fuel. This rises to 6% by 2030. The UK, the US and Japan all have a 10% mandate by 2030. In Asia, other major consumers, including China, India, South Korea and Singapore, all have mandates in place. Moreover, some airlines, including KLM, Air France, Delta and Japan Airlines have introduced voluntary targets, which in some cases exceed national mandates. In this Outlook, the assumption is that it is unlikely that all of these mandates will be achieved in all areas. This is not just due to political opposition, but also due to questions around cost, technology and the availability of feedstocks.

4.5 DoC liquids

As observed in previous Outlooks, with oil demand set to keep growing, and non-DoC liquids supply set to peak and plateau at around 60 mb/d from the early 2030s, the onus is on DoC producers to meet supply requirements in the long term. As a result, DoC total liquids, which averaged 50.6 mb/d in 2025, rises to 55.3 mb/d by 2030 and 64.5 mb/d by 2050 (Figure 4.16). This implies that the group's share of the global oil market is set to increase from 48% in 2025 to 52% in 2050.

Figure 4.16
DoC total liquids supply, mb/d



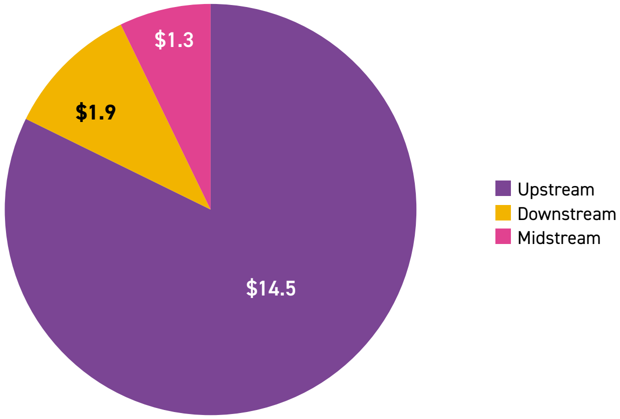
Source: OPEC.

4.6 Upstream investment requirements

Cumulative oil-related investment requirements to meet projected global demand add up to $17.7 trillion (US$2026) for the 2026–2050 period (Figure 4.17). This amount is slightly lower than projected in the WOO 2025, but the outlook horizon is also reduced by one year. In fact, for the upstream, investment requirements are adjusted upward slightly due to a higher projected supply in the long term, and a shift towards higher-cost resources. In particular, the lowered trajectory for US supply means that other non-DoC resources will be tapped, including, for instance, Canadian oil sands, but also oil from frontier areas, such as Argentina and new producers in Africa.

Upstream investment requirements are now assessed at $14.5 trillion, or $581 per annum, on average. Downstream needs are projected at $1.9 trillion, and midstream at $1.3 trillion. This means that average total per annum costs are now seen at $707 billion.

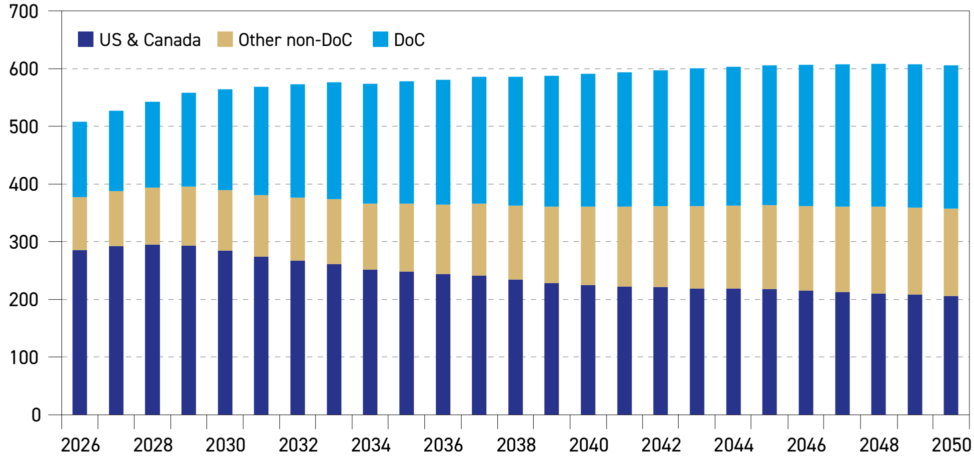
Figure 4.17
Cumulative oil-related investment requirements by segment, 2026–2050, $(2026) trillion



Source: OPEC.

Regionally, the long-term shift away from the US & Canada to both other non-DoC and DoC countries is more evident, given the re-evaluation of US supply perspectives. While investment requirements in 2026 in the US & Canada make up 56% of the total per annum cost, this shifts to only around 34% by the end of the outlook period. The respective shares of other non-DoC and DoC countries rise to 25% and 41%, respectively (Figure 4.18).

Figure 4.18
Annual upstream investment requirements, 2026–2050, $(2026) billion



Source: OPEC.

Refining outlook

Key takeaways

- In the medium term to 2030, around 4.9 mb/d of refining capacity additions are projected at the global level. Most of these are expected in developing countries and regions, led by India (1.2 mb/d), the Middle East (1 mb/d), China (1 mb/d), Africa (0.8 mb/d) and Other Asia-Pacific (0.6 mb/d). Only minor expansions are projected for other regions over the period.

- In the long term (2026–2050), global required refining capacity additions are projected at 19.3 mb/d, including debottlenecking expansion. Almost 80% of capacity additions will likely materialize before 2040, which is in line with global oil demand trends.

- Similar to the medium term, the majority of required long-term additions are set to be located in developing countries. The Asia-Pacific alone accounts for almost 60% of capacity additions, followed by Africa (16%) and the Middle East (10%).

- Relative to 2025, the global downstream market is likely to increasingly tighten over the medium-term horizon. The deficit between required and net potential refining capacity is set to steadily rise to more than 1.5 mb/d by 2030. This is due to strong oil demand growth, which is significantly higher compared to net refining capacity additions in the same period, especially in the Asia-Pacific.

- Refinery throughputs are expected to increase from around 83.5 mb/d in 2025 to 89.5 mb/d in 2030. Thereafter, growth is set to moderate, with global refinery runs climbing to nearly 96 mb/d in 2040 before reaching 97.2 mb/d by 2050. These global trends reflect diverging regional dynamics, with the US & Canada, Europe, and developed countries in the Asia-Pacific seeing declines. However, these are more than offset by robust increases across developing regions, including the Asia-Pacific, Africa, Latin America and the Middle East.

- As a result, global refinery utilization rates are expected to rise from around 80.8% in 2025 to almost 82.7% in 2030, supported by rising oil demand. They then start declining in the long term to settle at 79.5% by 2050, on the back of significant refining capacity additions.

- Announced refinery closures in the medium term are projected at just above 0.3 mb/d. Beyond 2030, further closures of around 4 mb/d will be required globally – mostly in developed regions – if refinery utilization rates are to be kept at sustainable levels.

- Secondary capacity additions over the outlook period are significant, with 20.7 mb/d of desulphurization, 10.9 mb/d of conversion capacities and 6.1 mb/d of gasoline upgrading units.

- In the long term, total required downstream sector investments are projected at almost $1.9 trillion. Nearly $600 billion will be required for refinery capacity additions and the expansion of existing units, while the remainder is for continuous maintenance and the replacement of existing capacity throughout the outlook period.

5

This chapter presents the downstream oil outlook for the period 2025–2050. It is fully consistent with the Outlook's projections on oil demand (Chapter 3) and liquids supply (Chapter 4). It examines various market drivers and factors that may impact the future global refining sector, underscoring challenges, uncertainties and opportunities. The analysis is conducted in two time frames, namely the medium term (2025–2030) and the long term (2030–2050). Additionally, in this year's Outlook, India is shown as a separate region due to its growing importance, meaning that Other Asia-Pacific now excludes both China and India.

Initially, the chapter focuses on recent downstream sector developments. This is then followed by an updated assessment of the current 'base' capacity for each major region that forms the foundation for all medium- and long-term projections. It is important to note that these projections are conducted according to two different methodologies.

Firstly, over the period of the medium term, refining capacity additions are assessed based on a thorough review of refinery projects and their progress by major downstream region. Based on this, the medium-term global and regional outlook for both primary and secondary capacity is developed. Secondly, based on global and regional oil demand and supply trends, long-term refining capacity additions (i.e. requirements for additions) are projected.

Additionally, the analysis in this chapter shows how the downstream market balance is expected to evolve in the medium and long term. This provides insights into regional market balances and respective utilization rates.

There are also discussions and forecasts for recent and near-term refinery closures. In the medium term, projections are based on announcements (firm closures) and an assessment of potential closures by 2030. Beyond 2030, the Outlook makes no explicit forecast for closures. It does, however, provide an indication of required refinery closures based on market balances and projected utilization rates.

This chapter also examines medium- and long-term secondary capacity additions. This includes projections for conversion capacities, i.e. FCC, coking and hydrocracking, as well as desulphurization capacity and octane units. Based on these secondary capacity additions, and the projected demand by product, the potential medium-term market balance is highlighted.

Finally, Chapter 5 also provides an outlook for global and regional investment requirements related to medium- and long-term capacity additions to the year 2050, as well as investments for continuous maintenance and replacement.

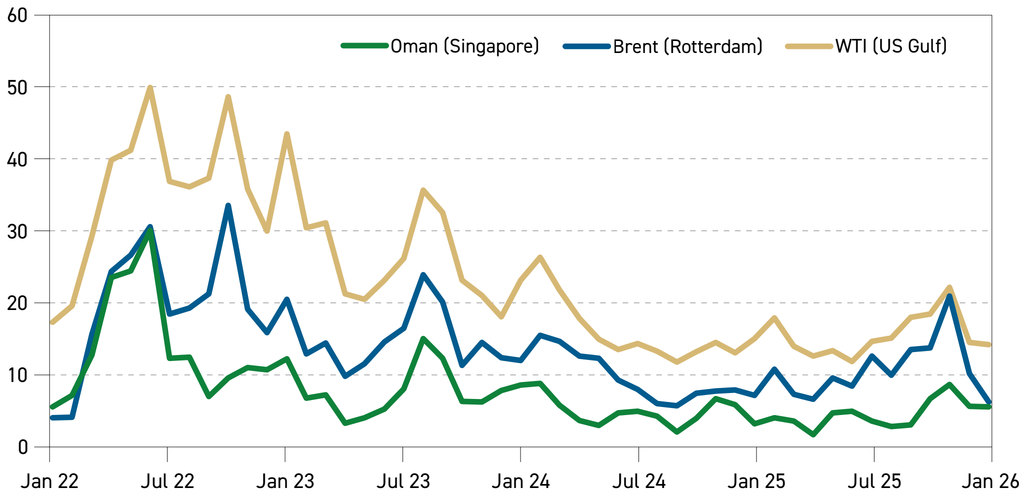
5.1 Existing refinery capacity

Recent developments in the downstream sector

The COVID-19 pandemic profoundly changed the downstream petroleum market. It moved from a relatively loose market when utilization rates reached their lowest level in ten years in 2019 to a depressed post-pandemic market, which became extremely tight in 2022 following the escalation of conflict in Eastern Europe.

The knock-on impact of this conflict was all-time high refining margins in 2022. For instance, US Gulf Coast refining margins against WTI, Rotterdam refining margins against Brent and Singapore's refining margins against Oman hit all-time high levels in June 2022 of \$50/b, \$31/b and \$30/b, respectively (Figure 5.1). This represents a clear illustration of the extreme tightness of the downstream market in mid-2022.

Figure 5.1
Selected refining margins, \$/b



Source: Argus Media, OPEC.

In early 2023, margins eased, partly due to rising utilization rates and a higher supply of products. Nevertheless, strong demand growth and continued refinery closures, primarily due to rising refinery costs (especially those related to energy and natural gas), as well as some long-lasting geopolitical challenges, supported margins and they remained higher relative to previous years.

In the second half of 2023, demand for transportation fuels increased significantly. It exceeded supply and stocks remained low. Middle distillates outperformed all other products, with a notable jump in their respective crack spreads. It led to profits per barrel jumping by about a third in Europe and the US. As a result, margins increased to \$36/b, \$24/b, \$15/b in the US, Europe and Asia, respectively, in August 2023.

Margins weakened throughout 2024 and 2025, as new supply arrived and growth in refined products demand eased (although it remained high relative to pre-pandemic levels). Several major new refineries ramped up during these two years, notably Dangote, Nigeria (650 tb/d), Al-Zour, Kuwait (615 tb/d), Dos Bocas, Mexico (340 tb/d), Duqm, Oman (approximately 230 tb/d,), as well as Yulong (400 tb/d) and Jieyong (200 tb/d) in China. The global demand growth slowdown was largely attributed to declining manufacturing activity in the US and slowing economic activity in China and Europe, and to some extent the increasing penetration of EVs and biofuels.

Margins rebounded again in late 2025 on the back of tightened product availability. Margins were tied to many factors, including geopolitical constraints, additional closures and unplanned end-of-year outages in Europe and the US. In November 2025, US Gulf Coast refining margins against WTI, Rotterdam refining margins against Brent and Singapore's refining margins against Oman reached \$22/b, \$21/b and \$8.7/b, respectively.

For 2026, global oil demand is set to increase by almost 1.4 mb/d, y-o-y, driven by the transportation sector, which will likely put additional pressure on the global refining system. While some relief should come from refining capacity additions, projections point to a tightening downstream market relative to 2025.

Base refinery capacity

This section provides a detailed update on base capacity assessments – distillation and secondary capacity, including condensate splitters – of refineries worldwide. It includes additions and expansions to existing refineries, new refineries that have come on stream, as well as closures that occurred in 2025.

Refineries, unless officially closed, are included in the database with their 'nameplate' capacity, although effective capacity may be identified as being well below nameplate level. Overall, it should be noted that no single data source for global and regional refinery capacities can be relied upon entirely. The quality and availability of capacity reporting varies by refinery, so there is always an element of determining a 'best estimate' for base capacity. This applies to primary capacity, secondary capacity, as well as new projects and closures.

Table 5.1 provides details on global base refining capacity by major downstream region, as well as by refining process. Total global distillation capacity was assessed at 103.3 mb/d as

Table 5.1
Global base refining capacity as of January 2026, mb/d

	US & Canada	Latin America	Africa	Europe	Russia & Other Eurasia	Middle East	China	India	Other Asia-Pacific	World
Distillation										
Crude oil (atmospheric)	19.2	8.2	4.3	13.8	8.5	11.6	17.9	5.2	14.6	103.3
Vacuum	9.0	3.6	1.1	5.9	3.4	3.3	7.6	2.2	3.4	39.5
Upgrading										
Coking	2.8	1.0	0.1	0.8	0.6	0.5	2.4	1.1	0.4	9.7
Catalytic cracking	5.9	1.7	0.4	2.1	1.0	1.1	4.5	1.1	2.4	20.1
Hydrocracking	2.6	0.2	0.4	2.1	1.1	1.3	2.6	0.8	1.1	12.1
Visbreaking	0.1	0.4	0.1	1.3	0.7	0.6	0.2	0.1	0.4	3.9
Solvent deasphalting	0.5	0.1	0.0	0.2	0.0	0.2	0.1	0.1	0.1	1.3
Octane units										
Reforming	4.0	0.7	0.6	2.2	0.9	1.4	2.5	0.5	2.1	14.9
Isomerization	0.8	0.2	0.1	0.6	0.3	0.5	0.2	0.1	0.3	3.2
Alkylation	1.4	0.2	0.1	0.2	0.1	0.1	0.2	0.1	0.3	2.7
Polymerization	0.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.1
MTBE/ETBE	0.0	0.0	0.0	0.1	0.0	0.0	0.3	0.0	0.0	0.6
Desulphurization										
Naphtha	5.0	0.9	0.6	2.7	1.0	2.1	2.5	0.3	2.5	17.6
Gasoline	3.1	0.5	0.2	0.7	0.3	0.4	1.5	0.2	1.1	8.0
Middle distillates	7.2	2.6	0.9	5.2	2.9	3.8	4.9	2.2	5.0	34.8
Heavy oil/Residual fuel	3.3	0.4	0.0	1.7	0.4	1.0	1.2	1.1	2.2	11.3

Source: OPEC.

of January 2026. This includes capacity additions and closures that occurred during 2025, as well as other necessary adjustments to the base capacity.

Close to 1.2 mb/d of new refining capacity came online in 2025. This mainly consisted of second-phase project completions at facilities that began operating a year earlier, as well as some major and minor capacity expansions at existing refineries. The non-OECD region accounts for almost the entirety of new additions, including the commissioning of the second phases of the Dangote, Dos Bocas and Yulong refineries.

Major expansions include Bapco's 135 tb/d Sitra refinery in Bahrain, the China National Offshore Oil Company's (CNOOC) 120 tb/d Daxie refinery in Ningbo, China, and Pertamina's 100 tb/d Balikpapan refinery in Indonesia. Furthermore, some smaller additions were recorded in Brazil, Georgia and Oman, and a minor expansion was seen at Chevron's Pasadena refinery in Texas, the US.

These additions were almost offset by capacity closures across the same year, predominantly in OECD countries. The total capacity decommissioned during 2025 is estimated at more than 1.1 mb/d.

In Europe, refinery rationalization in 2025 included the shutdown of the Grangemouth refinery in Scotland, where Petroineos permanently ceased crude processing of about 140 tb/d in April 2025. The site is now being converted into a fuel import terminal. In Germany, Shell and Holborn Europa Raffinera Gmbh closed crude distillation capacities at their Wesseling and Harburg refineries, respectively, as part of their conversion away from conventional fuels. In the US, LyondellBasell's Houston refinery completed its shutdown in February 2025, and Phillips 66's Wilmington refinery in Los Angeles ceased operations at the end of the year. In China, despite a period of broad industry rationalization taking place, aside from the shutdown of PetroChina's large 410 tb/d Dalian refinery in mid-2025, no other major permanent closures were widely reported across the year.

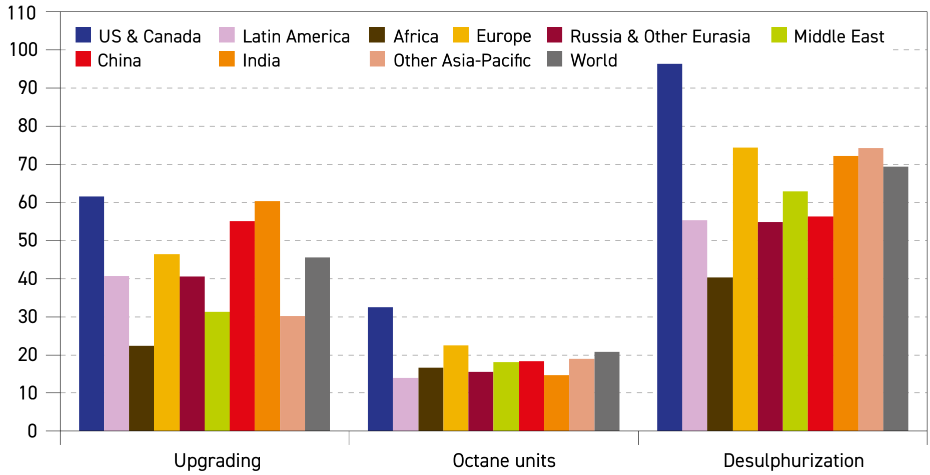
In addition to the capacity changes recorded in 2025, some adjustments were also made to the nameplate capacity database. This was to reflect commissioning or shutdowns from previous years that were officially confirmed in 2025; notably, Huaxing Petrochemical (120 tb/d) and Zhenghe Group (100 tb/d) refineries in Shandong province – formerly owned by Sinochem Group – were declared bankrupt by local courts in 2024 after prolonged weak margins and debt issues. They remained inactive through to the year-end. Following unsuccessful attempts to sell the assets, their permanent closure was confirmed in 2025 and they were consequently removed from the database.

Secondary capacity

Refining capacity analysis extends beyond primary crude distillation and often includes a range of secondary processing units. With the evolving product demand mix, the growing diversity of crude oil feedstocks, as well as environmental considerations and related regulations, refiners have been forced to change refinery configurations and adjust their strategies over time. Today, new refinery projects are mostly highly complex facilities that are capable of processing various types of crude and producing high-quality products. In addition, they are often integrated plants, which include petrochemical facilities and hydrogen production plants. Refinery complexity is typically assessed using the ratio of secondary processing capacity to atmospheric crude distillation capacity.

Figure 5.2 shows the ratio of existing secondary capacity to crude atmospheric distillation capacity as of January 2026. It highlights the complexity level of the refining sector at the global level and the variation between major regions. At the global level, vacuum distillation capacity accounted for 38% of distillation capacity, while upgrading capacity stood at around 46%, gasoline octane units at 21% and desulphurization at 69%. Although these figures appear broadly the same as last year's levels, there has been a steady increase in the complexity of the global refining system. This is reflected in the y-o-y increase in the proportion of secondary capacity relative to primary distillation capacity.

Figure 5.2
Secondary capacity relative to distillation capacity, January 2026, %



Source: OPEC.

Major downstream regions show significant variations in refinery complexity. The US & Canada continues to have the highest levels of upgrading, gasoline production and desulphurization relative to distillation. This reflects a traditionally complex refining system that has been enhanced over the past few decades, with refineries in this region forced to adapt to heavy and sour crudes coming from Canada, Latin America and the Middle East, as well as to strict regulations in terms of gasoline and diesel specifications. However, large, state-of-the-art refinery capacity additions, particularly in the Middle East, China and India, are raising overall secondary capacity relative to distillation in these regions too, with some countries narrowing the gap to the levels seen in the US & Canada.

For octane units, the US & Canada stands at the top of the list, with a ratio of more than 32% relative to distillation capacity. With gasoline demand being structurally dominant, the US & Canada has one of the world's largest fleets of FCC units, which produce a lot of gasoline-range material that needs octane upgrading, in addition to the massive catalytic reforming capacity to make high-octane reformate. In Europe, the ratio of octane units relative to distillation capacity is also relatively high at around 23%. This is linked to the presence of installed gasoline production capacities that existed before the region's car fleet dieselization.

The ratio of octane units to distillation capacity in the Middle East, China and Other Asia-Pacific is in the range of 18% to 19%. Latin America, India, Russia & Other Eurasia and Africa have lower ratios, lying between 14% and 17%. From a technological perspective, reforming is by far the most common technology used for gasoline production, accounting for more than 69% of global octane unit capacity, followed by isomerization (15%) and alkylation (13%).

For desulphurization capacity and associated sulphur recovery, levels vary considerably by region and largely depend on fuel standards and the quality of processed crudes. The US & Canada has by far the highest ratio of desulphurization relative to distillation capacity, at more than 96%. This reflects the ability to process a relatively heavier and more sour crude slate, from Latin America, the Middle East, and domestic Canadian barrels, while consistently meeting stringent low-sulphur product specifications for gasoline, diesel and jet fuel.

Other Asia-Pacific, Europe and India also show high desulphurization ratios, ranging between 72% and 74%, as their refineries process increasingly complex crude slates with consistently tighter low-sulphur product requirements. In Other Asia-Pacific and India, many facilities run imported medium-sour crudes and rely on extensive hydrotreating to deliver export-grade gasoline and diesel, while in Europe, stringent fuel-quality and environmental standards drive widespread hydrotreating across the barrel to meet ultra-low sulphur (ULS) fuels.

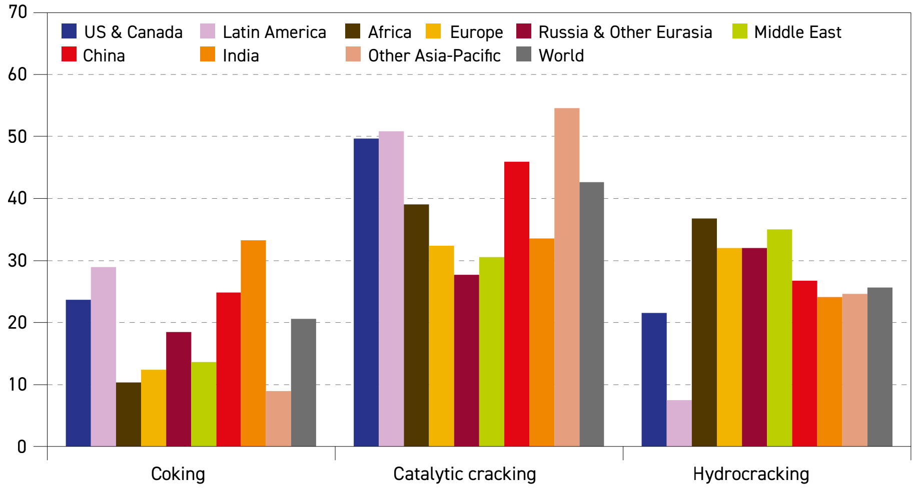
The Middle East has a ratio of 63%, which has increased in recent years, mirroring continued infrastructure upgrades in several countries to help boost exports. Regions such as China, Latin America and Russia & Other Eurasia have levels between 55% and 56%, while Africa stands at just 40%. Africa's low desulphurization to distillation capacity ratio is partly driven by the widespread use of sweeter domestic crude grades that inherently require less hydrotreating. Combined with a slower and uneven rollout of stringent low-sulphur fuel specifications across markets, this has reduced the incentive for large-scale investment in desulphurization units.

Finally, it is worth noting that regions with the highest levels of desulphurization relative to their distillation capacity also have the highest levels of sulphur recovery and hydrogen production capacities.

For the upgrading-to-distillation ratio, the US & Canada and India show by far the highest levels at 62% and 60%, respectively. China follows with 55%, reflecting the continued modernization of its refinery fleet. The ratio in Europe is also high, standing at almost 46%. It is important to mention that recent closures of the least complex refineries in Europe have contributed to raising the ratio of upgrading-to-distillation capacity in this region. Other regions have significantly lower upgrading ratios, with Latin America, Russia & Other Eurasia, Other Asia-Pacific and the Middle East in the 30–41% range. Africa still lags behind all the other regions and was assessed at around 22%. The commissioning of the Dangote refinery, however, has significantly improved the conversion ratio compared to previous years.

In terms of upgrading capacity from a technological perspective, regional preferences and technological deployments across regions also exist. Figure 5.3 shows upgrading capacity by selected technology relative to total upgrading capacity by region for January 2026. At the global level, catalytic cracking is the most widely used upgrading technology, accounting for 43% of global upgrading capacity. This is due to its efficiency in converting low-value, heavy crude oil fractions into lighter, high-value, and high-demand products, such as gasoline, and producing more with higher octane and valuable olefins.

Figure 5.3
Upgrading capacity by selected technology relative to total upgrading capacity by region, January 2026, %



Source: OPEC.

At the regional level, Other Asia-Pacific shows the highest ratio of catalytic cracking to total upgrading capacity, at a level of 55%. This reflects relatively lighter crude slates, gasoline-oriented demand patterns in several markets, and a large base of medium-scale, mid-complexity refineries where catalytic cracking units represent the most economical and flexible upgrading option. Latin America and the US & Canada also have about 50% each of their respective upgrading capacity devoted to this technology, followed by China with 46%. Except for Africa, with a 39% ratio of catalytic cracking to the region's upgrading capacity, the remaining regions stand in the range of around 30%. This distribution also reflects historical investment choices and the specificities of locally available crudes.

The second most widely used technology is hydrocracking, which accounted for almost 26% of the global upgrading capacity as of January 2026. Hydrocracking uses high pressure, hydrogen and catalysts to break down large and heavy oil molecules into lighter and more valuable products, such as jet fuel and diesel, while removing impurities (like sulphur and nitrogen) and saturating them to obtain clean and high-quality fuels, unlike carbon rejection methods (such as coking or FCC).

Africa has the highest ratio of hydrocracking to total upgrading capacity at 37%. This is due to a relatively low level of upgrading capacity, as well as recent large-scale hydrocracking capacity additions in Nigeria. The Middle East, Europe and Russia & Other Eurasia follow with ratios ranging between 32% and 35% relative to their total upgrading capacities. Other regions, such as China, Other Asia-Pacific, India and the US & Canada, are between 22% and 27%, while Latin America accounts for only 8%. This is due to the region's crude oil characteristics, which are less suitable for hydrocracking, as well as less developed infrastructure and less strict regulations regarding sulphur content.

When it comes to coking, India holds the highest levels in the upgrading mix at 33%. Indian refiners, such as Reliance and Nayara, have built deep-conversion, export-oriented refineries specifically to process heavier crude and convert large residue volumes into diesel and gasoline instead of producing low-value fuel oil. Latin America follows with 29%, while shares of coking in China and the US & Canada stand at 25% and 24%, respectively. The relatively high share in these three regions is attributed to the comparatively high availability of extra heavy crude. Finally, given the high levels of catalytic cracking and hydrocracking capacities in Other Asia-Pacific, this region holds the lowest share of coking capacities compared to other regions.

5.2 Distillation capacity outlook

This section explores refining capacity additions in the medium and long term. The analysis follows two different approaches. In the medium term (2026–2030), the refining capacity outlook is based on existing projects that are likely to come onstream within this time frame. The long-term outlook is built upon long-term demand and supply trends and represents the generic capacity required through to 2050 to ensure adequate product availability and market balance.

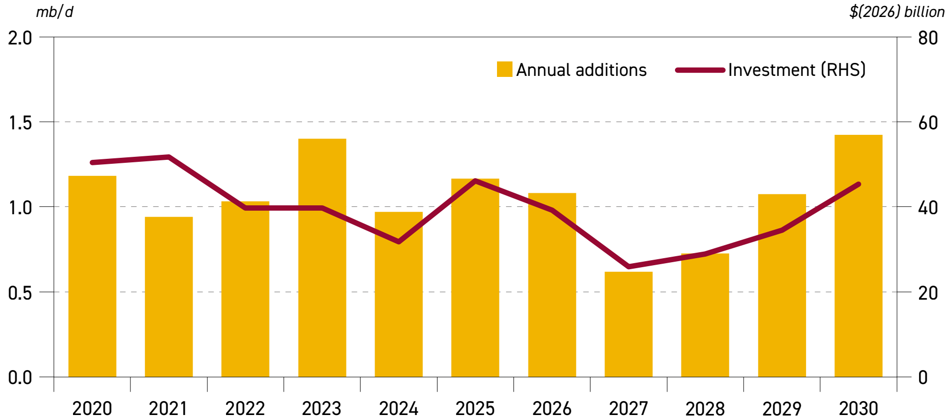
5.2.1 Medium-term distillation capacity additions

This section reviews medium-term downstream sector developments. It is based on a comprehensive and careful review of refining projects, including their respective status and progress. This includes new projects (greenfield refineries) and the expansion of existing units, covering both distillation and secondary capacities. These expansions are examined and profiled according to their likelihood of being implemented and commissioned within the medium-term period to 2030. It should be noted that these projections do not include continuous, small and under-the-radar additions (so-called 'creep' capacity), which refers to the natural increase in capacity at existing facilities that requires little or no capital expenditure.

Based on this methodology, global refining capacity additions between 2026 and 2030 are assessed at just above 4.9 mb/d. This figure reflects ongoing investments in the downstream sector to bring additional refining capacities onstream in response to rising oil demand.

Annual global distillation capacity additions and the estimated investment costs related to new refining projects for the period 2020–2030 are presented in Figure 5.4. Global annual

Figure 5.4
Annual distillation capacity additions and total project investment



Source: OPEC.

average capacity additions for the period 2026–2030 are estimated at 1 mb/d. Capacity additions in 2026 are seen at almost 1.1 mb/d, including many medium- and small-capacity refineries. More than half of this capacity is expected to materialize in India. Capacity additions are expected to drop for the years 2027 and 2028 to levels of 0.6 mb/d and 0.7 mb/d, respectively. China is projected to lead capacity additions in 2027, while Africa drives the expansion in 2028. In 2029, additions are anticipated to recover to almost 1.1 mb/d before continuing to rise to 1.4 mb/d in 2030. Additions in 2029 and 2030 are set to be dominated by the Middle East.

Investment costs follow the same trends as for capacity additions, with cumulative investment costs estimated at around $174 billion over the medium term.

Table 5.2 and Figure 5.5 show medium-term projections for refinery additions by region. Developing regions, mainly India, the Middle East, China, Africa and developing countries in the Asia-Pacific are set to drive almost all global refining capacity additions, with 94% of the expansion. These developments are consistent with regional oil demand trends. Demand is expected to grow strongly in these regions, driven by rising population growth and urbanization, as well as continued robust economic growth. Furthermore, efforts by some countries to reduce product imports and strategies adopted by other countries to boost product exports are also driving capacity expansion. These factors are reinforcing the long-standing trend of refining capacity migrating from developed to developing countries.

Table 5.2

Distillation capacity additions from existing projects by region, 2026–2030, mb/d

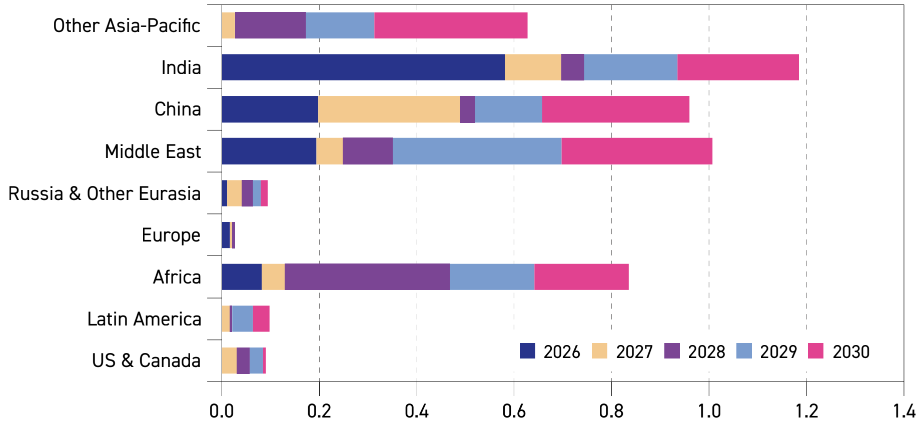
	US & Canada	Latin America	Africa	Europe	Russia & Other Eurasia	Middle East	China	India	Other Asia-Pacific	World
2026	0.0	0.0	0.1	0.0	0.0	0.2	0.2	0.6	0.0	1.1
2027	0.0	0.0	0.0	0.0	0.0	0.1	0.3	0.1	0.0	0.6
2028	0.0	0.0	0.3	0.0	0.0	0.1	0.0	0.0	0.1	0.7
2029	0.0	0.0	0.2	0.0	0.0	0.3	0.1	0.2	0.1	1.1
2030	0.0	0.0	0.2	0.0	0.0	0.3	0.3	0.2	0.3	1.4
2026–2030	**0.1**	**0.1**	**0.8**	**0.0**	**0.1**	**1.0**	**1.0**	**1.2**	**0.6**	**4.9**
Share (%)	*1.8*	*2.0*	*17.0*	*0.5*	*1.9*	*20.4*	*19.5*	*24.1*	*12.8*	*100*

Source: OPEC.

Total capacity additions of 4.9 mb/d are projected in the medium term, out of which 4.8 mb/d are expected to come online in regions outside the US & Canada and Europe. These projections include several small- to medium-scale projects, along with some large and state-of-the-art refineries that are integrated with petrochemical facilities.

India is projected to have the largest medium-term capacity additions among all regions, reflecting strong domestic demand growth and the country's strategy to strengthen its role

Figure 5.5
Distillation capacity additions from existing projects, 2026–2030, mb/d



Source: OPEC.

as a regional leading exporter of refined petroleum products. Additions in India are projected at around 1.2 mb/d, accounting for almost a quarter of global capacity additions throughout the medium-term period.

Furthermore, the Indian government's decision to boost refinery capacity from 260 mtpa to 300 mtpa in the coming years, further cements the country's position among the world's largest oil refining hubs. The move is intended to boost investment in the downstream sector at a time when refiners are accelerating their expansion into petrochemicals and exploring advanced pathways for bottom-of-the-barrel conversion to maximize value. This is part of a broader strategy to adapt to changing market dynamics, enhance competitiveness and position the country for sustained growth. It is expected that more than half of the additional capacity will come from expanding existing facilities, with the remainder delivered through new projects.

As part of the plans, nearly all major refiners are moving ahead with capacity expansions. Indian Oil Corporation (IOC) is expanding its Panipat, Gujarat, and Barauni refineries, with the aim of having the expansions operational in 2026. These projects aim to boost capacity and integrate petrochemical production to meet rising energy demands. Bharat Petroleum Corporation Limited (BPCL) plans to increase its overall refining capacity to 900 tb/d in the next five years. It aims to raise the capacity of its Kochi refinery in south India to 360 tb/d from its current 310 tb/d. Elsewhere, the new capacity of the Mumbai refinery is expected to increase to 320 tb/d, up from 240 tb/d currently, while the capacity of the Bina refinery is anticipated to increase by around 64 tb/d. In addition, BPCL aims to develop new petrochemical projects at its Kochi and Bina refineries by 2027 and 2028, respectively. Expansions of existing plants and investments in greenfield projects are also part of the strategies pursued by other companies, such as Hindustan Petroleum Corporation Limited (HPCL) and potentially Nayara Energy.

Not far behind India, the Middle East is set to have the second largest medium-term capacity expansion, at more than 1 mb/d. The region is set to see the start-up of several medium-sized projects that are currently under construction. IR Iran and Iraq are projected to host most of the additions. IR Iran has launched significant initiatives to enhance its oil refining capacity and optimize its global export strategy as part of its Seventh Development Plan. The plan sets major goals for refining and consumption optimization. By 2028, gasoline output is expected to reach 811 tb/d, diesel 817 tb/d and fuel oil 308 tb/d. Fuel quality is also projected to improve, with at least 75% of gasoline and diesel produced set to meet Euro-4 standards or higher.

Furthermore, Iraq has successfully increased its refining capacity to 1.3 mb/d and is steadily advancing towards achieving a refining capacity of 1.65 mb/d, in line with the government's goal. This is being achieved by upgrading Badji, Basra and Karbala facilities, alongside the development of new, large-scale refining facilities, such as the Fao integrated refining complex. Elsewhere, some minor expansions are also possible in Saudi Arabia, the UAE and Jordan.

In China, capacity additions are projected at around 1 mb/d between 2026 and 2030. Sinopec's 220 tb/d refinery project in Ningbo is likely to be commissioned during 2026, while Tahe refinery is still in the early stages of development, with commissioning scheduled to take place by 2029. Another prominent project is the joint venture between Saudi Aramco and Panjin Xincheng Industrial Group to develop 300 tb/d of processing capacity in Panjin. Construction began in late 2023 and the project is expected to be completed by 2027. In addition, the second phase of the Yulong refinery, which was commissioned last year, is progressing. It could also add another 400 tb/d towards the end of the medium-term period. Fujian Gulei Petrochemical Co. also has a world-scale facility under construction in Gulei. This is expected to come online by late 2030.

It is important to note that China's National Development and Reform Commission (NDRC) had previously announced a cap on the country's crude oil refining capacity at 20 mb/d by 2025. This is meant to reduce excess, improve utilization and align with emissions and industrial policy goals. According to the NDRC, every new refinery will have to have a capacity of at least 200 tb/d. Smaller, non-conforming refineries (40 tb/d or less) will likely be closed.

As part of a broad strategy to reduce reliance on imported refined petroleum products and to enhance national energy security, several countries in Africa are developing oil refining projects, with a number expected to come online within the medium term. In total, medium-term capacity additions in Africa are expected at around 0.8 mb/d, driven mostly by modular and small-scale capacity projects. Angola and Nigeria are expected to lead the expansions. Angola is set to see the inauguration of the first phase of its second refinery in Cabinda, developed by Sonangol, with a capacity of 30 tb/d. A second phase is scheduled for 2028, which will lift the refinery's capacity to 60 tb/d. Beyond Cabinda, Sonangol is also advancing two other refinery projects. The first is the 200 tb/d Lobito, currently under construction and slated to come online by 2028, while the second is in Soyo and is scheduled to commence operations by 2029.

Nigeria is playing a key role in developing the region's downstream sector. Following the Dangote refinery's ramp-up to full operational capacity in February 2026, the country is set to see the 200 tb/d Akwa Ibom refinery, as well as several other small, modular units, coming online in the next few years.

The Ugandan government is accelerating efforts to end its dependence on imported petroleum products, following the signing of new commercial agreements that are tied to its long-awaited 60 tb/d greenfield refinery in Hoima. The agreement between the Ugandan government and UAE-based Alpha MBM Investments LLC marks a major step towards a final investment decision (FID) that is expected in July 2026, with construction set to begin shortly after. In addition, several smaller modular refineries are in the pipeline across multiple countries, including the Republic of the Congo, Equatorial Guinea, Ghana and Uganda. In North Africa, new refining projects are underway in Algeria (Hassi Messaoud), Libya (Ubari) and Egypt (Sokhna).

In Other Asia-Pacific, capacity expansions are projected at more than 0.6 mb/d. These are located in countries like Indonesia, Pakistan, Bangladesh and Sri Lanka. In Indonesia, a key project is the greenfield project in Tuban, with a planned processing capacity of 300 tb/d. This project is being developed by Pertamina. In Pakistan, Saudi Aramco is helping to develop to a new refinery with a potential start-up in 2028. The Sri Lankan government has approved a refinery project with Sinopec near Hambantota Port, which is expected to process about 200 tb/d of crude. Bangladesh and Brunei could also see minor refining capacity additions in the medium term.

Latin America, Russia & Other Eurasia and the US & Canada are set to add around 0.1 tb/d each. In Latin America, Petróleo Brasileiro (Petrobras) is advancing the expansion of the Abreu e Lima Refinery in Pernambuco, Brazil. The facility currently has an operational unit processing about 130 tb/d and Petrobras has begun the construction of a second processing unit. When completed around 2029, the total processing capacity is expected to reach 260 tb/d. In the US, the Meridian Energy Group is continuing its expansion plans. The Davis refinery, located near Belfield, North Dakota, is under construction as a full-conversion crude oil refinery with an estimated capacity of 49 tb/d. Meridian has also identified a site in Winkler County, near Kermit, Texas, for another refinery project with a capacity of approximately 60 tb/d.

In Russia & Other Eurasia, minor expansions are expected in Azerbaijan, Georgia and Turkmenistan. Finally, an expansion of 30 tb/d is projected in Europe, reflecting a few small-scale projects in Türkiye.

It is worth noting that the medium-term outlook for new refining capacity is derived from a broad review of a long list of announced refinery projects, which account for more than 18 mb/d of capacity. This overall figure significantly overestimates the capacity expected to materialize between 2026 and 2030, as many of these projects remain at early conceptual, feasibility, or announcement stages. In addition, some of these projects may be delayed beyond the medium-term timeline, or even cancelled, for various reasons, including financing and technical challenges. Therefore, a degree of caution is required when elaborating the medium-term outlook for capacity additions from known refining projects.

The total projected capacity additions of 4.9 mb/d over the medium term corresponds to projects at different stages of development, albeit with a high or relatively high probability of materializing during this period. Of these, approximately 2.6 mb/d of capacity is either already under construction or approaching the construction phase, representing the projects with the highest likelihood of being completed within the medium-term time frame.

Beyond this, there are additional projects amounting to around 2.3 mb/d, most of which are not yet under construction and remain in the early stages of development. However, these projects have advanced sufficiently in terms of planning, financing and engineering to be classified as 'firm' medium-term additions in this Outlook. Despite this, a considerable degree of uncertainty remains regarding these projects, as they may face delays extending beyond the medium term or cancellation due to various factors.

5.2.2 Long-term distillation capacity additions

This section focuses on long-term refining capacity additions in the period 2026–2050 while taking into account the noted medium-term refinery capacity additions. It is also aligned with the underlying Outlook for oil demand and supply.

Table 5.3 and Figure 5.6 relate to projected distillation capacity additions by period. The time frame between 2026 and 2030 includes distillation capacity additions coming from the assessed refinery projects. It is important to emphasize that projected additions beyond 2030 are not linked to any specific refinery projects. Instead, they are projected as generic capacities required to meet long-term demand for refined products in all regions.

Table 5.3
Refinery distillation capacity additions by period, mb/d

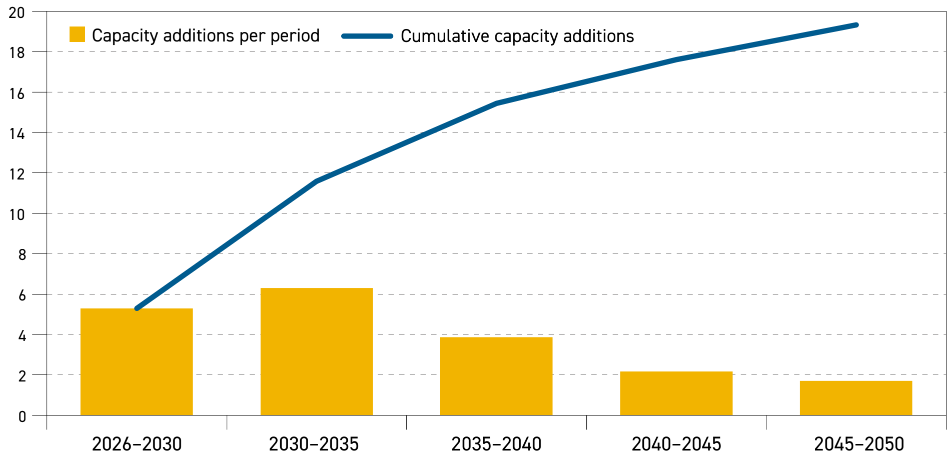
	Distillation capacity additions starting 2026			
	Assessed projects*	New units	Total	Annualized
2026–2030	4.9	0.4	5.3	1.1
2030–2035	0.0	6.3	6.3	1.3
2035–2040	0.0	3.9	3.9	0.8
2040–2045	0.0	2.2	2.2	0.4
2045–2050	0.0	1.7	1.7	0.3
	Cumulative distillation capacity additions			
	Assessed projects*	New units	Total	Annualized
2026–2030	4.9	0.4	5.3	1.1
2026–2035	4.9	6.7	11.6	1.2
2026–2040	4.9	10.5	15.4	1.0
2026–2045	4.9	12.7	17.6	0.9
2026–2050	4.9	14.4	19.3	0.8

* *Firm projects exclude additions resulting from capacity creep.*
Source: OPEC.

Capacity additions between 2026 and 2030 are estimated at 5.3 mb/d (including creep capacity), while in the long term an additional 14 mb/d of distillation capacity is required. Refining capacity is set to increase by 6.3 mb/d in the period 2030–2035, which is supported by rising demand in most developing regions. The rate of additions declines significantly thereafter and is estimated at around 3.9 mb/d in the 2035–2040 period, followed by 2.2 mb/d in the 2040–2045 period and 1.7 mb/d in the 2045–2050 period. This mirrors the slowdown in demand growth towards the end of the outlook horizon.

In total, distillation capacity additions between 2026 and 2050 are projected at 19.3 mb/d at the global level. These additions will more than offset potential refinery closures in developed regions that are expected due to a decline in oil demand in the long term.

Figure 5.6
Distillation capacity additions per period and cumulative, 2026–2050, mb/d



Source: OPEC.

In the initial period to 2030, annualized refining capacity additions (including creep capacity) are estimated at just below 1.1 mb/d p.a. The average annual additions between 2030 and 2035 are calculated at around 1.3 mb/d p.a., which marks a significant increase relative to the previous period. Average annual additions drop to around 0.8 mb/d p.a. between 2035 and 2040, before falling further to only 0.4 mb/d p.a. and 0.3 mb/d p.a. in the subsequent five-years periods of the Outlook. This translates into 60% of total capacity additions materializing during the 2026–2035 period and 80% occurring over the 2026–2040 horizon. This means that refinery capacity additions at the end of the outlook period will tend to be expansions of existing capacity, rather than new greenfield refineries.

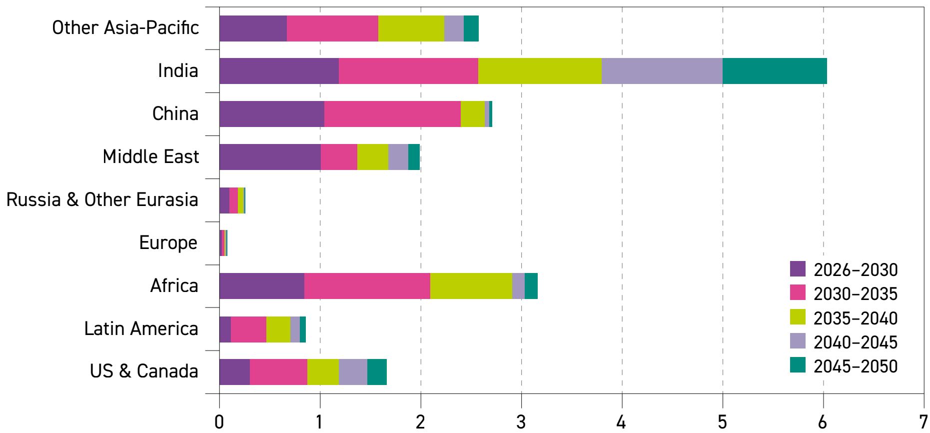
Regional additions

Global refining capacity additions for the period 2026–2050 are projected at around 19.3 mb/d. Of this, 5.3 mb/d (including creep capacity) is expected in the period between 2026 and 2030 and 14 mb/d in the period between 2030 and 2050. Following regional oil demand trends, the majority of the incremental refining capacity is expected to be added in developing regions, predominantly the Asia-Pacific, with India and China leading the way, the Middle East and Africa. These three regions are set to account for more than 85% of global refining capacity additions in the long term. Figure 5.7 depicts crude distillation capacity additions by region and by period.

India is expected to account for by far the largest share of global refinery capacity additions, with more than 6 mb/d projected to come online between 2026 and 2050. This is supported by robust oil demand growth, as well as the government's strategic objective of elevating India's role as a global refining and refined-products export hub, with long-term plans to scale capacity up to 450 mtpa.

The required refining capacity additions in Africa are estimated at nearly 3.2 mb/d over the long term. This is set to be driven primarily by domestic demand growth, as well as by efforts to reduce the region's heavy dependence on refined product imports, especially in West Africa. In this subregion of the continent, it is likely that most new refining capacity will largely rely on African crude oil supplies, whereas refineries on the east coast will likely continue to rely partly on imports from other regions. It is worth emphasizing that the implementation of large-scale projects remains challenging in Africa. Long project lead times, alongside financing and technical issues, continue to pose significant risks and these need to be addressed if the required refining capacity additions are to materialize.

Figure 5.7
Crude distillation capacity additions by region, 2026–2050, mb/d



Source: OPEC.

Refining capacity additions in China are estimated at 2.7 mb/d in the long term. The majority of the required capacity, or around 2.4 mb/d, is expected to come online in the period between 2026 and 2035. This is driven by oil demand growth, as well as the expected replacement of some old and inefficient refineries (mostly teapot plants) that might close due to more stringent regulations, unfavourable economics and government initiatives to limit the country's total refinery capacity to 20 mb/d. Minor additions are expected thereafter but will likely be limited to the expansion of existing facilities.

Long-term refining capacity additions in Other Asia-Pacific (excluding China and India) are assessed at close to 2.6 mb/d. This is driven almost entirely by developing countries in the region, where oil demand is projected to grow significantly given expected economic and population growth. Developing countries in this region are implementing import substitution policies aimed at replacing imported refined products with domestically produced supply, with governments focused on enhancing energy security and reducing exposure to disruptions in refined product supplies.

The Middle East is expected to add just around 2 mb/d of new distillation capacity over the entire outlook period. These additions, however, are front-loaded, with 1 mb/d expected to be commissioned by 2030. Beyond 2030, expansion of the region's refining system is set to slow, with additions of around 0.4 mb/d between 2030 and 2035, which then drops to 0.1 mb/d in the period from 2045–2050. Most of the additions beyond 2030 will likely be related to expansions of existing refineries, rather than the commissioning of new capacity. In recent years, the region has built several large, state-of-the-art refineries, which will help to meet expanding domestic demand and increase refined product exports to international markets.

Total refining capacity additions in the US & Canada are estimated at around 1.7 mb/d. Capacity additions include a range of continuous debottlenecking efforts at existing refineries, due to the large base capacity already installed. Additions are also expected to balance out the decommissioning of some older, inefficient plants. The availability of oil supply in this region supports the economics of the refining sector and its competitiveness in the international market.

Possible incremental refining capacity in Latin America is estimated at around 0.9 mb/d between 2026 and 2050. These additions are significantly lower relative to oil demand growth in the same period, which is projected at over 2.8 mb/d. To help meet that demand growth, Latin America may need to modernize some of its currently under-utilized refining capacities. Secondly, due to geography proximity and relatively low transportation costs, Latin America could also see increasing inflows of refined products from the US, if deficits arise.

Additions in Russia & Other Eurasia are minor. Total long-term refinery additions in the region are projected at almost 0.3 mb/d, consisting mostly of debottlenecking existing plants. Finally, in Europe – except for minor additions in the medium term – no new refining capacity is projected for the period beyond 2030, which is in line with the region's expected demand decline.

5.3 Refining sector market balance

This section focuses on the downstream market outlook by taking into consideration the outlined refining capacity development, regional oil demand and oil supply. The analysis covers the global downstream market, as well as specifics for major regions. The Outlook is divided into two medium- and long-term sub-sections that follow different approaches.

The medium-term outlook looks at refinery capacity net increments, considering refinery additions, as laid out in Section 5.2.1, and refinery closures, as presented in section 5.3.1, and compares them with the so-called 'call-on-refining' relative to the base year of 2025. In other words, this analysis shows how the market may change compared to the base year of this Outlook. The 'call-on-refining' is based on oil demand growth. It also considers demand for various non-refinery fuels, such as NGLs, CTLs, GTLs and biofuels.

The long-term outlook looks at modelling results over the period 2030–2050 and projects refinery throughputs and respective utilization rates at the regional level, including crude and product movements (see Chapter 6).

Medium-term global balance

Medium-term primary capacity additions are projected at 4.9 mb/d globally. On top of these additions, modelling results suggest further medium-term debottlenecking or 'creep' capacity additions of almost 0.4 mb/d. This is mostly in the US & Canada and parts of the Asia-Pacific due to the large number of existing refineries. Considering expected refinery closures in the medium term of around 0.3 mb/d, total net additions to global distillation capacity in the medium term are estimated at roughly 5 mb/d.

The methodology also assumes that new refining capacities may reach the maximum assumed utilization rate of 90% throughout the year. This is considered a reasonable assumption at the global level. Consequently, this provides insight into the potential incremental crude runs or potential refining capacity between 2026 and 2030. Furthermore, as this outlook is on an annual basis, the methodology attempts to capture uncertainties related to the start-up date of refining capacity within the year. This is why the calculation considers only one half of the current year (n) and one half of the previous year (n–1).

With respect to capacity closures, the potential loss of crude runs is calculated taking into account only one half of the current year (n) and one half of the next year (n+1), again to capture uncertainties related to the date of refining capacity closures within the year. With this approach, the cumulative net global potential refining capacity is set to reach levels of around 4.5 mb/d by 2030, when compared to 2025.

In the next step, the cumulative required incremental crude runs at the global and regional level are calculated. This is the so-called 'call-on-refining' and is based on demand patterns that consider non-refinery fuels, such as NGLs, biofuels, CTLs and GTLs, which bypass refinery processing. This section covers balances from the perspective of distillation capacity, crude runs and total demand without considering specific refined products that are discussed later.

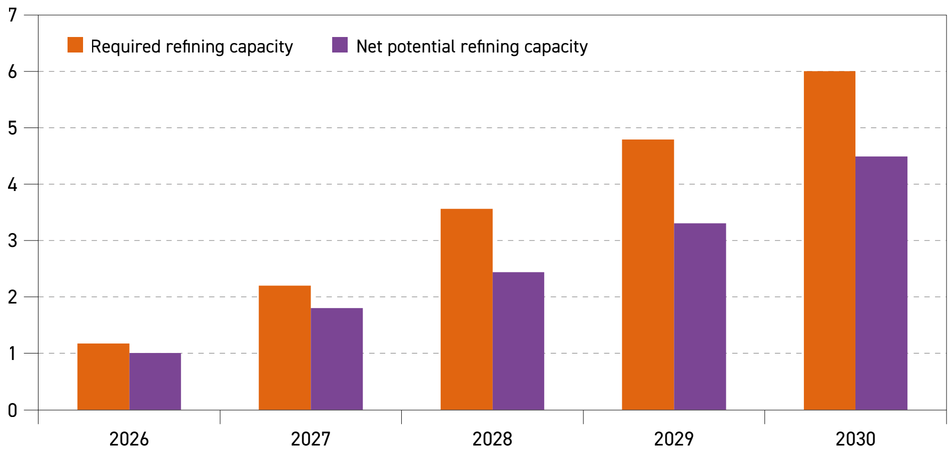
While medium-term global oil demand growth is estimated at almost 8.2 mb/d, the total required incremental crude runs are calculated at about 6 mb/d. In the final step, the potential incremental crude runs are compared with the cumulative incremental refined product demand at an annual level.

The analysis is done at the global level and for each of the major downstream regions. The resulting balances show the incremental net refining capacity compared to incremental refined product demand relative to the base year of 2025. This is a good indicator of the state and direction of the downstream market in the medium term, both globally and regionally.

Figure 5.8 provides a summary assessment of the global cumulative medium-term potential for net incremental distillation refining capacity based on project list and plant closures, compared to the required incremental product supply from refineries relative to the base year.

At the global level, the evolution of net incremental potential refining capacity and required refining capacity shows a tightening market throughout the medium term. At the start of the medium-term period, the required refining capacity is close to 1.2 mb/d. This is higher than the estimated potential net incremental capacity of 1 mb/d, resulting in a deficit of around 0.2 mb/d. However, the deficit widens significantly thereafter, on the back of strong oil demand growth, limited refining capacity additions and the closures set to occur in the initial years of the medium-term period. As a result, the gap between net incremental potential refining capacity and required refining capacity rises to more than 1.5 mb/d by 2030.

Figure 5.8
Additional global cumulative refinery crude runs, net potential* and required, mb/d**



* Net potential: based on expected distillation capacity expansion and closures.

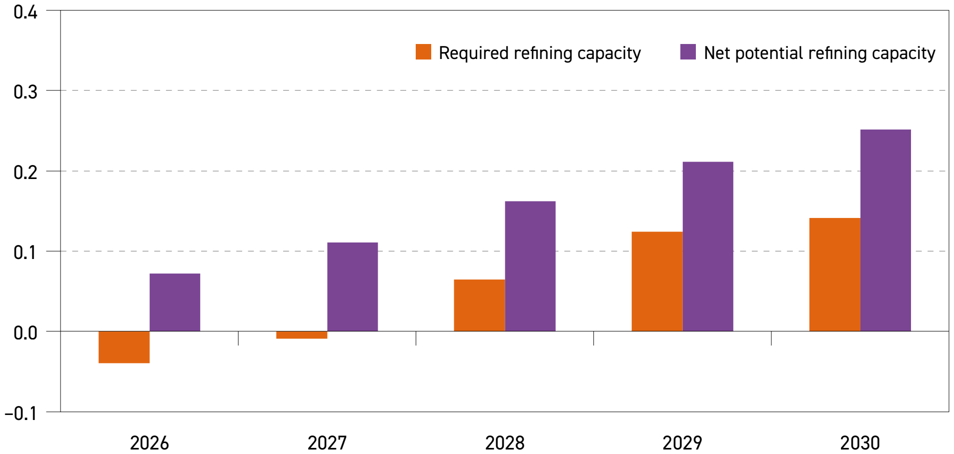
** Required: based on projected demand increases, assuming no change in refined products trade pattern.
Source: OPEC.

Medium-term regional balances

This section focuses on the regional medium-term balances. Figures 5.9 to 5.17 provide a comparison of data for all major downstream regions in the medium term.

Figure 5.9 shows the medium-term downstream market balance for the US & Canada relative to 2025. The net potential incremental refining capacity is expected to rise gradually to levels of around 0.3 mb/d over the medium term. This is a result of minor refinery expansions,

Figure 5.9
Additional cumulative crude runs in US & Canada, net potential and required, mb/d



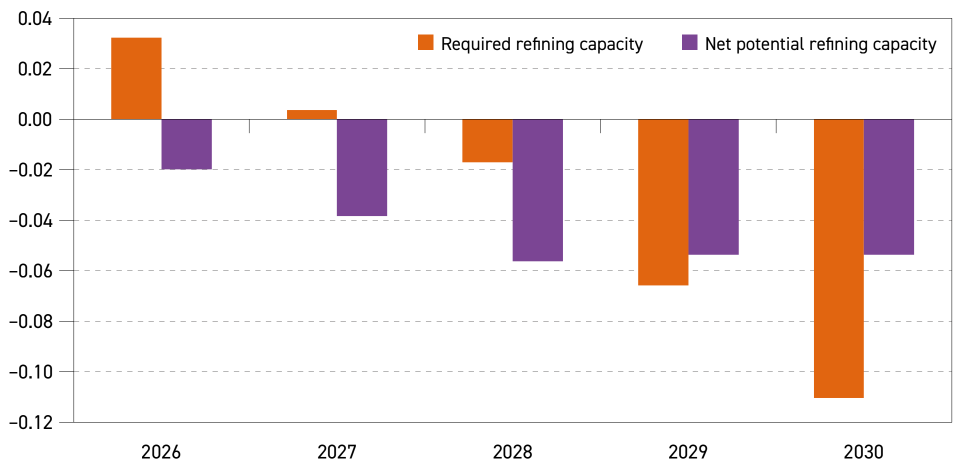
Source: OPEC.

combined with continuous debottlenecking expansions. At the same time, the required refining capacity increases in parallel with the potential refining capacity, which reaches more than 0.1 mb/d in 2030. Consequently, the medium-term downstream market in the US & Canada is expected to be mostly balanced throughout the outlook period, relative to the base year.

It is worth emphasizing that the US & Canada, with its complex and sophisticated refining system and ample domestic supply, is likely to remain competitive in the international downstream market in the medium and long term. This will help to keep utilization rates in this region elevated during the medium term, which is discussed later in this section.

Figure 5.10 shows the medium-term downstream market balance in Europe relative to 2025. It indicates a marginal drop in net cumulative potential refining capacity to 2028, reflecting capacity closures during this period coupled with minor capacity expansions. It then remains flat. Over the same period, the required incremental refining capacity is expected to fall by 0.1 mb/d between 2026 and 2030, in line with declining demand for refined products and rising non-refining products supply during this period. Consequently, Europe's downstream market is expected to be well-supplied relative to 2025.

Figure 5.10
Additional cumulative crude runs in Europe, net potential and required, mb/d

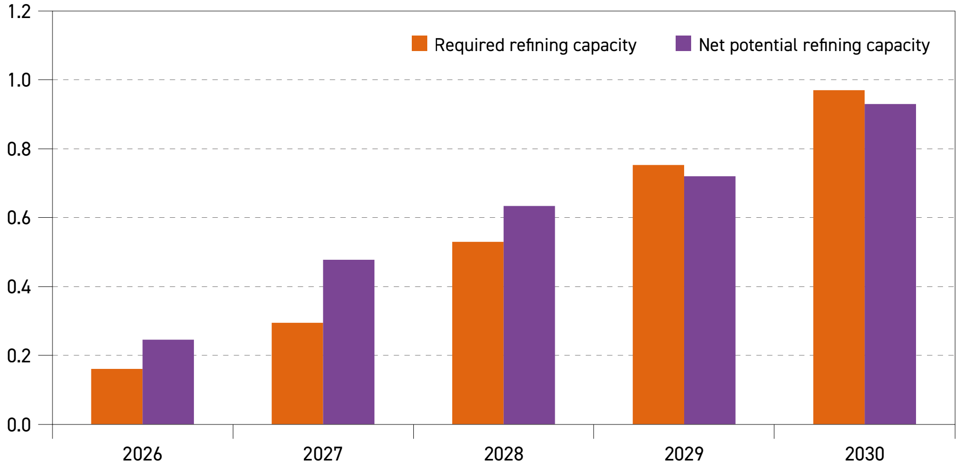


Source: OPEC.

Figure 5.11 highlights the medium-term downstream market balance for China. A surplus of almost 0.1 mb/d is observed in 2026 relative to 2025. Thereafter, net cumulative potential refining capacity is set to increase at around the same pace as cumulative required capacity through 2028, in line with refinery capacity additions and rising oil demand. As a result, China's downstream market is set to remain broadly balanced over this period relative to the base year. In the latter part of the medium term, the market is expected to tighten modestly in 2029 and 2030, as the pace of refining capacity additions slows while demand continues to grow.

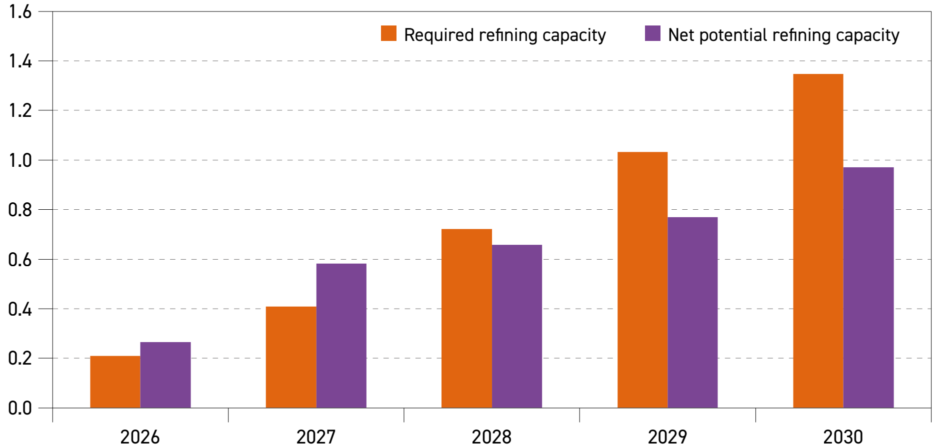
Figure 5.12 shows the medium-term downstream market balance for India. Required refining capacity is set to increase significantly from 0.2 mb/d in 2026 to more than 1.3 mb/d in 2030. This is based on strong oil demand growth in the country. At the same time, net potential

Figure 5.11
Additional cumulative crude runs in China, net potential and required, mb/d



Source: OPEC.

Figure 5.12
Additional cumulative crude runs in India, net potential and required, mb/d

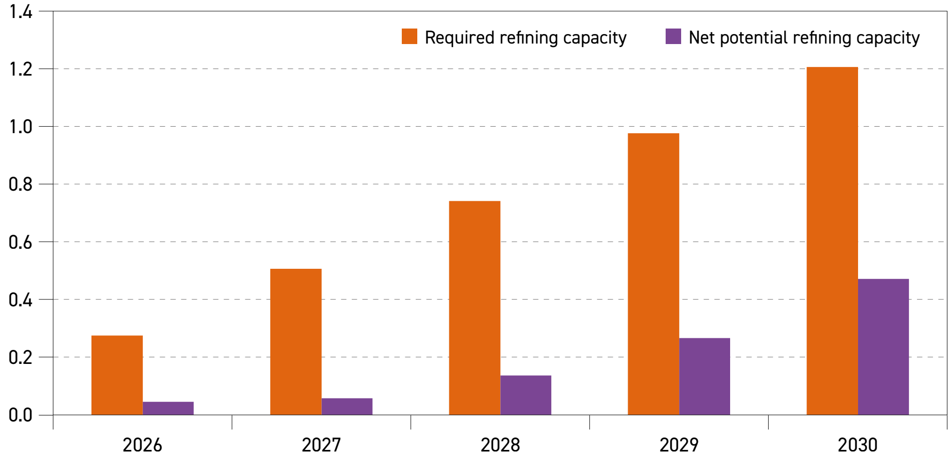


Source: OPEC.

incremental refining capacity increases from almost 0.3 mb/d in 2026 to around 1 mb/d in 2030. As a result, the Indian downstream market is expected to shift from a broadly balanced market in 2026 and 2027 to an increasingly tight market from 2028 onwards, as refining capacity additions grow at a slower pace than oil demand. The deficit is expected to grow from around 0.1 mb/d in 2028 to approximately 0.4 mb/d in 2030, relative to the base year. Consequently, India's downstream sector is likely to experience higher imports of non-refinery products and possibly reduced refinery product exports over this period, as it is already operating at a high utilization rate.

Figure 5.13 shows the medium-term downstream market balance for Other Asia-Pacific (excluding China and India). Required refining capacity increases strongly from just below

Figure 5.13
Additional cumulative crude runs in Other Asia-Pacific (excl. China and India), net potential and required, mb/d

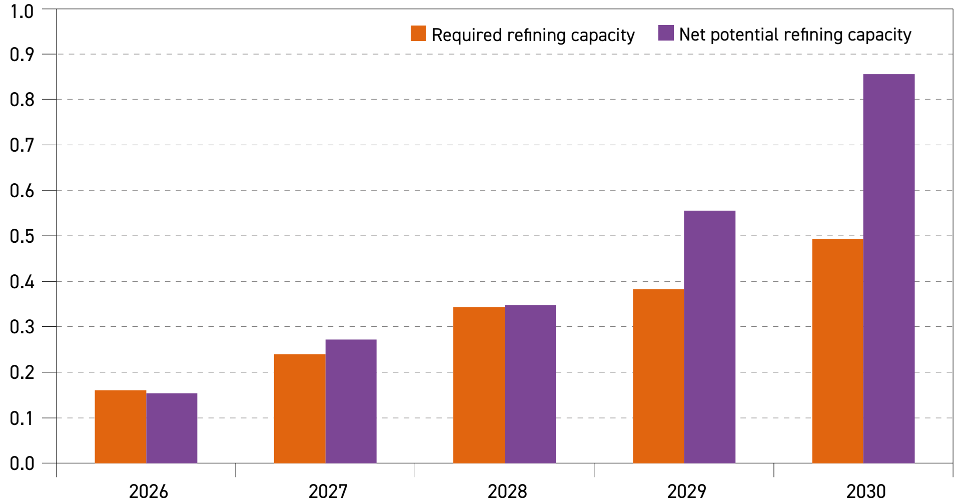


Source: OPEC.

0.3 mb/d in 2026 to over 1.2 mb/d in 2030. This is based on strong oil demand growth in the developing countries of this region. At the same time, net potential incremental refining capacity increases to 0.5 mb/d in 2030. This leads to a steady increase in the gap between required and potential refining capacity, from 0.2 mb/d in 2026 to around 0.7 mb/d in 2030, relative to 2025. Consequently, Other Asia-Pacific's downstream sector could see rising utilization rates and possibly higher product imports in the medium term.

Figure 5.14 shows the medium-term balance for the downstream market in the Middle East. Demand growth results in rising required incremental refining capacity relative to 2025. It increases from almost 0.2 mb/d in 2026 to around 0.5 mb/d in 2030. At the same time,

Figure 5.14
Additional cumulative crude runs in the Middle East, net potential and required, mb/d

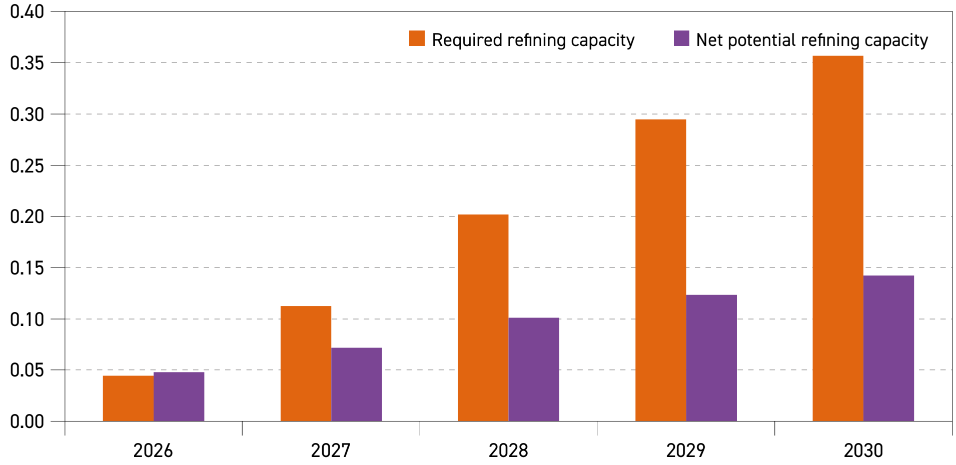


Source: OPEC.

net potential incremental refining capacity moves from below 0.2 mb/d in 2026 to almost 0.9 mb/d by 2030. During the 2026–2028 period, the market is set to be largely balanced relative to 2025. Thereafter, it is expected to shift into a well-supplied market, with a surging surplus on the back of strong refining capacity additions from 2029 onwards. The surplus is projected to rise from 0.2 mb/d in 2029 to 0.4 mb/d in 2030, leading to a possible increase in refining product exports.

In Russia & Other Eurasia, presented in Figure 5.15, cumulative required refining capacity reaches almost 0.4 mb/d by 2030, up from nearly zero in 2026. This reflects oil demand

Figure 5.15
Additional cumulative crude runs in Russia & Other Eurasia, net potential and required, mb/d



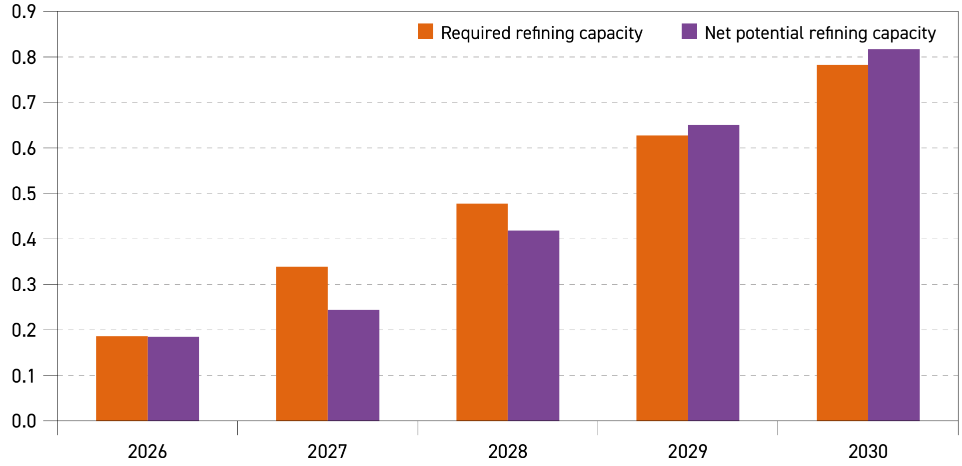
Source: OPEC.

growth in the region. Meanwhile, net potential refining capacity increases by around 0.1 mb/d between 2026 and 2030, in line with minor capacity expansions at existing plants. Consequently, the market in Russia & Other Eurasia is set to tighten gradually through the medium term, relative to the base year. This could lead to a possible decline in refinery product exports, as the downstream sector is largely linked to product exports and changes in the international downstream market.

Figure 5.16 highlights the downstream market balance for Africa between 2026 and 2030. Both required and net potential cumulative refining capacity are seen increasing from 0.2 mb/d in 2026 to 0.8 mb/d in 2030. Although the market is set to shift from a modest deficit in the early years of the medium term to a somewhat loose market relative to 2025, the African downstream market is set to remain broadly balanced throughout the medium-term period. This means that any delay in operationally commissioning the planned capacity additions could tighten the market and lead to higher product imports.

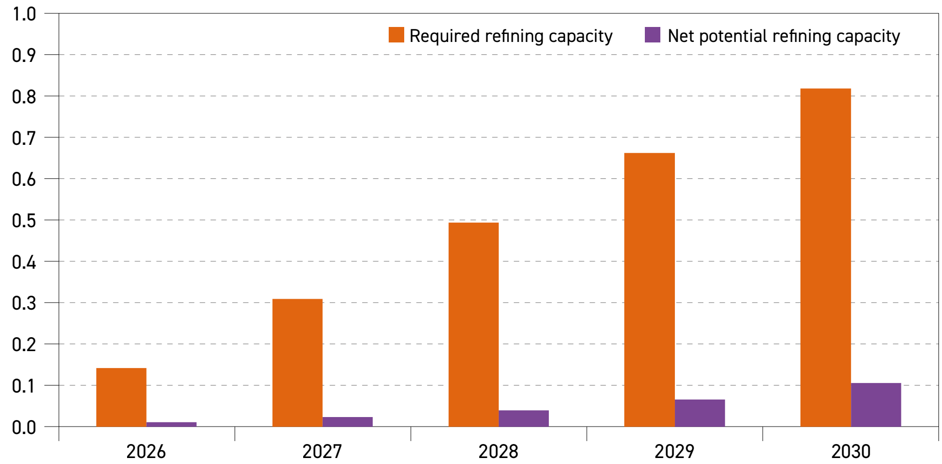
The downstream balance in Latin America is shown in Figure 5.17. The cumulative required refining capacity is projected to increase at just above 0.8 mb/d in 2030, up from over 0.1 mb/d in 2026. This is in line with rising oil demand. The net potential incremental capacity

Figure 5.16
Additional cumulative crude runs in Africa, net potential and required, mb/d



Source: OPEC.

Figure 5.17
Additional cumulative crude runs in Latin America, net potential and required, mb/d

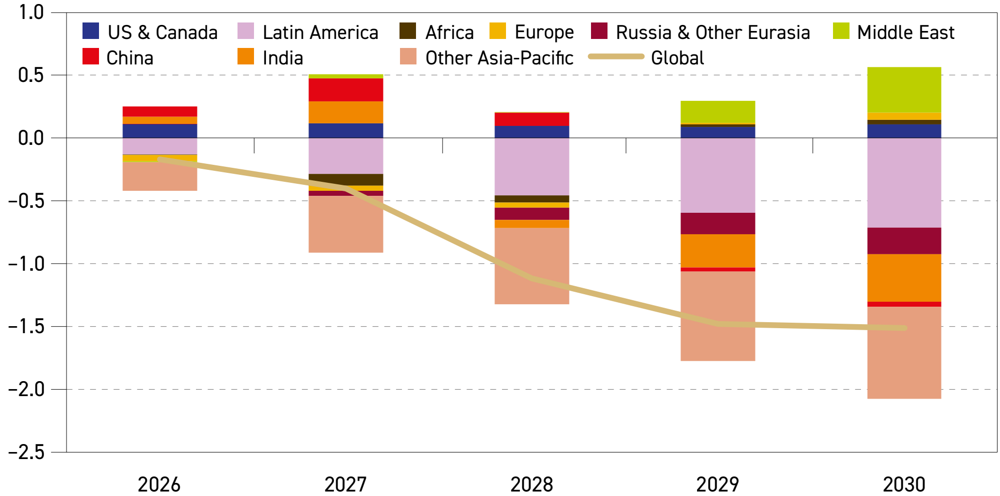


Source: OPEC.

is estimated to increase too, but at a much slower pace. It is seen reaching only 0.1 mb/d by 2030. Required refining capacity remains above net incremental potential capacity throughout the medium term, resulting in a deficit of above 0.7 mb/d in 2030. This demonstrates a tighter medium-term downstream market, potentially leading to higher refinery utilization rates and an increase in product imports.

Finally, an overall view of the cumulative downstream medium-term balance by region and globally, relative to the base year, is presented in Figure 5.18. This also shows the change in the gap between incremental potential and required refining capacity over the medium term. The largest deficit of refining capacity relative to requirements is expected in Other

Figure 5.18
Net cumulative regional refining potential surplus/deficits *versus* requirements, mb/d



Source: OPEC.

Asia-Pacific, due to relatively small capacity additions compared to oil demand growth in the medium term. Latin America is also set to see a large deficit in the same period, on the back of strong oil demand growth and very limited refining capacity additions. Despite having the highest capacity additions in the medium term, the downstream market in India is expected to experience a relatively high deficit driven by increased oil demand. Russia & Other Eurasia and China are expected to experience limited deficits towards the end of the period, relative to 2025. The cumulative deficit is set to increase steadily from just 0.4 mb/d in 2026 to reach levels of 2.1 mb/d in 2030.

On the other hand, a surplus of net refining capacity additions relative to requirements emerges in the Middle East, and to a lesser extent in other remaining regions, such as the US & Canada and Europe. The cumulative surplus for all these regions increases from just below 0.3 mb/d in 2026 to around 0.6 mb/d in 2030.

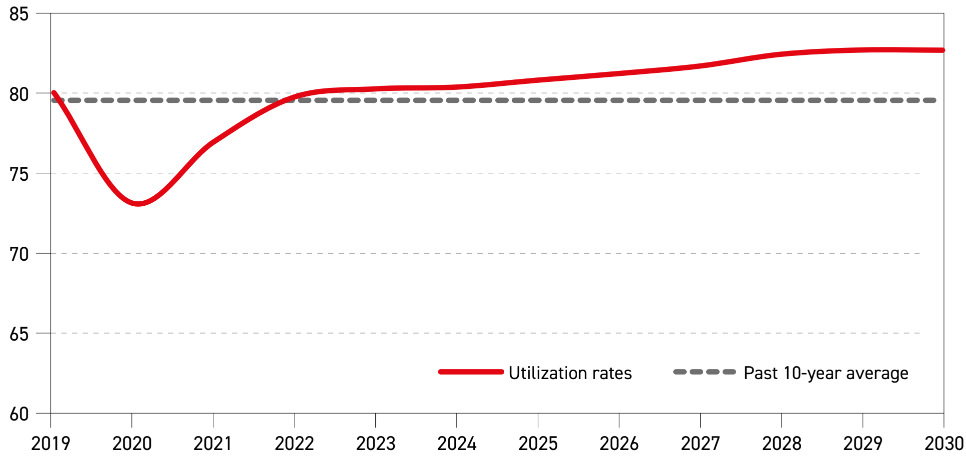
The resulting global balance shows an increasing deficit in net potential refining capacity relative to required capacity over most of the medium term. The deficit rises from 0.2 mb/d in 2026 to 1.5 mb/d in 2030. This signals a strong downstream market throughout this period, compared to the base year.

Medium-term refinery utilization and throughputs outlooks

This section discusses medium-term global refinery utilization and refinery throughputs. It underscores assumed global crude throughputs, the effects of historical and projected closures and assesses spare refining capacity over the medium term.

Global utilization rates in the period 2019–2030 are shown in Figure 5.19. Global utilization rates were at strong levels of around 80% in 2019, but these dropped to around 73% in 2020 due to the demand shock caused by the pandemic. The recovery has been gradual in subsequent years, reaching 76.9% in 2021 and almost 79.8% in 2022, before recovering to pre-pandemic levels of 80.3% in 2023, 80.4% in 2024 and around 80.8% in 2025.

Figure 5.19
Historical and projected global refinery utilization, 2019–2030, %



Source: OPEC.

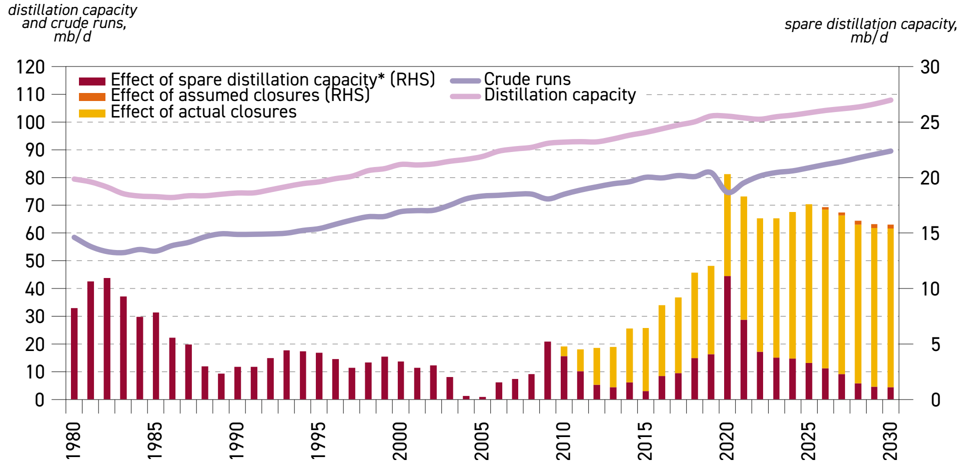
Global utilization rates are expected to continue to rise over the medium term, reaching 82.7% in 2030. This trend is underpinned by strong oil demand growth, insufficient refining capacity additions, as well as announced closures during the medium term. Moreover, any additional shutdowns beyond those officially announced, such as in the case of China's teapots refineries, would likely further increase global utilization rates. Additional uncertainty also relates to the timing of the estimated refining capacity additions expected to occur in the medium term. Any potential delays in their commissioning would inevitably push utilization rates even higher.

Global utilization rates are calculated based on nominal refining capacities, which are assumed to be fully available over the medium term. There is a significant discrepancy in regional utilization rates and many countries, particularly in Africa and Latin America, still operate old and inefficient refineries that have been running at relatively low levels for many years. Although an improvement is expected in the medium term, these rates are not expected to reach 70% (see Table 5.4). This implies that, to maintain a global utilization rate of 82.7% by 2030, refineries located in other regions will need to operate at levels significantly above 80%, especially those in India, the US & Canada and Europe.

Furthermore, other factors could limit the availability of actual refining capacity globally, raise utilization rates regionally and increase pressure on the downstream system. These factors include policies by some countries that restrict exports of refined products from their domestic refineries due to various reasons, including security of supply for specific products. In addition, geopolitical disruptions could affect the trade of crude oil and refined products. Logistical bottlenecks in pipelines, storage and ports also limit the ability to move products to market, leading to prolonged refinery downtime. Finally, stricter environmental regulations, carbon compliance costs and delays in refinery upgrading investments may result in refineries operating well below their available capacities.

Figure 5.20 shows the evolution of refinery crude and condensate throughputs, absolute nominal distillation capacity, spare distillation capacity, as well as the effect of actual and assumed closures, from 1980 to 2030.

Figure 5.20
Refining capacity and crude runs, 1980–2030



* Effective 'spare' capacity estimate based on assumed 84% utilization rate, accounting for already-closed capacity.
Source: OPEC.

It is worth noting that while oil demand grew by around 17 mb/d between 2010 and 2025, at the same time refinery runs increased by only 9.5 mb/d. This is due to shifts in the structure of global oil demand and the increasing share of demand attributed to non-refinery fuels, largely NGLs (which bypass refinery systems), but also biofuels, CTLs and GTLs. The increasing share of these liquids has led to a reduced share of refinery products in total oil demand. This has kept refining capacity additions lower relative to oil demand increments.

This trend is expected to continue in the medium term as global oil demand is projected to increase from around 105.1 mb/d in 2025 to 113.3 mb/d in 2030, an increase of 8.2 mb/d. At the same time, refinery runs are projected to rise by roughly 6 mb/d to reach 89.5 mb/d by the end of the medium-term period.

Figure 5.20 also shows the effects of actual and assumed refinery closures. Since 2010, more than 14 mb/d of refining capacity has been shut down for a variety of reasons. Almost half of that capacity, or around 6.3 mb/d, was put out of service between 2020 and 2025, mainly due to declining demand and margins during the pandemic. According to public announcements and plans, another 0.3 mb/d is expected to close in the medium term (2026–2030). In reality, closures not yet announced are likely to increase this number, particularly in China and Europe.

Another important factor that merits attention in assessing the downstream sector is the level of spare refining capacity. The latter is the difference between projected refinery runs and maximum global sustainable runs. Maximum sustainable global runs are assessed based on global refinery capacity, taking into consideration projected refinery additions and closures, and the maximum global utilization rate, estimated at around 84%. This is historically the highest observed level. Consequently, the level of spare capacity was at its highest level in 2020, at 11 mb/d, due to the collapse in oil demand caused by the pandemic. However, as demand recovered thereafter and significant capacity was closed, spare capacity dropped to below 3.3 mb/d in 2025. The global spare refining capacity level is expected to decline gradually to only around 1.1 mb/d in 2030, in line with rising refinery runs. This also includes the assumption that all projected refining capacity additions are built on time. Any delays could lower the level of spare capacity further or push utilization rates above projected levels.

Long-term balance for the refining sector

This section focuses on long-term crude and condensate throughputs, as well as long-term utilization rates at the global and regional level. These are based on modelling cases and in line with oil demand (Chapter 3) and liquid supply (Chapter 4) assumptions. Assumptions on medium-term refining capacity additions and refinery closures, as already discussed, are also an integral part of the modelling cases.

Table 5.4 illustrates projected refinery throughputs and respective utilization rates at the global level and for major downstream regions, by period from 2025 to 2050. While the assessment of utilization rates takes projected medium-term closures into account, it excludes the implied long-term closures.

Global refinery throughputs are projected to significantly increase from around 83.5 mb/d in 2025 to 89.5 mb/d in 2030 driven by strong medium-term demand growth. This is translated to annual average growth of around 1.2 mb/d p.a. Thereafter, global refinery runs are set to

Table 5.4
Crude unit throughputs and utilization rates, 2025–2050

Total crude unit throughputs *mb/d*										
	US & Canada	Latin America	Africa	Europe	Russia & Other Eurasia	Middle East	China	India	Other Asia-Pacific	Global
2025	18.1	4.8	2.4	11.5	6.6	8.9	14.8	5.4	11.1	83.5
2030	18.2	5.3	3.5	12.0	6.7	9.3	15.5	6.6	12.4	89.5
2035	18.6	6.0	4.6	11.7	6.8	9.6	16.3	7.7	12.8	94.1
2040	18.7	6.6	5.2	11.0	6.7	10.0	16.0	8.8	12.8	95.9
2045	18.4	6.8	5.5	10.3	6.6	10.5	15.8	9.8	12.9	96.7
2050	17.8	7.1	5.8	9.9	6.6	10.8	15.6	10.7	13.0	97.2
Crude unit utilizations *% of calendar day capacity*										
	US & Canada	Latin America	Africa	Europe	Russia & Other Eurasia	Middle East	China	India	Other Asia-Pacific	Global
2025	93.8	59.1	55.1	82.9	77.7	76.5	82.8	103.5	76.4	80.8
2030	94.0	64.2	68.8	87.8	78.0	73.4	82.0	102.2	81.4	82.7
2035	93.1	70.0	72.7	85.9	78.0	73.9	80.1	99.0	79.1	82.2
2040	92.0	74.6	72.5	80.2	77.0	75.5	78.1	97.5	76.3	81.0
2045	89.3	75.9	75.7	75.3	76.0	78.0	76.8	96.2	76.0	80.2
2050	85.7	78.5	77.6	72.5	75.3	79.1	75.5	95.2	75.8	79.5

Source: OPEC.

continue increasing, albeit at a slower pace, reaching levels of 94.1 mb/d in 2035, reflecting an annual average growth of more than 0.9 mb/d. Annual growth is then expected to continue to decline to approximately 0.4 mb/d p.a. and 0.2 mb/d p.a. in the periods 2035–2040 and 2040–2045, respectively. Finally, the annual average growth is estimated at just below 0.1 mb/d p.a. over the period between 2045 and 2050, resulting in global refinery throughputs of 97.2 mb/d by 2050. This is in line with the global demand growth slowdown, as well as the rising share of non-oil liquids (biofuels and synthetic fuels) and the increasing volume of NGLs that bypass refining.

At the same time, the global utilization rate is expected to rise from 80.8% in 2025 to more than 82.7% in 2030, highlighting a tightening downstream market. Post-2030, the utilization rate is expected to drop gradually to 79.5% by 2050. This is a result of rising refining capacity in developing countries, where demand is expected to grow. Declining utilization rates signal possible refinery closures in some regions in the long term.

Behind the global trend, there are specific and diverging regional trends. Refinery throughputs are expected to increase in developing regions, especially in the Asia-Pacific (including China and India), the Middle East, Africa, as well as Latin America. This will be partly offset by the long-term drop in all developed regions. In Russia & Other Eurasia, refinery throughputs are set to remain relatively flat throughout the outlook period.

In the US & Canada, refinery throughputs were assessed at 18.1 mb/d in 2025 and are expected to increase only marginally to 18.2 mb/d in 2030. Thereafter, they are set to rise to around 18.7 mb/d in the period to 2040. Despite the decline in domestic demand for oil, the region is expected to export significant volumes of refined products to other regions due to the availability of crude oil and its competitive refining system. However, in the last decade of the outlook, refinery throughputs are expected to drop to around 17.8 mb/d in 2050, reflecting declining regional demand. Refinery utilization rates in this region are set to increase slightly to 94% by 2030, up from 93.8% in 2025, before starting a gradual decline to settle at 85.7% in 2050. The utilization rate in 2050 is comparatively high at the global level, but still well below historical levels seen in the US & Canada. This could be an indication of limited refinery closures, and thus, some older and inefficient plants could be impacted in the long term.

Refinery throughputs in Europe are expected to increase slightly in the medium term, inching up from 11.5 mb/d in 2025 to 12 mb/d in 2030 following a slight demand increase. However, over the longer term, refinery throughputs in Europe are projected to decline from 12 mb/d in 2030 to just below 10 mb/d in 2050. This is due to declining oil demand and the lack of sufficient domestic oil supply. Consequently, the utilization rate is set to increase from 82.9% in 2025 to 87.8% in 2030, also supported by shrinking refinery capacity. Beyond 2030, utilization rates are expected to drop to 72.5% in 2050, which will almost certainly lead to refinery closures in the long term. Many refineries, however, could be repurposed for biofuels, synthetic fuel or hydrogen production, or serve as terminals.

In Russia & Other Eurasia, refinery runs are projected to increase marginally between 2025 and 2035 to 6.8 mb/d, followed by a minimal decline to its 2025 level of 6.6 mb/d by 2050. The major driver is peaking oil demand in this region after 2040. Accordingly, utilization rates in Russia & Other Eurasia are expected to increase marginally from almost 77.7% in 2025 to 78% by 2035, then start declining to 75.3% in 2050. This could lead to limited capacity closures, especially in the last decade of the forecast period.

Conversely, other regions show rising refinery throughputs over the forecast period. The largest increment in refinery runs is expected in India, on the back of strong oil demand growth and significant refinery capacity additions. Refinery throughputs in India are expected to increase steadily from about 5.4 mb/d in 2025 to 10.7 mb/d in 2050, representing growth of 5.3 mb/d. This growth is evenly distributed throughout the forecast period. However, refinery utilization rates are expected to drop gradually from around 104% in 2025 to 95.2% by 2050. This reflects significant refinery capacity expansion and assumes that older refineries will not be able to maintain their utilization rates at their highest levels, above 100%.

Africa is also set to see a significant increase in refinery crude runs, rising by 3.4 mb/d between 2025 and 2050. African refinery throughputs are set to expand strongly out to 2035, followed by gradual growth thereafter, in line with rising oil demand and increasing refining capacity. Refinery throughputs in the continent are set to rise from 2.4 mb/d in 2025 to 4.6 mb/d in 2035 and then continue increasing, albeit at a slower pace, to 5.8 mb/d by 2050. Fast refinery capacity expansions and the modernization of existing refineries is crucial for the continent to increase domestic throughputs and reduce its dependency on imported refined products. Based on these assumptions, Africa's refinery utilization rates are also expected to increase accordingly, moving up from around 55.1% in 2025 to around 77.6% in 2050. However, it is expected that Africa will be exposed to rising international competition. This is especially the case with the US and to a lesser extent Europe, where refiners are expected to increasingly turn to international downstream markets in the long term.

Latin America sees growth in refinery throughputs of 2.3 mb/d between 2025 and 2050. Refinery runs are set to increase significantly over the period to 2040, rising from 4.8 mb/d in 2025 to 6.6 mb/d in 2040, reflecting strong demand growth and rising domestic oil supplies. However, the increase is set to decelerate thereafter to reach 7.1 mb/d by 2050. Refinery utilization rates are projected to increase strongly from around 59.1% in 2025 to 78.5% in 2050. This assumes the modernization of existing refineries, in addition to projected refinery capacity expansion. This would enable the region to limit its dependency on refined product imports and profit from domestic crude oil supplies.

In the Middle East, refinery throughputs are projected to rise from 8.9 mb/d in 2025 to 10.8 mb/d in 2050. This will be driven by domestic demand growth and rising product exports to other regions. Available domestic crude oil supplies also support rising refinery throughputs. In line with rising throughputs, utilization rates are projected to gradually increase to 79.1% in 2050, up from 76.5% in 2025. Limited closures and the rationalization of older capacities in the long term cannot be excluded.

In Other Asia-Pacific, refinery throughputs are projected to increase strongly in the medium term, on the back of robust demand growth and, to a lesser extent, expected refinery capacity additions set to be commissioned over this time frame. Refinery runs are set to rise from 11.1 mb/d in 2025 to 12.4 mb/d in 2030. The growth in throughputs is then expected to decelerate gradually. Runs are set to reach 12.8 mb/d in 2035 before hitting 13 mb/d in 2050. Due to limited refinery capacity expansion, refinery utilization rates are set to increase strongly in the medium term, reaching 81.4% in 2030, up from 76.4% in 2025. Thereafter, a gradual decline is expected, mirroring significant refinery capacity additions. Utilization rates are seen at 75.8% in 2050.

Finally, China's refinery throughputs are expected to witness two distinct periods. Refinery runs are set to increase from 14.8 mb/d in 2025 to 16.3 mb/d by 2035, on the back of growing domestic oil demand. Runs are then seen declining gradually to around 15.6 mb/d by 2050. The combination of declining oil demand and a rising share of non-refinery products in the demand mix leads to declining refinery runs beyond 2035. Consequently, refinery utilization rates are expected to decrease gradually from 82.8% in 2025 to 80.1% in 2035, due to the significant increase in refinery additions. Thereafter, utilization rates continue falling to 75.5% in 2050. A potential implication of this development will be closures of older refining capacity, including teapots (see Section 5.4).

5.4 Refinery closures

This section discusses refinery closures in the medium and long term at the global and regional levels. Two different methodologies are applied in the analysis. Refinery closure projections in the medium term include firm and probable closures, largely based on announcements by companies and governments, and an analysis of refinery closures. In the long term (beyond 2030), the outlook is much more uncertain. The analysis is based on projections for regional utilization rates, and a conclusion is drawn on how many closures would be needed to keep regional utilization rates at technically and financially sustainable levels until the end of the outlook period.

Refinery closures in the medium term

Table 5.5 provides an overview of historical and projected refinery closures by major regions in the period to 2030.

Table 5.5
Recent and projected refinery closures by region, mb/d

	Total 2020–2025	2026	2027	2028	2029	2030	Total 2026–2030
US & Canada	1.8	0.1	0.0	0.0	0.0	0.0	0.1
Latin America	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Africa	0.3	0.0	0.0	0.0	0.0	0.0	0.0
Europe	1.3	0.1	0.0	0.1	0.0	0.0	0.2
Russia & Other Eurasia	0.2	0.0	0.0	0.0	0.0	0.0	0.0
Middle East	0.1	0.0	0.0	0.0	0.0	0.0	0.0
China	1.4	0.0	0.0	0.0	0.0	0.0	0.0
India	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Other Asia-Pacific	1.2	0.0	0.0	0.0	0.0	0.0	0.0
Total	**6.3**	**0.2**	**0.0**	**0.1**	**0.0**	**0.0**	**0.3**

Source: OPEC.

Prior to 2020, global refining capacity appeared to be approaching a phase of potential rationalization, following a ten-year high level of refinery spare capacity reached in 2019 (Figure 5.19), driven by significant additions of new state-of-the-art capacities and the rise of other energy sources in the global energy mix. The sudden drop in oil demand caused by the pandemic and the corresponding unfavourable economics in the downstream sector triggered a wave of refinery closures. It was primarily the older, less efficient, and less profitable facilities that were forced to cease operations.

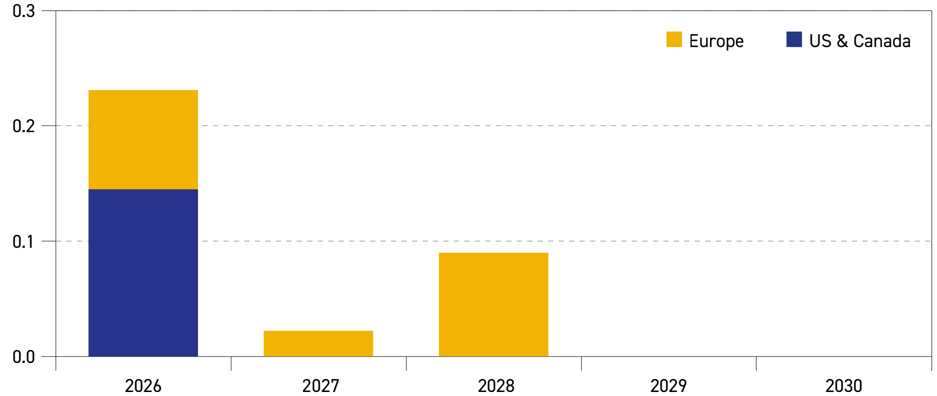
As a result, about 6.3 mb/d of refining capacity was closed between 2020 and 2025. Significant closures of around 3.9 mb/d occurred in developed countries (US & Canada, Europe and the developed countries in Other Asia-Pacific). The US & Canada experienced the largest volume of refinery capacity closures globally over the 2020–2025 period of around 1.8 mb/d – driven primarily by the US. The COVID-19 demand shock accelerated the shutdown of older, less complex refineries, particularly those facing high compliance costs and weak regional margins. China also closed a significant amount of refining capacity at around 1.4 mb/d over the same period, due to government policies aimed at closing old and inefficient teapot refineries and replacing them with new projects.

Closures in Europe reached almost 1.3 mb/d between 2020 and 2025, with steady refinery rationalization, driven by policy rather than sudden closures. Demand decline for gasoline and diesel, combined with high energy prices, carbon costs and tough competition, steadily undermined the economics of small, medium-sized and simple plants. Many refineries closed completely or were partially shut down, while others shifted towards biofuel production or terminal operations.

Other Asia-Pacific, led by developed economies such as Japan and Australia, closed 1.2 mb/d between 2020 and 2025, while Africa witnessed some refinery closures of around 0.3 mb/d, mainly attributed to several ageing plants in South Africa. Minor closures occurred in Russia & Other Eurasia and the Middle East.

Looking ahead, announced refinery closures amount to more than 0.3 mb/d in the medium-term timeline, primarily during 2026. Closures are set to occur mostly in Europe and the US & Canada. No closures have been announced or are expected in other regions in the period to 2030. Figure 5.21 depicts projected refinery closures by region for the period 2026 to 2030.

Figure 5.21
Projected refinery closures by region, mb/d



Source: OPEC.

Europe is set to account for almost 60% of global closures between 2026 and 2030. This includes the closure of Gelsenkirchen refinery in Germany in 2026, as well as the conversion of Lysekil refinery in Sweden and Petrobrazi in Romania by 2027 and 2028, respectively. In the US & Canada, Valero Energy announced the closure of its 145 tb/d Benecia Refinery in San Antonio by the second quarter of 2026.

It is also worth noting that refinery closures projected for the medium term in this outlook are significantly lower than the levels recorded in the period 2020 to 2025, when annual closures averaged around 1.1 mb/d, driven largely by the pandemic and related regional demand patterns. In contrast, closures in the coming years are expected to average below 0.1 mb/d annually, indicating that the major wave of rationalization has potentially subsided. Strong demand growth and improved refining margins have certainly helped keep many plants operating. It is important to note, however, that some refiners, especially those outside of Europe and the US, do not officially publish closure announcements, making it particularly difficult to anticipate capacity withdrawals from the market.

That said, beyond officially announced capacity closures, additional refinery capacities could be shut down in the medium term due to factors such as technical issues at ageing facilities, weak financial performance and evolving government policies. In China, for instance, several refineries – particularly smaller, independent 'teapot' facilities – are expected to cease operations. It is envisaged that they will be partly replaced by new, state-of-the-art integrated refineries. These modern plants, combined with government initiatives to increase energy efficiency and limit the country's refining capacity to 20 mb/d to boost utilization rates, are putting increasing pressure on older, less competitive refineries.

Refinery closures in the long term

As per the applied methodology, refinery closures in the long term (beyond 2030) are not explicitly projected. Instead, only so-called implied refinery closures are calculated and indicated, based on the long-term modelling results. In more detail, implied refinery closures are back-calculated while targeting a long-term sustainable average utilization rate at a regional level. While, it is assumed higher in India, this rate hovers around 80% in developed regions and China. However, it is different in other regions like Africa, Latin America and – to some extent – the Asia-Pacific and the Middle East.

The general assumption is that most of the implied closures will comprise simple, non-integrated and less efficient plants that will struggle to compete against complex and integrated plants once utilization rates start declining. It is important to note that long-term modelling cases already consider medium-term closures of around 0.3 mb/d.

Based on regional refinery utilization patterns in the long term, refining capacity of more than 4 mb/d could be closed globally if reasonable utilization rates are to be maintained, on top of the assessed medium-term closures. Due to the expected oil demand decline, stringent energy policies and rising competition, Europe could witness the largest share of closures in the long term, estimated at almost 2 mb/d. In this regard, it is important to note that, historically, Europe has been an important supplier of refined products to Africa. If the projected capacity additions in Africa do not materialize at the rate expected, this could support the European refining market and postpone closures to later dates.

In the US & Canada, the average utilization rate in 2050 is expected to remain above 85%. While this is comparatively high in international terms, it is still lower than the region's historical levels, which have at times exceeded 90%. This is why some closures of old and less complex units are possible in the long term. Due to the high level of competitiveness of this region's refineries in the global market, exports of refined products to other regions could increase significantly in the long term, which would help limit or delay possible additional closures in the region.

In China, some closures in the long term are possible and could include less efficient teapot refiners that could face challenges operating in a market dominated by large integrated plants. Some teapot plants were already shut at the beginning of this decade and replaced by large-scale refineries, many of which are integrated with petrochemicals. Furthermore, China's NDRC introduced several regulations and guidelines that could negatively affect the teapot sector, including potential closures in the medium and long term. These include limiting China's overall crude processing capacity to 20 mb/d and forcing the closure of small plants of less than 40 tb/d. China is also seeking to increase the energy efficiency of its refining sector. The benchmark level has been established at 8.5 kilograms of oil equivalent per tonne (kgoe/t), and all plants above this level should be phased out. These measures should help reduce the carbon footprint of the downstream industry in China and this represents a significant risk for small teapot plants. Overall, closures in China could exceed 1 mb/d of capacity beyond 2030.

Developed countries in Other Asia-Pacific could also see some closures. Significant capacity has already been shut in the region, particularly in Japan and to a lesser extent in Australia. While Japan has seen several closures in recent years, it has not announced specific closures for the medium term. However, the potential for closures in the medium and long term remains high. Japan's smaller and less complex refineries were built mainly to serve its domestic fuel needs and are also set to process lighter and sweeter crude oil grades, which are more expensive than heavier and sourer grades. The combination of higher yields of lower-value products, the use of more expensive crude oils and a decline in long term domestic oil demand, makes refiners in Japan less profitable and less competitive in global markets.

Finally, in Latin America and Africa, closures are possible throughout the outlook period. Both regions have many older and inefficient refineries, which operate at relatively low or even close-to-zero utilization rates. In addition, some countries have plans to build large-scale and sophisticated capacities to replace older ones. Other countries are trying to modernize existing refineries, but these efforts currently remain insufficient due to financial and technical issues. This is why some closures in these two regions can be expected in the long term.

5.5 Secondary capacity

Refining capacity is generally denoted by primary, i.e. distillation capacity. However, it is secondary capacity that includes conversion and product quality improvement units that are crucial for processing crude fractions into finished products that provide most of a refinery's 'value-added'. Secondary capacity provides flexibility to the refining system to meet the final product demand of a region, including seasonal and structural changes over time. The development of secondary capacity goes together with evolving refined product demand and product specifications, such as sulphur and aromatics content, as well as octane number standards. The development of secondary capacity also depends on the evolving crude slate, which changes over time. A heavier crude slate normally requires additional upgrading capacities, particularly if a large share of residual fuel is to be avoided.

The focus of this section is on secondary capacity additions in the medium and long term by major categories, namely conversion, desulphurization and octane units. Similar to distillation capacity, the Outlook provides projections for secondary capacity additions in the medium term based on existing projects, which are likely to be commissioned through 2030. The long-term outlook for secondary capacity additions is assessed based on the modelling results for anchor years, in terms of required capacity to meet regional and global product demand.

5.5.1 Medium-term secondary capacity additions

In terms of medium-term secondary capacity additions related to a list of specific projects that are under development, most projects are directly linked to distillation capacity additions that are estimated at 4.9 mb/d (excluding debottlenecking) in the period to 2030. Details on secondary capacity additions in the same period are shown in Table 5.6. These include 2.9 mb/d of conversion/upgrading capacity, 3.8 mb/d of desulphurization capacity and 1.1 mb/d of octane units.

Most of these additions are projected to be commissioned in the Middle East and Asia-Pacific (including China and India), as well as Africa. These regions account for around 90% of global

Table 5.6
Secondary capacity additions from existing projects, 2026–2030, mb/d

	By year		
	Conversion	Desulphurization*	Octane units
2026	0.7	1.1	0.2
2027	0.4	0.4	0.1
2028	0.4	0.6	0.3
2029	0.6	0.7	0.2
2030	0.7	1.0	0.3
	2.9	**3.8**	**1.1**
	By region		
	Conversion	**Desulphurization***	**Octane units**
US & Canada	0.0	0.1	0.0
Latin America	0.0	0.1	0.0
Africa	0.3	0.4	0.2
Europe	0.0	0.0	0.0
Russia & Other Eurasia	0.3	0.3	0.0
Middle East	0.4	1.1	0.2
China	0.7	0.6	0.2
India	0.7	0.8	0.2
Other Asia-Pacific	0.4	0.5	0.2
World	**2.9**	**3.8**	**1.1**

* *Desulphurization capacity in this table includes naphtha desulphurization.*
Source: OPEC.

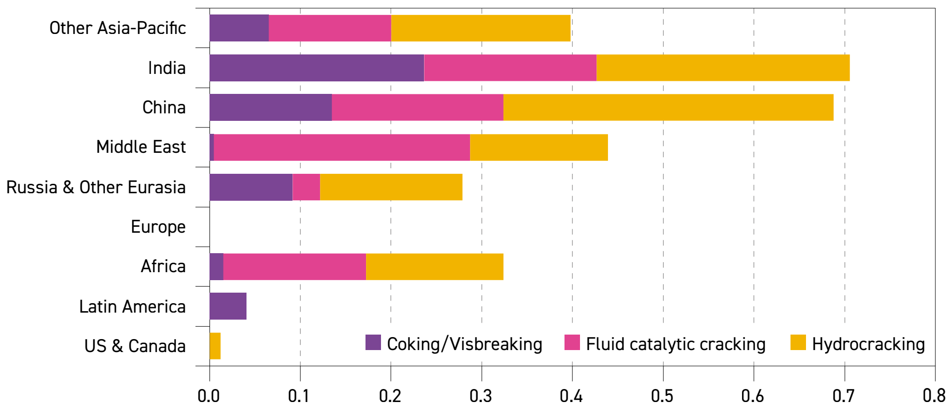
secondary capacity increments. This is somewhat lower than the share these regions have for distillation capacity additions. This is due to continuous additions of secondary capacity in other regions, related to upgrades and/or modernization of existing refineries, particularly in Russia & Other Eurasia.

It is worth noting that the rate of secondary capacity additions in relation to new distillation capacity is comparatively higher than the rate observed in the existing refinery system (see Table 5.1). This is mainly due to the large number of new refineries – especially in the Middle East and Asia-Pacific – that are highly complex and integrated and designed to process medium- and heavy-sour crude.

Conversion units

Figure 5.22 shows conversion capacity additions in the medium term by region and technology. At the global level, out of 2.9 mb/d of conversion unit additions, more than 1.3 mb/d (or 45%) are for hydrocracking units. The hydrocracker's inherent flexibility to handle a wide range of feedstocks, as well as its ability to produce ULS middle and light distillates, positions it as a key upgrading technology in modern refining configurations. In addition to hydrocracking, around 1 mb/d of FCC and around 0.6 mb/d of coking capacities are expected to be added in the period to 2030.

Figure 5.22
Conversion projects by region, 2026–2030, mb/d



Source: OPEC.

At a regional level, most medium-term conversion capacity additions are projected to occur in the Asia-Pacific (including China and India), Africa and the Middle East, and to an extent in Russia & Other Eurasia. India and China are set to lead the global expansion in conversion units, adding around 0.7 mb/d each. The Middle East is expected to commission more than 0.4 mb/d of additional conversion capacity, while Other Asia-Pacific and Africa are set to add 0.4 and 0.3 mb/d, respectively. Despite its low additions in distillation capacity, Russia & Other Eurasia is anticipated to see conversion capacity grow (mostly hydrocracking) by around 0.3 mb/d. This reflects the region's efforts to upgrade existing plants and reduce fuel oil output, especially in Russia. Finally, minor or no additions are expected in Latin America, the US & Canada and Europe.

Desulphurization units

In the medium term, around 3.8 mb/d of new desulphurization capacity is likely to be commissioned. The Asia-Pacific (including China and India), the Middle East, Africa and Russia & Other Eurasia combined are set to account for almost all global capacity additions, at around 95%, or 3.6 mb/d. The Middle East is set to lead global desulphurization capacity additions with 1.1 mb/d over the medium term. India and China are expected to add 0.8 mb/d and 0.6 mb/d of new capacity, respectively. At the same time, capacity additions in Other Asia-Pacific are estimated at almost 0.5 mb/d, while they are likely to amount to 0.4 mb/d and 0.3 mb/d in Africa and Russia & Other Eurasia, respectively. Only limited or no additions to desulphurization capacity are expected in Latin America, the US & Canada and Europe.

Looking at desulphurization capacity additions from the perspective of feedstock processing, 1.8 mb/d of capacity is dedicated to process middle distillates; around 1 mb/d is designated for naphtha processing, almost 0.7 mb/d for gasoline and the remainder for heavy streams (e.g. vacuum gasoil and residue).

Octane units

Around 1.1 mb/d of global octane unit capacity is likely to come online in the period between 2026 and 2030. This reflects growing gasoline demand, mostly in developing regions, as well as rising octane number specifications in some countries, including in some Eurasian and Latin American countries. Africa and the Middle East are expected to add more than 0.2 mb/d of octane unit capacity over the medium term, followed by China, India and Other Asia-Pacific with less than 0.2 mb/d each.

Additions in other regions are expected to be limited in the medium term due to a lack of gasoline demand growth, such as in Europe, or because significant octane unit capacities are already in place, as in the US & Canada.

From a technological perspective, the majority of octane unit additions – around 0.8 mb/d or 74% of the total – are expected to be from catalytic reforming. This will be accompanied by almost 0.2 mb/d of new isomerization capacity and more than 0.1 mb/d of added alkylation capacity. Capacity additions of MTBE/ethyl tertiary-butyl ether (ETBE) are expected to be limited in the medium term (less than 0.1 mb/d). The potential is focused on developing countries, especially in the Asia-Pacific.

5.2.2 Long-term secondary capacity additions

Long-term secondary capacity additions are set to be driven by the level and composition of oil demand, evolving product specifications, as well as crude oil quality. Many recent refinery additions have already included relatively large and complex units with high levels of upgrading, desulphurization and related secondary processing, generally with an increased focus on petrochemical integration. This is likely to continue in the future, in line with oil demand trends.

Global trends demonstrate strong demand growth for ethane/LPG and naphtha in the long term, supported by a continued focus on petrochemicals. Furthermore, global gasoline demand is expected to increase in the first part of the outlook period, followed by slower growth thereafter. This is mostly due to the offsetting effect between OECD and non-OECD demand trends. Middle distillates are also set to grow strongly, especially jet/kerosene. At the same time, the outlook sees increased supplies of medium- and heavy-sour crude, especially after 2030, raising requirements for additional upgrading and desulphurization capacities.

A few exceptions to the trend towards increased complexity should be highlighted. These are related to new condensate splitters, particularly in the Middle East. Condensate splitters tend to contain limited secondary processing, often related to the processing of light products such as naphtha and gasoline, and are centred on catalytic reforming, isomerization and hydrotreating. This trend could continue, as the share of condensates and NGLs in overall liquid supply is likely to increase.

In setting out to capture the outlook for global and regional refining, particularly future processing needs by type of unit, modelling is forced to contend with several challenges. One relates to the evolution of refinery process technology. This is assumed to be stable, with only gradual changes over time, mainly as catalysts slowly improve.

The emerging trend to increase petrochemical yields represents a second potential modelling challenge. While many existing refineries in the US and Europe have some degree of petrochemical capability, the number of large integrated refining plus petrochemical 'mega-projects' continues to rise, especially in the Middle East and the Asia-Pacific. Several of these new complexes are designed to produce a significant share of petrochemical feedstocks, often 40% or more. 'Crude-to-chemicals' technology represents a step forward in this direction. In addition to the flexibility and cost savings it provides, it reduces GHG emissions and the overall environmental impact by optimizing the conversion process and minimizing energy use.

Table 5.7 and Figure 5.23 show global secondary capacity requirements in addition to required primary capacity additions in the period to 2050. On top of 19.3 mb/d for distillation

Table 5.7
Global capacity requirements by process, 2026–2050, mb/d

	Existing projects to 2030*	Additional requirements		Total additions to 2050
		2030–2040	2040–2050	
Crude distillation	**4.9**	**10.5**	**3.9**	**19.3**
Conversion	**2.9**	**4.9**	**3.1**	**10.9**
Coking/Visbreaking	0.6	1.8	0.6	3.0
Catalytic cracking	1.0	1.0	0.9	2.9
Hydrocracking	1.3	2.1	1.6	5.0
Desulphurization**	**2.8**	**11.8**	**6.1**	**20.7**
Gasoline	0.7	2.4	1.1	4.2
Distillate	1.8	8.0	4.4	14.3
VGO/Resid	0.3	1.4	0.5	2.2
Octane units***	**1.1**	**3.4**	**1.6**	**6.1**
Catalytic reforming	0.8	2.0	1.3	4.1
Alkylation	0.1	1.3	0.3	1.6
Isomerization	0.2	0.0	0.0	0.2
MTBE	0.0	0.1	0.0	0.1

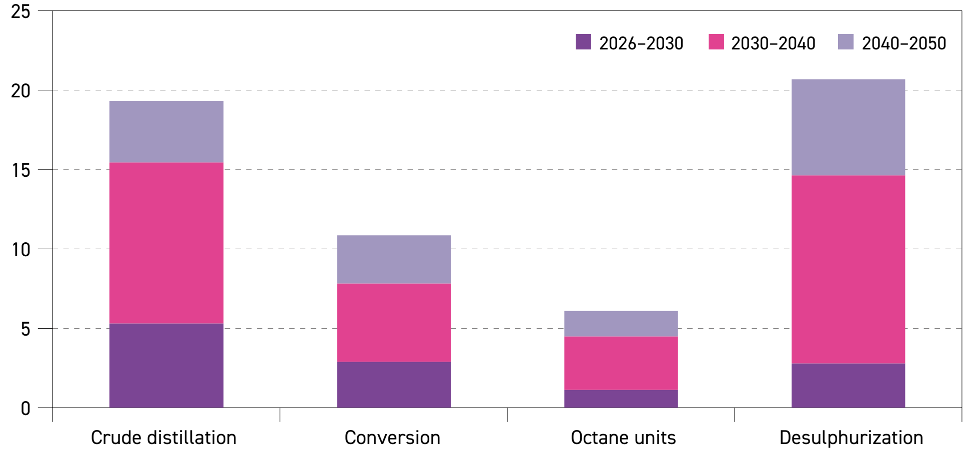
* *Existing projects exclude additions resulting from 'capacity creep'.*

** *Naphtha desulphurization not included.*

*** *New units only (excludes any revamping).*

Source: OPEC.

Figure 5.23
Global capacity requirements by process type, 2026–2050, mb/d



Source: OPEC.

capacity, there are requirements for around 10.9 mb/d of conversion capacity, 20.7 mb/d for desulphurization and 6.1 mb/d for octane units. These projections consider capacity additions assumed for the period 2026–2030, as already discussed.

In line with the medium-term trends, the rate of long-term secondary capacity additions in relation to new distillation capacity is higher than the rate observed in the existing refining system (see Section 5.1). Desulphurization capacity additions will account for the largest ratio, at 107%, followed by conversion units with 56%, and octane units with 31% relative to primary capacity additions.

Similar to distillation capacity expansions, the majority of new secondary capacity additions are expected to materialize in the first part of the forecast period. Thereafter, additions are set to be mostly limited to the expansion and modernization of existing plants, particularly towards the end of the long-term period.

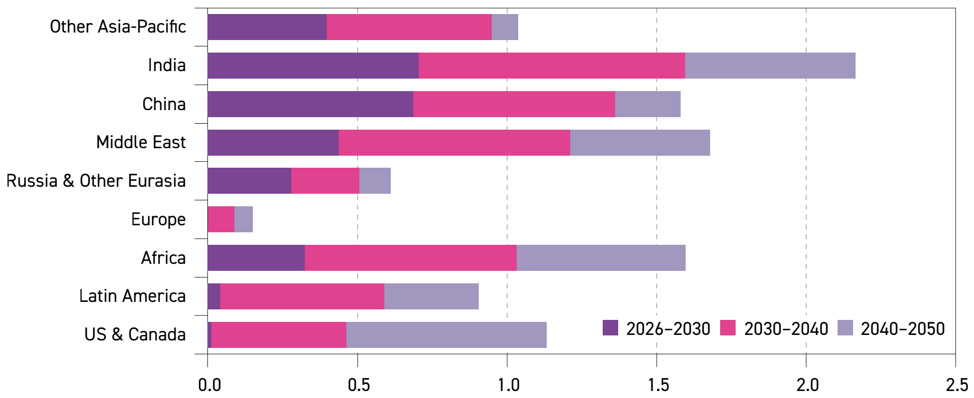
5

Conversion units

Figure 5.24 presents long-term conversion capacity requirements by region. In line with growing oil demand, the Asia-Pacific, the Middle East and Africa are expected to host most conversion capacity additions, accounting for 74% of the total. It is also important to note, however, that significant conversion capacity additions in other regions, such as the US & Canada, Latin America and to a lesser extent Russia & Other Eurasia, are expected, despite minimal distillation capacity additions. This reflects refiners' efforts in these regions, particularly in the US & Canada, to further upgrade their refining systems, enhance operational flexibility and improve overall competitiveness. Only limited additions are anticipated in Europe.

From a technological perspective, hydrocracking accounts for the largest increment, with 5 mb/d of the global required conversion capacity, as shown in Table 5.7. Hydrocracking is expected to remain the preferred upgrading option for many refiners. This is due to significant advancements in this technology, which can be tailored to adapt to different feed qualities and targeted products. It also reflects its ability to achieve deep conversion of bottom-barrel

Figure 5.24
Conversion capacity requirements by region, 2026–2050, mb/d



Source: OPEC.

streams to improve yields of light products, particularly petrochemical feedstocks, such as LPG and naphtha.

At the regional level, around 78% of hydrocracking capacity additions are expected in the Asia-Pacific (including China and India), the Middle East and Africa, totalling more than 3.9 mb/d over the long-term period. India is set to lead the expansion, with 1 mb/d alone, followed by the Middle East and China with 0.8 m/d each. Africa and Other Asia-Pacific will add 0.7 mb/d and 0.6 mb/d, respectively. Finally, due to upgrades of some existing refineries, Latin America, Russia & Other Eurasia and the US & Canada are likely to add around 0.3 mb/d of new hydrocracking capacity, each.

In the long term, around 2.9 mb/d of new FCC capacity will be added at the global level. FCC additions are primarily driven by gasoline demand, but also by light olefins. This is why most new FCC units are expected in developing regions, where gasoline demand is still likely to increase throughout the outlook period. At the same time, gasoline demand in developed countries is expected to peak in the coming years and then start declining in the long term, which is why almost no FCC additions are expected in these regions.

Similar to new hydrocracking capacities, developing regions are set to account for the majority of FCC capacity expansions in the period to 2050. Additions in India, China, the Middle East and Africa are in the range of 0.5 mb/d each. At the same time, Other Asia-Pacific is expected to add about 0.3 mb/d, followed by Latin America and the US & Canada with 0.2 mb/d each. Elsewhere, Russia & Other Eurasia is set to see minor expansions to its FCC capacity, of just below 0.1 mb/d, while no FCC capacity expansions are expected in Europe.

Finally, around 3 mb/d of coking/visbreaking (predominantly coking) capacity is required in the period to 2050. It should be noted that the modelling projections exclude upgraders of oil sands, as well as heavy Venezuelan and other crudes, as they employ projected volumes for crude streams delivered to the market, i.e. downstream of upgraders and blending.

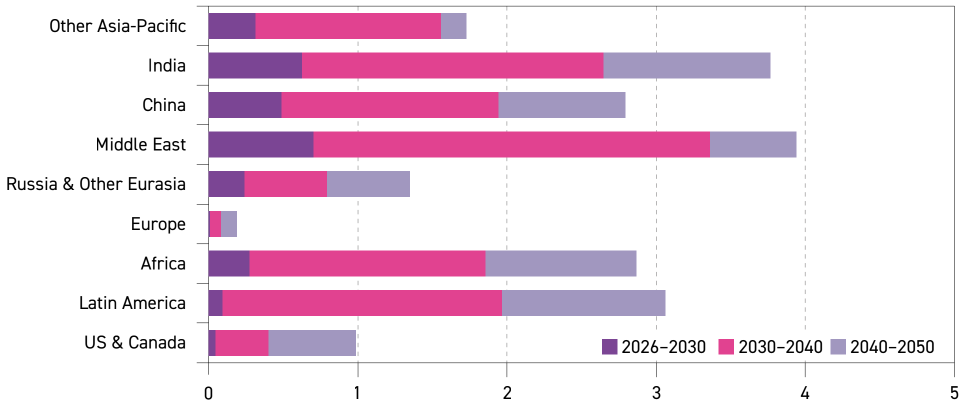
At the regional level, the Asia-Pacific is set to dominate coking/visbreaking capacity expansions, with more than 1 mb/d envisaged in the period between 2026 and 2050. Unlike other conversion technologies, significant capacity additions are also set to come from the US & Canada and Latin America, with 0.6 mb/d and 0.4 mb/d, respectively. Africa's coking additions over the outlook period are estimated at around 0.4 mb/d. Finally, roughly the same level of capacity additions is expected in the Middle East, Russia & Other Eurasia and China, at around 0.2 mb/d each.

Desulphurization units

Total desulphurization capacity requirements over the outlook period to 2050 are estimated at around 20.7 mb/d, somewhat higher than the required distillation capacity in the same period. This reflects the rising sulphur content of the average barrel, the shift to higher-quality fuels, as well as increasingly stringent environmental regulations related mostly to sulphur content in transportation fuels, especially in developing countries.

Figure 5.25 shows desulphurization capacity requirements by region and period. On top of 2.8 mb/d of global desulphurization capacity expected to be added during the period between 2026 and 2030, more than 11.8 mb/d of new capacity additions are projected from 2030 to 2040, followed by slower growth in the last decade of the outlook – with projected additions of 6.1 mb/d.

Figure 5.25
Desulphurization capacity requirements by region*, 2026–2050, mb/d



* *Projects and additions exclude naphtha desulphurization.*
Source: OPEC.

On a regional basis, the Middle East is set to lead additions with a substantial 4 mb/d increase throughout the forecast period, in line with the high sulphur content of local crude supplies. India is set to see an expansion of 3.8 mb/d. High desulphurization additions are in line with rising crude imports from the Middle East in the long term (see Chapter 6).

Desulphurization additions in Latin America are set to be significant and estimated to be almost 3.1 mb/d, driven by evolving ULS standards, as well as the relatively high sulphur content of a large share of domestic crude supplies. The additions are mostly related to the modernization of exiting refineries.

Even though a large share of Africa's crude supply has a relatively low sulphur content, new desulphurization capacity is expected to be significant, at around

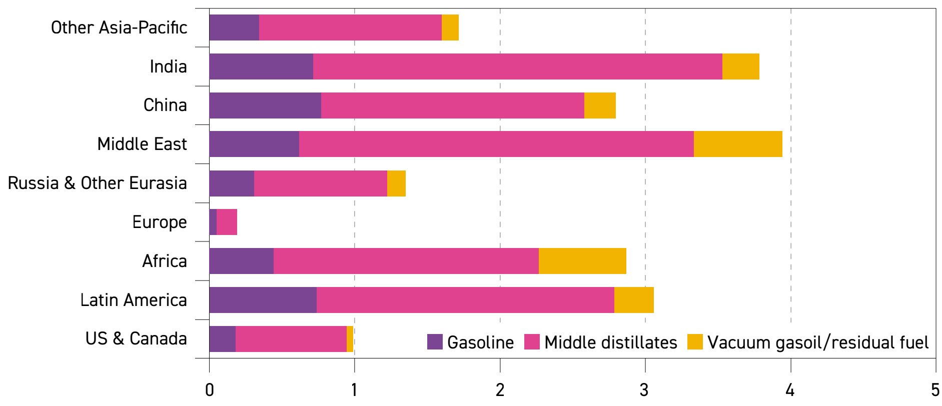
2.9 mb/d. This reflects the ongoing efforts of many African countries to strengthen fuel sulphur standards and bring them closer to those in place in other regions, particularly developed countries.

China is also set to witness significant desulphurization capacity additions, accounting for almost 2.8 mb/d, followed by Other Asia-Pacific at 1.7 mb/d. This is driven by rising crude imports from the Middle East and Latin America in the long term. Russia & Other Eurasia is set to add almost 1.3 mb/d to its existing desulphurization capacity.

Finally, the US & Canada, along with Europe, are expected to see additions of around 1 mb/d and 0.2 mb/d, respectively, mostly related to the modernization of existing capacities. The well-established desulphurization capacity in these regions, along with declining fuel demand in the long term, makes additional investments in this area unnecessary.

In terms of feedstock processing, desulphurization capacities related to middle distillates account for most required additions in the period to 2050. Figure 5.26 presents desulphurization capacity requirements by product and region between 2026 and 2050. At the global level, almost 14.3 mb/d of middle distillate (diesel and jet/kerosene) desulphurization capacity is expected to be added. Rising middle distillate demand throughout the outlook period and increasingly stringent regulations on sulphur levels in diesel (towards ULS standards) are the major drivers. The majority of the capacity additions are expected in the Asia-Pacific, the Middle East, Africa and Latin America.

Figure 5.26
Desulphurization capacity requirements by product and region*, 2026–2050, mb/d



* *Projects and additions exclude naphtha desulphurization.*
Source: OPEC.

Gasoline hydrotreating additions (excluding naphtha) are estimated at almost 4.2 mb/d and are projected to materialize mostly in developing regions, where gasoline demand is still expected to grow. This includes the Asia-Pacific, the Middle East, Latin America and Africa.

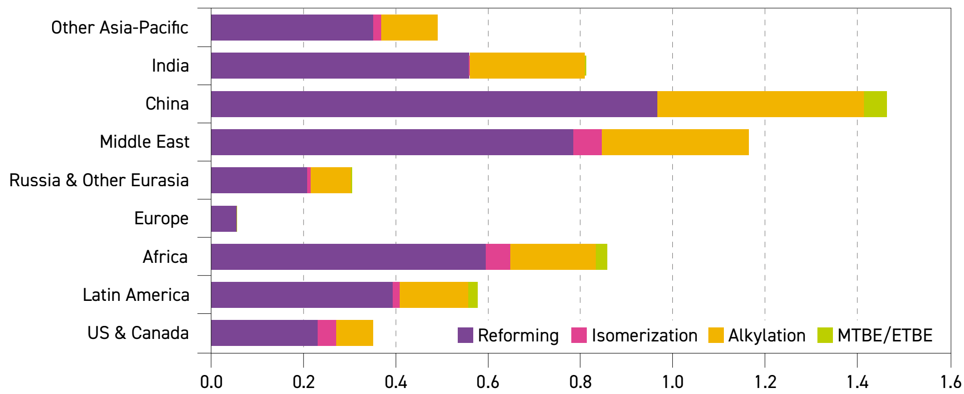
Vacuum gasoil/residual fuel desulphurization capacity is expected to grow by 2.2 mb/d in the long term. One of the drivers behind this trend is the rising demand for Very Low Sulphur Fuel

Oil (VLSFO) due to the IMO Sulphur Rule. Another reason is the substantial hydrocracking capacity additions (as shown in Table 5.7) that often require feedstock pre-treatment to remove impurities such as sulphur. Most additions are expected in the Middle East and the Asia-Pacific. Both regions predominantly process crude with a high sulphur content. Most of the remaining vacuum gasoil/residual fuel desulphurization capacity is required in Africa, Latin America and India, with some minor amounts in other regions.

Octane units

Figure 5.27 shows octane capacity requirements by process and region between 2026 and 2050. Around 6.1 mb/d of octane units will be required at the global level in the long term.

Figure 5.27
Octane capacity requirements by process and region, 2026–2050, mb/d



Source: OPEC.

At the regional level, around 2.8 mb/d, or 45% of these additions are projected for the Asia-Pacific (including China and India), supported by rising gasoline demand. Octane unit growth is also significant in the Middle East at 1.2 mb/d, reflecting the region's efforts to increase gasoline exports. Africa is set to add 0.9 mb/d over the long term, in line with growing distillation capacity. In Latin America, gasoline upgrading capacity is set to grow by 0.6 mb/d, reflecting efforts by some countries to improve gasoline octane specifications. Traditional gasoline markets – the US & Canada and Russia & Other Eurasia – are projected to add 0.4 mb/d and 0.3 mb/d of new octane units in the long term, respectively. Modest additions are expected in Europe over the forecast period.

Gasoline upgrading capacity additions are dominated by catalytic reforming, which is projected to expand by more than 4.1 mb/d in the period to 2050. The catalytic reforming converts low-octane naphtha into high-octane reformate, a key component in gasoline blending. This allows refineries to incorporate larger quantities of naphtha, including light streams derived from condensates into the gasoline blend while still meeting octane specifications. Alkylation accounts for 1.6 mb/d. Global isomerization and MTBE/ETBE additions will likely be only minor, at around 0.2 mb/d and 0.1 mb/d, respectively. Some markets in Asia still use MTBE as a gasoline enhancer and are the major drivers of these additions.

5.6 Refined products supply and demand balances

This section discusses refined products supply and demand balances at regional and global levels. It is important to note that while assessing the effects of capacity additions on specific products, refiners always have some limited flexibility to optimize their product slates. This will depend on seasonal demand patterns, changing market circumstances, demand patterns, as well as economics and the availability of feedstock. This can be done by changing feedstock composition (crude slate) and by adjusting process unit operating modes.

Table 5.8 presents an estimation of the cumulative potential incremental output of refined products resulting from the medium-term (2026–2030) downstream developments by major product category. This assumes a maximum utilization rate of 90% for the units expected to come online in the medium term and considers the potential loss of crude runs due to refinery closures in the same period. In addition, it is important to point out that the balance is relative to the base year of 2025. The refined products balances are also assessed in full compliance with the potential incremental throughputs shown in Section 5.3.

Table 5.8
Global cumulative potential for incremental product output*, 2026–2030, mb/d

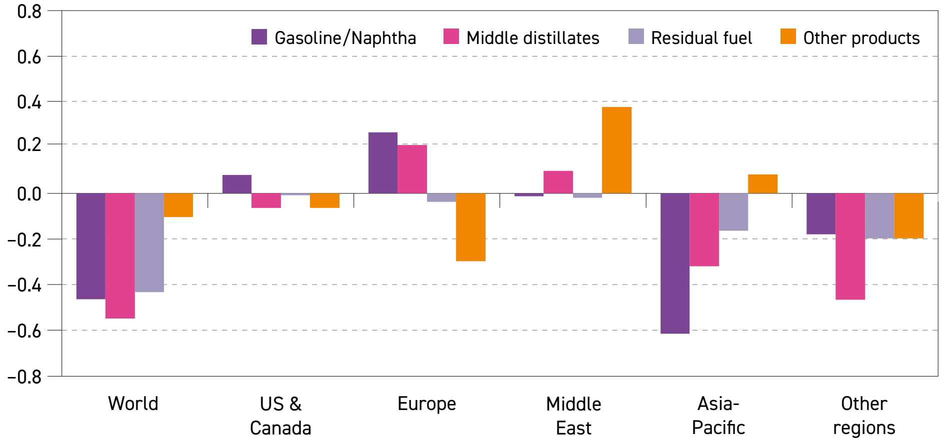
	Gasoline/ Naphtha	Middle distillates	Fuel oil	Other products	Total
2026	0.3	0.4	0.0	0.3	1.0
2027	0.5	0.7	-0.1	0.5	1.6
2028	0.7	1.0	-0.1	0.7	2.3
2029	0.9	1.4	-0.1	1.0	3.3
2030	1.2	2.0	0.1	1.3	4.6
Share (%)	*26.3*	*42.7*	*1.8*	*29.2*	*100*

* *Based on assumed 90% utilization rates for the new units.*
Source: OPEC.

The cumulative potential refining capacity in the period to 2030 is estimated at around 4.6 mb/d at the global level, when compared to 2025. More than 40% of new incremental production capacity is related to middle distillates (2 mb/d). Gasoline/naphtha incremental cumulative output potential is assessed at 1.2 mb/d in 2030, while the potential cumulative output of other products is at 1.3 mb/d. The potential output for fuel oil is at 0.1 mb/d. Once again, this reflects the expected strong secondary capacity additions linked to new, more complex projects, as well as to the ongoing modernization of some existing plants. These projects aim to yield more added-value products, such as gasoline, diesel and jet/kerosene, but also more petrochemical feedstocks like LPG and naphtha, by converting the bottom of the barrel into high-quality products.

Figure 5.28 presents the resulting balance by major product group and selected regions. It is calculated based on the difference between incremental potential output and projected demand growth. It is crucial to note that demand for refinery products is calculated based on total product demand and subtracting any non-refinery streams, including biofuels, CTLs, GTLs and NGLs. It is also important to mention that surpluses can be the result of declining demand.

Figure 5.28
Expected cumulative surplus/deficit* of incremental product output from existing refining projects, 2026–2030, mb/d



* *Declining product demand in some regions contributes to the surplus.*
Source: OPEC.

The global cumulative deficit is estimated at almost 1.6 mb/d by 2030, with all major products showing deficits. The largest deficits emerge for middle distillates and gasoline/naphtha, amounting to 0.5 mb/d each. The deficit for residual fuel is assessed at 0.4 mb/d, while there is a minor deficit in other products.

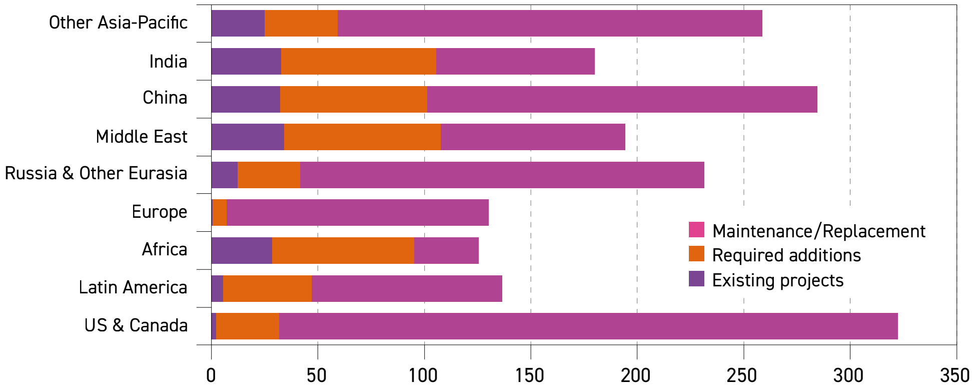
Regionally, the Asia-Pacific (including China and India) is expected to witness a deficit of around 1.0 mb/d by 2030 relative to the base year. In this region, except for other products, all major fuels show a deficit, especially gasoline/naphtha. This may lead to higher throughputs in these regions relative to 2025, and/or increased product imports if projected demand is to be met. The combined cumulative deficit in the remaining regions, including the US & Canada, could reach more than 1.1 mb/d by 2030.

On the other side, a product surplus is expected in the Middle East, at around 0.4 mb/d. This is in line with significant refinery capacity additions relative to oil demand growth in the region, which could potentially lead to an increase in refined products exports. Finally, Europe shows a total surplus of more than 0.1 mb/d, on the back of rising non-refinery products supply, despite expected refinery rationalizations.

5.7 Investment requirements

This section provides details related to downstream investment requirements by region and by category, as shown in Figure 5.29. The investment requirements are grouped into three categories. The first category includes investment costs related to announced refining projects in the medium term (Section 5.2.1). Investment costs in this category are based on available and reported information to the extent possible. The second category encompasses investment requirements for new refinery projects in the period 2030–2050. As these projects are generic, estimates are based on unit refining capacity costs at the regional level. The third and final category is related to necessary continuous replacement and maintenance

Figure 5.29
Refinery investments by region, 2026–2050, $ (2026) billion



Source: OPEC.

CAPEX throughout the entire outlook period. In total, downstream investment requirements are estimated at almost $1.9 trillion over the entire outlook period.

For the first category, a total investment cost of $174 billion is estimated. The largest share of medium-term CAPEX is in developing regions, namely the Asia-Pacific, the Middle East and Africa, in line with refining capacity addition trends. These regions combined are expected to invest around $153 billion in the medium term. Investments in the Middle East account for $34 billion, followed by China and India with around $33 billion, each. Other Asia-Pacific and Africa are likely to invest around $29 billion and $25 billion in the downstream sector in the medium term, respectively.

Investments in Russia & Other Eurasia and Latin America are estimated at around $12 billion and $5 billion, respectively. Elsewhere, modest investments are projected for the US & Canada, while no investments are seen in Europe.

In the period post-2030, global investment requirements for new refinery projects and expansions are evaluated at $423 billion. The largest share is expected in India and the Middle East, estimated at $73 billion each, followed by China and Africa, estimated at $69 billion and $67 billion, respectively. This reflects the significant refining capacity requirements in these regions, as discussed in Section 5.2.2 and Section 5.5.2. In Latin America, required downstream investments are projected at around $42 billion.

In the US & Canada, between 2030 and 2050, investments are estimated at close to $30 billion. In this region, secondary capacity expansions support downstream investments. This partly relates to a gradual change in refinery feedstock, with the average crude barrel becoming heavier due to potential additional volumes from Canada, Latin America and the Middle East. Similar investment amounts are expected in Russia & Other Eurasia – most of which are set to be dedicated to the expansion of secondary capacity. In Europe, downstream-related investments are estimated at below $7 billion over the same period.

Finally, maintenance requirements and the necessary continuous replacement of installed refining capacity are estimated at almost $1.3 trillion for the period 2026–2050. To assess this category, it is assumed that the annual capital needed for capacity maintenance and replacement is around 2% of the cost of the installed base capacity. The US & Canada is expected to lead the way for global maintenance investments at around $290 billion cumulatively, followed by Other Asia-Pacific at almost $200 billion. Russia & Other Eurasia and China also have relatively significant replacement costs of around $190 billion and $183 billion, respectively. Investments required for maintenance and continuous replacement in these regions reflect the size of the installed base refining capacity.

Oil movements

Key takeaways

- Global interregional oil trade (crude oil, condensate and refined products) is estimated at over 55 mb/d in 2025. By 2030, trade is projected to increase significantly to above 62 mb/d, with a gradual increase thereafter to 69 mb/d in 2050.
- Interregional crude and condensate trade stood at around 37.3 mb/d in 2025. By 2030, it is expected to increase to 41.8 mb/d, supported by rising oil demand in major consuming regions. After 2030, the rate of growth slows markedly but volumes still rise to a level of 47.8 mb/d by 2050.
- Total oil product trade was assessed at 18 mb/d in 2025 and is expected to rise above 21 mb/d in 2050. This is in line with increasing demand in the Asia-Pacific and growing exports from the US & Canada and the Middle East.
- The Middle East will expand its role as the world's largest exporting region for crude and condensate, growing from 17.8 mb/d in 2025 to 28.7 mb/d in 2050. Exports to the Asia-Pacific region are set to account for over 80% of the Middle East's total exports throughout the entire outlook period.
- Crude and condensate exports from Latin America are likely to increase strongly from 4.5 mb/d in 2025 to 7.4 mb/d in 2050, driven by robust regional oil supply growth. The majority of crude and condensate flows from Latin America are destined for the Asia-Pacific and the US & Canada.
- Exports of crude and condensates from the Russia & Other Eurasia region, which in 2025 stood at 5.6 mb/d, are projected to gradually decline to a level of 4.5 mb/d in 2050. The Asia-Pacific is anticipated to be the top destination for these exports.
- In Africa, overall exports of crude and condensate were estimated at 5 mb/d in 2025. The region's volume of exports is expected to expand slightly to 5.2 mb/d in 2030 before gradually declining to 4.3 mb/d in 2050.
- Exports of crude and condensates from the US & Canada – excluding flows between the two countries – were assessed at 4.2 mb/d in 2025. Export volumes are projected to rise to over 4.5 mb/d in 2030 and remain at this level out to 2035. In the second half of the outlook, exports are expected to decline noticeably to 2.9 mb/d in 2050.
- In terms of crude and condensate imports, the Asia-Pacific is the largest region by far in this outlook, with imports assessed at 25.1 mb/d in 2025. In the years out to 2030, imports are projected to grow to 28.3 mb/d before increasing to 34.4 mb/d by 2050.
- Crude and condensate imports to Europe were estimated at 8.8 mb/d in 2025. The overall level of imports is anticipated to remain consistent until 2035, before gradually declining in the second half of the outlook period to reach 7.8 mb/d in 2050, in line with the region's declining oil demand trend.

The movement of oil between countries and regions is an essential aspect of a balanced global oil market. Flows of crude oil, condensate and refined products move between producers and consumers to offset supply shortages and surpluses. A well-functioning midstream sector ensures adequate supply to the global downstream sector, providing consumers with flexibility and helping to reduce demand and supply shocks.

This chapter examines the main trends related to the trade movements of crude oil and condensate, as well as refined products, between major regions. Projections are based on the assumptions and modelling results discussed throughout this Outlook, including oil demand (Chapter 3), supply (Chapter 4) and refining (Chapter 5).

6.1 Recent developments

Global oil movements are enabled by vital infrastructure, including pipelines, ports and railways. As oil demand and supply trends evolve, the continuous development of existing and new infrastructure is important for maintaining oil trade. When considering the outlook for global trade movements, existing and planned infrastructure represents a key determinant of potential export capacity.

International market access and export flexibility are especially impacted by infrastructure development, including long-distance oil pipelines, coastal terminals and berthing capacity for shipments of crude oil, refined products and other oil liquids.

The world's oil pipeline infrastructure continues to expand, with thousands of kilometres of pipeline under construction in 2026 and even more planned in the coming years. Much of this expansion will serve domestic needs or facilitate the movement of oil within each of the geographical regions covered in this chapter. However, a smaller number of projects aim to facilitate the addition of new barrels to the global market and thus could have an impact on global trade.

For instance, Uganda is racing towards its first commercial oil production in 2026, with new infrastructure allowing this supply to reach the global market. Construction of the East African Oil Pipeline is well underway and is expected to be completed in 2026. At full capacity, it will take 246 tb/d of oil from Uganda to the Tanzanian coast for export.

Elsewhere, in Argentina, major midstream projects are underway to bring increased oil production from the prolific Vaca Muerta shale formation to the South Atlantic coast. This supports growth in the country's overall export capacity in line with expected increases in production capacity. The Vaca Muerta Sur Oil Pipeline began construction in early 2025, and phase 1 of the project could be completed by the end of 2026, providing an initial capacity of 180 tb/d. The capacity of the pipeline is then expected to increase in the coming years, enabling the potential transport of 700 tb/d to a new terminal in Punta Colorada that is capable of loading very large crude carriers (VLCCs).

The existing Oldelval pipeline network is also expected to expand further. In April 2025, the completion of the Duplicar expansion project brought the pipeline that runs to the South Atlantic port of Puerto Rosales up to a capacity of 540 tb/d. In addition, the Duplicar Norte project started construction in November 2025, aiming to boost the northern region of the Vaca Muerta formation by adding an additional 220 tb/d of capacity. Of note, there is also potential for further expansions after it becomes fully operational in 2027.

One of the most significant recent trends in oil movements was the emergence of the US as a major exporter of oil in the mid-2010s. The US Gulf Coast is estimated to have sufficient infrastructure to support a crude export capacity of around 6 mb/d, and although the rate of infrastructure development has slowed, there are still a number of projects worth monitoring.

As trade flows change over time, so too does the need for appropriate port facilities. From 2022, crude oil exports to Europe increased rapidly and overtook Asia as the top destination for US exports. If trade to Asia from the US was to gather momentum, this would likely increase the need for larger tankers than those typically used for transport to Europe. On the US Gulf Coast, there have been a number of proposals for deepwater ports to load oil onto larger vessels. This would add to the country's only existing deepwater port in Louisiana.

One of these proposals concerns the Sea Port Oil Terminal in Texas, which has a proposed export capacity of 2 mb/d. However, having received a deepwater port license from the previous administration in April 2024, the development company advised in February 2025 that there was not enough interest in the project to make it a commercial success. Elsewhere, the 1 mb/d Texas GulfLink Deepwater Port project made some progress in 2025, receiving permits and the backing of investors from Japan worth $2.1 billion. Applications are also being made for the Blue Marlin project and the Bluewater Texas Terminal, which have a proposed capacity of around 2 mb/d, respectively.

Moving further inland, the 750 tb/d Dakota Access Pipeline continues to bring crude oil out of the Bakken region. In this context, a long-awaited environmental impact statement was released in December 2025 that recommended that the Dakota Access Pipeline keep operating, albeit with recommendations also made regarding a crossing under the Missouri River, which had been the focus of concerns about impacts on the local water supply. The pipeline's future nevertheless remains uncertain due to ongoing legal challenges. Irrespective of the outcome, the Bakken region should have enough rail capacity – estimated at around 1 mb/d – to at least mitigate potential drops in pipeline flows.

Elsewhere, in Canada, the country's production of heavy crude has been rising, with the vast majority of Canada's oil production exported to the US via a well-established network of existing pipelines and railways. Despite increasing production, public resistance and political opposition, often resulting in legal challenges, have constrained the development of new infrastructure projects such as the 830 tb/d Keystone XL Pipeline, which was stopped by the US government in 2021. In 2025, indications from both Canada and the US hinted at a potential revival of the idea, but no formal steps have been taken since.

Separately, Canadian company Enbridge has approved expansion projects for the next few years on existing pipelines to the US that could add an additional 250 tb/d from 2027. If Canada's oil production continues to grow as expected, further expansions and new infrastructure may be required to avoid bottlenecks. If the pipeline system operates at capacity, volumes transported over rail to the US could increase, although this is less efficient and more costly.

Aside from US-bound routes, only the Trans Mountain Pipeline – with a capacity of 890 tb/d – currently moves crude from the province of Alberta to the Pacific Coast for export. Important export volumes have been shipped to the Asia-Pacific region since it started operating in May 2024, and China has become a top destination. Trans Mountain Corporation, owned by the Government of Canada, has announced that it is exploring optimization projects that could add another 360 tb/d to the pipeline's capacity in the next four or five years.

In terms of other possible Canadian projects, a previous project proposal to add pipeline capacity in Western Canada, the Northern Gateway pipeline, was cancelled by the federal government in 2016. However, the current government is seeking to boost major infrastructure across the country while diversifying exports and, towards this end, has signed an agreement with Alberta to develop a proposal for a new pipeline. The project is at a very preliminary stage and will likely face significant opposition, including from the province of British Columbia, through which the pipeline would run. On the country's east coast, a much smaller volume of crude oil is currently being exported to non-US destinations. The province of Ontario has also commissioned a feasibility study, to be completed in the course of 2026, on a potential pipeline to bring oil from Alberta eastwards to Ontario's export infrastructure.

Generally, economics and long-term contracts drive the movement of crude oil, condensates and refined products through the global midstream and downstream sectors. However, geopolitics also has the potential to redraw global trade routes for any commodity, including oil.

The impact of the conflict in Eastern Europe continues to be felt since trade flows were first altered in 2022. The EU, UK and the US continue to maintain the existing ban on seaborne Russian crude imports and in January 2026 a ban on the import of products refined from Russian crude to the EU came into force. Having declined at the start of 2023, the flow of crude through the Druzhba pipeline network to Hungary and Slovakia has continued despite intermittent disruptions and some proposals to include pipeline flows in the ban on Russian crude imports. While it is unclear how the overall situation might be resolved, it is unlikely that flows will return to pre-conflict levels. The assumption in this outlook, therefore, is that there will be a lasting effect on the mix of imports into Europe, with Russian crude predominantly flowing to China and other Asian countries, as well as other destinations.

6.2 Crude oil, condensate and refined product movements

This section discusses future global crude oil, condensate and refined product movement trends. It reflects the outlook presented in the previous chapters of this Outlook, while aiming to resolve regional supply and demand imbalances for both crude and products.

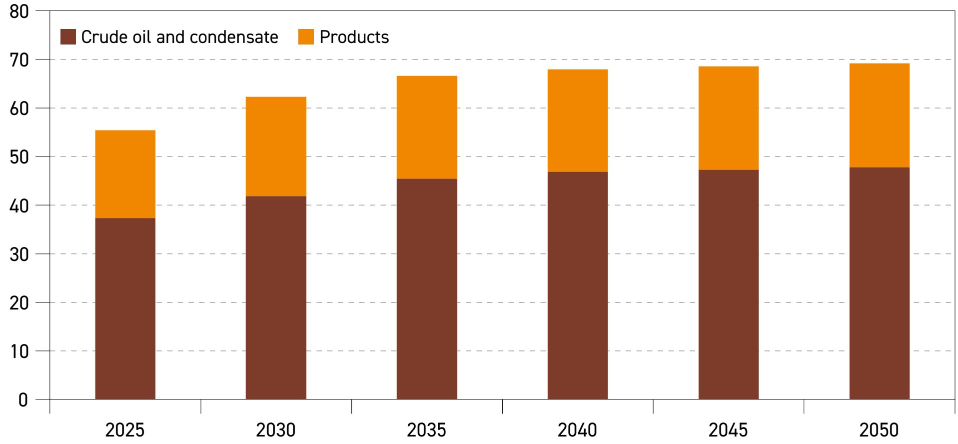
As highlighted in the previous section, factors such as geopolitics and trade policies can have an impact on both current and future trade patterns. In addition, recent years have seen policymakers continuing to focus on energy security, including crude oil supply. In the downstream sector, this can materialize as efforts to increase refining capacity in high-demand countries (e.g. in Asia) in order to secure the supply of products for domestic markets. The refining outlook in Chapter 5 shows a continuation of this trend in the long term.

This means that changes in global oil trade, especially those affecting the short term, are particularly difficult to predict, as they do not always provide for the most economical trade flows. That being said, modelling can provide a helpful indication of crude and product flow trends over the long term based on the economics of trade between regions. Modelling for this Outlook is largely based on an optimization procedure that seeks to minimize global costs across the entire refining/transport supply system, in accordance with existing and additional refining capacity, logistical options and costs. It also considers various constraints, including trade barriers.

A combination of factors need to be considered for an outlook on global oil movements, including: oil demand trends; the production and quality of crude and non-crude streams; product quality specifications and related changes; refining sector availability and configurations; potential trade barriers or policy-driven incentives; the capacity and economics of existing transport infrastructure, such as ports, tankers, pipelines and railways; ownership interests; term contracts; and price levels and differentials. It is worth noting that interregional trade, seaborne transport accounts for the largest share of projected oil movements, while pipeline trade remains limited.

Figure 6.1 illustrates the size of oil movements across the globe in 2025 and as projected until 2050. Overall, when considering crude oil, condensate and products, over 55 mb/d are estimated to have navigated the world's trade routes in 2025 between the major regions covered in this chapter. No trade movements within each of these regions are detailed.

Figure 6.1
Interregional crude oil, condensate and products exports, 2025–2050, mb/d



Source: OPEC.

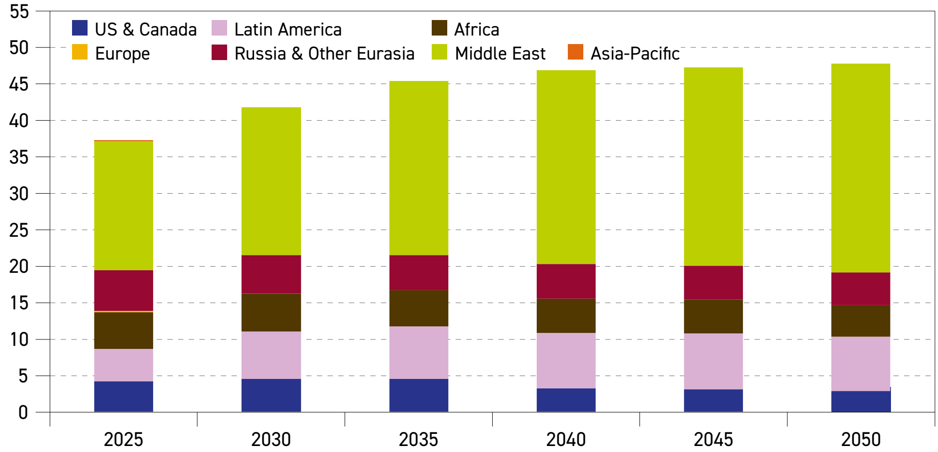
In line with increasing oil demand as discussed in Chapter 3, growth in interregional exports is anticipated as regions balance their demand needs with availability of supply. In 2030, global oil trade is projected to reach 62 mb/d. In the following decade, further growth is expected to drive trade volumes to almost 67 mb/d in 2035 and just under 68 mb/d in 2040. Between 2040 and 2050, global trade growth is expected to slow, reflecting a global trend of slowing oil demand growth and growing refinery capacity in developing countries that is anticipated to reduce the volumes available for export. By 2050, global crude, condensate and product exports are expected to total around 69 mb/d.

Of note, the biggest share, 67%, of global oil movements in 2025, was dedicated to crude and condensate trade. It is anticipated that this figure will grow to over 69% by 2050.

Crude oil and condensate movements

Figure 6.2 provides a global view of the world's crude and condensate exports across the long term. Again, only movements between major regions are shown and intra-trade movements within those regions are not taken into account. The outlook covers crude and condensates, including oil sands and synthetic crudes, but excludes other liquids, such as biofuels, synthetic fuels, CTLs, GTLs and NGLs.

Figure 6.2
Global crude and condensate exports by origin*, 2025–2050, mb/d



* *Only trade between major regions is considered, intratrade is excluded.*
Source: OPEC.

In 2025, global crude oil and condensate trade totalled 37.3 mb/d, up from the year before. Looking ahead, global exports are projected to grow by 10.5 mb/d from 2025 to 2050. Initially, exports will rise more sharply and reach 41.8 mb/d in 2030. In the long term, the rate of growth slows markedly but volumes rise to a level of 47.8 mb/d in 2050. An important driver of this upward trend is rising demand in the Asia-Pacific, including India, China and other Asian countries, with local supply in these regions declining at the same time.

As a major region of long-term supply growth, the Middle East will expand its role as the world's largest crude and condensate exporting region, growing across the entire outlook period from 17.8 mb/d in 2025 to 28.7 mb/d in 2050. This represents a total increase of nearly 11 mb/d.

Latin America is also expected to see an overall increase in exports of 3 mb/d between 2025 and 2050, cementing its long-term role as the second-largest exporting region. As the region's supply expands, much of the increase in exports comes over the first five years of the outlook, moving from 4.5 mb/d in 2025 to 6.5 mb/d in 2030. Exports continue growing strongly in the long term, with a small decline towards the end of the outlook to reach 7.4 mb/d in 2050.

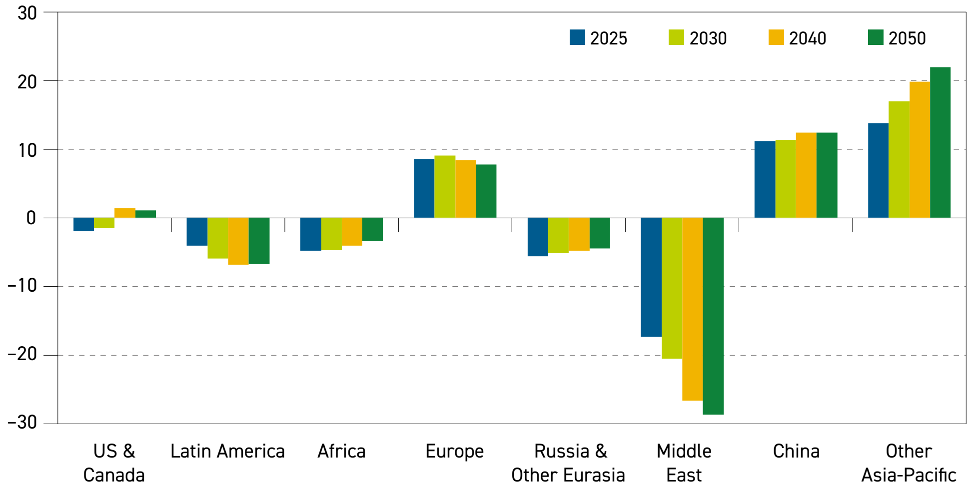
The US & Canada is also expected to see an initial increase in exports, rising from 4.2 mb/d in 2025 to 4.5 mb/d in 2030. Thereafter, in line with declining supply from the US, regional exports will gradually but noticeably decline in the coming decades to reach 2.9 mb/d by 2050.

Russia & Other Eurasia, as well as Africa, are expected to see gradual declines in export volumes across the outlook to 2050, albeit for different reasons. In 2025, exports from Russia & Other Eurasia were around 5.6 mb/d. By 2030, this is expected to drop slightly to

5.3 mb/d before falling to 4.5 mb/d in 2050 due to declining supply. In Africa, after a small increase from 5 mb/d to 5.2 mb/d between 2025 and 2030, exports start to decline as the domestic use of crude and condensate increases. Export volumes are expected to reach 4.3 mb/d in 2050.

Figure 6.3 illustrates net crude and condensate imports globally. Net importers can be clearly identified as Europe, China and Other Asia-Pacific. The latter was the largest net importer in 2025 at 13.8 mb/d, with volumes projected to grow significantly over the outlook period before reaching close to 22 mb/d by 2050. China's net imports were estimated at 11.2 mb/d in 2025, and while the trend is not as robust as Other Asia-Pacific, net imports are expected to rise to 12.4 mb/d in 2050. Taken as a whole, the Asia-Pacific region is by far the largest importer of crude and condensate, both today and through to mid-century.

Figure 6.3
Regional net crude and condensate imports, 2025, 2030, 2040 and 2050, mb/d



Source: OPEC.

In Europe, net imports were estimated at 8.6 mb/d in 2025, a figure that is expected to rise to 9.1 mb/d in 2030. Over the long term, however, net imports decline on the back of lower oil demand in this region. In 2050, net imports to Europe are projected to be around 7.8 mb/d.

Latin America, Africa, Russia & Other Eurasia and the Middle East were all net exporters in 2025 and will remain so through to 2050, albeit with different export trajectories. The Middle East was the largest net exporter in 2025 at 17.4 mb/d and its role as an exporter is projected to expand extensively to 28.7 mb/d in 2050 – representing a difference of over 11 mb/d.

Latin American net exports will also rise but not through the entire outlook to 2050. After a sharp increase from 4 mb/d in 2025 to 5.9 mb/d in 2030, the region's growth in net exports will slow to 6.8 mb/d in 2040 before dropping slightly to 6.7 mb/d in 2050.

The other net exporters, Africa and Russia & Other Eurasia, are expected to see a gradually declining trend in net exports after 2025. Africa's net exports decrease from under 4.8 mb/d in 2025 to 3.4 mb/d in 2050. Net exports from Russia & Other Eurasia fall from 5.6 mb/d in 2025 to 4.5 mb/d in 2050.

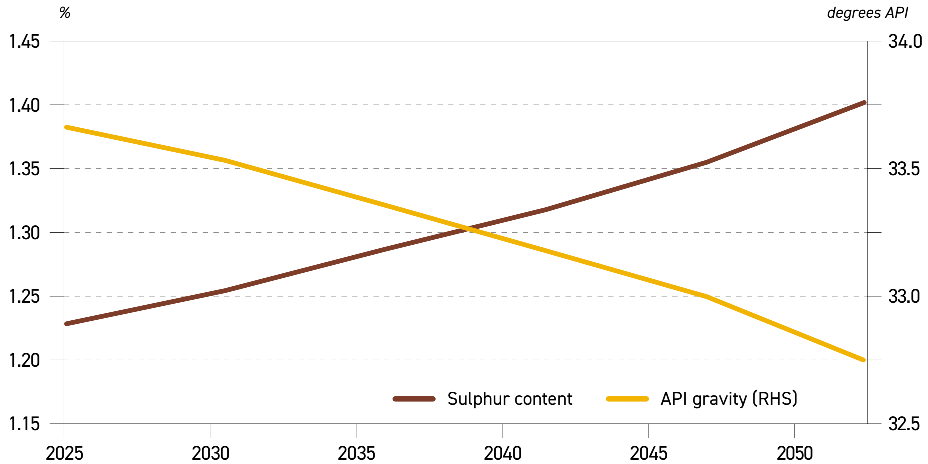
The US & Canada began the outlook period as a net exporter of crude and condensate (just under 2 mb/d in 2025), but the region transitions to a net importer in the long term, with net imports projected at 1.1 mb/d in 2050. The principal driver of this transition is declining supply levels in the US & Canada.

Figure 6.3 makes it clear that interregional trade in crude and condensate is currently dominated by the Middle East and the Asia-Pacific, a trend that is set to continue. The Middle East accounted for almost 48% of exports in 2025 and this share is expected to grow in the future, reaching 60% in 2050. Conversely, the Asia-Pacific region is estimated to have imported 67% of available crude and condensate in 2025; by 2050, this share is projected to reach 72%. As a result, the trade link between these two regions will only grow in importance in the future.

Global oil supply trends will also have an impact on the quality of the average barrel in terms of API gravity and sulphur content, as seen in Figure 6.4. Estimated at around 1.2% in 2025, the sulphur content of the average barrel could reach 1.4% by 2050. This is in line with a growing supply of crude grades from the Middle East, Canada and some countries in Latin America, which traditionally have a higher sulphur content.

The other trend outlined in Figure 6.4 relates to a decrease in the API gravity of the average barrel over the outlook period. In 2025, the average barrel was estimated at 33.7° API. The change in the remainder of this decade is expected to be the most gradual, reaching 33.5° API in 2030. In the following decade, more medium and heavy crude will come from the Middle East, Africa and Latin America, while the supply of light barrels from the US will decline. This will lead to a slightly larger incremental decline in API over the long term to a level of 33° API in 2045. In the final period of the outlook, the decline in API gravity steepens, reaching 32.8° API in 2050. These API changes will be reflected in crude oil exports over the long term, amid rising exports from the Middle East and Latin America, as well as increased supplies of Canada's heavy sour crude.

Figure 6.4
Global average API gravity and sulphur content, 2025–2050



Source: OPEC.

Refined product movements

As illustrated in Figure 6.1, interregional trade in oil products accounted for a 33% share of all global oil trade in 2025. This is equivalent to 18 mb/d. Product movements are less than crude and condensate movements partly because of the goal of many consuming countries to have domestic refining capacity to serve local demand for products, especially in regions with growing demand. As a result, crude and condensates account for most trade, especially over long distances.

As discussed in Chapter 5, not all regions have sufficient capacity to meet their product demand needs and instead turn to the global market to satisfy their needs. In other cases, large crude oil producers – including several countries in the Middle East – choose to expand their refining capacity and replace a portion of their crude exports with product exports.

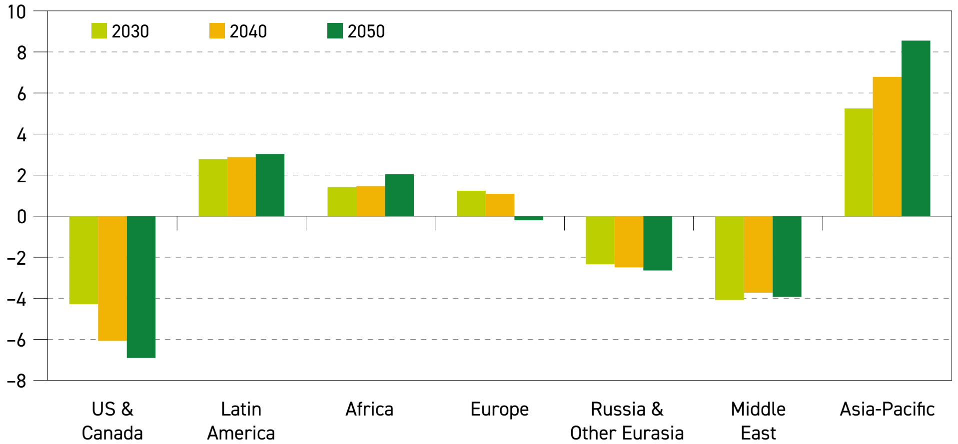
Products trade is projected to increase across the outlook period, reaching 20 mb/d in 2030 before gradually rising to over 21 mb/d in 2050. Key factors behind this trend are demand growth and existing and new refining capacity. Naturally, should less or more refinery capacity be added when compared to the assumptions of this outlook, product flows will shift accordingly.

Figure 6.5 shows the projected net product imports of several major regions. The Asia-Pacific has the highest level of net product imports, which is also expected to grow strongly from 5.3 mb/d in 2030 to 8.6 mb/d in 2050. On a smaller scale, net imports to Latin America and Africa are also projected to increase, growing in Latin America from 2.8 mb/d in 2030 to 3 mb/d in 2050, and expanding in Africa from 1.4 mb/d in 2030 to 2.1 mb/d in 2050.

Over the course of the outlook period, Europe is expected to change from being a net importer of products (1.2 mb/d in 2030) to a net exporter (0.2 mb/d in 2050).

The remaining regions are expected to be net exporters that provide sufficient exports to meet the needs of the above net importers. The US & Canada, in particular, will be the largest exporter of products by the end of the long term – increasing from 4.3 mb/d in 2030 to

Figure 6.5
Regional net product imports, 2030, 2040 and 2050, mb/d



Source: OPEC.

6.9 mb/d in 2050. Incremental increases in net exports from Russia & Other Eurasia will be much smaller, with net exports projected at 2.3 mb/d in 2030 and 2.6 mb/d in 2050. Net exports from the Middle East are expected to be 4.1 mb/d in 2030 and decline only marginally to under 4 mb/d in 2050.

6.3 Regional crude oil and condensate exports

This section and figures 6.6–6.10 provide an outlook for crude oil and condensate exports by each major exporting region. This is in line with the projections presented in Chapter 4, as changes in long-term supply influence the origin of crude and condensate movements.

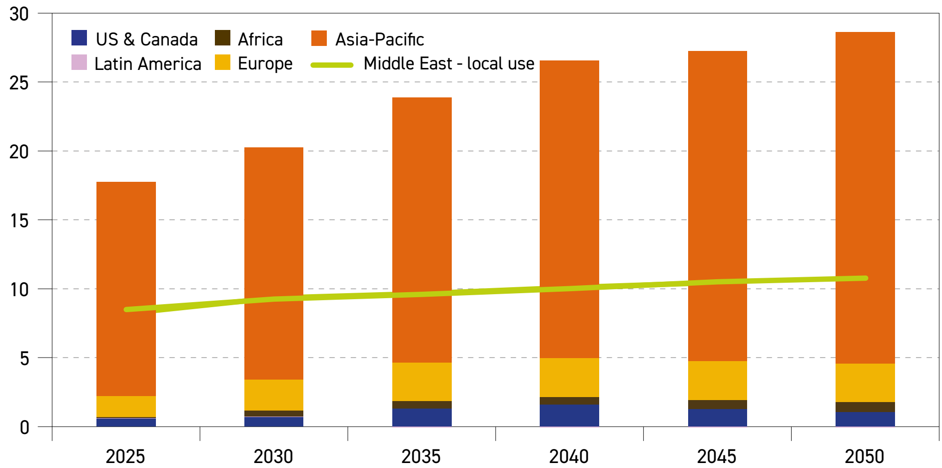
Crude and condensate exports from the Middle East are estimated at around 17.8 mb/d in 2025 (Figure 6.6). In the medium term, exports are projected to increase to 20.3 mb/d by 2030 and continue growing strongly through the 2030s. In the long term, growth continues but at a slower pace, leading exports to reach 28.7 mb/d in 2050. In comparison to exports, local use of crude and condensate grows at a more moderate pace, increasing from 8.5 mb/d to 10.8 mb/d in 2050. This reflects the region's growing refining capacity and increasing domestic product demand and exports.

At 15.6 mb/d, the Asia-Pacific region took the vast majority of crude and condensate exports from the Middle East in 2025. In line with the expected increase in demand in the Asia-Pacific, the region will account for over 80% of the Middle East's exports throughout the entire outlook period. In 2030, 16.9 mb/d is expected to flow to the Asia-Pacific before growing further over the long term to reach 24.1 mb/d by 2050.

In 2025, Europe was the second largest destination for Middle East exports, with the region importing 1.5 mb/d of crude and condensate that is well suited for the quality requirements of European countries. Exports are projected to grow to 2.2 mb/d in 2030

Figure 6.6

Crude and condensate exports from the Middle East by major destination (and local use), 2025–2050, mb/d



Source: OPEC.

before plateauing across the long-term period, with 2.8 mb/d expected to be exported to Europe in 2050.

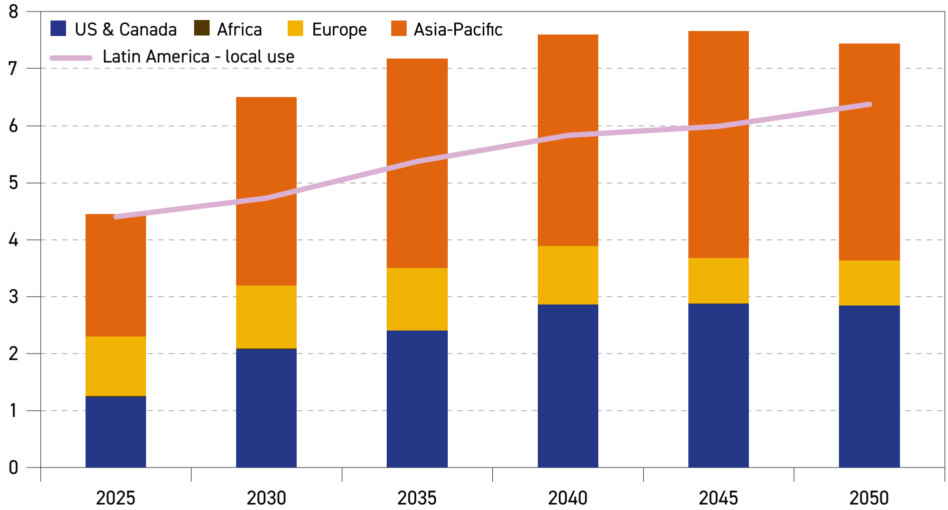
Although the volumes involved are small, the US & Canada represents another growing destination for exports, with this trend influenced by changing demand dynamics in the region over the medium and long term. In 2025, around 0.6 mb/d was exported from the Middle East. These exports will increase over the medium-term period and into the 2030s, before declining to a level of 1 mb/d in 2050.

Exports to Africa from the Middle East are currently very minor, estimated at 0.04 mb/d in 2025. However, the Middle East is anticipated to play a larger role in helping the continent meet its incremental demand growth in the future. Exports to Africa will reach 0.5 mb/d in 2030 before increasing to over 0.7 mb/d in 2050.

In Latin America (Figure 6.7), strong supply growth will facilitate the export of more crude and condensate to the global market. In 2025, the region exported 4.5 mb/d, which is expected to increase sharply over the next few years to 6.5 mb/d in 2030 and beyond into the 2040s, reaching a peak of around 7.7 mb/d in 2045. Towards the end of the long term, as supply growth slows and local use of crude and condensate continues to increase, a small decrease in exports is expected, bringing exports to 7.4 mb/d in 2050. The increase in exports comes even as the region's supply is used to meet its own growing oil demand. From 4.4 mb/d in 2025, local use in Latin America will increase throughout the entire outlook period, reaching 6.4 mb/d in 2050 on the back of expected increases in refinery throughputs in the region.

In 2025, the Asia-Pacific, Europe and the US & Canada received relatively similar shares of Latin American crude and condensate. Moving forward, exports to Europe are expected to decline gradually from 1.0 mb/d in 2025 to settle at 0.8 mb/d in the final part of the long-term period and in 2050.

Figure 6.7
Crude and condensate exports from Latin America by major destination (and local use), 2025–2050, mb/d



Source: OPEC.

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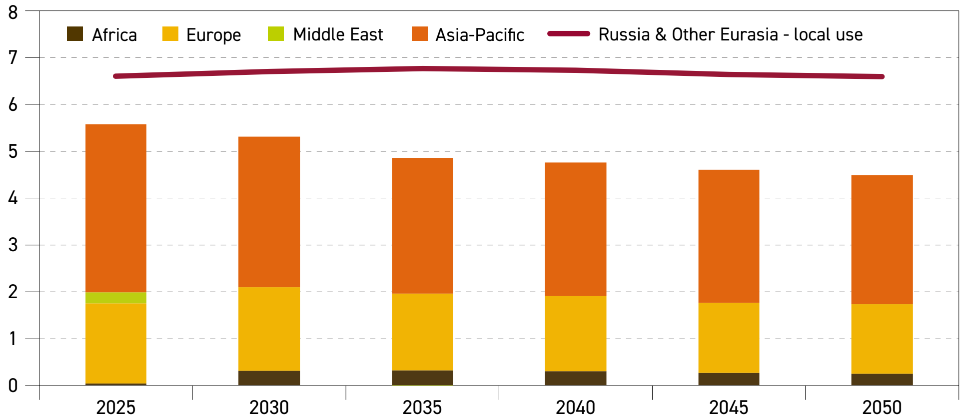
In contrast, exports to the Asia-Pacific jump from 2.1 mb/d in 2025 to 3.3 mb/d in 2030, helping to meet this region's expanding demand. In the long term, although the rate of growth moderates, exports still rise to almost 4 mb/d in 2045 before falling slightly to 3.8 mb/d in 2050.

Exports to the US & Canada are expected to follow an upward trend through the medium term and up to 2040, with refineries, especially in the US, well configured to utilize medium and heavy crude from Latin America. Exports to the US & Canada will rise from 1.2 mb/d in 2025 to a plateau of around 2.9 mb/d after 2040, finishing the outlook period at 2.8 mb/d in 2050.

Figure 6.8 provides the outlook for exports from Russia & Other Eurasia, which in 2025 stood at 5.6 mb/d. It is projected that exports will gradually decline to a level of 4.5 mb/d in 2050. This trend is influenced by declining supply at the same time as local use for product exports remains steady – starting and finishing the outlook period at 6.6 mb/d.

The Asia-Pacific is anticipated to be the top destination for exports of crude and condensate, equivalent to 3.6 mb/d in 2025. A decline is expected until a plateau is reached at around 2.9 mb/d for most of the long term. In 2050, exports to the Asia-Pacific are expected to be 2.7 mb/d. Exports to Europe are also expected to be lower in the future. Estimated at 1.7 mb/d in 2025, flows to Europe are projected to see a gradual decline throughout the outlook period before reaching 1.5 mb/d in 2050.

Figure 6.8
Crude and condensate exports from Russia & Other Eurasia by major destination (and local use), 2025–2050, mb/d



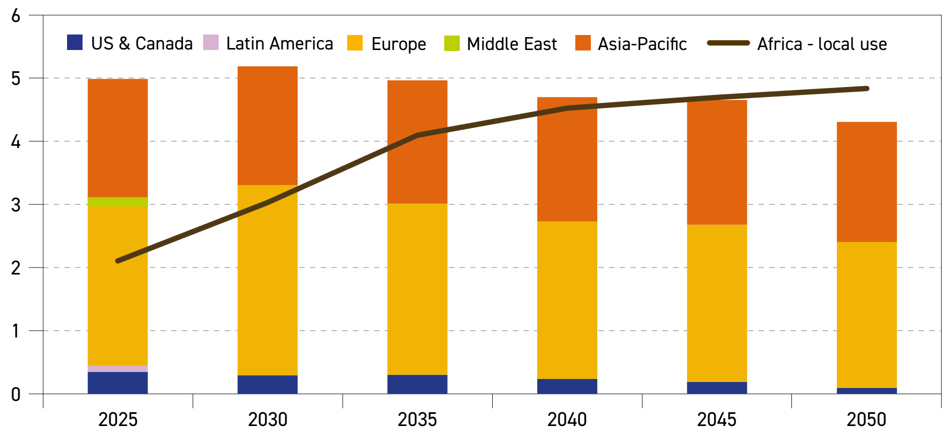
Source: OPEC.

Within Russia & Other Eurasia, uncertainty surrounds the direction of future flows from Russia. Prior to 2022, Europe was the top destination for exports, well supported by existing infrastructure. However, the ongoing EU and UK embargoes on seaborne Russian crude imports have reduced flows to Europe, and the European Commission is also proposing to permanently ban all Russian oil imports. The latter would affect remaining pipeline supplies to Hungary and Slovakia, in particular. As a result of these actions, Russia has seen some flows re-routed to the Asia-Pacific, while Europe has also increased imports from other countries in this region, including Azerbaijan and Kazakhstan.

Smaller volumes are expected to be exported to Africa, increasing to around 0.3 mb/d by 2030 and only marginally changing over the long-term period.

In Africa, overall exports of crude and condensate are estimated at 5 mb/d in 2025. The volume of exports is expected to grow slightly to 5.2 mb/d in 2030 (Figure 6.9). In the years to 2050, a gradual decline is expected until the volume of exports reaches 4.3 mb/d in 2050. The primary factor in this trend will be a significant increase in the local use of crude, as required capacity additions to Africa's refining sector come online. As a result, local use of crude and condensate grows from 2.1 mb/d in 2025 to 3 mb/d in 2030, before continuing at the same pace and reaching 4.1 mb/d in 2035. Thereafter, the rate of increase slows, but local use continues to grow to 4.8 mb/d in 2050. If refining capacity additions are delayed, it is possible that more crude could be freed up for export. Conversely, if capacity is brought online sooner, or if additional projects are added in the long term, exports from Africa could decrease.

Figure 6.9
Crude and condensate exports from Africa by major destination (and local use), 2025–2050, mb/d



Source: OPEC.

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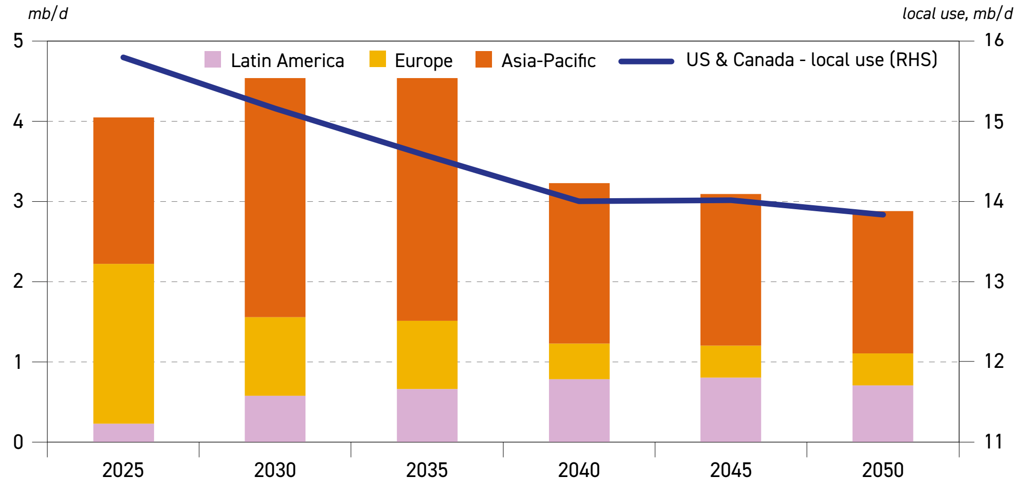
Europe received the largest share of African crude and condensate exports in 2025, at 2.5 mb/d. After increasing to 3 mb/d in 2030, exports to Europe will then slowly decline to 2.3 mb/d in 2050 – a trend driven by declining demand in the European region.

From just under 1.9 mb/d in 2025, exports to the Asia-Pacific will vary only marginally throughout the outlook, finishing in 2050 at 1.9 mb/d. The overall drop in Africa's export volumes limits its ability to meet growing oil demand in the Asia-Pacific.

Exports to other regions were limited in 2025. Only minor exports to the US & Canada are projected through the medium term, dropping from 0.4 mb/d in 2025 to under 0.1 mb/d in 2050, as the US & Canada makes use of imports from Latin America to lower costs.

Figure 6.10 shows crude and condensate exports from the US & Canada over the long term. As the region with the largest demand for oil, and as a major exporter of products, the US & Canada allocates a large share of its own supply to local use. Specifically, it is estimated

Figure 6.10
Crude and condensate exports from US & Canada by major destination (and local use), 2025–2050



Source: OPEC.

that 15.8 mb/d of crude and condensate was used locally in 2025. This figure is expected to decrease to 13.8 mb/d in 2050, with declining oil demand and refinery runs in the region supporting this trend in the long term.

In comparison, exports from the US & Canada were assessed at 4 mb/d in 2025. Export volumes are projected to rise to over 4.5 mb/d in 2030 and remain at this level to 2035. In the second half of the outlook, exports are expected to decline noticeably to 2.9 mb/d in 2050.

In 2025, exports to Europe and the Asia-Pacific were 2 mb/d and 1.8 mb/d, respectively. However, future export trends to these two destinations differ greatly. Although exports to Europe have been boosted in recent years by EU efforts to diversify its imports, it is expected that exports will decline to a level of 1 mb/d by 2030 and continue to drop until reaching 0.4 mb/d at the end of the outlook period. One important reason for this is that US tight crude oil is not best suited for Europe's refining sector and European demand structure.

Exports to the Asia-Pacific are expected to grow considerably to 3 mb/d in 2030 and remain around this level until 2035. US tight oil provides a major share of these exports, with these light-sweet volumes favoured by the region's growing refining sector. The aforementioned increase is also aided by growing exports from the west coast of Canada, which so far have been shipped to Asia. However, after 2035, a gradually declining supply base is expected to lead to a drop in exports from the US & Canada. By 2050, exports to the Asia-Pacific are projected at 1.8 mb/d.

Exports to Latin America are projected to grow from only 0.2 mb/d in 2025 to 0.6 mb/d in 2030, before increasing to 0.8 mb/d by 2040. The import of light-sweet crude from the US is also attractive for the local refining sector, which partly consists of old and relatively simple refineries. As overall exports from the US & Canada decline towards the end of the outlook, exports to Latin America are expected to be 0.7 mb/d in 2050.

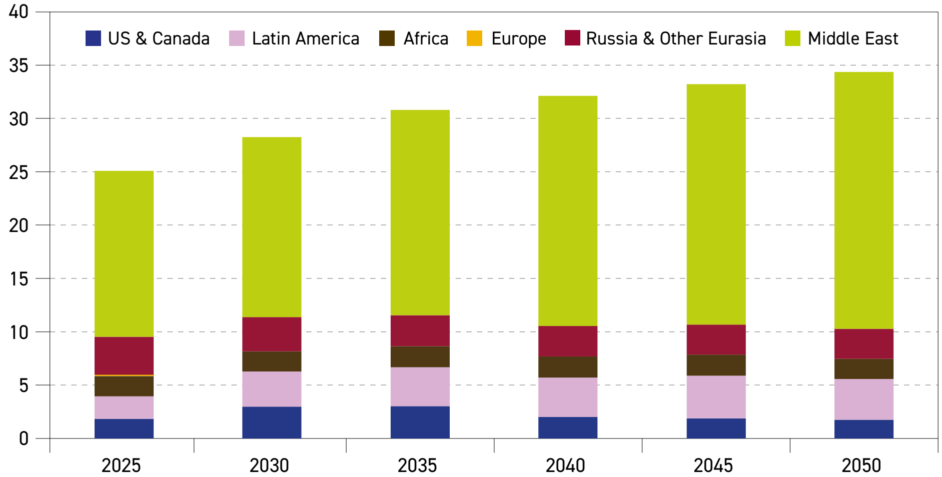
6.4 Regional crude oil and condensate imports

Figures 6.11, 6.12 and 6.13 show crude and condensate imports for the three largest importing regions, namely the Asia-Pacific, the US & Canada and Europe.

The Asia-Pacific is the largest region by far in terms of crude and condensate imports in this outlook (Figure 6.11). Unlike the US & Canada, as well as Europe, the trends for imports into the Asia-Pacific are positive throughout the entire period from 2025 to 2050.

Total crude and condensate imports to the Asia-Pacific were assessed at 25.1 mb/d in 2025. In the years to 2030, imports are projected to grow to 28.3 mb/d, before continuing to increase to 34.4 mb/d in 2050. This large increase in imports is expected to be needed to meet the region's rising oil demand, especially as local supply is projected to decline over the long term.

Figure 6.11
Crude and condensate imports to the Asia-Pacific by origin, 2025–2050, mb/d



Source: OPEC.

For the Asia-Pacific region, the most essential source of imports in 2025 was the Middle East, which accounted for 15.6 mb/d of crude and condensate, equivalent to approximately 62% of all imports. The importance of supply from the Middle East grows further in the long term, reaching 16.9 mb/d by 2030 and 24.1 mb/d in 2050, equivalent to a 70% share of total crude and condensate imports.

Remaining import volumes come from a mix of other major exporting regions. Russia & Other Eurasia will continue to play an important role, especially due to the pipeline infrastructure that connects Kazakhstan and Russia to the Asian country with the largest oil demand – China. Although imports from Russia & Other Eurasia were estimated at around 3.6 mb/d in 2025, this figure is projected to decline gradually over time – reaching 2.8 mb/d in 2050 – as supply from this exporting region starts to decline. As previously mentioned, geopolitical uncertainty continues to surround flows from Russia; depending on future developments, volumes could eventually shift back towards Europe and away from Asia.

Imports from both Latin America and the US & Canada grow noticeably from 2025 to 2030, as these regions' available supply helps to meet growing demand in the Asia-Pacific. That said, both regions' long-term trends differ. After growing from 1.8 mb/d in 2025, imports from the US & Canada reach a high point at around 3 mb/d in the 2030s before seeing a significant decline that leads imports to drop back to 1.8 mb/d in 2050. In contrast, Latin America's role is longer lived and more pronounced. Imports from Latin America grow from 2.1 mb/d in 2025 to a high of around 4 mb/d by 2045 before declining slightly to 3.8 mb/d in 2050.

Imports from Africa to the Asia-Pacific are projected to remain stable, starting and ending the outlook period at 1.9 mb/d, with only a slight variation in the middle of the period to a maximum of 2 mb/d. Ultimately, this reflects the limited export potential for Africa as its local use of crude oil increases.

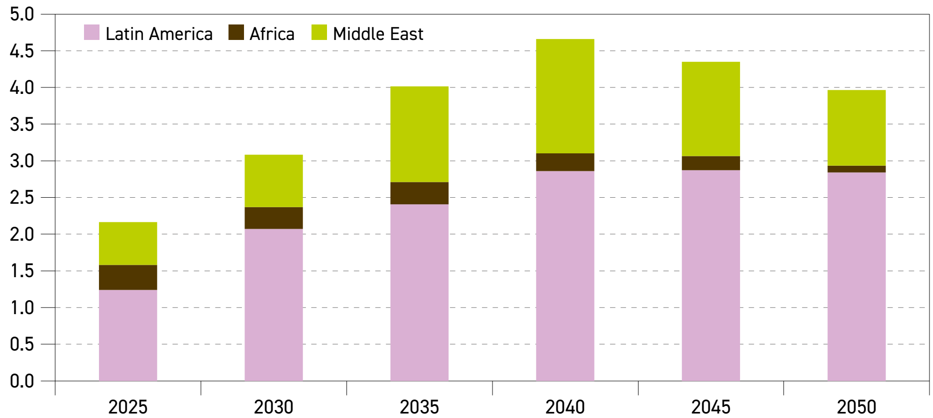
Figure 6.12 shows crude and condensate imports to the US & Canada, which stood at 2.2 mb/d in 2025. With some growth in oil demand in the coming years, imports are expected to grow to 3.1 mb/d in 2030. Thereafter, as regional refining requirements increase and regional supply gradually declines, imports are projected to grow to almost 4.7 mb/d in 2040. In the last decade of the outlook period, imports are expected to fall to around 4 mb/d in 2050.

Latin America is anticipated to continue being the key provider for the US & Canada, due to its favourable location and crude quality. Imports are projected to rise from 1.2 mb/d in 2025 to 2.8 mb/d in 2050, with a slight decline seen in the last five years of the outlook period.

The Middle East provided an estimated 0.6 mb/d of crude and condensate in 2025. However, with strong growth in supply anticipated from the Middle East in the long term, more imports are expected to come to the US & Canada from the region. These are expected to reach 1.6 mb/d in 2040 before declining to around 1 mb/d in 2050.

In 2025, the US & Canada imported less than 0.4 mb/d from Africa. In the long term, these imports are expected to play only a very minor role due to Latin America being a more competitive source of imports. By 2050, imports from Africa are expected to drop below 0.1 mb/d.

Figure 6.12
Crude and condensate imports to the US & Canada by origin, 2025–2050, mb/d

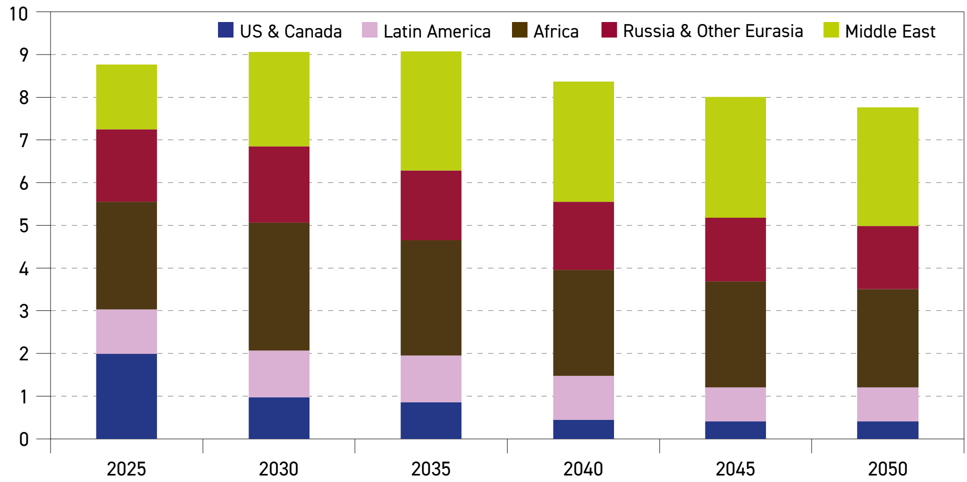


Source: OPEC.

The outlook for crude and condensate imports to Europe is shown in Figure 6.13. Total imports were estimated at 8.8 mb/d in 2025, sourced from a diverse group of major exporting regions. Until 2035, the overall level of imports is anticipated to remain consistent, before a gradual decline in the second half of the outlook period sees the region reaching 7.8 mb/d in 2050, in line with its declining demand trend.

Moving forward, Figure 6.13 highlights the increasing role of imports from the Middle East in offsetting declining supply to Europe from other regions. Imports from the Middle East will grow from 1.5 mb/d in 2025 to 2.2 mb/d in 2030, before reaching a plateau of around 2.8 mb/d for the remainder of the long term.

Figure 6.13
Crude and condensate imports to Europe by origin, 2025–2050, mb/d



Source: OPEC.

The largest source of imports to Europe in 2025, 2.5 mb/d, was from Africa. African imports to Europe are expected to expand to 3 mb/d in 2030, helping to offset declines from other sources, including the US & Canada. In the long term, however, Africa's declining export capacity will limit trade with Europe and volumes will decline to 2.3 mb/d in 2050.

Imports from Russia & Other Eurasia have already been reduced from significantly higher historical levels due to the EU's ban on seaborne Russian crude imports. In 2025, imports were at around 1.7 mb/d, primarily to non-EU countries. Moving forward, imports are anticipated to decline slowly to less than 1.5 mb/d by 2050. It is worth emphasizing the uncertainty in these projections, however, which do not make assumptions on geopolitical developments.

Imports from Latin America are projected to decline from 1 mb/d in 2025 to 0.8 mb/d in 2030. Imports will continue to decline in the long term but will reach a plateau of 0.8 mb/d between 2045 and 2050.

The US & Canada played an important role in providing around 2 mb/d of crude and condensate to Europe in 2025 to help EU countries diversify their supply. However, this level of imports is not anticipated to remain this high in the long term, declining to 1 mb/d by 2030 and 0.4 mb/d by 2050.

Energy scenarios

Key takeaways

- Two alternative scenarios to the Reference Case used in past editions of the Outlook, namely the 'Technology-Driven Scenario' (TDS) and the 'Equitable Growth Scenario' (EGS), have proven useful in exploring the range of uncertainty facing the global energy system. Thus, recalibrated versions of the same scenarios are used in this Outlook to provide an overview of possible future oil and energy demand levels.

- The TDS assumes faster and larger investments in advanced energy technologies than the Reference Case. These investments result in significant fuel substitution and energy efficiency improvements, leading to lower primary energy demand and a changed energy mix. Moreover, the TDS includes a greater deployment of carbon removal technologies, such as stationary and mobile CCUS and DAC technologies, especially in the second part of the forecast period.

- Primary energy demand in the TDS expands at a slower pace than in the Reference Case, reaching 340 mboe/d by 2050, lower by 43 mboe/d. The share of non-hydrocarbons in this scenario is projected to steadily expand to 45% by 2050.

- Global oil demand in this scenario gradually departs from the Reference Case and starts declining after 2035. Nonetheless, it remains in the range of 110 mb/d to 115 mb/d for most of the forecast period before declining to slightly below 107 mb/d by 2050. This is more than 17 mb/d lower *vis-à-vis* Reference Case demand.

- The cumulative effect of the faster penetration of EVs and ICE efficiency improvements assumed in the TDS could lower road transport oil demand by as much as 8 mb/d by 2050, compared to the Reference Case. Significant potential for lower demand also exists in the industrial and aviation sectors.

- By contrast, the EGS assumes moderately faster economic growth in developing countries, compared to the Reference Case, coupled with greater efforts to improve electricity access and eradicate energy poverty. At the same time, it assumes that countries adopt differentiated approaches as to how and when they achieve emissions reduction targets in light of national circumstances. This scenario results in higher long-term demand for energy, in general, and oil, in particular.

- Primary energy demand in the EGS rises to almost 400 mboe/d by 2050, which is around 17 mboe/d higher than in the Reference Case. This incremental energy demand is almost entirely met by solar PV, oil and gas, while some growth in other energy sources is broadly offset by lower coal demand compared to the Reference Case.

- Oil demand in the EGS tops 121 mb/d by 2035 and continues expanding to 131 mb/d in 2050. Compared to the Reference Case, this is almost 7 mb/d higher by 2050. The largest potential for higher oil demand in the EGS is in the industrial, residential and road transport sectors, each in the range of up to 2 mb/d by 2050.

- These scenarios emphasize the critical role of oil and gas in meeting future energy demand and empowering human progress and prosperity.

Recent years have seen the rise and fall of overly ambitious scenarios envisaging an extremely fast global energy system transformation and a corresponding steep decline in energy-related emissions. These scenarios tended to reflect wishful thinking rather than solid data, viability checks or realism. 'Net zero cases', in particular, proved misleading to policymakers and were often detrimental to investments in key energy sources.

Moreover, they also widened the possible demand range for major energy sources beyond any reasonable scale. Many of these scenarios unfortunately sowed confusion rather than fostering a clearer understanding of complex energy issues and failed to provide realistic guidance for decision-makers. This experience highlights the importance of providing scenarios that are well designed and address critical issues with realistic assumptions, while offering a meaningful range of potential pathways for specific energy sources and the global energy system.

In this respect, the scenarios provided in recent editions of the Outlook proved to be a useful guide regarding potential departures of primary energy sources from the projections outlined in the Reference Case. It is important that this analysis is reassessed regularly to reflect updates to a number of uncertainties that exist around all key parameters driving Reference Case projections. These uncertainties pertain to the economic outlook, policy shifts, the pace of technological advancement, political and geopolitical developments, investment priorities and consumer behaviour, among others.

Therefore, the two alternative scenarios to the Reference Case used in last year's edition of the Outlook have been updated. These aim to present the range of different but viable outcomes and their impact on energy demand, the future energy mix and oil demand, specifically. Notably, the underlying narrative of the alternative scenarios remains largely unchanged from last year.

The first, the **'Technology-Driven Scenario' (TDS)**, starts with the same basic socio-economic assumptions on global population demographics and economic development as adopted in the Reference Case. However, it assumes a quicker pace and larger investments in advanced energy technologies. These investments result in both significant fuel substitutions and energy efficiency improvements, leading to lower primary energy demand and a changed energy mix.

The main reason for larger fuel substitution in this scenario is faster electrification across most sectors of consumption. This encompasses faster EV penetration; coal, oil and gas substitution by electricity in industrial processes; and electrification of the residential sector, which, in turn, all result in higher electricity demand. An important part of this scenario relates to how this higher electricity demand is met. It is assumed that further investments flow into a faster expansion of renewable electricity, mainly solar PV and wind, nuclear power and partially more efficient gas turbines, which, to a large extent, substitute for inefficient coal-based electricity generation.

Moreover, the faster adoption of more efficient technologies will lower primary energy demand without fuel substitution. Key examples include more efficient engine technology, more efficient power generation of various kinds, advances in the residential sector – including housing insulation and appliances – as well as the industrial use of energy. The net effect is reduced energy demand, including demand for oil, and a significantly different energy mix compared to the Reference Case.

Besides efficiency improvements and fuel substitution, the TDS also assumes a greater deployment of carbon removal technologies, such as stationary and mobile CCUS and DAC technologies, especially in the second part of the forecast period. Moreover, an expansion in hydrogen use, and a more rapid shift towards adopting a Circular Carbon Economy (CCE) framework, are also part of this scenario.

The key component underpinning the second scenario, the **'Equitable Growth Scenario' (EGS)**, is moderately faster economic growth in developing countries, compared to the Reference Case, coupled with efforts to improve electricity access and eradicate energy poverty. At the same time, this scenario assumes that countries adopt differentiated approaches as to how and when they achieve emissions reduction targets in the light of national circumstances and priorities.

Stronger long-term economic growth, especially in Africa, India, and developing countries in Asia and Latin America, goes hand in hand with higher levels of industrialization and urbanization, which then lead to a larger middle class and improved living conditions for billions of people in the EGS. While the problem of energy poverty in the widest sense is not fully resolved in this scenario, it is alleviated. Indeed, the EGS assumes that at least minimum access to modern energy is provided to everybody and in an affordable manner.

The EGS also assumes a quicker transition (to modern energy sources in these countries, compared to the Reference Case, including renewable energy, oil, gas and nuclear power, especially in the second half of the outlook period. However, considering the challenges in collective efforts to reduce global emissions, the implementation differentiates in light of diverse capabilities and circumstances. In this scenario, national development needs are prioritized over global issues due to wider protectionism and unilateralism holding sway. While the above issues are largely apparent in emerging economies, there is nevertheless also a spillover effect into OECD countries due to higher consumption and trade.

This chapter subsequently presents and compares the results of these two alternative scenarios relative to the Reference Case. The focus is on highlighting the impact of the various narratives adopted in these scenarios on future energy demand, the sectoral and regional implications, as well as the changing energy mix, in general, and oil demand, in particular.

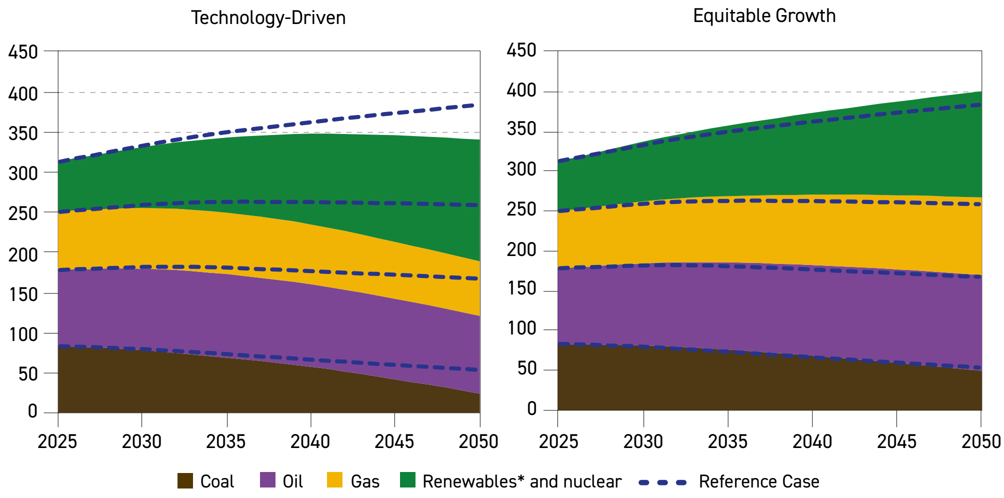
7.1 Energy demand and mix

Reflecting recent changes in respective regional policies, as well as prospects for demographic, economic and technological developments, Reference Case projections foresee steady growth in global primary energy demand over the entire forecast period. This is expected to rise from 312 mboe/d in 2025 to almost 383 mboe/d in 2050. Apart from coal, which is expected to peak in the next few years and start declining thereafter, demand for all other primary energy sources is set to continue growing.

Driven by strong growth in demand for electricity, nuclear and renewable electricity sources are set to expand significantly, led by solar and wind power generation. Combined, these energy sources are forecast to add more than 51 mboe/d to the global energy system, providing more than 70% of incremental energy demand to 2050. Oil and gas demand will also continue growing in the Reference Case, with each adding close to 20 mboe/d between 2025 and 2050, more than offsetting losses in coal demand.

These major trends outlined in the Reference Case also prevail in the alternative scenarios in the initial years of the forecast period, though momentum for a larger trend divergence in the latter period gradually builds. This is clearly visible in Figure 7.1, which shows that primary energy demand in both the TDS and EGS only gradually departs from the levels indicated in the Reference Case. By way of explanation, it is expected that policymakers' focus for the next few years will be on energy security, stabilizing economic growth, fostering international trade and responding to defence issues, rather than prioritizing investments in improving energy efficiency and energy access, and mitigating poverty in developing countries.

Figure 7.1
Global primary energy demand in the Reference Case and in alternative scenarios, 2025–2050, mboe/d



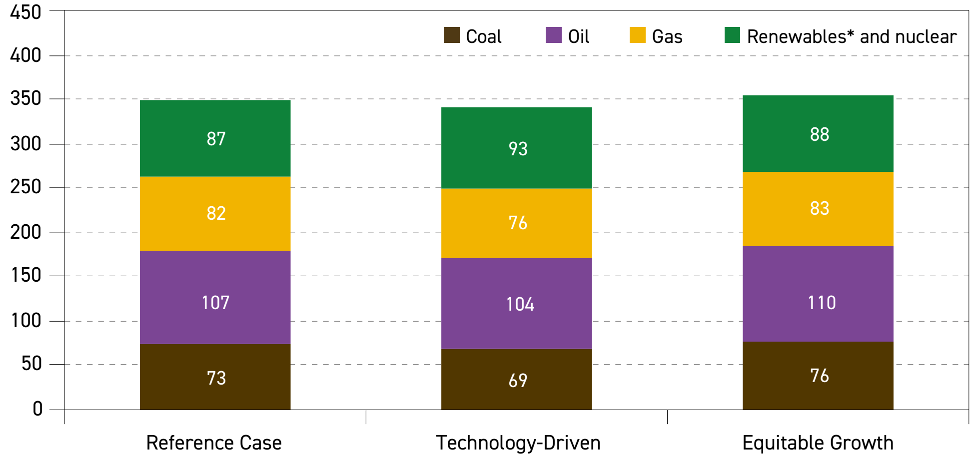
* *Renewables include hydro, biomass, wind, solar and others.*
Source: OPEC.

That said, the impact of the differing alternative scenario narratives on the outlook starts to become much more noticeable around 2035, as summarized in Figure 7.2. By then, the demand difference between the Reference Case and alternative scenarios is in the range of 7 mboe/d each, despite the fact that, except for coal in the TDS, all primary energy sources are forecast to continue growing between 2025 and 2035, albeit at differing rates.

The average energy demand growth in the Reference Case is expected to be around 1.1% p.a. over the next ten years. This growth, however, will be just 0.9% p.a. in the TDS due to energy savings and a changing energy mix in the transformation sector due to the faster penetration of advanced technologies. Therefore, energy demand in this scenario is set to reach just 342 mboe/d by 2035 – almost 7 mboe/d lower than in the Reference Case.

Coal will be especially affected in this scenario, as its decline accelerates to almost 2% p.a. on average, falling to below 69 mboe/d by 2035 from almost 83 mboe/d in 2025. This level is around 4 mboe/d lower than the corresponding demand in the Reference Case. Demand for oil and gas is also adversely affected relative to the Reference Case by 2035, reaching 3.6 mboe/d and 5.6 mboe/d, respectively.

Figure 7.2
Global primary energy demand in the Reference Case and in alternative scenarios, 2035, mboe/d



* *Renewables include hydro, biomass, wind, solar and others.*
Source: OPEC.

However, around half of the demand decline for hydrocarbons is offset by growing demand for renewable electricity sources and, to a lesser extent, nuclear energy. In the period to 2035, these energy sources are expected to grow by 4% p.a. on average, adding more than 30 mboe/d to the global energy mix in the TDS, almost 7 mboe/d more than estimated in the Reference Case. The largest part of this growth is attributable to solar PV and wind.

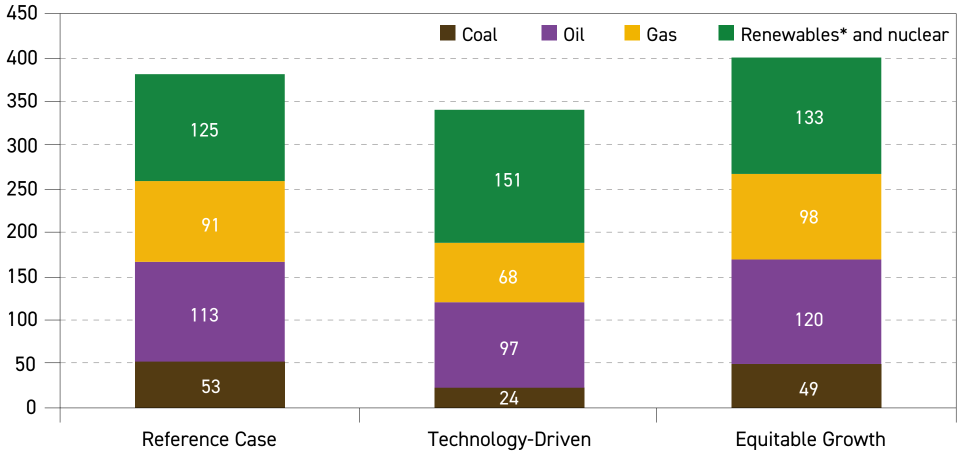
Turning to the EGS, global energy demand in this scenario is expected to be almost 8 mboe/d higher than in the Reference Case by 2035. In contrast to the TDS, all energy sources in the EGS broadly follow the demand trajectories outlined in the Reference Case, albeit at somewhat higher levels, reaching a combined demand level of more than 356 mboe/d in 2035.

The largest increase, almost 3 mboe/d compared to the Reference Case, occurs in coal. This reflects its availability, especially in developing Asian countries, where it is mainly used to meet incremental electricity demand, alongside higher consumption in the industrial sector. Higher demand growth for coal is supplemented by other fuels too. By 2035, oil demand is projected to be more than 2 mboe/d higher, while gas and the combined demand for renewables and nuclear increase by another 2 mboe/d and 1 mboe/d, respectively.

Moving beyond 2035, diverging trends from the Reference Case emerge relatively quickly, impacting the level of future demand in the alternative scenarios, as well as the related energy mix. This is demonstrated in Figure 7.3, which shows that the demand gap between the TDS and the Reference Case widens to 43 mboe/d by 2050, while the corresponding change for the EGS sees an expansion of 17 mboe/d. Significantly lower energy demand in the TDS is a result of the dual impact of the faster deployment of more efficient technologies. The direct impact of these technologies progressively limits final energy consumption growth.

Moreover, there will be an equally important indirect impact of these technologies, namely the higher electrification of major consumption sectors. For example, the increased use of

Figure 7.3
Global primary energy demand in the Reference Case and in alternative scenarios, 2050, mboe/d



* *Renewables include hydro, biomass, wind, solar and others.*
Source: OPEC.

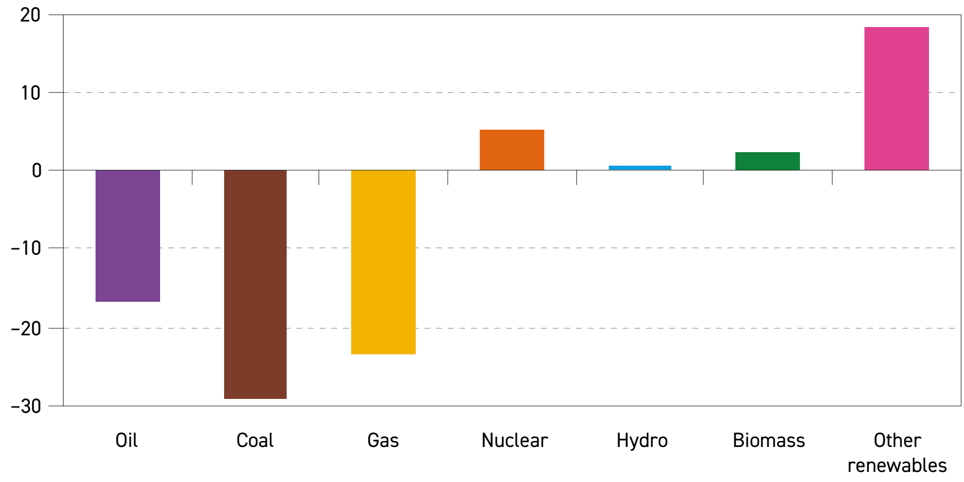
electric arc furnaces; capturing exhaust heat from furnaces; the use of industrial heat pumps; AI-based energy management systems; and induction and resistance heating systems could all significantly increase electricity demand in the industry sector, while reducing overall energy demand. Similarly, the faster penetration of EVs, electric boilers and heat pumps would have comparable effects on road transportation and the residential sector.

The implied higher electrification of the global energy system in the TDS, in turn, provides an opportunity for more efficient electricity supply. This scenario assumes that most of the additional demand is met by solar PV and wind power, followed by nuclear energy. A higher share of the former two electricity sources significantly contributes to lower energy demand when measured in terms of primary energy. Moreover, growth in electricity generation from renewables and nuclear primarily displaced coal-fired power due to its low level of efficiency. To a lesser extent, it will also reduce natural gas use, alongside efficiency improvements in gas turbines and the substitution of gas in industrial and residential applications.

Figure 7.4 provides a quantification of these trends with respect to the changing energy mix in this scenario by 2050. Clearly, coal is the most adversely impacted fuel in the TDS as its demand continues to decline throughout the forecast period, falling to a level of 24 mboe/d in 2050. This is around 29 mboe/d lower than the corresponding Reference Case demand and almost 59 mboe/d lower than the observed demand in 2025. In OECD countries, specifically, this would mean the almost complete elimination of future coal consumption.

Gas demand is also expected to undergo significant changes, though its demand pattern differs from coal. Natural gas consumption maintains steady momentum throughout the rest of this decade, tracking the Reference Case closely before reaching a broad plateau of roughly 76 mboe/d across much of the 2030s. While demand in the industry and residential sectors gradually weakens over this period, this decline is largely offset by continued growth in the power sector. This is due to gas-fired generation remaining critical for system stability,

Figure 7.4
Change in primary energy demand between the Technology-Driven Scenario and the Reference Case in 2050, mboe/d



Source: OPEC.

helping to balance intermittent renewables and supply incremental electricity through highly efficient turbine technologies.

In the 2040s, the trajectory shifts. As solar PV, wind, and nuclear energy assume a larger share of power generation and battery storage expands, the structural need for gas diminishes. Its function becomes increasingly concentrated in providing baseload capacity, while it is also frequently paired with CCUS to limit emissions. By mid-century, global gas demand falls below 68 mboe/d, around 4 mboe/d below 2025 levels. Relative to the Reference Case, this implies a cumulative reduction of close to 24 mboe/d by 2050.

Oil demand in the TDS follows a similar trajectory to gas, with oil growing over the next ten years before slowly declining over the remainder of the forecast period. The focus on better energy efficiency in this scenario supports moderate oil demand growth out to 2035, especially due tothe penetration of advanced technology in the OECD and China. Oil demand reaches almost 104 mboe/d by 2035, around 4 mboe/d lower than in the Reference Case.

Subsequently, global oil demand starts to decline in the years thereafter and reaches 97 mboe/d by 2050, as the impact of technology moderates demand growth in developing countries. This represents a difference of almost 17 mboe/d compared to the Reference Case. More details on these oil demand trends, including sectoral breakdown, are provided in Section 7.2.

Finally, the main beneficiaries of the changes assumed in the TDS will be renewable electricity sources. As mentioned earlier, due to accelerated electrification across all sectors, combined demand for renewables and nuclear electricity in this scenario is projected to increase from less than 63 mboe/d in 2025 to more than 151 mboe/d by 2050, almost 27 mboe/d higher than the corresponding level in the Reference Case. The largest part of this additional growth, almost 20 mboe/d, is captured by solar PV and wind, while nuclear energy contributes an additional 5 mboe/d.

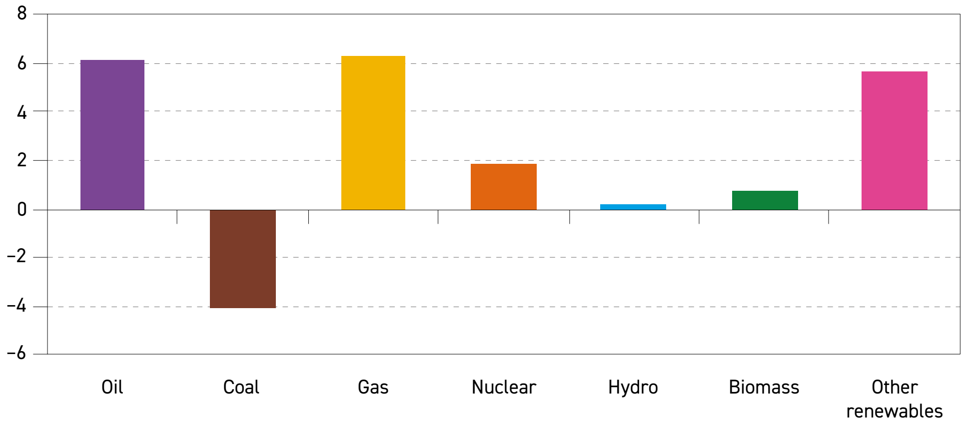
Significant long-term changes in the level of global energy demand and its composition are also expected in the EGS. While the TDS indicates a possible lower boundary for future energy demand, the EGS hints at the upper range as energy demand in this scenario continues to expand faster than in the Reference Case.

Another difference between these two alternative scenarios stems from the fact that the likely trend departure in the EGS is concentrated in developing regions – mainly in Africa, Other Asia and India – where the potential for faster economic and demand growth, as well as the need to alleviate energy poverty, is greatest. The driving force behind higher energy demand in this scenario will likely be higher electricity, gas and oil demand in the residential and industrial sectors, supplemented by the commercial and agricultural sectors. In turn, this will necessitate some changes in the power sector as well.

Major long-term changes for primary fuels in the EGS, compared to the Reference Case, are provided in Figure 7.5. The only fuel for which demand is expected to fall below Reference Case levels by 2050 is coal, despite its demand moving slightly above the Reference Case for most of the forecast period. This pattern is primarily driven by developments in China and the OECD, where higher GDP allows for the implementation of additional measures to reduce emissions, thus further lowering coal demand in the long term (which, even in the Reference Case, will be on a declining trajectory by then). Moreover, the broader availability of renewable electricity, including in developing countries, also contributes to a faster decline in coal demand. As a result, global coal demand is projected to drop below 50 mboe/d by 2050, more than 4 mboe/d lower than in the Reference Case.

Figure 7.5

Change in primary energy demand between the Equitable Growth Scenario and the Reference Case in 2050, mboe/d



Source: OPEC.

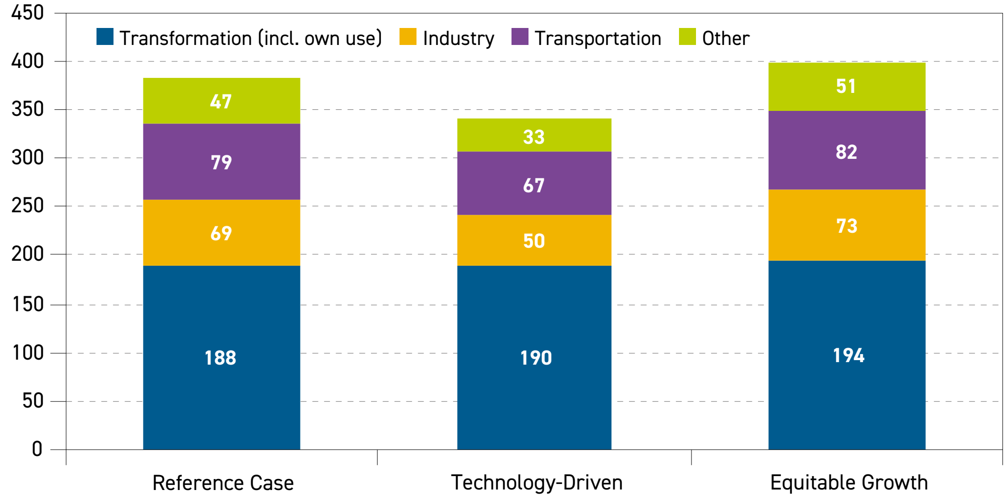
All other energy sources in the EGS are expected to grow faster than in the Reference Case, pushing global primary energy demand by 2050 to a level almost 17 mboe/d higher than in the Reference Case. This incremental energy demand primarily comes from oil, gas and 'other renewables', with each increasing by around 6 mboe/d more than in the Reference

Case. Some demand increases are also projected for nuclear, biomass and hydroelectricity, albeit on a much lower scale, as the combined contribution of these energy sources accounts for less than 3 mboe/d by 2050.

From a sectoral perspective, higher oil demand in the EGS mainly comes from the residential and industrial sectors in developing countries, and partly from the transportation sector. Since these sectors are also the primary sources of incremental growth in the Reference Case, the example of oil demonstrates that the key trends in the EGS also broadly follow Reference Case developments. Indeed, as presented in Figure 7.6, EGS energy demand in all major sectors increases almost proportionally with the overall demand change. The largest incremental demand is projected for the transformation sector, which is expected to be around 5 mboe/d higher than demand in 2050 in the Reference Case. This is driven by higher electricity demand in this scenario.

However, this relatively small change in overall demand only masks larger changes in the fuel composition within this sector. In the period to around 2035, the focus of the EGS is on providing additional electricity to those either lacking or having only limited access. During this period, all options for higher electricity generation are used, including coal, which stays above levels projected in the Reference Case. In the latter period, however, the decline in coal demand is expected to accelerate and fall below Reference Case levels as sufficient capacity from solar, wind, gas and nuclear power is built. This results in electricity being generated in a significantly different way by 2050. Moreover, energy demand is also projected to increase in sectors such as industry, transportation and others. However, the level of incremental demand in these sectors is somewhat lower, in the range of 3 mboe/d to 4 mboe/d, compared to the Reference Case.

Figure 7.6
Global energy demand* by sector in the Reference Case and in alternative scenarios, 2050, mboe/d



* *Sectoral energy demand in this figure is measured from the primary energy side. Therefore, electricity is included in the transformation sector and is excluded from industry, transport and other sectors.*

Source: OPEC.

7

It is important to note that the sectoral breakdown presented in Figure 7.6 considers energy demand from the primary side, meaning that all sources of electricity are included in the transformation sector. To avoid double counting, electricity demand is then excluded from the industrial, transportation and other sectors, which then provides a somewhat distorted picture of the relative importance of these sectors. Nevertheless, despite this shortcoming, the approach helps to demonstrate the impact of the various narratives on the structural changes in sectoral energy consumption between the scenarios.

These structural changes are even more pronounced in the TDS. While the EGS narrative is largely one of adding energy demand, the key feature of the TDS is the substitution of oil, gas and coal by electricity, supplemented by energy savings from the use of more efficient technologies, such as more advanced ICEs, improved gas furnaces and improved building codes. Moreover, the complexity of the sectoral changes in this scenario goes beyond these major shifts, as some fuel substitution is also set to take place from coal to gas (mainly in the power and industrial sectors), from oil to gas (mainly in the industrial sector) and from biomass to oil (in the residential sector).

The primary effect of these changes is a significant demand reduction in all sectors of final consumption compared to the Reference Case. The largest demand drop is expected in the industrial sector, as it provides some 'low-hanging fruit' for better efficiency and emissions reductions. Global energy demand in this sector could potentially be 18 mboe/d lower than the corresponding level in the Reference Case by 2050.

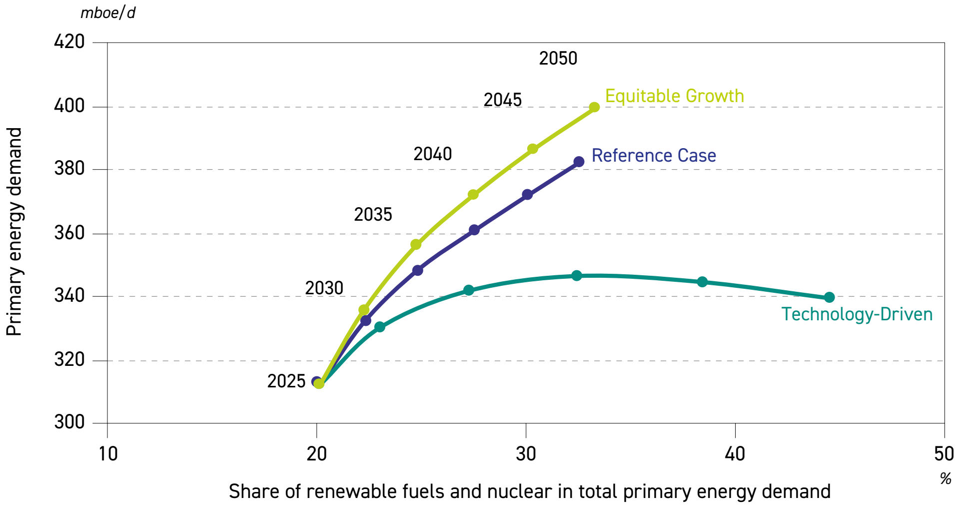
A smaller but still significant demand drop of around 15 mboe/d, compared to the Reference Case, is also projected for the residential, commercial and agricultural sectors, which are combined into the 'other sectors' category in Figure 7.6. The primary means of achieving such reductions will likely include more efficient lighting, heating and cooling systems, the application of better insulation and stricter building codes, as well as more efficient vehicles in the agricultural sector, among others.

There is also considerable potential to reduce energy demand in the transportation sector. Starting with road transportation, this could be achieved through a combination of faster EV penetration and the proliferation of more efficient ICEs. Further electrification of rail transport, especially in developing countries, and efficiency improvements in aviation and marine transport, could also play a role in lowering energy demand. The combined effect of these measures is estimated at 12 mboe/d by 2050, bringing global energy demand in this sector to 67 mboe/d in the TDS, compared to 79 mboe/d in the Reference Case.

A small change between energy demand in the TDS and the Reference Case is projected in the transformation sector. As shown in Figure 7.6, the TDS is 2 mboe/d higher by 2050 on the back of offsetting larger electricity generation by a significantly more efficient way of generation, especially when measured from the primary energy side. At the same time, this sector is set to see the largest change with respect to its energy mix as large amounts of inefficient coal use is substituted by renewables and efficient gas-powered electricity, and supplemented by nuclear electricity. Moreover, the TDS also assumes a significant increase in CCUS capacity, which helps to meet higher electricity demand without increasing CO_2 emissions.

As expected, trends outlined in the previous paragraphs for specific sectors will also have significant impacts on the overall energy system under the alternative scenarios. These are presented in Figure 7.7 by linking primary energy demand and the share of non-hydrocarbons

Figure 7.7
Global energy system in the Reference Case and in alternative scenarios, 2025–2050



Source: OPEC.

in the respective energy mixes. The figure shows that the TDS and EGS evolve in diverse ways and represent fundamentally different energy systems by the end of the outlook period.

In the EGS, the share of non-hydrocarbons only marginally varies from the Reference Case over the next ten years and remains generally comparable even in the second part of the forecast period, with the overall difference not exceeding 1 percentage point. This means that, from a structural point of view, the two pathways are broadly aligned and follow the same trajectory.

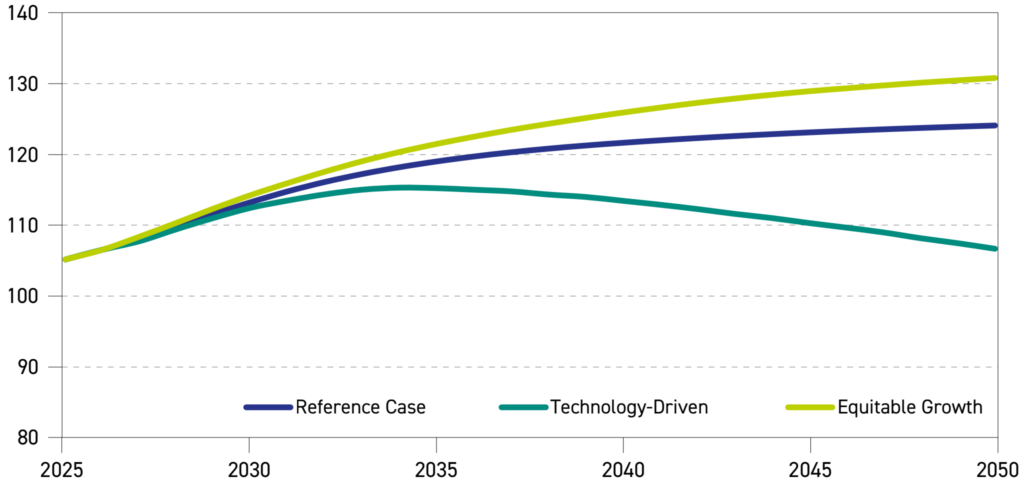
However, this is not the case in the TDS where assumed efficiency improvements and the faster penetration of renewable energy sources result in a significantly faster-expanding share of non-hydrocarbons. Indeed, their combined share exceeds 27% already by 2035 and continues growing to almost 45% by 2050. This compares to around 25% and 33% in the Reference Case, respectively.

7.2 Oil demand

In previous parts of this chapter, all figures were expressed on an energy content basis, given the need to use a common unit when dealing with various energy sources. This sub-section now expresses oil demand to the commonly used volumetric basis in b/d. Furthermore, for better comparability – as oil demand expressed in barrels per day typically also includes other liquids blended with refined products, such as biofuels, GTLs and CTLs – corresponding adjustments were also made (hence, the reference to 'liquids' demand in Figure 7.8 through Figure 7.11).

The outlook for global liquids demand in the Reference Case, as well as the two alternative scenarios, is presented in Figure 7.8. Demand in both the EGS and the Reference Case is generally closely aligned, especially in the first half of the outlook period, and both continue to grow until 2050. By contrast, liquids demand in the TDS gradually departs from the Reference Case and starts declining after 2035. This trajectory gradually opens up a demand difference compared to the Reference Case of close to 4 mb/d in 2035, which expands to 13 mb/d in 2045 and more than 17 mb/d in 2050.

Figure 7.8
Global liquids demand in the Reference Case and in alternative scenarios, 2025–2050, mb/d

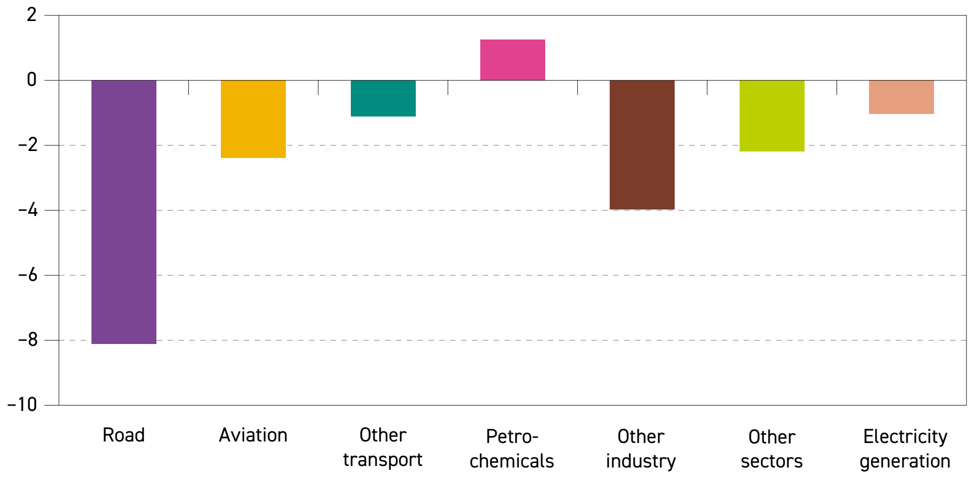


Source: OPEC.

This demand difference increases steadily because of the gradual penetration of more efficient technology and ongoing oil substitution in all key sectors of consumption, with the exception of petrochemicals. It is worth emphasizing that, concurrently with the above evolution, the TDS also assumes a steady expansion of deployed CCUS, as well as, increasingly, the emergence of a CCE framework. The combined effect of these two developments is a reduction in the need for a much larger substitution of oil and gas in the global energy mix, particularly in so-called hard-to-abate sectors. As a result, beyond 2050, any assumed decline in oil demand would almost certainly be much less pronounced, potentially stabilizing for many years to come.

More detail on the key oil demand differences between the TDS and the Reference Case at a sectoral level is visible in Figure 7.9. By 2050, the TDS is 17.5 mb/d lower than in the Reference Case, with road transportation having the most potential for both oil substitution and efficiency improvements. Specifically, the foremost option for oil substitution is through a higher penetration of EVs. In the Reference Case, the EV fleet is projected to increase by 565 million vehicles over the forecast period, reaching more than 630 million by 2050. In the TDS, a larger share of EVs is assumed, especially in China and the OECD, with the global fleet reaching almost 1 billion vehicles by 2050. This results in more than 4 mb/d of oil demand being replaced. On top of this, assumed further efficiency improvements in ICE vehicles could result in a further oil demand reduction of nearly 4 mb/d by 2050, compared to the Reference Case.

Figure 7.9
Difference in liquids demand between the Technology-Driven Scenario and the Reference Case by sector in 2050, mb/d



Source: OPEC.

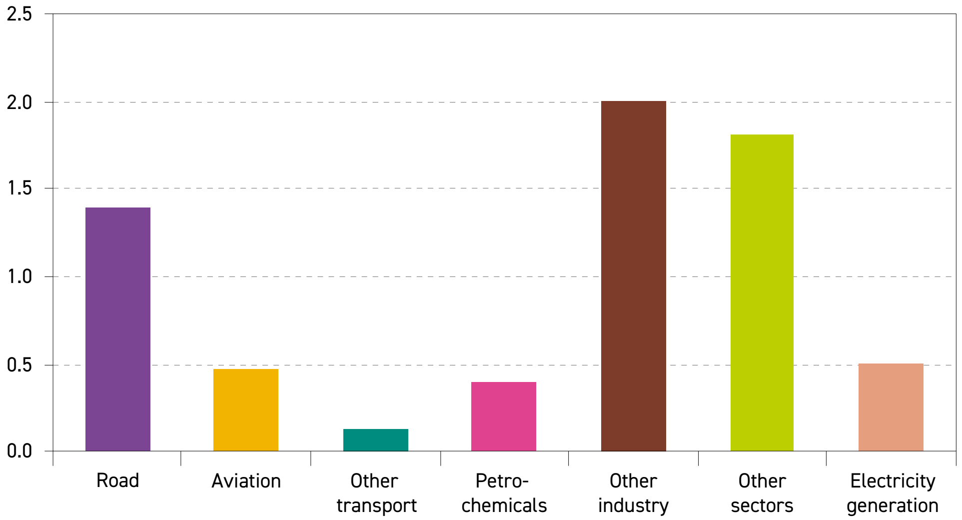
In the industrial sector, the potential demand reduction is projected at 4 mb/d by 2050. Compared to the Reference Case, some of this is related to oil being further substituted by electricity and natural gas. Another part of this effect stems from the use of more efficient industrial furnaces.

Similar effects are possible in the aviation sector, resulting in oil demand potentially being around 2.4 mb/d lower than the Reference Case. This is primarily due to more efficient aircraft but is also partly the result of some electrification in the sector, particularly regarding smaller aircraft operating on shorter routes.

In the remaining sectors, lower oil demand is mainly due to the substitution of oil by gas and electricity. The petrochemical sector is the main exception, as oil demand here is expected to increase in the TDS compared to the Reference Case. In this instance, however, oil demand is only around 1.2 mb/d higher than in the Reference Case. This increase is largely due to stronger demand for petrochemical products in car manufacturing, the construction sector, intensified agriculture and higher demand for solar PV panels and wind turbines. At the same time, some of this higher demand is offset by the increased reuse of plastics within the CCE framework.

Liquids demand in the EGS is expected to surpass Reference Case levels consistently over the entire outlook period, supported by higher economic growth and increased efforts to improve energy and electricity access, especially in developing countries. The gap between these two pathways gradually opens up to 2.5 mb/d by 2035 and further to almost 7 mb/d by 2050. By then, liquids demand in the EGS reaches almost 131 mb/d. As presented in Figure 7.10, higher demand will be spread across all major consumption sectors, with the largest additional demand (compared to the Reference Case) occurring in the 'other industry' (+2 mb/d), 'other sectors' (+1.8 mb/d, mainly in residential sector) and road transportation (+1.4 mb/d) sectors.

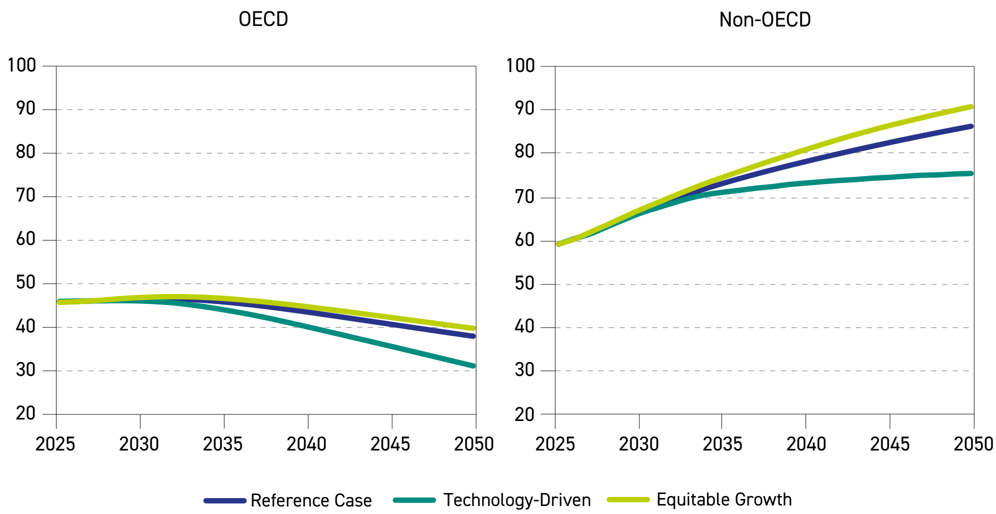
Figure 7.10
Difference in liquids demand between the Equitable Growth Scenario and the Reference Case by sector in 2050, mb/d



Source: OPEC.

Different narratives and driving forces that underpin major changes in alternative scenarios will have a varying impact on sectoral oil demand but also on its regional distribution. Having reviewed related changes at the sectoral level in figures 7.9 and 7.10, Figure 7.11 provides a high-level summary of regional trends, breaking down OECD and non-OECD oil demand separately.

Figure 7.11
Liquids demand in the Reference Case and in alternative scenarios, 2025–2050, mb/d



Source: OPEC.

This figure shows that a large part of the incremental demand in the EGS materializes in non-OECD countries on the back of faster economic growth leading to a larger middle class and accelerating urbanization and industrialization, especially in Africa, India and Other Asia. An integral part of this development is improved mobility and availability, as well as better affordability of modern energy sources, including LPG. The net effect is that non-OECD oil demand in the EGS expands from 59 mb/d in 2025 to almost 91 mb/d in 2050, which is about 5 mb/d higher than in the Reference Case.

Upside potential for OECD oil demand in the EGS is rather limited. As mentioned in the scenario description, faster GDP growth in the EGS is biased towards developing countries and only slightly impacts OECD countries. Moreover, under the policy setup assumed in the Reference Case and the EGS, higher GDP in these countries allows for a slightly faster adoption of advanced technologies and more rapid growth of renewable energy compared to the Reference Case, thus partially offsetting potentially higher oil demand. The same impact is also visible for gas and coal demand. Therefore, OECD liquids demand is expected to be less than 2 mb/d higher compared to the corresponding level in the Reference Case by 2050.

In contrast to the EGS, regional liquids demand changes in the TDS are much more amplified. This is particularly the case for the non-OECD, where projections indicate that oil demand could potentially be as much as 11 mb/d lower by 2050, compared to the Reference Case. A significant part of this demand decline would come from China due to its fast-growing number of EVs, the potential for further efficiency improvements and oil substitution by electricity. This would especially be the case in the initial years of the forecast period, when a major part of the non-OECD demand reduction is linked to developments in China.

In the second part of the forecast period, the above-mentioned factors gradually impact other developing countries too. Indeed, non-OECD demand growth in the TDS decelerates significantly and starts plateauing somewhere around 76 mb/d towards the end of the forecast period, as opposed to continued growth seen in the EGS and the Reference Case.

OECD oil demand in the TDS is also significantly impacted. In addition, it is expected to be on a declining trajectory in the long term in the Reference Case too. An emphasis on oil substitution by electricity and gas, combined with the efficiency improvements in the road transportation and industrial sectors assumed in the TDS, consistently pushes this trajectory to lower levels. The difference to Reference Case levels evolves relatively fast, reaching almost 2 mb/d by 2035 before progressing to around 7 mb/d by 2050, representing a demand reduction of almost 20%.

Focus on Africa

Key takeaways

- Africa's population is projected to expand from just over 1.5 billion in 2025 to almost 2.5 billion by 2050. The overall increase is around 917 million. By mid-century, Africa's projected population is set to represent nearly one-quarter of the global population.

- Africa's GDP is expected to grow strongly, with average annual growth of 4.4% between 2025 and 2050, supported by rapid population growth, urbanization and gradual economic diversification. This compares with average annual GDP growth rates of 3.6% for the non-OECD and 2.9% for the world. Africa is set to be the only region to witness a continued acceleration in growth over the outlook period.

- Across Africa, energy policy is driven by the imperatives of energy security, closing the vast energy access gap affecting hundreds of millions of people and pursuing a modern, climate-resilient development path. This has spurred a diverse range of national strategies aimed at deploying technologies across all energy sources.

- Meeting Africa's increasing energy demand will require all sources of energy over the medium to long term. Energy demand is expected to increase from 17.5 mboe/d in 2025 to 29.3 mboe/d by 2050, representing an average growth rate of 2.1% p.a.

- Demand for natural gas is set to rise to 6.5 mboe/d in 2050 from 3 mboe/d in 2025. Its share in the energy mix is expected to increase to over 22% by the end of the outlook period, driven by the availability of abundant resources and its lower-carbon footprint. Demand for coal is expected to increase gradually to reach 2.3 mboe/d by 2050, up from 2.1 mboe/d in 2025, while nuclear energy is expected to remain limited, reaching only 0.1 mboe/d by 2050.

- Demand for renewable energy (including biomass, wind, solar, hydro and others) is projected to increase steadily to reach 11.7 mboe/d in 2050. Biomass is set to account for the largest share, reaching 8.8 mboe/d, despite policy challenges and the gradual expansion of other energy solutions, particularly in the residential sector.

- Oil demand in Africa is expected to register an increase of around 4.3 mb/d to reach 9.2 mb/d by 2050. Demand patterns vary across countries, reflecting differences in economic expansion.

- Africa's role in world oil supply continues to expand as the continent focuses on expanding E&P activities to increase hydrocarbon output. Liquids supply is set to increase steadily over the coming decades, with non-DoC African countries contributing an additional 0.7 mb/d to global supply growth over the outlook. This growth is primarily driven by developments in countries such as Uganda, Côte d'Ivoire, Senegal and Namibia.

- Africa's projected refinery capacity additions are set to be significant. Distillation capacity is expected to increase by about 3.2 mb/d between 2026 and 2050, supporting efforts to reduce refined product import dependence, enhance intra-African trade and strengthen regional supply security.

Africa is one of the richest regions in terms of commodity resources in the world, given its huge reserves of oil, natural gas and other critical minerals. In addition to its abundant natural resources, the continent is also characterized by dynamic demographic developments, a young population, a growing workforce and significant potential for industrial expansion and urbanization, which are expected to lead to an increase in energy demand.

Africa is expected to have by far the largest population growth, increase in working-age population and expansion in urbanization of all regions over the outlook period. In developed regions around the world, increased urbanization has historically led to higher demand for modern energy, and this is expected to be the case in Africa as well. Across the continent, the population living in cities has increased rapidly in recent years, and this trend is set to continue in the future.

Given these developments, access to reliable and affordable modern energy is essential to support the projected level of urbanization, drive economic development and help improve the livelihoods of over 900 million people who currently do not have access to modern cooking facilities and 600 million people who have no access to electricity across the continent.

This chapter provides insights into four broad areas of assumptions: population and demographics, economic growth, energy policies, and technology and innovation. It also interweaves key aspects of energy and oil demand, oil supply and refining in Africa.

8.1 Population and demographics

Africa's current and projected population growth can be viewed as a key asset that could enable the continent to become a major driver of global economic growth. Projections indicate that Africa's population is set to expand from just over 1.5 billion in 2025 to almost 2.5 billion by 2050. The overall increase is around 917 million. This expansion implies that, by 2050, over a quarter of the world's population will reside in Africa.

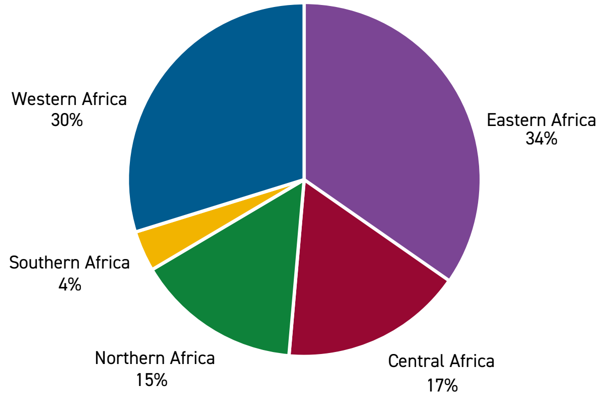
While all regions are set to witness population growth, there are disparities across the continent. At the regional level, Table 8.1 and Figure 8.1 show that the population is expected to

Table 8.1
Africa's population by region, millions

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
Eastern Africa	513	579	647	716	786	855	342
Central Africa	220	254	290	329	370	412	192
Northern Africa	276	297	317	336	355	373	97
Southern Africa	74	78	82	85	89	92	18
Western Africa	467	519	574	628	683	735	268
Africa	**1,550**	**1,727**	**1,910**	**2,096**	**2,282**	**2,467**	**917**

Source: UN.

Figure 8.1
African share of population by region, 2050, %



Source: UN.

remain heavily skewed towards Eastern and Western Africa. These two regions are projected to continue to account for the largest share of the continent's population by mid-century. They will also witness the largest growth over the forecast period. About 855 million people, or approximately 34% of the total population, are set to be located in Eastern Africa, while about 735 million people, or approximately 30% of the total population, are expected to be concentrated in Western Africa. More broadly, Sub-Saharan Africa is forecast to witness the largest growth.

By 2050, Central Africa and Northern Africa are projected to have populations of approximately 412 million and 373 million, respectively, which is equivalent to 17% and 15% of the continent's total population. Southern Africa is forecast to have a population level of about 92 million by 2050, or approximately 4% of the continent's total population.

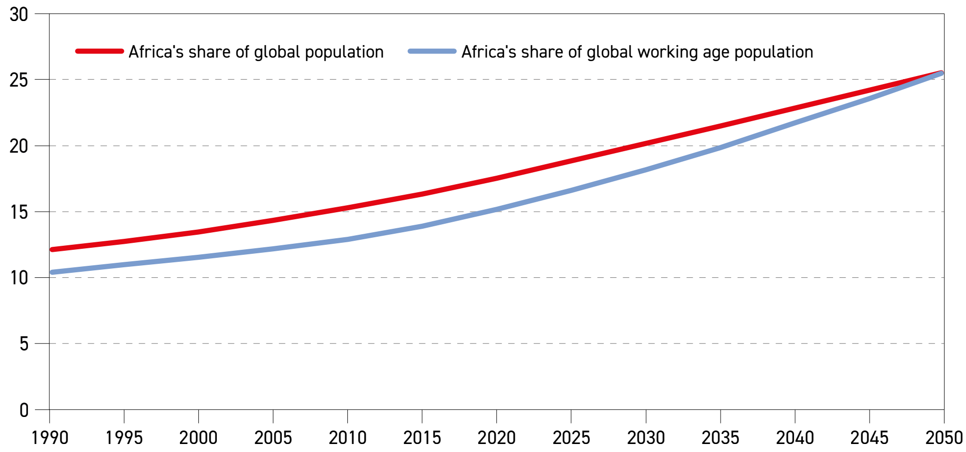
By country, population growth is set to be concentrated mostly in Nigeria, the Democratic Republic of Congo (DRC), Ethiopia, Egypt and Tanzania. Nigeria, in particular, is projected to be the fifth largest country in terms of population by 2050. The highest percentage growth rates are expected in South Sudan, Niger, Angola and Benin.

8.1.1 Working-age population

Africa is the youngest and fastest expanding continent in terms of population. The result is a growing working-age population that could support future global growth and productivity. The rapid expansion of the working-age population, defined as individuals between 15 and 64 years old, means Africa is set to add about 667 million people to the global workforce between 2025 and 2050.

Figure 8.2 shows that Africa's share of the global working-age population is set to be about 25% by 2050. This translates to around 1.56 billion people, although the demographic profile is unevenly distributed. For instance, some countries in Northern Africa and Southern Africa may see a decline in the working-age population in the longer term due to lower fertility rates, while countries in Western Africa and Eastern Africa are expected to witness a significant increase in the working-age population.

Figure 8.2
Africa's share of global population and working-age population, 1990–2050, %

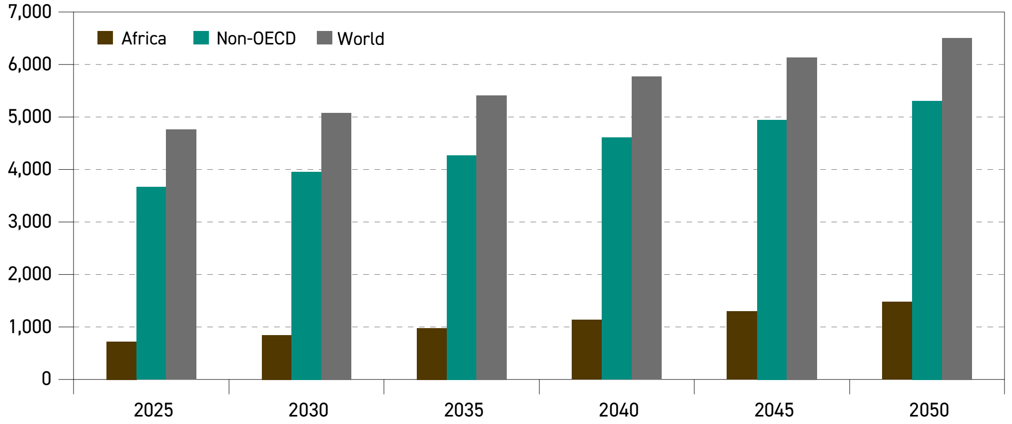


Source: UN.

8.1.2 Urbanization

Africa's urban population growth is faster than every other region, a trend that is also connected to its overall population growth and expanding working-age population. The continent's urban population is expected to double from 720 million in 2025 to almost 1.5 billion by 2050 (Figure 8.3).

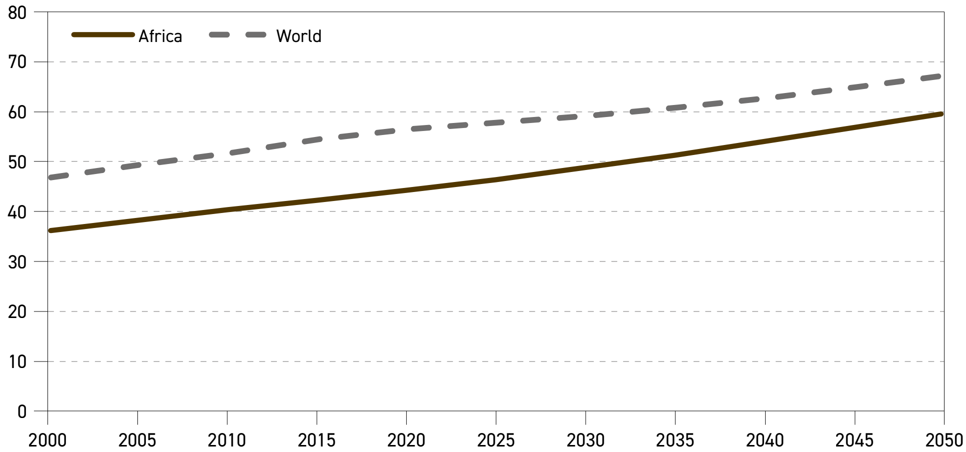
Figure 8.3
Urban population of selected regions, millions



Source: UN.

Consequently, the continent's urban population rate is projected to increase from around 46% in 2025 to about 60% by 2050 (Figure 8.4). This has the potential to drive the expansion of megacities and foster the emergence of new urban corridors.

Figure 8.4
Urbanization rate, 2000–2050, %



Source: UN

Although the continent's expected urbanization level remains below the global average of around 67% by 2050, there is potential for further expansion. For example, through higher fertility rates within cities and increased migration from rural communities to urban centres in search of access to essential services, including healthcare, education and infrastructure, such as electricity and potable water, which may be lacking in rural communities.

However, this expansion may be concentrated more in some countries than others. For instance, Nigeria has the highest urban population rate, and its cities, particularly Lagos, are experiencing rapid expansion. Lagos is projected to become one of the world's largest megacities, with its population growth driven by both a high natural increase and sustained rural-to-urban migration. Kinshasa (DRC) is also expected to see a major expansion as a megacity, reflecting the high population growth of Central Africa.

In addition, East African cities like Addis Ababa (Ethiopia), Nairobi (Kenya) and Dar es Salaam (Tanzania) are witnessing rapid urban expansion, often forming interconnected urban systems that drive regional economic activity. Cities like Cairo (Egypt) in Northern Africa are also set to continue to grow due to urbanization. However, the pace of growth is generally slower than in Sub-Saharan Africa.

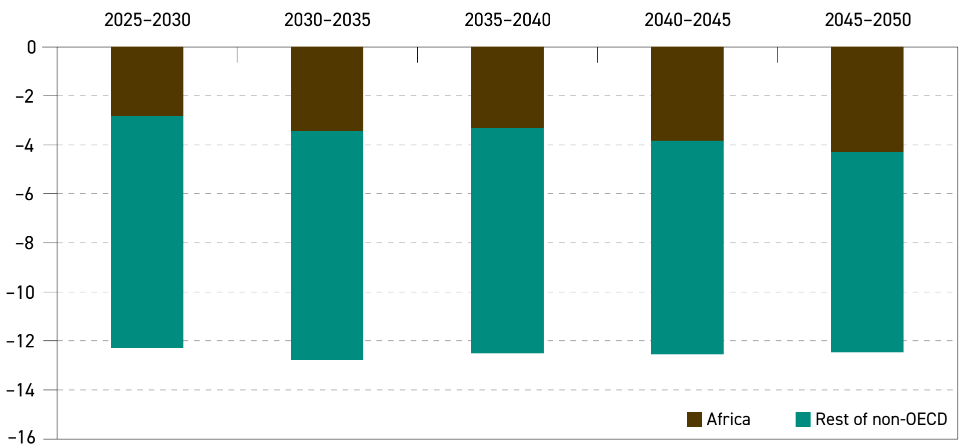
8.1.3 Migration

Migration in Africa is a key component of its demographic landscape. It is generally driven by economic opportunities and the search for better living conditions. Most African migration is predominantly intra-African, between countries on the continent, rather than to other world regions. The predominant migration trend in Africa is rural-to-urban or country-to-country migration within the same region. For example, people move from landlocked countries to coastal economies, or from politically fragile states to more stable countries.

Globally, medium-term migration patterns are sensitive to contemporary geopolitical events that could lead to inter-regional movements of people. In addition, long-term projections see movements of people from non-OECD to OECD countries.

Africa's medium-term migration outlook indicates a net movement of 2.8 million people out of the continent before increasing to 3.5 million in the period from 2030 to 2035. Although this movement is set to decline between 2035 and 2040, it is expected to increase again, reaching 3.8 million between 2040 and 2045 and 4.3 million in the final five-year period of the forecast period (Figure 8.5).

Figure 8.5
Net migration by region, millions



Source: UN.

8.2 Economic growth

The medium- to long-term economic trajectory for the African continent, including an appreciation of key trends, is a vital component in assessing the continent's future energy outlook.

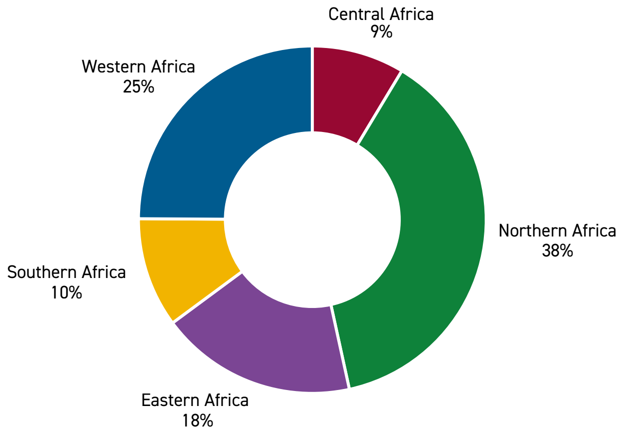
Economies on the African continent have in the past benefitted from commodity-driven expansions, improved macroeconomic policies, economic diversification and some regional integration, although growth has sometimes been cyclical and subject to external factors, such as global financial conditions, periodic volatility and commodity price swings.

Over recent decades, despite periodic fluctuations, the continent has continued to maintain positive, albeit moderate real GDP growth of around 3% annually. This demonstrates resilience in the face of periodic domestic and global shocks.

However, the nature of economic growth in Africa varies significantly from one region to another, reflecting differences in economic structure, resource endowments and the degree of integration with global markets.

Northern Africa is the largest contributor to GDP on the continent and is expected to make up 38% of the continent's economy by 2025 It is also the fastest growing African region in terms of economic growth, has a relatively high level of GDP per capita and is well integrated into European trade and investment systems. Egypt, Algeria, Morocco, Libya and Tunisia are the leading regional economies. Growth is driven by tourism, investment in infrastructure, energy exports and production ties with Europe.

Figure 8.6
Share of total African GDP by region, 2025, %



Source: IMF/EIU/Oxford Economics/World Bank/National Sources/OPEC.

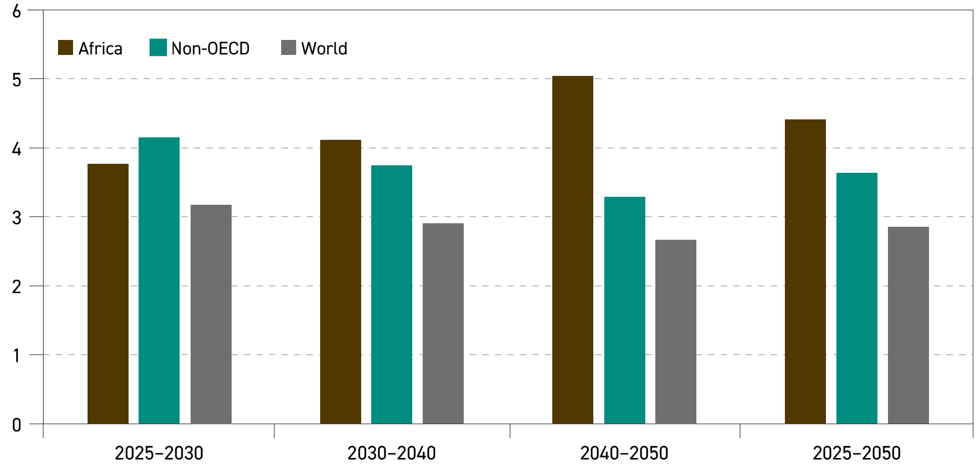
In 2025, Western Africa had the second largest share of Africa's GDP at 25%. Major countries like Nigeria, Côte d'Ivoire and Senegal have performed strongly in terms of average GDP expansion, due to their focus on hydrocarbons and agricultural production, basic infrastructure development, and emerging industries such as textiles and power generation.

Eastern Africa is one of the fastest growing regions on the continent with an 18% share of total GDP in 2025. Several countries in this region, including Ethiopia, Kenya, Rwanda and Tanzania, have recently recorded higher growth rates due to government spending, service sector expansion and improved business environments. Rwanda and Tanzania have also witnessed sustained growth in agriculture, tourism and information technology. Elsewhere, Southern and Central Africa account for 10% and 8%, respectively, of total GDP on the continent.

Figure 8.7 shows that in the long term, Africa's GDP growth rate trajectory is expected to rise to around 5% between 2040 and 2050, driven by strong demographic expansion. This corresponds to an average annual growth rate of 4.4% over the entire outlook period, compared with average annual GDP growth rates of 3.6% for the non-OECD and 2.9% for the world. Africa is set to be the only region to witness continued growth acceleration over the outlook period.

As already noted, the continent's expected demographic dividend out to 2050 has enormous potential for global growth dynamics. However, sustaining these growth levels will require addressing key issues, including infrastructure investment, inclusive growth strategies, structural and governance reforms and human capital development.

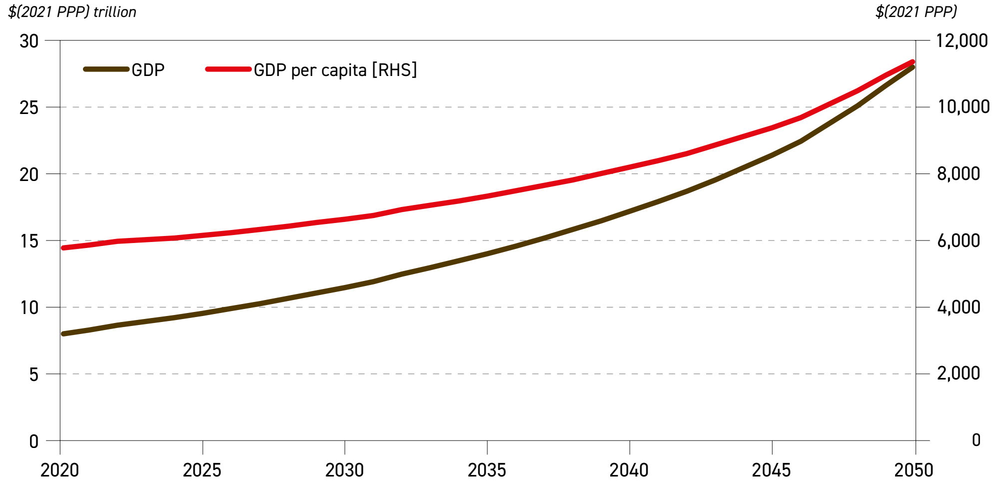
Figure 8.7
Long-term annual GDP growth rate (in real terms), % p.a.



Source: OPEC.

Figure 8.8 shows that although Africa's economic history over the past few decades is one of moderate but steady expansion, it does not automatically translate into rising incomes per person. Per capita income growth has been slower than in other regions, meaning that improvements in average living standards have been more limited than what the overall average growth figures suggest.

Figure 8.8
African GDP and GDP per capita, 2020–2050



Source: OPEC.

8.3 Energy policies

Across Africa, energy policy is driven by the imperatives of energy security, closing a vast energy access gap for hundreds of millions of people and simultaneously pursuing a modern, climate-resilient development path. This has led to a diverse range of national strategies aimed at deploying both conventional and low-emissions energy solutions.

For this reason, the long-term strategy of many African countries involves striking a balance between sustaining hydrocarbon production – particularly oil and gas – and addressing environmental concerns. This approach also aligns with the broader vision of advocating for sustainable practices within the hydrocarbon industry and emphasizing the continued importance of oil and gas in the global energy mix.

It is also vital to note that many African countries are simultaneously critical players in the global emissions reduction agenda. By leveraging new technology, improving energy efficiency and investing in mitigation initiatives, many have established a framework for lower-emission hydrocarbon production and the scaled deployment of renewables, where conditions allow.

Regionally, energy policy in **Northern Africa** is guided by the development of hydrocarbon resources, as well as the expansion of renewable energy sources for both domestic and export purposes. This is clearly evident in countries such as Algeria, Egypt, Libya and Morocco. Energy policies include the development of oil, natural gas and green hydrogen for domestic consumption and export, as well as the expansion of other renewable energy sources, such as hydro, solar and wind. Additionally, Egypt's Nuclear Energy Expansion Plan, announced in 2023, is part of a broader road map to scale up nuclear power as a cornerstone of long-term energy security and emissions reduction goals.

In **Southern Africa**, South Africa plays a key role in the region's energy policy, given its extensive industrial base and its heavy dependence on coal for power generation. However, the country does have plans to shift away from coal with the Just Energy Transition Partnership initiative. Moreover, South Africa's energy sector is undergoing a major transformation driven by the Electricity Regulation Amendment Act, which came into effect in January 2025. This landmark reform opens the electricity market to competitive generation, encouraging private investment in renewable energy and transmission infrastructure. Furthermore, South Africa's Green Transport Strategy, launched in 2023, presents a national framework for advancing sustainable mobility through the development of electric public transportation systems, including buses and railways.

Elsewhere, Zambia's Renewable Energy Strategy 2035, unveiled in 2023, sets a bold target of achieving a 75% share of renewables in the national energy mix by 2035. This vision is underpinned by major investments across a diversified portfolio of solar, wind, geothermal and hydroelectric projects, including the development of 463 MW of solar, 500 MW of wind, 130 MW of geothermal and 880 MW of large-scale hydro capacity.

The **Eastern African** region has seen a number of plans and projects launched in recent years. Rwanda's Wind Energy Development Plan, launched in 2022, outlines a strategic effort to install 500 MW of wind capacity by 2027, reinforcing the country's commitment to diversifying its energy mix and reducing dependence on hydropower, which remains vulnerable to seasonal variability. The plan prioritizes investment in wind farms and supporting infrastructure, particularly in high-potential regions such as the Eastern Province, with flagship projects such as the Gishoma Wind Farm, a 100 MW facility, being developed in collaboration with private investors.

Elsewhere, Kenya's Nuclear Research and Training Program, launched in 2022, is a foundational step towards building national capacity in nuclear energy through the development of skilled human resources and technical expertise. Additionally, in the road transportation sector, Kenya's National Electric Mobility scheme, launched in 2022, presents a comprehensive framework to accelerate the adoption of EVs and hybrid technologies as part of the country's broader transition towards a low-carbon economy. Simultaneously, the KenGen Geothermal Development Project targets 5 GW of geothermal capacity by 2030, with the Olkaria expansion adding 140 MW, strengthening the role of geothermal energy in Kenya's 90% renewable energy mix. In addition, the Solar Power Expansion Program, launched in 2022, seeks to add 1 GW of solar capacity by 2025 through large-scale and rooftop installations, supported by financial incentives and regulatory reforms.

Launched in April 2024, the World Bank's PRIME programme is a strategic initiative designed to modernize Ethiopia's electricity sector by expanding grid infrastructure, improving financial sustainability and unlocking renewable energy potential through private sector engagement. Backed by a $522 million International Development Association (IDA) credit in its first phase, the programme focuses on critical investments to support the country's rapid electrification and long-term energy resilience.

In **Western Africa**, hydrocarbon-producing countries such as Nigeria play a critical role in the region as both major energy producers and consumers, given their large populations. Against this backdrop, Nigeria's main focus is on advancing its oil and gas infrastructure. For instance, the country's 'Decade of Gas' initiative, announced in 2021, aims to drive gas utilization, minimize gas flaring and expand LNG exports to support economic growth.

Other countries like Ghana, Senegal and Niger are also expanding the role of natural gas in power generation and industrial applications sectors. For instance, Ghana's infrastructure enhancements, particularly through the expansion of the Western Corridor Gas Infrastructure Development Project, are designed to improve the processing and transportation of natural gas from offshore fields to onshore facilities. This is vital for ensuring a stable domestic gas supply, reducing reliance on imported fuels and contributing to lower GHG emissions. Additionally, its *Petrochemical Industry Development Plan*, launched in 2022, outlines a transformative strategy to strengthen domestic production, reduce import dependency and embed sustainable practices within the sector.

In **Central Africa**, the DRC has plans to diversify its current energy portfolio, with a focus on gas monetization, hydropower and other energy sources like solar. It aims to double its power generation to 1,500 MW by 2030. Elswhere, Equatorial Guinea plans to further develop its oil and gas portfolio and meet GHG emissions reduction targets of 35% and 50% by 2030 and 2050, respectively.

Meanwhile, Gabon's energy diversification strategy is focused on the progressive replacement of diesel-fired power plants with natural gas-fuelled thermal power generation facilities. This strategic shift, is anticipated to yield substantial reductions in GHG emissions. Other countries in the region, such as Angola, have plans to develop their hydrocarbon resources, particularly oil and LNG.

8.4 Energy demand outlook in Africa

Africa remains one of the most underdeveloped energy markets globally, with energy poverty constituting a significant barrier to economic development. Across the continent, over 600 million people still lack access to electricity and nearly a billion remain without access to clean cooking, relying on traditional biomass that has negative long-term effects on household health and well-being, especially for women and children.

Today, access to electricity in some regions in Africa also remains limited. It is often unaffordable and unreliable, largely due to frequent power outages caused by weak distribution and transmission networks that cause power supply instability for households. It should be noted that energy poverty is not only a social challenge related to basic electricity access, it is also a constraint on unlocking economic growth through industrialization, improved healthcare, job creation, better education outcomes, digital access and overall human development.

This helps explain why energy use per capita on the continent still ranks among the lowest globally, especially in Sub-Saharan Africa, despite the continent being endowed with vast natural energy resources that remain underutilized in meeting increasing demand.

It is vital for Africa that these resources are fully utilized. This is essential to improve access to affordable and reliable energy for those who continue to go without, and to support industrialization and broader development. Energy security, alongside emissions reductions, will be fundamental to the continent's long-term prosperity.

For Africa to sufficiently meet its increasing energy demand, all sources of energy, especially oil and gas, will be required in the medium to long term. Energy demand is expected to increase from 17.5 mboe/d to 29.3 mboe/d by 2050, an average growth rate of 2.1% p.a.

Table 8.2 provides an overview of Africa's energy demand outlook by fuel type. It shows an increase in demand for all energy sources between 2025 and 2050, with oil and gas anticipated to increase significantly, in line with strong demand for reliable and affordable energy.

Oil demand is set to increase from 4.6 mboe/d in 2025 to 8.7 mboe/d in 2050, an increase of more than 4 mboe/d over the outlook period. Oil's share in the fuel mix is set to increase from just over 26% in 2025 to around 30% by 2050. More details on oil demand, supply and refining are presented in sections 8.5, 8.6 and 8.7, respectively.

Strong demand growth is also expected for **natural gas**, driven by the availability of abundant resources and its comparatively lower-emissions footprint. Natural gas is set to play a critical role in providing cleaner-burning fuels, especially for cooking in households, as well as for industrial use and power generation. Natural gas demand is forecast to rise to 6.5 mboe/d in 2050 from the current level of 3 mboe/d, while its share in the energy mix is expected to increase to over 22% by the end of the outlook period.

Some major projects are set to increase natural gas export capacity. This includes the Nigeria LNG (Train 7) expansion that will add between 22–30 mtpa. Additionally, in Mozambique, the Coral South Floating LNG (FLNG) facility commissioned in June 2022 and the Coral North FLNG in pre-development phase, as well as the Greater Tortue Ahmeyim on the border of Senegal and Mauritania, represent ongoing gas project expansions. Furthermore, the proposed Trans-Saharan Gas Pipeline (TSGP) project, currently in the implementation phase, is designed to transport about 30 billion cubic metre of natural gas from Nigeria through Niger to Algeria's export pipelines and onto European markets. This transcontinental natural gas infrastructure project will enhance energy cooperation between the participating countries, while also strengthening Africa's role in global gas supply.

Table 8.2
Africa's primary energy demand by fuel type, 2025–2050

	Levels mboe/d						Growth mboe/d	Growth % p.a.	Fuel share %	
	2025	2030	2035	2040	2045	2050	2025–2050	2025–2050	2025	2050
Oil	4.6	5.4	6.2	7.0	7.9	8.7	4.1	2.6	26.5	29.7
Gas	3.0	3.5	4.1	4.8	5.6	6.5	3.6	3.2	17.0	22.3
Coal	2.1	2.2	2.2	2.2	2.3	2.3	0.2	0.4	12.0	7.8
Nuclear	0.0	0.0	0.0	0.1	0.1	0.1	0.0	1.8	0.2	0.2
Renewables	7.7	8.4	9.2	10.0	10.8	11.7	4.0	1.7	44.3	40.0
Biomass	*7.2*	*7.6*	*8.0*	*8.3*	*8.5*	*8.8*	*1.6*	*0.8*	*41.4*	*30.0*
Solar	*0.0*	*0.1*	*0.2*	*0.4*	*0.6*	*0.8*	*0.8*	*12.6*	*0.2*	*2.8*
Wind	*0.1*	*0.1*	*0.1*	*0.2*	*0.2*	*0.3*	*0.2*	*6.4*	*0.3*	*0.9*
Hydro	*0.3*	*0.4*	*0.5*	*0.6*	*0.7*	*0.8*	*0.5*	*3.9*	*1.7*	*2.6*
*Others**	*0.1*	*0.2*	*0.3*	*0.6*	*0.8*	*1.1*	*0.9*	*9.3*	*0.7*	*3.6*
Africa	**17.5**	**19.5**	**21.8**	**24.2**	**26.6**	**29.3**	**11.8**	**2.1**	**100**	**100**

* *Includes geothermal, tidal and wave.*
Source: OPEC.

The use of **coal** in Africa remains relevant and economically significant, especially for industrialized economies with vast reserves, although its dominance is gradually being replaced by renewable energy sources and natural gas. South Africa leads coal consumption in Africa, with its main focus on coal-fired power generation. Demand for coal in Africa is expected to increase only gradually over the outlook period, reaching 2.3 mboe/d by 2050, from 2.1 mboe/d in 2025.

Nuclear energy offers a low-emissions, high-capacity option for Africa in the long term, particularly for industrialization and baseload electricity. Currently, however, only South Africa's Koeberg nuclear energy project is operating as a commercial nuclear plant and any further deployment across the region is expected to remain limited. Indeed, nuclear energy is expected to increase to a mere 0.1 mboe/d by 2050. Countries like South Africa, Egypt, Ghana, Kenya, Uganda, Nigeria and Morocco are nevertheless actively involved in nuclear energy projects, albeit these projects are at different stages of development. For example, Egypt has four reactors under construction that are expected to add around 4.5 GW of capacity when commissioned.

Demand for **renewable energy** (including biomass, solar, wind, hydro and others) is projected to increase steadily to reach 11.7 mboe/d in 2050. **Biomass** remains the largest renewable energy source, widely used in households to supply heat and cooking fuels. Despite policy challenges and the gradual expansion of alternative energy solutions, particularly in the residential sector, biomass demand is expected to increase, reaching 8.8 mboe/d by 2050, up from 7.2 mboe/d in 2025. Traditional biomass is currently used in the form of charcoal and firewood for residential energy in countries like the DRC, Tanzania, Nigeria and Ethiopia. Universal access to clean cooking is still a long way off, although many countries are introducing LPG and household kerosene to support clean cooking on the continent.

For Africa, the use of biomass in its traditional form as a subsistence fuel represents energy poverty; however, in its modern scalable form, it represents a renewable energy solution. In this form, it can help foster rural development and industrial growth that could lead to economic transformation across Africa.

Solar and **wind** are also set to contribute to overall demand growth, supported by strong policy backing. Solar energy is the fastest-growing renewable energy source in Africa, with the continent having some of the highest solar irradiation levels globally. Much of the continent consistently receives sunlight year-round, making solar energy an abundant energy source. The potential for wind energy is more limited than solar, and although its average annual growth rate is significant, its overall contribution remains relatively small. By 2050, wind is expected to have the second-lowest demand level after nuclear. Solar and wind energy demand in Africa is expected to reach 0.8 mboe/d and 0.3 mboe/d by 2050, respectively.

Hydro accounts for a significant share of electricity generation in Africa compared to solar and wind, and is often referred to as the backbone of grid power in many countries, particularly those with vast reserves of water. Moreover, due to hydro's relatively low cost in large-scale generation once related infrastructure is in place, countries generally keep it as a long-term power generation source. Although hydropower alone is not sufficient for the continent's future power generation needs, it is evolving into a core component of Africa's renewable energy mix. Hydro energy demand is projected to increase from 0.3 mboe/d in 2025 to about 0.8 mboe/d by 2050.

The use of **Others** under renewable sources, such as geothermal, is also set to increase over the outlook period, reaching 1.1 mboe/d in 2050, with an average growth rate of around 9.3% p.a. between 2025 and 2050.

8.4.1 Final electricity consumption

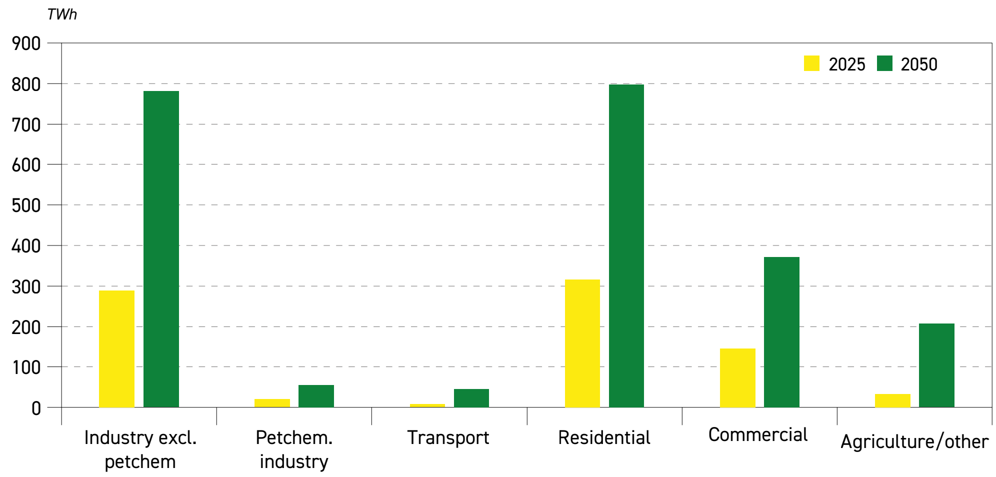
This section focuses on TFC of electricity, discussing recent trends and providing a long-term outlook for the continent.

Total final electricity consumption is projected to rise from more than 800 TWh in 2025 to around 2,250 TWh by 2050, an increase of nearly 180%, driven by growing electrification and economic development across the continent, as well as the expansion of the middle class.

From a sectoral perspective, Figure 8.9 shows Africa's projected TFC by sector between 2025 and 2050. Strong growth is expected across the continent, driven not only by population and economic growth, but also by increasing electrification of end-use sectors, including the residential, industry, commercial and agricultural sectors, as well as petrochemicals and transport.

The relatively low level of electricity consumption illustrates how electricity use in Africa lags behind other regions due to a combination of factors that include economic, infrastructural and policy challenges. The main challenge in terms of the continent's power sector is that many African nations remain without access to sustainable and reliable energy.

Figure 8.9
Africa's final electricity demand by sector, 2025 and 2050, TWh



* *Total final electric consumption (TFC) does not include own use and transmission/distribution losses and, therefore, is not equal to electricity generation.*
Source: OPEC.

The largest demand for electricity is in the residential sector, with demand expected to increase to about 797 TWh by 2050. This is largely driven by continued urbanization and improved livelihoods over the medium to long term. The second largest demand for electricity is in the industrial sector (excluding petrochemicals), with consumption estimated at around 781 TWh in 2050. This can be attributed to industrialization and manufacturing expansion in the coming decades, as Africa's population increases.

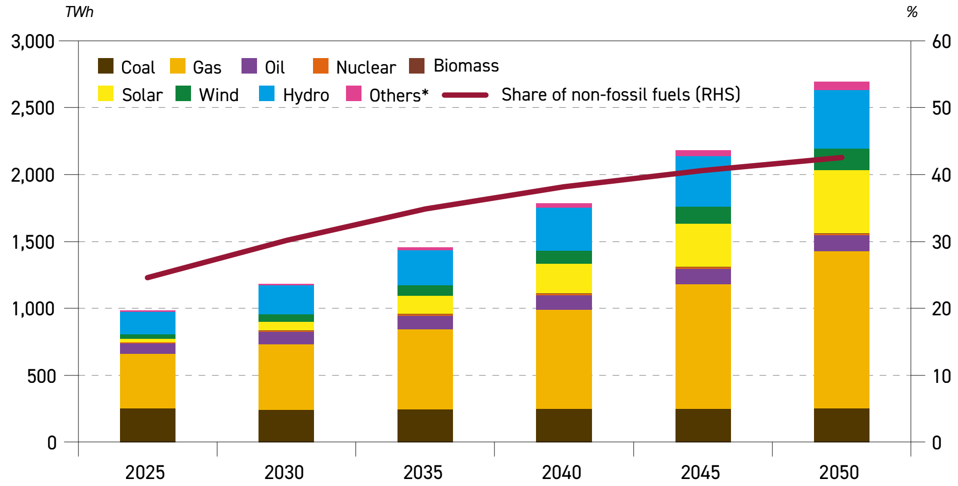
Commercial and agriculture/other sectors are also set to witness significant consumption, reaching over 370 TWh and around 206 TWh by 2050, respectively, driven by increased electrification and the adoption of more advanced technologies. Demand in the petrochemical sector is expected to more than double, reaching about 54 TWh. This is due to the energy intensive nature of petrochemical production, as well as increased processing of feedstock and a broader range of petrochemical products, all of which raise electricity consumption. Finally, consumption in the transportation sector is set to increase from just below 6 TWh in 2025 to almost 45 TWh by 2050, driven by the growing uptake of electric mobility, including rising EV sales and increased electrification of rail transport in the coming decades.

8.4.2 Electricity generation

This subsection focuses on electricity generation, including final electricity consumption, own use and transmission/distribution losses.

Figure 8.10 shows Africa's electricity generation by major fuel through to 2050. Total generation is projected to increase from around 981 TWh in 2025 to about 2,690 TWh in 2050, an increase of close to 175% over the outlook period. By far the largest increase in the generation mix comes from gas, which nearly triples to just under 1,200 TWh by 2050. Gas is set to continue to play a key role due to its cost competitiveness and lower emissions footprint. After gas, coal is the next largest fuel source in 2025, providing 253 TWh of electricity. The outlook for coal is almost flat over the long term, with minor fluctuations before returning to its 2025 level of around 253 TWh in 2050.

Figure 8.10
Africa's electricty generation by fuel, 2025–2050



* *Includes geothermal, tidal and wave.*
Source: OPEC.

Solar is also set to contribute significantly to electricity generation growth, with generation expected to reach 471 TWh by 2050, up from only 23 TWh in 2025. This is mainly driven by vast resources, favourable energy policies and a declining generation cost. Hydropower is expected to remain an important part of the electricity mix, increasing from 168 TWh in 2025 to around 441 TWh in 2050. Electricity generation from wind is projected to increase from 34 TWh in 2025 to 160 TWh by 2050.

The use of oil in electricity generation is set to increase from 80 TWh in 2025 to 119 TWh by 2050, as oil remains useful in Africa's power generation mix due to its availability and ease of deployment, especially during peak load hours. The use of biomass in electricity generation is expected to remain broadly stable at a low level of around 2 TWh.

8.5 Oil demand trends in Africa

Overall, African oil demand has remained relatively stable over the past decade. In 2024, total demand for oil was 4.6 mb/d. In 2025, total demand increased by 0.3 mb/d to reach 4.9 mb/d (Table 8.3).

Africa's refined products demand is mainly satisfied by imports from the rest of the world. This is due to a lack of refining capacities in Africa and the overall supply strategy between the continent's regions. Except in some rare cases, such as Algeria, almost all African oil producers are net product importers. Gasoil/diesel registered the highest

Table 8.3
Africa's oil demand by sector, 2000–2025, mb/d

	2000	2005	2010	2015	2020	2025	Growth 2000–2025
Road	1.0	1.3	1.7	2.2	2.3	2.6	1.5
Aviation	0.2	0.2	0.2	0.2	0.1	0.2	0.1
Rail/waterways	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Marine bunkers	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Transportation	**1.4**	**1.6**	**2.1**	**2.6**	**2.5**	**3.0**	**1.6**
Petrochemicals	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Other industry	0.4	0.5	0.6	0.6	0.9	0.7	0.3
Industry	**0.4**	**0.6**	**0.6**	**0.6**	**0.9**	**0.8**	**0.4**
Res./Comm./Agric.	0.4	0.4	0.5	0.5	0.5	0.6	0.3
Electr. generation	0.2	0.3	0.3	0.5	0.3	0.5	0.3
Other uses	**0.6**	**0.7**	**0.8**	**1.0**	**0.8**	**1.1**	**0.5**
Africa	**2.4**	**2.9**	**3.5**	**4.2**	**4.2**	**4.9**	**2.5**

Source: OPEC.

demand level at 1.9 mb/d in 2025, corresponding to a 40% share, with gasoline following at 1.3 mb/d.

Diesel and gasoline remain the most-used refined products, collectively making up around 65% of total demand in 2025. This is linked to consumption in the transportation sector. As shown in Table 8.4, this sector, including vehicles, trucks, buses and rail, represents more than half of the total oil demand.

Table 8.4
Africa's oil demand by product, 2000–2025, mb/d

	2000	2005	2010	2015	2020	2025	Growth 2000–2025
Ethane/LPG	0.2	0.3	0.3	0.4	0.5	0.6	0.3
Naphtha	0.0	0.0	0.1	0.0	0.0	0.1	0.0
Gasoline	0.6	0.7	0.9	1.1	1.2	1.3	0.7
Light products	**0.8**	**1.0**	**1.2**	**1.5**	**1.7**	**1.9**	**1.1**
Jet/kero	0.3	0.3	0.3	0.3	0.2	0.3	0.0
Gasoil/diesel	0.8	1.0	1.4	1.8	1.9	1.9	1.1
Middle distillates	**1.1**	**1.3**	**1.6**	**2.0**	**2.0**	**2.3**	**1.2**
Residual fuel	0.4	0.5	0.5	0.5	0.3	0.5	0.1
Other products	0.1	0.1	0.2	0.2	0.2	0.2	0.2
Heavy products	**0.5**	**0.6**	**0.6**	**0.6**	**0.6**	**0.7**	**0.2**
Africa	**2.4**	**2.9**	**3.5**	**4.2**	**4.2**	**4.9**	**2.5**

Source: OPEC.

8.5.1 Oil demand outlook for Africa

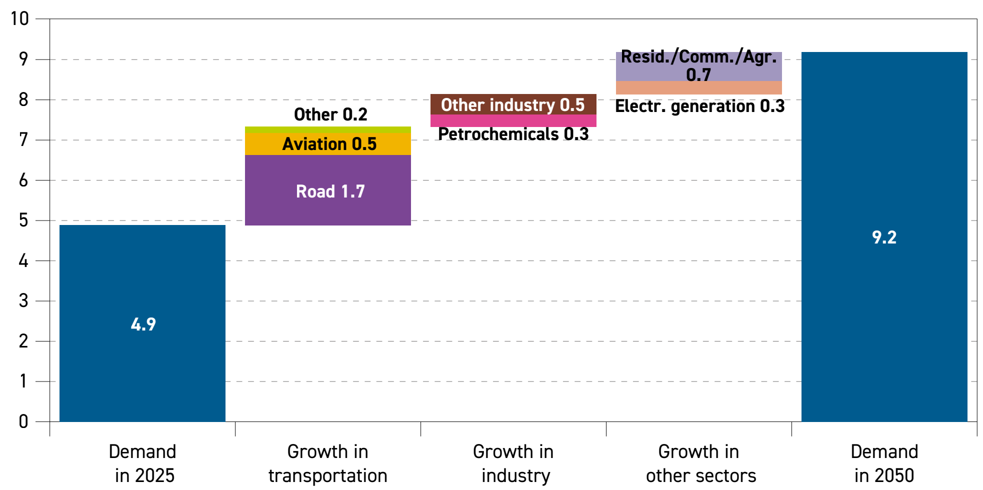
Africa's oil demand is expected to increase significantly over the coming decades, driven by the continent's dynamic demographic development, young population, growing workforce, abundant natural resources and strong potential for industrial expansion and urbanization. With these attributes, oil demand in the region is expected to register an increase of around 4.3 mb/d to reach 9.2 mb/d by 2050 (Table 8.5 and Figure 8.11). This demand will vary across countries, with some nations' consumption higher than others due to differences in base levels and expected future economic expansion.

Table 8.5
Africa's oil demand by sector, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
Road	2.6	2.9	3.3	3.6	4.0	4.3	1.7
Aviation	0.2	0.3	0.4	0.5	0.7	0.8	0.5
Rail/waterways	0.1	0.1	0.1	0.1	0.1	0.0	0.0
Marine bunkers	0.1	0.1	0.2	0.2	0.2	0.3	0.2
Transportation	**3.0**	**3.4**	**3.9**	**4.4**	**4.9**	**5.4**	**2.4**
Petrochemicals	0.1	0.1	0.2	0.2	0.3	0.4	0.3
Other industry	0.7	0.9	1.0	1.1	1.2	1.2	0.5
Industry	**0.8**	**1.0**	**1.2**	**1.3**	**1.5**	**1.6**	**0.8**
Res./Comm./Agric.	0.6	0.8	0.9	1.0	1.2	1.4	0.7
Electr. generation	0.5	0.5	0.6	0.7	0.7	0.8	0.3
Other uses	**1.1**	**1.3**	**1.5**	**1.7**	**1.9**	**2.2**	**1.0**
Africa	**4.9**	**5.7**	**6.5**	**7.4**	**8.3**	**9.2**	**4.3**

Source: OPEC.

Figure 8.11
Africa's oil demand growth by sector, 2025–2050, mb/d



Source: OPEC.

From a sectoral perspective, the demand for oil in Africa shows that the sector with the highest demand level, as well as the largest incremental demand over the forecast period, is transportation. This sector is expected to account for nearly 60% of the total demand increase, adding about 2.4 mb/d between 2025 and 2050. This substantial growth is mainly in the road transportation sector due to the expectation that vehicle fleets will more than double in size by 2050 and as living standards improve on the back of expected economic growth and the vehicle fleet continues to be dominated by ICE vehicles. The expansion of EVs in Africa is expected to occur at a slow pace.

In the category of 'other uses', incremental oil demand in Africa is set to be split between the residential/commercial/agriculture sector (0.7 mb/d, mainly LPG and domestic kerosene) and electricity generation (0.3 mb/d). In contrast to other regions, where oil use for electricity generation is typically declining, it is set to continue to expand in Africa. This is due to the need for decentralized power generation in many places, including those still lacking access to electricity.

Another interesting observation for Africa relates to the limited use of oil in the petrochemical industry, currently at around 0.1 mb/d in 2025. This is anticipated to increase substantially to 0.4 mb/d in 2050, which is still nevertheless much lower than in other high-growth regions.

The industrial sector (excluding petrochemicals) is also expected to see a substantial demand increase as energy becomes more affordable and accessible in Africa. This also helps create a more industrialized continent with an integrated market that processes raw materials locally. As a result, oil demand in this sector is set to rise to about 1.2 mb/d by 2050, adding around 0.5 mb/d of incremental demand over the forecast period.

Table 8.6 provides an overview of Africa's oil demand outlook from the perspective of refined products. The largest demand increase, of about 1.5 mb/d between 2025 and 2050, is projected

Table 8.6
Africa's oil demand by product, 2025–2050, mb/d

	2025	2030	2035	2040	2045	2050	Growth 2025–2050
Ethane/LPG	0.6	0.7	0.8	0.9	1.0	1.1	0.6
Naphtha	0.1	0.1	0.2	0.2	0.3	0.4	0.3
Gasoline	1.3	1.5	1.6	1.8	1.9	2.1	0.8
Light products	**1.9**	**2.2**	**2.5**	**2.9**	**3.2**	**3.6**	**1.7**
Jet/kero	0.3	0.4	0.5	0.6	0.8	0.9	0.6
Gasoil/diesel	1.9	2.2	2.6	2.9	3.1	3.4	1.5
Middle distillates	**2.3**	**2.6**	**3.1**	**3.5**	**3.9**	**4.3**	**2.1**
Residual fuel	0.5	0.5	0.6	0.7	0.7	0.8	0.4
Other products	0.2	0.3	0.3	0.4	0.4	0.4	0.2
Heavy products	**0.7**	**0.8**	**0.9**	**1.0**	**1.1**	**1.2**	**0.5**
Africa	**4.9**	**5.7**	**6.5**	**7.4**	**8.3**	**9.2**	**4.3**

Source: OPEC.

for gasoil/diesel, as this product is supported by a strong increase in the commercial vehicle and passenger car fleet, as well as expanding industrial and agricultural activity in the region. Jet/kero demand is forecast to rise by around 0.6 mb/d, bringing demand for middle distillates to around 4.3 mb/d by 2050.

Moreover, expansion in the transportation sector, as well as growth in the residential and industrial sectors, is set to support demand for gasoline, ethane/LPG and naphtha. This leads to demand for light products rising to about 3.6 mb/d by 2050, with demand for gasoline showing the largest incremental increase of around 0.8 mb/d. Somewhat lower, but still significant, demand increases are projected for fuel oil and other products. These heavy products are expected to add around 0.5 mb/d to oil demand from 2025 to 2050.

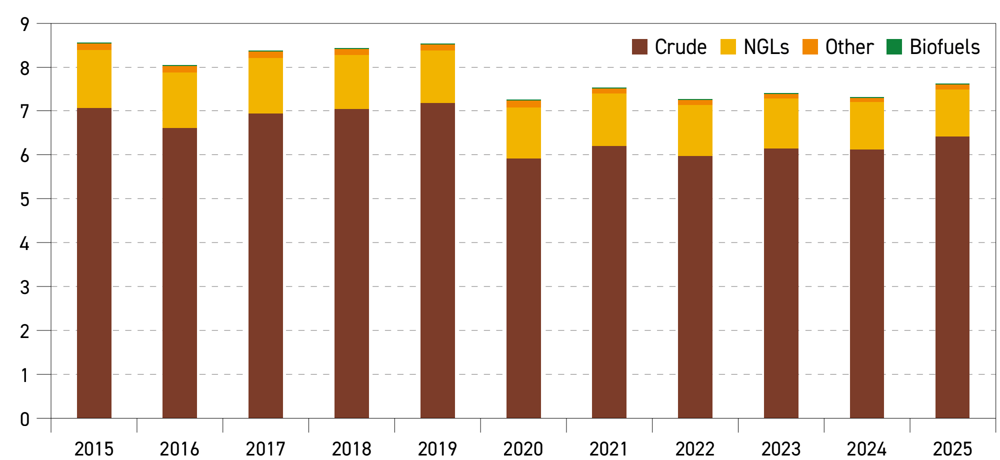
8.6 Liquids supply outlook

Africa is a major source of liquids supply. It holds substantial oil reserves and contributes significantly to global production thanks to established oil companies and emerging players. Historically, Africa, including producers that are OPEC Members and part of the DoC, has accounted for roughly 8% of the global liquids supply in the past decade. This corresponds to an average of 7.8 mb/d (2015–2025) with the following breakdown: 83% crude oil, 15% NGLs and 2% other liquids.

Looking ahead, Africa's role in world oil supply is expected to continue to expand as the continent's focus is on expanding E&P activities in the medium to long term to increase hydrocarbon output that will help meet increasing regional and global demand.

Considering this expansion, liquids supply in Africa is set to increase steadily over the coming decades, with non-DoC African countries contributing an additional 0.7 mb/d to global supply growth by 2050. This growth is set to be primarily driven by new discoveries and ongoing developments in countries like Uganda, Côte d'Ivoire, Senegal and Namibia.

Figure 8.12
Liquids production in Africa, 2015–2025, mb/d



Source: OPEC.

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As shown in Table 8.7, non-DoC Africa's projected liquids supply is set to rise from 2.3 mb/d in 2025 to reach a level of around 2.8 mb/d by 2030 and then rise more modestly thereafter. Over 0.1 mb/d of this growth is set to come from Angola, Africa's largest non-DoC producer, which is forecast to see total supply rise to 1.3 mb/d by the mid-2030s. Incremental barrels will come from ramp-ups at the Agogo, Begonia and CLOV clusters, while the major 80 tb/d Kaminho block operated by TotalEnergies is set to come onstream in 2028.

Table 8.7
Non-DoC Africa liquids production outlook, 2025-2050, mb/d

	2025	2030	2035	2040	2045	2050	Change 2025–2050
Angola	1.1	1.2	1.3	1.3	1.3	1.2	0.1
Chad	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Egypt	0.5	0.5	0.5	0.5	0.4	0.4	–0.1
Ghana	0.1	0.1	0.1	0.1	0.1	0.1	0.0
South Africa	0.1	0.1	0.1	0.1	0.1	0.1	0.0
Africa other*	0.3	0.8	1.0	1.0	1.1	1.1	0.8
Total non-DoC Africa	**2.3**	**2.8**	**3.0**	**3.1**	**3.1**	**3.0**	**0.7**

* *Africa other includes Cameroon, Côte d'Ivoire, Namibia, Niger, Senegal, Uganda and smaller producers, and excludes African DoC producers.*

Source: OPEC.

The largest increment, however, is forecast to be driven by a host of new start-ups, including from emerging producers. Uganda is set to lead this development, with first barrels expected from the major 230 tb/d Lake Albert projects in 2026. TotalEnergies is expected to bring the 190 tb/d Tilenga field online later this year, to be followed by another 40 tb/d tranche operated by CNOOC, likely in 2027. The projects, and their associated ramp-up, are being coordinated with the commissioning of the major East African Crude Oil Pipeline, which will transport exports to the coast. It is an almost 1,500 km journey to the port of Tanga in Tanzania.

In Western Africa, established producer Côte d'Ivoire has benefited from ramp-ups at the large Baleine field. In 2028, phase 3, with another 80 tb/d of capacity, is due to come onstream. Namibia, an up-and-coming producer, is also set to emerge as a key player in the next decade, if recent discoveries in the Orange Basin, such as the Venus oil field, are developed. Venus has not yet received a final investment decision, but asset swaps in late December 2025 between potential operator TotalEnergies and Galp, which holds rights to another large discovery, the Mopane field, arguably make development more likely due to risk-sharing. Mopane is likely to bring on around 120 tb/d of capacity, Venus 200 tb/d or more, both from the early 2030s. At the same time, additional finds and strong resource potential suggest that the development of key anchor fields could trigger a stream of projects being realized.

In addition, several other countries on the continent are emerging as new exploration frontiers. In Western Africa, buoyed by Eni's Calao discovery in Côte d'Ivoire, which further strengthened the country's deepwater potential, first wells have been drilled in São Tomé and Príncipe, in areas close to recent discoveries. Additionally, TotalEnergies has committed to exploration offshore Liberia, Cameroon has held a new offshore licensing round, while Ghana is repositioning itself as an attractive investment proposition.

Elsewhere, other countries are enhancing their existing production capabilities. For instance, in Southern Africa, the liquids production landscape is undergoing significant transformation, while Eastern Africa is emerging as a significant player in the oil and gas sector, supported by new discoveries, infrastructure development and regional collaboration.

More generally, across emerging producer nations, policy environments are gradually improving, with governments introducing fiscal incentives, clearer regulatory frameworks and local content strategies to attract foreign direct investment. Improved regional cooperation is also signalling growing alignment between national oil policies and infrastructure development.

8.7 Refining outlook for Africa

This section focuses on Africa's downstream sector. It discusses recent developments and examines various market drivers and factors that may impact the future of Africa's refining sector, including challenges, uncertainties and opportunities. Similar to Chapter 5, the analysis is conducted in two different time frames, namely the medium term (2026–2030) and long term (2026–2050). It is important to note that this section is fully aligned with the assumptions of the Outlook.

For decades, Africa's refining sector has been operating below its full potential due to various factors, including underinvestment, poor utilization and maintenance, resulting in infrastructural bottlenecks, backlogs and feedstock supply issues. The continent, however, is advancing rapidly and entering a pivotal phase in the sector's development, driven by recent capacity expansions and growing investment in the sector.

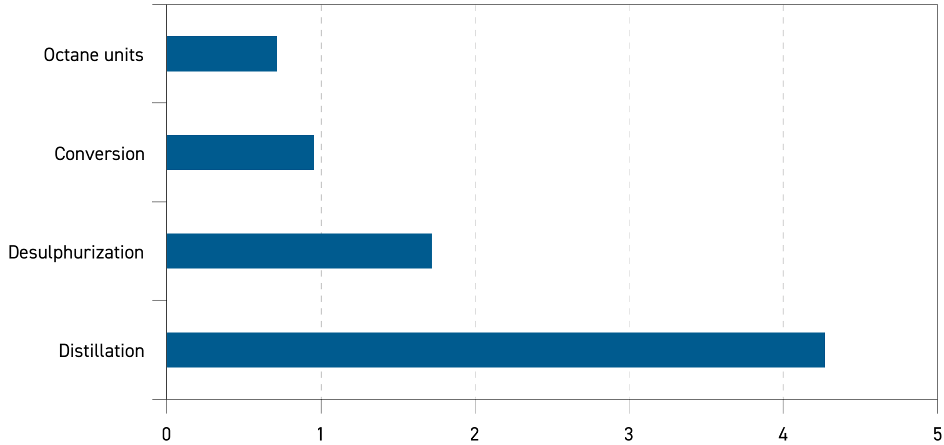
The continent's most prominent greenfield project over the past decade is the Dangote refinery in Nigeria, with a capacity of 650 tb/d, making it the largest single-train refinery in the world. The unit began processing crude into diesel, naphtha and jet fuel in January 2024, followed by gasoline production in September of the same year. The refinery reached its full design capacity in February 2026. It is the first refinery globally to achieve full nameplate capacity in a single-train configuration of this magnitude.

Figure 8.13 provides base refining capacity by major refining processes on the continent. The total distillation capacity is assessed at 4.3 mb/d as of January 2026. This represents an increase of 8.2% compared to January 2025, driven entirely by the commissioning of the second phase of the Dangote refinery. Installed refining capacity remains highly concentrated, as three countries, namely, Nigeria, Egypt and Algeria, account for approximately 77% of Africa's refining capacity, while the remaining 23% is distributed across other countries, such as South Africa, Libya, Ghana, Côte d'Ivoire and others.

It is important to note that refining capacity in Africa is the lowest among all global regions. This is a reflection of the historical structure of many countries, which has been based on the export of crude oil and the import of refined products, rather than on the local processing of crude oil.

In terms of secondary capacity, desulphurization, conversion and octane units account for 1.7 mb/d, 1 mb/d and 0.7 mb/d, respectively. Overall, secondary capacity had increased by

Figure 8.13
Africa's base refining capacity as of January 2026, mb/d



Source: OPEC.

13.5% at the start of 2026, compared to January 2025, again due to the commissioning of the Dangote refinery. It is also important to highlight that Africa's refining system is the least complex among all regions, as many African refineries have few advanced units such as catalytic cracking, hydrocracking or coking. This is primarily due to refineries being designed to process predominantly light crude oil, less stringent product specifications and historical underinvestment.

Looking ahead, many African countries have ambitious plans to develop their refining industry capabilities, in line with the region's expanding oil demand driven by population growth, urbanization and robust economic growth. Efforts by some countries to reduce product imports and enhance national energy security are also supporting this momentum.

An assessment of an extensive list of announced projects across the continent results in capacity additions of more than 0.8 mb/d likely to be commissioned over the medium term. This could lift the continent's overall capacity to 5.1 mb/d by 2030, an increase of 20% compared to the 2025 level. These additions relate to projects that are under construction or approaching the construction phase and have a high or relatively high probability of being completed during this period. In addition, some projects that are not yet under construction but are sufficiently advanced in terms of planning, financing and engineering, are also considered. Figure 8.14 presents the expected medium-term capacity additions per year.

A major part of the medium-term expansion is set to come from Angola and Niger. The former is expected to add 90 tb/d at the Cabinda refinery, 200 tb/d at Lobito and additional capacity at a refinery in Soyo later in the medium-term period. As the Dangote refinery is now fully commissioned, additional capacity in Nigeria is anticipated to come from the 200 tb/d Akwa Ibom refinery, as well as from other smaller-scale projects.

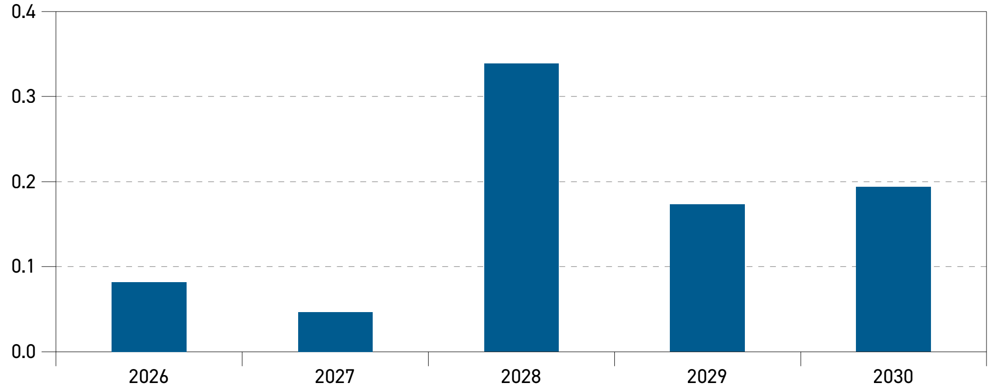
Several smaller modular refineries are also in the pipeline across multiple countries, including the Republic of the Congo, Equatorial Guinea, Ghana and Uganda. In Northern Africa, new refining projects are underway in Algeria (Hassi Messaoud), Libya (Ubari) and

Egypt (Sokhna). Overall, capacity additions in Africa are set to be driven mostly by modular and small-scale capacity projects.

Taking these developments into consideration, the analysis of the cumulative required capacity compared to the potential cumulative refining capacity in Africa shows a relatively balanced downstream market throughout the medium-term period, relative to 2025 (see Figure 8.14). It should be noted, however, that the continent's overall refining capacity is expected to remain far below oil demand, as the region's market is already under pressure. This situation will likely be exacerbated by the low utilization rates of existing units. This means that any delay in operationally commissioning the planned capacity additions could tighten the market and lead to higher product imports. Figure 8.15 shows refining capacity compared to oil demand between 2025 and 2030.

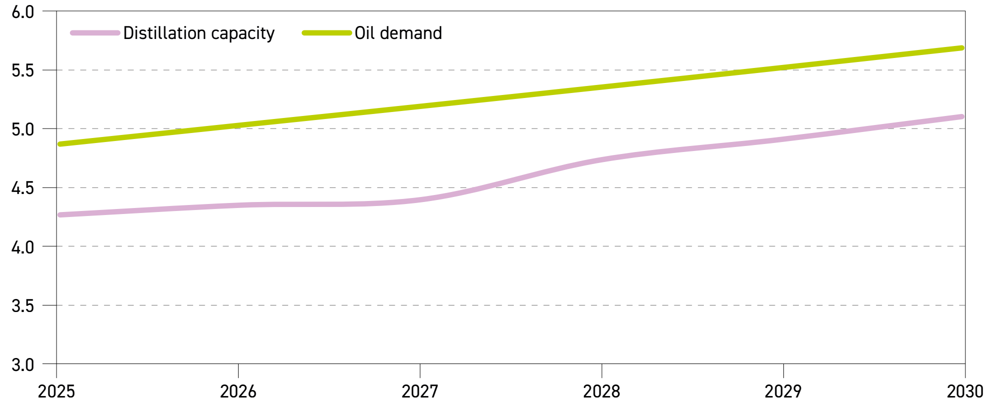
In the long term, and based on oil demand trends, liquid supply, trade patterns, as well as assumptions on medium-term additions, refining capacity requirements in Africa are assessed at almost 3.2 mb/d between 2026 and 2050.

Figure 8.14
Africa's distillation capacity additions, 2026–2030, mb/d



Source: OPEC.

Figure 8.15
Africa's oil demand and refining capacity, 2025–2030, mb/d



Source: OPEC.

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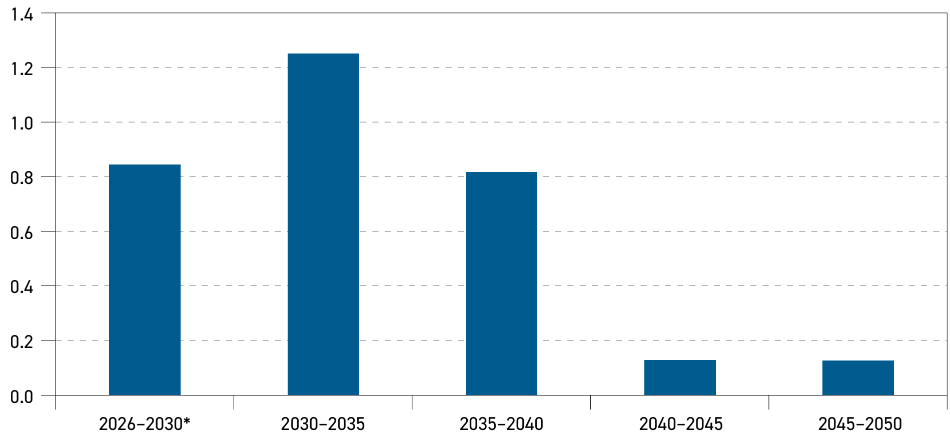
Figure 8.16 depicts Africa's crude capacity distillations in the long term. It is clear that the majority of capacity expansions are set to materialize over the period between 2026 and 2040, driven by strong oil demand growth and efforts to reduce refined product imports. In contrast, only minor additions are expected over the last decade of the forecast period and are likely limited to some capacity expansions. This is mainly due to rising competition given expected exports from the US & Canada and Europe.

Figure 8.16
Africa's distillation capacity additions per period, 2026–2050, mb/d



* *Firm projects including additions resulting from capacity creep.*
Source: OPEC.

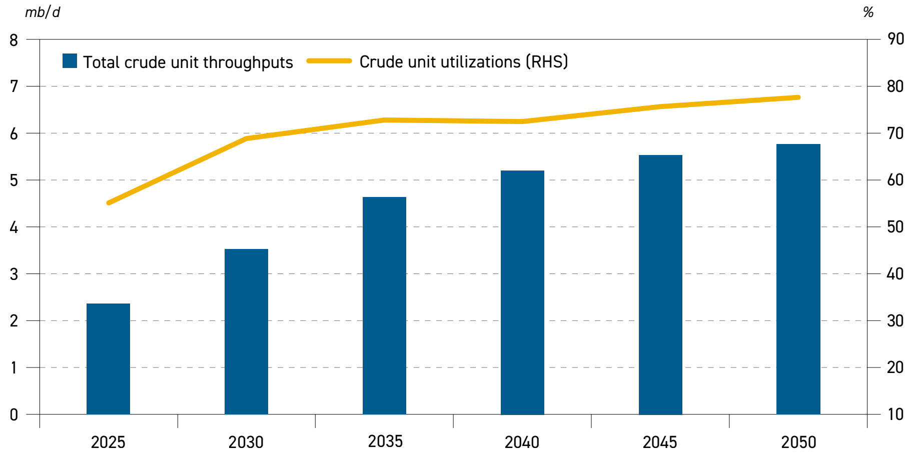
Figure 8.17 shows long-term crude unit throughputs and utilization rates in Africa. The trend of increasing crude unit throughputs is especially pronounced in the medium term, increasing from 2.4 mb/d in 2025 to 3.5 mb/d in 2030. It is also clear over the long term that throughputs are projected to reach 5.8 mb/d by 2050. Crude unit utilization is also expected to rise strongly, with utilization rates jumping from 55% in 2025 to 69% in 2030. Thereafter, the utilization rates increase further to reach 78% by 2050.

This suggests that refinery utilization rates in Africa are expected to improve significantly over the long term, increasing by about 23 percentage points between 2025 and 2050. Nevertheless, they are projected to remain among the lowest globally, largely because some older refineries are likely to continue operating at very low utilization rates.

From the secondary capacity perspective, this outlook indicates a gradual expansion over the outlook period, supported by the commissioning of new complex refineries and the upgrading of existing facilities, particularly in terms of desulphurization and conversion units. By 2030, identified projects could add around 0.3 mb/d of desulphurization capacity, 0.3 mb/d of conversion capacity and over 0.2 mb/d of octane units. These additions reflect the need to improve fuel quality and more efficiently convert heavy fractions of crude oil into lighter products, such as gasoline and diesel.

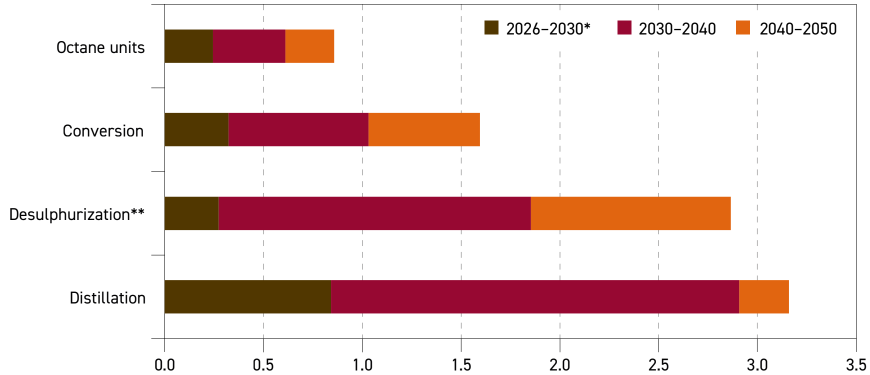
This expansion is also part of a broader trend towards increasing the complexity of African refineries. Historically, many of the continent's refineries were designed with relatively simple configurations, dominated by primary distillation, sometimes with few or no deep

Figure 8.17
Africa's crude unit throughputs and utilization rates, 2025–2050



Source: OPEC.

Figure 8.18
Africa's refining capacity requirements by process type, 2026–2050, mb/d



* *Firm projects including additions resulting from capacity creep.*
** *Projects and additions exclude naphtha desulphurization.*
Source: OPEC.

conversion units. New projects and upgrades are expected to gradually correct this structure by integrating more technologies, such as catalytic cracking, hydrocracking and desulphurization units, thereby increasing clean fuel yields and reducing the production of heavy residues.

Long-term prospects also indicate that the expansion of secondary capacity in Africa will be closely linked to oil demand growth and trends, increasingly stringent product specifications and the objective of reducing dependence on refined product imports. Secondary capacity additions are set to increase significantly in the period between 2030 and 2040, in line with rising distillation capacity additions.

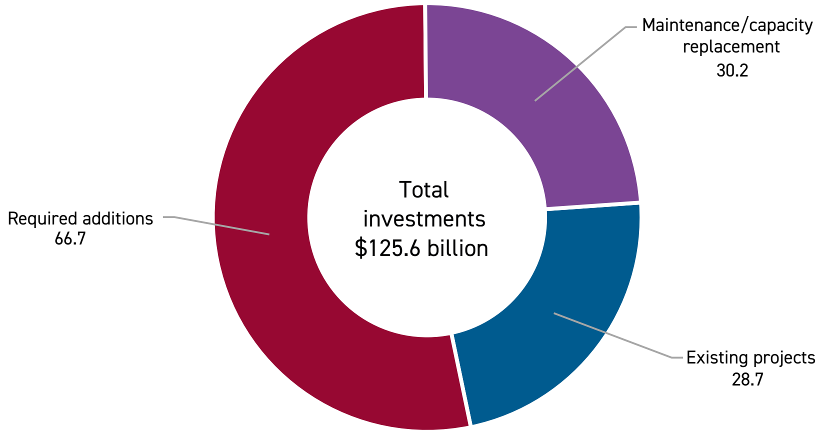
Desulphurization capacity is expected to expand the most, adding 1.6 mb/d over this period. This reflects the need for refiners to produce ULS fuels, as many countries have already set goals to lower the sulphur content threshold in gasoline and diesel to 10 parts per million (ppm). Conversion capacity additions are estimated at 0.7 mb/d between 2030 and 2040, driven by the need to meet growing refined products demand and to improve fuel quality in response to market requirements and government regulations. Finally, Africa is set to see its octane unit capacity growing by around 0.4 mb/d in the same period, on the back of rising demand for gasoline and increasing octane number specifications in some countries.

In the last decade of the outlook period, secondary capacity additions are anticipated to continue increasing, albeit at a slower pace. Desulphurization, conversion and octane upgrader unit capacities are set to expand by around 1 mb/d, 0.6 mb/d and 0.2 mb/d, respectively.

Investments in large refineries and modular projects should enable the continent to process more crude oil locally and strengthen its role in regional petroleum product flows. Several major projects in Western and Central Africa exemplify this strategy of developing more integrated petroleum value chains.

In addition, for Africa to realize its goal of increasing value from hydrocarbon resources and reduce its reliance on the importation of refined products, it will require significant infrastructure development, coordinated financing, policy support, expanded capital markets, improved private-public partnerships and large-scale development of modular refineries to fast-track deliveries.

Figure 8.19
Refinery investments in Africa by category, 2026–2050, $(2026) billion



Source: OPEC.

Total investment requirements in the refining sector in Africa are estimated at $125.6 billion, out of which $28.7 billion is devoted to projects to 2030 (Figure 8.19). Over $66.7 billion is required between 2030 and 2050 for new capacity additions to the refining sector and $30.2 billion is needed for maintenance requirements and the necessary continuous replacement of installed refining capacity.

Annex A
Abbreviations

AFV	Alternative fuel vehicle
AI	Artificial intelligence
APG	ASEAN Power Grid
API	American Petroleum Institute
ASEAN	Association of Southeast Asian Nations
bcm	Billion cubic metres
BESS	Battery energy storage system
BEVs	Battery electric vehicles
BPCL	Bharat Petroleum Corporation Limited
CAGR	Compound annual growth rate
CBAM	Carbon Border Adjustment Mechanism
CBDR-RC	Common but differentiated responsibilities and respective capabilities
CCE	Circular Carbon Economy
CCUS	Carbon capture, utilization and storage
CDM	Clean Development Mechanism
CfD	Contracts for Difference
CHPS	Clean Hydrogen Portfolio Standard
CO_2	Carbon dioxide
CORSIA	Carbon Offsetting and Reduction Scheme for International Aviation
CTLs	Coal-to-liquids
DoC	Declaration of Cooperation
DLE	Dry low emissions
EEZ	Exclusive economic zone
EGS	Equitable Growth Scenario
EOR	Enhanced oil recovery
EPA	Environmental Protection Agency
EPL	Energy Profits Levy
ETBE	Ethyl tertiary-butyl ether
ETS	Emissions Trading System
EU ETS	European Union's Emissions Trading System
EVs	Electric vehicles
FCC	Fluid catalytic cracking
FCEV	Fuel cell electric vehicle
FIP	Feed-in Premiums
FIT	Feed-in Tariffs
FLNG	Floating liquefied natural gas
FPSO	Floating Production, Storage and Offloading
FYP	Five-Year Plan
GCF	Green Climate Fund
GDP	Gross Domestic Product
GEA-4	Green Energy Auction
GGA	Global Goal on Adaptation

GHG	Greenhouse gas
GoM	Gulf of Mexico
GST	Global Stocktake
GTLs	Gas-to-liquids
GW	Gigawatt
HPCL	Hindustan Petroleum Corporation Limited
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
ICE	Internal combustion engine
IDA	International Development Association
IMO	International Maritime Organization
INC	Intergovernmental Negotiating Committee
IOC	Indian Oil Corporation
IOCs	International oil companies
IoT	Internet of Things
IPCC	Intergovernmental Panel on Climate Change
kgoe/t	Kilograms of oil equivalent per tonne
LDES	Long-duration electricity storage
LNG	Liquefied Natural Gas
LPG	Liquefied Petroleum Gas
LSS6	Large Scale Solar 6
LTAG	Long-Term Aspirational Goal
LTMS PIP	Lao PDR-Thailand-Malaysia-Singapore Power Integration Project
mb/d	Million barrels per day
mboe/d	Million barrels of oil equivalent per day
MEPC	Marine Environment Protection Committee
MTBE	Methyl tertiary butyl ether
mtpa	Million tonnes per annum
MW	Megawatt
MWe	Megawatt electric
NAPs	National Adaptation Plans
NDCs	Nationally Determined Contributions
NDRC	National Development and Reform Commission
NETR	National Energy Transition Roadmap
NEVs	New energy vehicles
NGLs	Natural Gas Liquids
NGVs	Natural Gas Vehicles
NO_x	Nitrogen oxides
OBBBA	One Big Beautiful Bill Act
p.a.	Per annum

pp	Percentage points
PDH	Propane Dehydrogenation
PDP8	Power Development Plan 8
PLF	Passenger load factor
ppm	Parts per million
PV	Photovoltaic
RED	Renewable Energy Directive
RFNBOs	Renewable Fuels of Non-Biological Origin
RPK	Revenue Passenger Kilometres
SAF	Sustainable Aviation Fuel
SDG	Sustainable Development Goal
SMRs	Small modular reactors
Solar ATAP	Solar Accelerated Transition Action Programme
tb/d	Thousand barrels per day
TDS	Technology-Driven Scenario
TFC	Total final consumption
TSGP	Trans-Saharan Gas Pipeline
TWh	Terawatt hour
ULS	Ultra-low sulphur
UNEA	United Nations Environment Assembly
UNDESA	United Nations Department of Economic and Social Affairs
UNFCCC	United Nations Framework Convention on Climate Change
VLCCs	Very large crude carriers
VLSFO	Very low sulphur fuel oil
VMOS	Vaca Muerta Oil Sur
WOO	World Oil Outlook (OPEC)
WTO	World Trade Organization

Annex B
Regional definitions for energy and oil demand

OECD

OECD Americas
Canada
Chile
Mexico
United States of America
(including US dependencies)

OECD Europe
Austria
Belgium
Czechia
Denmark
Estonia
Finland
France
Germany
Greece
Hungary
Iceland
Ireland
Italy
Latvia
Lithuania
Luxembourg
Netherlands
Norway
Poland
Portugal
Slovakia
Slovenia
Spain
Sweden
Switzerland
Türkiye
United Kingdom

OECD Asia-Pacific
Australia
Japan
Korea (Republic of)
New Zealand

NON-OECD COUNTRIES

Latin America
Anguilla
Antigua and Barbuda
Argentina
Aruba
Bahamas
Barbados
Belize
Bermuda
Bolivia (Plurinational State of)
Brazil
British Virgin Islands
Cayman Islands
Colombia
Costa Rica
Cuba
Dominica
Dominican Republic
Ecuador
El Salvador
French Guiana
Grenada
Guadeloupe
Guatemala
Guyana
Haiti
Honduras
Jamaica
Martinique
Montserrat
Netherlands Antilles
Nicaragua
Panama
Paraguay
Peru
St. Kitts and Nevis
St. Lucia
St. Pierre et Miquelon
St. Vincent and the Grenadines
Suriname
Trinidad and Tobago
Turks and Caicos Islands
Uruguay
Venezuela

Middle East
Bahrain
IR Iran
Iraq
Jordan
Kuwait
Lebanon
Oman
Qatar
Saudi Arabia
Syrian Arab Republic
United Arab Emirates
Yemen

Africa
Algeria
Angola
Benin
Botswana
Burkina Faso
Burundi
Cameroon (United Republic of)
Cape Verde
Central African Republic
Chad
Comoros
Congo
Congo (Democratic Republic of the)
Côte d'Ivoire
Djibouti
Egypt
Equatorial Guinea
Eritrea
Eswatini
Ethiopia
Gabon
Gambia (The)
Ghana
Guinea
Guinea-Bissau
Kenya
Lesotho
Liberia
Libya
Madagascar
Malawi
Mali
Mauritania
Mauritius
Morocco
Mozambique
Namibia
Niger
Nigeria
Réunion
Rwanda
São Tomé and Príncipe
Senegal
Seychelles
Sierra Leone
Somalia
South Africa
South Sudan
Sudan
Tanzania
Togo
Tunisia
Uganda
Western Sahara
Zambia
Zimbabwe

India
India

China
People's Republic of China

Other Asia
Afghanistan
American Samoa
Bangladesh
Bhutan
Brunei Darussalam
Cambodia
China, Hong Kong SAR
Cook Islands
Fiji
French Polynesia
Indonesia
Kiribati
Korea (Democratic People's Republic of)
Lao People's Democratic Republic
Malaysia
Maldives
Micronesia (Federated States of)
Mongolia
Myanmar
Nauru
Nepal
New Caledonia
Niue
Pakistan
Papua New Guinea
Philippines
Samoa
Singapore
Solomon Islands
Sri Lanka
Thailand

Timor-Leste
Tonga
Tuvalu
Vanuatu
Vietnam
Other other Asia

Russia
Russian Federation

Other Eurasia
Armenia
Azerbaijan
Belarus
Georgia
Kazakhstan
Kyrgyzstan
Moldova (Republic of)
Tajikistan
Turkmenistan
Ukraine
Uzbekistan

Other Europe
Albania
Bosnia and Herzegovina
Bulgaria
Croatia
Cyprus
Gibraltar
Malta
Montenegro
Romania
Serbia
Republic of North Macedonia

Note: For Chapter 4 'Liquids supply' the DoC regional grouping is shown, which includes the following countries: Algeria, Azerbaijan, Bahrain, Brunei, Congo, Equatorial Guinea, Gabon, IR Iran, Iraq, Kazakhstan, Kuwait, Libya, Malaysia, Mexico, Nigeria, Oman, Russia, Saudi Arabia, Sudan, South Sudan, the United Arab Emirates and Venezuela.

Annex C
Regional definitions for oil refining and trade

US & Canada
Canada
United States of America
(including US dependencies)

Latin America
Anguilla
Antigua and Barbuda
Argentina
Aruba
Bahamas
Barbados
Belize
Bermuda
Bolivia (Plurinational State of)
Brazil
British Virgin Islands
Cayman Islands
Chile
Colombia
Costa Rica
Cuba
Dominica
Dominican Republic
Ecuador
El Salvador
French Guiana
Grenada
Guadeloupe
Guatemala
Guyana
Haiti
Honduras
Jamaica
Martinique
Mexico
Montserrat
Netherlands Antilles
Nicaragua
Panama
Paraguay
Peru
St. Kitts and Nevis
St. Lucia
St. Pierre et Miquelon
St. Vincent and The Grenadines
Suriname
Trinidad and Tobago
Turks And Caicos Islands
United States Virgin Islands
Uruguay
Venezuela

Africa
Algeria
Angola
Benin
Botswana
Burkina Faso
Burundi
Cameroon (United Republic of)
Cape Verde
Central African Republic
Chad
Comoros
Congo
Congo (Democratic Republic of)
Côte d'Ivoire
Djibouti
Egypt
Equatorial Guinea
Eritrea
Eswatini
Ethiopia
Gabon
Gambia (The)
Ghana
Guinea
Guinea-Bissau
Kenya
Lesotho
Liberia
Libya
Madagascar
Malawi
Mali
Mauritania
Mauritius
Morocco
Mozambique
Namibia
Niger
Nigeria
Réunion
Rwanda
São Tomé and Príncipe
Senegal
Seychelles
Sierra Leone
Somalia
South Africa
South Sudan
Sudan
Tanzania
Togo
Tunisia
Uganda
Western Sahara
Zambia
Zimbabwe

Europe
Albania
Austria
Belgium
Bosnia and Herzegovina
Bulgaria
Croatia
Cyprus
Czechia
Denmark
Estonia
Finland
France
Germany
Gibraltar
Greece
Hungary
Iceland
Ireland
Italy
Latvia
Lithuania
Luxembourg
Malta
Montenegro
Netherlands
Norway
Poland
Portugal
Republic of North Macedonia
Romania
Serbia
Slovakia
Slovenia
Spain
Sweden

Russia & Other Eurasia
Armenia
Azerbaijan
Belarus
Georgia
Kazakhstan
Kyrgyzstan
Moldova (Republic of)
Russian Federation
Tajikistan
Turkmenistan
Ukraine
Uzbekistan

Middle East
Bahrain
IR Iran
Iraq
Jordan
Kuwait
Lebanon
Oman
Qatar
Saudi Arabia
Syrian Arab Republic
United Arab Emirates
Yemen

China
People's Republic of China

Other Asia-Pacific
Afghanistan
American Samoa
Australia
Bangladesh
Bhutan
Brunei Darussalam
Cambodia
Cook Islands

Fiji
French Polynesia
Indonesia
Japan
Kiribati
Korea (Democratic People's Republic of)
Korea (Republic of)
Lao People's Democratic Republic
Malaysia
Maldives
Micronesia (Federated States of)
Mongolia
Myanmar
Nauru
Nepal
New Caledonia
New Zealand
Niue
Pakistan
Papua New Guinea
Philippines
Samoa
Singapore
Solomon Islands
Sri Lanka
Thailand
Timor-Leste
Tonga
Tuvalu
Vanuatu
Vietnam
Other other Asia

Annex D
Regional definitions for Africa

Central Africa
Angola
Cameroon
Central African Republic
Chad
Congo
Congo, Dem Rep
Equatorial Guinea
Gabon
Sao Tome / Principe

Eastern Africa
Burundi
Comoros
Djibouti
Eritrea
Ethiopia
Kenya
Madagascar
Malawi
Mauritius
Mozambique
Rwanda
Seychelles
Somalia
South Sudan
Uganda
Tanzania
Zambia
Zimbabwe

Northern Africa
Algeria
Egypt
Libya
Morocco
Sudan
Tunisia

Southern Africa
Botswana
Eswatini
Lesotho
Namibia
South Africa

Western Africa
Benin
Burkina Faso
Cape Verde
Ivory Coast
Gambia, The
Ghana
Guinea
Guinea-Bissau
Liberia
Mali
Mauritania
Niger
Nigeria
Senegal
Sierra Leone
Togo

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